GRAND CANYON NATIONAL PARK
WORLD HERITAGE SITE
REHABILITATE NORTH RIM INFRASTRUCTURE
VOLUME 6 – NORTH RIM WATER TREATMENT

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 3	PIPELINES - NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES - ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 6	NORTH RIM WATER TREATMENT
VOLUME 7	NOT USED
VOLUME 8	NORTH RIM WASTEWATER COLLECTION SYSTEM
VOLUME 9	POWER DISTRIBUTION - NORTH RIM
VOLUME 10	POWER DISTRIBUTION - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	060G001	GENERAL	SHEET INDEX
2	060G002	GENERAL	SHEET INDEX CONTINUED
3	060C101	GENERAL	OVERALL SITE PLAN
4	060L101	GENERAL	LANDSCAPE PLAN
5	060S001	GENERAL	NOTES
6	060S002	GENERAL	STANDARD DETAILS
7	060S003	GENERAL	TYPICAL CMU DETAILS
8	060D601	GENERAL	PROCESS DESIGN CRITERIA
9	060D602	GENERAL	TREATMENT PROCESS FLOW DIAGRAM
10	060D603	GENERAL	WASTE PROCESS FLOW DIAGRAM
11	060D604	GENERAL	HYDRAULIC PROFILE
12	060D605	GENERAL	PIPE SCHEDULE
13	060E601	GENERAL	POWER DISTRIBUTION FUNCTIONAL DIAGRAM
14	060Y601	GENERAL	SCADA NETWORK BLOCK DIAGRAM
15	061X101	SITEWORK	DEMOLITION PLAN
16	061X102	SITEWORK	TREE REMOVAL AND PROTECTION PLAN
17	061X103	SITEWORK	TREE REMOVAL TABLES
18	061C101	SITEWORK	NORTH RIM WTP GESC PLAN
19	061C102	SITEWORK	GRADING AND PAVING PLAN - WTP
20	061C103	SITEWORK	GRADING AND PAVING PLAN - WTP
21	061C104	SITEWORK	GRADING AND PAVING POINT TABLES
22	061C105	SITEWORK	OVERALL YARD PIPING PLAN
23	061C301	SITEWORK	GRADING AND PAVING SECTIONS WTP
24	061C401	SITEWORK	YARD PIPING ENLARGED PLAN 1
25	061C402	SITEWORK	YARD PIPING ENLARGED PLAN 2 - WTP
26	061C403	SITEWORK	YARD PIPING ENLARGED PLAN 3 - FINISHED WATER TANKS
27	061C404	SITEWORK	YARD PIPING ENLARGED PLAN 4 - FINISHED WATER TANKS
28	061C501	SITEWORK	CIVIL DETAILS
29	061C502	SITEWORK	CIVIL DETAILS
30	061C503	SITEWORK	CIVIL DETAILS
31	061C801	SITEWORK	YARD PIPING PLAN AND PROFILES 12" FW FROM FW TANKS
32	061C802	SITEWORK	YARD PIPING PLAN AND PROFILES 8" FW (DISTRIBUTION)
33	061C803	SITEWORK	YARD PIPING PLAN AND PROFILES 4" FW TO FW TANKS
34	061C804	SITEWORK	YARD PIPING PLAN AND PROFILES 4" FW (TO BORE)
35	061C805	SITEWORK	YARD PIPING PLAN AND PROFILES 2" FW
36	061C806	SITEWORK	YARD PIPING PLAN AND PROFILES 8" FW (BYPASS)
37	061C807	SITEWORK	YARD PIPING PLAN AND PROFILES 8" RW FROM TANK
38	061C808	SITEWORK	YARD PIPING PLAN AND PROFILES 4" RW (FROM BORE)
39	061C809	SITEWORK	YARD PIPING PLAN AND PROFILES 6" DR
40	061C810	SITEWORK	YARD PIPING PLAN AND PROFILES 4" DR
41	061C811	SITEWORK	YARD PIPING PLAN AND PROFILES 6" SS
42	061C812	SITEWORK	YARD PIPING PLAN AND PROFILES 6" OVF (TANKS)
43	061C813	SITEWORK	YARD PIPING PLAN AND PROFILES 6" OVF (WTF)
44	061C814	SITEWORK	YARD PIPING PLAN AND PROFILES 8" FW (TO CAMPGROUND)
45	061S101	SITEWORK	PROCESS WASTE TANK FOUNDATION PLAN
46	061S102	SITEWORK	COMMUNICATION TOWER FOUNDATION PLAN AND SECTION
47	061S301	SITEWORK	PROCESS WASTE TANK SECTIONS
48	061E101	SITEWORK	ELECTRICAL OVERALL SITE PLAN
49	061E401	SITEWORK	ELECTRICAL SITE PLAN - WTP
50	061E402	SITEWORK	RADIO ANTENNA AND TOWER SITE PLAN
51	061E631	SITEWORK	ELECTRICAL DUCT BANK SECTIONS AND SCHEDULES
52	061Y501	SITEWORK	WTP TELECOMMUNICATION TOWER
53	061Y620	SITEWORK	NPS TELECOM CABINET LAYOUT
54	062F001	WATER TREATMENT	CODE ANALYSIS
55	062F002	WATER TREATMENT	CODE ANALYSIS (CONT.)
56	062F101	WATER TREATMENT	GROUND LEVEL LIFE SAFETY PLAN
57	062S100	WATER TREATMENT	FOUNDATION PLAN
58	062S101	WATER TREATMENT	GROUND LEVEL PLAN
59	062S102	WATER TREATMENT	UPPER LEVEL PLAN
60	062S103	WATER TREATMENT	ROOF FRAMING PLAN
61	062S301	WATER TREATMENT	SECTIONS
62	062S302	WATER TREATMENT	SECTIONS
63	062S303	WATER TREATMENT	SECTIONS
64	062S304	WATER TREATMENT	SECTIONS
65	062S305	WATER TREATMENT	SECTIONS
66	062S501	WATER TREATMENT	DETAILS
67	062S502	WATER TREATMENT	DETAILS
68	062S503	WATER TREATMENT	DETAILS
69	062S504	WATER TREATMENT	DETAILS
70	062S601	WATER TREATMENT	COLUMN AND FOOTING SCHEDULES
71	062S602	WATER TREATMENT	WIND LOADING DIAGRAMS - ROOFS AND WALLS
72	062S603	WATER TREATMENT	SNOW LOADING DIAGRAMS - ROOF
73	062S604	WATER TREATMENT	ROOF JOISTS LOADING TABLE AND DIAGRAM
74	062A101	WATER TREATMENT	GROUND LEVEL PLAN
75	062A102	WATER TREATMENT	ROOF PLAN - LOWER ROOF
76	062A103	WATER TREATMENT	ROOF PLAN - UPPER ROOF
77	062A104	WATER TREATMENT	REFLECTED CEILING PLAN
78	062A105	WATER TREATMENT	INTERIOR FINISH PLAN
79	062A201	WATER TREATMENT	EXTERIOR ELEVATIONS
80	062A202	WATER TREATMENT	EXTERIOR ELEVATIONS
81	062A301	WATER TREATMENT	BUILDING SECTIONS
82	062A302	WATER TREATMENT	BUILDING SECTIONS
83	062A303	WATER TREATMENT	WALL SECTIONS
84	062A304	WATER TREATMENT	WALL SECTIONS
85	062A401	WATER TREATMENT	ENLARGED PLANS AND INTERIOR ELEVATIONS
86	062A402	WATER TREATMENT	LABORATORY FURNISHINGS KEY SHEET
87	062A403	WATER TREATMENT	LABORATORY FURNISHINGS
88	062A901	WATER TREATMENT	3D REPRESENTATION
89	062A902	WATER TREATMENT	3D REPRESENTATION
90	062D101	WATER TREATMENT	GROUND LEVEL PLAN
91	062D301	WATER TREATMENT	REACTION TANK AND ULTRAFILTRATION SKID SECTION
92	062D302	WATER TREATMENT	ULTRAFILTRATION SKIDS AND CT TANK SECTION
93	062D303	WATER TREATMENT	REACTION TANK AND CIP TANK SECTION
94	062D304	WATER TREATMENT	RAW WATER AND FINISHED WATER PIPING SECTIONS
95	062D305	WATER TREATMENT	REACTION TANK AND CT TANK SECTIONS
96	062D306	WATER TREATMENT	DISTRIBUTION PUMPING SECTION
97	062D307	WATER TREATMENT	DISTRIBUTION PIPING SECTION
98	062D308	WATER TREATMENT	CHEMICAL ROOM SECTIONS
99	062D401	WATER TREATMENT	PROCESS FLOOR ENLARGED PLAN
100	062D402	WATER TREATMENT	PIPE TRENCH ENLARGED PLAN
101	062D403	WATER TREATMENT	CHEMICAL DRUM ROOM ENLARGED PLANS
102	062D501	WATER TREATMENT	REACTION TANK DETAILS
103	062D502	WATER TREATMENT	INSTRUMENTATION PANEL DETAILS
104	062D901	WATER TREATMENT	PROCESS ROOM ISOMETRIC VIEW
105	062D902	WATER TREATMENT	ACIDS CHEMICAL ROOM ISOMETRIC VIEWS
106	062D903	WATER TREATMENT	BASES CHEMICAL ROOM ISOMETRIC VIEWS
107	062D904	WATER TREATMENT	PROCESS ROOM UF MASTER SKID ISOMETRICS
108	062D905	WATER TREATMENT	PROCESS ROOM UF REMOTE SKID ISOMETRICS
109	062D906	WATER TREATMENT	PROCESS ROOM BOOSTER PUMPS ISOMETRIC
110	062D907	WATER TREATMENT	COMPRESSED AIR, CHEMICAL, AND SAMPLE LINE ISOMETRIC
111	062M101	WATER TREATMENT	HVAC PLAN
112	062M102	WATER TREATMENT	HVAC PIPING PLAN
113	062M301	WATER TREATMENT	SECTIONS
114	062M302	WATER TREATMENT	SECTIONS
115	062M601	WATER TREATMENT	SCHEDULES
116	062M602	WATER TREATMENT	SCHEDULES
117	062M701	WATER TREATMENT	HVAC CONTROLS
118	062M702	WATER TREATMENT	HVAC CONTROLS
119	062P101	WATER TREATMENT	WATER PIPING OVERALL PLAN
120	062P102	WATER TREATMENT	DRAINAGE AND VENT OVERALL PLAN
121	062P401	WATER TREATMENT	ENLARGED PLANS
122	062P501	WATER TREATMENT	DETAILS
123	062P601	WATER TREATMENT	SCHEDULES

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

Registered Professional Engineer CERTIFICATE NO. 41809 RYAN RHOADES Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. STILLMAN DRAWN: N. LYNCH TECH. REVIEW: M. GOUGH DATE: AUG 2026	SUB SHEET NO. 060 G001	TITLE OF SHEET GENERAL SHEET INDEX NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 1 OF 192

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GRAND CANYON NATIONAL PARK

WORLD HERITAGE SITE

REHABILITATE NORTH RIM INFRASTRUCTURE

100% DRAFT CONSTRUCTION DOCUMENTS

VOLUME 6 – NORTH RIM WATER TREATMENT

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
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VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
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VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
124	062P602	WATER TREATMENT	SCHEDULES
125	062P901	WATER TREATMENT	WATER PIPING ISOMETRICS
126	062P902	WATER TREATMENT	DOMESTIC WASTE AND VENT ISOMETRICS
127	062P903	WATER TREATMENT	PROCESS WASTE AND VENT ISOMETRICS
128	062E111	WATER TREATMENT	GROUND LEVEL POWER PLAN - ADMIN
129	062E112	WATER TREATMENT	GROUND LEVEL POWER PLAN - PROCESS
130	062E121	WATER TREATMENT	GROUND LEVEL LIGHTING & RECEPTACLE PLAN - ADMIN
131	062E122	WATER TREATMENT	GROUND LEVEL LIGHTING & RECEPTACLE PLAN - PROCESS
132	062E131	WATER TREATMENT	GROUND LEVEL SPECIAL SYSTEMS PLAN - ADMIN
133	062E132	WATER TREATMENT	GROUND LEVEL SPECIAL SYSTEMS PLAN - PROCESS
134	062E401	WATER TREATMENT	LIGHTING & RECEPTACLE ENLARGED PLAN
135	062E601	WATER TREATMENT	ELECTRIC MOTOR CONTROL CENTER ONE-LINE DIAGRAM
136	062E602	WATER TREATMENT	MCC ELEVATIONS
137	062E603	WATER TREATMENT	ELECTRICAL PANELBOARD SCHEDULES
138	062E604	WATER TREATMENT	ELECTRICAL PANELBOARD SCHEDULES
139	062E606	WATER TREATMENT	NR WTP PLC CP ONE-LINE DIAGRAM 1
140	062E607	WATER TREATMENT	NR WTP PLC CP ONE-LINE DIAGRAM 2
141	062E608	WATER TREATMENT	NR WTP CHEMICAL RIO CP ONE-LINE DIAGRAM
142	062E610	WATER TREATMENT	ELECTRICAL MISCELLANEOUS ONE-LINE DIAGRAMS
143	062E612	WATER TREATMENT	MOTOR CONTROL DIAGRAMS
144	062E632	WATER TREATMENT	LUMINAIRE SCHEDULES
145	062E633	WATER TREATMENT	LUMINAIRE SCHEDULES
146	062E634	WATER TREATMENT	LUMINAIRE SCHEDULES
147	062Y401	WATER TREATMENT	SERVER ROOM ENLARGED PLAN
148	062Y601	WATER TREATMENT	RAW WATER TANKS P&ID
149	062Y602	WATER TREATMENT	RAW WATER METERING AND PUMPING P&ID
150	062Y603	WATER TREATMENT	REACTION TANKS P&ID
151	062Y604	WATER TREATMENT	ULTRAFILTRATION SYSTEM P&ID
152	062Y605	WATER TREATMENT	DISINFECTION AND CHLORINE CONTACT TANK P&ID
153	062Y606	WATER TREATMENT	FINISHED WATER TANK PUMPS P&ID
154	062Y607	WATER TREATMENT	FINISHED WATER TANKS P&ID
155	062Y608	WATER TREATMENT	FINISHED WATER METERING AND PUMP 1 P&ID
156	062Y609	WATER TREATMENT	FINISHED WATER METERING AND PUMP 2 P&ID
157	062Y610	WATER TREATMENT	COAGULANT & CAUSTIC SYSTEMS P&ID
158	062Y611	WATER TREATMENT	SODIUM HYPOCHLORITE SYSTEMS P&ID
159	062Y612	WATER TREATMENT	SODIUM BISULFITE & CITRIC ACID SYSTEMS P&ID
160	062Y613	WATER TREATMENT	DRUM CHEMICAL SYSTEM P&ID
161	062Y614	WATER TREATMENT	PROCESS WASTE TANK P&ID
162	062Y615	WATER TREATMENT	MISCELLANEOUS SYSTEMS P&ID
163	062Y620	WATER TREATMENT	WTP NETWORK RACK
164	062Y901	WATER TREATMENT	SERVER ROOM ISOMETRIC
165	063S101	RAW WATER TANKS	UNDERSLAB FLOOR PLAN
166	063S102	RAW WATER TANKS	ABOVE SLAB FLOOR PLAN
167	063S103	RAW WATER TANKS	DOME ROOF PLAN
168	063S301	RAW WATER TANKS	STRUCTURAL SECTIONS
169	063D101	RAW WATER TANKS	UNDERSLAB PIPING PLAN
170	063D102	RAW WATER TANKS	ABOVE SLAB PIPING PLAN
171	063D103	RAW WATER TANKS	DOME ROOF PLAN
172	063D301	RAW WATER TANKS	SECTIONS
173	063E101	RAW WATER TANKS	ABOVE SLAB FLOOR PLAN RW TANKS
174	063E601	RAW WATER TANKS	CATHODIC PROTECTION PLAN AND SECTION
175	063E602	RAW WATER TANKS	CATHODIC PROTECTION DETAILS
176	064D101	FINISHED WATER TANKS	TANK A FLOOR PLAN
177	064D102	FINISHED WATER TANKS	TANK B FLOOR PLAN
178	064D103	FINISHED WATER TANKS	TANK A ROOF PLAN
179	064D104	FINISHED WATER TANKS	TANK B ROOF PLAN
180	064D301	FINISHED WATER TANKS	TANK A SECTION
181	064D302	FINISHED WATER TANKS	TANK B SECTION
182	064D401	FINISHED WATER TANKS	ENLARGED VAULT PLANS
183	064D501	FINISHED WATER TANKS	DETAILS
184	064D502	FINISHED WATER TANKS	DETAILS
185	064D901	FINISHED WATER TANKS	SEQUENCING PLAN OVERVIEW
186	064D902	FINISHED WATER TANKS	SEQUENCING PLAN PHASE 1
187	064D903	FINISHED WATER TANKS	SEQUENCING PLAN PHASE 2
188	064D904	FINISHED WATER TANKS	SEQUENCING PLAN PHASE 3
189	064E101	FINISHED WATER TANKS	ELECTRICAL OVERALL SITE PLAN
190	064E601	FINISHED WATER TANKS	CATHODIC PROTECTION PLAN AND SECTION
191	064E602	FINISHED WATER TANKS	CATHODIC PROTECTION DETAILS
192	065Y501	RAW WATER TANKS	SOUTH RIM TANK C COMMUNICATIONS

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

Registered Professional Engineer (Civil) CERTIFICATE NO. 41809 RYAN RHOADES Date Signed 5/1/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. STILLMAN CADD N. LYNCH TECH. REVIEW: GOUGH/GLOVER DATE: AUG 2026	SUB SHEET NO. 060 G002	TITLE OF SHEET GENERAL SHEET INDEX CONTINUED NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 2 OF 192

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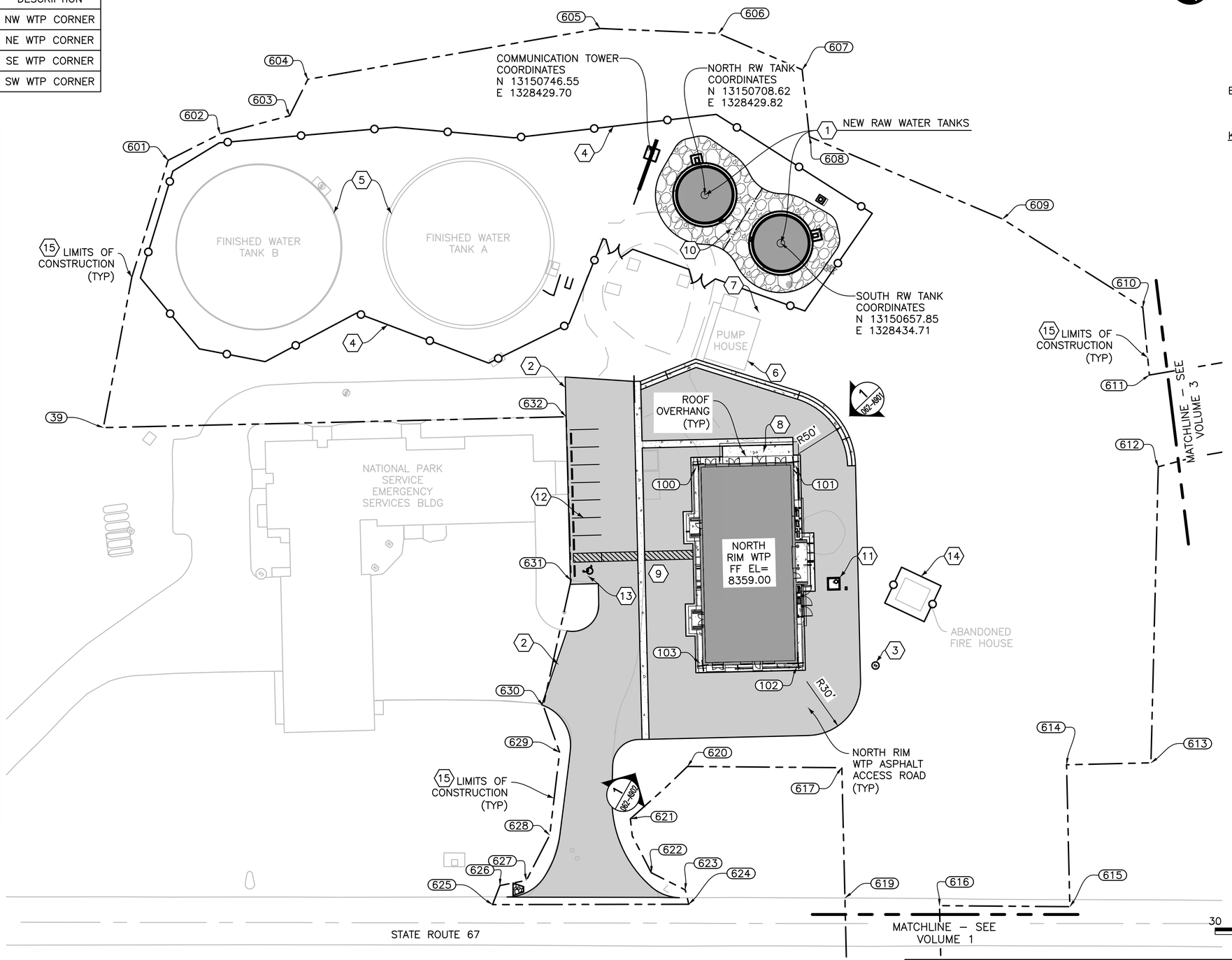
NORTH RIM WTP POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
100	13150617.88	1328307.11	NW WTP CORNER
101	13150576.06	1328341.28	NE WTP CORNER
102	13150503.93	1328253.01	SE WTP CORNER
103	13150545.74	1328218.84	SW WTP CORNER

LIMITS OF CONSTRUCTION		
POINT NO.	NORTHING	EASTING
39	13150896.37	1328116.62
601	13150960.42	1328258.56
602	13150946.37	1328288.47
603	13150921.65	1328320.74
604	13150926.18	1328343.37
605	13150813.32	1328467.75
606	13150758.77	1328506.71
607	13150708.66	1328519.63
608	13150682.12	1328492.22
609	13150567.23	1328522.63
610	13150473.66	1328531.85
611	13150447.28	1328504.01
612	13150411.40	1328466.14
613	13150311.67	1328331.33
614	13150348.70	1328301.11
615	13150297.88	1328238.82
616	13150356.45	1328193.94
617	13150448.10	1328221.61
619	13150401.39	1328164.45
620	13150517.52	1328168.41
621	13150524.78	1328125.19
622	13150496.68	1328108.38
623	13150475.55	1328112.51
624	13150468.97	1328107.34
625	13150556.55	1328038.88
626	13150559.85	1328050.01
627	13150549.85	1328061.43
628	13150555.14	1328089.84
629	13150579.56	1328130.30
630	13150603.06	1328145.41
631	13150634.31	1328211.15
632	13150693.40	1328282.48



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SEE VOLUME 0 FOR STAGING AREAS AND TEMPORARY ACCESS ROAD.

KEY NOTES: (X)

1. NEW TANKS SHALL BE 150,000 GALLONS EACH, WELDED STEEL, 32' ID, 30' TALL, WITH 10' WIDE GRAVEL PAD AROUND PERIMETER.
2. NEW ASPHALT TO TIE INTO EXISTING ACCESS DRIVE (TYP.)
3. NEW 48" MANHOLE.
4. NEW 8' TALL FENCE
5. EXISTING STEEL FINISHED WATER TANKS, 40' TALL X 90' DIA. CONTRACTOR SHALL REMOVE 1"–2" OF SEDIMENT FROM FLOOR, POWERWASH INTERIOR, PROVIDE THIRD PARTY CONDITION ASSESSMENT, RE–COAT INTERIOR AND EXTERIOR, REPLACE VENTS, REPLACE VISUAL LEVEL INDICATORS, AND REPLACE SEALS ON SIDE ACCESS HATCHES
6. PRESERVE AND PROTECT IN PLACE.
7. RADIO REPEATER OPERATION MUST BE MAINTAINED THROUGHOUT CONSTRUCTION
8. NEW 6" THICK CONCRETE PAD. QUANTITY = 8 CY (APPROX.).
9. NEW 4" ASPHALTIC CONCRETE OVER 8" OF MOISTURE CONDITIONED AND COMPACTED ADOT CLASS 2 AGGREGATE BASE. ASPHALTIC CONCRETE QUANTITY = 640 TONS (APPROX.). ADOT CLASS 2 AGGREGATE BASE QUANTITY = 885 TONS (APPROX.).
10. NEW AGGREGATE PAVING TANK ACCESS, 8" THICK. QUANTITY = 90 CY (APPROX.).
11. PROCESS VAULT.
12. RESTRIPE PER EXISTING PARKING STRIPING.
13. ADA PARKING STALL. SEE DETAIL 04/061C502 FOR DETAIL.
14. INSTALL 94 LF OF RIGID CHAINLINK FENCE 3' OFFSET FROM EXIST STRUCTURE.
15. AVAILABLE CONSTRUCTION LIMITS ARE SHOWN IN THE LIMITS OF CONSTRUCTION TABLE.

REGISTERED PROFESSIONAL ENGINEER – CERTIFICATE NO. 75262 – GREGORY YOUNG – Date Signed 5/11/26 – ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	G. YOUNG	060 C101	GENERAL OVERALL SITE PLAN	113 182752
	CADD P. OXBORROW			PMIS/PKG NO. 318719
	TECH. REVIEW: GOUGH/LUSTER			SHEET 3 OF 192
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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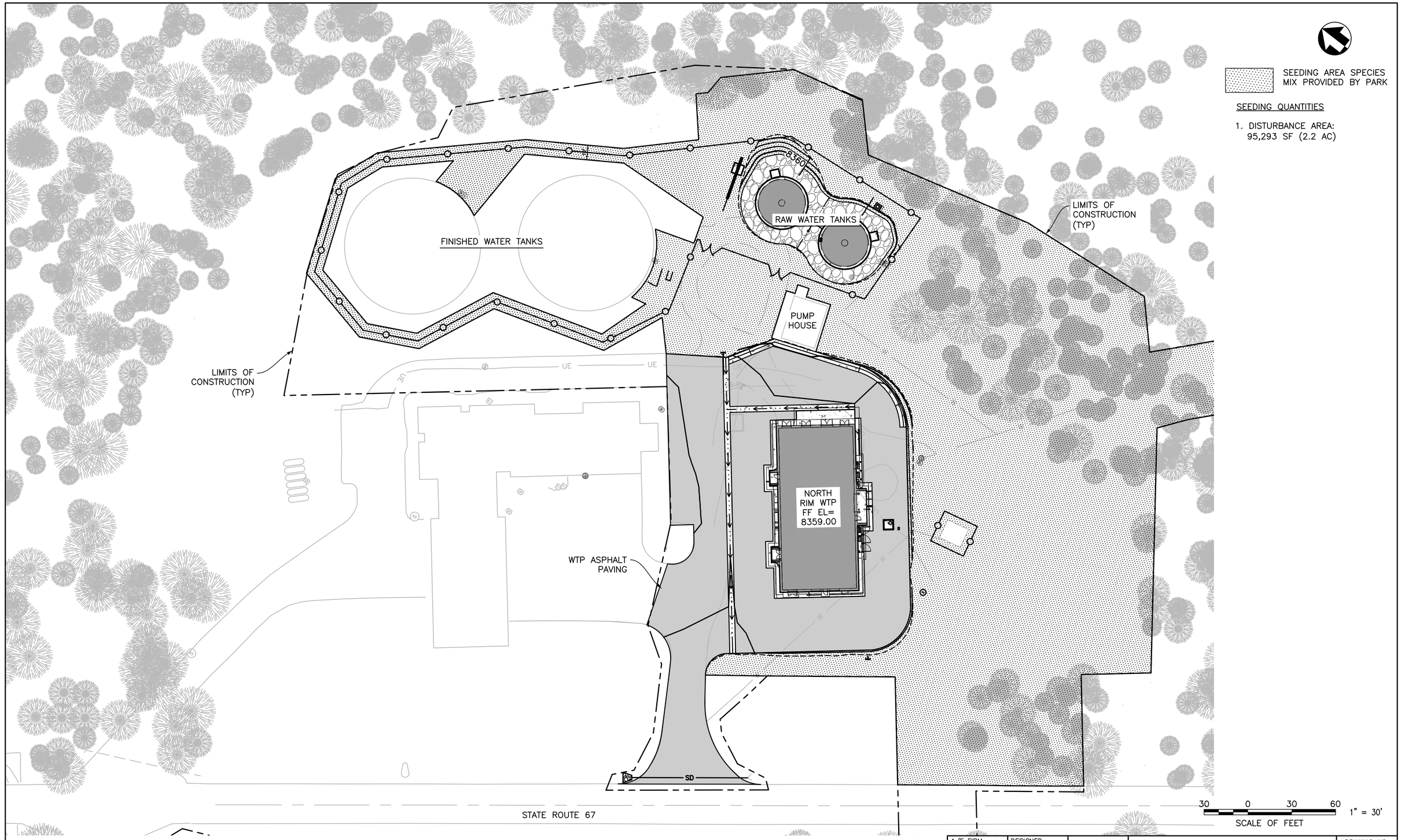

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — State of ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 060 L101	TITLE OF SHEET GENERAL LANDSCAPE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 4 OF 192

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GENERAL

G1. SCOPE
THE NOTES ON THIS SHEET AND THE STANDARD STRUCTURAL DETAILS ARE GENERAL AND APPLY TO THE ENTIRE PROJECT WHETHER SPECIFICALLY CALLED OUT OR NOT, EXCEPT WHERE THERE ARE SPECIFIC INDICATIONS TO THE CONTRARY ON STRUCTURAL SHEETS. IF THERE ARE QUESTIONS, THEY SHALL BE SUBMITTED TO THE CONTRACTING OFFICER AND ANSWERED IN WRITING PRIOR TO CONSTRUCTION.

G2. APPLICABLE SPECIFICATIONS AND CODES
A. INTERNATIONAL BUILDING CODE, IBC 2024, WITH APPLICABLE EDITIONS OF THE CODE REFERENCED STANDARDS.
B. AUTHORITY HAVING JURISDICTION: NATIONAL PARK SERVICE (WIND, SNOW, AND SEISMIC DESIGN CRITERIA IN ACCORDANCE WITH THE MORE CONSERVATIVE OF COCONINO COUNTY, AZ AND NATIONAL PARK SERVICE.)
C. CODE REQUIREMENTS FOR ENVIRONMENTAL ENGINEERING CONCRETE STRUCTURES, ACI 350-20.
D. BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE, ACI 318-19.
E. MINIMUM DESIGN LOADS AND ASSOCIATED CRITERIA FOR BUILDINGS AND OTHER STRUCTURES, ASCE 7-22.
F. AMERICAN INSTITUTE OF STEEL CONSTRUCTION STEEL CONSTRUCTION MANUAL, 16TH ED.
G. STEEL DECK INSTITUTE SPECIFICATION AND LOAD TABLES.
H. RCSC'S SPECIFICATION FOR STRUCTURAL JOINTS USING HIGH-STRENGTH BOLTS.
I. AMERICAN WELDING SOCIETY, AWS D1.1.
J. ASTM MATERIAL STANDARDS AS NOTED.
K. BUILDING CODE REQUIREMENTS FOR MASONRY STRUCTURES, TMS 402-2022
L. ALUMINUM DESIGN MANUAL, ADM-2020
M. STANDARD FOR STEEL DECK, ANSI/SDI SD-2022
N. STANDARD SPECIFICATIONS FOR K-SERIES, LH-SERIES, AND DLH-SERIES OPEN WEB STEEL JOISTS AND FOR JOISTS GIRDERS, SJI 100-2020
O. THE MASONRY SOCIETY BUILDING CODE REQUIREMENTS AND SPECIFICATIONS FOR MASONRY STRUCTURES, TMS 402/602-2022.

G3. DESIGN CRITERIA
1. APPLIES TO ALL STRUCTURES (UNO)
A. DEAD LOAD:
1. ACTUAL TRIBUTARY STRUCTURE WEIGHT
2. SUPERIMPOSED DEAD LOAD: 20 PSF AT ROOF
B. LIVE LOAD:
1. WALKWAYS, STAIRS, GRATING: 100 PSF
2. SLAB ON GRADE: 250 PSF
3. STORAGE ABOVE CEILING: 20 PSF
4. ROOF: 20 PSF (NOT REDUCIBLE)
C. WIND:
1. ULTIMATE WIND SPEED (V_{ult}): 111 MPH
2. TORNADO SPEED (V_T): NOT APPLICABLE PER ASCE 7 FIGURES 32.1-1 AND 32.1-2
3. NOMINAL WIND SPEED (V_{ASD}): 86 MPH
4. RISK CATEGORY: IV
5. EXPOSURE: C
6. INTERNAL PRESSURE COEFFICIENT (GC_{pi}): ±0.18
7. DESIGN WIND PRESSURE: 23 PSF
8. COMPONENTS AND CLADDING PRESSURES: SEE 062S602
D. SEISMIC:
1. WATER TREATMENT FACILITY:
a. RISK CATEGORY: IV
b. IMPORTANCE FACTOR (I_e): 1.50
c. SPECTRAL RESPONSE ACCELERATION, S_S: 0.41g
d. SPECTRAL RESPONSE ACCELERATION, S_1: 0.11g
e. SITE CLASS: C
f. SEISMIC DESIGN CATEGORY: C
g. SPECTRAL RESPONSE ACCELERATION, S_{DS}: 0.3g
h. SPECTRAL RESPONSE ACCELERATION, S_{D1}: 0.11g
i. BASIC SEISMIC FORCE RESISTING SYSTEM: INTERMEDIATE REINFORCED MASONRY SHEAR WALLS
j. ANALYSIS PROCEDURE: EQUIVALENT LATERAL FORCE
k. DESIGN BASE SHEAR: 165 KIP
l. SEISMIC RESPONSE COEFFICIENT (C_s) = 0.129
m. RESPONSE MODIFICATION FACTOR (R) = 3.5
n. OVERSTRENGTH FACTOR = 2.5
E. SNOW LOAD:
1. GROUND SNOW LOAD (p_g) = 256 PSF
2. FLAT ROOF SNOW LOAD (p_f) = 197 PSF
3. EXPOSURE FACTOR (C_e) = 1.0
4. RISK CATEGORY = IV
5. SLOPED ROOF FACTOR (C_s) = 1.0
6. THERMAL FACTOR (C_t) = 1.1
7. DRIFT SURCHARGE (p_d) = 150 PSF
8. WIDTH OF DRIFT (w) = 12'-8"
9. WINTER WIND PARAMETER (W_2) = 0.35
10.FROST DEPTH = 36 INCHES
F. ICE LOADS:
1. ICE THICKNESS: 0.02 IN
2. ICE THICKNESS HEIGHT FACTOR: 1.00
3. DESIGN ICE THICKNESS: 0.02 IN
4. DESIGN WEIGHT OF ICE: 0.09 PSF
G. FLOOD CRITERIA: NOT APPLICABLE
H. ROOF RAIN LOAD
1. 15-MINUTE PRECIPITATION INTENSITY: 8.6 IN/HR
2. 60-MINUTE PRECIPITATION INTENSITY: 3.58 IN/HR

GENERAL

G4. THE FOLLOWING NON-CONTRACTUAL GEOTECHNICAL REPORT WAS DEVELOPED FOR THIS PROJECT AND IS THE BASIS OF THIS STRUCTURAL DESIGN:
GEOTECHNICAL FIRM NAME: LITHOS ENGINEERING
ADDRESS: 2750 S. WADSWOTH BLVD., SUITE D-200 DENVER, CO 80227
REPORT NUMBER: 21186
REPORT DATE: JANUARY 16, 2024
MAXIMUM ALLOWABLE SOIL BEARING = 3,000 PSF
COEFFICIENT FRICTION OF SLIDING = 0.35
ACTIVE LATERAL EARTH PRESSURE = 42 PCF
AT-REST LATERAL EARTH PRESSURE = 63 PCF
PASSIVE LATERAL EARTH PRESSURE = 200 PCF

G5. SAFETY
SAFETY AND STRUCTURE STABILITY DURING CONSTRUCTION ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR. STRUCTURES HAVE BEEN DESIGNED TO RESIST THE DESIGN LIVE LOADS ONLY AS A COMPLETED STRUCTURE.

G6. OPENINGS
OPENINGS FOR PIPES, DUCTS, CONDUITS, ETC. ARE NOT ALL SHOWN ON THE STRUCTURAL DRAWINGS. COORDINATE AND PROVIDE OPENINGS AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

G7. SPECIAL INSPECTIONS
SPECIAL INSPECTIONS ARE REQUIRED IN ACCORDANCE WITH CHAPTER 1 AND CHAPTER 17 OF THE IBC. PAYMENT FOR THESE INSPECTIONS IS NOT THE RESPONSIBILITY OF THE CONTRACTOR IN ACCORDANCE WITH THE REQUIREMENTS OF SPECIFICATION SECTION 01 40 00. THE CONTRACTOR SHALL PROVIDE FOR FULL ACCESS TO THE WORK BY THE SPECIAL INSPECTOR AND SHALL PROVIDE FOR THESE INSPECTIONS IN THE OVERALL CONSTRUCTION SCHEDULE. THE SPECIAL INSPECTION PROGRAM SHALL BE IN ACCORDANCE WITH SPECIFICATION SECTION 01 40 00 AND "STATMENT OF STRUCTURAL TESTS AND SPECIAL INSPECTIONS" DOCUMENTS INCLUDED AS ATTACHMENTS TO SPECIFICATION SECTION 01 40 00.

G8. STANDARD DETAILS
THE STANDARD DETAILS IN VOLUME 00 DEPICT TYPICAL DETAILING TO BE USED ON THIS PROJECT. IF CONDITIONS ARE NOT EXPLICITLY SHOWN ON THE DRAWINGS THEY SHALL BE MADE SIMILAR TO THE STANDARD DETAILS. OBTAIN APPROVAL OF CONTRACTING OFFICER IN WRITING FOR SIMILAR CONDITIONS PRIOR TO CONSTRUCTION.

G9. THE CONTACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND ELEVATIONS OF EXISTING CONSTRUCTION AS REQUIRED TO COORDINATE NEW CONSTRUCTION. SUBMIT REQUIRED CHANGES FOR APPROVAL.

G10. CONTRACTOR TO SUBMIT FOR REVIEW ALL EQUIPMENT SIZES, OPERATING WEIGHTS, VIBRATION FORCES, SUPPORT LOCATIONS, ALONG WITH ANY FLOOR OPENINGS, NOTCHES, AND RECESSES REQUIRED BY SUCH EQUIPMENT. CONCRETE SUPPORT PADS AND/OR FRAMING REQUIRED TO SUPPORT SAID EQUIPMENT SHALL NOT BE FABRICATED AND PLACED UNTIL THE CONCRETE SUPPORT PADS AND/OR FRAMING IS APPROVED TO SUPPORT THE EQUIPMENT.

CONCRETE

C1. DESIGN STRENGTHS:
F'c = 4,500 PSI (UNO)
F'c = 5,000 PSI (WATER BEARING AND PRECAST CONCRETE)
Fy = 60,000 PSI REINFORCEMENT BARS - ASTM A615, GR 60

C2. CONCRETE COVER
UNLESS OTHERWISE NOTED, PROVIDE CONCRETE COVER FOR REINFORCING AS FOLLOWS:
CONCRETE DEPOSITED AGAINST EARTH: 3"
ALL OTHER: 2"
WATERSTOPS: 3 1/2"
SEE DRAWINGS FOR EXCEPTIONS

C3. SEE SPECIFICATIONS FOR REINFORCING PLACEMENT REQUIREMENTS.

C4. REFER TO OTHER DISCIPLINE DRAWINGS PRIOR TO CONSTRUCTION FOR EMBEDDED ITEMS AND PENETRATIONS NOT SHOWN ON STRUCTURAL DRAWINGS. AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

C5. PROVIDE 3/4" CHAMFERS AT ALL EXPOSED EDGES. NOT ALL CHAMFERS MAY BE SHOWN ON DRAWINGS.

C6. FIELD ADJUST REINFORCING AT OPENINGS AND EMBEDDED ITEMS AS INDICATED.

C7. ANCHOR BOLTS NOT SPECIFIED BY ENGINEER SHALL BE DESIGNED AND CERTIFIED BY A REGISTERED PROFESSIONAL ENGINEER LICENSED IN THE STATE OF ARIZONA, RETAINED BY THE CONTRACTOR, IN ACCORDANCE WITH APPLICABLE PROJECT AND CODE REQUIREMENTS. SUBMIT AS A SHOP DRAWING FOR REVIEW AND APPROVAL BY THE CONTRACTING OFFICER. COORDINATE LOCATION, SIZE AND EMBEDMENT PRIOR TO CASTING CONCRETE.

CONCRETE (CONTINUED)

C8. CONTINUOUS WATERSTOP SHALL BE INSTALLED IN JOINTS SUBJECT TO STATIC WATER PRESSURE.

C9. ABSOLUTELY NO WELDING OF REINFORCING BARS OR TORCHING TO BEND REINFORCING BARS SHALL BE ALLOWED.

C10. CONTRACTOR SHALL SUBMIT A CONCRETE PLACEMENT PLAN PER SPECIFICATIONS IDENTIFYING JOINT TYPES, JOINT LOCATIONS AND CONCRETE PLACEMENT SEQUENCE.

C11. ALL CAST IN PLACE AND POST-INSTALLED ANCHORS INDICATED IN THE STRUCTURAL DOCUMENTS SHALL COMPLY WITH CHAPTER 17 OF ACI 318, APPENDIX E OF ACI 350, AND CHAPTER 19 OF THE IBC. ALL EXPANSION AND ADHESIVE ANCHORS SHALL HAVE THE ICC REPORT SHOWING EQUIVALENT LOAD CAPACITY. SUBMIT AND INSTALL PER THE ICC EVALUATION REPORT.

MASONRY

M1. DESIGN STRENGTHS:
MASONRY DESIGN STRENGTH: F'm= 2,000 PSI
REINFORCING STEEL - ASTM A615: Fy = 60,000 PSI
GROUT - ASTM C476: F'c = 2,000 PSI
MORTAR - ASTM C270, TYPE S: F'c = 2,000 PSI
JOINT REINFORCING - TRUSS TYPE

M2. GROUT FOR FILLING MASONRY CAVITIES TO BE COARSE GROUT UNO, MAXIMUM COARSE AGGREGATE SIZE IS 3/8 INCH.

M3. GROUT POURS SHALL NOT EXCEED 4 FEET IN HEIGHT UNLESS CLEANOUTS ARE PROVIDED IN THE BOTTOM COURSE OF THE CELL(S) TO BE GROUTED AND WRITTEN PERMISSION IS OBTAINED FOR HIGH LIFT GROUTING.

M4. RESTRICTED BAR ANCHORAGE:
IN CASES WHERE REINFORCING BARS CANNOT BE EXTENDED AS FAR AS REQUIRED, THE BARS SHALL EXTEND AS FAR AS POSSIBLE AND END IN STANDARD HOOK. SHOW ON SHOP DRAWINGS AND HIGHLIGHT WITH A BOX TO BRING TO ENGINEER'S ATTENTION.

M5. ANCHOR BOLTS:
ALL EXPANSION AND ADHESIVE ANCHORS SHALL HAVE THE ICC REPORT SHOWING EQUIVALENT LOAD CAPACITY. SUBMIT AND INSTALL PER THE ICC EVALUATION REPORT.

M6. IF BOND BEAMS AT INTERSECTING WALLS ARE SHOWN ON THE DRAWINGS TO MEET AT DIFFERENT ELEVATIONS, EXTEND REIFORCING OF BOTH BOND BEAMS AROUND INTERSECTING CORNER NOT LESS THAN 4 FEET IN EACH DIRECTION.

M7. LINTEL BLOCKS SHALL NOT BE USED AS BOND BEAM BLOCKS EXCEPT AT OPENINGS WHERE BOND BEAMS AND LINTELS COINCIDE.

STEEL

S1. DESIGN STRENGTHS:
WIDE FLANGE AND TEES - ASTM A992: Fy=50 KSI
PIPES - ASTM A53, GRADE B: Fy=35 KSI
STAINLESS STEEL, TYPE 304 OR 316 AND TYPE 304L AND 316L WHERE WELDING IS REQUIRED: Fy=33 KSI
HSS SECTIONS, ASTM A500 GRADE C: Fy=50 KSI
ALL OTHER PLATES AND SHAPES, ASTM A36: Fy=36 KSI
ANCHOR RODS, ASTM F1554 GRADE 36: Fy=36 KSI

S2. DIMENSIONS:
TO CENTERLINES OF COLUMNS AND BEAMS, TOP SURFACES OF BEAMS AND TUBES AND BACKS OF CHANNELS AND ANGLES UNO.

S3. ELEVATIONS:
TOP OF STEEL REFERS TO TOP SURFACE OF MEMBER OR FLANGE UNO.

S4. WHEN FILLET WELD SIZE IS NOT INDICATED, PROVIDE MAXIMUM WELD SIZE BASED ON MATERIAL THICKNESS IN ACCORDANCE WITH AISC SPECIFICATIONS.

S5. ALL BOLTED STRUCTURAL CONNECTIONS ARE BEARING TYPE CONNECTIONS UNLESS OTHERWISE SPECIFIED TO BE SLIP-CRITICAL. PROVIDE LOAD INDICATING WASHERS AT SLIP-CRITICAL CONNECTIONS.

S6. CONFORM TO AISC 360, STEEL CONSTRUCTION MANUAL AND AISC 341, SEISMIC DESIGN MANUAL.

ALUMINUM

A1. STRUCTURAL ALUMINUM YIELD STRENGTHS
STRUCTURAL ALUMINUM: Fy=35 KSI
STRUCTURAL ALUMINUM IS ALLOY 6061-T6 UNO

A2. DIMENSIONS:
TO CENTERLINES OF COLUMNS AND BEAMS, TOP SURFACES OF BEAMS AND TUBES AND BACKS OF CHANNELS AND ANGLES UNO.

A3. ELEVATIONS:
TOP OF ALUMINUM REFERS TO TOP SURFACE OR FLANGE OF MEMBER UNO.

ALUMINUM (CONTINUED)

A4. WHEN FILLET WELD SIZE IS NOT INDICATED, PROVIDE MAXIMUM WELD SIZE FOR THE MATERIAL THICKNESS IN ACCORDANCE WITH THE LATEST EDITION OF THE "ALUMINUM DESIGN MANUAL" BY THE ALUMINUM ASSOCIATION.

A5. ALUMINUM IN CONTACT WITH DISSIMILAR MATERIALS OR CONCRETE: CONTACT SURFACES SHALL BE PROVIDED WITH GALVANIC SEPERATION PER SPECIFICATIONS.

STEEL BAR JOIST:

SJ1. STEEL JOISTS ARE SIZED AS SHOWN ON THE DRAWINGS AND DESIGNED BY THE CONTRACTOR FOR THE LOADINGS INDICATED.

SJ2. STEEL JOISTS AND BRIDGING SHALL BE DESIGNED BY THE MANUFACTURER TO RESIST UPLIFT LOAD INDICATED ON THE DRAWINGS. BRIDGING IS NOT SHOWN ON PLAN.

SJ5. ADJUST JOIST BRIDGING DESIGN AND DETAILING WHERE MECHANICAL DUCTS OR OTHER OBSTRUCTIONS RUN BETWEEN THE JOISTS. SEE "TYPICAL INTERRUPTION OF BRIDGING AT DUCTWORK DETAIL"

STEEL DECK:

SD1. THE DESIGN, FABRICATION, AND ERECTION OF METAL DECKING SHALL BE IN ACCORDANCE WITH THE CURRENT EDITION OF THE SDI SPECIFICATIONS AND THE SDI DIAPHRAGM MANUAL

SD2. STEEL ROOF DECK IS TO BE A STRUCTURAL DIAPHRAGM AND SHALL BE FASTENED AS SHOWN ON THE DRAWINGS.

SD3. STEEL FLOOR DECK IS AS INDICATED ON PLANS DESIGNED FOR THE DEAD AND LIVE LOADS INDICATED.

SD4. THE PLANS INDICATE DECK SPAN DIRECTION.

SD5. SUSPENDED CEILINGS, LIGHT FIXTURES, DUCTS, AND OTHER UTILITIES SHALL NOT BE SUPPORTED FROM THE STEEL DECK.

EARTHWORK:

E1. ALL EARTHWORK AND SITE PREPERATION SHALL BE PERFORMED IN STRICT ACCORDANCE WITH THE SPECIFICATIONS. ALL FOUNDATION EXCAVATIONS SHALL BE OBSERVED AND APPROVED BY THE LICENSED GEOTECHNICAL ENGINEER.

E2. ANY EXISTING FILLS OR UNSUITABLE SOILS AS DETERMINED BY THE GEOTECHNICAL ENGINEER SHALL BE EXCAVATED AND REPALCED WITH PROPERLY COMPACTED GRANULAR FILL PER SPECIFICATIONS.

E3. CONTRACTOR SHALL EXERCISE EXTREME CARE WHEN EXCAVATING OR GRADING ADJACENT TO EXISTING STRUCTURES SO AS TO NOT DAMAGE OR UNDERMINE FOUNDATIONS, WALLS, SLABS, UTILITIES, ETC.

E4. DO NOT EXCAVATE BELOW THE BEARING ELEVATION OF ANY COMPLETED FOOTING NOR ANY CLOSER TO THE FOOTING THAN A SLOPE OF TWO HORIZONTAL (MEASURED FROM EDGE OF FOOTING TO NEAREST POINT IN EXCAVATION) TO ONE VERTICAL.

DELEGATED DESIGN SUBMITTALS:

D1. DEFINITION:

DELEGATED DESIGN SUBMITTALS ARE ITEMS FOR WHICH THE FINAL DESIGN OF A SPECIFIC ELEMENT OF THE PERMANENT CONSTRUCTION WORK EXPRESSLY DELEGATED TO A PROPERLY LICENSED AND REGISTERED DESIGN PROFESSIONAL RETAINED OR EMPLOYED BY THE CONTRACTOR, SUBCONTRACTOR, SUPPLIER OF ANY TIER. "DELEGATED DESIGN" APPLIES ONLY TO PERMANENT WORK THAT WILL BE PART OF THE COMPLETED PROJECT AS A FUNCTIONING WHOLE.

D2. DELEGATED DESIGN LIST:

THE FOLLOWING IS A LIST OF DELEGATED DESIGN SUBMITTALS THAT ARE EXPECTED TO CONTAIN STRUCTURAL CALCULATIONS OF SAFETY-RELATED SYSTEM INFORMATION FOR REVIEW. THESE SUBMITTALS APPLY TO DELEGATED DESIGNED SYSTEMS PRIOR TO INSTALLATION OF THE INDICATED STRUCTURAL ELEMENT, EQUIPMENT, DISTRIBUTION SYSTEM, COMPONENT, OR ITS ANCHORAGE. THE CONTRACTOR SHALL SUBMIT THE REQUIRED ENGINEER CERTIFICATION, SUPPORTING DATA, AND DRAWINGS FOR REVIEW AND ACCEPTANCE BY THE CONTRACTING OFFICER.

D3. DELEGATED DESIGN SUBMITTALS INCLUDE:
1. GUARDRAIL
2. STAIRS
3. FRP GRATING AND ASSOCIATED SUPPORTS AND CONNECTIONS
4. LADDERS
5. PRECAST CONCRETE ELEMENTS.
6. OPEN WEB ROOF JOISTS
7. CANOPIES AND AWNINGS
8. LIGHT GAUGE STUD WALLS

Registered Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed 10/31/2025 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK CADD R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 060 S001	TITLE OF SHEET GENERAL NOTES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 5 OF 192

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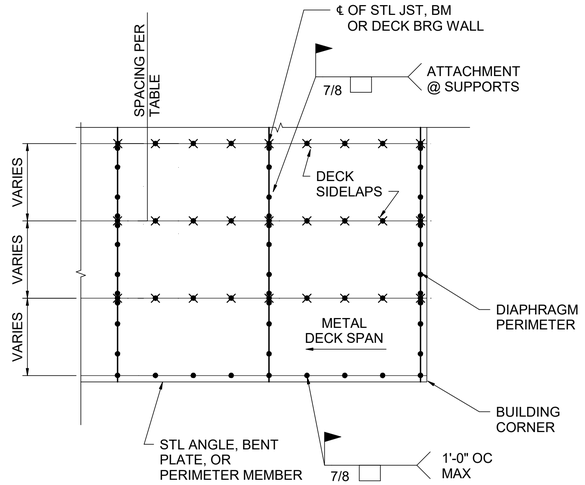
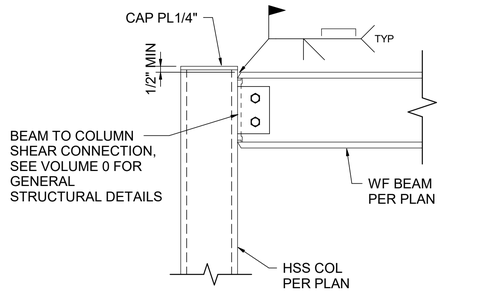


DECK ATTACHMENT SCHEDULE

TYPE	ATTACHMENT @ SUPPORTS	SIDELAP FASTENER SPACING	REMARKS
2.0DA20 GRADE 40 20 GAUGE	24.5/4: 7/8" DIA. ARC SPOT WELDS AT EACH LOW FLUTE	#12 SDST SCREWS @ 6" OC	ATTACHMENT @ EDGES TO BLOCKERS, SAME AS SUPPORT
1.5B18 GRADE 50 18 GAUGE	36/5: #12 SDST SCREWS	#10 SDST SCREWS @12" O.C.	ATTACHMENT @ EDGES TO BLOCKERS, SAME AS SUPPORT
1.5B18 GRADE 50 18 GAUGE	36/7: #12 SDST SCREWS	#10 SDST SCREWS @8" O.C.	ATTACHMENT @ EDGES TO BLOCKERS, SAME AS SUPPORT. USE PATTERN BTWN GRIDS 1 AND 2.

1 060S002 TYPICAL ROOF DECK
NOT TO SCALE



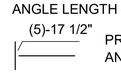
NOTE:
1. CONNECTION SHOWN THUS ON PLAN:

2 060S002 TYPICAL BEAM TO HSS COLUMN MOMENT CONNECTION
NOT TO SCALE

MIN. WELD PLATE SCHEDULE		
NOM. BM. DEPTH "D"	ANGLE LENGTH	NO. STUD ROWS "N"
8, 10, 12	5 1/2"	1
14, 16, 18	8 1/2"	2

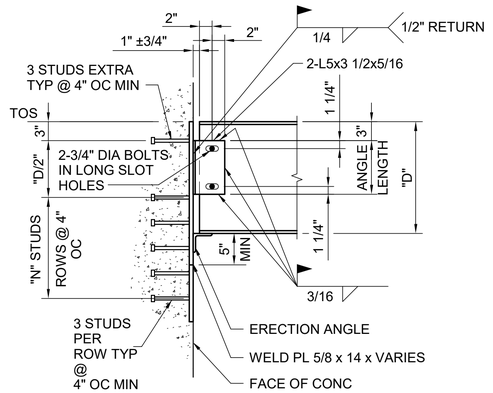
NOTES:

1. ALL STUDS SHALL BE 5/8" DIA x 0'-6" FINAL LENGTH HEADED STUDS.
2. PROVIDE MINIMUM WELD PLATE CONNECTIONS AS INDICATED IN ABOVE SCHEDULE TYPICAL. WHERE STUD ROWS NOTED IN PARENTHESES FOLLOWED BY ANGLE LENGTH ARE INDICATED, I.E.

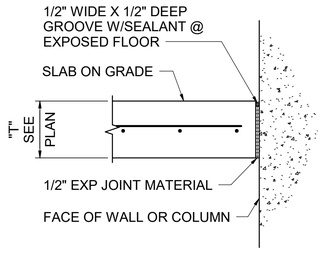


PROVIDE NUMBER OF ROWS AND ANGLE LENGTH AS INDICATED.

3. CONNECTION INDICATED THUS ON PLAN.



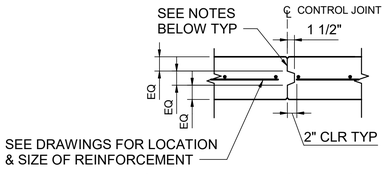
3 060S002 BEAM TO CONCRETE WALL CONNECTION
NOT TO SCALE



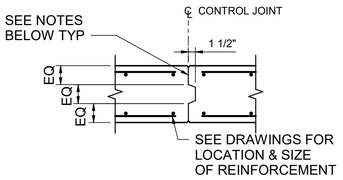
NOTES:

1. AT SLAB-ON-GRADE.

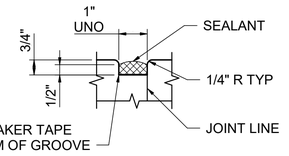
4 060S002 TYPICAL ISOLATION JOINT (IJ)
NOT TO SCALE



NON-WATER BEARING WALL OR SLAB LESS THAN 10" THICK



NON-WATER BEARING SLAB OR WALL 10" THICK OR GREATER

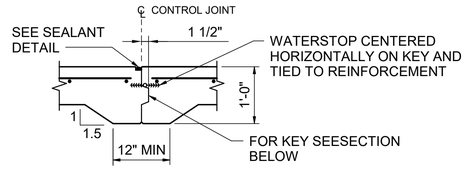


SEALANT DETAIL

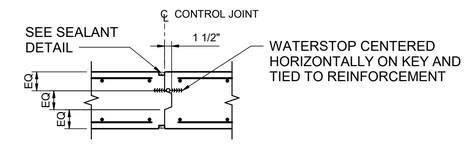
NOTES:

1. COAT CONCRETE JOINT SURFACES WITH BOND BREAKER COMPOUND.
2. FOR WALLS, FORM ALL JOINT EDGES AT 1/2" CHAMFER.
3. FOR SLABS, EDGE TOP OF EXPOSED SLAB JOINT AT 1/4" RADIUS.
4. FOR UNDERSIDE OF EXPOSED SLABS, FORM JOINT EDGES AT 1/2" CHAMFER.

5 060S002 CONTROL JOINT (CLJ)
NOT TO SCALE



WATER BEARING SLAB LESS THAN 12" THICK



PROVIDE SINGLE REINF MAT FOR WALLS WHEN THICKNESS IS LESS THAN 10". REINFORCING AS SPECIFIED

WATER BEARING SLAB OR WALL

Registered Professional Engineer (Civil)
CERTIFICATE NO.
72480
COLIN MULDERICK
State Standard Construction
ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	C. MULDERICK	060 S002	GENERAL STANDARD DETAILS	113 182752
	CADD R. NELSON			PMIS/PKG NO. 318719
	TECH. REVIEW: L. MENDOZA			SHEET 6 OF 192
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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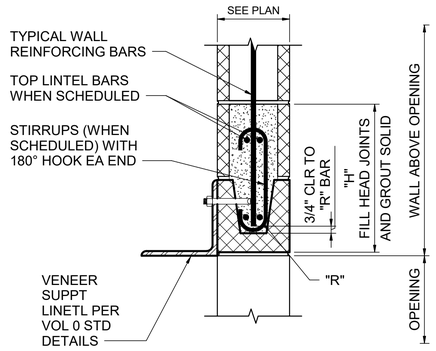
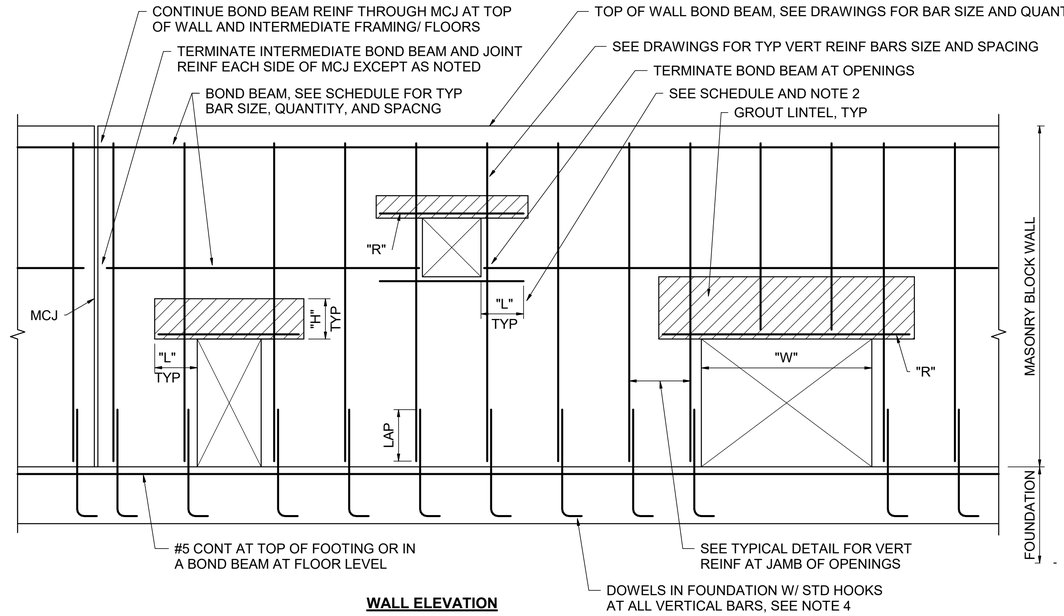
LINTEL REINFORCEMENT SCHEDULE						
MARK	"W"	"R"	"L"	"H"	STIRRUPS	REMARKS
ML-1	1'-4" THRU 3'-4"	(1) #5	25"	8"	-	-
ML-2	1'-4" THRU 3'-4"	(2) #5	25"	16"	-	-
ML-3	3'-5" THRU 6'-3"	(2) #5	25"	24"	-	-
ML-4	6'-4" THRU 10'-0"	(2) #6	30"	32"	-	-
ML-5	10'-1" THRU 12'-0"	(2) #7	35"	32"	#3@16"	-
ML-5	PER PLAN	(2) #6	30"	32"	#3@16"	-



LINTEL AT OPENING

LINTEL NOTES:

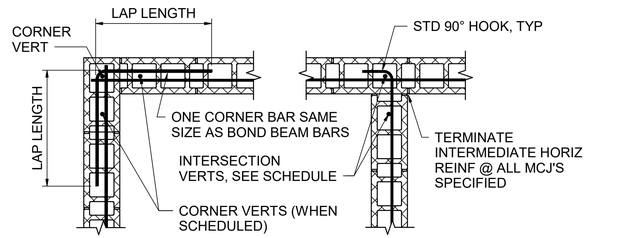
1. OPENINGS 8" OR LESS WIDE MAY OCCUR WITHOUT LINTEL REINFORCING AS LONG AS NO REINFORCINGS IS INTERUPTED.
2. ML-1 TO BE USED ONLY AT NON-LOAD BEARING SITUATIONS.
3. SEE DRAWINGS FOR LINTEL TYPES. WHERE LINTEL TYPES ARE NOT SHOWN, PROVIDE LINTELS FROM THE ABOVE SCHEDULE BASED ON THICKNESS OF WALL AND MAX CLEAR.
4. OPENING WIDTH AND VERIFY LINTEL TYPE W/ CONTRACTING OFFICER PRIOR TO CONSTRUCTION.



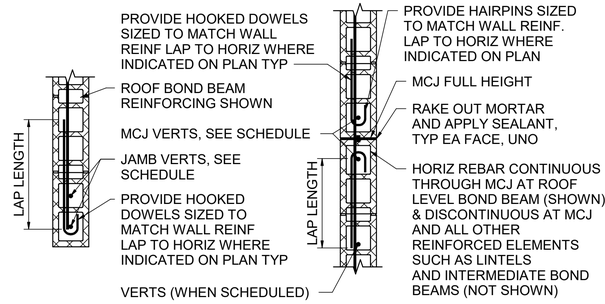
WALL ELEVATION

WALL ELEVATION NOTES:

1. ONLY LINTEL GROUT IS SHOWN. GROUT SOLID ALL REINFORCED CELLS. SEE CMU WALL REINFORCING SCHEDULE.
2. WHERE OTHER DRAWING DETAILS INDICATE SOLID MASONRY SILL, PLACE BOND BEAM REINF IN FIRST COURSE BELOW SOLID CMU.
3. PROVIDE BOND BEAM AT ALL ELEVATED FLOORS WITH SAME REINFORCING AS TOP OF WALL, UNO.
4. STRAIGHT BARS EMBEDDED ONE CONCRETE LAP LENGTH INTO CONC FOUNDATION MAY BE USED AT CONTRACTOR'S OPTION.
5. LINTEL REINFORCING AND BARS PASSING THROUGH "H" SHALL NOT BE SPLICED.
6. SHORE LINTEL MINIMUM 7 DAYS AFTER GROUTING OR UNTIL GROUT ATTAINS FULL DESIGN STRENGTH.



CORNER **INTERSECTION**



JAMB OR END OF WALL **MASONRY CONTROL JOINT (MCJ)**

NOTES:

1. FOR REINFORCING, SEE CMU WALL REINFORCING SHEDULE.
2. INDICATES LOCATION OF VERTICAL BARS AT CENTERLINE OF WALL, UNO IN SCHEDULE.
3. EXTEND MCJ FULL HEIGHT OF MASONRY BOND BEAM.
4. LIMIT DISTANCE BETWEEN MCJ TO MAX 24'-0". SEE DRAWINGS FOR LOCATIONS.
5. HORIZONTAL JOINT REINFORCING NOT SHOWN.
6. MODIFY BAR CONFIGURATION SHOWN AS REQUIRED WHERE TWO VERTICAL REINFORCING BARS ARE SHOWN ON THE SCHEDULE.
7. SEE ARCHITECTURAL DRAWINGS FOR MCJ DETAILS AND REMAINDER OF JOINT DETAILS.

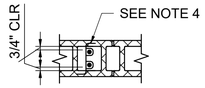
2 060S003 CMU WALL REINFORCING

NOT TO SCALE

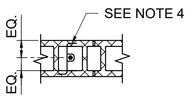
	STRUCTURE / WALL LOCATION	TYPICAL BEARING WALLS	CHEMICAL STORAGE ROOM
TYPE	REINFORCING LOCATION	12" WALL THK	8" WALL THK
VERTS	VERTICAL BARS	(2)#6 @ 16" OC DOUBLE CENTERED	#5 @ 16" OC
	JAMBS/ENDS/MCJ	(2)#6, SEE DETAIL 2 THIS SHEET	(1)#5, SEE DETAIL 2 THIS SHEET
	CORNERS / INTERSECTIONS	(2)#6, SEE DETAIL 2 THIS SHEET	(1)#5, SEE DETAIL 2 THIS SHEET
HORIZ	TOP OF WALL BOND BEAMS	(2)#6, 16" TALL BOND BEAMS	(2)#5
	INTERMEDIATE BOND BEAMS	(2)#6 @4'-0" VERT SPACING	(1)#5 @4'-0" VERT SPACING
	HORIZ JOINT REINF	9 GAGE WIRE, TRUSS TYPE JOINT REINF @ 16" OC BETWEEN BOND BEAMS	9 GAGE WIRE, TRUSS TYPE JOINT REINF @ 16" OC BETWEEN BOND BEAMS
	BELOW OPENINGS	(2)#6, SEE DETAIL 1 THIS SHEET	(1)#5, SEE DETAIL 1 THIS SHEET
	LINTELS	SEE CMU REINFORCEMENT ELEVATION DETAIL 1 THIS SHEET	SEE CMU REINFORCEMENT ELEVATION DETAIL 1 THIS SHEET

NOTES:

1. PROVIDE THE ABOVE SCHEDULED MINIMUM WALL REINFORCING IN ALL CMU, UNO. SEE CMU WALL REINFORCEMENT DETAIL.
2. GROUT ONLY THE CELLS WITH REINFORCING.
3. MAINTAIN MINIMUM 3" X 3" CONTINUOUS VERTICAL CELL AT EACH REBAR. PLACE WALLS TO MAXIMUM 4'-0" HEIGHT BEFORE GROUTING.
4. PROVIDE WIRE REBAR POSITIONERS TO HOLD BARS IN PLACE.
5. STOP GROUT POUR 1/2" BELOW TOP OF COURSE AT EACH GROUT LIFT.



DOUBLE CENTERED



SINGLE CENTERED

3 060S003 CMU REINFORCING SCHEDULE

NOT TO SCALE

MASONRY LAP SPLICE LENGTHS: f'm=2,000 psi, fy=60,000 psi				
BAR SIZE	8" BLOCK		12" BLOCK	
	BAR @ CL	BAR @ EDGE	BAR @ CL	BAR @ EDGE
#3	19"	19"	19"	19"
#4	25"	31"	25"	28"
#5	31"	48"	31"	43"
#6	57"	-	52"	-

WHEN REQD SPLICE LENGTH EXCEEDS 4'-0", USE HIGH LIFT GROUTING WITH NO SPLICES OR USE MECHANICAL TENSION SPLICES WITH LOW LIFT GROUTING.

4 060S003 CMU REINFORCING LAP SCHEDULE

NOT TO SCALE

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	C. MULDERICK	060 S003	GENERAL TYPICAL CMU DETAILS	113 182752
	DRAWN: R. NELSON			PMIS/PKG NO. 318719
	TECH. REVIEW: L. MENDOZA			SHEET 7 OF 192
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_062_NWTP_S.rvt
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TREATMENT FLOW RATES

RAW WATER FEED RATE (MIN/MAX)	22 - 108 GPM
NET FINISHED WATER PRODUCTION RATE (MIN/MAX)	20 - 100 GPM
MINIMUM RECOVERY RATE (OVERALL PROCESS)	92%

RAW WATER TANKS

TYPE	WELDED STEEL
NUMBER	2
TANK VOLUME (EA)	150,000 GAL
TANK VOLUME (TOTAL)	300,000 GAL
DETENTION TIME AT MAX PUMPING RATE	46 HR

RAW WATER PUMPS

NUMBER OF PUMP	1
PUMP TYPE	INLINE VERTICAL
DRIVE TYPE	VFD
DESIGN CAPACITY	108 GPM
TDH AT DESIGN CAPACITY	9.5 FT
HORSEPOWER	0.5 HP

REACTION TANKS

TYPE	CONICAL BOTTOM CARBON STEEL
NUMBER	2
TANK VOLUME (EA)	2500 GAL
TANK VOLUME (OPERATING RANGE)	1020 GAL
DETENTION TIME (MIN/MAX)	20 - 100 MIN
MIXER TYPE	ELECTRIC, IMPELLER
DRIVE TYPE	VFD
NUMBER PER TANK	1
G VALUE	60 - 250 SEC-1
HORSEPOWER	5 HP

ULTRAFILTRATION MEMBRANE TREATMENT SYSTEM

MEMBRANE SYSTEM	
NUMBERS OF TRAINS/SKIDS	2 (1 DUTY, 1 STANDBY)
FILTRATE CAPACITY PER TRAIN (EA) (MIX/MAX)	20 - 100 GPM
MAXIMUM FLUX RATE @ 20°C	20 GFD
MAXIMUM TRANSMEMBRANE PRESSURE LOSS	25 PSI
FEED PUMPS	
NUMBER	2 (1 PER SKID)
PUMP TYPE	CLOSE COUPLED CENTRIFUGAL
DRIVE TYPE	VFD
DESIGN CAPACITY	100 GPM
TDH AT DESIGN CAPACITY	150 FT
HORSEPOWER	15 HP
CIP SOLUTION TANK	
NUMBER	1
VOLUME	500 GAL
TANK HEATER	25 KW
CIP CIRCULATION PUMPS	
NUMBER	1 (MASTER SKID ONLY)
PUMP TYPE	END SUCTION, CENTRIFUGAL
DRIVE TYPE	MOTOR
DESIGN CAPACITY	20 GPM
TDH AT DESIGN CAPACITY	30 FT
HORSEPOWER	0.5 HP
SCREENS	
NUMBER	2 (1 PER SKID)
SCREEN SIZE	200 MICRON
TYPE OF CLEANING SYSTEM	SELF-CLEANING, REMAINING IN SERVICE DURING CLEANING

FINISHED WATER AND HIGH DEMAND PUMPS

FW TANK PUMPS	
NUMBER OF PUMPS	2 (1 DUTY, 1 STANDBY)
PUMP TYPE	INLINE VERTICAL
DRIVE TYPE	VFD
DESIGN CAPACITY (EA)	100 GPM
TDH AT DESIGN CAPACITY	37 FT
HORSEPOWER	2.0 HP
UPPER ZONE BOOSTER PUMPS	
NUMBER OF PUMPS	3 (2 DUTY, 1 STANDBY)
PUMP TYPE	INLINE VERTICAL
DRIVE TYPE	VFD
DESIGN CAPACITY (EA)	45 GPM
TDH AT DESIGN CAPACITY	104 FT
HORSEPOWER	3.0 HP
HIGH DEMAND PUMP	
NUMBER OF PUMPS	3 (2 DUTY, 1 STANDBY)
PUMP TYPE	INLINE VERTICAL
DRIVE TYPE	VFD
DESIGN CAPACITY (EA)	1,000 GPM
TDH AT DESIGN CAPACITY	104 FT
HORSEPOWER	40 HP
HYDRO-PNEUMATIC TANKS (BLADDER STYLE)	
NUMBER	2
TANK VOLUME (EA)	185 GAL
TANK VOLUME (TOTAL)	370 GAL
OPERATIONAL VOLUME (EFFECTIVE)	35 GAL
RATED OPERATING PRESSURE	100 PSI

FINISHED WATER TANKS

TYPE	WELDED STEEL
NUMBER	2
TANK VOLUME (EA)	2,000,000 GAL
TANK VOLUME (TOTAL)	4,000,000 GAL

PROCESS WASTE TANK

TYPE	PRECAST CONCRETE
DIMENSIONS	6 FT W X 6 FT L X 12 FT H
OPERATING VOLUME	550 GAL

CT TANK

DESIGN FLOW RATE	100 GPM
VIRUS INACTIVATION	4 LOG (MIN)
REQUIRED CT	8 @ 5˚C
BAFFLING FACTOR	0.15
CHLORINE CONCENTRATION	1.2 MG/L
STORAGE VOLUME AT MAX DEPTH	5240 GAL
MINIMUM VOLUME REQUIRED FOR CT	4600 GAL
DEPTH IN TANK FOR CT	8 FT

CHEMICAL FEED SYSTEMS

COAGULANT

TYPE	LIQUID
ACTIVE STRENGTH	48% (ALUM) 50% (ACH)
STORAGE	
TYPE OF STORAGE	55 GAL DRUM
TOTAL WORKING VOLUME	110 GAL
NUMBER OF OPERATING DRUMS	2
CHEMICAL METERING PUMPS	
APPLICATION POINT	REACTION TANK FEED
DOSAGE (TYP)	5 MG/L
NUMBER OF CONTRACTOR SUPPLIED PUMPS	1
NUMBER OF VENDOR SUPPLIED PUMPS	0
CONTRACTOR SUPPLIED PUMP CAPACITY	0.004 - 0.10 GPH

CAUSTIC SODA

TYPE	LIQUID
ACTIVE STRENGTH	25%
STORAGE	
TYPE OF STORAGE	55 GAL DRUM
TOTAL WORKING VOLUME	110 GAL
NUMBER OF OPERATING DRUMS	2
CHEMICAL METERING PUMPS	
APPLICATION POINT	UF CIP SYSTEM (BY MFR)
APPLICATION POINT	UPSTREAM OF CT TANK
DOSAGE (TYP)	3 MG/L
NUMBER OF CONTRACTOR SUPPLIED PUMPS	1
NUMBER OF VENDOR SUPPLIED PUMPS	1
CONTRACTOR SUPPLIED PUMP CAPACITY	0.01 - 0.10 GPH

SODIUM HYPOCHLORITE

TYPE	LIQUID
ACTIVE STRENGTH	12.5%
STORAGE	
TYPE OF STORAGE	300 GAL TOTE
TOTAL WORKING VOLUME	300 GAL
NUMBER OF OPERATING TOTES	1
CHEMICAL METERING PUMPS	
APPLICATION POINT	UF CIP SYSTEM (BY MFR)
APPLICATION POINT	REACTION TANK FEED
APPLICATION POINT	UPSTREAM OF CT TANK FINISHED WATER PUMP
APPLICATION POINT	FEED
DOSAGE (TYP)	2.5 MG/L
NUMBER OF CONTRACTOR SUPPLIED PUMPS	4
NUMBER OF VENDOR SUPPLIED PUMPS	1
CONTRACTOR SUPPLIED PUMP CAPACITY	0.004 - 0.14 GPH

SODIUM BISULFITE

TYPE	LIQUID
ACTIVE STRENGTH	38%
STORAGE	
TYPE OF STORAGE	55 GAL DRUM
TOTAL WORKING VOLUME	55 GAL
NUMBER OF OPERATING DRUMS	1
CHEMICAL METERING PUMPS	
APPLICATION POINT	UF CIP SYSTEM (BY MFR)
NUMBER OF CONTRACTOR SUPPLIED PUMPS	0
NUMBER OF VENDOR SUPPLIED PUMPS	1

CITRIC ACID

TYPE	LIQUID
ACTIVE STRENGTH	50%
STORAGE	
TYPE OF STORAGE	55 GAL DRUM
TOTAL WORKING VOLUME	55 GAL
NUMBER OF OPERATING DRUMS	1
CHEMICAL METERING PUMPS	
APPLICATION POINT	UF CIP SYSTEM (BY MFR)
NUMBER OF CONTRACTOR SUPPLIED PUMPS	0
NUMBER OF VENDOR SUPPLIED PUMPS	1

HYDROCHLORIC ACID

TYPE	LIQUID
ACTIVE STRENGTH	31%
STORAGE	
TYPE OF STORAGE	55 GAL DRUM
TOTAL WORKING VOLUME	55 GAL
NUMBER OF OPERATING DRUMS	1
CHEMICAL METERING PUMPS	
APPLICATION POINT	UF CIP SYSTEM (BY MFR)
NUMBER OF CONTRACTOR SUPPLIED PUMPS	0
NUMBER OF VENDOR SUPPLIED PUMPS	1

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. STILLMAN DRAWN: D. DEATERLA TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 060 D601	TITLE OF SHEET GENERAL PROCESS DESIGN CRITERIA NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 8 OF 192

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GENERAL NOTES

A. ONLY FLOW CONTROL VALVES ARE SHOWN.

B. SEE THE 062Y SHEETS AS REFERENCED FOR DETAILED PROCESS AND INSTRUMENTATION DIAGRAMS.

KEY NOTES (X)

1. EQUIPMENT PROVIDED BY UF MEMBRANE MANUFACTURER.
2. PUMP NOT USED UNDER NORMAL OPERATING CONDITIONS.

LEGEND

- TREATMENT PROCESS FLOW
- WASTE PROCESS FLOW
- AIR SUPPLY
- P&IDs
- BUILDING

INSIDE NORTH RIM WTP BUILDING

062 Y601

RAW WATER TANKS (QTY 2)

FROM RSPH

062 Y602

RAW WATER FLOW CONTROL VALVE

RAW WATER PUMP DISCHARGE FLOW CONTROL VALVE

RAW WATER PUMP (2)

RAW WATER FLOW METER

COAG

HYPO

062 Y603

REACTION TANKS

062 Y604

UF MEMBRANE SKIDS (1)

SCREEN, TYP.

FEED PUMP, TYP.

MEMBRANE MODULES, TYP.

CIT

HYPO

HCL

CA

SBS

(1)

UF CIP SYSTEM

(1)

COMPRESSOR

062 Y605

FINISHED WATER FLOW METER

HYPO

CA

CHLORINE CONTACT TANK (CT TANK)

062 Y606

FW TANK PUMPS (1 DUTY, 1 STANDBY)

HYPO

062 Y607

FINISHED WATER TANKS (QTY 2)

COMBINED BACKFEED FLOW METER

062 Y608

SUPAI TUNNEL BACKFEED FLOW METER

BACKFEED TO SUPAI TUNNEL

BACKFEED TO INNER CANYON

LOWER ZONE DISTRIBUTION SYSTEM FLOW METER

TO DISTRIBUTION SYSTEM LOWER ZONE

HIGH DEMAND PUMPS (2 DUTY, 1 STANDBY)

062 Y609

UPPER ZONE DISTRIBUTION SYSTEM FLOW METER

TO DISTRIBUTION SYSTEM UPPER ZONE

UPPER ZONE BOOSTER PUMPS (1 DUTY, 2 STANDBY)

HYDRO PNEUMATIC TANK

HYDRO PNEUMATIC TANK

TO PROCESS WASTE TANK

TO PROCESS WASTE TANK

TO PROCESS WASTE TANK

COAGULANT AND CAUSTIC SYSTEMS	062 Y610
SODIUM HYPOCHLORITE SYSTEMS	062 Y611
SODIUM BISULFITE AND CITRIC ACID SYSTEMS	062 Y612
HYDROCHLORIC ACID SYSTEM	062 Y613

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. STILLMAN DRAWN: D. DEATERLA TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 060 D602	TITLE OF SHEET GENERAL TREATMENT PROCESS FLOW DIAGRAM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 9 OF 192

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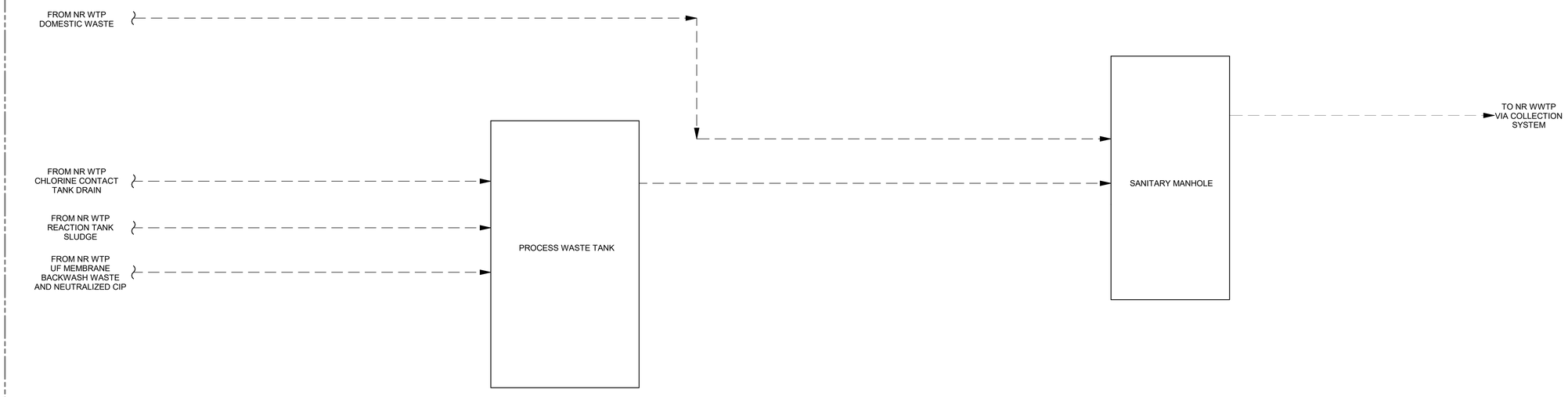
GENERAL NOTES

A. SEE THE 062Y SHEETS AS REFERENCED FOR DETAILED PROCESS AND INSTRUMENTATION DIAGRAMS.

LEGEND

- TREATMENT PROCESS FLOW
- WASTE PROCESS FLOW
- AIR SUPPLY
- P&IDs
- BUILDING

062 Y614



Registered Professional Engineer (Civil)
CERTIFICATE NO.
75827
JENNIFER
STILLMAN
Date Signed 5/1/2026
ARIZONA, U.S.A.

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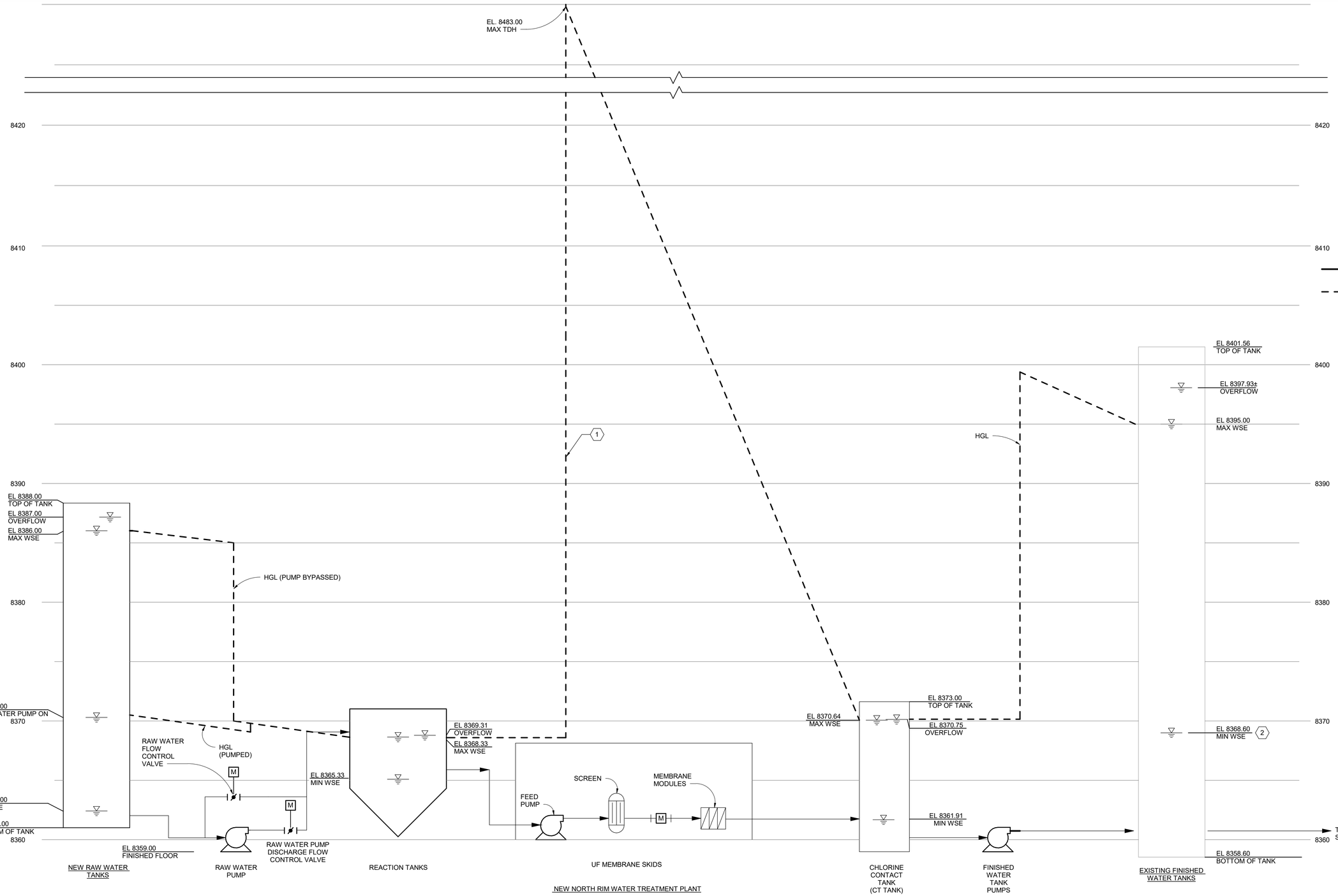

GENERAL NOTES

A. HYDRAULIC GRADELINE IS BASED ON 100 GPM PRODUCTION RATE TO FINISHED WATER TANKS.

B. THIS PROFILE IS CONCEPTUAL IN NATURE. ONLY THE FLOW PATH IS SHOWN AND NOT THE ACTUAL PIPE CONFIGURATION.

C. RAW WATER PUMP TO OPERATE ONLY WHEN THE WATER LEVEL IN THE RAW WATER TANK FALLS BELOW 10 FEET (WSE 8371.00').

KEY NOTES (X)

1. HGL SHOWN AT MAXIMUM TRANSMEMBRANE PRESSURE.
2. MINIMUM WSE IS SET TO ACHIEVE NPSHR OF THE HIGH DEMAND PUMPS

LEGEND

CONCEPTUAL FLOW PATH

HYDRAULIC GRADELINE

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/14/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. STILLMAN DRAWN: D. DEATERLA TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 060 D604	TITLE OF SHEET GENERAL HYDRAULIC PROFILE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 11 OF 192

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GENERAL NOTES

A. THIS SCHEDULE APPLIES TO INTERIOR PROCESS PIPING INSIDE THE NORTH RIM WTP. SEE CIVIL SHEETS FOR BURIED PIPING REQUIREMENTS.

B. USE HARD PIPE FOR CHEMICAL PIPING ROUTED ON WALLS AND USE TUBING FOR CHEMICAL PIPING ROUTED ABOVE PERSONNEL WALKING PATHS. SEE DRAWINGS FOR EXACT LOCATIONS.

C. CONTAINMENT PIPING FOR TUBING.

D. COLOR CODING TO BE APPLIED TO CONTAINMENT PIPING WHERE PROVIDED.

E. TUBING SIZE IS INNER DIAMETER.

PIPE SERVICE	DESCRIPTION	SPECIFICATION SECTION	SIZE	MATERIAL	CLASS OR WALL THICKNESS	INTERIOR LINING	EXTERIOR COATING	PRESSURE TESTING REQUIREMENTS			COLOR CODING (NOTE D)
AIR	COMPRESSED AIR	40 05 23	≤2"	SST	SCH 40	NONE	NONE	MEDIUM: AIR	DURATION: 6 HRS	TEST PRESSURE: 220 PSI	DARK GREEN BANDS
CA	CAUSTIC SODA	40 05 31	1/2" 1/2" (NOTE E) 1.5"	CPVC (NOTE B) TEFLON TUBING (NOTE B) PVC (NOTE C)	SCH 80 0.062" SCH 80	NONE	NONE	MEDIUM: WATER MEDIUM: WATER NONE	DURATION: 6 HRS DURATION: 6 HRS NONE	TEST PRESSURE: 50 PSI TEST PRESSURE: 50 PSI NONE	YELLOW AND GREEN BANDS
CIP	CLEAN-IN-PLACE	40 05 31	≤6"	CPVC	SCH 80	NONE	NONE	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 50 PSI	ORANGE AND BLUE BANDS
CIT	CITRIC ACID	40 05 31	1/2" 1/2" (NOTE E) 1.5"	PVC (NOTE B) TEFLON TUBING (NOTE B) PVC (NOTE C)	SCH 80 0.062" SCH 80	NONE	NONE	MEDIUM: WATER MEDIUM: WATER NONE	DURATION: 6 HRS DURATION: 6 HRS NONE	TEST PRESSURE: 50 PSI TEST PRESSURE: 50 PSI NONE	ORANGE AND RED BANDS
COAG	COAGULANT	40 05 31	1/2"	PVC	SCH 80	NONE	NONE	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 50 PSI	ORANGE BANDS
DR	BACKWASH WASTE/CIP DRAIN	40 05 24	6"	DIP	CLASS 53	EPOXY	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 2 HRS	TEST PRESSURE: 75 PSI	LIGHT BROWN COATING
FLT	FILTRATE	40 05 19	≥4"	DIP	CLASS 53	CEMENT	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 75 PSI	LIGHT BLUE COATING
HCL	HYDROCHLORIC ACID	40 05 31	1/2" 1/2" (NOTE E) 1.5"	PVC (NOTE B) TEFLON TUBING (NOTE B) PVC (NOTE C)	SCH 80 0.062" SCH 80	NONE	NONE	MEDIUM: WATER MEDIUM: WATER NONE	DURATION: 6 HRS DURATION: 6 HRS NONE	TEST PRESSURE: 50 PSI TEST PRESSURE: 50 PSI NONE	YELLOW AND RED BANDS
HYPO	SODIUM HYPOCHLORITE	40 05 31	1/2" 1/2" (NOTE E) 1.5"	CPVC (NOTE B) TEFLON TUBING (NOTE B) PVC (NOTE C)	SCH 80 0.062" SCH 80	NONE	NONE	MEDIUM: WATER MEDIUM: WATER NONE	DURATION: 6 HRS DURATION: 6 HRS NONE	TEST PRESSURE: 50 PSI TEST PRESSURE: 50 PSI NONE	YELLOW BANDS
RW1	RAW WATER	40 05 19	≥4"	DIP	CLASS 53	CEMENT	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 50 PSI	DARK GREEN COATING
SBS	SODIUM BISULFITE	40 05 31	1/2" 1/2" (NOTE E) 1.5"	PVC (NOTE B) TEFLON TUBING (NOTE B) PVC (NOTE C)	SCH 80 0.062" SCH 80	NONE	NONE	MEDIUM: WATER MEDIUM: WATER NONE	DURATION: 6 HRS DURATION: 6 HRS NONE	TEST PRESSURE: 50 PSI TEST PRESSURE: 50 PSI NONE	BLUE AND RED BANDS
SLG	SLUDGE	40 05 24	4"	DIP	CLASS 53	CEMENT	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 25 PSI	DARK BROWN COATING
SMP	SAMPLE	40 05 31	≥1/2"	PVC	SCH 80	NONE	NONE	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 50 PSI	MAGENTA BANDS
UFF	ULTRAFILTRATION FEED	40 05 19	≥4"	DIP	CLASS 53	CEMENT	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 75 PSI	ORANGE COATING
W1	FINISHED WATER	40 05 24 40 05 19	<4" ≥4"	STL DIP	SCH 40 CLASS 53	NONE CEMENT	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 150 PSI	DARK BLUE COATING

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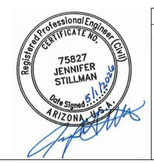
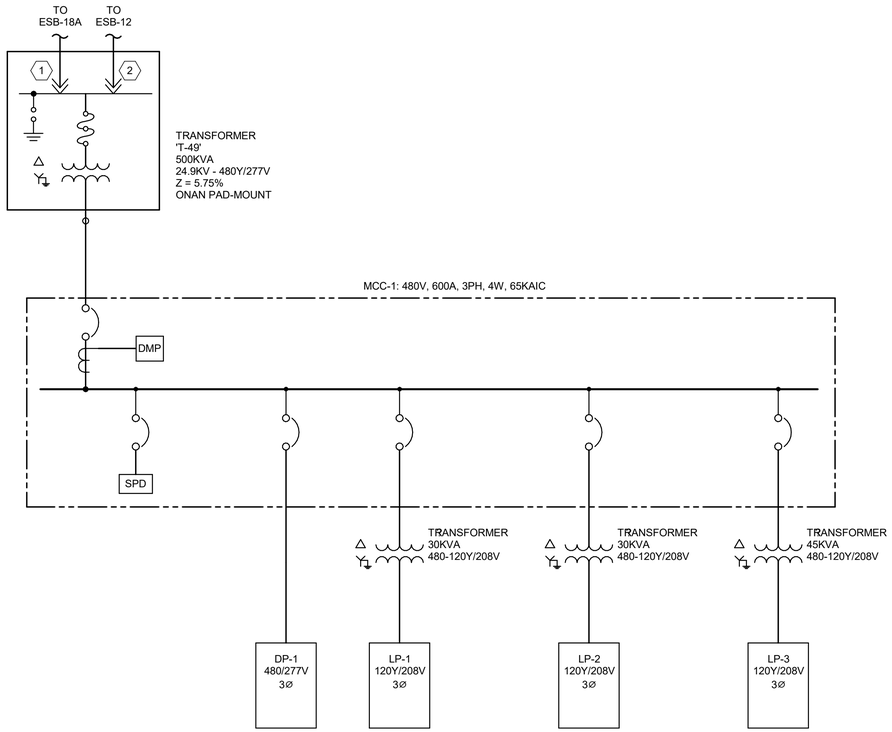


A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD CADD D. DEATERLA TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 060 D605	TITLE OF SHEET GENERAL PIPE SCHEDULE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 12 OF 192	



GENERAL NOTES

A. SITE-SPECIFIC STANDBY POWER IS NOT PROVIDED AT THE WATER TREATMENT PLANT SITE. BACKUP POWER IS PROVIDED BY THE CENTRAL BANK OF STANDBY GENERATORS AT THE POWER DISTRIBUTION CENTER.

B. REFER TO SHEETS 062E603 AND 062E604 FOR PANELBOARD SCHEDULES. ALL CIRCUITING AND DISCONNECTING INFORMATION IS CONTAINED WITHIN ONE LINE DIAGRAMS AND PANEL SCHEDULES.

C. REFER TO SHEET 091E602 FOR MEDIUM VOLTAGE ONE-LINE DIAGRAM.

KEYNOTES: ⟨X⟩

1 TO ELECTRICAL SPLICE BOX ESB-18A, APPROX. 20FT DIRECT-BURIED 25kV ARMORED ALUMINUM CABLE.

2 TO ELECTRICAL SPLICE BOX ESB-12 NEAR POWER DISTRIBUTION CENTER. APPROX. 2000FT DIRECT-BURIED 25kV ARMORED ALUMINUM CABLE.

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO.
58037
TRAVIS T. MOORE
Date Signed 2/18/2026
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAFTED: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 060 E601	TITLE OF SHEET GENERAL POWER DISTRIBUTION FUNCTIONAL DIAGRAM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 13 OF 192

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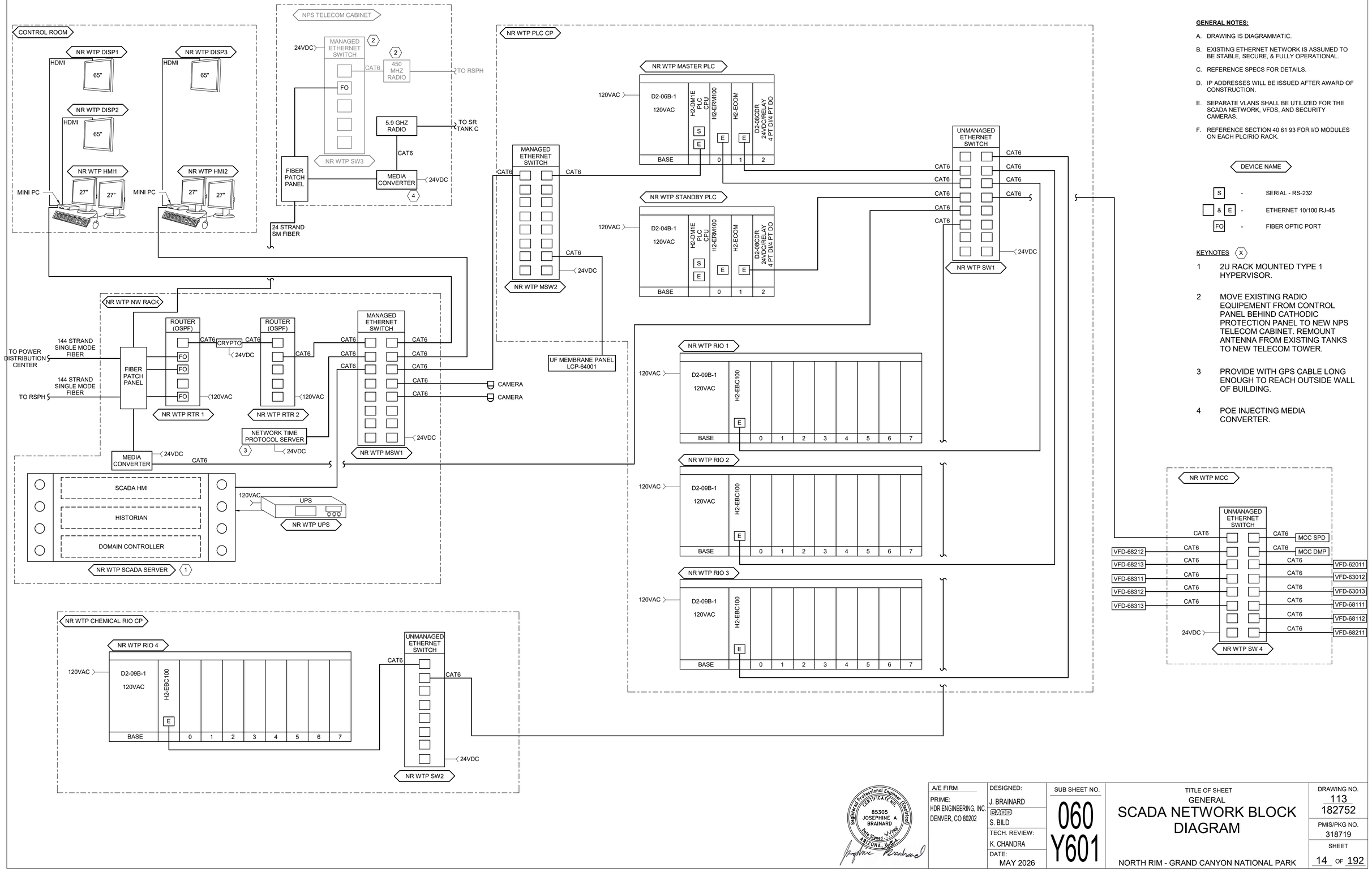

GENERAL NOTES:

A. DRAWING IS DIAGRAMMATIC.

B. EXISTING ETHERNET NETWORK IS ASSUMED TO BE STABLE, SECURE, & FULLY OPERATIONAL.

C. REFERENCE SPECS FOR DETAILS.

D. IP ADDRESSES WILL BE ISSUED AFTER AWARD OF CONSTRUCTION.

E. SEPARATE VLANS SHALL BE UTILIZED FOR THE SCADA NETWORK, VFDS, AND SECURITY CAMERAS.

F. REFERENCE SECTION 40 61 93 FOR I/O MODULES ON EACH PLC/RIO RACK.

DEVICE NAME

S - SERIAL - RS-232

& E - ETHERNET 10/100 RJ-45

FO - FIBER OPTIC PORT

KEYNOTES (X)

1 2U RACK MOUNTED TYPE 1 HYPERVISOR.

2 MOVE EXISTING RADIO EQUIPEMENT FROM CONTROL PANEL BEHIND CATHODIC PROTECTION PANEL TO NEW NPS TELECOM CABINET. REMOUNT ANTENNA FROM EXISTING TANKS TO NEW TELECOM TOWER.

3 PROVIDE WITH GPS CABLE LONG ENOUGH TO REACH OUTSIDE WALL OF BUILDING.

4 POE INJECTING MEDIA CONVERTER.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 4/27/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: S. BILD TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 060 Y601	TITLE OF SHEET GENERAL SCADA NETWORK BLOCK DIAGRAM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 14 OF 192

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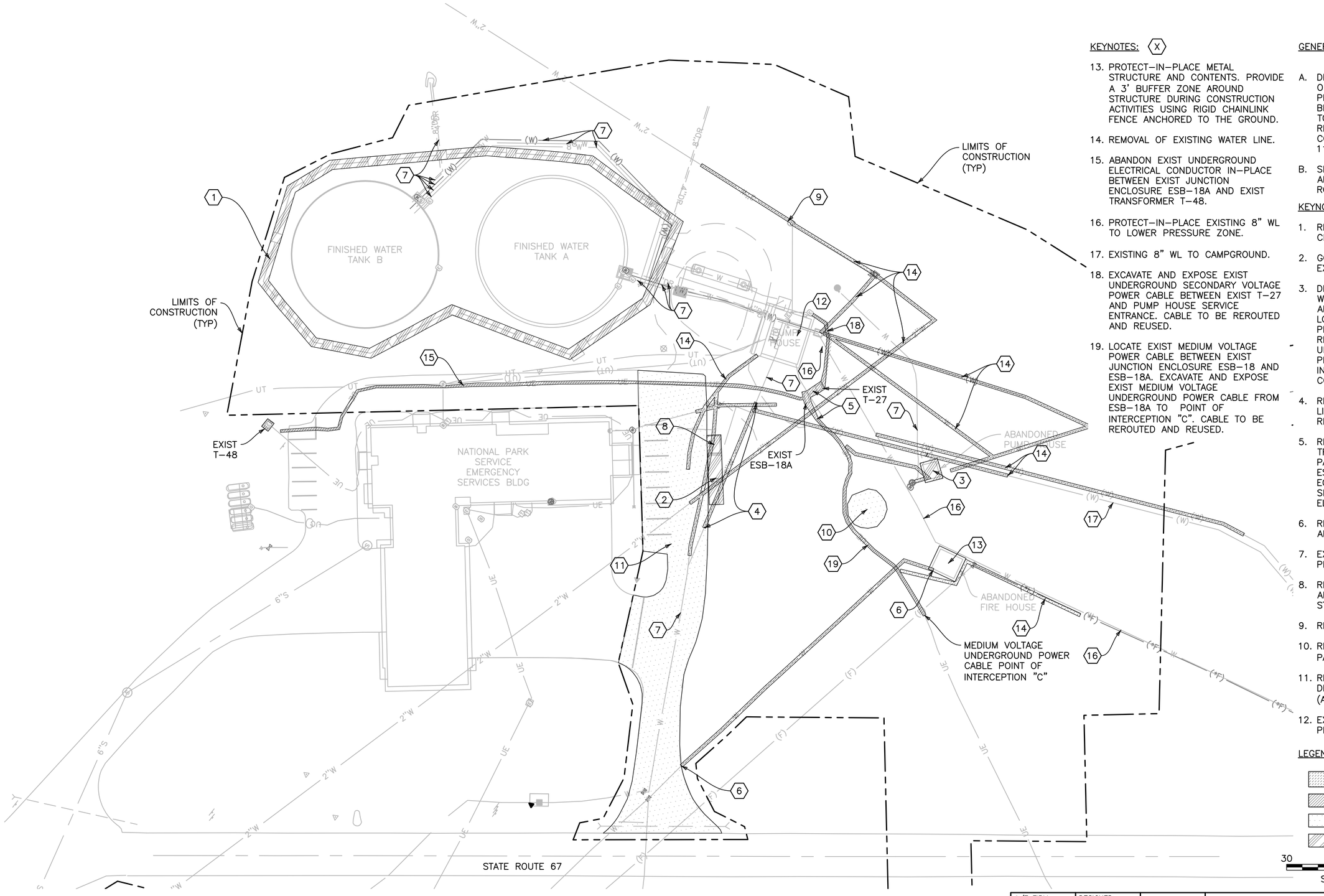


KEYNOTES: (X)

13. PROTECT–IN–PLACE METAL STRUCTURE AND CONTENTS. PROVIDE A 3' BUFFER ZONE AROUND STRUCTURE DURING CONSTRUCTION ACTIVITIES USING RIGID CHAINLINK FENCE ANCHORED TO THE GROUND.
14. REMOVAL OF EXISTING WATER LINE.
15. ABANDON EXIST UNDERGROUND ELECTRICAL CONDUCTOR IN–PLACE BETWEEN EXIST JUNCTION ENCLOSURE ESB–18A AND EXIST TRANSFORMER T–48.
16. PROTECT–IN–PLACE EXISTING 8" WL TO LOWER PRESSURE ZONE.
17. EXISTING 8" WL TO CAMPGROUND.
18. EXCAVATE AND EXPOSE EXIST UNDERGROUND SECONDARY VOLTAGE POWER CABLE BETWEEN EXIST T–27 AND PUMP HOUSE SERVICE ENTRANCE. CABLE TO BE REROUTED AND REUSED.
19. LOCATE EXIST MEDIUM VOLTAGE POWER CABLE BETWEEN EXIST JUNCTION ENCLOSURE ESB–18 AND ESB–18A. EXCAVATE AND EXPOSE EXIST MEDIUM VOLTAGE UNDERGROUND POWER CABLE FROM ESB–18A TO POINT OF INTERCEPTION "C". CABLE TO BE REROUTED AND REUSED.

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SEE VOLUME 0 FOR STAGING AREAS AND TEMPORARY ACCESS ROAD.

KEYNOTES: (X)

1. REMOVAL OF 664 LF EXISTING CHAIN LINK FENCE.
2. GOVERNMENT TO RELOCATE EXISTING STORAGE CABINETS.
3. DEMOLISH AND REMOVE SMALL WOOD STRUCTURE, APPURTENANCES, AND CONCRETE PAD. DISPOSAL LOCATION IS OUTSIDE OF THE PARK PER STATE AND LOCAL REGULATIONS. EXISTING UNDERGROUND PIPING SHALL BE PROTECTED IN PLACE AND REMAIN IN SERVICE UNTIL SUBSTANTIAL COMPLETION.
4. REMOVAL OF 90 LF EXISTING FW LINE. SEE SHEET 061C105 FOR REROUTING OF LINE.
5. REMOVE EXIST PADMOUNT TRANSFORMER T–27 AND PADMOUNT JUNCTION ENCLOSURE ESB–18A. REMOVE EXIST EQUIPMENT ENCLOSURE SLEEVES. SEE SHEET 061E101 FOR ELECTRICAL WORK.
6. REMOVAL OF 215 LF EXISTING ABANDONED WATER LINE.
7. EXISTING WATER LINE TO REMAIN. PROTECT–IN–PLACE.
8. REMOVAL OF EXISTING ABSORBENT BIN LOCATED NEXT TO STORAGE CABINETS.
9. REMOVAL OF EXISTING CMP SUMP.
10. REMOVAL OF EXISTING CONCRETE PAD. QUANTITY= 53.7 SY (APPROX).
11. REMOVAL OF EXISTING ASPHALT DRIVEWAY. QUANTITY= 1263.6 SY (APPROX).
12. EXISTING BUILDING TO REMAIN. PROTECT–IN–PLACE.

LEGEND:

- PIPE DEMOLITION
- ELECTRICAL DEMOLITION
- PAVEMENT DEMOLITION
- GENERAL DEMOLITION

30 0 30 60 1" = 30'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — Date Signed — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 061 X101	TITLE OF SHEET SITEWORK DEMOLITION PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 15 OF 192

4/18/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\061–X101.dwg

KEY NOTES: (X)

1. SEE VOLUME 01 AND 03 FOR ADDITIONAL LIMITS OF TREE REMOVAL, OUTSIDE OF LIMITS SHOWN ON THIS SHEET, VOLUME 06.

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SEE 061X103 FOR TREE REMOVAL TABLE. CONTRACTOR TO CONFIRM THE NUMBER, SIZE AND CALIPER OF TREES TO BE REMOVED.

C. APPROXIMATE AREA OF TREE REMOVAL SHOWN ON THIS SHEET = 1.34 ACRES.

D. PROVIDE TREE PROTECTION FOR ALL TREES TO REMAIN WITHIN 10 LF OFFSET FROM THE LIMITS OF TREE REMOVAL. TREE PROTECTION MAY BE PROVIDED WITH FENCING AND PROTECTION MEASURES ALONG THE LENGTH OF THE LIMITS OF DISTURBANCE OR ON AN INDIVIDUAL TREE BASIS.

E. CONTRACTOR SHALL ASSESS AND IDENTIFY TREES THAT QUALIFY FOR RELOCATION WITHIN THE LIMITS OF TREE REMOVAL. THESE TREES SHALL BE A MINIMUM OF 5–INCH DBH, SINGLE STEM WITH A HEALTHY DOMINANT LEADER, AND HAVE NO BROKEN LIMBS. THE CONTRACTOR SHALL CONFIRM 70% OF THE TREES THAT MEET THESE CRITERIA AS TREES TO BE RELOCATED IN THE TREE LOCATION AREAS DESIGNATED ON THE LANDSCAPE PLANS. NPS WILL REVIEW AND APPROVE FINAL TREE SELECTION PRIOR TO SITE DISTURBANCE. REFER TO TREE RELOCATION SPECIFICATION FOR TREE REMOVAL AND INSTALLATION.

F. SEE SPECIFICATION 01 11 00 FOR SLASH AND TREE REMOVAL REQUIREMENTS.

G. PROTECT–IN–PLACE TREES NOT INDICATED FOR REMOVAL WITHIN LIMITS OF CONSTRUCTION. CONTRACTOR SHALL INSTALL MULCH WITHIN TREE DRIP LINES TO PROTECT ROOT SYSTEMS. LIMIT HEAVY EQUIPMENT ACTIVITY WITHIN TREE DRIP LINES.

30 0 30 60
SCALE OF FEET 1" = 30'

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — Date Signed — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: JUNE 2026	SUB SHEET NO. 061 X102	TITLE OF SHEET SITEWORK TREE REMOVAL AND PROTECTION PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 16 OF 192

8/18/2026 Green, Jake C:\Users\jgreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-X102.dwg

Tree #	Quantity	Tree Type	Diameter (in)
1	1	PINE	10
2	1	PINE	10
3	1	PINE	15
4	1	PINE	12
5	1	PINE	10
6	1	PINE	30
7	1	PINE	14
8	1	PINE	14
9	1	PINE	12
10	1	PINE	10
11	1	PINE	11
12	1	PINE	10
13	1	PINE	15
14	1	PINE	10
15	1	PINE	14
16	1	PINE	24
17	1	PINE	10
18	1	PINE	12
19	1	PINE	17
20	1	PINE	10
21	1	PINE	12
22	1	PINE	17
23	1	PINE	14
24	1	PINE	28
25	1	PINE	16
26	1	PINE	11
27	1	PINE	10
28	1	PINE	11
29	1	PINE	10
30	1	PINE	10
31	1	PINE	10
32	1	PINE	14
33	1	PINE	10
34	1	PINE	16
35	1	PINE	10
36	1	PINE	10
37	1	PINE	10
38	1	PINE	10
39	1	PINE	10
40	1	PINE	10
41	1	PINE	24
42	1	PINE	10
43	1	PINE	10
44	1	PINE	10
45	1	PINE	13
46	1	PINE	11
47	1	PINE	10
48	1	PINE	13
49	1	PINE	10
50	1	PINE	11
51	1	PINE	12
52	1	PINE	10
53	1	PINE	10
54	1	PINE	28
55	1	PINE	10
56	1	PINE	17
57	1	PINE	10
58	1	PINE	10
59	1	PINE	10
60	1	PINE	12
61	1	PINE	10
62	1	PINE	10
63	1	PINE	10
64	1	PINE	10
65	1	PINE	10
66	1	PINE	10
67	1	PINE	10
68	1	PINE	10
69	1	PINE	10
70	1	PINE	10
71	1	PINE	11
72	1	PINE	10
73	1	PINE	10
74	1	PINE	10
75	1	PINE	10

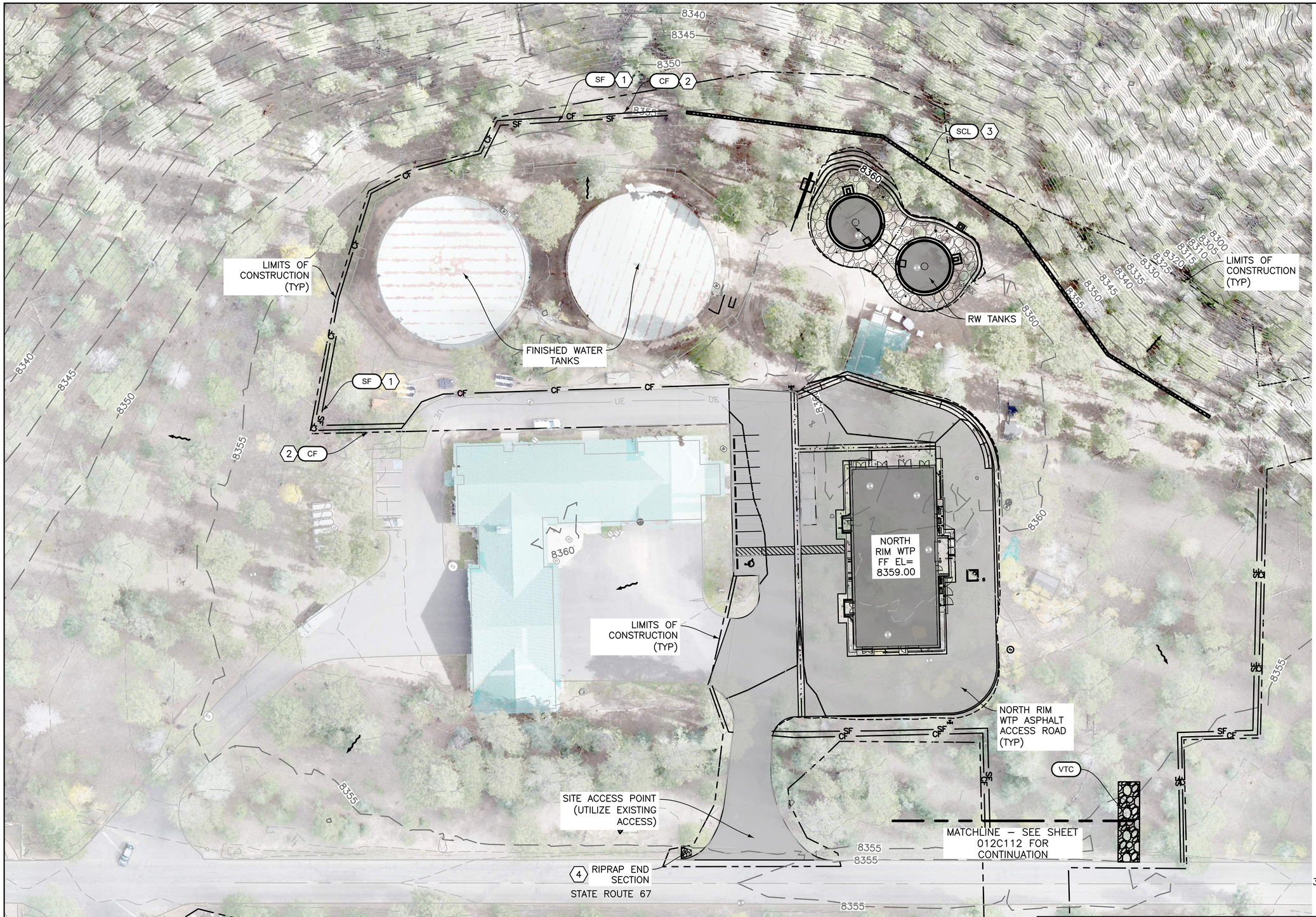
Tree #	Quantity	Tree Type	Diameter (in)
76	1	PINE	18
77	1	PINE	10
78	1	PINE	10
79	1	PINE	18
80	1	PINE	11
81	1	PINE	30
82	1	PINE	10
83	1	PINE	13
84	1	PINE	13
85	1	PINE	14
86	1	PINE	10
87	1	PINE	10
88	1	PINE	10
89	1	PINE	23
90	1	PINE	13
91	1	PINE	13
92	1	PINE	11
93	1	PINE	18
94	1	PINE	16
95	1	PINE	19
96	1	PINE	20
97	1	PINE	19
98	1	PINE	22
99	1	PINE	12
100	1	PINE	12
101	1	PINE	13
102	1	PINE	14
103	1	PINE	10
104	1	PINE	30
105	1	PINE	10
106	1	PINE	10
107	1	PINE	6
108	1	PINE	6
109	1	PINE	28
110	1	PINE	10
111	1	PINE	10
112	1	PINE	11
113	1	PINE	12
114	1	PINE	8
115	1	PINE	10
116	1	DEC	6
117	1	DEC	6
118	1	DEC	10
119	1	PINE	12
120	1	PINE	11
121	1	PINE	12
122	1	PINE	15
123	1	PINE	12
124	1	PINE	6
125	1	PINE	24
126	1	PINE	9
127	1	PINE	20
128	1	PINE	10
129	1	PINE	12
130	1	PINE	12
131	1	DEC	7
132	1	DEC	6
133	1	DEC	6
134	1	PINE	6
135	1	PINE	12
136	1	PINE	11
137	1	PINE	7
138	1	PINE	12
139	1	PINE	9
140	1	PINE	25
141	1	PINE	12
142	1	PINE	12
143	1	PINE	14
144	1	PINE	14
145	1	PINE	11
146	1	PINE	8
147	1	DEC	17
148	1	PINE	18
149	1	PINE	6
150	1	PINE	6
151	1	PINE	14
152	1	PINE	9

Tree #	Quantity	Tree Type	Diameter (in)
153	1	PINE	8
154	1	PINE	6
155	1	PINE	6
156	1	PINE	10
157	1	PINE	8
158	1	PINE	11
159	1	PINE	18
160	1	PINE	24
161	1	PINE	28
162	1	PINE	15
163	1	PINE	10
164	1	PINE	17
165	1	PINE	11
166	1	PINE	11
167	1	DEC	6
168	1	DEC	6
169	1	DEC	6
170	1	PINE	12
171	1	PINE	15
172	1	PINE	11
173	1	PINE	11
174	1	PINE	12
175	1	PINE	15
176	1	PINE	12
177	1	PINE	24
178	1	PINE	18
179	1	PINE	15
180	1	PINE	6
181	1	PINE	12
182	1	DEC	8
183	1	PINE	15
184	1	PINE	12
185	1	PINE	11
186	1	PINE	7
187	1	PINE	16
188	1	PINE	10
189	1	PINE	17
190	1	PINE	15
191	1	PINE	10
192	1	PINE	12
193	1	PINE	11
194	1	PINE	20
195	1	PINE	13
196	1	PINE	14
197	1	PINE	10
198	1	PINE	10
199	1	PINE	10
200	1	PINE	10
201	1	PINE	10
202	1	PINE	12
203	1	PINE	12
204	1	PINE	9
205	1	PINE	7
206	1	PINE	10
207	1	PINE	11
208	1	PINE	16
209	1	PINE	12
210	1	PINE	12
211	1	PINE	12
212	1	PINE	8
213	1	PINE	11
214	1	PINE	13
215	1	PINE	30
216	1	PINE	7
217	1	PINE	14
218	1	PINE	13
219	1	PINE	19
220	1	PINE	17
221	1	PINE	10

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — Date Signed 5/1/26 — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: J. GLOVER DATE: JUNE 2026	SUB SHEET NO. 061 X103	TITLE OF SHEET SITEWORK TREE REMOVAL TABLES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 17 OF 192

8/18/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-X103.dwg



GENERAL NOTES:

A. FOR EROSION CONTROL NOTES AND DETAILS SEE VOLUME 0 SHEETS 002C507 – 002C510

KEYNOTES: (X)

1. SILT FENCE
   1.1. QUANTITY = 658 LF (VOLUME 6 TOTAL)
2. CONSTRUCTION FENCE
   2.1. QUANTITY = 1295 LF (VOLUME 6 TOTAL)
3. SEDIMENT CONTROL LOG
   3.1. QUANTITY = 400 LF (VOLUME 6 TOTAL)
4. FOR RIPRAP END SECTION DETAIL SEE SHEET 05/061C502.

LEGEND:

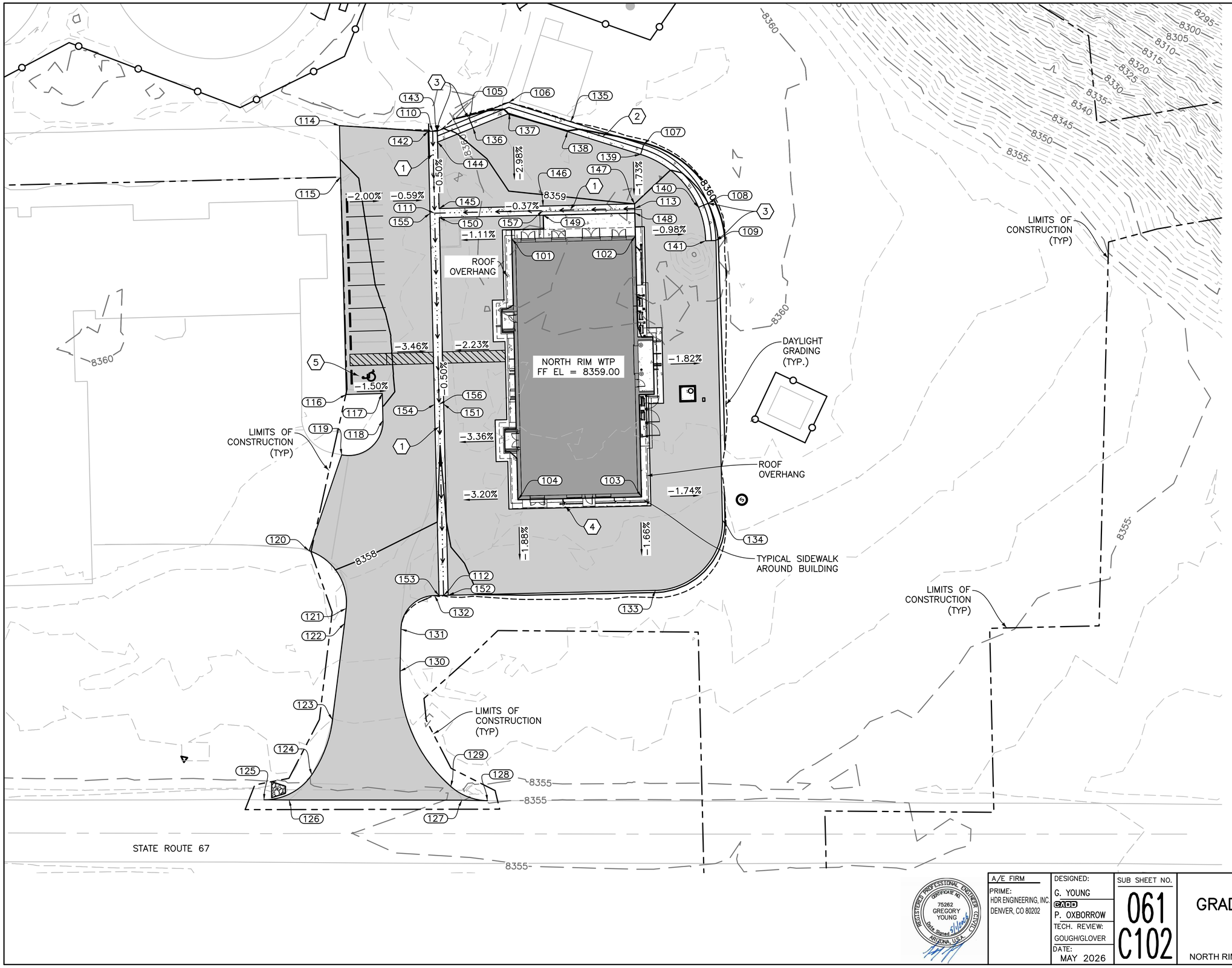
- SF — SF — SILT FENCE. SEE 01/002C509
- CF — CF — CONSTRUCTION FENCING 01/002C508
- VTC — VEHICLE TRACKING CONTROL 01/002C510
- SCL — SEDIMENT CONTROL LOG 03/002C509
- FLOW ARROW

30 0 30 60 1" = 30'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 061 C101	TITLE OF SHEET SITEWORK NORTH RIM WTP GESC PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 18 OF 192

4/18/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C101.dwg



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. ESTIMATED NET EARTHWORK VOLUME FOR THE WTP SITE IS: 292 CY (CUT) (APPROX)

C. SEE SPECIFICATIONS, DIVISION 31 – EARTHWORK FOR MORE INFORMATION ON SITE GRADING.

D. SEE SHEET 060L101 FOR LANDSCAPE PLAN.

E. SEE SHEET 061C104 FOR POINT TABLE.

KEY NOTES: (X)

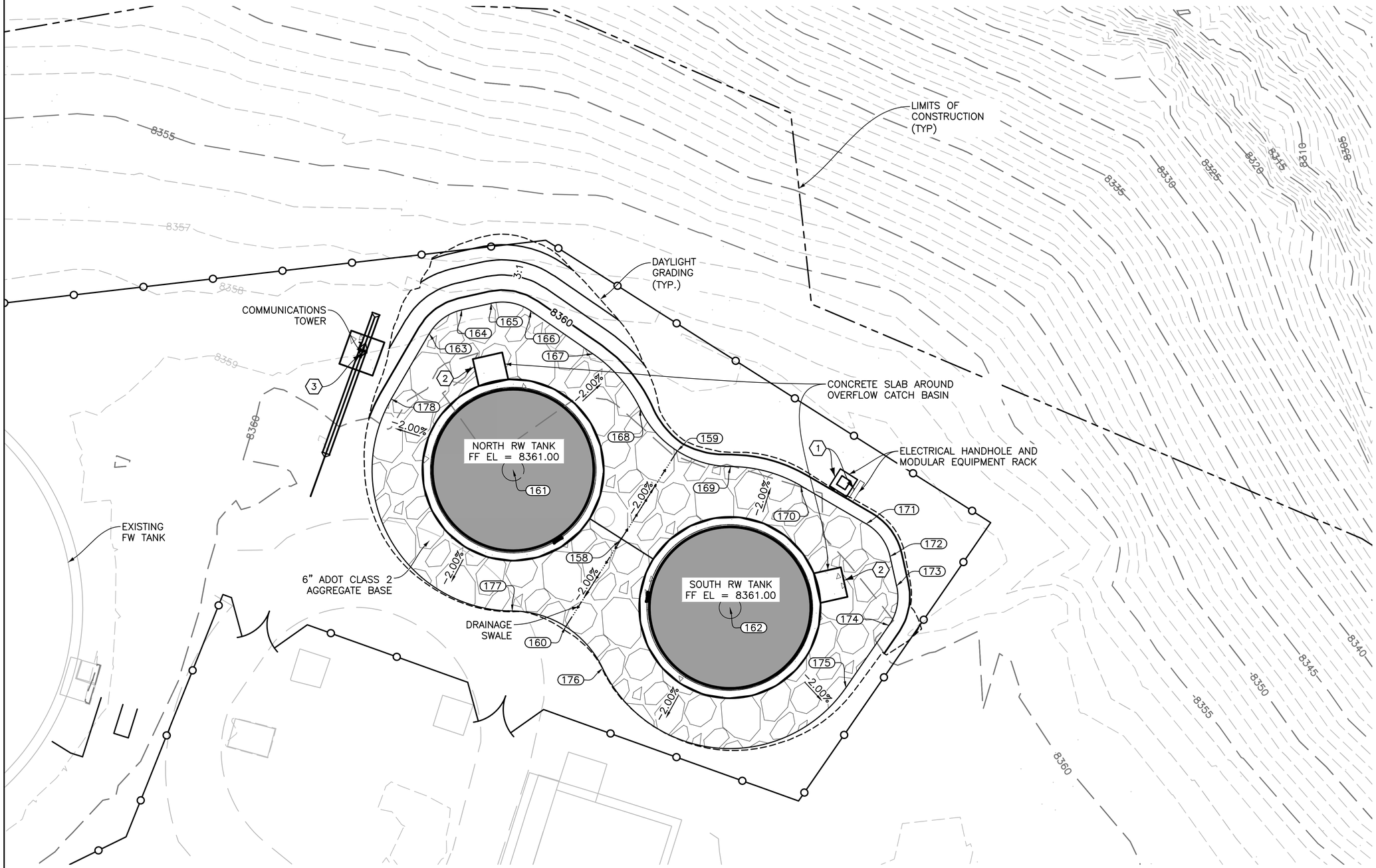
1. CONCRETE GUTTER PAN. SEE DETAIL 01/061C502 FOR DETAIL.
2. CONCRETE MOUNTABLE CURB. SEE DETAIL 02/061C502 FOR DETAIL.
3. CONCRETE CURB TRANSITION. SEE DETAIL 03/061C502 FOR DETAIL.
4. TYPICAL SIDEWALK SLOPED AT 2.0% AWAY FROM WTP. SEE DETAIL 03/002C506.
5. ADA PARKING STALL. SEE DETAIL 04/061C502 FOR DETAIL.

20 0 20 40
SCALE OF FEET 1" = 20'

REGISTERED PROFESSIONAL ENGINEER – CERTIFICATE NO. 75262 – GREGORY YOUNG – Date Signed – ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C102	TITLE OF SHEET SITEWORK GRADING AND PAVING PLAN – WTP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 19 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C102.dwg



GENERAL NOTES:

A. REFERENCE SHEET 061C102 FOR ADDITIONAL GENERAL NOTES.

B. SEE SHEET 061C104 FOR POINT TABLE.

KEY NOTES: (X)

1. SEE ELECTRICAL SHEETS FOR ADDITIONAL INFORMATION.
2. SEE PROCESS SHEETS FOR ADDITIONAL INFORMATION.
3. SEE STRUCTURAL SHEETS FOR ADDITIONAL INFORMATION.

10 0 10 20
SCALE OF FEET
1" = 10'

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — Date Signed — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C103	TITLE OF SHEET SITEWORK GRADING AND PAVING PLAN – RW TANKS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 20 OF 192

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POINT TABLE				
POINT NO.	NORTHING	EASTING	ELEVATION	DESCRIPTION
101	13150617.88	1328307.11	8359.00	BLDG CORNER
102	13150576.06	1328341.28	8359.00	BLDG CORNER
103	13150503.93	1328253.01	8359.00	BLDG CORNER
104	13150545.74	1328218.84	8359.00	BLDG CORNER
105	13150664.91	1328337.22	8359.56	BEGIN CURB AND GUTTER
106	13150655.56	1328353.33	8360.45	ANGLE POINT CURB AND GUTTER
107	13150597.71	1328374.98	8359.76	PC CURB AND GUTTER
108	13150559.77	1328371.99	8358.98	END CURB AND GUTTER/BEGIN CURB TRANSITION
109	13150547.10	1328361.31	8358.65	END CURB TRANSITION
110	13150673.99	1328322.98	8358.69	BEGIN FLOWLINE GUTTER PAN
111	13150651.34	1328295.26	8358.51	INTERSECTION FLOWLINE GUTTER PAN
112	13150545.68	1328165.96	8357.67	END FLOWLINE GUTTER PAN
113	13150583.66	1328350.58	8358.83	BEGIN FLOWLINE GUTTER PAN
114	13150707.61	1328299.90	8358.70	EDGE OF PAVEMENT
115	13150693.40	1328282.48	8359.16	ANGLE POINT EDGE OF PAVEMENT
116	13150633.11	1328209.11	8359.40	ANGLE POINT EDGE OF PAVEMENT
117	13150620.95	1328219.10	8359.08	ANGLE POINT EDGE OF PAVEMENT
118	13150613.53	1328210.01	8358.82	PC EDGE OF PAVEMENT
119	13150618.50	1328187.07	8358.73	PT EDGE OF PAVEMENT
120	13150603.96	1328145.25	8358.26	PC EDGE OF PAVEMENT
121	13150575.74	1328135.32	8357.67	PT EDGE OF PAVEMENT
122	13150572.25	1328129.76	8357.48	ANGLE POINT EDGE OF PAVEMENT
123	13150550.58	1328093.33	8355.97	PC EDGE OF PAVEMENT
124	13150542.96	1328068.81	8355.04	PCC EDGE OF PAVEMENT
125	13150552.70	1328047.33	8354.70	PT EDGE OF PAVEMENT
126	13150544.00	1328054.00	8354.80	ANGLE POINT EDGE OF PAVEMENT
127	13150484.40	1328100.30	8355.04	ANGLE POINT EDGE OF PAVEMENT
128	13150475.49	1328106.98	8354.99	PC EDGE OF PAVEMENT
129	13150491.93	1328102.03	8355.03	PCC EDGE OF PAVEMENT
130	13150540.33	1328128.14	8356.77	PT EDGE OF PAVEMENT
131	13150551.29	1328142.88	8357.30	PC EDGE OF PAVEMENT
132	13150548.75	1328163.45	8357.80	PT EDGE OF PAVEMENT
133	13150474.04	1328224.50	8358.31	PC EDGE OF PAVEMENT
134	13150469.80	1328266.71	8358.37	PT EDGE OF PAVEMENT
135	13150629.04	1328363.25	8360.42	CURB AND GUTTER
136	13150661.02	1328334.96	8359.13	EDGE OF GUTTER/PAVEMENT
137	13150652.47	1328349.68	8360.02	EDGE OF GUTTER/PAVEMENT
138	13150627.46	1328359.04	8359.99	EDGE OF GUTTER/PAVEMENT
139	13150596.13	1328370.76	8359.32	EDGE OF GUTTER/PAVEMENT
140	13150561.98	1328368.07	8358.55	EDGE OF GUTTER/PAVEMENT

POINT TABLE				
POINT NO.	NORTHING	EASTING	ELEVATION	DESCRIPTION
141	13150550.58	1328358.46	8358.69	EDGE OF CURB TRANSITION/PAVEMENT
142	13150675.54	1328321.71	8358.85	EDGE OF GUTTER PAN/PAVEMENT
143	13150672.44	1328324.25	8358.85	EDGE OF GUTTER PAN/CURB TRANSITION
144	13150669.40	1328320.52	8358.83	EDGE OF GUTTER PAN/CURB TRANSITION
145	13150651.06	1328298.08	8358.68	EDGE OF GUTTER PAN/PAVEMENT
146	13150615.90	1328326.81	8358.85	EDGE OF GUTTER PAN/PAVEMENT
147	13150584.92	1328352.13	8359.00	EDGE OF GUTTER PAN/PAVEMENT
148	13150582.39	1328349.03	8359.00	EDGE OF GUTTER PAN/CONCRETE PAD
149	13150613.37	1328323.72	8359.00	EDGE OF GUTTER PAN/CONCRETE PAD
150	13150648.53	1328294.98	8358.66	EDGE OF GUTTER PAN/PAVEMENT
151	13150596.96	1328231.88	8358.26	EDGE OF GUTTER PAN/PAVEMENT
152	13150544.13	1328167.23	8357.84	EDGE OF GUTTER PAN/PAVEMENT
153	13150547.22	1328164.69	8357.84	EDGE OF GUTTER PAN/PAVEMENT
154	13150600.06	1328229.35	8358.26	EDGE OF GUTTER PAN/PAVEMENT
155	13150652.89	1328294.00	8358.67	EDGE OF GUTTER PAN/PAVEMENT
156	13150598.51	1328230.61	8358.09	FLOWLINE GUTTER PAN
157	13150614.63	1328325.27	8358.68	FLOWLINE GUTTER PAN
158	13150683.23	1328431.73	8360.99	HP DRAINAGE SWALE
159	13150685.22	1328452.83	8360.59	END DRAINAGE SWALE
160	13150681.25	1328411.69	8360.59	END DRAINAGE SWALE
161	13150708.62	1328429.82	8361.00	CENTER NORTH RW TANK
162	13150657.85	1328434.71	8361.00	CENTER SOUTH RW TANK
163	13150738.38	1328440.95	8360.80	PC EDGE OF GRAVEL ROAD
164	13150736.18	1328448.65	8360.80	PT EDGE OF GRAVEL ROAD
165	13150732.37	1328453.28	8360.80	PC EDGE OF GRAVEL ROAD
166	13150725.23	1328456.90	8360.80	PT EDGE OF GRAVEL ROAD
167	13150710.24	1328457.77	8360.80	PC EDGE OF GRAVEL ROAD
168	13150696.09	1328454.86	8360.80	PRC EDGE OF GRAVEL ROAD
169	13150674.93	1328456.89	8360.80	PRC EDGE OF GRAVEL ROAD
170	13150661.60	1328462.46	8360.80	PT EDGE OF GRAVEL ROAD
171	13150646.72	1328464.47	8360.80	PC EDGE OF GRAVEL ROAD
172	13150639.02	1328462.27	8360.80	PT EDGE OF GRAVEL ROAD
173	13150634.39	1328458.46	8360.80	PC EDGE OF GRAVEL ROAD
174	13150630.77	1328451.32	8360.80	PT EDGE OF GRAVEL ROAD
175	13150629.90	1328436.33	8360.80	PC EDGE OF GRAVEL ROAD
176	13150670.38	1328409.67	8360.80	PRC EDGE OF GRAVEL ROAD
177	13150691.53	1328407.63	8360.80	PRC EDGE OF GRAVEL ROAD
178	13150736.36	1328426.07	8360.80	PT EDGE OF GRAVEL ROAD

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
State of ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C104	TITLE OF SHEET SITEWORK GRADING AND PAVING POINT TABLES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 21 OF 192

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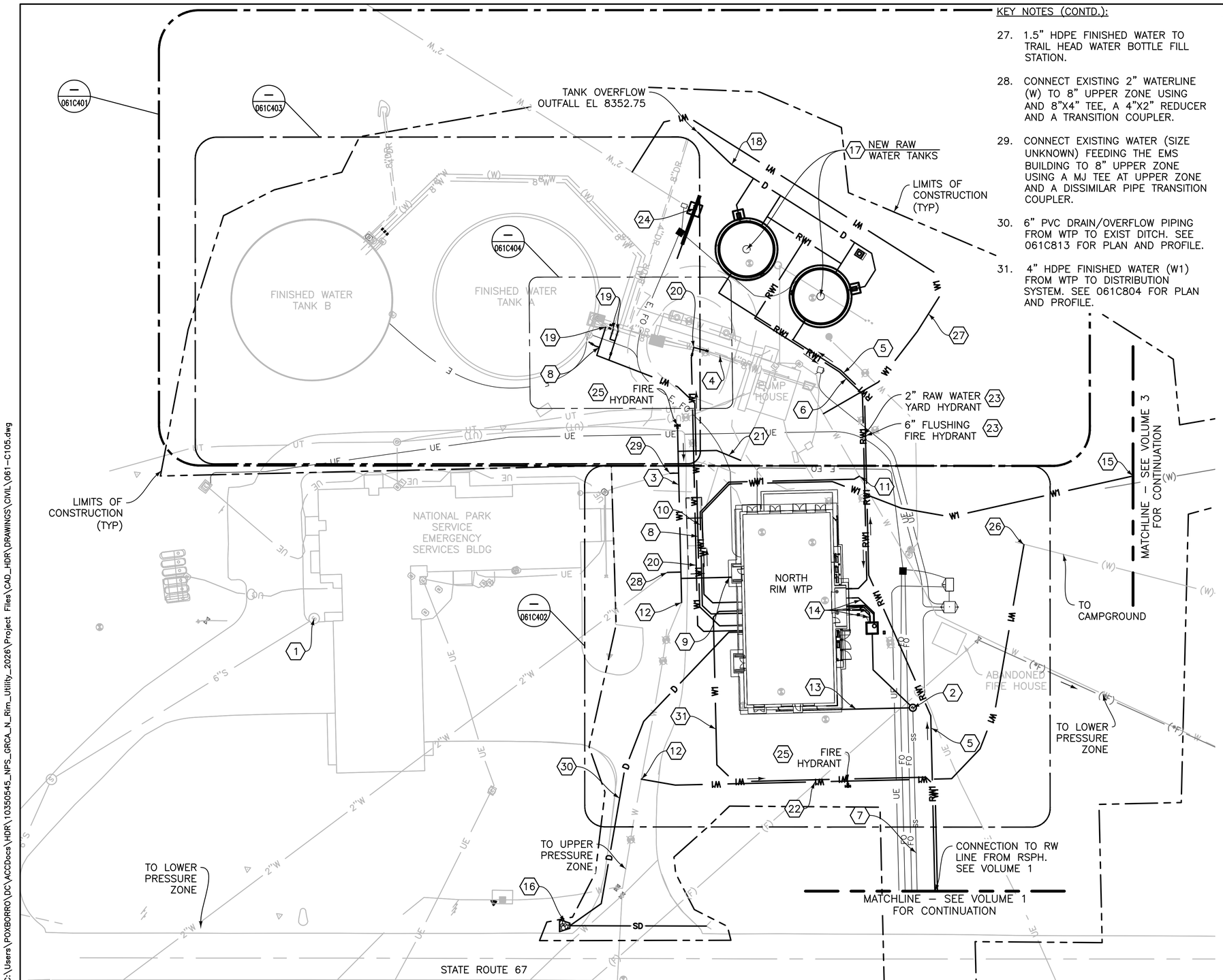

KEY NOTES (CONTD.):

27. 1.5" HDPE FINISHED WATER TO TRAIL HEAD WATER BOTTLE FILL STATION.
28. CONNECT EXISTING 2" WATERLINE (W) TO 8" UPPER ZONE USING AND 8"X4" TEE, A 4"X2" REDUCER AND A TRANSITION COUPLER.
29. CONNECT EXISTING WATER (SIZE UNKNOWN) FEEDING THE EMS BUILDING TO 8" UPPER ZONE USING A MJ TEE AT UPPER ZONE AND A DISSIMILAR PIPE TRANSITION COUPLER.
30. 6" PVC DRAIN/OVERFLOW PIPING FROM WTP TO EXIST DITCH. SEE 061C813 FOR PLAN AND PROFILE.
31. 4" HDPE FINISHED WATER (W1) FROM WTP TO DISTRIBUTION SYSTEM. SEE 061C804 FOR PLAN AND PROFILE.

KEY NOTES (CONTD.):

11. CONNECT TO EXISTING LOWER PRESSURE ZONE LINE.
12. CONNECT TO EXISTING UPPER PRESSURE ZONE LINE.
13. 4" SANITARY SEWER FROM WTP TO PROCESS WASTE LIFT STATION. SEE 061C811 FOR PLAN AND PROFILE.
14. (2)6" AND (2)4" DRAIN (D) PIPES FROM WTP TO PROCESS WASTE TANK. IN COMMON TRENCH. SEE 061C809 & 061C810 FOR PLAN AND PROFILE.
15. 2" HDPE CONNECT TO 2" COMPOSITE PIPELINE FOR POTABLE WATER BACKFEED TO SUPAI TUNNEL. SEE 061C805 FOR PLAN AND PROFILE AND VOLUME 3, SHEET 031C103 FOR CONTINUATION.
16. REPLACE EXISTING 18" CMP CULVERT WITH NEW 18" RCP. INSTALL RIPRAP END SECTION PER DETAIL IN THIS VOLUME.
17. TANKS SHALL BE 150,000 GALLONS EACH, WELDED STEEL, 32' ID, 30' TALL, WITH 10' WIDE GRAVEL PAD AROUND PERIMETER.
18. 6" PVC DRAIN/OVERFLOW PIPING FROM RAW WATER TANKS. PROVIDE RIPRAP EROSION CONTROL AT DAYLIGHT LOCATION. SEE 061C812 FOR PLAN AND PROFILE.
19. 8" HDPE FINISHED WATER (W1) FROM FW TANKS TO WTP. CONNECT TO EXISTING 8" CIP FW USING TRANSITION PIPE COUPLER.
20. 4" HDPE FINISHED WATER (W1) FROM WTP TO FW TANKS. CONNECT TO EXISTING 8" CIP FW USING TRANSITION PIPE COUPLER. SEE 061C803 FOR PLAN AND PROFILE.
21. 8" HDPE FINISH WATER TEMP BYPASS. CONNECT TO EXISTING WATERLINE. SEE SHEET 061C806 FOR PLAN AND PROFILE.
22. 8" HDPE FINISHED WATER (W1) TO FIRE HYDRANT AND CAMPGROUND. SEE 061C814 FOR PLAN AND PROFILE.
23. COORDINATE FLUSHING FIRE HYDRANT COLOR WITH COR.
24. COMMUNICATION TOWER. SEE ELECTRICAL AND STRUCTURAL SHEETS FOR MORE INFORMATION.
25. SEE DETAIL 02/002C503 FOR FIRE HYDRANT INFORMATION.
26. CONNECT TO EXISTING 8" WATERLINE TO CAMPGROUND USING AN 8"X8" DI MJ TEE.

GENERAL NOTES:

A. CONTRACTOR TO FIELD VERIFY DEPTH AND LOCATION OF ALL EXISTING UTILITIES.
B. ALL YARD PIPING & APPURTENANCES SHALL BE FULLY RESTRAINED.
C. ALL PIPE CONNECTIONS BETWEEN WTP AND YARD PIPING SHALL BE MADE USING A DISSIMILAR MJ PIPE TRANSITION COUPLER.
D. NRWTP PROCESS PIPING (SEE SPECIFICATION DIVISION 40) EXTENDS 2–FT FROM OUTSIDE OF BUILDING FOOTER. REFER TO SPECIFICATION DIVISION 33 FOR ALL OTHER PIPING OUTSIDE BUILDING.

KEY NOTES: (X)

1. EXISTING CALMING MANHOLE FOR EMS BUILDING LIFT STATION AND FORCEMAIN. FLOW TRANSITIONS TO GRAVITY DOWNSTREAM FROM THIS MANHOLE.
2. 48" MANHOLE. SEE VOLUME 1 FOR PLAN AND PROFILE OF 6" SANITARY SEWER FROM DOWNSTREAM OF THIS MANHOLE TO COLLECTION SYSTEM.
3. 8"/6" WATER TO FIRE HYDRANT & TEMP BYPASS. SEE 061C806 FOR PLAN AND PROFILE. REROUTED FW LINE TO AVOID NEW WTP.
4. ABANDON EXISTING WATERLINES IN–PLACE BETWEEN EXISTING PUMPHOUSE AND TANKS/DISTRIBUTION SYSTEM CONNECTIONS.
5. 4" HDPE RAW WATER (RW1) PIPE FROM RSPH TO RW TANKS. SEE 061C808 FOR PLAN AND PROFILE.
6. 8" HDPE RAW WATER (RW1) FROM RW TANKS TO WTP. SEE 061C807 FOR PLAN AND PROFILE.
7. 6" PVC SANITARY SEWER (SS). SEE VOLUME 1 FOR DETAILS.
8. 12" HDPE FINISHED WATER (W1) FROM FW TANKS TO WTP. CONNECT TO PROPOSED 8" PVC FW USING TRANSITION PIPE COUPLER. SEE 061C801 FOR PLAN AND PROFILE.
9. 8" HDPE FINISHED WATER (W1) FROM WTP TO UPPER PRESSURE ZONE DISTRIBUTION (PUMPED). SEE 061C802 FOR PLAN AND PROFILE.
10. 8" HDPE FINISHED WATER (W1) FROM WTP TO LOWER PRESSURE ZONE DISTRIBUTION (HEAD PRESSURE). SEE 061C802 FOR PLAN AND PROFILE.

30 0 30 60
SCALE OF FEET
1" = 30'

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — United States — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 061 C105	TITLE OF SHEET SITEWORK OVERALL YARD PIPING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 22 OF 192

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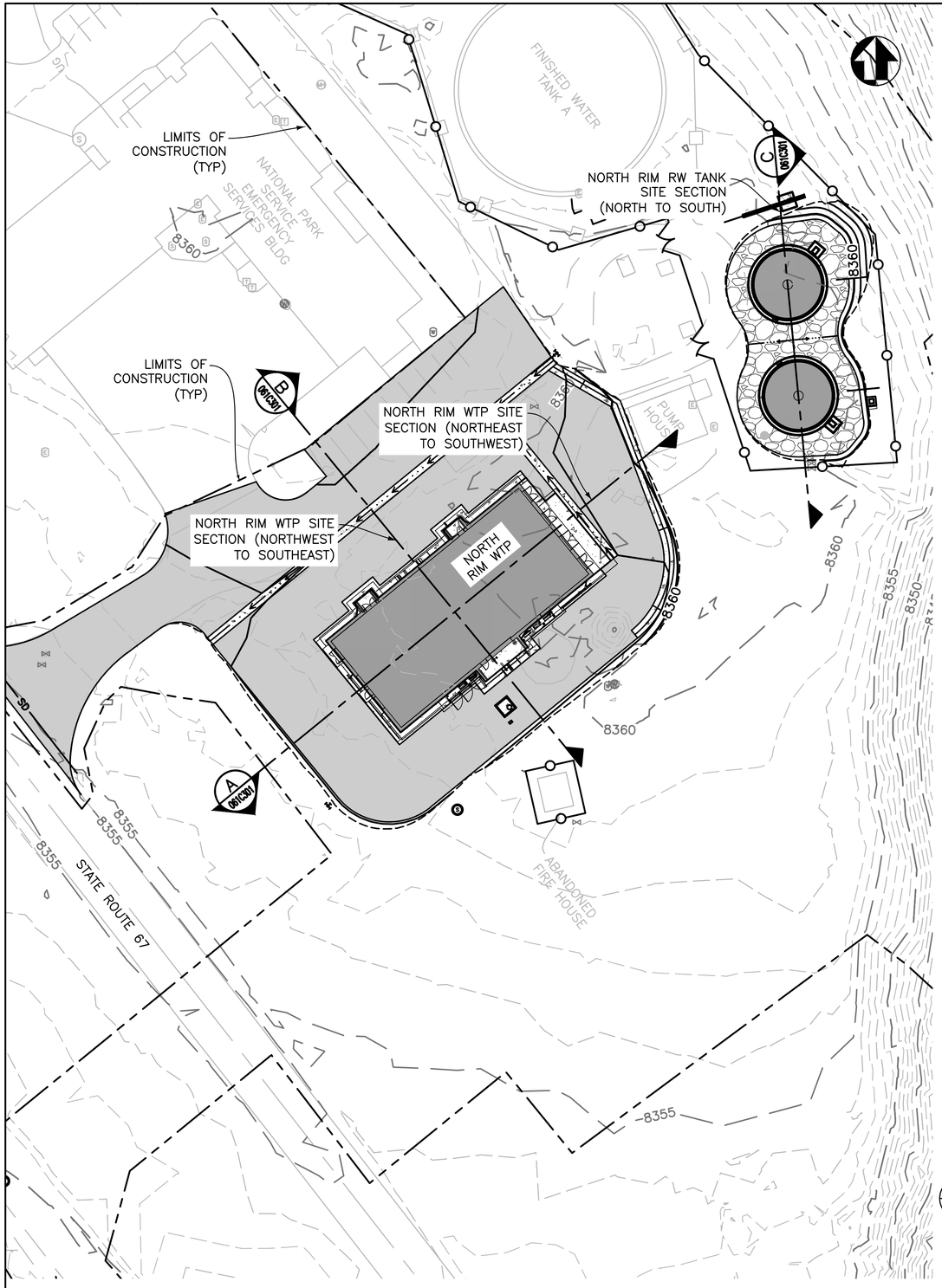
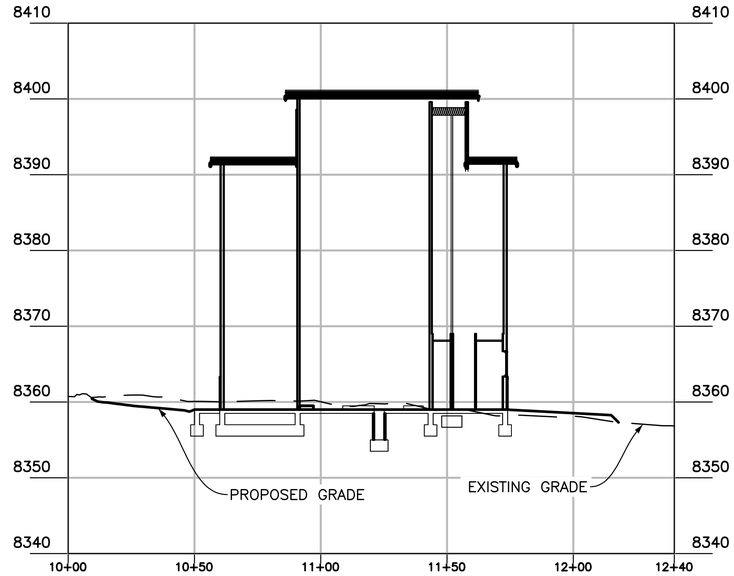
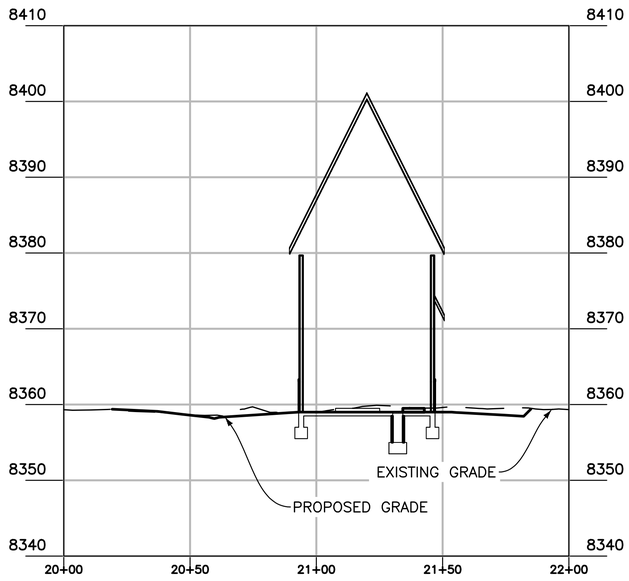


PLAN VIEW
SCALE: 1"=30'

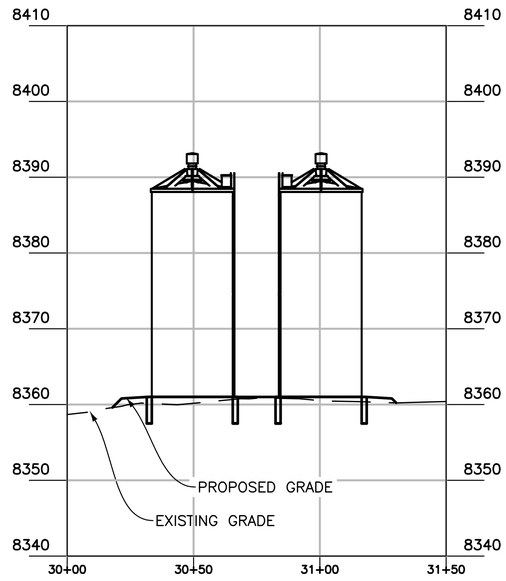
30 0 30 60
SCALE OF FEET



A 061C301 NORTH RIM WTP SITE SECTION (NORTHEAST TO SOUTHWEST)
SCALE: H: 1"=30', V: 1"=10'



B 061C301 NORTH RIM WTP SITE SECTION (NORTHWEST TO SOUTHEAST)
SCALE: H: 1"=30', V: 1"=10'



C 061C301 NORTH RIM RW TANK SITE SECTION (NORTH TO SOUTH)
SCALE: H: 1"=30', V: 1"=10'

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75262 GREGORY YOUNG DATE SIGNED 5/11/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 061 C301	TITLE OF SHEET SITEWORK GRADING AND PAVING SECTIONS WTP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 23 OF 192

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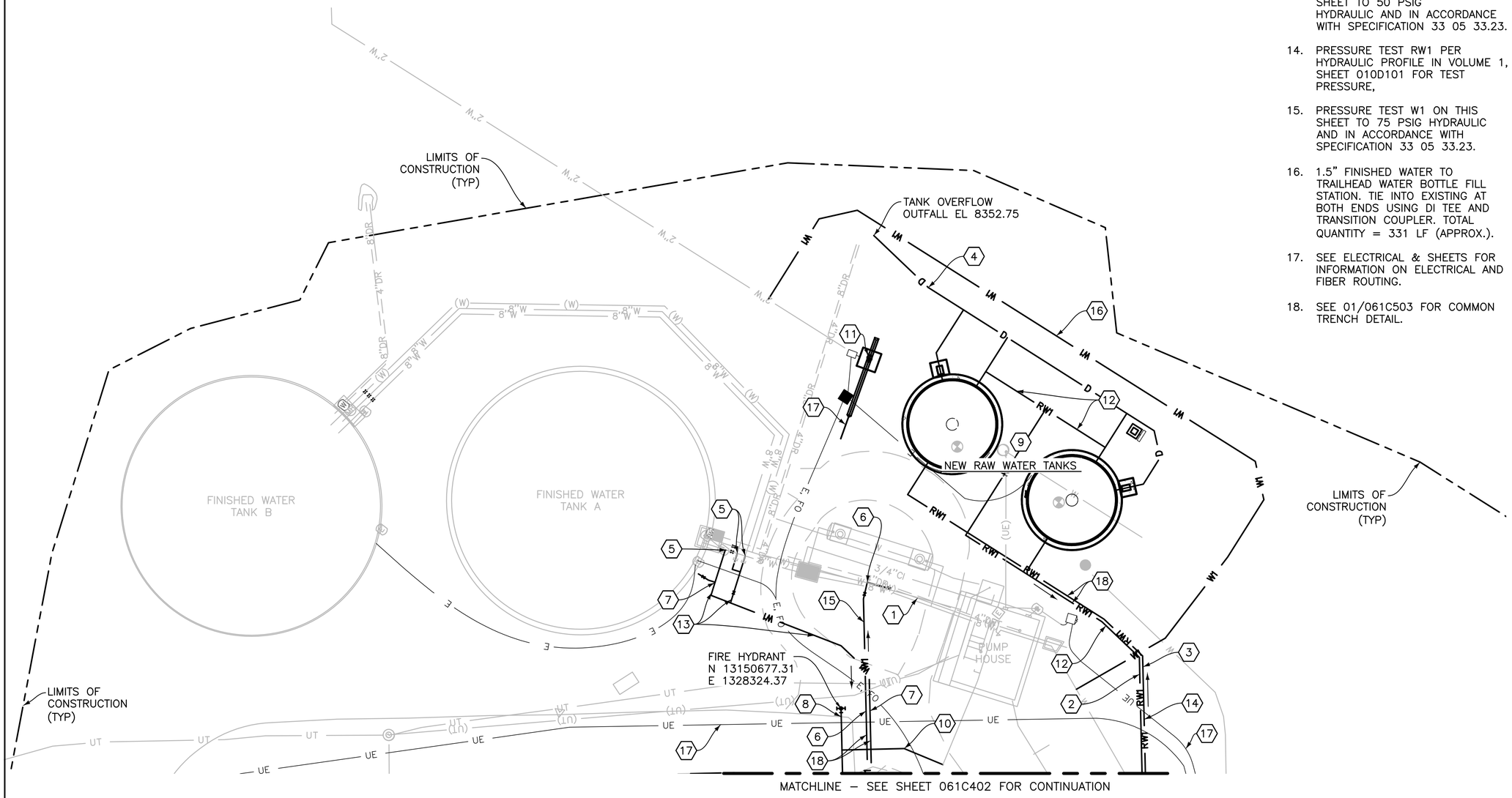


KEY NOTES: (X)

11. COMMUNICATIONS TOWER. SEE ELECTRICAL, STRUCTURAL, & I&C SHEETS FOR INFORMATION.
12. PRESSURE TEST RW1 ON THIS SHEET TO 25 PSIG HYDRAULIC AND IN ACCORDANCE WITH SPECIFICATION 33 05 33.23.
13. PRESSURE TEST W1 ON THIS SHEET TO 50 PSIG HYDRAULIC AND IN ACCORDANCE WITH SPECIFICATION 33 05 33.23.
14. PRESSURE TEST RW1 PER HYDRAULIC PROFILE IN VOLUME 1, SHEET 010D101 FOR TEST PRESSURE,
15. PRESSURE TEST W1 ON THIS SHEET TO 75 PSIG HYDRAULIC AND IN ACCORDANCE WITH SPECIFICATION 33 05 33.23.
16. 1.5" FINISHED WATER TO TRAILHEAD WATER BOTTLE FILL STATION. TIE INTO EXISTING AT BOTH ENDS USING DI TEE AND TRANSITION COUPLER. TOTAL QUANTITY = 331 LF (APPROX.).
17. SEE ELECTRICAL & SHEETS FOR INFORMATION ON ELECTRICAL AND FIBER ROUTING.
18. SEE 01/061C503 FOR COMMON TRENCH DETAIL.

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTH AND LOCATION OF ALL EXISTING UTILITIES.

C. CONTRACTOR CAN HORIZONTALLY REALIGN SIMILAR UTILITY TYPES FOR INSTALLATION WITHIN A SHARED TRENCH.

KEY NOTES: (X)

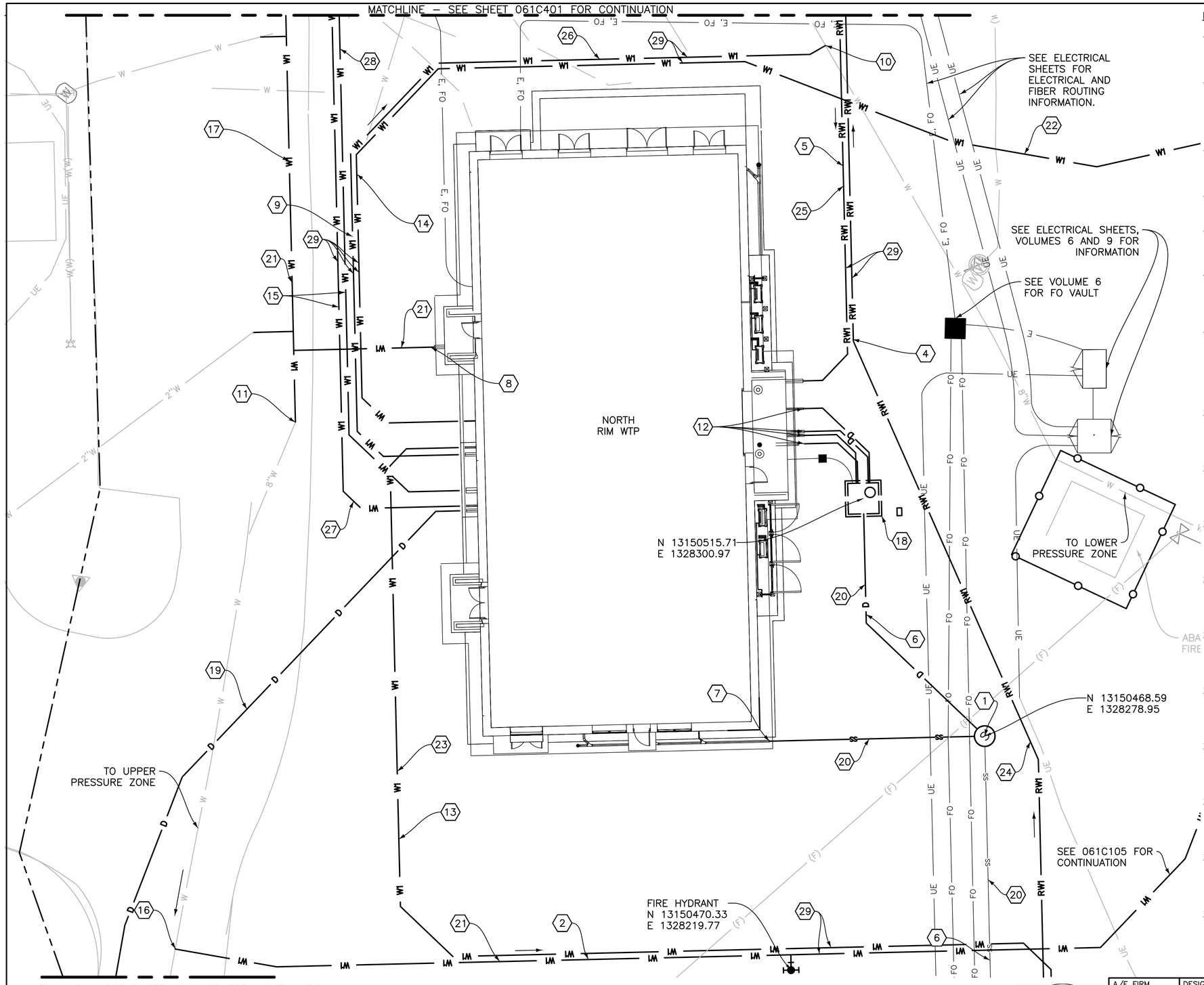
1. ABANDON EXISTING WATERLINES IN–PLACE BETWEEN EXISTING PUMPHOUSE AND TANKS/DISTRIBUTION SYSTEM CONNECTIONS.
2. 8" HDPE RAW WATER (RW) FROM RW TANKS TO WTP. SEE SHEET 061C402 FOR QUANTITIES.
3. 4" HDPE RAW WATER (RW) FROM RSPH TO RW TANKS. SEE SHEET 061C402 FOR QUANTITIES.
4. 4" PVC DRAIN/OVERFLOW PIPING FROM RAW WATER TANKS. PROVIDE RIPRAP EROSION CONTROL AT DAYLIGHT LOCATION. TOTAL QUANTITY = 140 LF (APPROX.).
5. 8" HDPE FINISHED WATER (FW) FROM FW TANKS TO WTP. CONNECT TO EXISTING 8" CIP FW USING TRANSITION PIPE COUPLER. TOTAL QUANTITY = 59 LF (APPROX.)
6. 4" HDPE FINISHED WATER (FW) FROM WTP TO FW TANKS. CONNECT TO EXISTING 8" CIP FW USING TRANSITION PIPE COUPLER. INSTALL 8"X4" HDPE REDUCER. TOTAL QUANTITY = 177 LF (APPROX.).
7. 12" HDPE FINISHED WATER (FW) FROM FW TANKS TO WTP. CONNECT TO EXISTING 8" CIP FW USING TRANSITION PIPE COUPLER. TOTAL QUANTITY = 248 LF (APPROX.)
8. 6" HDPE WATER TO FIRE HYDRANT. SEE SHEET 061C402 FOR QUANTITIES.
9. SEE PROCESS SHEETS FOR PIPING BETWEEN TANKS
10. 8" TEMPORARY BYPASS TO UPPER PRESSURE ZONE. CONTRACTOR SHALL CAP AND ABANDON ONCE WTP HAS BEEN COMMISSIONED.

20 0 20 40 1" = 20'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75262 GREGORY YOUNG ARIZONA, U.S.A. COUNTY OF COCONINO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 061 C401	TITLE OF SHEET SITEWORK YARD PIPING ENLARGED PLAN 1 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 24 OF 192

4/18/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C401.dwg



KEY NOTES (CONTD.):

13. 4" HDPE FINISHED WATER (W1) FROM WTP TO DISTRIBUTION SYSTEM.
14. 2" HDPE CONNECT TO EXISTING N RIM PIPELINE FOR POTABLE WATER BACKFEED TO SUPAI TUNNEL. TOTAL QUANTITY = 5 LF (APPROX.)
15. SEE SHEET 061C401 FOR FINISH WATER (W1) INFORMATION.
16. CONNECT TO EXISTING UPPER PRESSURE ZONE LINE.
17. 8" HDPE FINISHED WATER (W1) TO UPPER PRESSURE ZONE AND HYDRANT.
18. PROCESS WASTE TANK, SEE DETAIL SHEET 061C501. SEE 061 STRUCTURAL SHEETS FOR STRUCTURAL DETAILS.
19. 6" PVC DRAIN (D) LINE FOR PLANT OVERFLOW. OUTLET PIPE TO ROADSIDE DITCH. INSTALL INLINE CHECK VALVE AT DOWNSTREAM END, BEFORE OUTLET.
20. PRESSURE TEST SAN ON THIS SHEET TO 25 PSIG HYDRAULIC AND IN ACCORDANCE WITH SPECIFICATION 33 31 11.
21. PRESSURE TEST W1 ON THIS SHEET TO 100 PSIG HYDRAULIC AND IN ACCORDANCE WITH SPECIFICATION 33 05 33.23
22. SUPAI TUNNEL REST AREA WATER LINE. SEE HYDRAULIC PROFILE VOLUME 3 FOR TEST PRESSURE.
23. PRESSURE TEST RW1 PER HYDRAULIC PROFILE IN VOLUME 1, SHEET 010D101 FOR TEST PRESSURE.
24. PRESSURE TEST RW1 PER HYDRAULIC PROFILE IN VOLUME 1, SHEET 010D101 FOR TEST PRESSURE.
25. PRESSURE TEST RW1 ON THIS SHEET TO 25 PSIG HYDRAULIC AND IN ACCORDANCE WITH SPECIFICATION 33 05 33.23.
26. PRESSURE TEST W1 ON THIS SHEET TO 50 PSIG HYDRAULIC AND IN ACCORDANCE WITH SPECIFICATION 33 05 33.23.
27. PRESSURE TEST W1 ON THIS SHEET TO 75 PSIG HYDRAULIC AND IN ACCORDANCE WITH SPECIFICATION 33 05 33.23.
28. PRESSURE TEST W1 ON THIS SHEET TO 50 PSIG HYDRAULIC AND IN ACCORDANCE WITH SPECIFICATION 33 05 33.23.
29. SEE 01/061C503 FOR COMMON TRENCH DETAIL.

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTH AND LOCATION OF ALL EXISTING UTILITIES.

C. SEE ELECTRICAL AND I&C SHEETS FOR UE, E, AND FO INFORMATION.

D. FOR PIPE CONNECTIONS AT WTP TIE–INS. SEE DETAIL 02/061C503.

KEY NOTES: (X)

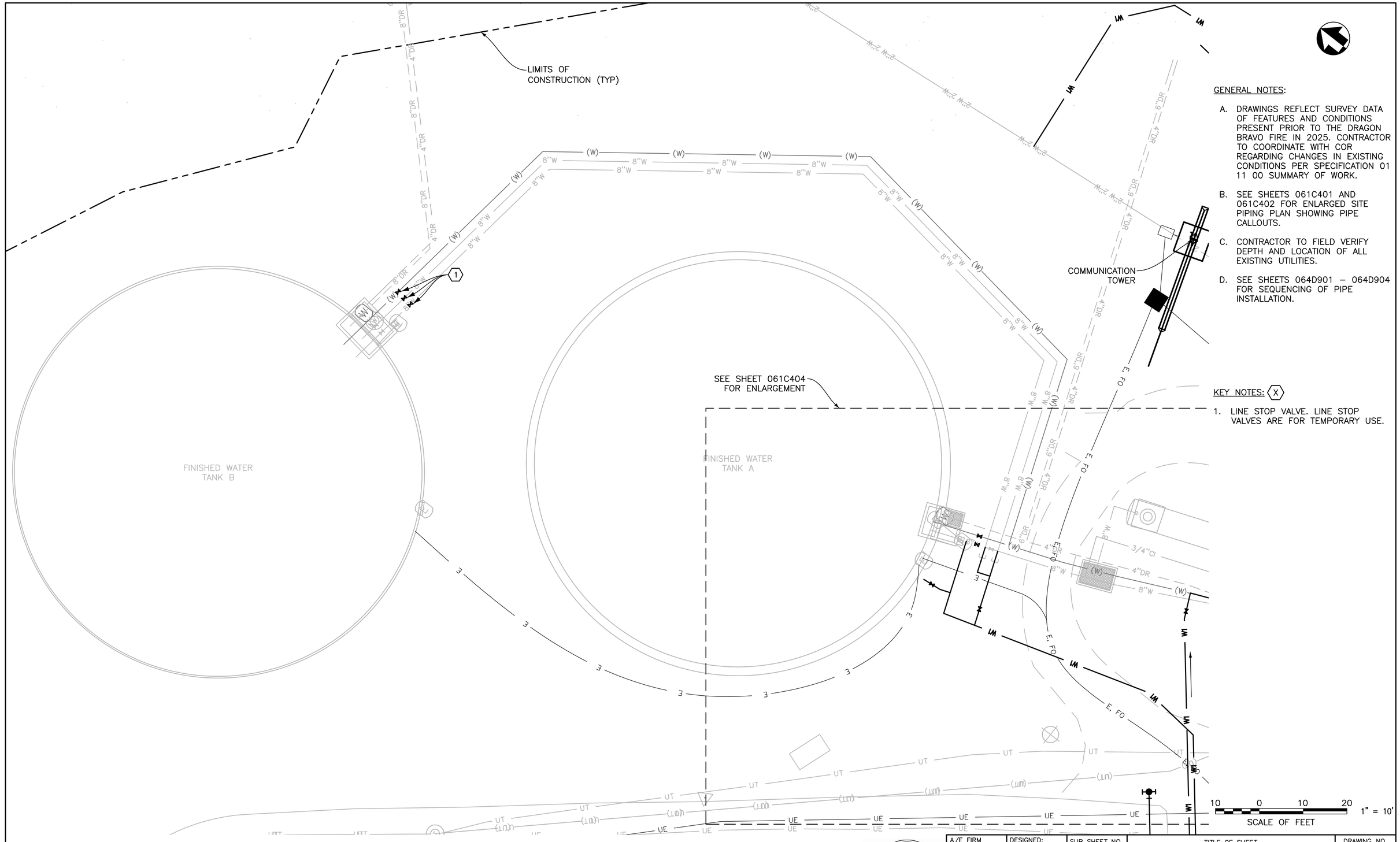
1. 48" MANHOLE. SEE VOLUME 1 FOR SANITARY SEWER PLAN AND PROFILE.
2. 6" WATER (W1) TO FIRE HYDRANT. TOTAL QUANTITY = 169 LF (APPROX.).
3. ABANDON EXISTING WATERLINES IN–PLACE BETWEEN EXISTING PUMPHOUSE AND TANKS/DISTRIBUTION SYSTEM CONNECTIONS.
4. 4" HDPE RAW WATER (RW1) PIPE FROM RSPH TO RW TANKS. TOTAL QUANTITY = 430 LF (APPROX.).
5. 8" HDPE RAW WATER (RW1) FROM RW TANKS TO WTP. TOTAL QUANTITY = 232 LF (APPROX.).
6. 6" PVC SANITARY SEWER (SS). QUANTITY = 120 (APPROX.). SEE VOL 01 FOR INFORMATION.
7. 4" SANITARY SEWER FROM WTP TO MANHOLE QUANTITY = 52 LF (APPROX.).
8. 8" HDPE FINISHED WATER (W1) FROM WTP TO UPPER PRESSURE ZONE DISTRIBUTION (PUMPED). TOTAL QUANTITY = 30 LF (APPROX.)
9. 8" HDPE FINISHED WATER (W1) FROM WTP TO LOWER PRESSURE ZONE DISTRIBUTION (HEAD PRESSURE). TOTAL QUANTITY = 206 LF (APPROX.)
10. CONNECT TO EXISTING LOWER PRESSURE ZONE LINE.
11. CONNECT TO EXISTING UPPER PRESSURE ZONE LINE.
12. (2)6" AND (2)4" PVC DRAIN (D) PIPES FROM WTP TO PROCESS WASTE TANK. IN COMMON TRENCH. TOTAL QUANTITY = 75 LF (APPROX.)

10 0 10 20
SCALE OF FEET
1" = 10'

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
Date Signed 5/1/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 061 C402	TITLE OF SHEET SITEWORK YARD PIPING ENLARGED PLAN 2 WTP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 25 OF 192

4/18/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\061–C402.dwg



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SEE SHEETS 061C401 AND 061C402 FOR ENLARGED SITE PIPING PLAN SHOWING PIPE CALLOUTS.

C. CONTRACTOR TO FIELD VERIFY DEPTH AND LOCATION OF ALL EXISTING UTILITIES.

D. SEE SHEETS 064D901 – 064D904 FOR SEQUENCING OF PIPE INSTALLATION.

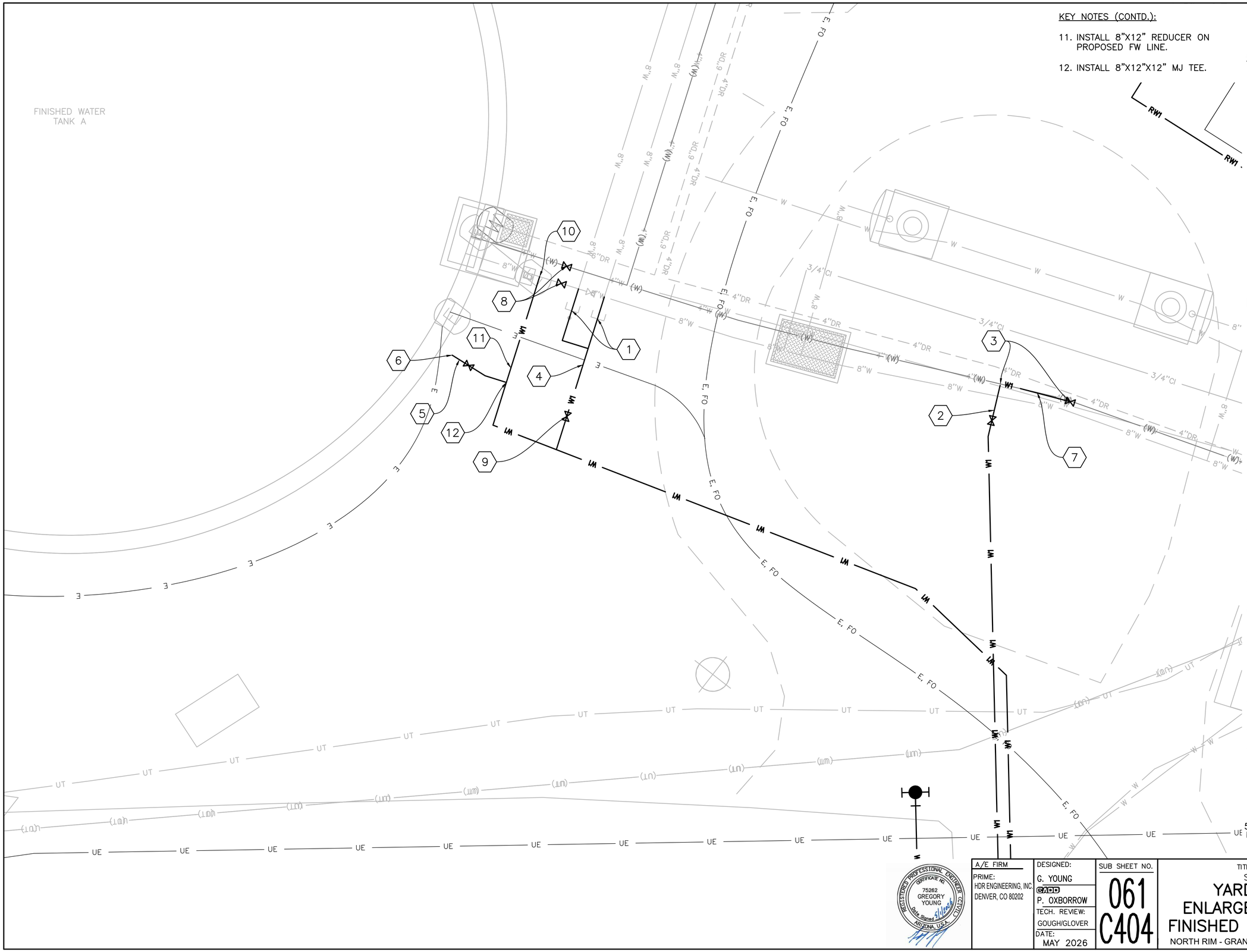
KEY NOTES: (X)

1. LINE STOP VALVE. LINE STOP VALVES ARE FOR TEMPORARY USE.

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
Date Signed
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C403	TITLE OF SHEET SITEWORK YARD PIPING ENLARGED PLAN 3 FINISHED WATER TANKS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 26 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\103505445_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C403.dwg



KEY NOTES (CONTD.):

11. INSTALL 8"X12" REDUCER ON PROPOSED FW LINE.

12. INSTALL 8"X12"X12" MJ TEE.

RW1 RW1

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SEE SHEETS 061C401 AND 061C402 FOR ENLARGED SITE PIPING PLAN SHOWING PIPE CALLOUTS.

C. CONTRACTOR TO FIELD VERIFY DEPTH AND LOCATION OF ALL EXISTING UTILITIES.

D. SEE SHEETS 064D901 – 064D904 FOR ADDITIONAL DETAILS ON SEQUENCING OF PIPE INSTALLATION.

KEY NOTES: (X)

1. 8" HDPE FINISHED WATER (FW) FROM FW TANKS TO WTP. CONNECT TO EXISTING 8" CIP FW USING TRANSITION PIPE COUPLER.

2. 4" HDPE FINISHED WATER (FW) FROM WTP TO FW TANKS. CONNECT TO NEW 4" MJ TEE. RESTRAIN JOINTS.

3. CUT AND REMOVE 8 LF (APPROX) EXISTING CI PIPE. INSTALL 4"X4" MJ TEE & 4" MJ GV ON EX 4" CI PIPE ENDS. CONNECT WITH 4" HDPE (6.5 LF APPROX, CTF) AND 4" MJ SOLID SLEEVE. ADD 4" MJ SOLID SLEEVE HALFWAY IN BETWEEN TEE AND VALVE. RESTRAIN ALL JOINTS.

4. 12" HDPE FINISHED WATER (FW) FROM FW TANKS TO WTP. CONNECT TO PROPOSED 8" HDPE FW USING TRANSITION PIPE COUPLER.

5. 8" STEEL FINISHED WATER (FW) FROM FW TANKS TO WTP.

6. CONNECT 8" STEEL FINISHED WATER (FW) TO 8" HDPE FINISHED WATER (FW) USING TRANSITION PIPE COUPLER.

7. INSTALL MJ PLUG TO VALVE WHEN LINE TO PUMPHOUSE IS ABANDONED.

8. LINE STOP VALVE. LINE STOP VALVES ARE FOR TEMPORARY USE.

9. INSTALL 12" GATE VALVE ON PIPE.

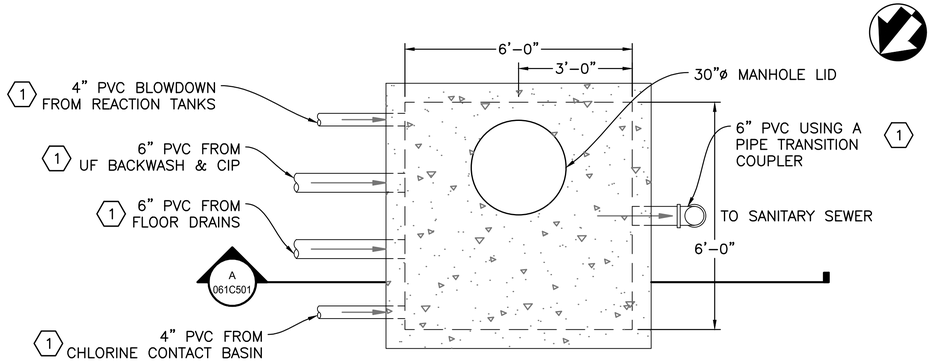
10. CUT AND REMOVE EXISTING CI PIPE UP TO LINE STOP VALVE. INSTALL 8" MJ ELBOW ON EX 8" CI PIPE END. CONNECT 8" HDPE FW TO ELBOW. RESTRAIN JOINTS.

5 0 5 10 1" = 5'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 75262 GREGORY YOUNG Date Signed 5/1/2026 ARIZONA, U.S.A.

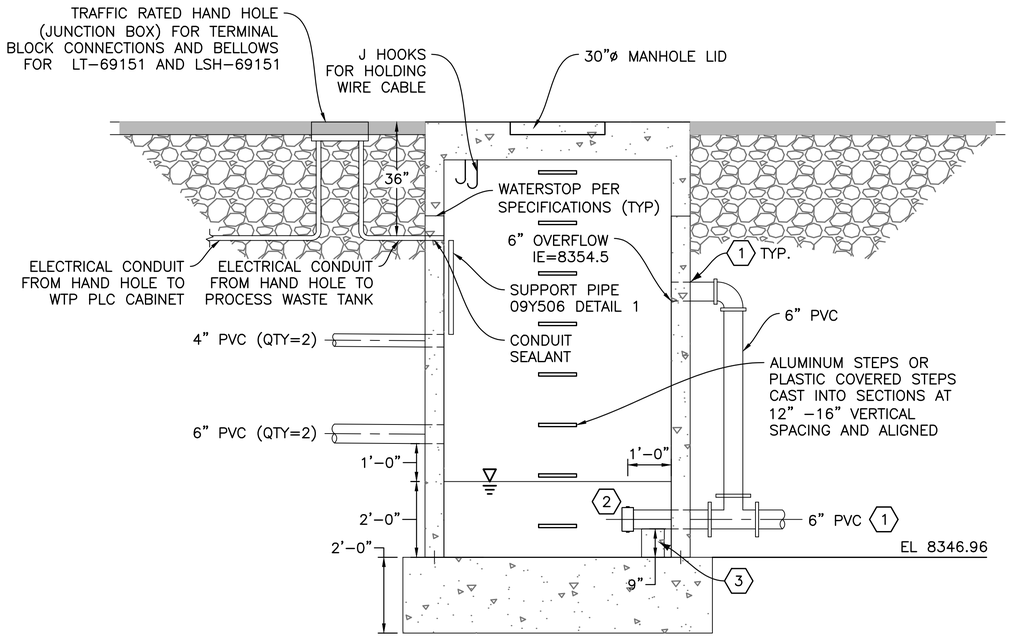
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C404	TITLE OF SHEET SITEWORK YARD PIPING ENLARGED PLAN 4 FINISHED WATER TANKS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 27 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C404.dwg



1 / 061C501 NORTH RIM PROCESS WASTE TANK PLAN VIEW

NTS



A / 061C501 NORTH RIM PROCESS WASTE TANK SECTION VIEW

NTS

GENERAL NOTES:

A. SEE SHEETS 061C809 – 061C811 FOR PIPE INVERT ELEVATIONS.

B. SEE SHEET 062Y614 FOR INFORMATION ON LEVEL INSTRUMENTATION INSIDE TANK. SEE DETAIL 1/009Y502 FOR SIMILAR MOUNTING DETAIL INSIDE TANK.

C. SEE SHEET 061S101 AND 061S301 FOR STRUCTURAL DETAILS.

KEY NOTES: (X)

1. UNDERGROUND PIPE PENETRATION. SEE DETAIL 1/005D505.

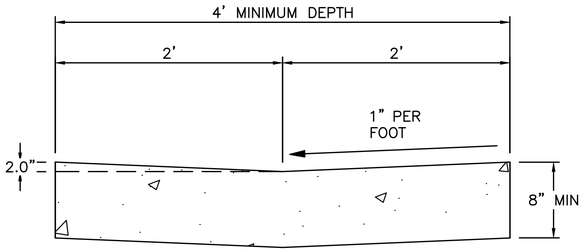
2. FLANGE ADAPTOR WITH BOLTED 316 SS ORIFICE PLATE AT END OF PIPE. ORIFICE OPENING DIA. SHALL BE 2".

3. CONCRETE PEDESTAL SUPPORT. SEE DETAIL 3/005D501.

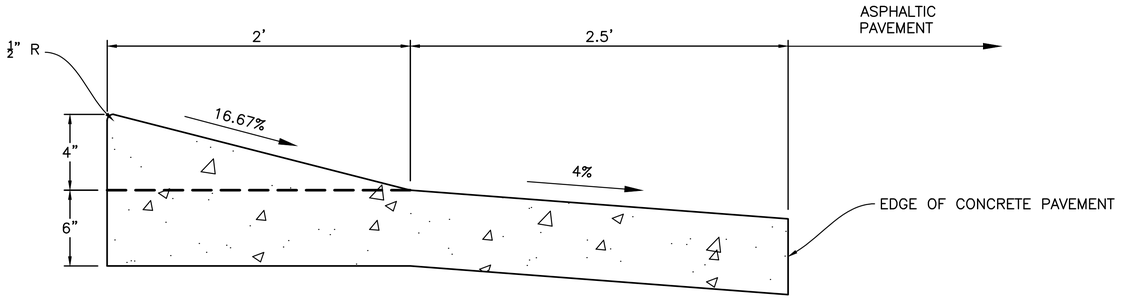
REGISTERED PROFESSIONAL ENGINEER – CIVIL – CERTIFICATE NO. 75262 GREGORY YOUNG – Date Signed 5/1/2026 – ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. SCHUMACHER DRAWN: J. GREEN TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C501	TITLE OF SHEET SITEWORK CIVIL DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 28 OF 192

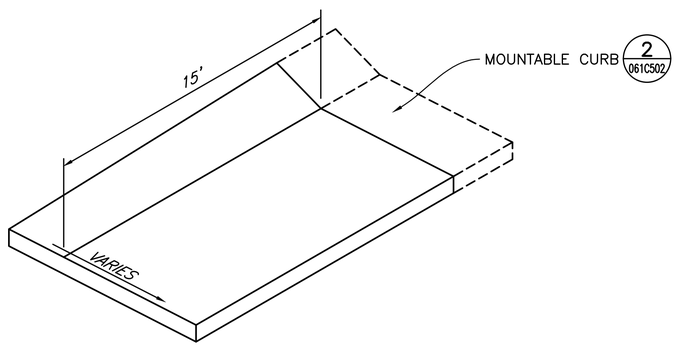
4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C501.dwg



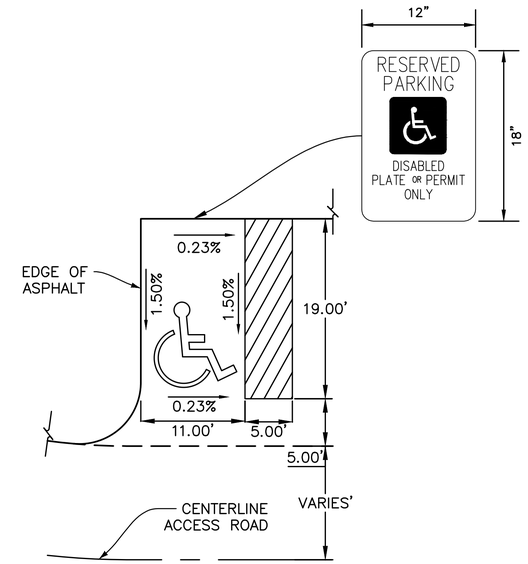
1 061C102 GUTTER PAN DETAIL
NTS



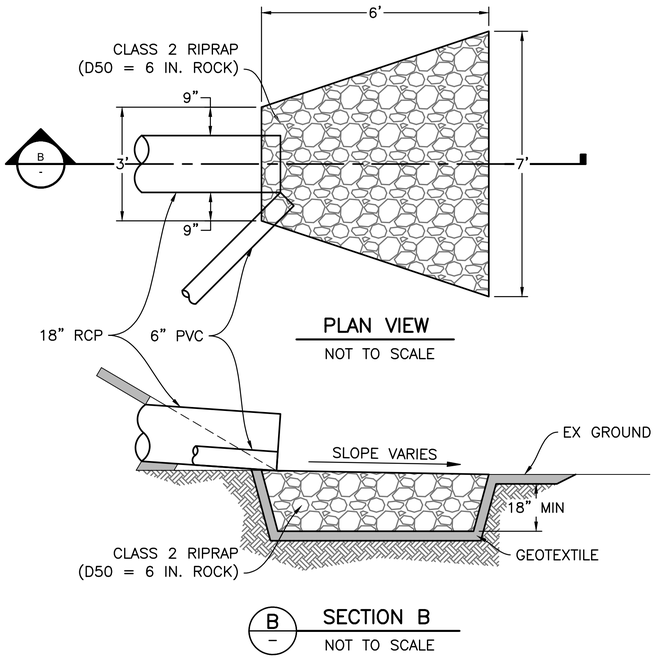
2 061C102 MOUNTABLE CURB DETAIL
NTS



3 061C102 CURB TRANSITION DETAIL
NTS



4 061C502 ADA VAN ACCESSIBLE PARKING STALL DETAIL
NTS



5 061C101 RIPRAP APRON – CULVERT WITHOUT STANDARD END SECTION
NOT TO SCALE

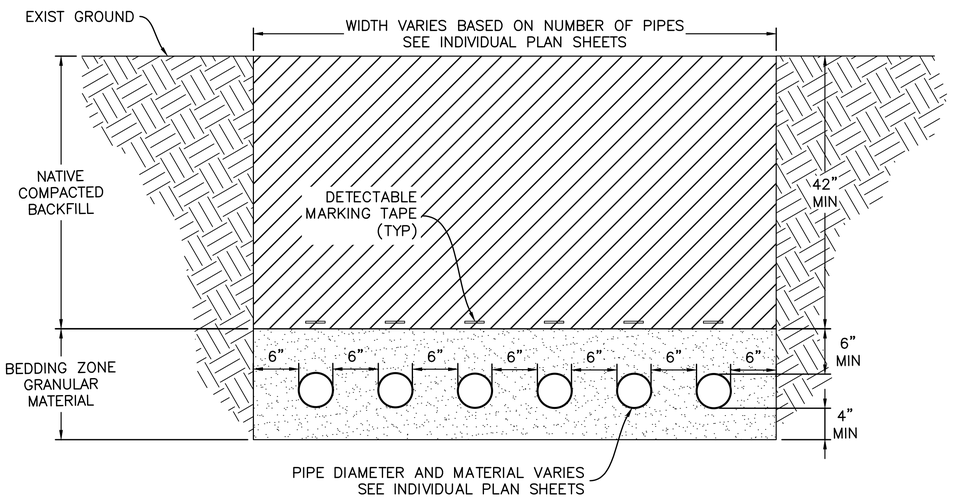
NOTES:

1. EXCAVATION FOR PLACEMENT OF RIPRAP WILL NOT BE MEASURED FOR PAYMENT.
2. FURNISH GEOTEXTILE CONFORMING TO SPEC 31 23 00 SECTION 2.1 D.

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75262 GREGORY YOUNG ARIZONA, U.S.A. CIVIL

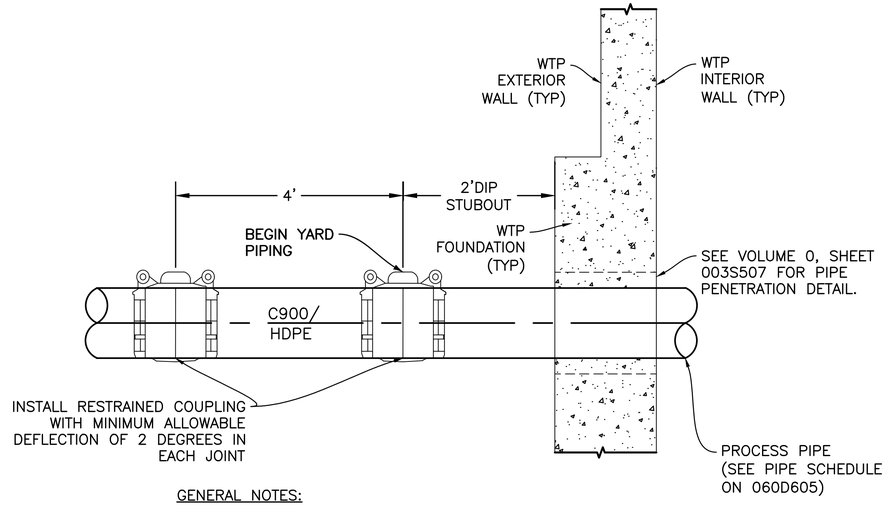
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C502	TITLE OF SHEET SITEWORK CIVIL DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 29 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C502.dwg



1 / 061C503 **COMMON TRENCH DETAIL**

NTS



GENERAL NOTES:

1) CONFIRM RESTRAINED COUPLINGS ARE COMPATIBLE BETWEEN DIP CLASS 350 TO HDPE 4710 DR7 AND PVC C900 DR25, INCLUDING OD OF DISSIMILAR MATERIALS.

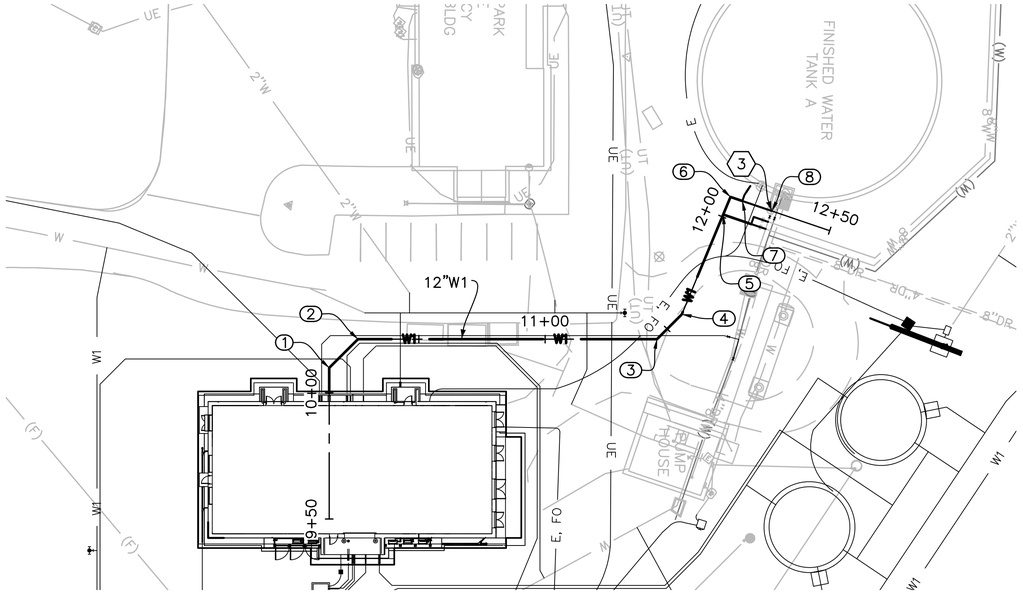
2 / 061C402 **TRANSITION COUPLER/DYNAMIC MOVEMENT DETAIL**

NTS

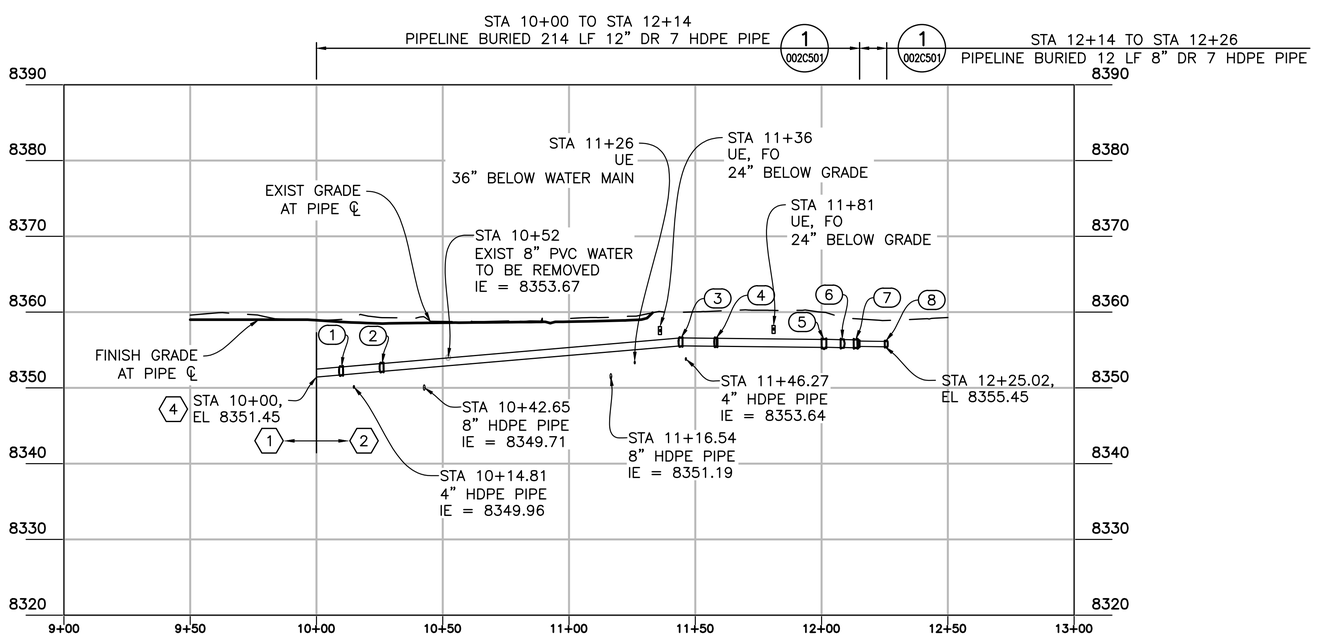
REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 75262 GREGORY YOUNG Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD A. DALL TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C503	TITLE OF SHEET SITEWORK CIVIL DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 30 OF 192

4/18/2026 Young, Gregory C:\Users\gryoung\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C503.dwg



PLAN VIEW
1"=30'



PROFILE VIEW

POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
1	13150586.43	1328247.54	45° HORIZONTAL BEND
2	13150602.41	1328249.15	45° HORIZONTAL BEND
3	13150677.32	1328340.82	45° HORIZONTAL BEND
4	13150691.18	1328342.21	22.5° HORIZONTAL BEND
5	13150732.10	1328329.89	TEE
6	13150739.33	1328327.72	90° HORIZONTAL BEND
7	13150741.06	1328332.37	12"X8" TEE W/ 12"X8" REDUCER
8	13150745.37	1328343.89	8" TEE

GENERAL NOTE:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

C. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8-16-2021.

D. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

E. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002C102.

F. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES (X)

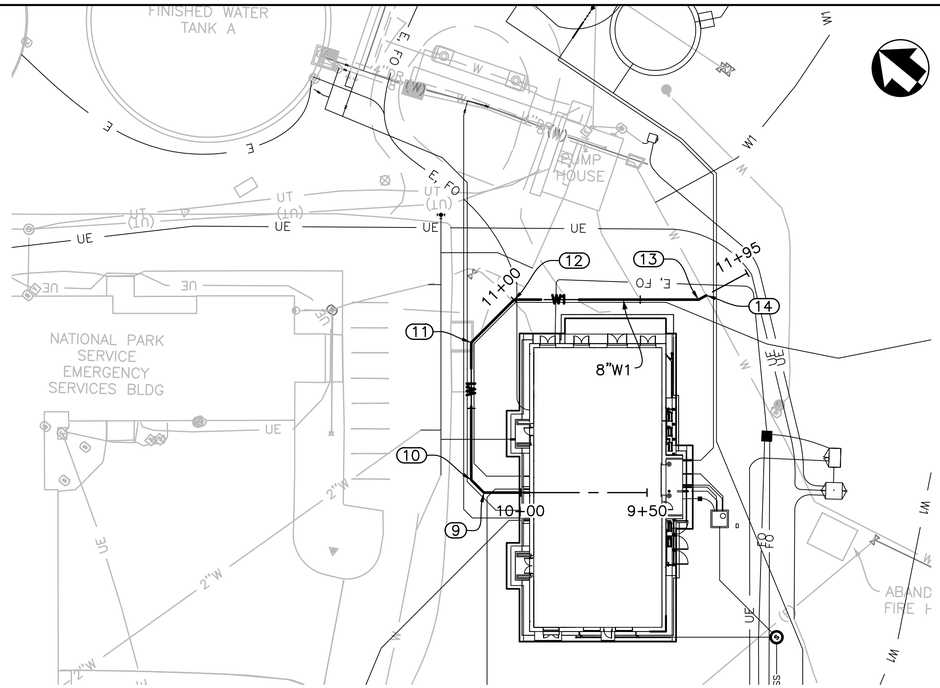
1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.
3. TIE 12" WATER LINE TO EXIST 8" WATER USING A TEE FITTING.
4. SEE DETAIL 02/061C503 FOR PIPE TRANSITION INFORMATION.

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

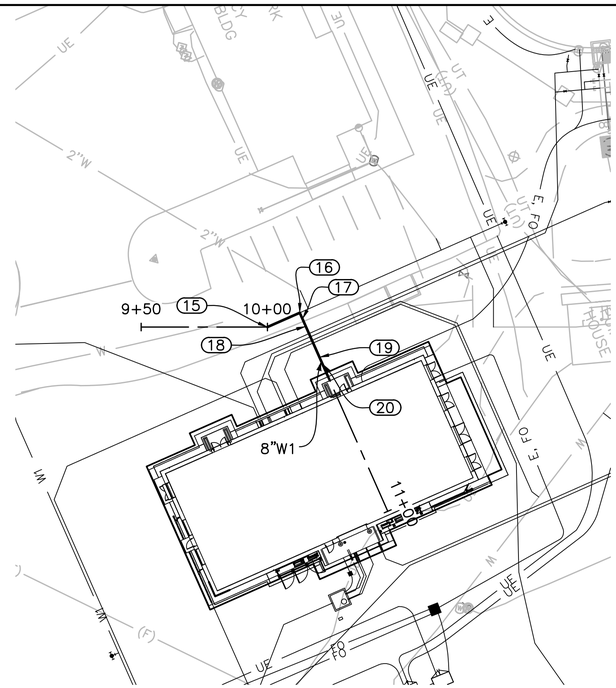
REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
State Banner
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C801	TITLE OF SHEET SITEWORK YARD PIPING PLAN AND PROFILE 12" FW FROM FW TANKS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 31 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C801.dwg

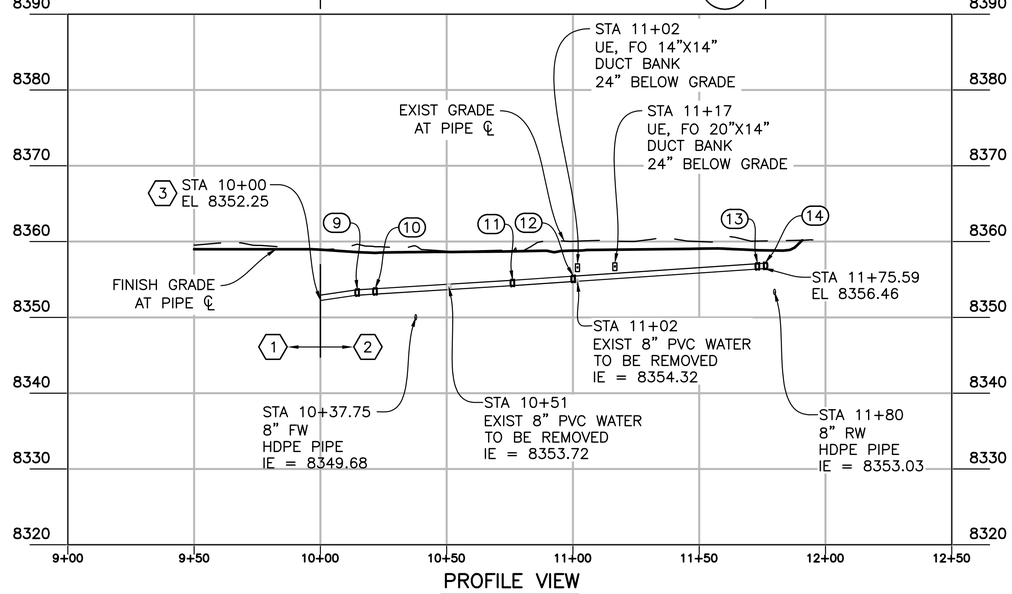


PLAN VIEW
1"=30'



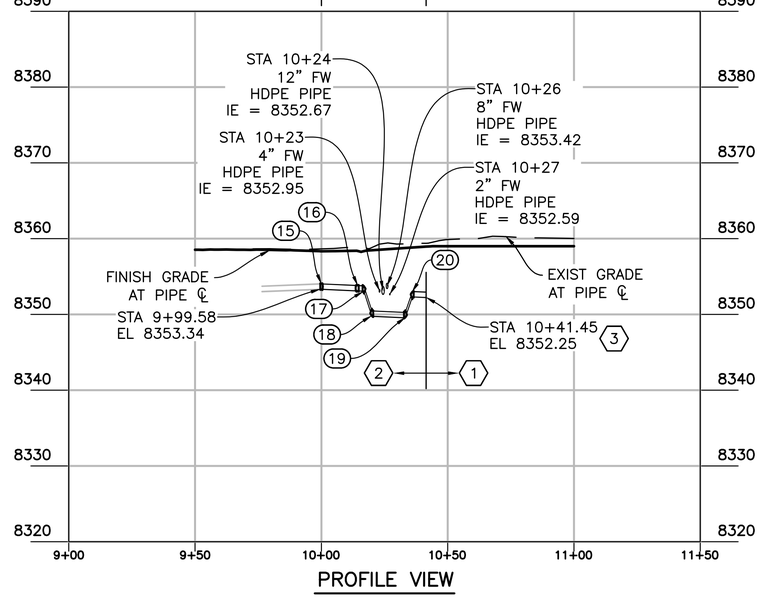
PLAN VIEW
1"=30'

STA 10+00 TO STA 11+77
PIPELINE BURIED 77 LF 8" DR 7 HDPE PIPE (1 / 002C501)



PROFILE VIEW

STA 10+00 TO STA 10+41
PIPELINE BURIED 41 LF 8" DR 7 HDPE PIPE (1 / 002C501)



PROFILE VIEW

GENERAL NOTE:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

C. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8-16-2021.

D. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

E. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002-C102.

F. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES (X)

1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.

2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.

3. SEE DETAIL 02/061C503 FOR PIPE TRANSITION INFORMATION.

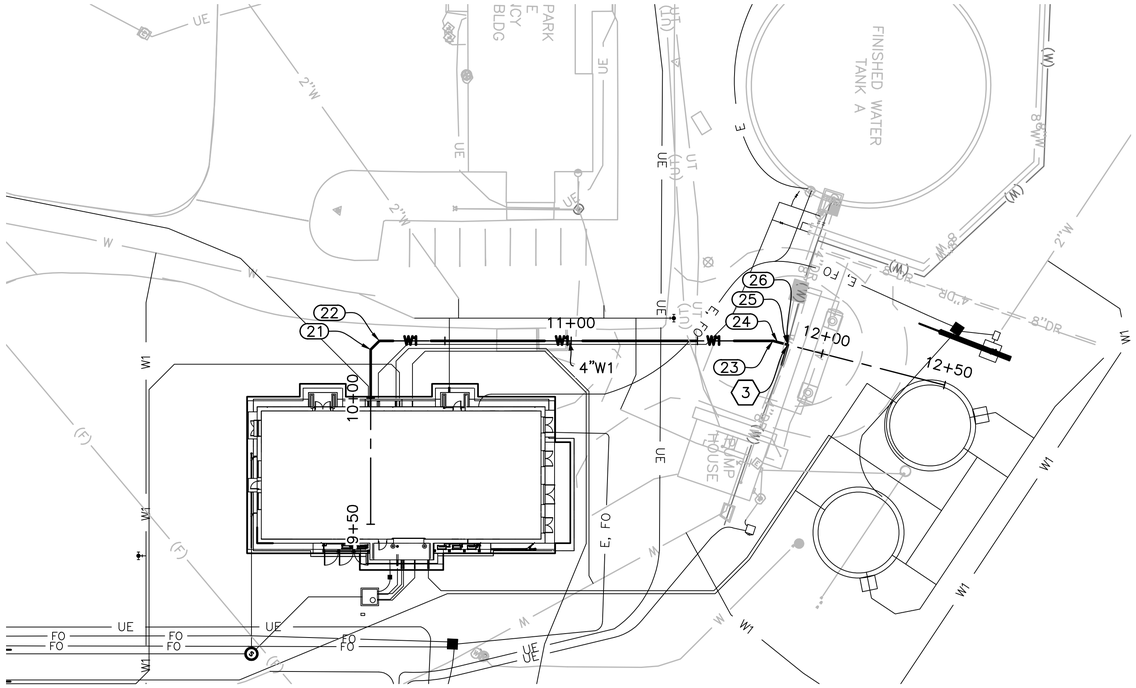
POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
9	13150594.55	1328249.97	45° HORIZONTAL BEND
10	13150601.59	1328250.67	45° HORIZONTAL BEND
11	13150636.01	1328292.80	45° HORIZONTAL BEND
12	13150633.51	1328316.43	45° HORIZONTAL BEND
13	13150577.10	1328362.89	22.5° HORIZONTAL BEND
14	13150575.79	1328366.11	TEE
15	13150612.20	1328244.66	22.5° HORIZONTAL BEND
16	13150621.07	1328255.51	TEE
17	13150619.01	1328257.20	45° VERTICAL BEND
18	13150616.48	1328259.27	45° VERTICAL BEND
19	13150606.14	1328267.72	45° VERTICAL BEND
20	13150603.92	1328269.53	45° VERTICAL BEND

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 75262 GREGORY YOUNG Date Signed 5/1/2026 ARIZONA, U.S.A.

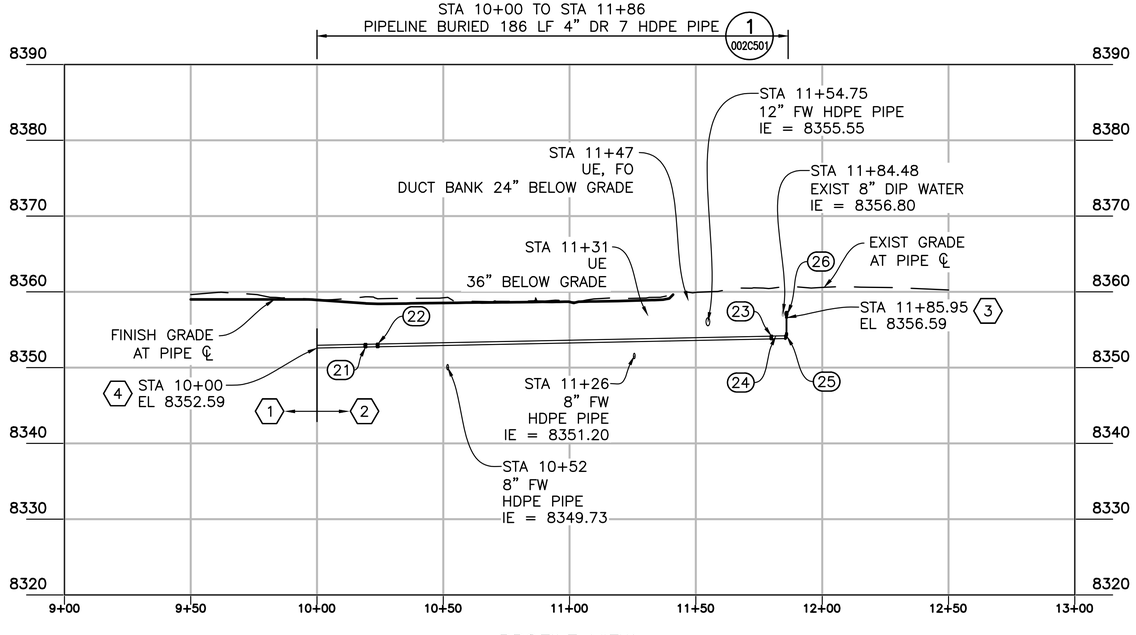
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C802	TITLE OF SHEET SITEWORK YARD PIPING PLAN AND PROFILE 8" FW (DISTRIBUTION) NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 32 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C802.dwg



PLAN VIEW

1"=30'



PROFILE VIEW

POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
21	13150591.75	1328239.30	45° HORIZONTAL BEND
22	13150596.56	1328239.78	45° HORIZONTAL BEND
23	13150695.19	1328360.48	22.5° HORIZONTAL BEND
24	13150695.96	1328362.14	VALVE
25	13150697.73	1328365.99	90° VERTICAL BEND
26	13150697.73	1328365.99	TEE

GENERAL NOTE:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

C. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED AND SAFE SITE UTILITY SERVICES, LLC, 8-16-2021.

D. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

E. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002-C102.

F. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES (X)

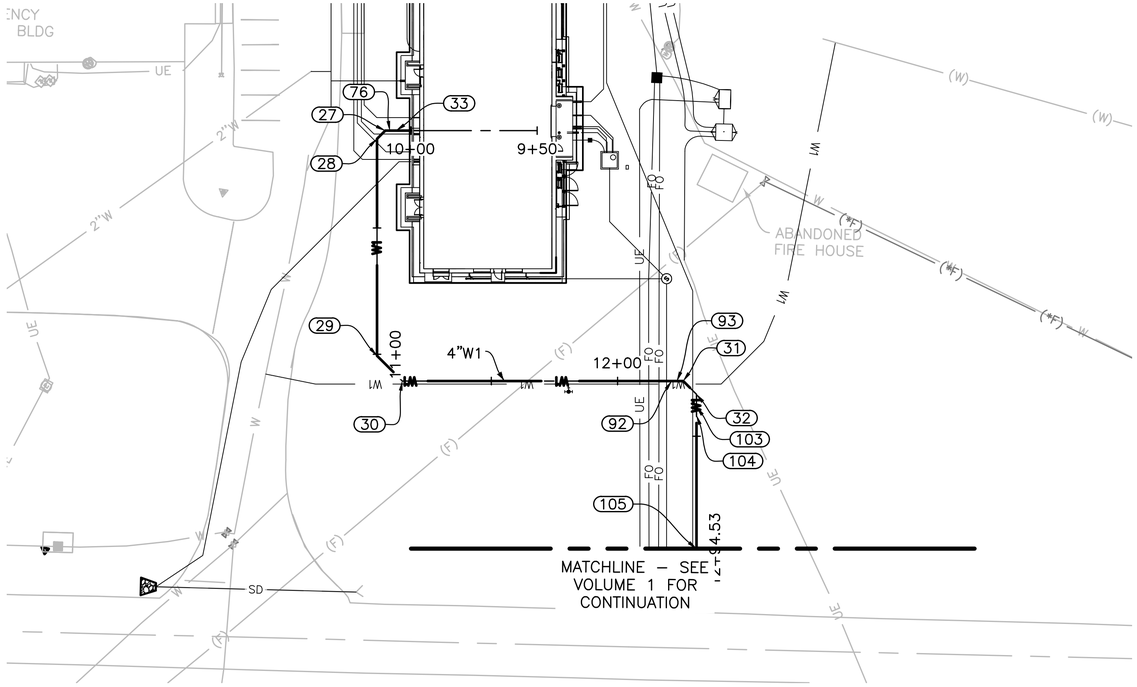
1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.
3. SEE 061C401 FOR INFORMATION ON CONNECTIONS TO EXIST SYSTEM, FINISHED WATER TANKS. SEE PROCESS FOR CONSTRUCTION SEQUENCING INFORMATION.
4. SEE DETAIL 02/061C503 FOR PIPE TRANSITION INFORMATION.

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
Date Signed 5/1/2026
ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	G. YOUNG	061 C803	SITEWORK YARD PIPING PLAN AND PROFILE 4" FW TO FW TANKS NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
	CADD P. OXBORROW			PMIS/PKG NO. 318719
	TECH. REVIEW: GOUGH/GLOVER			SHEET 33 OF 192
	DATE: MAY 2026			

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C803.dwg



PLAN VIEW

1"=30'

POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
27	13150591.95	1328253.87	45° HORIZONTAL BEND
28	13150592.41	1328249.31	45° HORIZONTAL BEND
29	13150538.04	1328182.78	45° HORIZONTAL BEND
30	13150523.97	1328181.36	45° HORIZONTAL BEND
31	13150437.84	1328251.74	45° HORIZONTAL BEND
32	13150430.46	1328251.00	45° HORIZONTAL BEND
33	13150588.57	1328256.63	45° VERTICAL BEND
76	13150590.58	1328254.98	45° VERTICAL BEND
92	13150441.40	1328248.84	45° VERTICAL BEND
93	13150439.79	1328250.15	45° VERTICAL BEND
103	13150427.29	1328247.13	45° VERTICAL BEND
104	13150425.21	1328244.58	45° VERTICAL BEND
105	13150391.78	1328203.67	4" FW END

GENERAL NOTE:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

C. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8–16–2021.

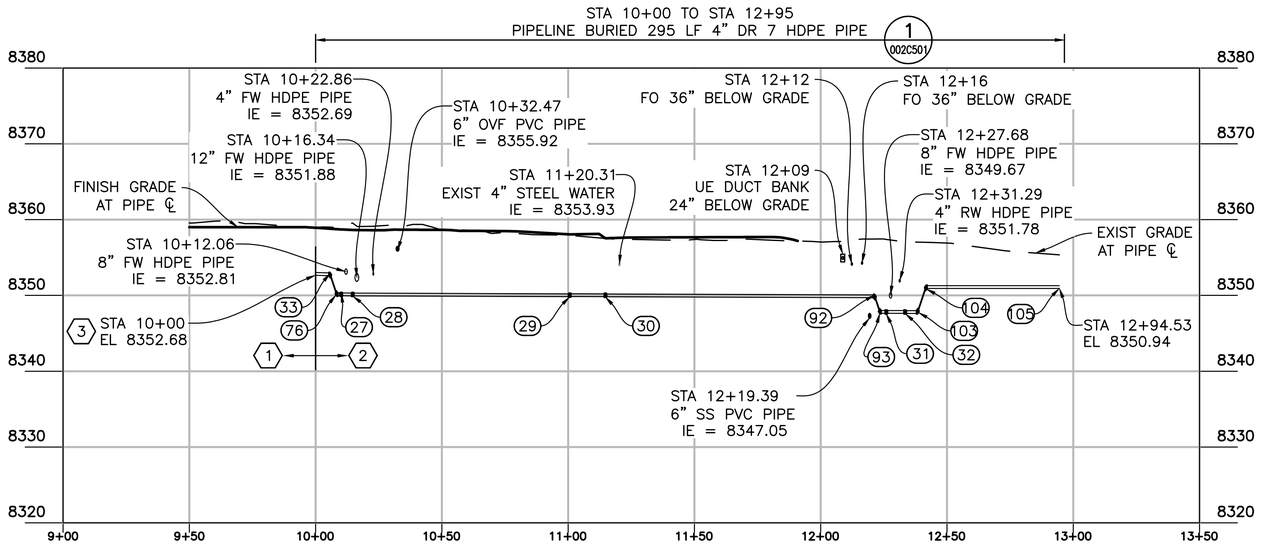
D. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

E. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002–C102.

F. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES (X)

1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.
3. SEE DETAIL 02/061C503 FOR PIPE TRANSITION INFORMATION.



PROFILE VIEW

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75262 — GREGORY YOUNG — STATE OF ARIZONA, U.S.A. — COUNTY CO

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	G. YOUNG	061 C804	SITEWORK YARD PIPING PLAN AND PROFILE 4" FW (TO BORE) NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
	CADD P. OXBORROW			PMIS/PKG NO. 318719
	TECH. REVIEW: GOUGH/GLOVER			SHEET 34 OF 192
	DATE: MAY 2026			

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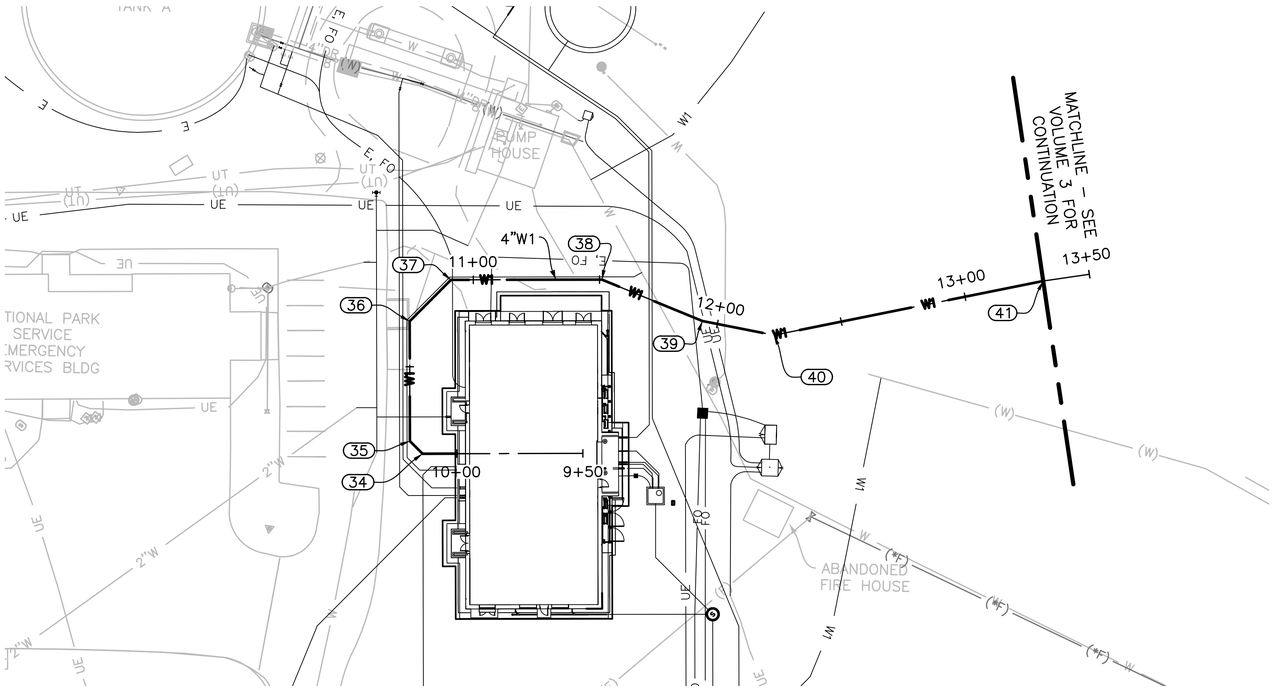
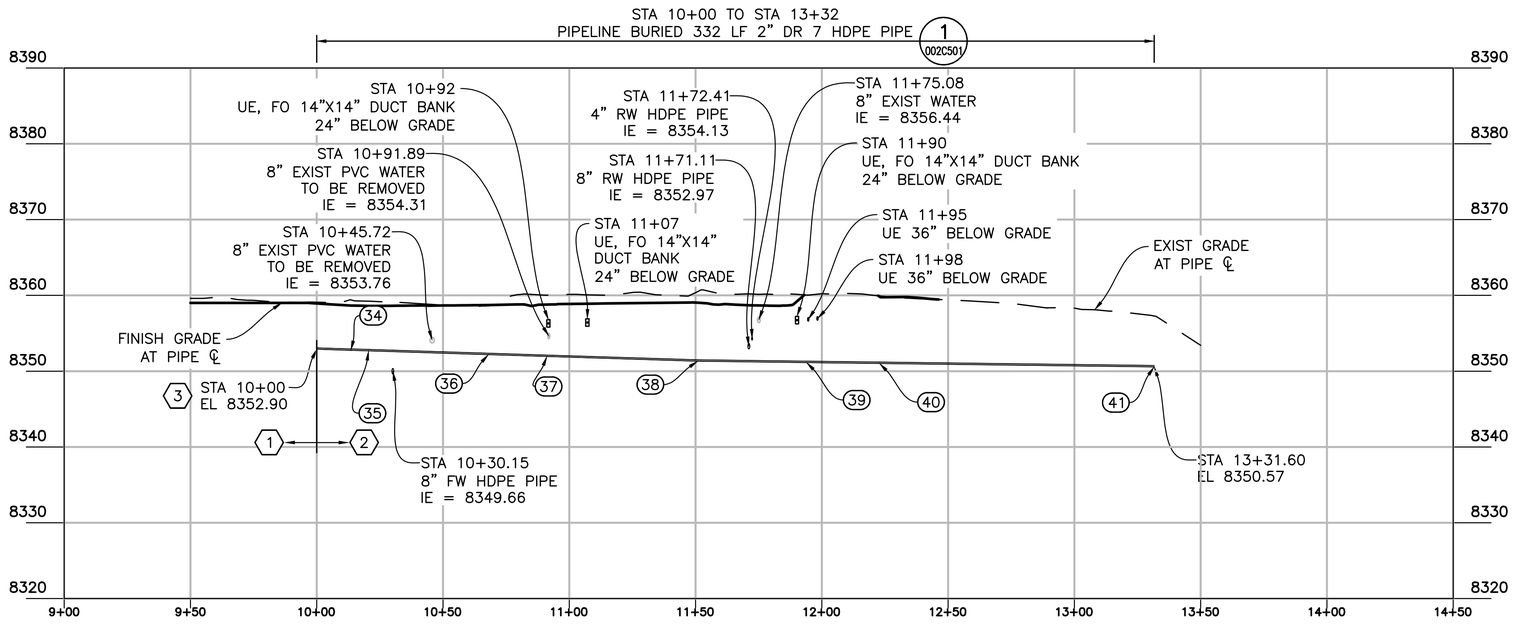


PLAN VIEW

1"=30'



PROFILE VIEW

POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
34	13150597.82	1328255.70	45° HORIZONTAL BEND
35	13150604.85	1328256.41	45° HORIZONTAL BEND
36	13150634.87	1328293.14	45° HORIZONTAL BEND
37	13150632.55	1328316.16	45° HORIZONTAL BEND
38	13150586.26	1328353.99	22.5° HORIZONTAL BEND
39	13150545.03	1328366.40	11.25° HORIZONTAL BEND
40	13150519.55	1328379.95	22.5° HORIZONTAL BEND
41	13150450.38	1328463.88	2.95° HORIZONTAL DEFLECTION

GENERAL NOTE:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

C. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8–16–2021.

D. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

E. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002–C102.

F. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES (X)

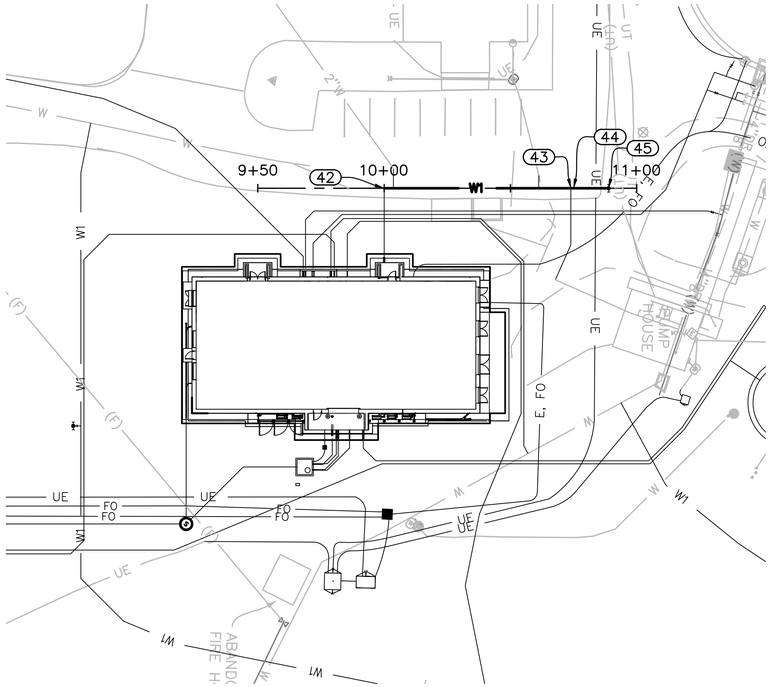
1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.
3. SEE DETAIL 02/061C503 FOR PIPE TRANSITION INFORMATION.

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

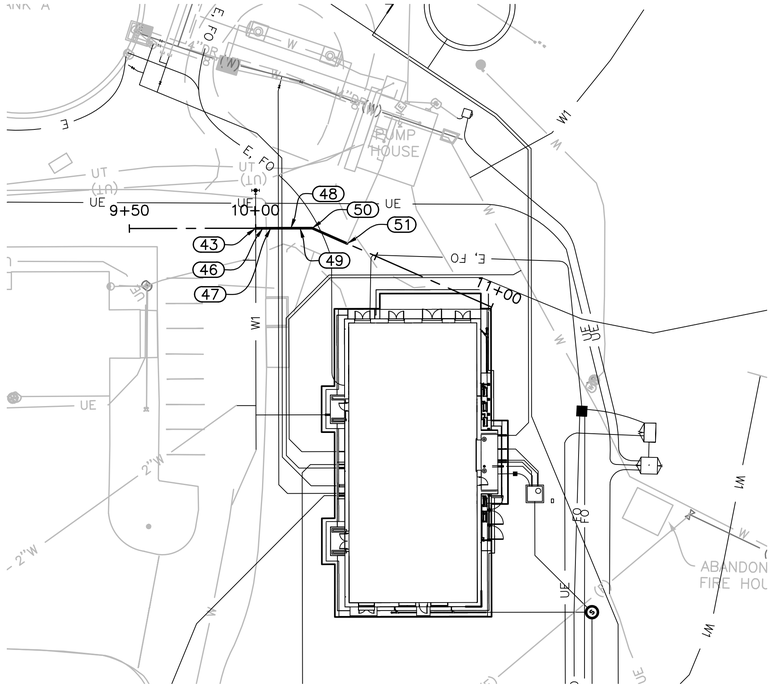
REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
United States
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C805	TITLE OF SHEET SITEWORK YARD PIPING PLAN AND PROFILE 2" FW NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 35 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C805.dwg



PLAN VIEW



PLAN VIEW

GENERAL NOTE:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

C. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED BY THE CONTRACTOR AND SAFE SITE UTILITY SERVICES, LLC, 8-16-2021.

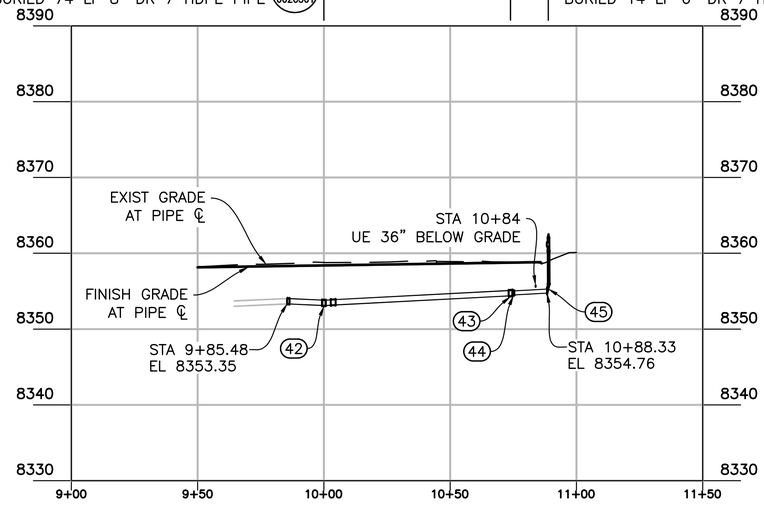
D. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

E. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002-C102.

F. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

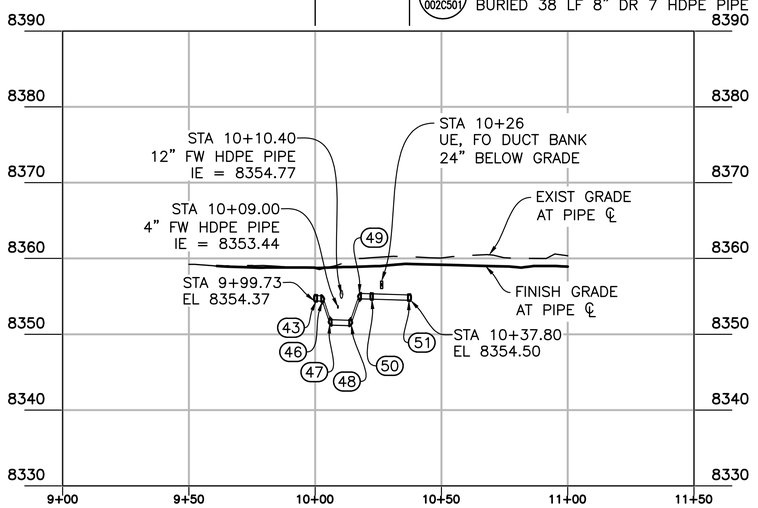
STA 10+00 TO STA 10+74 PIPELINE (1 / 002C501)
BURIED 74 LF 8" DR 7 HDPE PIPE

STA 10+74 TO STA 10+88 PIPELINE (1 / 002C501)
BURIED 14 LF 6" DR 7 HDPE PIPE



PROFILE VIEW

STA 10+00 TO STA 10+38 PIPELINE (1 / 002C501)
BURIED 38 LF 8" DR 7 HDPE PIPE



PROFILE VIEW

POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
42	13150621.05	1328255.53	TEE
43	13150667.82	1328312.76	TEE
44	13150668.53	1328313.63	REDUCER
45	13150677.30	1328324.35	HYDRANT
46	13150665.58	1328314.57	45° VERTICAL BEND
47	13150663.17	1328316.55	45° VERTICAL BEND
48	13150657.00	1328321.59	45° VERTICAL BEND
49	13150654.19	1328323.88	45° VERTICAL BEND
50	13150650.49	1328326.91	22.5° HORIZONTAL BEND
51	13150635.91	1328330.94	TEE

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
Date Signed 5/1/2026
ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	G. YOUNG	061 C806	SITEWORK YARD PIPING PLAN AND PROFILE 8" FW (BYPASS) NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
	DRAWN: P. OXBORROW			PMIS/PKG NO. 318719
	TECH. REVIEW: GOUGH/GLOVER			SHEET 36 OF 192
	DATE: MAY 2026			

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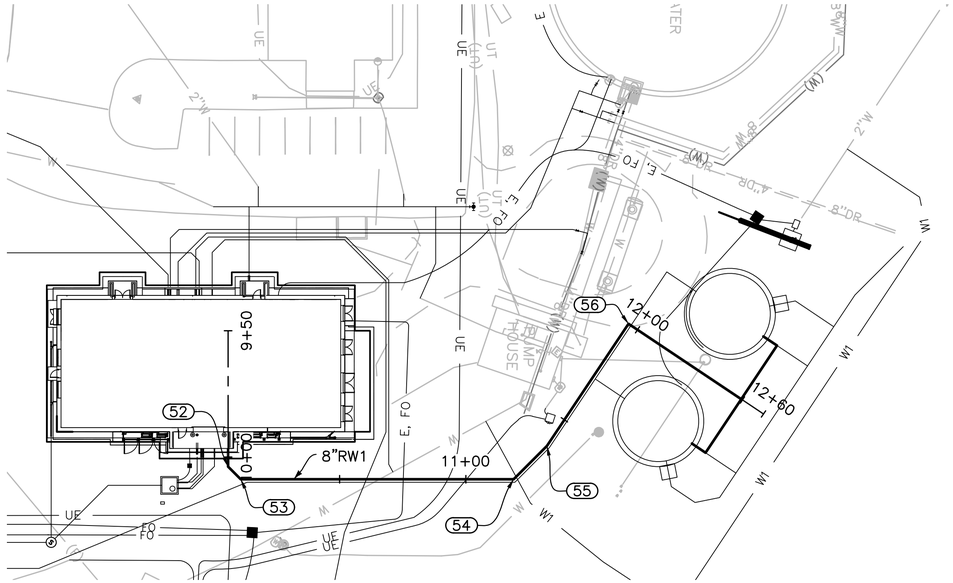
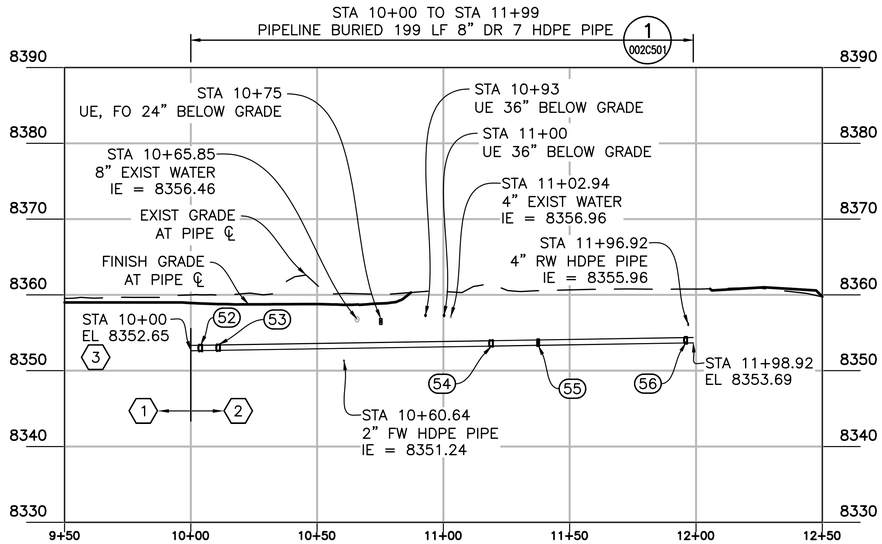


PLAN VIEW
1"=30'



PROFILE VIEW

POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
52	13150536.14	1328314.23	45° HORIZONTAL BEND
53	13150535.43	1328321.27	45° HORIZONTAL BEND
54	13150603.80	1328404.92	45° HORIZONTAL BEND
55	13150622.34	1328406.79	11.25° HORIZONTAL BEND
56	13150680.24	1328401.21	90° HORIZONTAL BEND

GENERAL NOTE:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

C. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8–16–2021.

D. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

E. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002–C102.

F. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS.

KEYNOTES ⟨X⟩

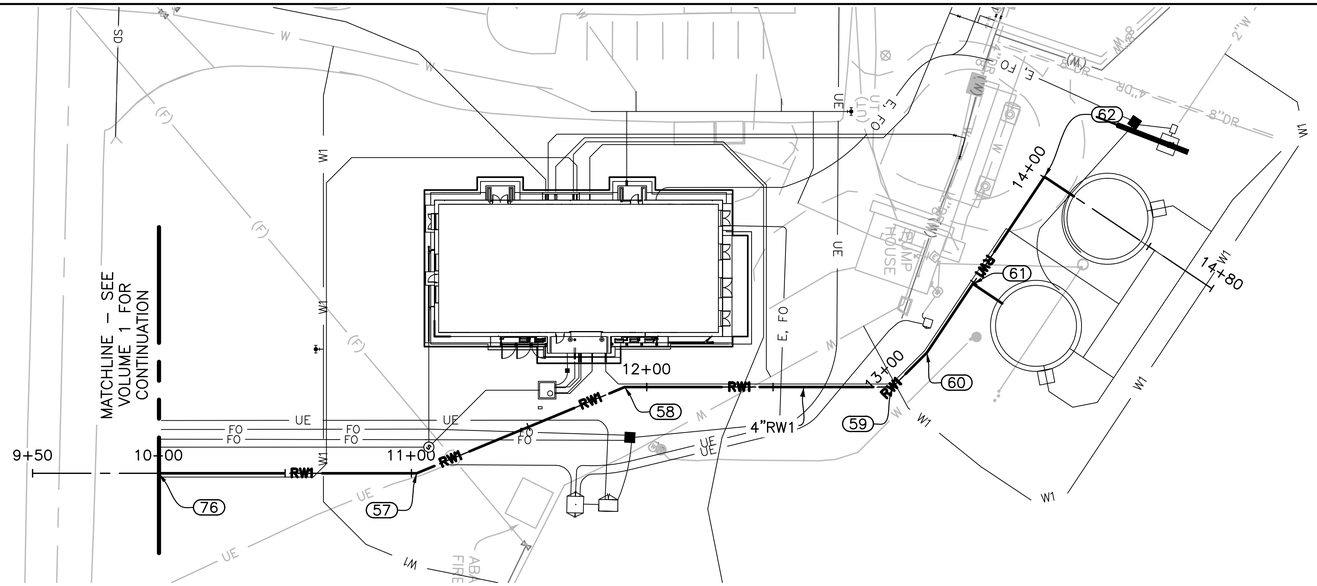
1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.
3. SEE DETAIL 02/061C503 FOR PIPE TRANSITION INFORMATION.

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
Date Signed 5/11/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C807	TITLE OF SHEET SITEWORK YARD PIPING PLAN AND PROFILE 8" RW FROM TANK NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 37 OF 192	

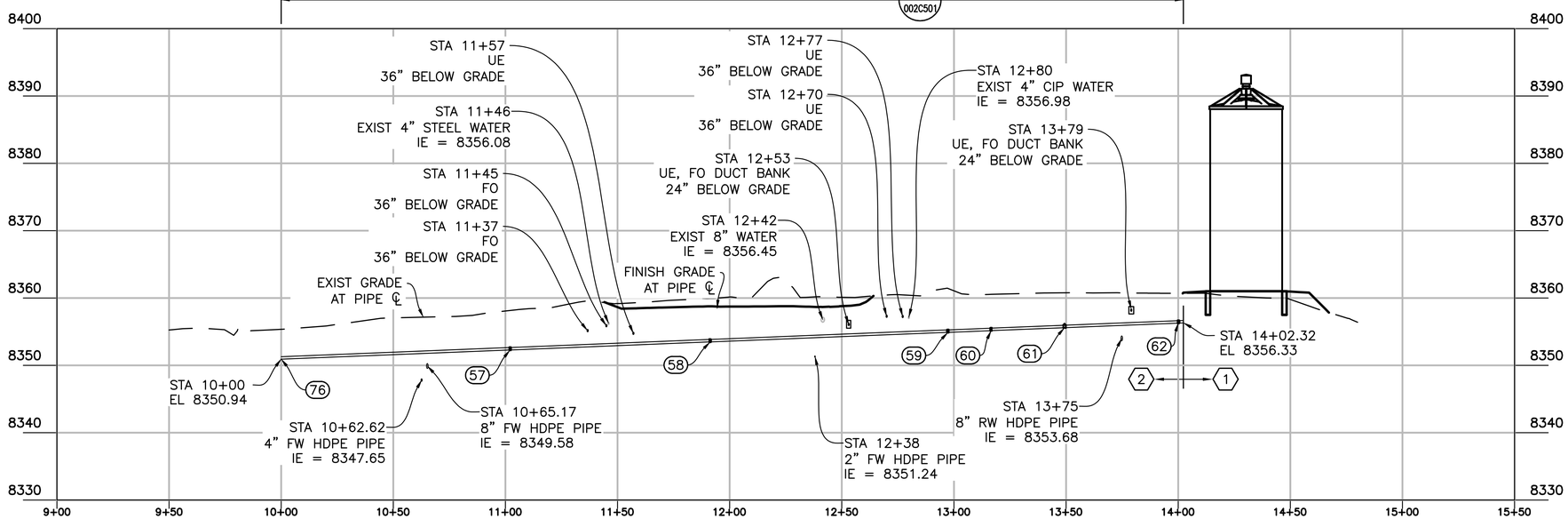
4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C807.dwg



PLAN VIEW

1"=30'

STA 10+00 TO STA 14+02
PIPELINE BURIED 402 LF 4" DR 7 HDPE PIPE (1 / 002C501)



PROFILE VIEW

GENERAL NOTE:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

C. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8–16–2021.

D. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

E. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002–C102.

F. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES (X)

1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.

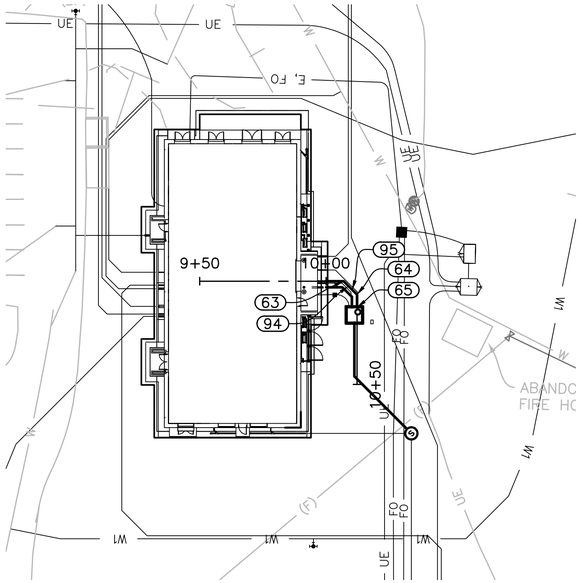
POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
57	13150457.54	1328281.77	22.5° HORIZONTAL BEND
58	13150536.09	1328323.97	22.5° HORIZONTAL BEND
59	13150603.18	1328406.07	45° HORIZONTAL BEND
60	13150622.34	1328407.99	11.25° HORIZONTAL BEND
61	13150654.97	1328404.85	TEE
62	13150705.74	1328399.95	90° HORIZONTAL BEND
76	13150392.94	1328202.72	4" RW END

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

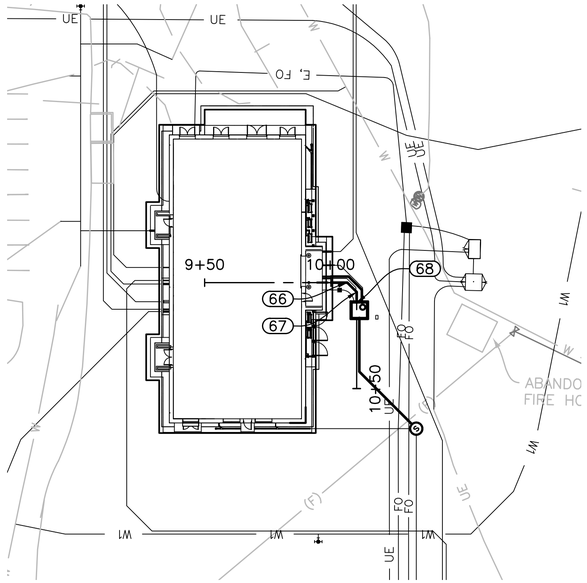
REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75262 GREGORY YOUNG State of Arizona ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C808	TITLE OF SHEET SITEWORK YARD PIPING PLAN AND PROFILE 4" RW (FROM BORE) NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 38 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061–C808.dwg

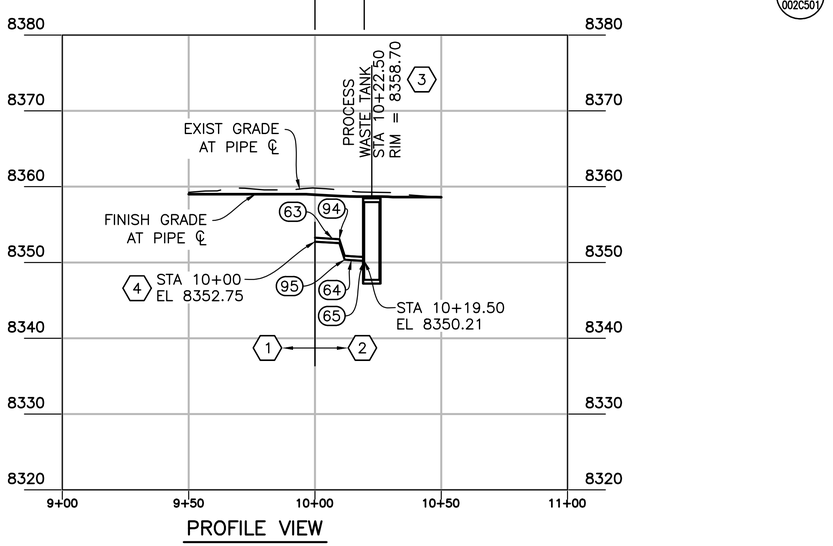


PLAN VIEW
1"=30'



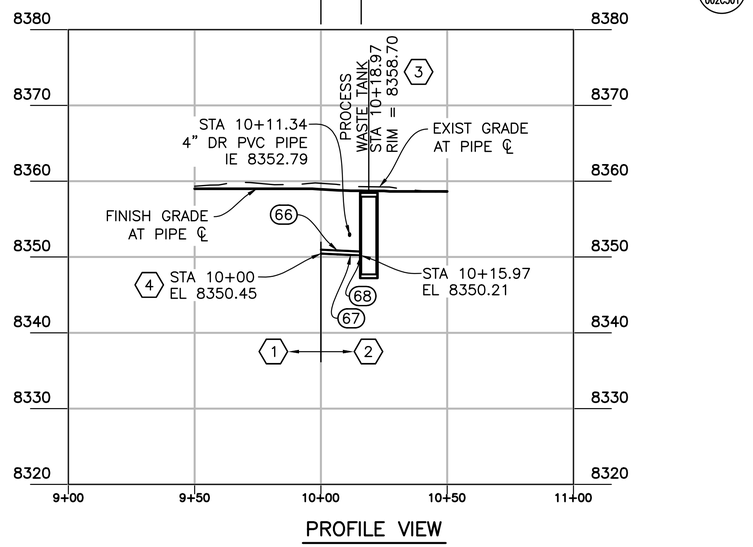
PLAN VIEW
1"=30'

STA 10+00 TO STA 10+20
PIPELINE BURIED 20 LF 6" C900 DR 25 PVC PIPE (1) 002C501



PROFILE VIEW

STA 10+00 TO STA 10+16 PIPELINE
BURIED 16 LF 6" C900 DR 25 PVC PIPE (1) 002C501



PROFILE VIEW

GENERAL NOTE:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

C. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8-16-2021.

D. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

E. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002-C102.

F. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES (X)

1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.
3. SEE 061C501 FOR ADDITIONAL PROCESS WASTE TANK INFORMATION.
4. SEE DETAIL 02/061C503 FOR PIPE TRANSITION INFORMATION.

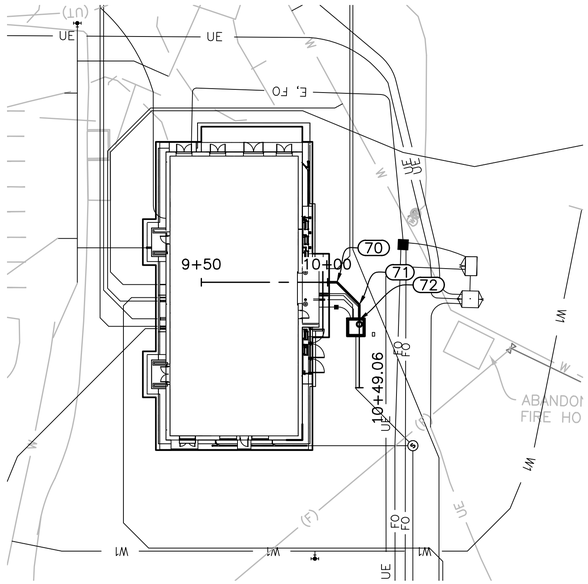
POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
63	13150527.41	1328308.61	45° HORIZONTAL BEND
64	13150520.03	1328307.87	45° HORIZONTAL BEND
65	13150516.82	1328303.94	CONNECT TO PROCESS WASTE TANK
66	13150526.30	1328306.44	45° HORIZONTAL BEND
67	13150521.06	1328305.91	45° HORIZONTAL BEND
68	13150518.39	1328302.65	CONNECT TO PROCESS WASTE TANK
94	13150524.77	1328308.35	45° VERTICAL BEND
95	13150522.68	1328308.14	45° VERTICAL BEND

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 75262 GREGORY YOUNG Date Signed 5/1/2026 ARIZONA, U.S.A.

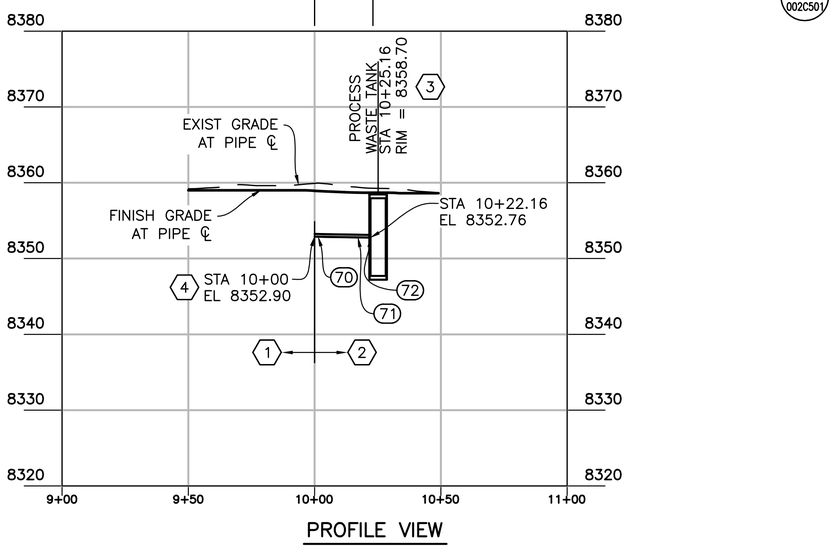
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C809	TITLE OF SHEET SITEWORK YARD PIPING PLAN AND PROFILE 6" DR NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 39 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C809.dwg

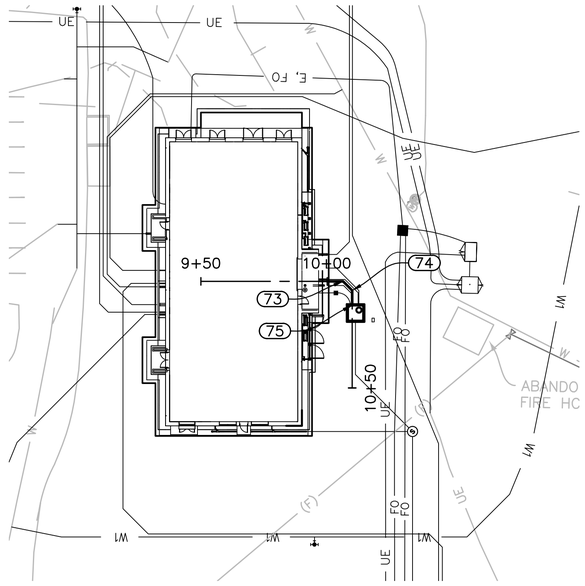


PLAN VIEW
1"=30'

STA 10+00 TO STA 10+23
PIPELINE BURIED 23 LF 4" C900 DR 25 PVC PIPE (1) 002C501

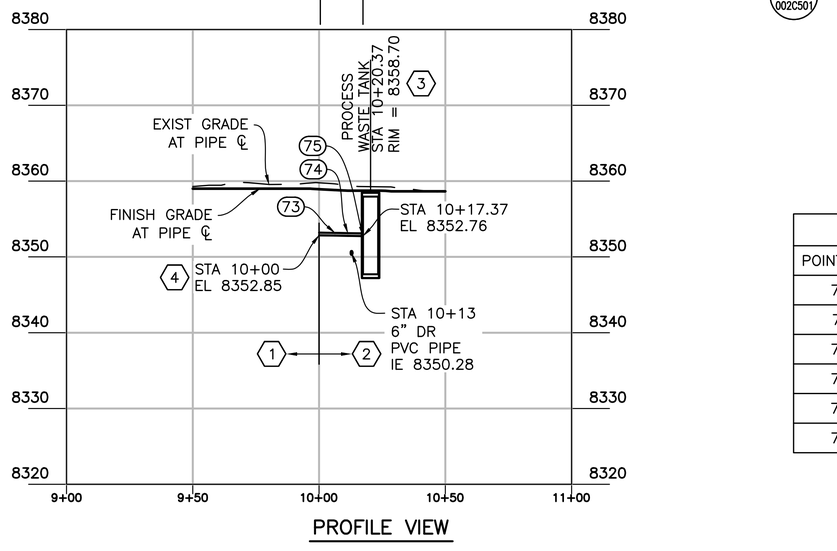


PROFILE VIEW



PLAN VIEW
1"=30'

STA 10+00 TO STA 10+17
PIPELINE BURIED 17 LF 4" C900 DR 25 PVC PIPE (1) 002C501



PROFILE VIEW

GENERAL NOTE:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

C. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8-16-2021.

D. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

E. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002-C102.

F. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES (X)

1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.
3. SEE 061C501 FOR ADDITIONAL PROCESS WASTE TANK INFORMATION.
4. SEE DETAIL 02/061C503 FOR PIPE TRANSITION INFORMATION.

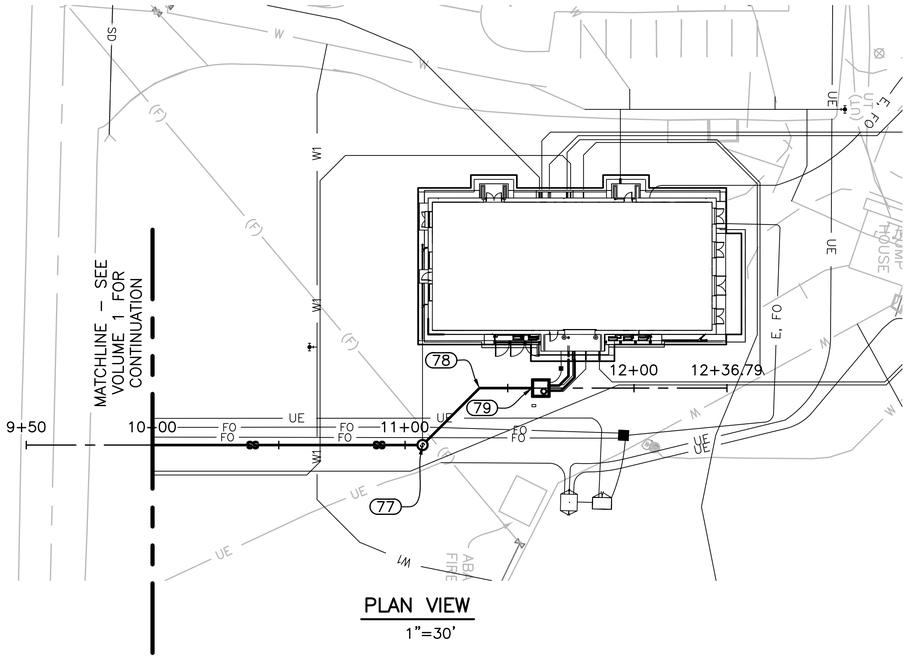
POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
70	13150532.80	1328309.98	45° HORIZONTAL BEND
71	13150520.34	1328308.73	45° HORIZONTAL BEND
72	13150516.58	1328304.13	CONNECT TO PROCESS WASTE TANK
73	13150527.48	1328307.59	45° HORIZONTAL BEND
74	13150522.39	1328307.08	45° HORIZONTAL BEND
75	13150518.62	1328302.46	CONNECT TO PROCESS WASTE TANK

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

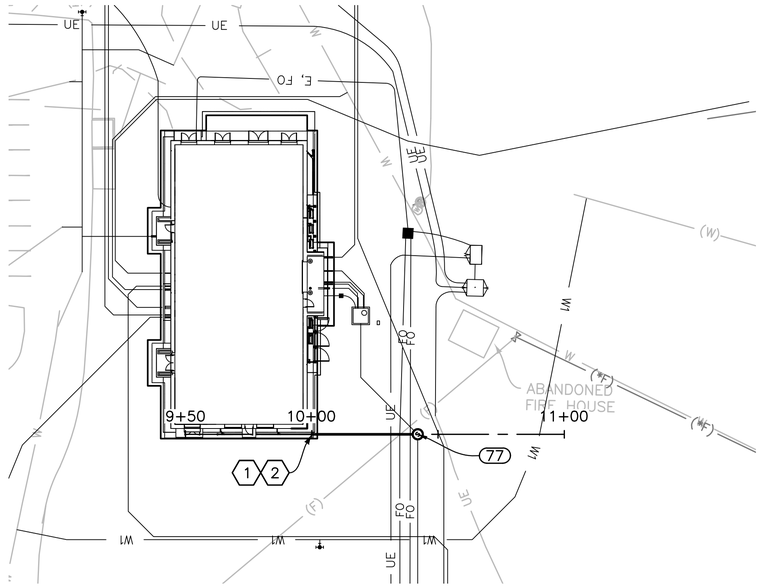
REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
STATE OF ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C810	TITLE OF SHEET SITEWORK YARD PIPING PLAN AND PROFILE 4" DR NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 40 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C810.dwg

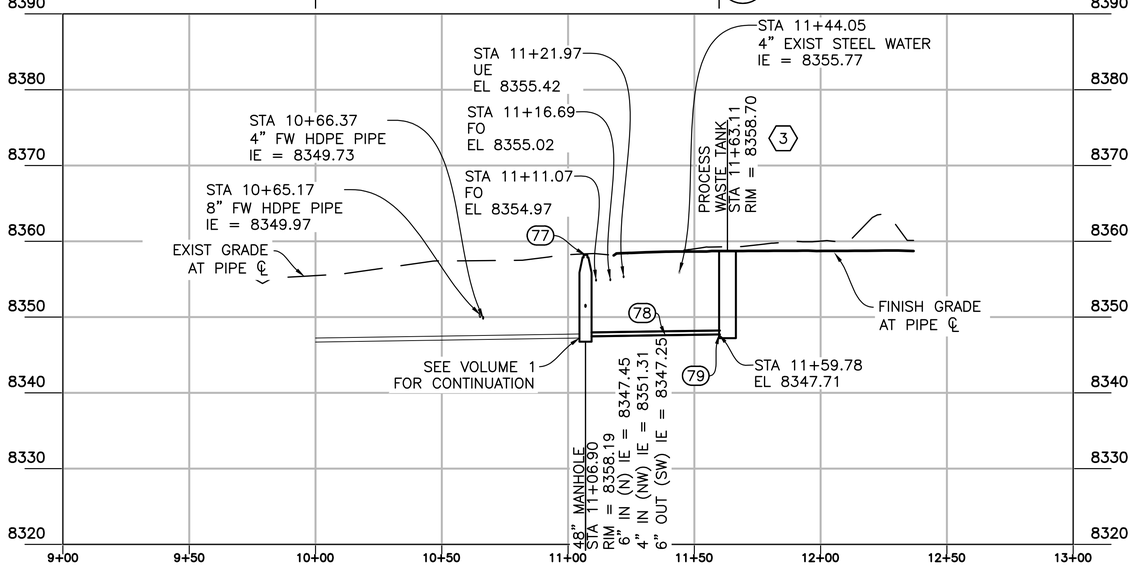


PLAN VIEW
1"=30'



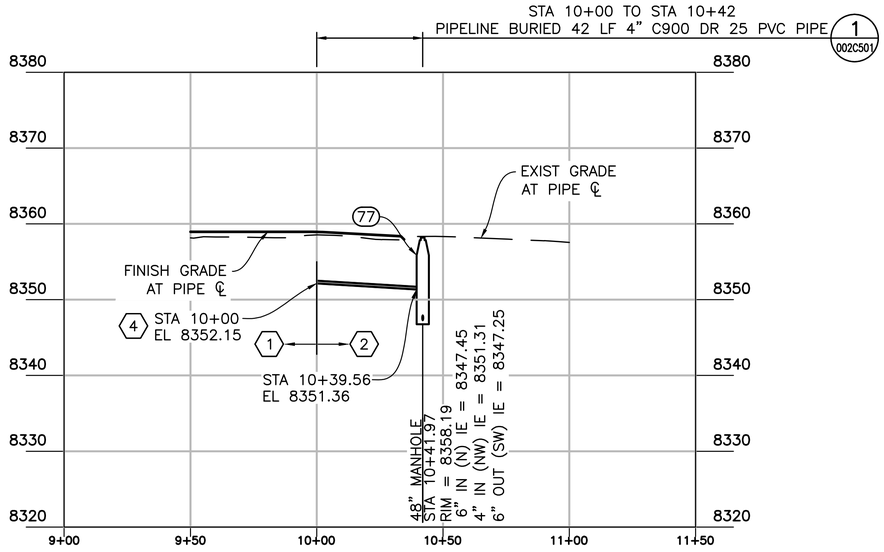
PLAN VIEW
1"=30'

STA 10+00 TO STA 11+60
PIPELINE BURIED 160 LF 6" C900 DR 25 PVC PIPE (1) 002C501



PROFILE VIEW

STA 10+00 TO STA 10+42
PIPELINE BURIED 42 LF 4" C900 DR 25 PVC PIPE (1) 002C501



PROFILE VIEW

POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
77	13150468.59	1328278.95	SSMH
78	13150500.33	1328282.14	45° HORIZONTAL BEND
79	13150513.50	1328298.26	CONNECT TO PROCESS WASTE TANK

GENERAL NOTE:

A. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

B. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8-16-2021.

C. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

D. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002-C102.

E. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES (X)

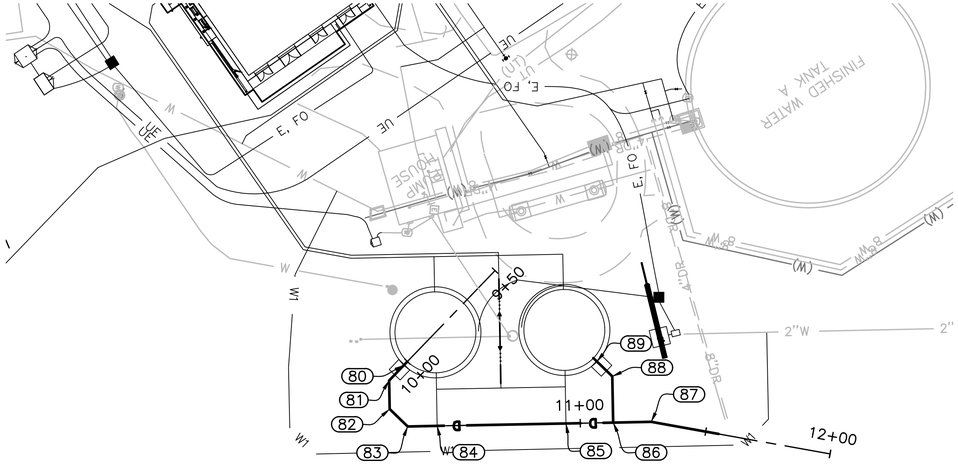
1. SEE PLUMBING SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.
3. SEE 061C501 FOR ADDITIONAL PROCESS WASTE TANK INFORMATION.
4. SEE DETAIL 02/061C503 FOR PIPE TRANSITION INFORMATION.

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

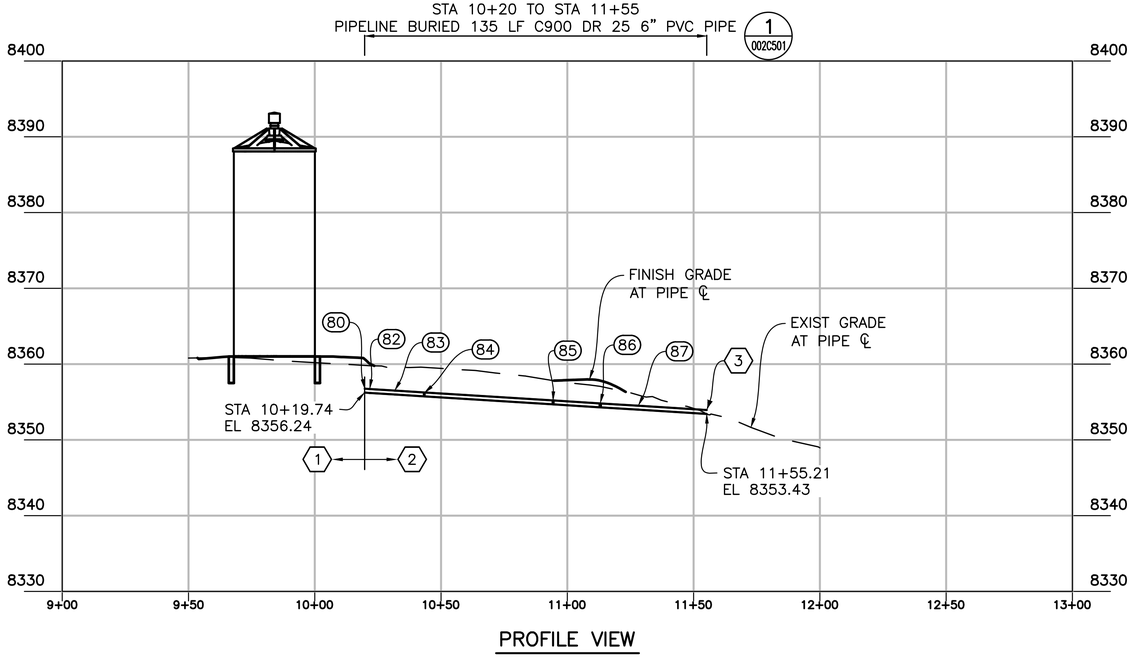
REGISTERED PROFESSIONAL ENGINEER – CERTIFICATE NO. 75262 – GREGORY YOUNG – Date Signed 5/1/2026 – ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C811	TITLE OF SHEET SITEWORK YARD PIPING PLAN AND PROFILE 6" SS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 41 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C811.dwg



PLAN VIEW
1"=30'



PROFILE VIEW

POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
80	13150646.40	1328448.60	CONNECT TO TANK
81	13150641.04	1328455.11	45° HORIZONTAL BEND
82	13150642.12	1328466.36	45° HORIZONTAL BEND
83	13150649.76	1328472.66	45° HORIZONTAL BEND
84	13150661.40	1328471.54	TEE
85	13150712.17	1328466.65	TEE
86	13150730.77	1328464.85	TEE
87	13150745.76	1328463.41	11.25° HORIZONTAL BEND
88	13150729.02	1328446.63	45° HORIZONTAL BEND
89	13150722.51	1328441.26	CONNECTION TO TANK

GENERAL NOTE:

A. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

B. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8-16-2021.

C. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

D. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002-C102.

E. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES ⟨X⟩

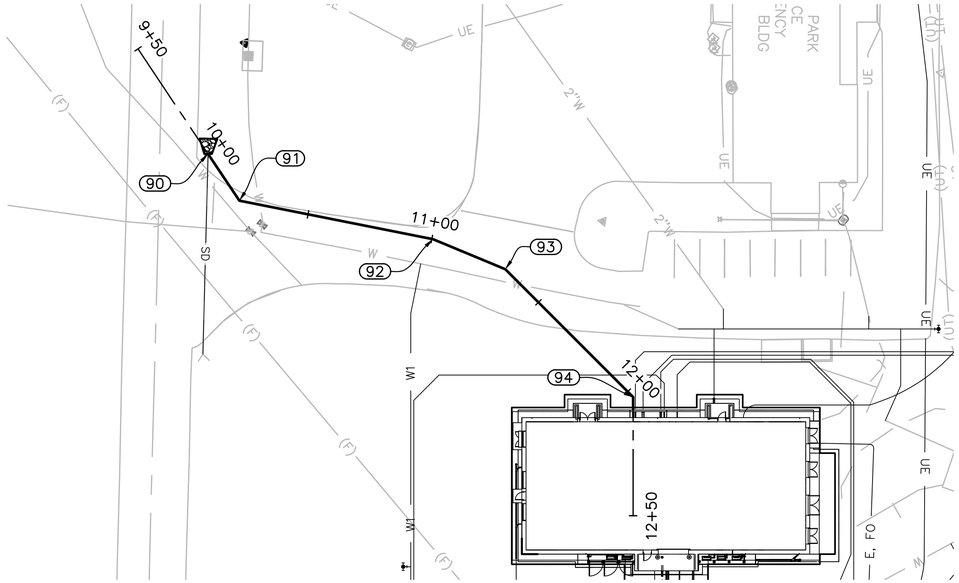
1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.
3. RAW WATER DRAIN OUTLET SHALL EXTEND TO EXIST CATCH BASIN FOR THE EXIST FRESH WATER TANKS.

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
Date Signed
ARIZONA, U.S.A.

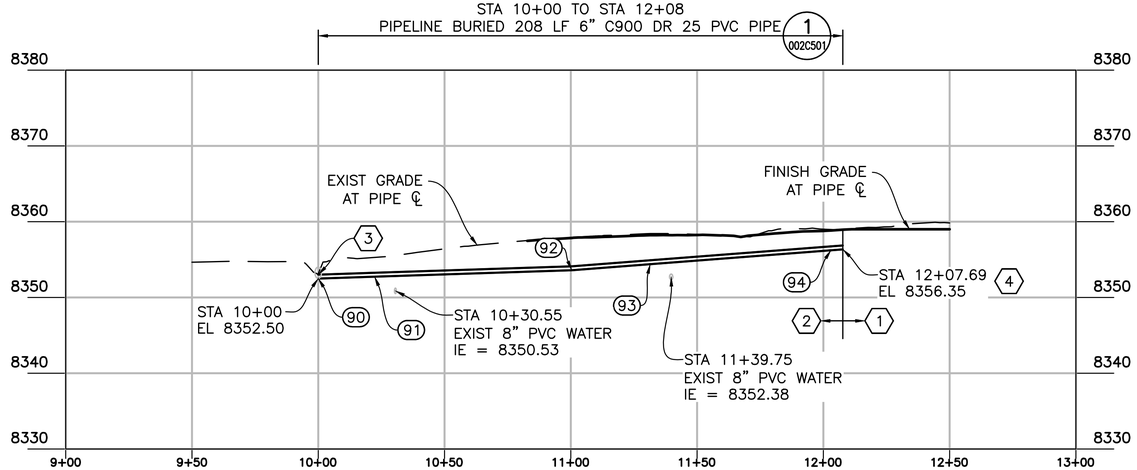
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	061 C812	SITEWORK YARD PIPING PLAN AND PROFILE 6" OVF (TANKS) NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 42 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C812.dwg



PLAN VIEW
1"=30'

STA 10+00 TO STA 12+08
PIPELINE BURIED 208 LF 6" C900 DR 25 PVC PIPE (1 002C501)



PROFILE VIEW

POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
90	13150547.98	1328056.55	OVF END / CHECK VALVE
91	13150541.47	1328077.88	45° HORIZONTAL BEND
92	13150578.15	1328146.86	11.25° HORIZONTAL BEND
93	13150587.10	1328176.57	22.5° HORIZONTAL BEND
94	13150579.93	1328247.78	45° HORIZONTAL BEND

GENERAL NOTE:

A. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

B. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8-16-2021.

C. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

D. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002-C102.

E. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES (X)

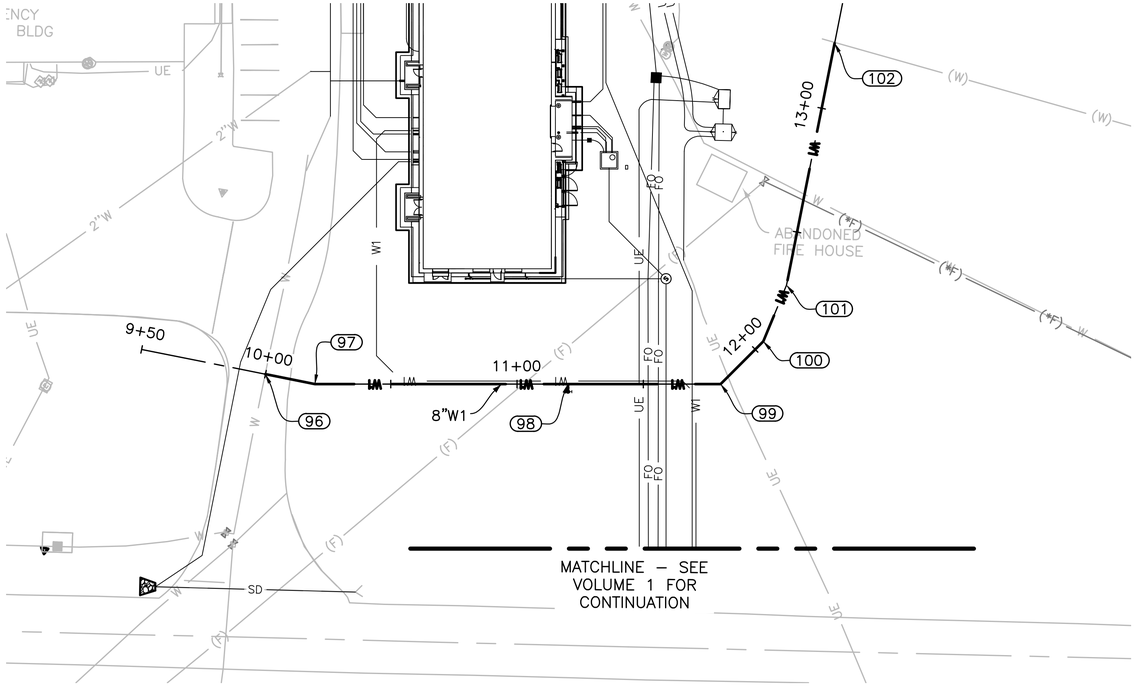
1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.
3. RAW WATER DRAIN OUTLET SHALL EXTEND TO EXIST DITCH. INSTALL FLAPGATE VALVE AT PIPE TERMINATION.
4. SEE DETAIL 02/061C503 FOR PIPE TRANSITION INFORMATION.

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

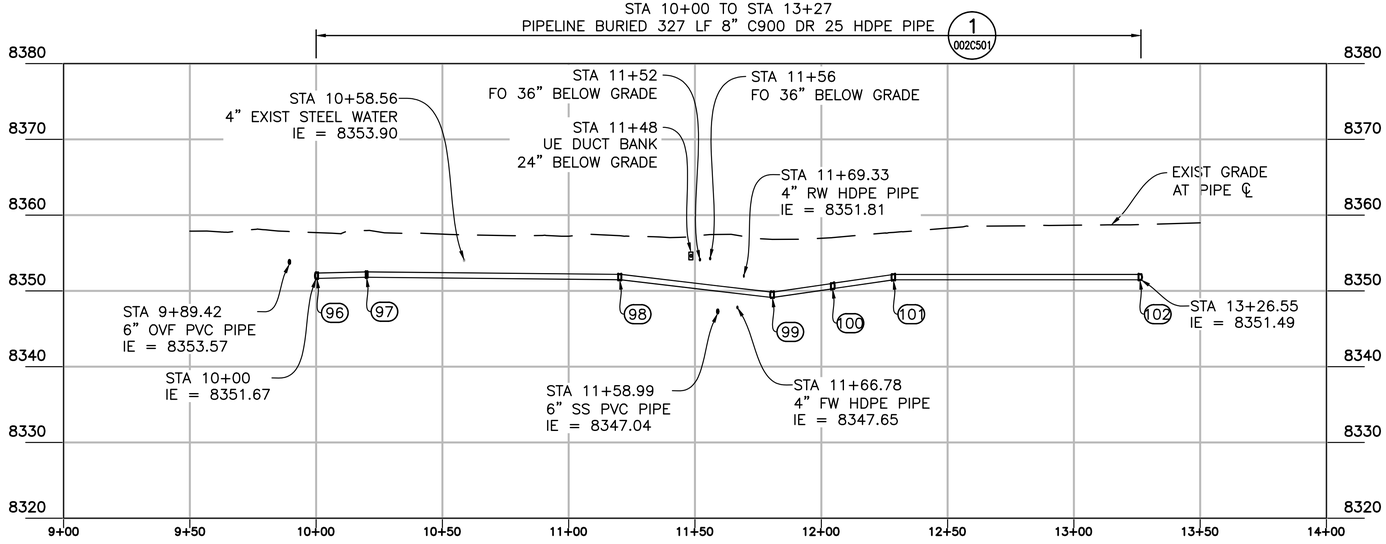
REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 75262 GREGORY YOUNG Date Signed 5/11/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD P. OXBORROW TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 061 C813	TITLE OF SHEET SITEWORK YARD PIPING PLAN AND PROFILE 6" OVF (WTP) NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 43 OF 192

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061-C813.dwg



PLAN VIEW
1"=30'



PROFILE VIEW

POINT TABLE			
POINT NO.	NORTHING	EASTING	DESCRIPTION
96	13150567.57	1328149.26	8"X8"X8" TEE
97	13150549.84	1328158.67	11.25° HORIZONTAL BEND
98	13150472.21	1328222.10	8"X8"X6" FIRE HYDRANT TEE
99	13150425.47	1328260.30	45° HORIZONTAL BEND
100	13150423.08	1328284.08	22.5° HORIZONTAL BEND
101	13150430.02	1328307.15	11.25° HORIZONTAL BEND
102	13150475.89	1328393.39	8"X8"X8" TEE

GENERAL NOTE:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR TO FIELD VERIFY DEPTHS AND LOCATIONS OF ALL EXISTING UTILITIES.

C. ABBREVIATION PH ON ALL THE UTILITY PLAN AND PROFILES STANDS FOR POTHOLE INFORMATION THAT WAS PROVIDED SAFE SITE UTILITY SERVICES, LLC, 8–16–2021.

D. ALL MECHANICAL JOINTS AND FITTINGS SHALL BE DUCTILE IRON AND ALL JOINTS SHOULD BE RESTRAINED.

E. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0, 002–C102.

F. LENGTHS OF PIPE SHOWN IN PROFILE REFLECT CALCULATED SLOPE LENGTHS. CONTRACTOR RESPONSIBLE FOR VERIFYING LENGTHS PRIOR TO MATERIAL PROCUREMENT.

KEYNOTES (X)

1. SEE PROCESS SHEETS FOR CONTINUATION. REFERENCE SPECIFICATION DIVISION 40 FOR PROCESS PIPING.

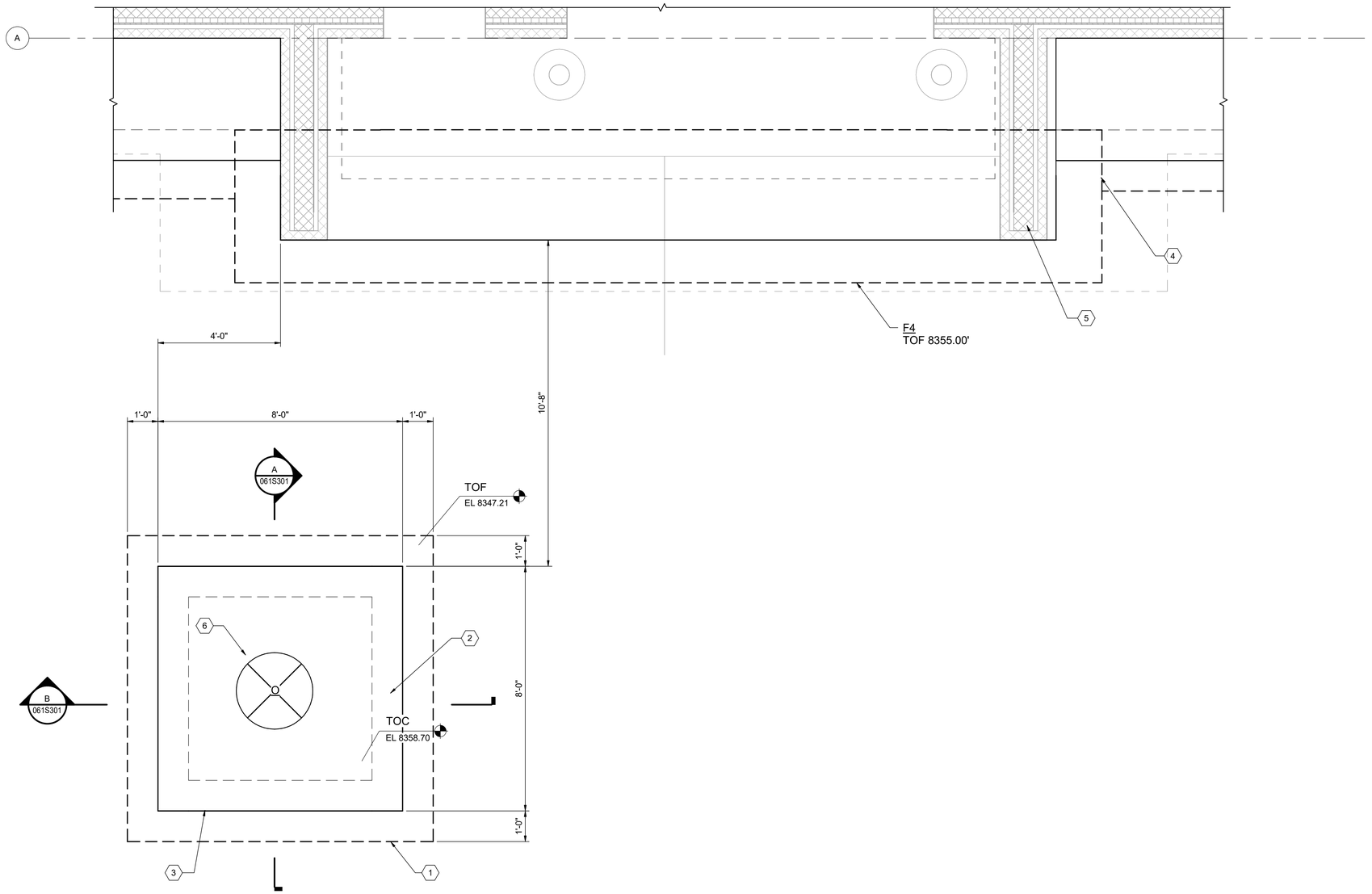
2. REFERENCE SPECIFICATION DIVISION 33 FOR SITE YARD PIPING.

30 0 30 60 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
Date Signed 5/11/26
ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	G. YOUNG	061 C814	SITEWORK YARD PIPING PLAN AND PROFILE 8" FW (TO CAMPGROUND) NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
	CADD P. OXBORROW			PMIS/PKG NO. 318719
	TECH. REVIEW: GOUGH/GLOVER			SHEET 44 OF 192
	DATE: MAY 2026			

4/17/2026 Oxborrow, Paul C:\Users\POXBORRO\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\061–C814.dwg



1 061S101 PROCESS WASTE TANK FOUNDATION PLAN

SCALE A

GENERAL NOTES

A. REFER TO SHEETS 060S001 THRU 060S003 FOR GENERAL STRUCTURAL NOTES AND TYPICAL DETAILS.

B. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING. EQUIPMENT, ANCHORS. CONDUITS, HOUSEKEEPING PADS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTURAL ELEMENTS THAT MAY NOT BE SHOWN.

C. CXX DENOTES STEEL COLUMNS. REFER TO SHEET 062S601 FOR STEEL COLUMN, BASE PLATE.

D. TOP OF CONCRETE FOOTING ELEVATIONS NOTED THUS: EL. 8355.00.

E. FX DENOTES SPREAD FOOTING. REFER TO SHEET 062S601 FOR FOOTING SCHEDULE.

F. O DENOTES OPENING IN CONCRETE LID.

KEYNOTES

1 24" CONCRETE MAT REINFORCED WITH #8@9" O.C. EACH WAY, TOP AND BOTTOM.

2 12" CONCRETE WALL REINFORCED WITH #8@9" O.C. VERTICAL EACH FACE AND #8@10" O.C. HORIZONTAL EACH FACE.

3 12" CONCRETE LID REINFORCED WITH #8@9" O.C. TOP AND BOTTOM EACH WAY.

4 BOTTOM STEP FOOTING PER VOLUME 0 TYPICAL DETAILS.

5 12" CMU BEARING WALLS. SEE TYPICAL DETAILS FOR REINFORCING SIZE AND SPACING. SEE ARCH DRAWINGS FOR EXTENTS OF EXTERIOR FINISHES.

6 PROVIDE ADDITIONAL BARS AT RE-ENTRANT CORNERS PER VOLUME 0 TYPICAL DETAILS.

SCALE A 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Colo.) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed: 04/27/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK CADD R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 061 S101	TITLE OF SHEET SITEWORK PROCESS WASTE TANK FOUNDATION PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 45 OF 192

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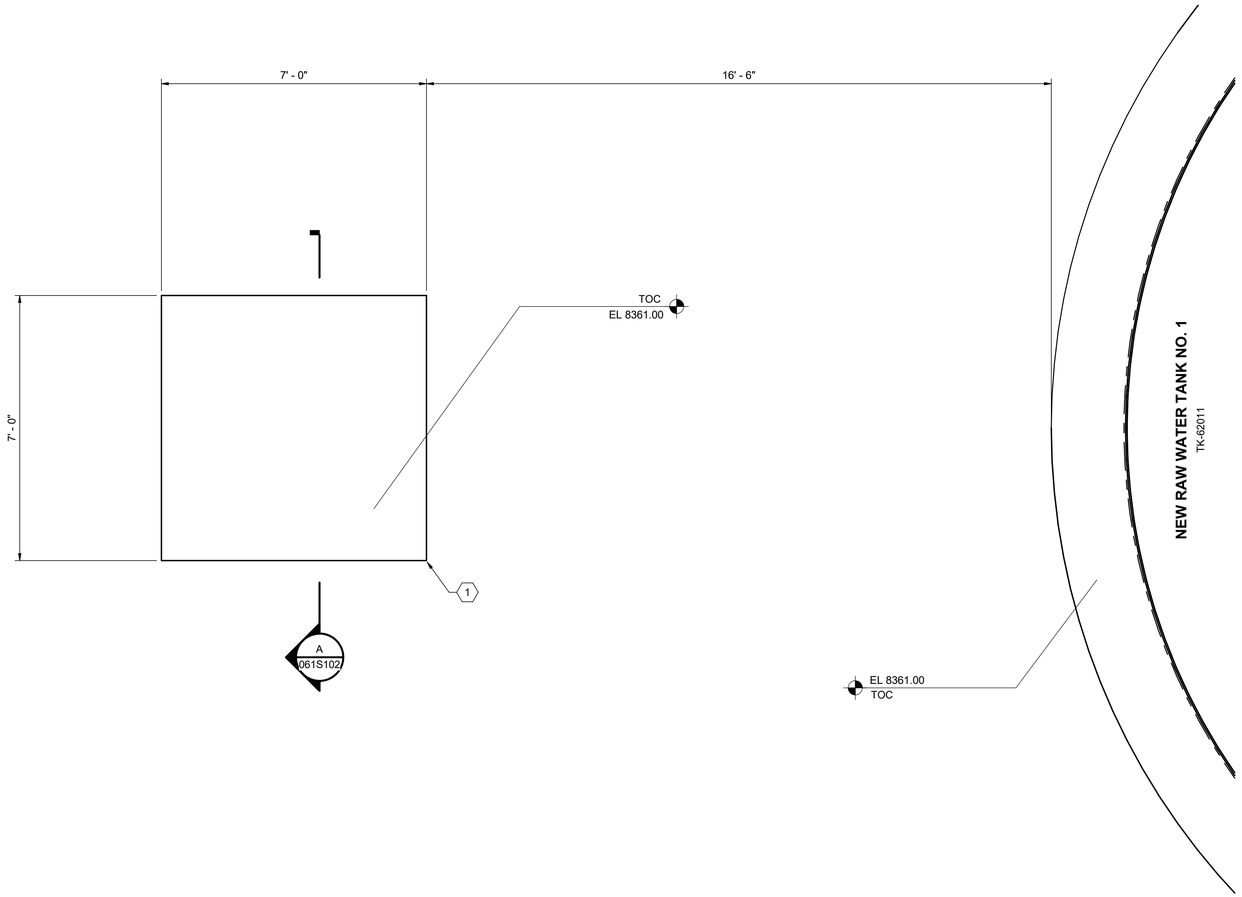
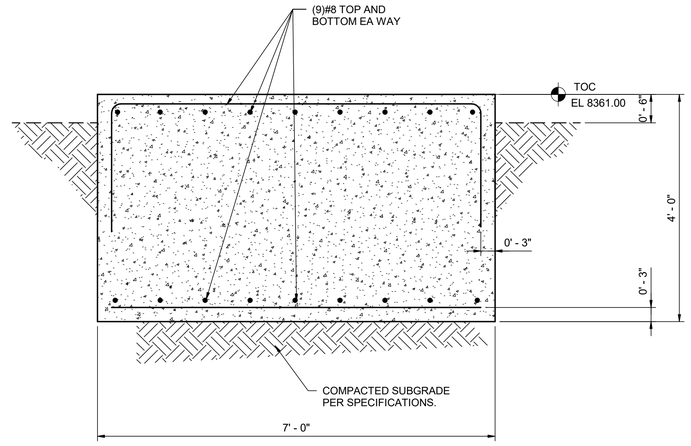


1 061S102 COMMUNICATION TOWER FOUNDATION PLAN
SCALE (A)



A 061S102 COMMUNICATION TOWER CONCRETE PAD
SCALE (B)

GENERAL NOTES

A. REFER TO SHEETS 060S001 FOR GENERAL STRUCTURAL NOTES.

B. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING, EQUIPMENT, ANCHORS, CONDUITS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTURAL ELEMENTS THAT MAY NOT BE SHOWN.

KEYNOTES

1 48" THICK CONCRETE PAD FOR COMMUNICATIONS TOWER.

SCALE (B) 1 0 1 2 3 — 3/4" = 1'-0"
SCALE OF FEET

SCALE (A) 2 0 2 4 — 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil)
CERTIFICATE NO.
72480
COLIN MULDERICK
Date Signed 10/31/25
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 061 S102	TITLE OF SHEET SITEWORK COMMUNICATION TOWER FOUNDATION PLAN AND SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 46 OF 192

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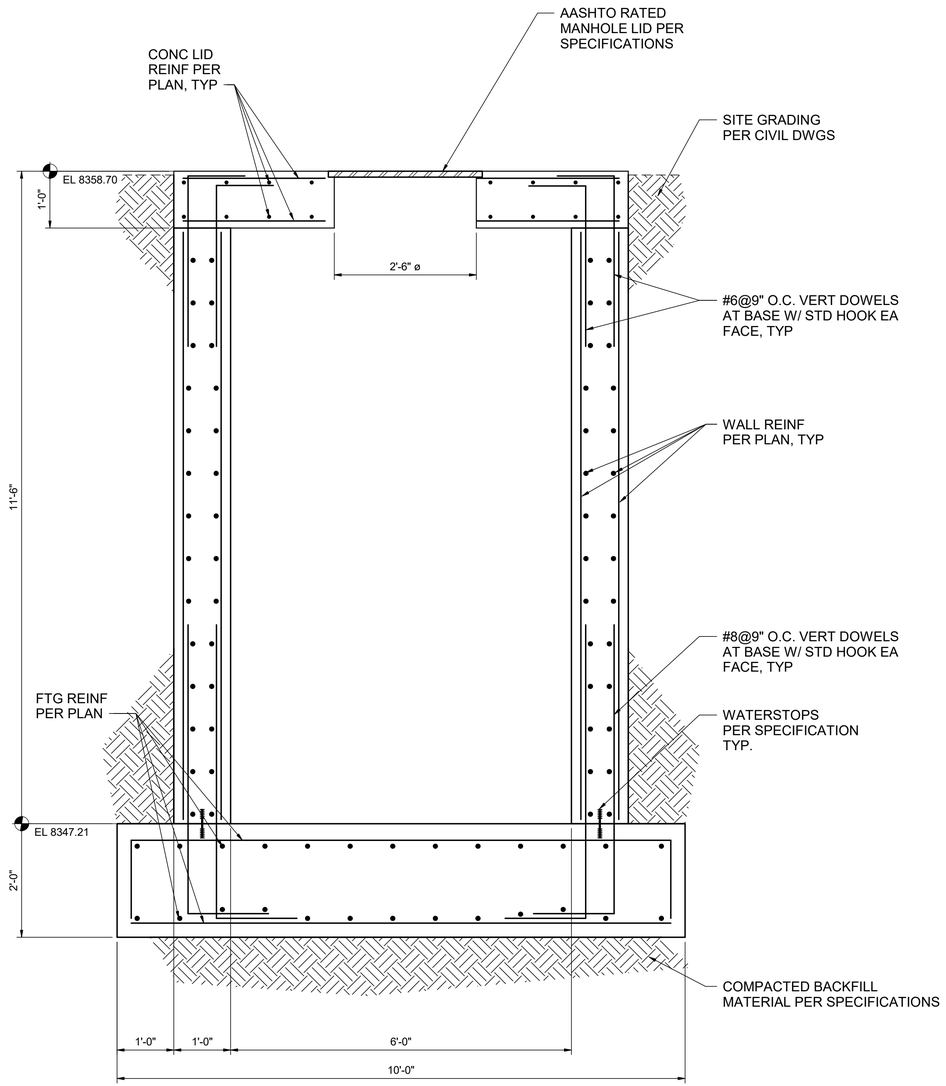
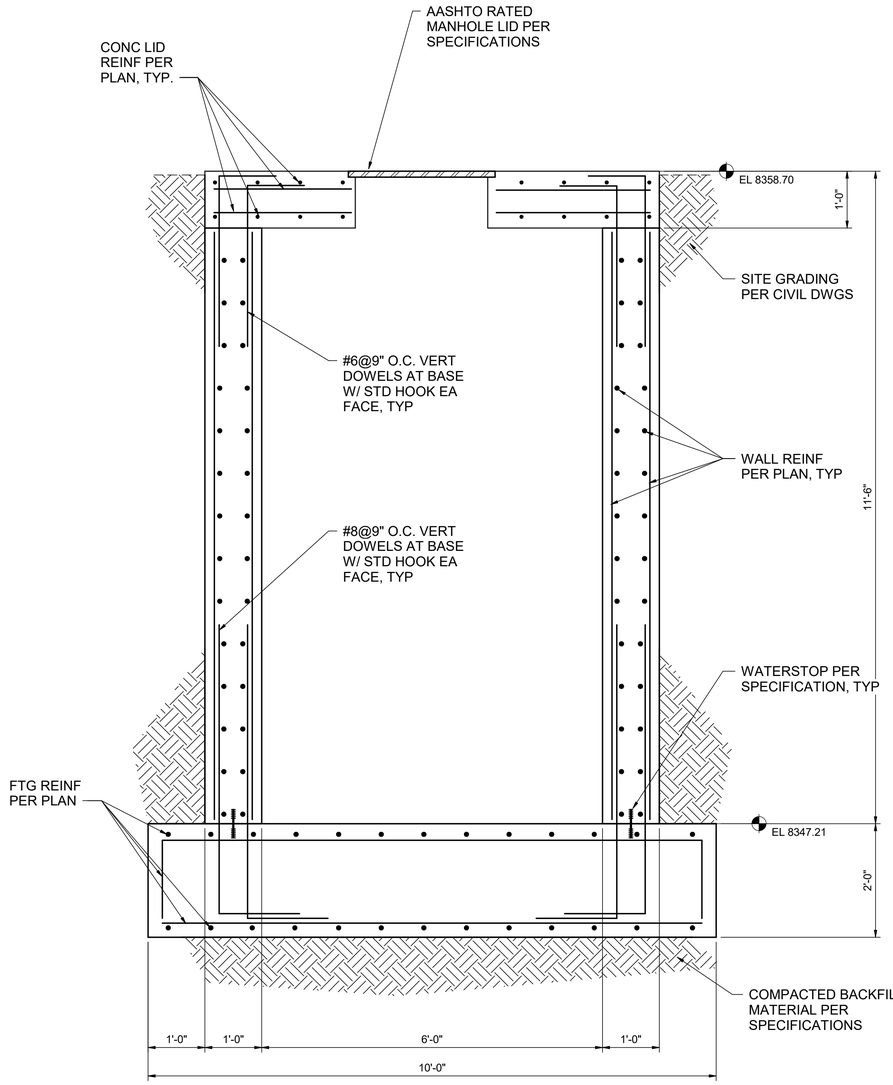


A 061S301 PROCESS WASTE TANK SECTION
SCALE A



B 061S301 PROCESS WASTE TANK SECTION
SCALE A

SCALE A 1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Pro) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed 05/07/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 061 S301	TITLE OF SHEET SITEWORK PROCESS WASTE TANK SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 47 OF 192

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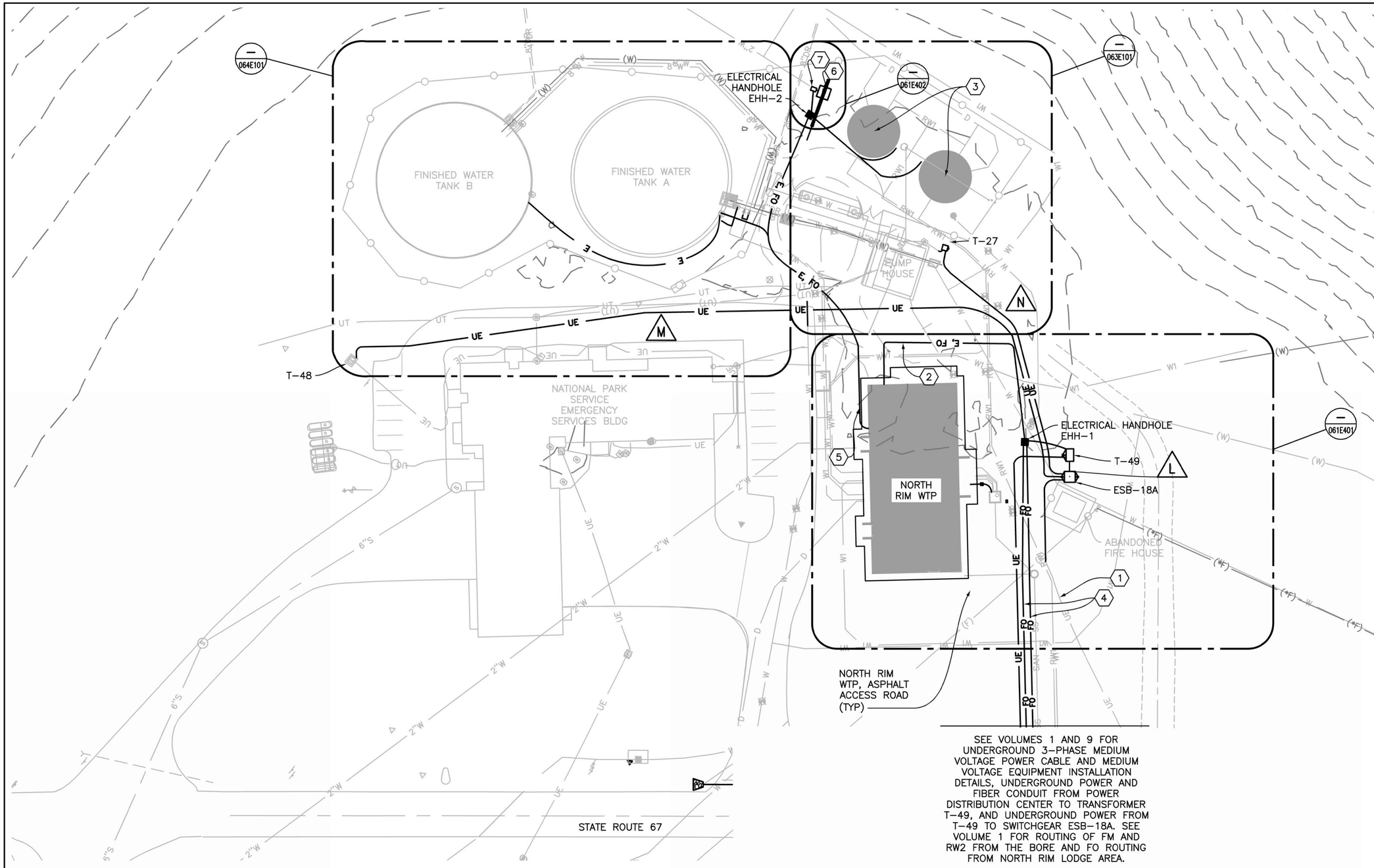


GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. ELECTRICAL DEMO IS SHOWN ON SHEET 061X101.

C. SEE SHEET 091E502 FOR MEDIUM VOLTAGE POWER CABLE SCHEDULE.

D. SEE VOLUME 0 SHEET 008E509 FOR PADMOUNTED JUNCTION AND TRANSFORMER DETAILS.

E. SEE VOLUME 0 SHEET 008E510 FOR CONDUCTOR TRENCHINNG, ELBOWS AND CONNECOTRS, VAULT GROUND DETAILS, AND PADMOUNTED SWTICHGEAR DETAILS.

KEY NOTES: (X)

1. RELOCATE EXISTING UNDERGROUND ELECTRICAL UTILITY LINES TO CLEAR NEW WATER TREATMENT PLANT BUILDING FOOTPRINT.
2. CONCRETE–ENCASED DUCT BANK FROM TRANSFORMER T–49 TO WTP ELECTRICAL ROOM.
3. LOCATION FOR NEW RAW WATER TANKS.
4. INSTALL 144–STRAND SM FIBER IN ORANGE TUBE OF NEW DIRECT–BURIED 7–WAY MICRODUCT. TRANSITION TO 2 INCH CONDUIT AT EHH–1
5. INSTALL 24–STRAND SM FIBER IN 2 INCH CONDUIT
6. TELECOMMUNICATIONS ANTENNA
7. TELECOMMUNICATIONS PANEL

30 0 30 60 1" = 30'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL CERTIFICATE NO. 75014 TOU VANG Date Signed 4/17/2026

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 4/17/2026 ARIZONA, USA

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S–DELGADO CADD A. S–DELGADO TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 061 E101	TITLE OF SHEET SITEWORK ELECTRICAL OVERALL SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 48 OF 192

4/17/2026 Salerno–Delgado, Antonio C:\Users\ASALERNODELG\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\061–E101.dwg

CONTINUED ON 063E101

NORTH RIM
WATER TREATMENT
PLANT

ELECTRICAL HANDHOLE
EHH–1

ELECTRICAL
HANDHOLE
EHH–3

PROCESS
WASTE TANK
TK–69111

LT
69151

LSHH
69151

T49

ESB–18A

MEDIUM VOLTAGE
UNDERGROUND POWER
CABLE POINT OF INTERCEPT
"C"

EXIST MEDIUM
VOLTAGE
UNDERGROUND POWER
CABLE TO ESB–18 AT
GAS STATION

CONTINUED ON
061E101

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. ELECTRICAL DEMO IS SHOWN ON SHEET 061X101.

KEY NOTES: (X)

1. RELOCATE EXISTING UNDERGROUND ELECTRICAL UTILITY LINES TO CLEAR NEW WATER TREATMENT PLANT BUILDING FOOTPRINT.

2. REINFORCED CONCRETE–ENCASED DUCT BANK FROM TRANSFORMER T–49 TO WTP ELECTRICAL ROOM. REFER TO DETAIL 1/008E501

3. INSTALL 144–STRAND SM FIBER IN ORANGE TUBE OF NEW DIRECT–BURIED 7–WAY MICRODUCT. TRANSITION TO 2 INCH CONDUIT AT EHH–1.

4. TRAFFIC–RATED ELECTRICAL HANDHOLE. INSTALL WASTE TANK INSTRUMENTATION TERMINAL BLOCKS AND INTRINSICALLY SAFE DEVICES IN HANDHOLE. ROUTE WASTE TANK INSTRUMENTATION CABLE IN DIRECT BURIED 2 INCH PVC–80 CONDUIT.

5. INSTALL 500KVA, 24.9kV DELTA – 480GY/277V PADMOUNT TRANSFORMER (T–49). INSTALL 200 AMP FEED THRU BUSHINGS ON TRANSFORMER BUSHING. INSTALL 200 AMP SURGE ARRESTER ELBOWS. INSTALL AND TERMINATE 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE FROM ESB–12 AND ESB–18A WITH 200 AMP LOADBREAK ELBOWS

6. INSTALL 25kV 4 WAY (2 SOURCE – 2 UNPROTECTED TAPS) PADMOUNT SWITCHGEAR. INTERCEPT AND TERMINATE EXIST MEDIUM VOLTAGE POWER CABLE FROM ESB–18 WITH 200 AMP LOADBREAK ELBOWS ON SOURCE POSITION (NORMAL CLOSED). INSTALL AND TERMINATE 25kV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE TO T–27 AND T–48 WITH 200 AMP LOADBREAK ELBOWS ON TAP POSITIONS. INSTALL AND TERMINATE MEDIUM VOLTAGE POWER CABLE TO T–49 ON SOURCE POSITION WITH 200 AMP LOADBREAK ELBOWS. LEAVE TAP TO T–49 AS NORMAL OPEN.

7. LOCATE EXIST MEDIUM VOLTAGE POWER CABLE BETWEEN EXIST JUNCTION ENCLOSURE ESB–18A LOCATION AND POINT OF INTERCEPT "C". PULL BACK EXIST CABLE TO POINT OF INTERCEPT "C". REROUTE AND TERMINATE EXIST MEDIUM VOLTAGE POWER CABLE TO SWITCHGEAR ESB–18A. ADJUST YARD PIPING. EXIST MEDIUM VOLTAGE POWER CABLE LENGTH SHALL BE SUFFICIENT ENOUGH TO REACH EQUIPMENT CABLE TERMINATION BUSHINGS AND HAVE AT MINIMUM 5 FT OF CABLE SLACK AT EQUIPMENT BASEMENT.

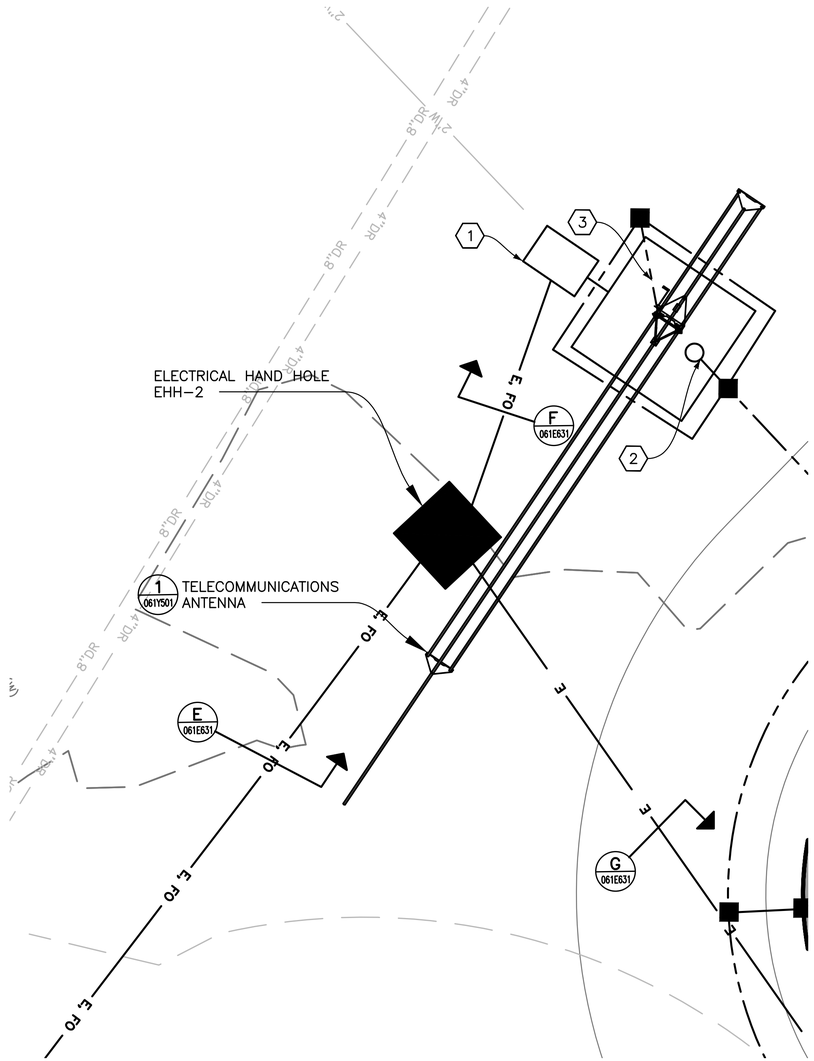
10 0 10 20
SCALE OF FEET
1" = 10'

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO. 75014
TOU VANG
Date Signed 1/16/2026
ARIZONA, U.S.A.

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO. 58037
TRAVIS T. MOORE
Date Signed 1/16/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S–DELGADO DRAWN: A. S–DELGADO TECH. REVIEW: P. DAVILA DATE: JUNE 2026	SUB SHEET NO. 061 E401	TITLE OF SHEET SITEWORK ELECTRICAL SITE PLAN – WTP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 49 OF 192

8/19/2026 Green, Jake C:\Users\jgreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\061-E401.dwg



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. ELECTRICAL DEMO IS SHOWN ON SHEET 061X101.

C. RADIO TOWER SHOWN IN HORIZONTAL POSITION FOR REFERENCE. THE RADIO TOWER WILL BE POSITIONED VERTICALLY UNDER NORMAL CONDITIONS.

KEY NOTES: (X)

1. NPS TELECOM CABINET. REFER TO SHEET 061Y620 FOR ADDITIONAL DETAILS.

2. #4/0 AWG CONNECTION TO LIGHTNING PROTECTION SYSTEM DOWNLEAD AT TOP OF ANTENNA TOWER. COORDINATE EXACT DOWNLEAD LOCATION AND REQUIRED SERVICE LOOP WITH LIGHTNING PROTECTION SYSTEM DESIGNER. REFER TO SPECIFICATION SECTION 26 41 13 FOR LIGHTNING PROTECTION SYSTEM REQUIREMENTS.

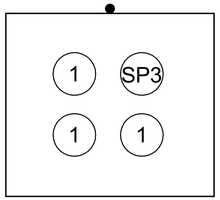
3. #4/0 AWG BOND TO RADIO TOWER. CONNECT TOWER GROUND RING TO ADJACENT WATER TANK GROUND RING.

10 0 10 20
SCALE OF FEET
1" = 10'

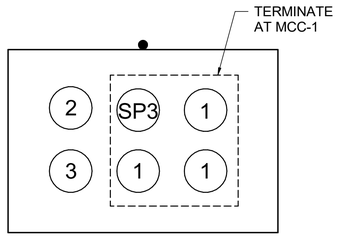
REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL)
CERTIFICATE NO
58037
TRAVIS T. MOORE
Date Signed 7/14/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO CADD A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: JUNE 2026	SUB SHEET NO. 061 E402	TITLE OF SHEET RAW WATER TANKS RADIO ANTENNA AND TOWER SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 50 OF 192

8/11/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10414450_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\ELEC\061-E402.dwg

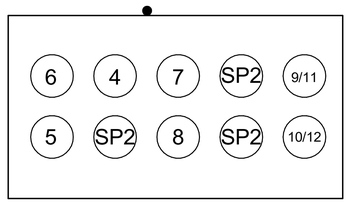


A 061E631 DUCT BANK SECTION TRANSFORMER
NOT TO SCALE

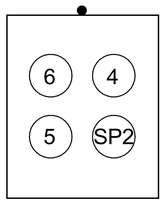


B 061E631 DUCT BANK SECTION MCC-1
NOT TO SCALE

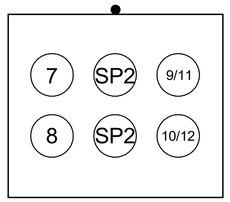
DUCT BANK SCHEDULE - NORTH RIM WATER TREATMENT PLANT			
CONDUIT NUMBER	SIZE & TYPE	CIRCUIT NUMBER	REMARKS
1	3" PVC-40	T-49:1	MCC-1 FEEDER
2	2" PVC-40	144-STRAND FIBER OPTIC CABLE	FIBER CONNECTION TO/FROM RSPH
3	2" PVC-40	144-STRAND FIBER OPTIC CABLE	FIBER CONNECTION TO/FROM NR GENERATOR BUILDING
4	2" PVC-40	LP-1:19	FINISHED WATER TANK HEAT TRACE SUPPLY
5	2" PVC-40	PLC-1:I22, I23	FINISHED WATER TANK LEVEL INSTRUMENTS
6	2" PVC-40	PLC-I1:C52	HEAT TRACE PANEL TEMPERATURE SENSOR
7	2" PVC-40	LP-1:4	TELECOM PANEL FEEDER
8	2" PVC-40	24-STRAND FIBER OPTIC CABLE	FIBER CONNECTION TO/FROM TELECOM PANEL
9	2" PVC-40	PLC-1:I1	RAW WATER TANK 62051 INSTRUMENTS
10	2" PVC-40	PLC-1:C1	RAW WATER TANK 62051 CONTROL
11	2" PVC-40	PLC-1:I2	RAW WATER TANK 62052 INSTRUMENTS
12	2" PVC-40	PLC-1:C2	RAW WATER TANK 62052 CONTROL
SP2	2" PVC-40	SPARE (EMPTY)	
SP3	3" PVC-40	SPARE (EMPTY)	



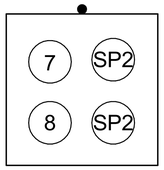
C 061E631 DUCT BANK SECTION TANK INSTRUMENTATION
NOT TO SCALE



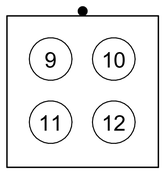
D 061E631 DUCT BANK SECTION FINISHED WATER TANKS
NOT TO SCALE



E 061E631 DUCT BANK SECTION RAW WATER INSTRUMENTATION
NOT TO SCALE



F 061E631 DUCT BANK SECTION FIBER CABINET
NOT TO SCALE



G 061E631 DUCT BANK SECTION RAW WATER TANK INSTRUMENTATION
NOT TO SCALE

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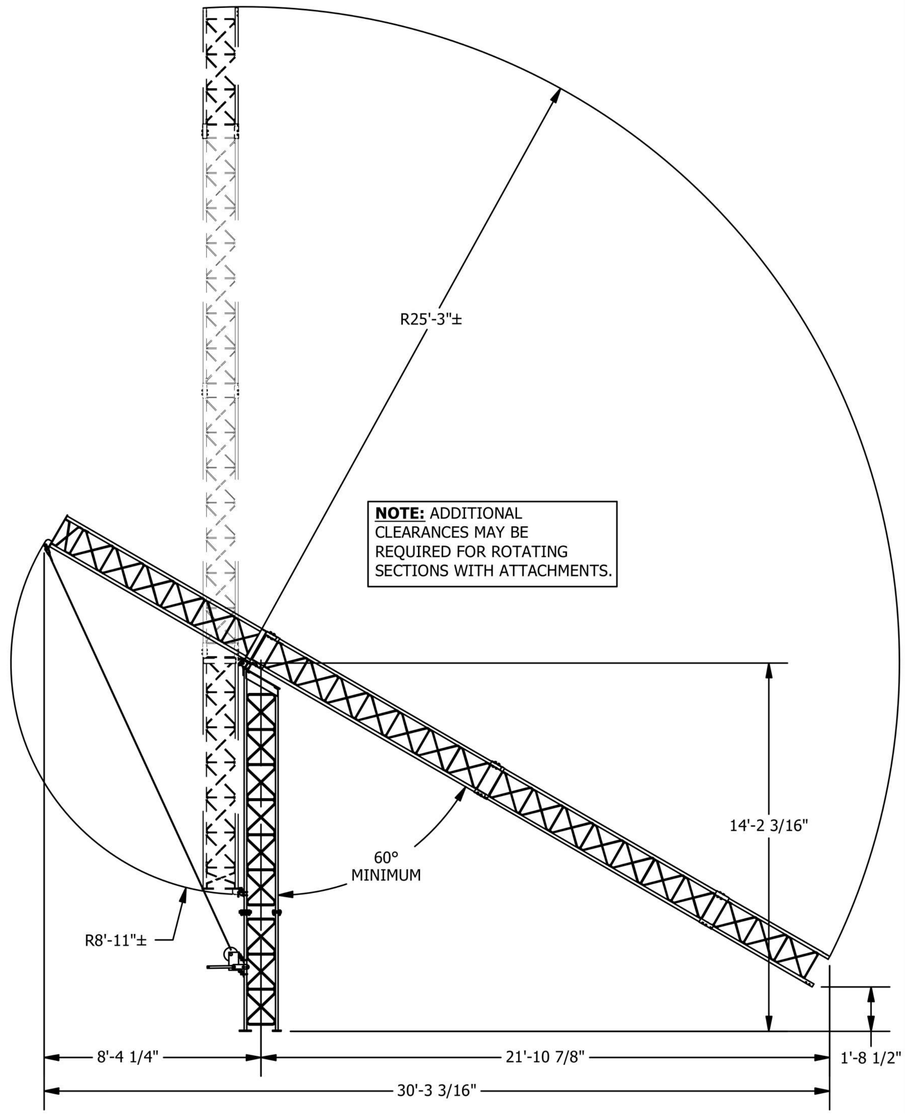
8/12/2026 9:06:45 PM

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO. 58037 TRAVIS T. MOORE
ARIZONA U.S.A.

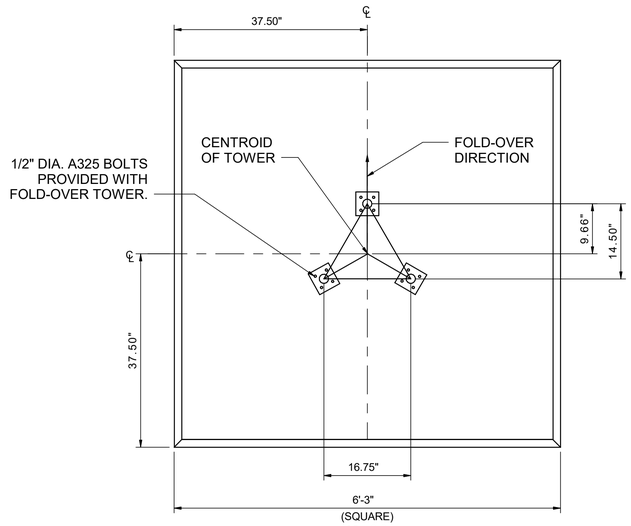
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 061 E631	TITLE OF SHEET SITEWORK ELECTRICAL DUCT BANK SECTIONS AND SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 51 OF 192

GENERAL NOTE:

A. THIS DRAWING IS FOR DIAGRAMMATICAL PURPOSES ONLY, AND ALL STRUCTURAL REFERENCES SHOWN ARE FROM THE MANUFACTURER'S CUT SHEETS. ALL STRUCTURAL REQUIREMENTS MUST BE CONFIRMED BY THE CONTRACTOR'S STRUCTURAL ENGINEER PRIOR TO CONSTRUCTION AND ARE NOT THE RESPONSIBILITY OF HDR.



1 061Y501 TELECOMMUNICATION TOWER DIAGRAM ROHN 55FK-40
NOT TO SCALE



PLAN

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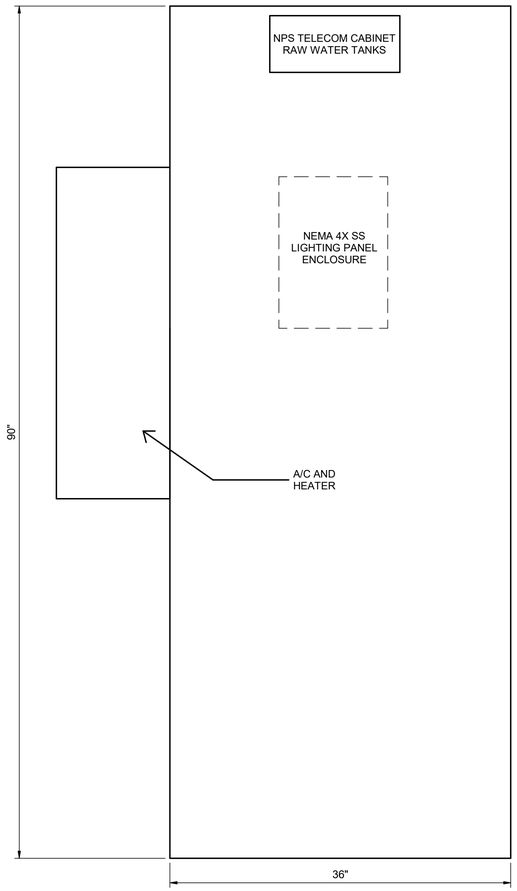
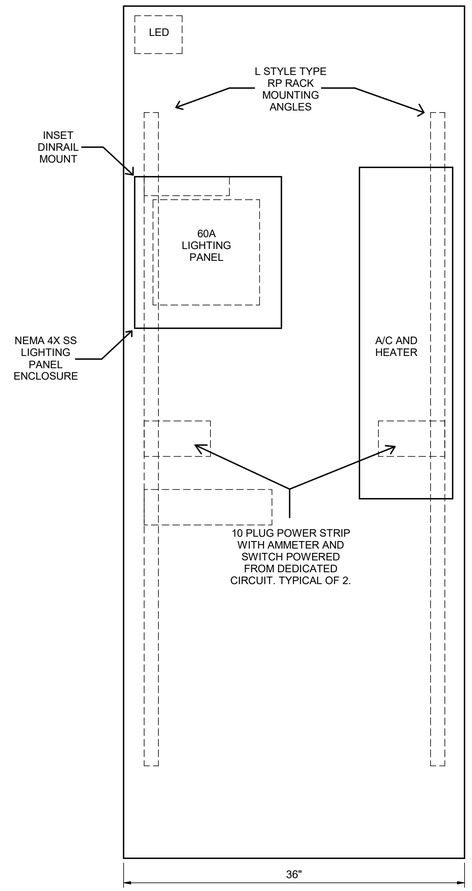
Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 4/27/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD CADD S. BILD TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 061 Y501	TITLE OF SHEET WATER TREATMENT WTP TELECOMMUNICATION TOWER NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 52 OF 192



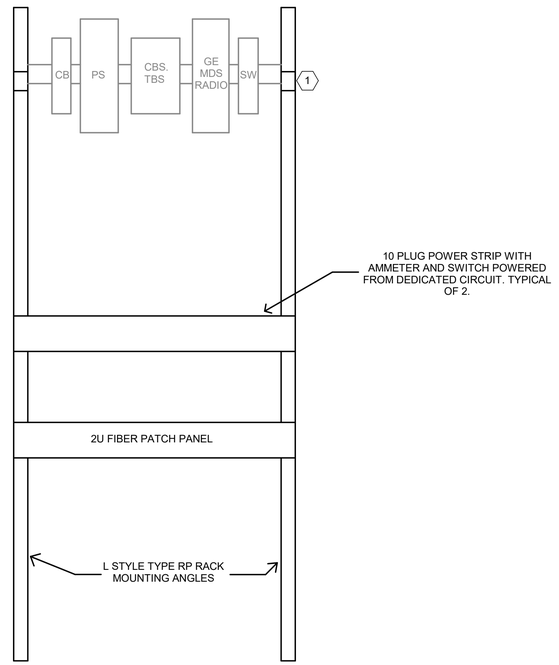
1 063Y620 **NPS TELECOM CABINET FRONT EXTERIOR ELEVATION**

NOT TO SCALE



2 063Y620 **NPS TELECOM CABINET SIDE EXTERIOR ELEVATION**

NOT TO SCALE



3 063Y620 **NPS TELECOM CABINET FRONT INTERIOR ELEVATION**

NOT TO SCALE

KEYNOTES

1

Registered Professional Engineer (Electrical)
CERTIFICATE NO.
85305
JOSEPHINE A
BRAINARD
Date Signed 4/27/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAFTED: R.D. CLARK TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 061 Y620	TITLE OF SHEET SITEWORK NPS TELECOM CABINET LAYOUT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 53 OF 192

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CODE REVIEW ANALYSIS

1.0 INTRODUCTION

THE FOLLOWING CODE REVIEW NARRATIVE IS PROVIDED TO SERVE AS A BASIS OF UNDERSTANDING FOR THE DEVELOPMENT OF THE DESIGN, DRAWINGS AND SPECIFICATIONS FOR CONSTRUCTION OF THE NEW NORTH RIM WATER TREATMENT PLANT (WTP) BUILDING IN THE GRAND CANYON (GRCA) NATIONAL PARK. THE FOLLOWING ANALYSIS REVIEWS THE APPLICABLE BUILDING AND LIFE SAFETY CODES.

THE AUTHORITY HAVING JURISDICTION (AHJ) IS THE NATIONAL PARKS SYSTEM (NPS), INCLUDING THE DENVER SERVICE CENTER (DSC).

1.1 DESIGN CODES*

BUILDING CODE	2024 INTERNATIONAL BUILDING CODE
FIRE CODES	2024 INTERNATIONAL FIRE CODE
	NPS STRUCTURAL FIRE REFERENCE MANUAL 58
MECHANICAL CODE	2024 INTERNATIONAL MECHANICAL CODE
PLUMBING CODE	2024 INTERNATIONAL PLUMBING CODE
FUEL GAS CODE	2024 INTERNATIONAL FUEL GAS CODE
ENERGY CONSERVATION CODES	2024 INTERNATIONAL ENERGY CONSERVATION CODE
	2016 ASHRAE 90.1
ELECTRICAL CODE	2023 NATIONAL ELECTRICAL CODE (NFPA 70)
WILDLAND-URBAN INTERFACE	2024 INTERNATIONAL WILDLAND-URBAN INTERFACE CODE (IWUIC)
STANDARD FOR THE INSTALLATION OF SPRINKLER SYSTEMS	2022 NFPA 13
STANDARD FOR THE INSPECTION, TESTING, AND MAINTENANCE OF WATER-BASED FIRE PROTECTION SYSTEMS	2023 NFPA 25
NATIONAL FIRE ALARM CODE	2022 NFPA 72

CODE DATA TABLE

2.0 GRAND CANYON NATIONAL PARK - NEW NORTH RIM WATER TREATMENT PLANT [62]

2.1 DESCRIPTION

THE FOLLOWING CODE REVIEW NARRATIVE IS PROVIDED TO SERVE AS A BASIS OF UNDERSTANDING FOR THE DEVELOPMENT OF THE DESIGN, DRAWINGS AND SPECIFICATIONS FOR THE NEW NORTH RIM WATER TREATMENT PLANT (WTP) IN THE GRAND CANYON (GRCA) NATIONAL PARK.

2.2 GENERAL BUILDING DATA SUMMARY

OCCUPANCY CLASSIFICATION(S)	MIXED USE; NON-SEPARATED GROUP B: BUSINESS (ADMIN SPACES) GROUP F-1: MODERATE HAZARD INDUSTRIAL (PROCESS AND MECH/ELEC EQUIPMENT SPACES)
BUILDING AREA	5,620 SF
FIRE AREA	5,620 SF
HEIGHT	AVERAGE BUILDING HEIGHT 31'-6" (1 STORIES ABOVE GRADE)
TYPE OF CONSTRUCTION	II-B (NON-COMBUSTIBLE)
FIRE SEPARATION DISTANCE	FSD = 11'
FIRE PROTECTION SYSTEMS	FIRE EXTINGUISHERS FIRE HYDRANT
OCCUPANT LOAD (TOTAL)	52 (SEE 2.10 FOR CALCULATION)

2.3 USE & OCCUPANCY CLASSIFICATION:		IBC	IFC
OCCUPANCY GROUP(S):	MIXED USE; NON-SEPARATED GROUP B: BUSINESS GROUP F-1: MODERATE HAZARD INDUSTRIAL	§304, AND §306	-
ACCESS. OCCUPANCY (<10%)	GROUP S-1: STORAGE	§304	-
CLASSIFICATION OF HAZARDS	HEALTH HAZARDS: CORROSIVES, IRRITANTS	§307	-
HAZARDOUS MATERIALS	SODIUM HYDROXIDE, CITRIC ACID, LIQUID ALUM, HYDROCHLORIC ACID, SODIUM BISULFITE, SODIUM HYPOCHLORITE, ORTHOPHOSPHATE	§307	-

CODE DATA TABLE (CONTINUED)

2.4 SPECIAL REQUIREMENTS:		IBC	IFC
AREAS WITH HAZARDOUS MATERIALS	BASE AND ACID DRUM STORAGE AREAS	§414	-
EMERGENCY EYE WASH / EMERGENCY SHOWER (ESEW) STATIONS	ESEW: HYDROCHLORIC ACID, SODIUM HYPOCHLORITE, ALUMINUM SULFATE, SODIUM HYDROXIDE, SODIUM BISULFITE, CITRIC ACID EW: NUTREAT™ 1700	[29 CODE OF FEDERAL REGULATIONS, OSHA 1910.151]	-

2.4.1 HAZARDOUS MATERIALS CLASSIFICATIONS

MATERIAL NAME AND QUANTITY PRESENT; LOCATION AND ANY RELEVANT DETAILS	HAZARD CLASSIFICATIONS (MAXIMUM ALLOWABLE QUANTITY, OR MAQ); IF OVER MAQ, OCCUPANCIES RESULTING	CONCENTRATION	C.A.S. # (HMEX)
SODIUM HYPHCHLORITE (BLEACH) LIQUID; 300 GALLON TOTE; BASES DRUM STORAGE AREA	PHYSICAL HAZARD: N/A HEALTH HAZARD: CORROSIVE (500 GALLON MAQ); EXCEEDING MAQ CREATES HIGH-HAZARD GROUP H-4 OCCUPANCY CATEGORY 1: SEVERE SKIN BURNS AND EYE DAMAGE	13% SOLUTION	SODIUM HYPOCHLORITE 7681-52-9
SODIUM HYDROXIDE (CAUSTIC SODA) LIQUID; 110 GALLONS (TWO (2) 55 GALLON DRUMS); BASES DRUM STORAGE AREA	PHYSICAL HAZARD: WATER REACTIVE CLASS 1 (NOT LIMITED) HEALTH HAZARD: CORROSIVE (500 GALLONS); EXCEEDING MAQ CREATES GROUP H-4 OCCUPANCY	25% SOLUTION	CAUSTIC SODA 1310-73-2
LIQUID ALUMINUM SULFATE (ALUM) LIQUID; 220 GALLONS (FOUR (4) 55 GALLON DRUMS); ACID DRUM STORAGE AREA	PHYSICAL HAZARD: N/A HEALTH HAZARD: CORROSIVE (500 GALLONS); EXCEEDING MAQ CREATES GROUP H-4 OCCUPANCY	48% SOLUTION	10043-01-3
HYDROCHLORIC ACID LIQUID; 55 GALLON DRUMS; ACID DRUM STORAGE AREA	PHYSICAL HAZARD: N/A HEALTH HAZARD: CORROSIVE (500 GALLONS); EXCEEDING MAQ CREATES GROUP H-4 OCCUPANCY	31% SOLUTION	7647-01-0
SODIUM BISULFITE LIQUID; 55 GALLONS (ONE (1) 55 GALLON DRUM); ACID DRUM STORAGE AREA	PHYSICAL HAZARD: N/A HEALTH HAZARD: CORROSIVE (500 GALLONS); EXCEEDING MAQ CREATES GROUP H-4 OCCUPANCY	30-60% SOLUTION	7631-90-5
CITRIC ACID LIQUID; 55 GALLONS (ONE (1) 55 GALLON DRUM); ACID DRUM STORAGE AREA	PHYSICAL HAZARD: N/A HEALTH HAZARD: CORROSIVE (500 GALLONS); EXCEEDING MAQ CREATES GROUP H-4 OCCUPANCY	50% SOLUTION	CITRIC ACID 77-92-9
ORTHOPHOSPHATE LIQUID; 55 GALLONS (ONE (1) 55 GALLON DRUM); CHEMICAL DRUM STORAGE AREA	PHYSICAL HAZARD: N/A HEALTH HAZARD: IRRITANT (NO MAQ);	28.5% SOLUTION	PROPRIETARY MIXTURE

2.4.2 REQUIREMENTS:		IBC	IFC
HIGH HAZARD CLASSIFICATION	*H-4: SODIUM HYDROXIDE, LIQUID, ALUMINUM SULFATE, SODIUM BISULFITE, SODIUM HYPOCHLORITE, CITRIC ACID*	*§307.5, §307.6*	-
PERMISSIBLE STORAGE (MAQ)	*SEE 2.4.1 ABOVE FOR QUANTITIES PRESENT & MAQ'S*	*TABLE §307.1(1) TABLE §307.1(2)*	*TABLE 5003.1.1(1) TABLE 5003.1.1(2)*
CONTROL AREA FIRE BARRIER	N/A (SEE 2.7 FOR OCCUPANCY SEPARATION)	*TABLE 414.2.2*	-
MIXED OCCUPANCY	NOT ALLOWED FOR GROUP H	*§508.2.4 EX. 1 §508.3.3 EX. 1*	-
MAXIMUM CONTROL AREAS	N/A (SEE 2.7 FOR OCCUPANCY SEPARATION)	*TABLE 414.2.2*	-
GENERAL SAFETY PRECAUTIONS	*REQUIRED*	-	*§5003.9*
SPILL CONTROL	*REQUIRED: > 55 GALLON CAPACITY OF INDIVIDUAL VESSEL, >1,000 GALLON AGGREGATE CAPACITY*	-	*§5004.2.1 §5005.1.3*
SECONDARY CONTAINMENT, DRAINAGE METHODS	*REQUIRED: LIQUIDS: > 55 GALLON CAPACITY OF INDIVIDUAL VESSEL, >1,000 GALLON AGGREGATE CAPACITY; SOLIDS: >550 POUNDS OR >10,000 POUNDS AGGREGATE CAPACITY*	-	*TABLE 5004.2.2 TABLE 5005.2.1.4*
VENTILATION	*REQUIRED PER IMC [1 CFM / SF MINIMUM]*	-	*§5004.3*
INCOMPATIBLE MATERIALS & CORROSIVITY	*MATERIAL CORROSIVITY - SEE MSDS SHEETS FOR DETAILS;*	-	*§5004.4*
	SODIUM HYDROXIDE: WATER-REACTIVE (EXOTHERMIC) AND INCOMPATIBLE WITH ACIDS, REACTIVE COMPOUNDS, STRONG BASES, AND OXIDIZING AGENTS		
	LIQUID ALUMINUM SULFATE: IS A IRRITANT INCOMPATIBLE WITH STRONG OXIDIZING AGENTS, STRONG ACIDS, STRONG BASES, AND METALS		
	SODIUM HYPOCHLORITE: CORROSIVE AND INCOMPATIBLE WITH STRONG ACIDS AND BASES, OXIDIZING AGENTS		
	HYDROCHLORIC ACID: BASES, AMINES, ALKALI METALS , PERMANGANATES, FLUORINE, METAL ACETYLIDES, HEXALITHIUM DISILICIDE		
	CITRIC ACID: METAL NITRATES (POTENTIALLY EXPLOSIVE REACTION), ALKALI CARBONATES AND BICARBONATES, POTASSIUM TARTRATE. WILL CORRODE COPPER, ZINC, ALUMINUM AND THEIR ALLOYS.		
	SODIUM BISULFITE: CORROSIVE AND INCOMPATIBLE WITH OXIDIZING AGENTS, ACIDS, METALS AND COMBUSTIBLE MATERIALS		
AUTOMATIC SPRINKLER SYSTEMS	*REQUIRED FOR INDOOR STORAGE EXCEEDING THE MAQ (GROUP H); SEE 2.9*	-	*§5004.5*
EXPLOSION CONTROL	*N/A*	-	*§5004.6*
EMERGENCY & STANDBY POWER	*REQUIRED FOR VENTILATION, TEMPERATURE CONTROL, EMERGENCY ALARM SYSTEMS, ETC.*	-	*§5004.7 §5005.1.5*
LIMIT CONTROLS	*REQUIRED FOR TEMPERATURE AND/OR PRESSURE, AS APPLICABLE*	-	*§5004.8 §5005.1.4*
EMERGENCY ALARMS	*A MANUAL EMERGENCY ALARM SYSTEM IS REQUIRED FOR ALL GROUP H OCCUPANCIES (SEE 2.9)*	*§415.5*	*§5004.9*
SUPERVISION AND MONITORING	*REQUIRED: REQUIRED SYSTEMS SHALL BE SUPERVISED/MONITORED; APPROVED CENTRAL STATION OR APPROVED LOCAL AUDIBLE/VISUAL*	*§415.5*	*§5004.10*
NONCOMBUSTIBLE FLOOR	*REQUIRED*	-	*§5004.12*

2.4.2 HAZARDOUS MATERIALS - SODIUM BISULFITE		IBC	IFC
GAS DETECTION	SULFUR DIOXIDE GAS DETECTION IS REQUIRED FOR SODIUM BISULFITE STORAGE. THIS IS DUE TO THE FACT THAT SODIUM BISULFITE, WHEN EXPOSED TO AIR, LOSES SOME SULFUR DIOXIDE WHICH GRADUALLY OXIDIZES TO SULFATE. BOTH ACIDIFICATION AND HEAT ACCELERATE THE RELEASE OF SULFUR DIOXIDE FUMES. SULFUR DIOXIDE IS CONSIDERED A TOXIC GAS.	§916	§916
EMERGENCY AND STANDBY POWER	STANDBY OR EMERGENCY POWER SHALL BE PROVIDED OR THE GAS DETECTION SYSTEM SHALL INITIATE A TROUBLE SIGNAL AT AN APPROVED LOCATION IF THE POWER SUPPLY IS INTERRUPTED.	§916.5	§916.5
SYSTEM ACTIVATION	SULFUR DIOXIDE GAS DETECTION ALARM SHALL BE INITIATED WHERE ANY SENSOR DETECTS A CONCENTRATION OF SULFUR DIOXIDE GAS EXCEEDING ONE-HALF OF THE IMMEDIATELY DANGEROUS TO LIFE AND HEALTH. THE SCADA PANEL SHALL MONITOR THE SULFUR DIOXIDE GAS SENSOR AND UPON AN ALARM IT WILL ACTIVATE AUDIBLE AND VISIBLE ALARMS TO INITIATE EVACUTATION AND SEND A SIGNAL TO SCADA FOR HAZMAT RESPONSE INITIATION. THE GAS DETECTION ALARM STROBES SHALL HAVE AMBER LENSES. SEE I&C SPECIFICATIONS FOR SULFUR DIOXIDE GAS SENSOR DETAILS.	§916.8	§916.8

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. COOPER CADD J. BELON TECH. REVIEW: M. SHUMPERT DATE: MAY 2026	SUB SHEET NO. 062 F001	TITLE OF SHEET WATER TREATMENT CODE ANALYSIS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 54 OF 192	

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CODE DATA TABLE (CONTINUED)

2.5 BUILDING HEIGHTS & AREAS:		IBC	IFC
ACTUAL HEIGHT & AREA	31'-6" AND 1 STORY TOTAL AREA OF 5,620 SF	-	-
ALLOWABLE AREA	GROUP B: 23,000 SF GROUP F-1: 15,500 SF	TABLE 506.2	
ALLOWABLE HEIGHT	UNSPRINKLERED GROUP B: 55' AND 2 STORIES UNSPRINKLERED GROUP F-1: 55' AND 3 STORIES	TABLE 504.3, AND TABLE 504.4	-
MEZZANINE AREA (%)	N/A	§505.2	-
EQUIPMENT PLATFORM AREA (%)	N/A	§505.3	-
SPRINKLER INCREASE	N/A	§506.3	-
FRONTAGE INCREASE	NOT NEEDED NOR TAKEN	§506.3	-
UNLIMITED AREA BUILDINGS	N/A	§507	-
MIXED-USE & OCCUPANCY	MIXED USE, NON-SEPARATED GROUP B & F-1	§508	-
INCIDENTAL USES	N/A	§509	-
SPECIAL PROVISIONS	N/A	§510	-
2.6 TYPES OF CONSTRUCTION:		IBC	IFC
CONSTRUCTION TYPE	II-B	§602	-
GENERAL BUILDING FIRE-RESISTANCE RATINGS (PRIMARY STRUCTURAL FRAME, BEARING AND NONBEARING WALLS AND PARTITIONS, FLOOR, ROOF)	0-HR	TABLE 601	-
FIRE SEPARATION DISTANCE (FSD)	FSD = 11'	TABLE 705.5	-
FIRE-RESISTANCE RATING REQUIREMENTS FOR EXTERIOR WALLS BASED UPON FSD	0-HR	TABLE 705.5	-
2.7 FIRE & SMOKE PROTECTION:		IBC	IFC
SHAFT ENCLOSURES FIRE-RESISTANCE RATINGS	N/A (NO SHAFT ENCLOSURES)	§707.3.1, AND §713.4	-
VERTICAL EXIT ENCLOSURE FIRE-RESISTANCE RATINGS	N/A (NO RATED EXIT ECLOSURES)	§707.3.2, AND §1023.1	-
CONTROL AREA SEPARATIONS	A MINIMUM OF 1-HR FIRE BARRIER IS REQUIRED BETWEEN EACH CONTROL AREA (ACIDS & BASES) AND THE REST OF THE BUILDING	§414.2, AND TABLE 414.2.2	-
OCCUPANCY SEPARATIONS	N/A (MIXED OCCUPANCY, NO SEPARATIONS)	§508, AND TABLE 508.4	-
OPENING PROTECTIVES	FIRE-RATED BARRIERS, PARTITIONS, ASSEMBLIES, ETC.	§716	-
DUCT AND AIR TRANSFER OPENINGS	FIRE-RATED BARRIERS, PARTITIONS, ASSEMBLIES, ETC.	§717	-
UNPROTECTED EXTERIOR WALL OPENINGS	NO LIMIT	§705.9.1 EXCEPTION 2, AND TABLE 705.9	-
CORRIDORS	NO RATING REQUIRED; GROUP B WITH OCCUPANT LOAD NOT GREATER THAN 30	§1020	-
2.8 INTERIOR FINISHES:		IBC	IFC
CEILING & WALL (ROOMS AND ENCL. SPACES)	UNSPRINKLERED GROUP B: CLASS A, B, OR C UNSPRINKLERED GROUP F-1: CLASS A, B, OR C	TABLE 803.13	-
CEILING & WALL (EXIT ENCLOSURES)	UNSPRINKLERED GROUP B: CLASS A UNSPRINKLERED GROUP F-1: CLASS A OR B		
CEILING & WALL (CORRIDORS)	UNSPRINKLERED GROUP B: CLASS A OR B UNSPRINKLERED GROUP F-1: CLASS A, B, OR C		
FLOOR	UNSPRINKLERED GROUP B: CLASS I OR II UNSPRINKLERED GROUP F-1: DOC FF-1 "PILL TEST" (CPSC 16 CFR PART 1630) OR WITH ASTM D2859	§804.4	-

CODE DATA TABLE (CONTINUED)

2.9 FIRE PROTECTION SYSTEMS:		IBC	IFC
FIRE PROTECTION SYSTEM FEED MAIN - HEAT TRACE	WHERE THE PIPING FOR THE FIRE PROTECTION SYSTEM FEED MAIN IS SUBJECT TO FREEZING, A UL LISTED HEAT-TRACING SYSTEM AND CONTROLLER SHALL BE PERMITTED TO BE PROTECTED AGAINST FREEZING TEMPERATURES. THE ELECTRIC SUPERVISION, OF THE HEAT-TRACING SYSTEM, SHALL REPORT TO SCADA FOR ALARMING CONDITIONS. THE DESIGN OF THE HEAT TRACE AND HEAT TRACE CONTROLLER IS DELEGATED TO THE CONTRACTOR. CONTRACTOR SHALL PROVIDE A CODE COMPLIANT SYSTEM MEETING THE REQUIREMENTS OF HEAT TRACE AND A CONTROLLER THAT IS BEING USED FOR A FIRE PROTECTION SYSTEM PER SPECIFICATION SECTION 40 41 13 AS WELL AS ALL LOCAL AND NATIONAL CODES.	-	[NFPA 13 §16.4.1]
AUTOMATIC SPRINKLER SYSTEM	NOT REQUIRED FOR GROUP F-1 FIRE AREAS UNDER 12,000 SF	§903.2.4	§903.2.4
PORTABLE FIRE EXTINGUISHERS	REQUIRED; PROVIDED THROUGHOUT	§906.1	§906.1
FIRE ALARM AND DETECTION	NOT REQUIRED; SMOKE DETECTORS IN ELECTRICAL ROOMS ARE FOR ASSET PROTECTION ONLY. NOT FOR FIRE & LIFE SAFETY. SEE SHEET 062E132.	§907.2.4	§907.2.4
FIRE-FLOW REQUIREMENT	1,500 GPM (@20 PSI) FOR 2 HOURS	-	§B105, AND TABLE B105.1(2)
MINIMUM NUMBER OF FIRE HYDRANTS	1 HYDRANTS; SEE CIVIL SITE UTILITY PLANS FOR LOCATIONS OF FIRE HYDRANTS	-	TABLE C102.1
2.10 MEANS OF EGRESS:		IBC	IFC
CUMMULATIVE OCCUPANT LOAD	52 OCCUPANTS	§1004, AND TABLE 1004.5	-
INDUSTRIAL	4,150 SF @ 1:100 SF (GROSS) = 42		
BUSINESS	1,470 SF @ 1:150 SF (GROSS) = 10		
EGRESS CAPACITY PER OCCUPANT LOAD	0.3 (STAIRS) & 0.2 (OTHER)	§1005.3	-
EXIT ACCESS - COMMON PATH	UNSPRINKLERED GROUP B: 100 FT UNSPRINKLERED GROUP F-1: 75 FT	TABLE 1006.2.1	-
EXIT ACCESS - TRAVEL DISTANCE	UNSPRINKLERED GROUP B: 200 FT UNSPRINKLERED GROUP F-1: 200 FT	TABLE 1017.2	-
MIN. EXIT DOORS	2 FOR MOST SPACES; SEE LIFE SAFETY PLANS FOR EXITING	§1006	-
ACCESSIBLE MEANS OF EGRESS	REQUIRED FOR GROUP B OCCUPANCIES	§1009	-
EXIT DISCHARGE; ACCESS TO PUBLIC WAY	PUBLIC WAY	§1028.5	-
2.11 ACCESSIBILITY:		IBC	IFC
EQUIPMENT SPACES	EQUIPMENT SPACES ARE EXEMPT; REQUIRED FOR GROUP B OCCUPANCIES	§1103.2.9 [2010 ADA §203.5]	-
2.12 WILDLAND-URBAN INTERFACE SUMMARY:			
CRITICAL FIRE WEATHER FREQUENCY	2-7 DAYS PER ANNUM		-
SLOPE	<40% WITHIN 100' OF THE BUILD SITE		-
FUEL MODEL	LIGHT FUEL		-
FIRE HAZARD SEVERITY	MODERATE HAZARD		TABLE 502.1
DEFENSIBLE SPACE	NOT LESS THAN 30'		§603, TABLE 603.2
WATER SUPPLY	CONFORMING		§404
IGNITION RESISTANCE CONSTRUCTION	IR 3; MUST COMPLY WITH §503 & §505		TABLE 503.1

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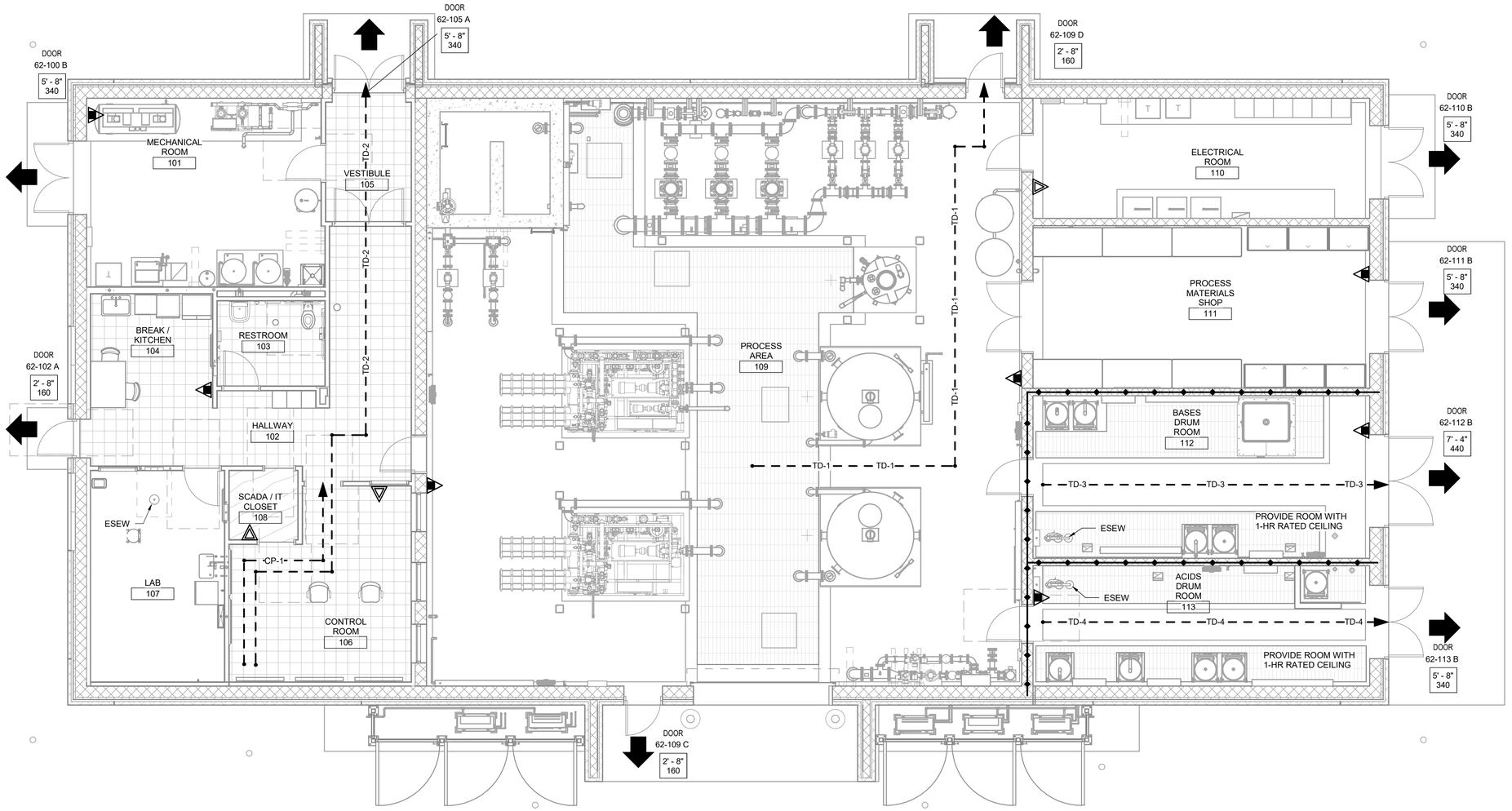
Registered Professional Engineer - State of Arizona
CERTIFICATE NO. 67918
CHRISTOPHER DEWITT
Signed 8-7-26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. COOPER CADD J. BELON TECH. REVIEW: M. SHUMPERT DATE: MAY 2026	SUB SHEET NO. 062 F001	TITLE OF SHEET WATER TREATMENT CODE ANALYSIS (CONT.) NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 55 OF 192



1 062F101 GROUND LEVEL LIFE SAFETY PLAN

LIFE SAFETY NOTES:

1. FOR EXIT LIGHTS AND EMERGENCY LIGHTING, SEE ELECTRICAL LIGHTING DRAWINGS (E SERIES).

LIFE SAFETY SYMBOLS:

EXIT DOOR TAG

A — CLEAR DOOR WIDTH [FEET - INCHES]
B — OCCUPANT LOAD CAPACITY [NO.OF PEOPLE]

FIRE EXTINGUISHER

CLEAN AGENT FIRE EXTINGUISHER

LINE TYPES FOR MEANS OF EGRESS:

COMPONENT — SYMBOL

TRAVEL DISTANCE — TD

COMMON PATH — CP

EXIT DISCHARGE

EXIT DISCHARGE ARROW

WALL RATING LEGEND

1 HOUR FIRE BARRIER

MEANS OF EGRESS ARRANGEMENT DISTANCE SCHEDULE - BUILDING SUMMARY

EGRESS PATH ID	EGRESS DISTANCE PROVIDED	EGRESS DISTANCE MAXIMUM PERMITTED
CP-1	22' - 6"	100' - 0"
TD-1	53' - 0"	200' - 0"
TD-2	60' - 0"	200' - 0"
TD-3	30' - 0"	200' - 0"
TD-4	30' - 0"	200' - 0"

MEANS OF EGRESS WIDTH CAPACITY

EXIT ID	PROVIDED EGRESS CAPACITY - LEVEL COMPONENTS & RAMPS: EGRESS CAPACITY FACTOR	CLEAR WIDTH	OCCUPANT LOAD CAPACITY
62-100 B	0.2	5' - 8"	340
62-102 A	0.2	2' - 8"	160
62-105 A	0.2	5' - 8"	340
62-109 C	0.2	2' - 8"	160
62-109 D	0.2	2' - 8"	160
62-110 B	0.2	5' - 8"	340
62-111 B	0.2	5' - 8"	340
62-113 B	0.2	5' - 8"	340

SCALE (A) 4 0 4 8 12 — 3/16" = 1'-0"
SCALE OF FEET

Registered Professional Engineer in the State of Arizona — CERTIFICATE NO. 67918 — STUART GRAHAM — Signed 5-1-26 — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. COOPER DRAWN: J. BELON TECH. REVIEW: M. SHUMPERT DATE: MAY 2026	SUB SHEET NO. 062 F101	TITLE OF SHEET WATER TREATMENT GROUND LEVEL LIFE SAFETY PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 56 OF 192

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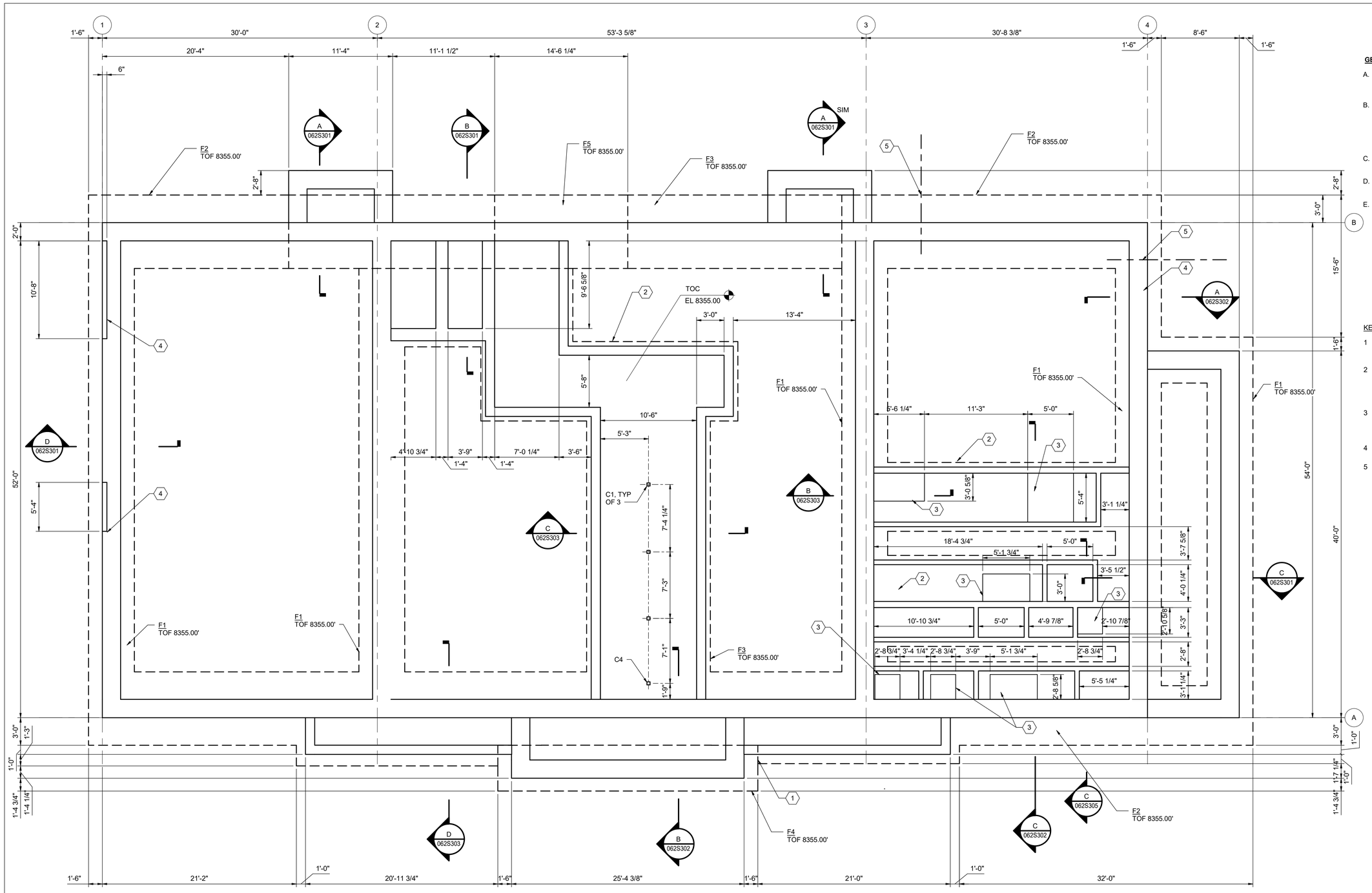

1 062S100 FOUNDATION PLAN
SCALE (A)

SCALE (A) 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

GENERAL NOTES

A. REFER TO SHEETS 062S001 THRU 062S003 FOR GENERAL STRUCTURAL NOTES AND TYPICAL DETAILS.

B. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING. EQUIPMENT, ANCHORS. CONDUITS, HOUSEKEEPING PADS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTURAL ELEMENTS THAT MAY NOT BE SHOWN.

C. CXX DENOTES STEEL COLUMNS. REFER TO SHEET 062S601 FOR STEEL COLUMN, BASE PLATE.

D. TOP OF CONCRETE FOOTING ELEVATIONS NOTED THUS: EL. 8355.00.

E. FX DENOTES SPREAD FOOTING. REFER TO SHEET 062S601 FOR FOOTING SCHEDULE.

KEYNOTES

1 BOTTOM STEP FOOTING PER VOLUME 0 TYPICAL DETAILS.

2 18" THICK CONTINUOUS CONCRETE MAT SLAB. REINFORCE WITH #6@12" O.C. TOP AND BOTTOM EACH WAY. SHORTER BARS TO BE PLACED IN THE OUTER MOST LAYER.

3 CONCRETE PEDESTAL BELOW DRUM SCALES. COORDINATE LOCATIONS WITH PROCESS DRAWINGS.

4 DOOR STOOP LEDGE.

5 APPROXIMATE CENTERLINE OF BELOW GRADE ELECTRICAL DUCTBANK. SEE STD DETAIL 6/003S507 FOR PIPING THROUGH STEM WALLS AND FOUNDATIONS.

Registered Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed: 04/27/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAFTED: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S100	TITLE OF SHEET WATER TREATMENT FOUNDATION PLAN NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 57 OF 192

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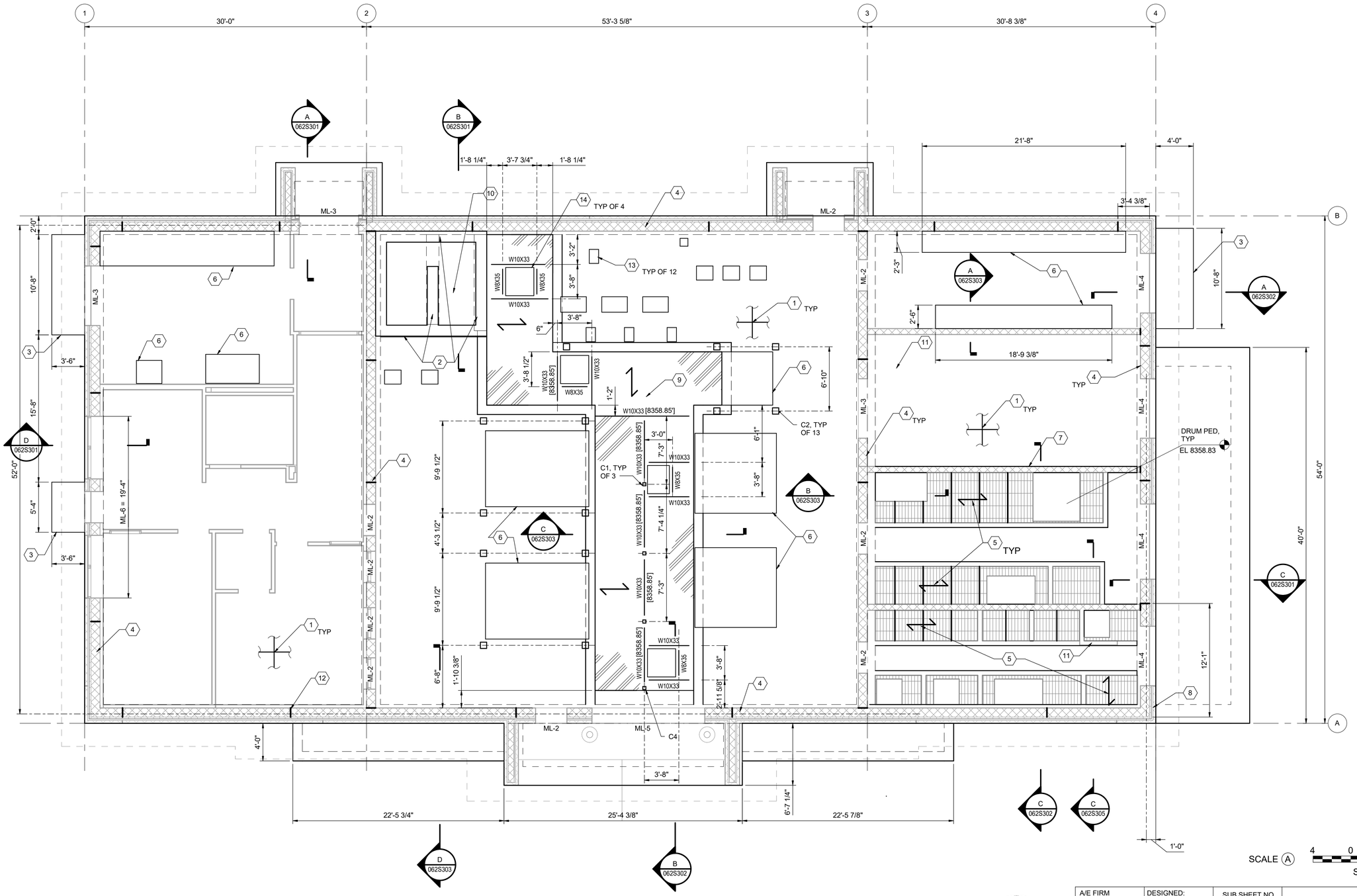

1 062S101 GROUND LEVEL PLAN

SCALE A

SCALE A 4 0 4 8 12 3/16" = 1'-0"

SCALE OF FEET

GENERAL NOTES

A. REFER TO SHEETS 062S001 THRU 062S003 FOR GENERAL STRUCTURAL NOTES AND TYPICAL DETAILS.

B. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING, EQUIPMENT, ANCHORS, CONDUITS, HOUSEKEEPING PADS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTURAL ELEMENTS THAT MAY NOT BE SHOWN.

C. CXX DENOTES STEEL COLUMNS. REFER TO SHEET 062S601 FOR STEEL COLUMN, BASE PLATE, AND BLOCKOUT SCHEDULE. REFER TO STANDARD DETAILS FOR ISOLATION JOINTS AROUND COLUMN BASES.

D. TOP OF CONCRETE ELEVATIONS NOTED THUS: EL. 8359.00.

E. ML-X - INDICATES CMU LINTEL MARK ML-X. SEE 1/060S003 FOR REINFORCING SCHEDULE. NOT ALL LINTELS SHOWN ON PLAN. CONTRACTOR SHALL COORDINATE ALL OPENINGS IN CMU WALL WITH OTHER DISCIPLINES.

F. W10x33 [8358.85'] - INDICATES W10x33 STEEL BEAM WITH TOP OF STEEL ELEVATION AT 8358.85'.

KEYNOTES

1 6" CONCRETE SLAB-ON-GRADE REINFORCED WITH #5@9" O.C. EACH WAY, PLACED MID-DEPTH. PLACE CONTROL JOINTS AT 12'-0" O.C. MAX, SEE DETAIL 5 ON SHEET 060S002. TOP OF SLAB PER PLAN. PROVIDE VAPOR BARRIER BELOW ADMIN AREA BETWEEN GRID LINES 1 AND 2, AND A AND B PER SPECIFICATIONS.

2 14" CONCRETE WALL REINFORCED WITH #9@6" O.C. HORIZONTAL AND VERTICAL, EACH FACE. WATER BEARING WALLS TO BE CONSTRUCTED AND LEAK TESTED PRIOR TO PLACEMENT OF CMU WALLS. REFER TO SPECIFICATION 01 45 25 FOR LEAK TEST REQUIREMENTS.

3 CONCRETE STOOP. SEE VOLUME 0 TYPICAL DETAILS FOR THICKNESS AND POURDOWN DEPTHS.

4 12" CMU BEARING WALLS. SEE TYPICAL DETAILS FOR REINFORCING SIZE AND SPACING. SEE ARCH DRAWINGS FOR EXTENTS OF EXTERIOR FINISHES.

5 1 1/2" MINIMUM FRP GRATING. LINES ON THE PLANS REPRESENT THE GRATING SUPPORT BEAM LAYOUT TO BE USED FOR BID PURPSOES ONLY. CONTRACTOR SHALL SIZE THE BEAMS, SUPPORTS, ATTACHMENTS AND CONNECTIONS, AND SHALL SUBMIT CALCULATIONS AND DRAWINGS SIGNED AND SEALED BY A PROFESSIONAL ENGINEER LICENSED IN THE STATE OF ARIZONA FOR APPROVAL PRIOR TO CONSTRUCTION. FRP BEAMS SHALL BE DESIGNED TO SUPPORT A 300 GALLON TANK (3,000 LBS MIN.) FOR ADDITIONAL INFORMATION, SEE SPECIFICATION SECTION 06 82 00.

6 SHALLOW EQUIPMENT PAD. SEE VOLUME 00 TYPICAL DETAILS FOR THICKNESS AND REINFORCING DETAILS. COORDINATE DIMENSIONS WITH EQUIPMENT SUPPLIED.

7 8" CMU BEARING WALLS. SEE TYPICAL DETAILS FOR REINFORCING SIZE AND SPACING. SEE ARCH DRAWINGS FOR EXTENTS OF INTERIOR FINISHES.

8 PROVIDE ISOLATION JOINT PER DETAIL 4 ON SHEET 060S002.

9 1 3/4" THICK, NONPUNCHED ALUMINUM PLANKS. PLANK SPAN DIRECTION AS INDICATED PER PLAN. COORDINATE PIPING LOCATIONS WITH PROCESS DRAWINGS.

10 12" THICK CONCRETE SLAB BELOW CT TANK. REINFORCE WITH #6@12" O.C. TOP AND BOTTOM EACH WAY.

11 INTERIOR NON-LOAD BEARING CMU WALL. COORDINATE PLACEMENT WITH ARCHITECTURAL DRAWINGS.

12 MASONRY CONTROL JOINT IN CMU WALL. COORDINATE WITH ARCHITECTURAL DRAWINGS AND PROJECT SPECIFICATIONS.

13 EQUIPMENT PADS PER VOLUME 00 DETAILS. COORDINATE PLACEMENT WITH MECHANICAL PROCESS DRAWINGS.

14 PUNCHED ALUMINUM PLANKS FOR REMOVABLE PANEL FOR ACCESS TO PIPE TRENCH BELOW.

Registered Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed: 04/27/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S101	TITLE OF SHEET WATER TREATMENT GROUND LEVEL PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 58 OF 192

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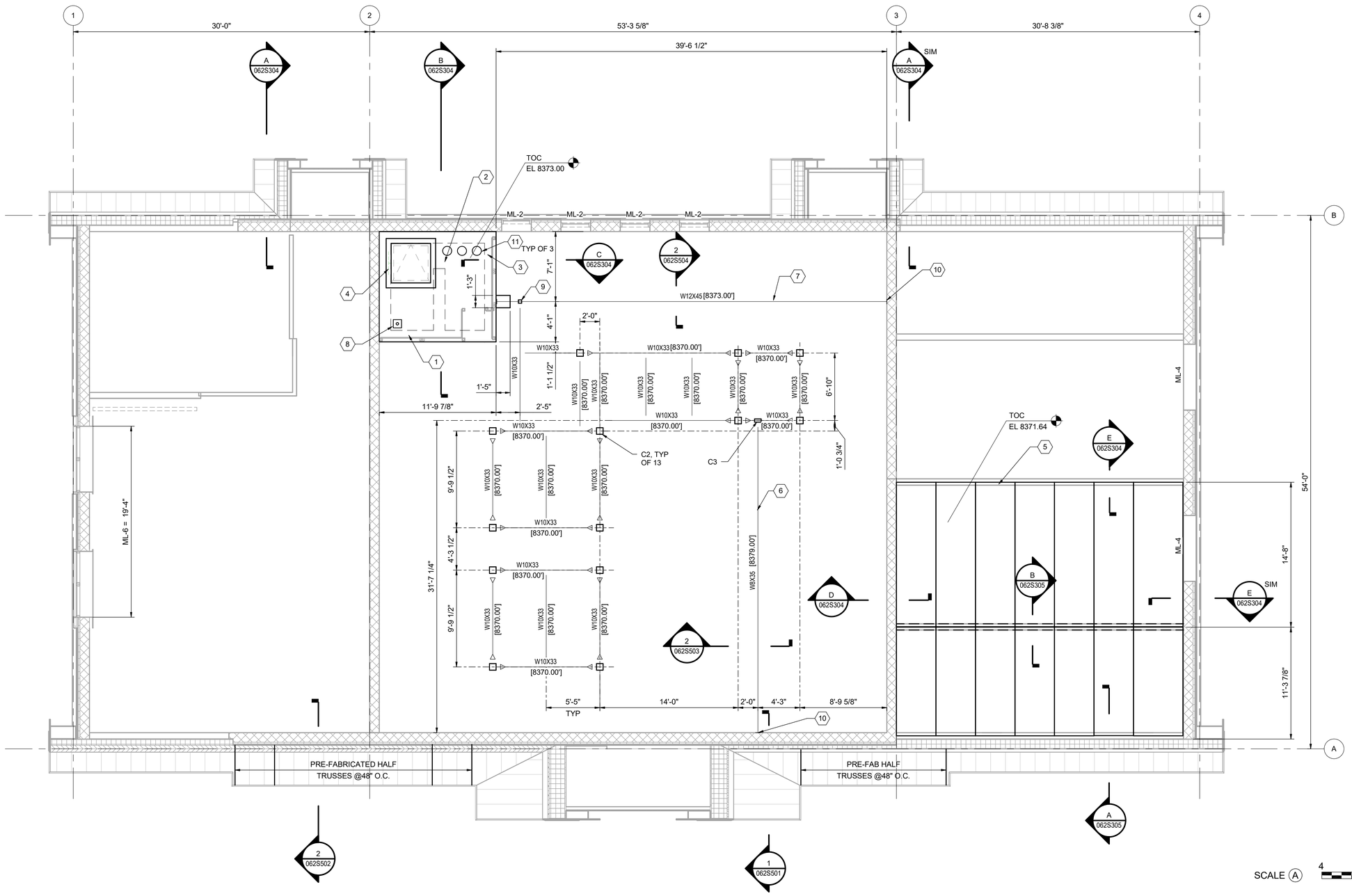

A 062S102 UPPER LEVEL PLAN

SCALE (A)

SCALE (A) 4 0 4 8 12 3/16" = 1'-0"

SCALE OF FEET

GENERAL NOTES

A. REFER TO SHEETS 062S001 THRU 062S003 FOR GENERAL STRUCTURAL NOTES AND TYPICAL DETAILS.

B. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING. EQUIPMENT, ANCHORS. CONDUITS, HOUSEKEEPING PADS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTURAL ELEMENTS THAT MAY NOT BE SHOWN.

C. CXX DENOTES STEEL COLUMNS. REFER TO SHEET 062S601 FOR STEEL COLUMN AND BASE PLATE SCHEDULE.

D. W10x33 [8370.00'] - INDICATES W10x33 STEEL BEAM WITH TOP OF STEEL ELEVATION AT 8370.00'.

E. ML-X - INDICATES CMU LINTEL MARK ML-X. SEE 1/060S003 FOR REINFORCING SCHEDULE. NOT ALL LINTELS SHOWN ON PLAN. CONTRACTOR SHALL COORDINATE ALL OPENINGS IN CMU WALL WITH OTHER DISCIPLINES.

KEYNOTES

1 ALUMINUM GUARDRAIL AT TANK LID. REFER TO STANDARD DETAILS AND SPECIFICATIONS FOR MEMBER SIZES AND ANCHORAGE.

2 8" THICK CONCRETE LID REINFORCED WITH #6@9" O.C. TOP AND BOTTOM, EACH WAY.

3 14" CONCRETE WALL REINFORCED WITH #9@6" O.C. HORIZONTAL AND VERTICAL, EACH FACE. WATER BEARING WALLS TO BE CONSTRUCTED AND LEAK TESTED PRIOR TO PLACEMENT OF CMU WALLS. REFER TO SPECIFICATION 01 45 25 FOR LEAK TEST REQUIREMENTS.

4 ACCESS HATCH. COORDINATE PLACEMENT WITH PROCESS AND ARCHITECTURAL. PROVIDE ADDITIONAL BARS AT RE-ENTRANT CORNERS PER VOLUME 0 TYPICAL DETAIL.

5 HOLLOW CORE PLANK LAYOUT IS BASED ON 4 FOOT WIDE HOLLOWCORE PLANK WITH A 8-INCH DEPTH CONSIDERING 15 FOOT SPAN.

6 TOP BRACING AT 4'-0" O.C. PER PARALLEL MONORAL CONNECTION DETAIL IN STANDARD DETAILS.

7 TOP BRACING AT 4'-0" O.C. PER PERPENDICULAR MONORAIL CONNECTION IN STANDARD DETAILS.

8 SHALLOW EQUIPMENT PAD. SEE VOLUME 00 TYPICAL DETAILS FOR THICKNESS AND REINFORCING DETAILS. COORDINATE DIMENSIONS WITH EQUIPMENT SUPPLIED.

9 HSS4x4x3/8 POST.

10 BEAM TO CMU CONNECTION PER VOLUME 00 STANDARD DETAILS.

11 PIPE PENETRATIONS THROUGH CONCRETE LID. SEE PROCESS SHEETS.

Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK State of Arizona

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK CADD R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S102	TITLE OF SHEET WATER TREATMENT UPPER LEVEL PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 59 OF 192

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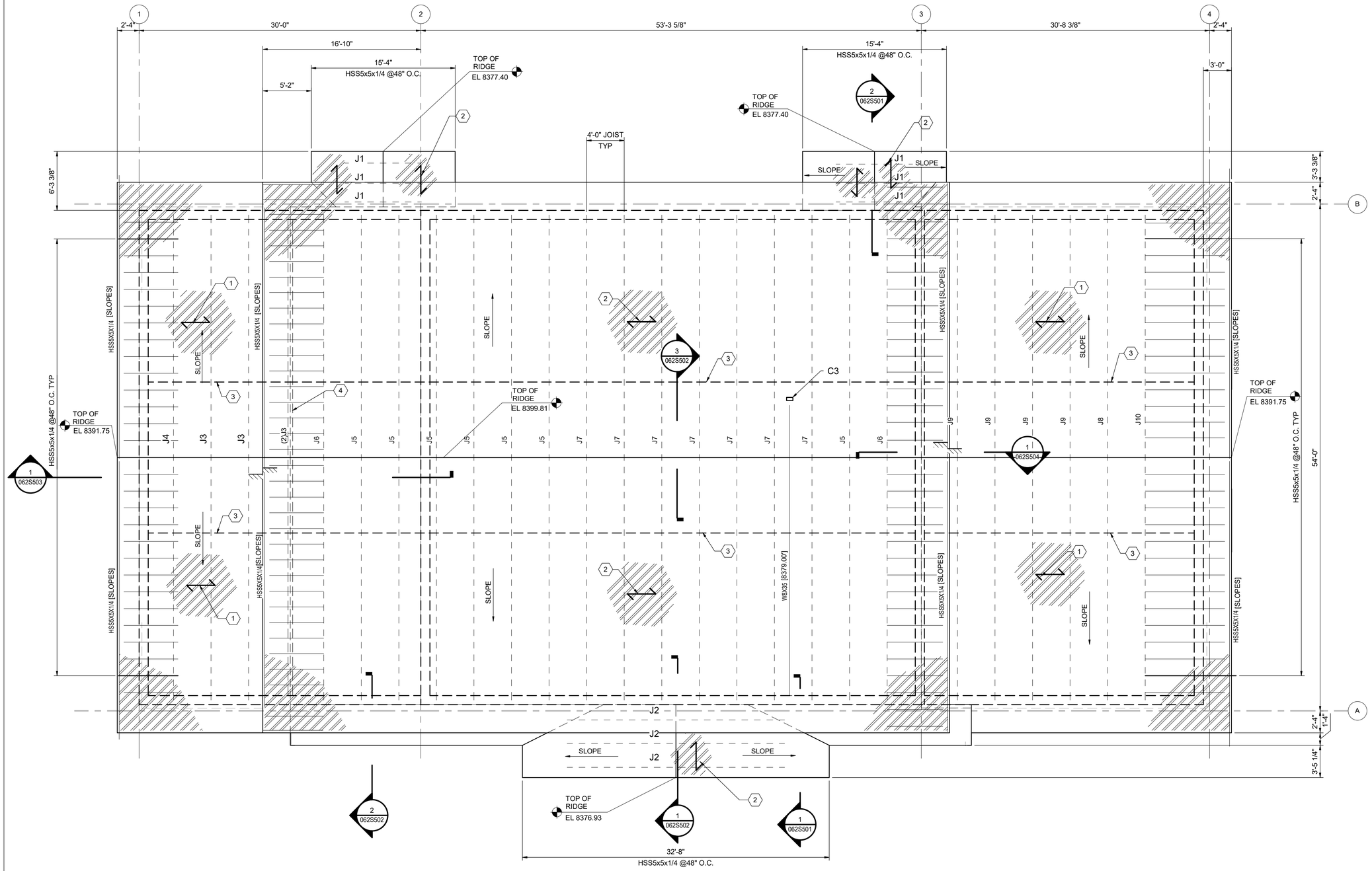

3 062S103 ROOF FRAMING PLAN
SCALE A

SCALE A 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

GENERAL NOTES

A. REFER TO SHEETS 0602S001 THRU 0602S003 FOR GENERAL STRUCTURAL NOTES AND TYPICAL DETAILS.

B. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING. EQUIPMENT, ANCHORS. CONDUITS, HOUSEKEEPING PADS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTURAL ELEMENTS THAT MAY NOT BE SHOWN.

C. CXX DENOTES STEEL COLUMNS. REFER TO SHEET 062S601 FOR STEEL COLUMN SCHEDULE.

D. TOP OF STEEL ELEVATIONS NOTED THUS: EL. XXX.XX.

E. - INDICATES SPAN DIRECTION OF DECK.

F. J1 DENOTES ROOF JOIST TYPE J1. REFER TO ROOF JOIST LOADING DIAGRAM AND TABLE ON SHEET 062S604.

KEYNOTES

1 TYPE 1.5B18 STEEL DECK. DECK SHEETS SHALL BE CONTINUOUS OVER THREE OR MORE SUPPORTS. SEE 1/060S002 FOR ADDITIONAL DECK FASTENING INFORMATION.

2 TYPE 2.0DA20 ACOUSTICAL STEEL DECK. DECK SHEETS SHALL BE CONTINUOUS OVER THREE OR MORE SUPPORTS. SEE 1/060S002 FOR ADDITIONAL DECK FASTENING INFORMATION.

3 JOIST BRIDGING PER MANUFACTURER'S GUIDELINES. SEE GENERAL NOTES AND SPECIFICATIONS FOR ADDITIONAL INFORMATION.

4 LOAD BEARING LIGHT GAUGE STUD WALL. REFER TO SPECIFICATIONS FOR DELEGATED DESIGN PROCEDURES OF LIGHT GAUGE STUDS WALL.

Registered Professional Engineer (Colo.) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed: 04/27/26

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK CADD R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S103	TITLE OF SHEET WATER TREATMENT ROOF FRAMING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 60 OF 192

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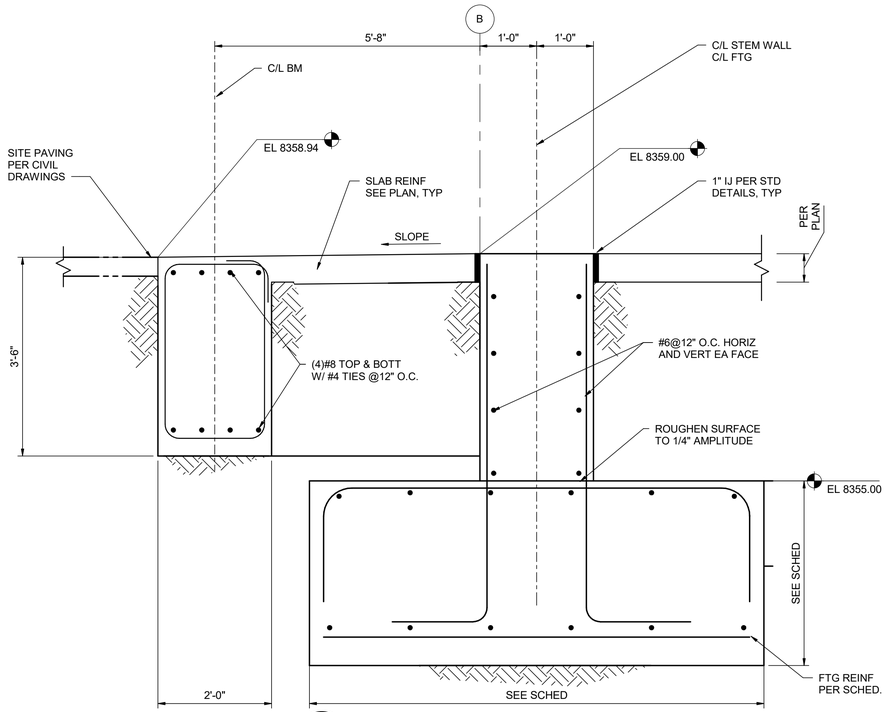
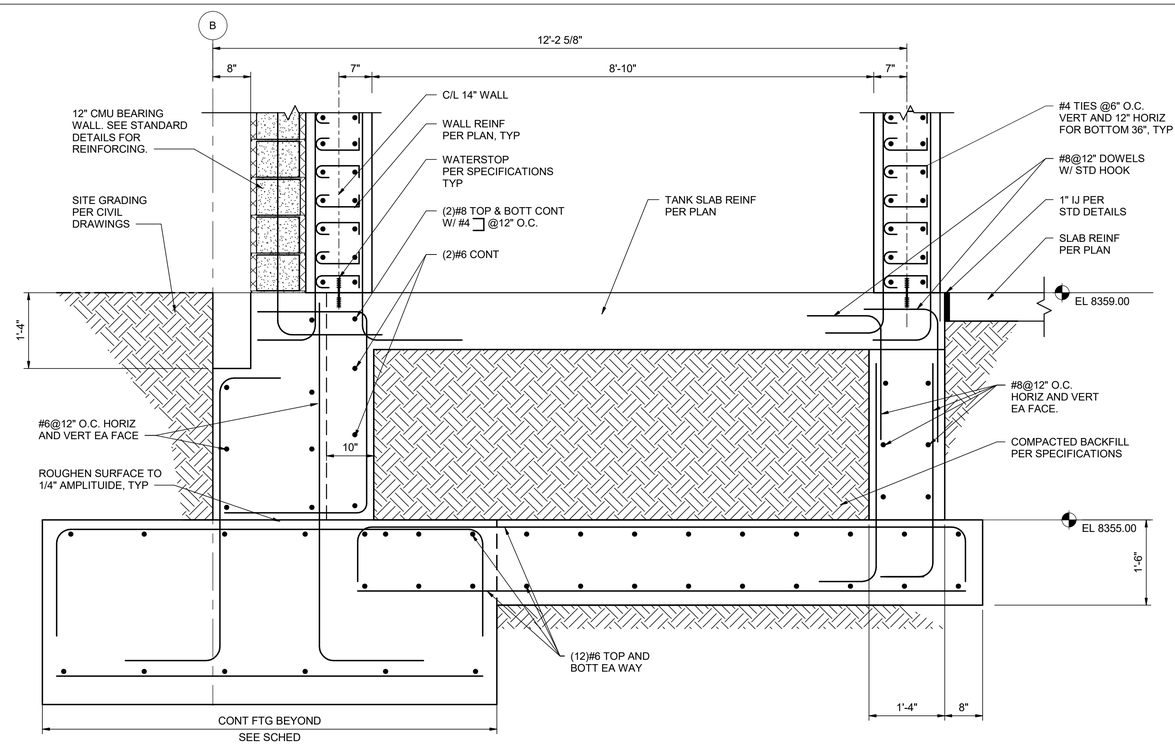
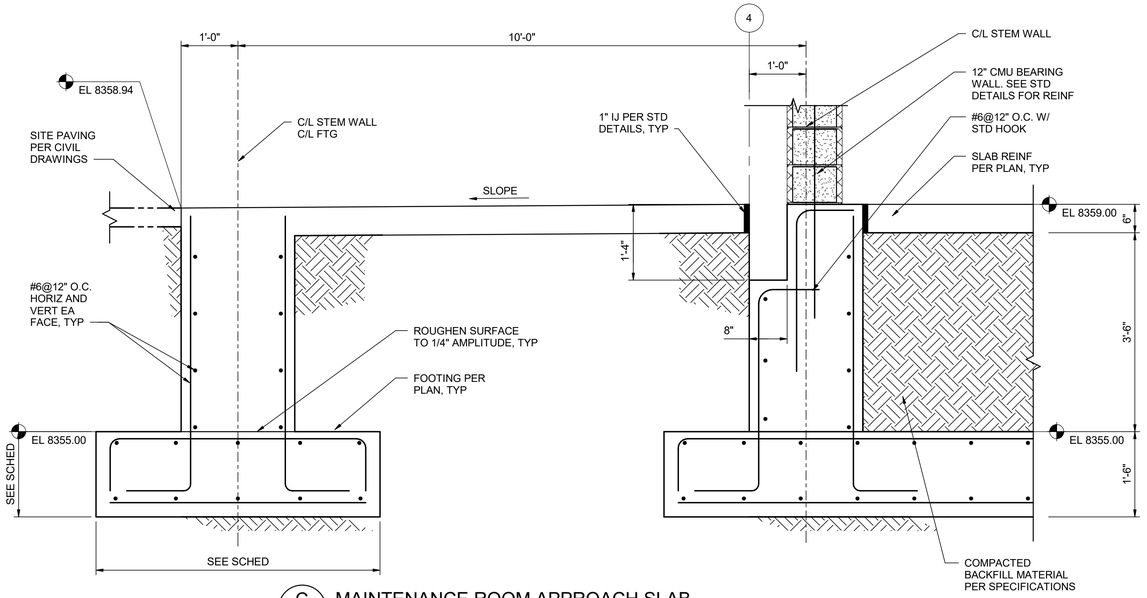


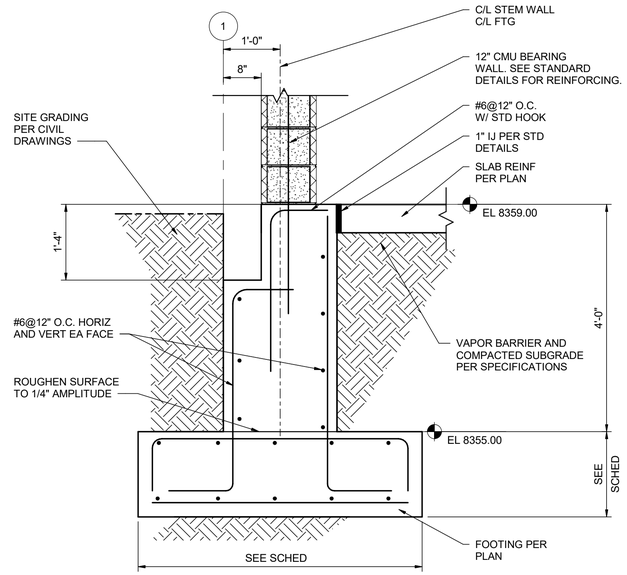
A 062S301 ENTRANCE SLAB
SCALE A



B 062S301 CT TANK BASE SUPPORT
SCALE A



C 062S301 MAINTENANCE ROOM APPROACH SLAB
SCALE A



D 062S301 SECTION AT ADMIN AREA WALL
SCALE A

SCALE A
1 0 1 2 3
SCALE OF FEET
3/4" = 1'-0"

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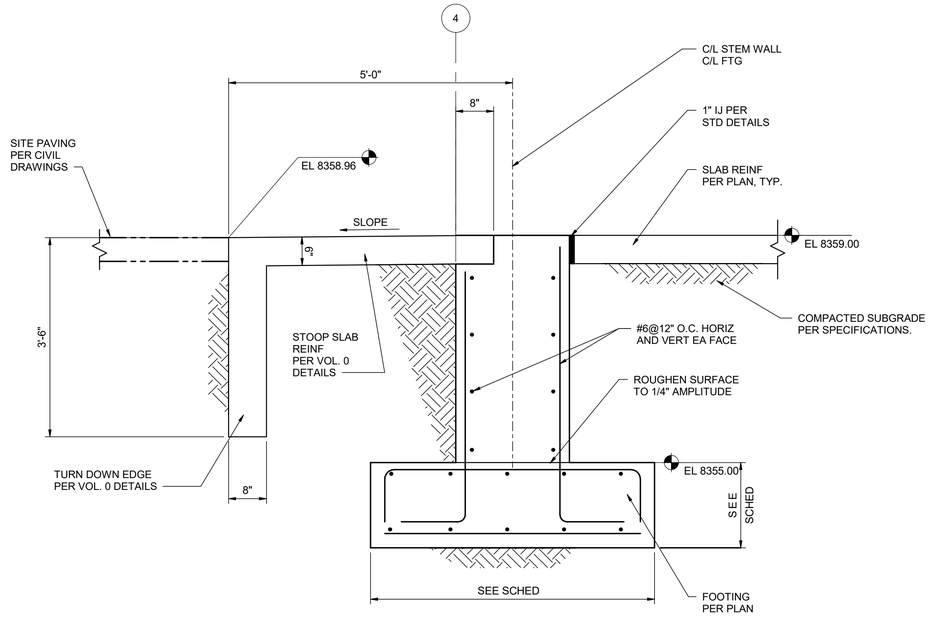
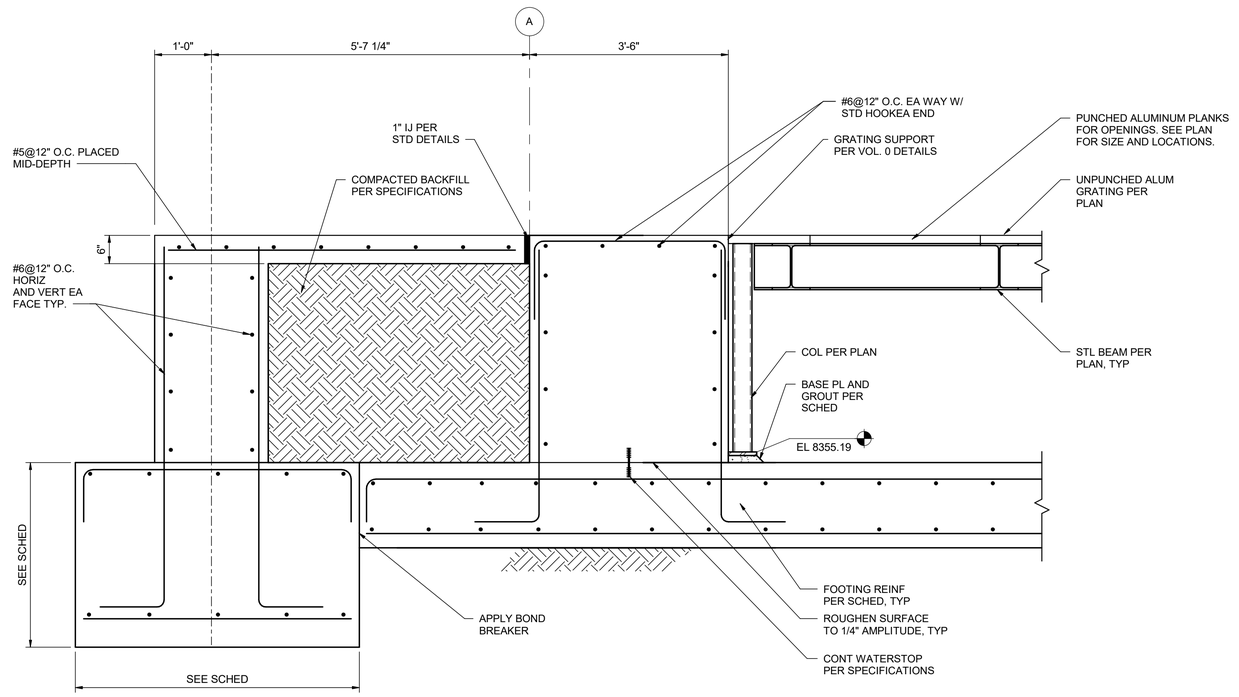
Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK State of ARIZONA

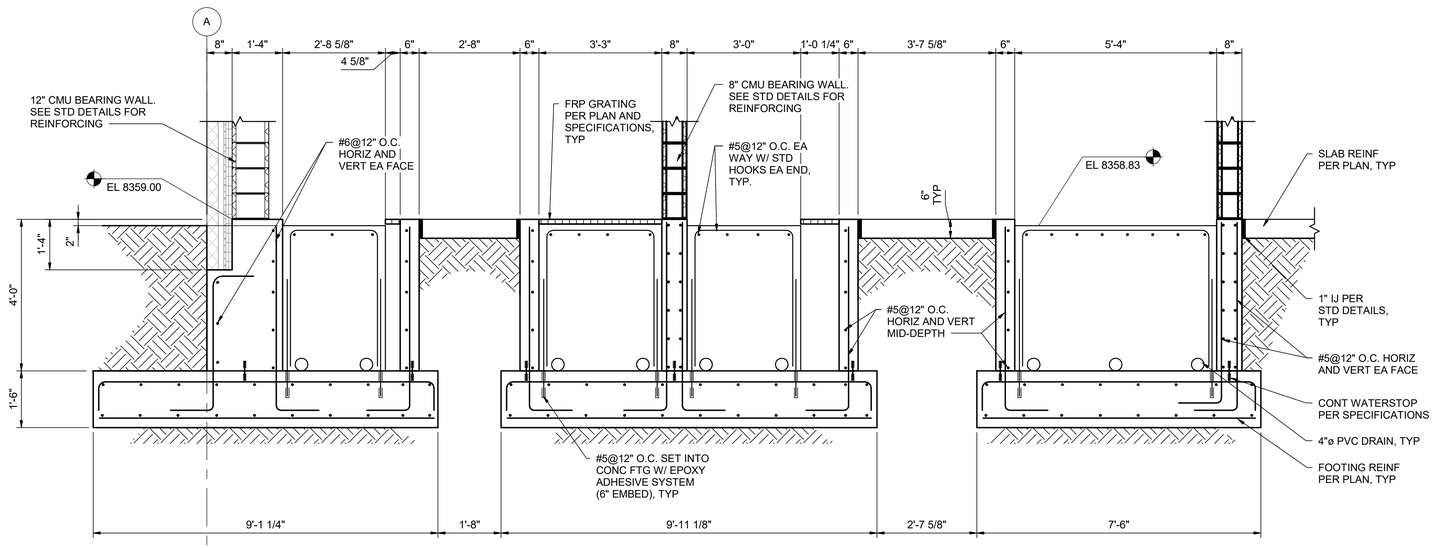
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S301	TITLE OF SHEET WATER TREATMENT SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 61 OF 192



A 062S302 ELECTRICAL ROOM STOOP
SCALE (A)



B 062S302 SECTION AT PROCESS AREA WALL
SCALE (A)



C 062S302 CHEMICAL DRUM ROOM CONTAINMENT BASINS
SCALE (B)

SCALE (B) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

SCALE (A) 1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

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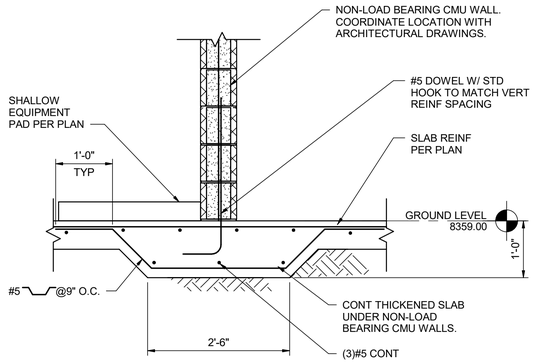
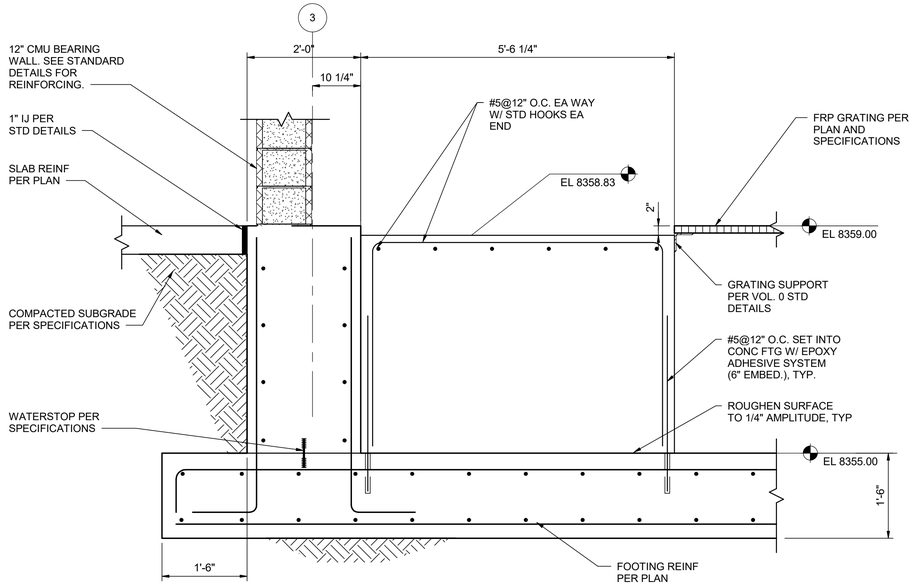
Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK State of ARIZONA

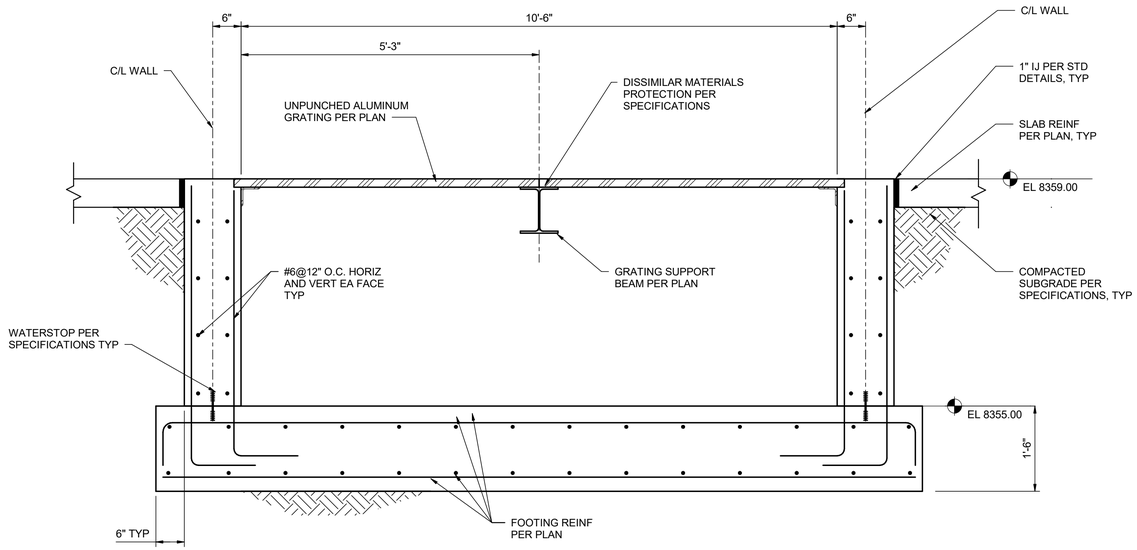
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S302	TITLE OF SHEET WATER TREATMENT SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 62 OF 192



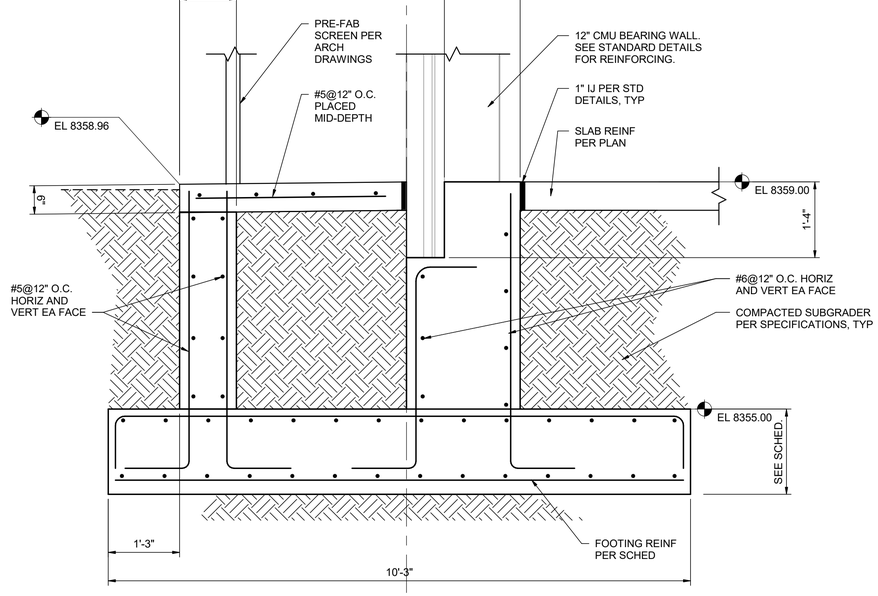
A 062S303 SECTION AT NON-LOAD BEARING CMU WALL
SCALE A



B 062S303 INTERIOR SECTION AT SUMP TRENCH TANK SUPPORT
SCALE A



C 062S303 INTERIOR SECTION AT TRENCH DRAIN
SCALE A



D 062S303 MECHANICAL EQUIPMENT PAD AND FOUNDATION
SCALE A

SCALE A 1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Original) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed 10/31/2025 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S303	TITLE OF SHEET WATER TREATMENT SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 63 OF 192

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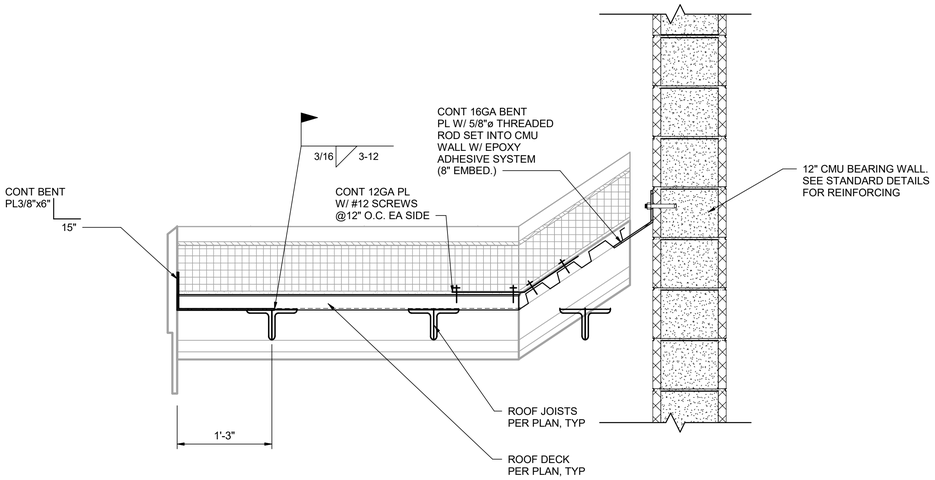
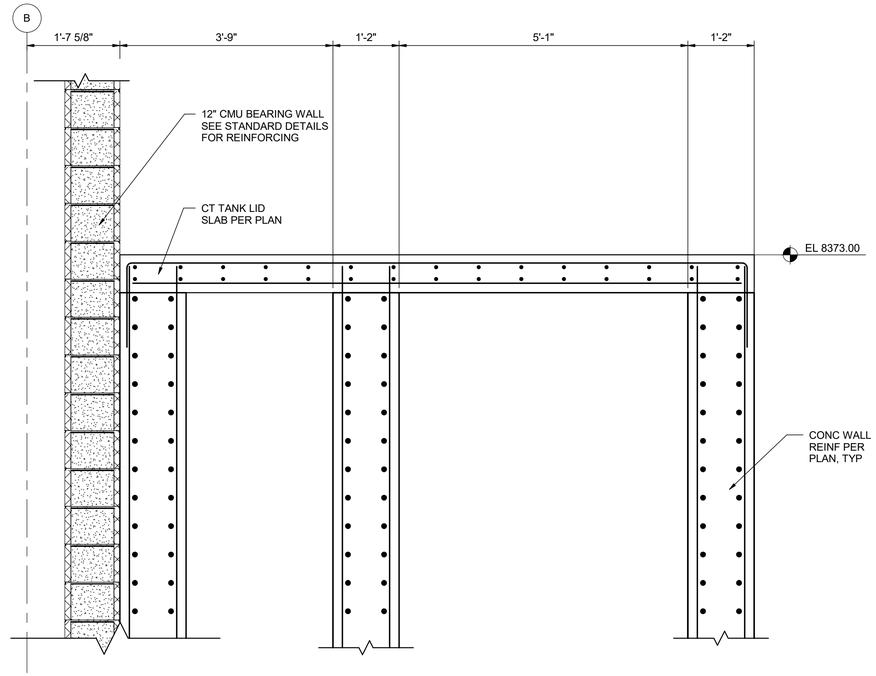
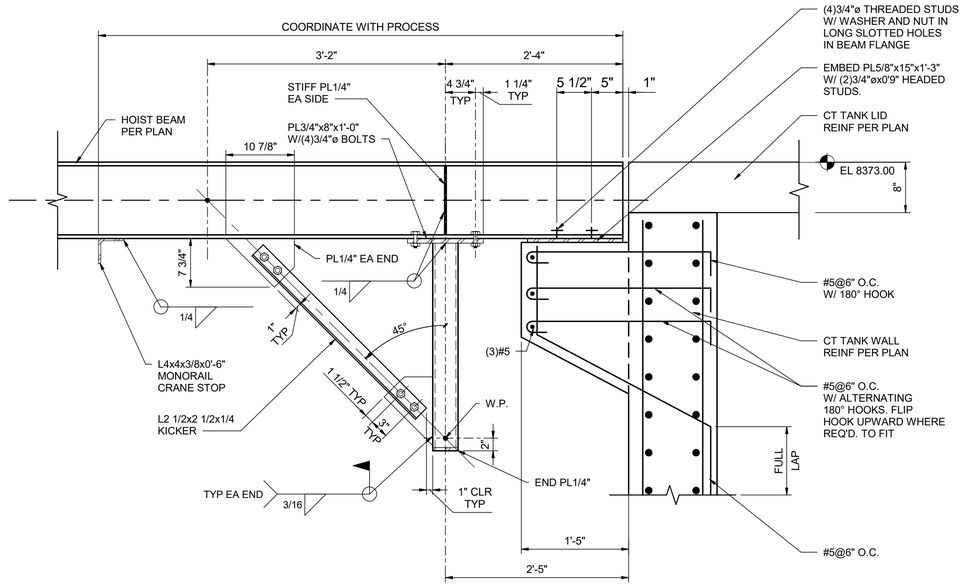


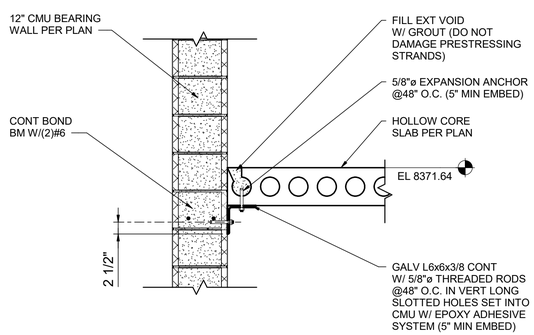
A 062S304 UPPER LEVEL FRAMING AT ENTRANCE
SCALE (A)



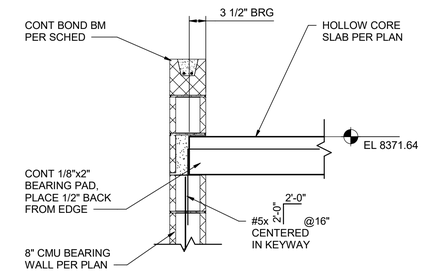
B 062S304 CT TANK LID FRAMING
SCALE (A)



C 062S304 HOIST BEAM CORBEL SUPPORT
SCALE (A)



D 062S304 HOLLOW CORE CONNECTION AT CMU
SCALE (A)



E 062S304 HOLLOW CORE BEARING DETAIL
SCALE (A)

SCALE (A) 1 0 1 2 1" = 1'-0"
SCALE OF FEET

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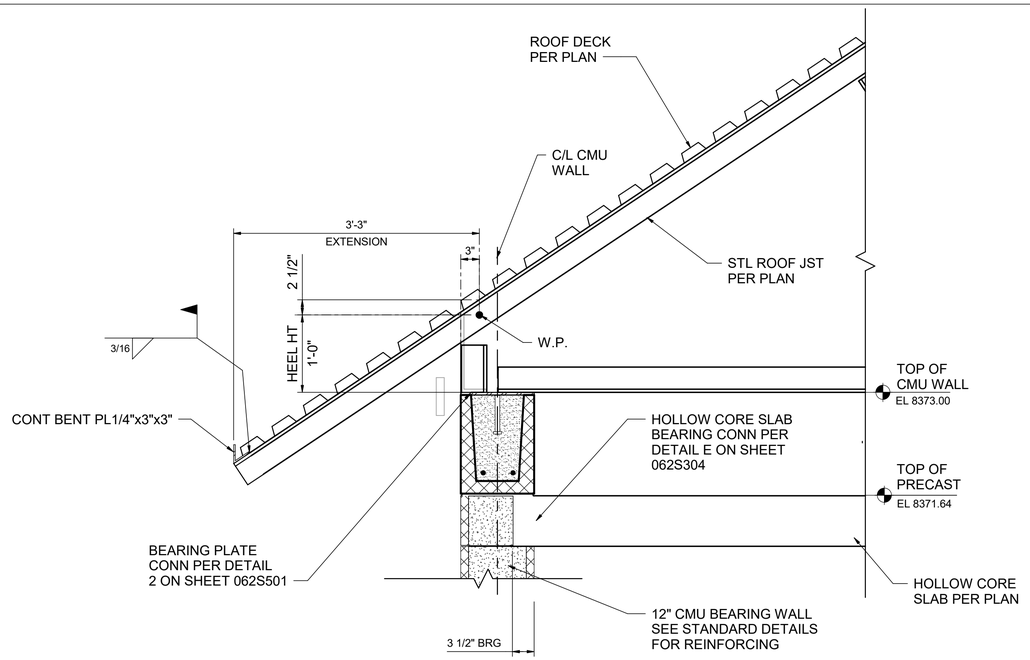
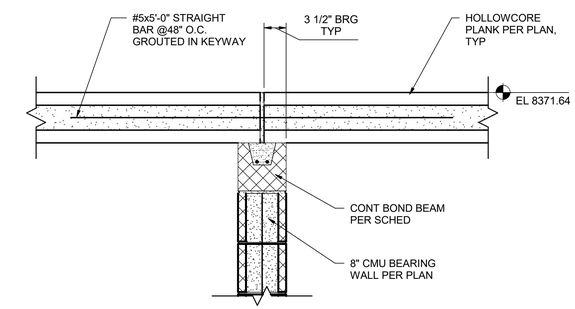
Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK State of ARIZONA

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S304	TITLE OF SHEET WATER TREATMENT SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 64 OF 192



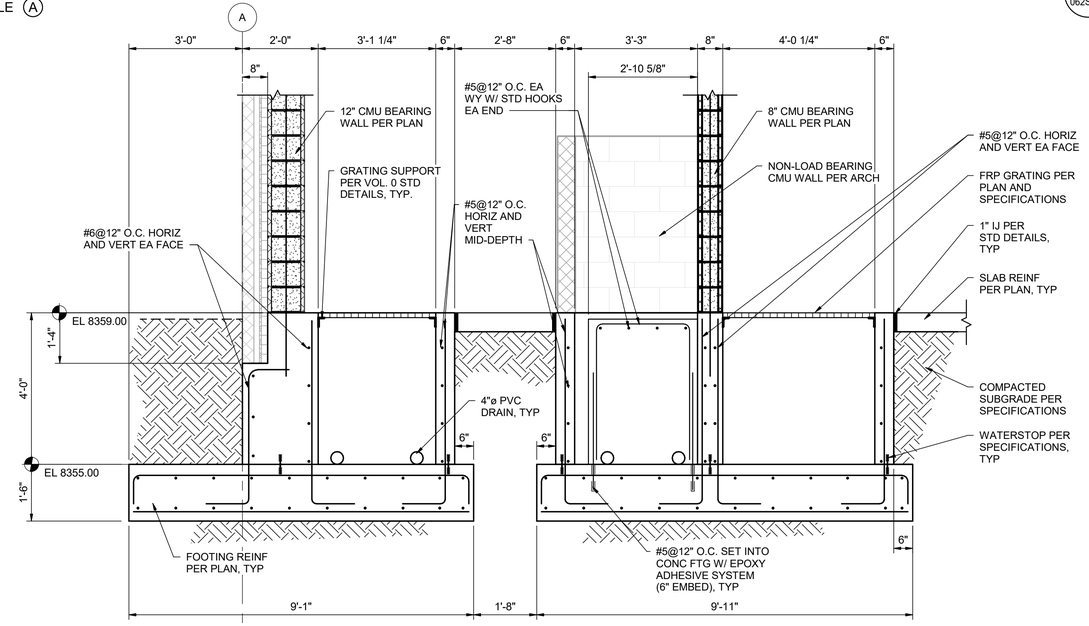
A 062S305 ROOF JOIST END BEARING

SCALE (A)



B 062S305 HOLLOW CORE BEARING AT INTERIOR CMU WALL

SCALE (A)



C 062S305 CHEMICAL DRUM ROOM CONTAINMENT BASINS

SCALE (B)

SCALE (B) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

SCALE (A) 1 0 1 2 1" = 1'-0"
SCALE OF FEET

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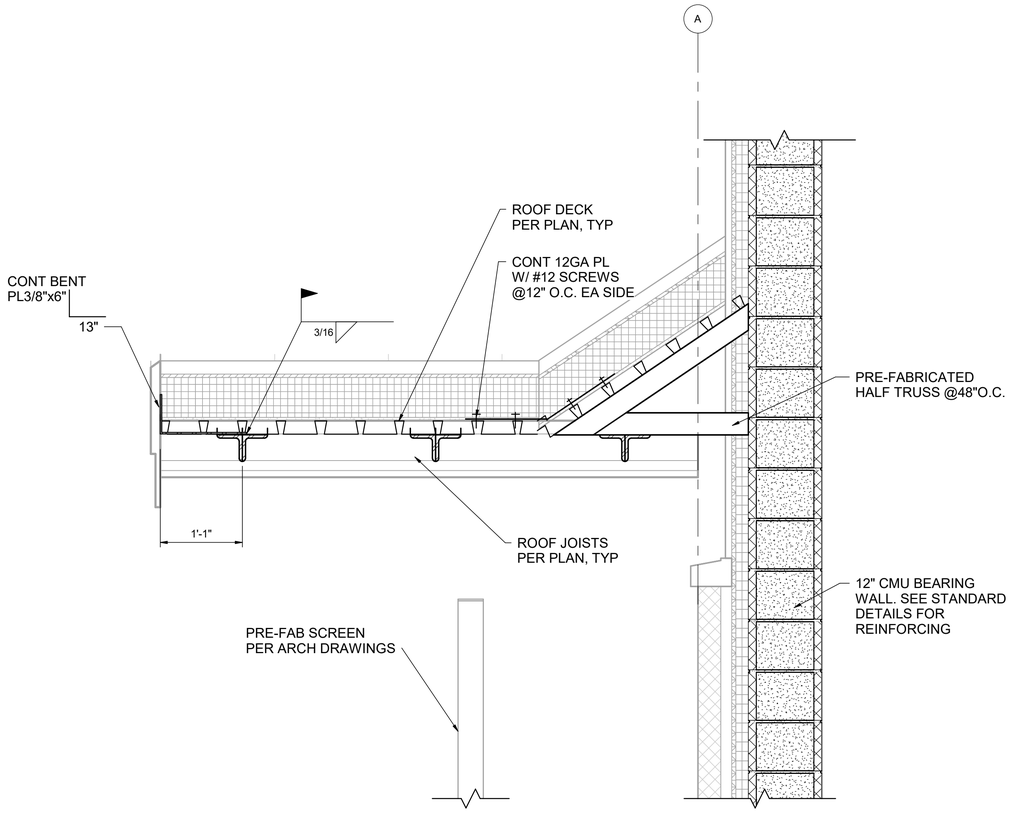
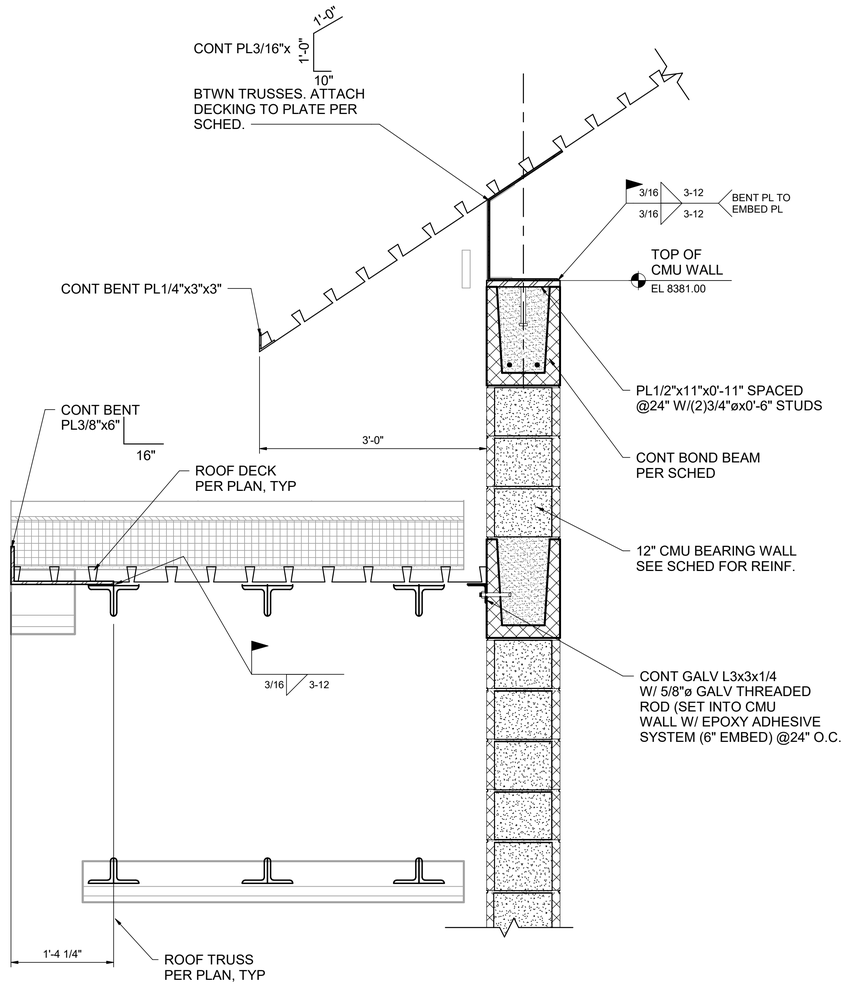
Registered Professional Engineer (Corp)
CERTIFICATE NO.
72480
COLIN MULDERICK
State Serving Humanity
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S305	TITLE OF SHEET WATER TREATMENT SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 65 OF 192



1 062S501 ROOF SECTION AT ROOF TRANSITION
SCALE (A)



2 062S501 PROCESS AREA ENTRANCE CANOPY
SCALE (A)

SCALE (A) 1 0 1 2 1" = 1'-0"
SCALE OF FEET

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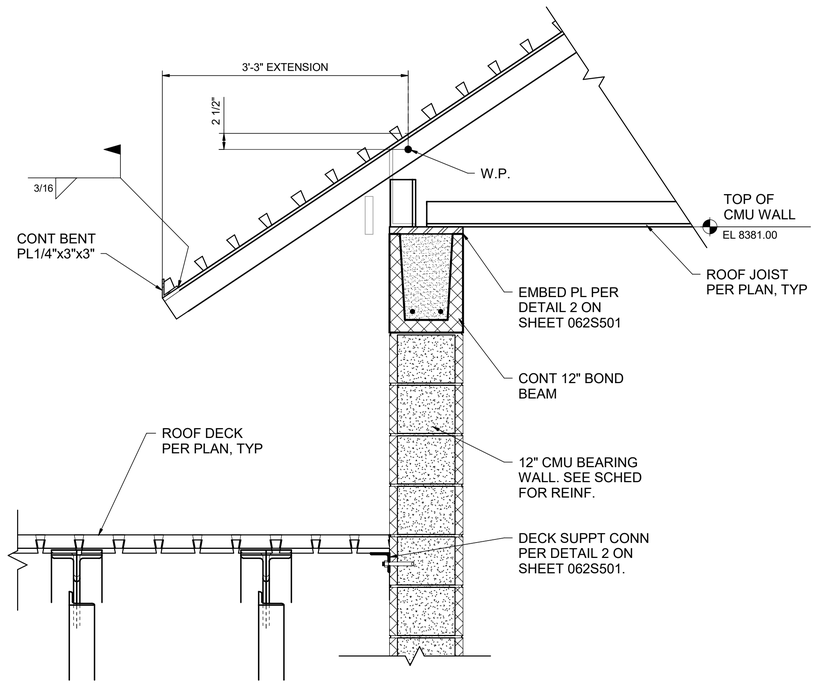
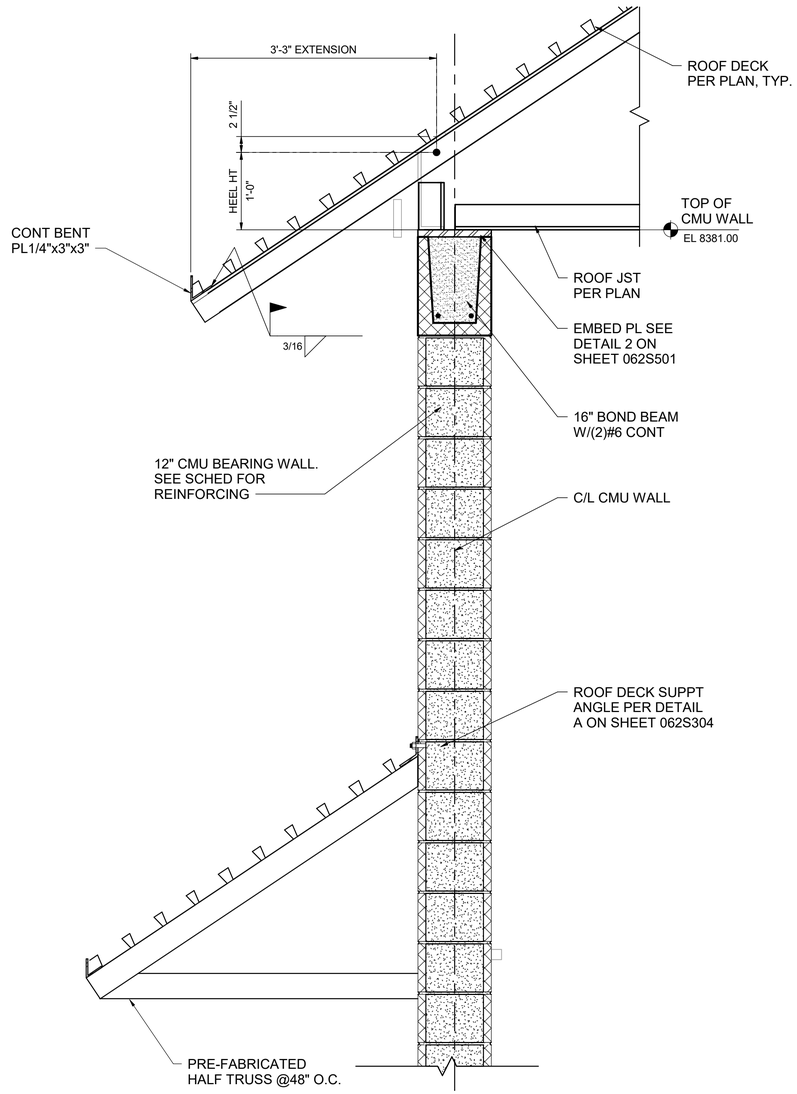
Registered Professional Engineer (Civil)
CERTIFICATE NO.
72480
COLIN MULDERICK
Date Signed 04/27/2026
ARIZONA, U.S.A.

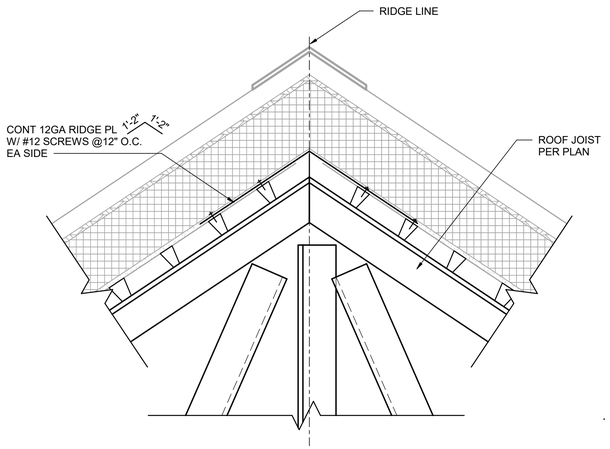
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S501	TITLE OF SHEET WATER TREATMENT DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 66 OF 192



1 062S502 SECTION AT PROCESS AREA ROOF
SCALE (A)



2 062S502 SECTION AT OVERFRAMING
SCALE (A)



NOTES:
1. TYPICAL AT RIDGE. SIMILAR PLATE AT 1 1/2" METAL DECK.

3 062S502 SECTION AT RIDGE
SCALE (B)

SCALE (B) 12 6 0 12 1-1/2" = 1'-0"
SCALE OF FEET

SCALE (A) 1 0 1 2 1" = 1'-0"
SCALE OF FEET

Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed 10/31/2025 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S502	TITLE OF SHEET WATER TREATMENT DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 67 OF 192

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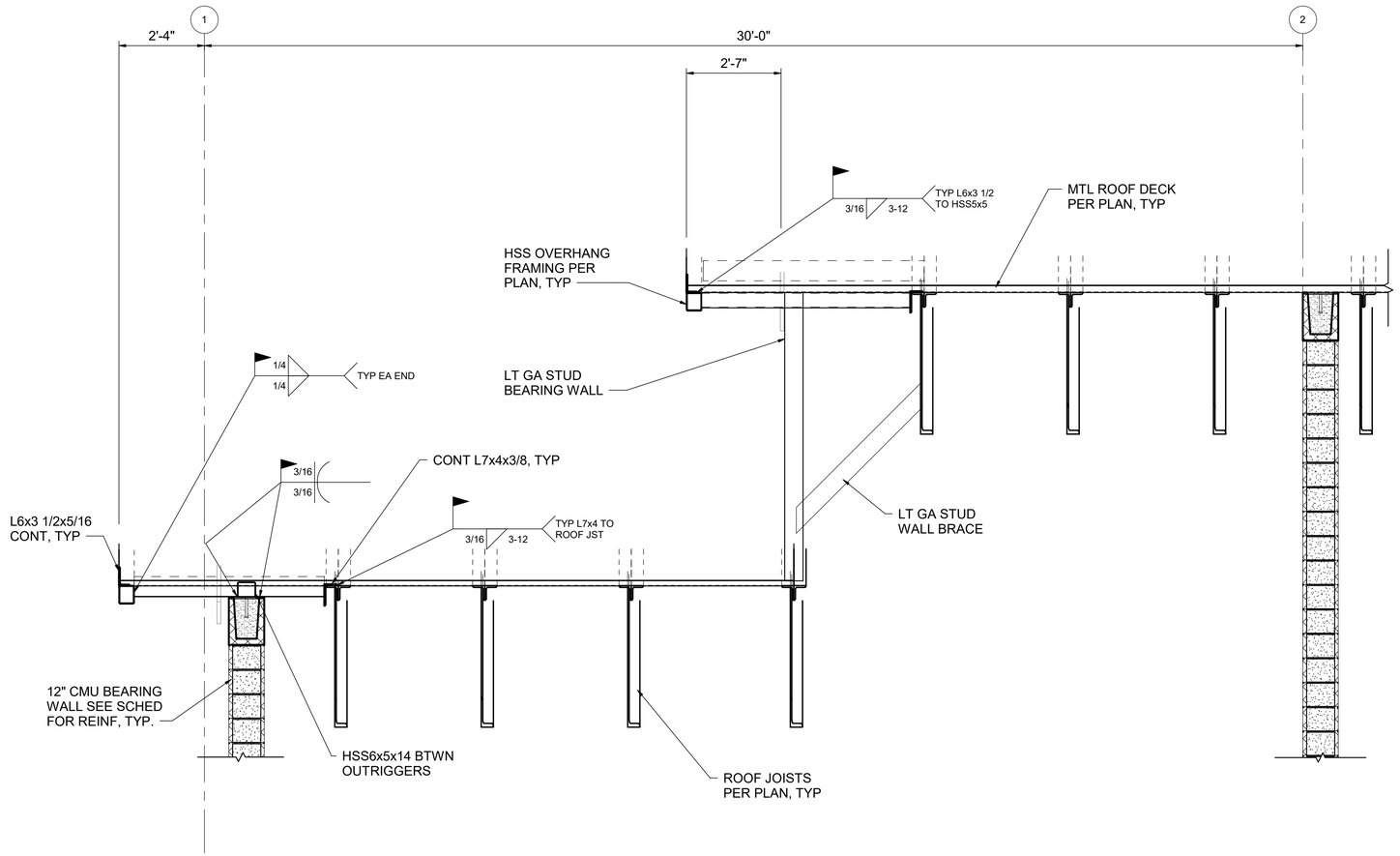
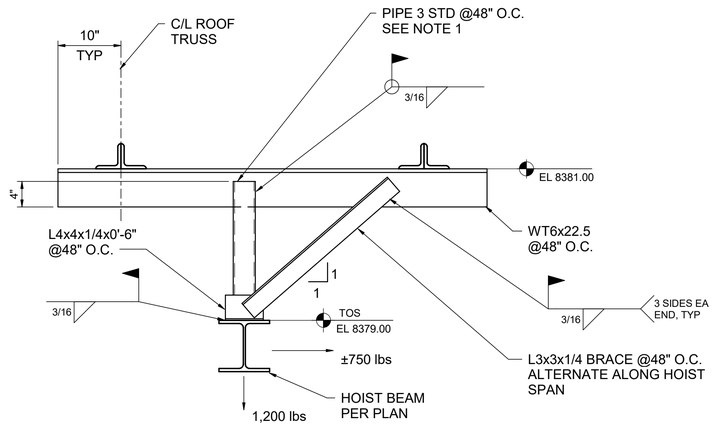


1 062S503 SECTION AT UPPER LEVEL ROOF
SCALE (A)



2 062S503 HOIST BEAM PARALLEL TO ROOF JOISTS
SCALE (B)

NOTES:
1. DO NOT LOCATE PIPE CLOSER THAN 4" FROM END OF MONRAIL SUPPORT BEAM.
2. VERTICAL AND LATERAL LOADING INDICATED ABOVE TO BE INCLUDED INTO ROOF JOIST DESIGN ANALYSIS. SEE SHEET 062S604 FOR ADDITIONAL INFORMATION.

SCALE (B) 1 0 1 2 — 1" = 1'-0"
SCALE OF FEET

SCALE (A) 2 0 2 4 — 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer, State of Arizona
CERTIFICATE NO. 72480
COLIN MULDERICK
Date Signed 05/07/2026

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S503	TITLE OF SHEET WATER TREATMENT DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 68 OF 192

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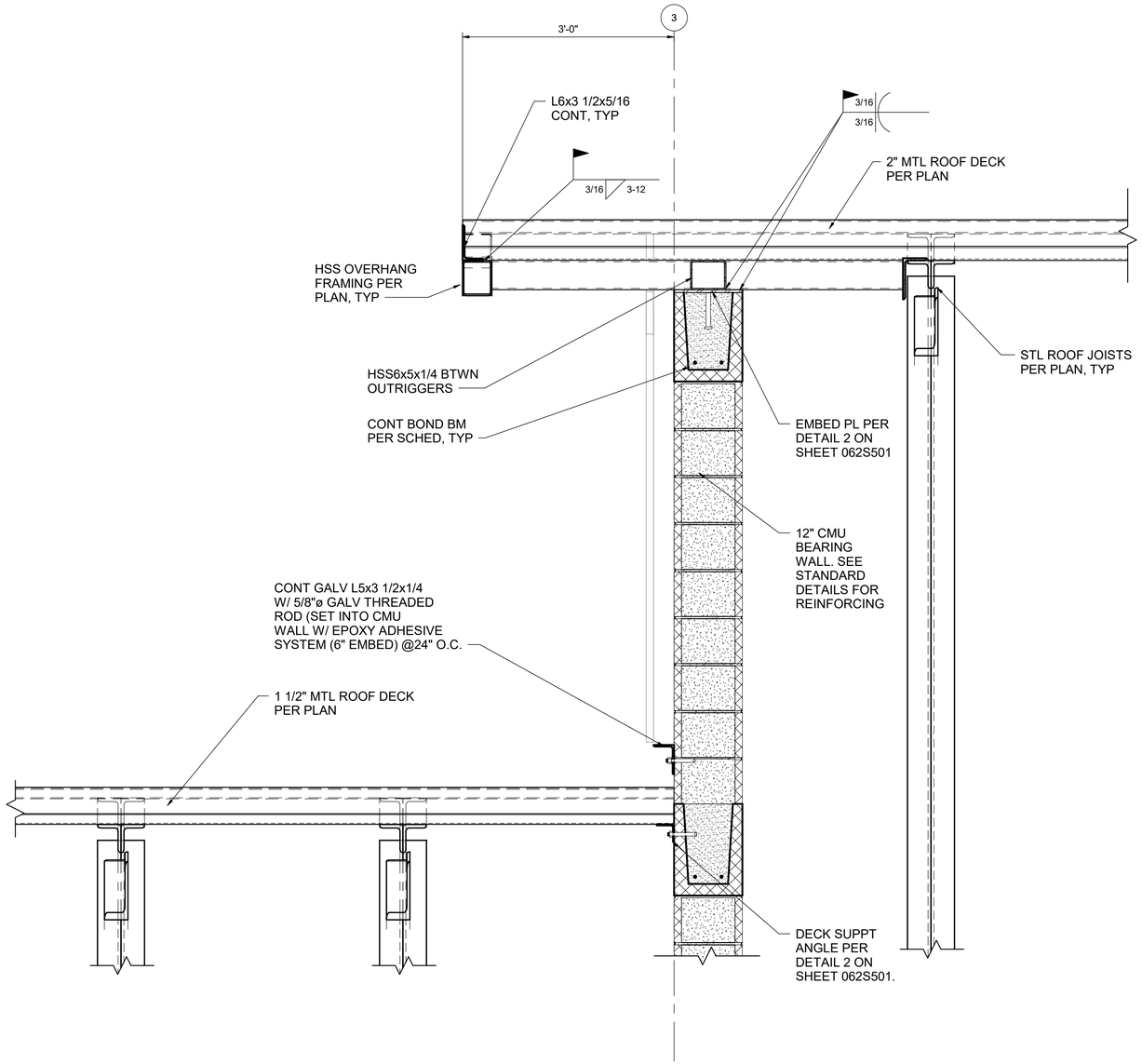
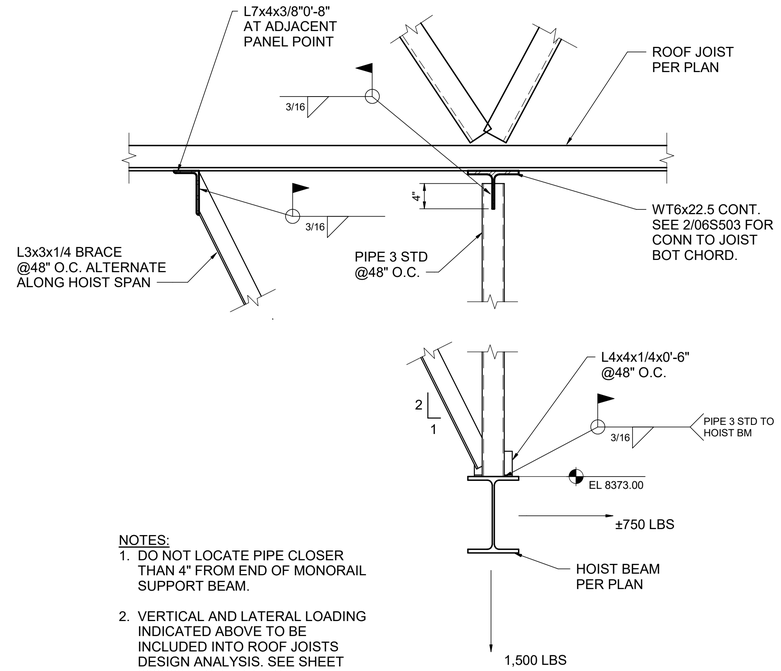


1 062S504 SECTION AT ROOF TRANSITION
SCALE A



NOTES:
1. DO NOT LOCATE PIPE CLOSER THAN 4" FROM END OF MONORAIL SUPPORT BEAM.
2. VERTICAL AND LATERAL LOADING INDICATED ABOVE TO BE INCLUDED INTO ROOF JOISTS DESIGN ANALYSIS. SEE SHEET 062S604 FOR ADDITIONAL INFORMATION.

2 062S504 HOIST BEAM PERPENDICULAR TO JOISTS
SCALE A

SCALE A 1 0 1 2 1" = 1'-0"
SCALE OF FEET

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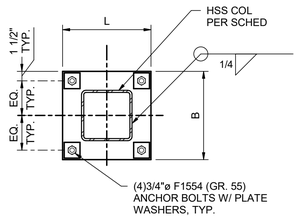
Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed 05/01/2026 ARIZONA, U.S.A.

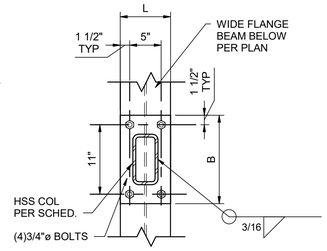
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S504	TITLE OF SHEET WATER TREATMENT DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 69 OF 192

STRUCTURAL COLUMN AND BASEPLATE SCHEDULE								
COLUMN MARK	COLUMN SIZE	TOP OF COLUMN EL.	BASE PLATE TYPE	THICKNESS	L	B	TOP OF BASE PLATE EL.	REMARKS
C1	HSS4x4x1/4	8358.85'	TYPE 1	3/4"	10"	10"	8355.19'	
C2	HSS8x8x1/4	8370.00'	TYPE 1	3/4"	14"	14"	8359.19'	
C3	HSS8x4x1/2	8379.00'	TYPE 2	3/4"	8"	14"	8370.06'	
C4	HSS4x4x1/4	8358.85'	TYPE 3	3/4"	6"	14"	8355.19'	

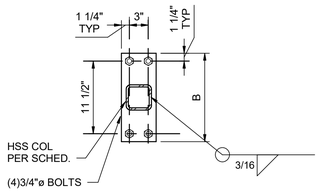
FOOTING SCHEDULE					
MARK	L	W	D	REINFORCING	REMARKS
F1	CONT	5'-0"	18"	(3)#6 LONG T&B; #6@12" O.C. TRANS T&B	
F2	CONT	8'-0"	18"	(6)#6 LONG T&B; #6@12" O.C. TRANS T&B	
F3	CONT	8'-0"	39"	(10)#6 LONG T&B; #6@6" O.C. TRANS T&B	
F4	CONT	5'-0"	36"	(9)#6 LONG T&B; #6@12" O.C. TRANS T&B	
F5	CONT	8'-0"	51"	(14)#6 LONG T&B; #6@12" O.C. TRANS T&B	REFER TO VOL. 00 FOR UNDERSLAB PIPE ENCASEMENT



TYPE 1



TYPE 2



TYPE 3

1 BASE PLATE DETAILS
062S601
NOT TO SCALE

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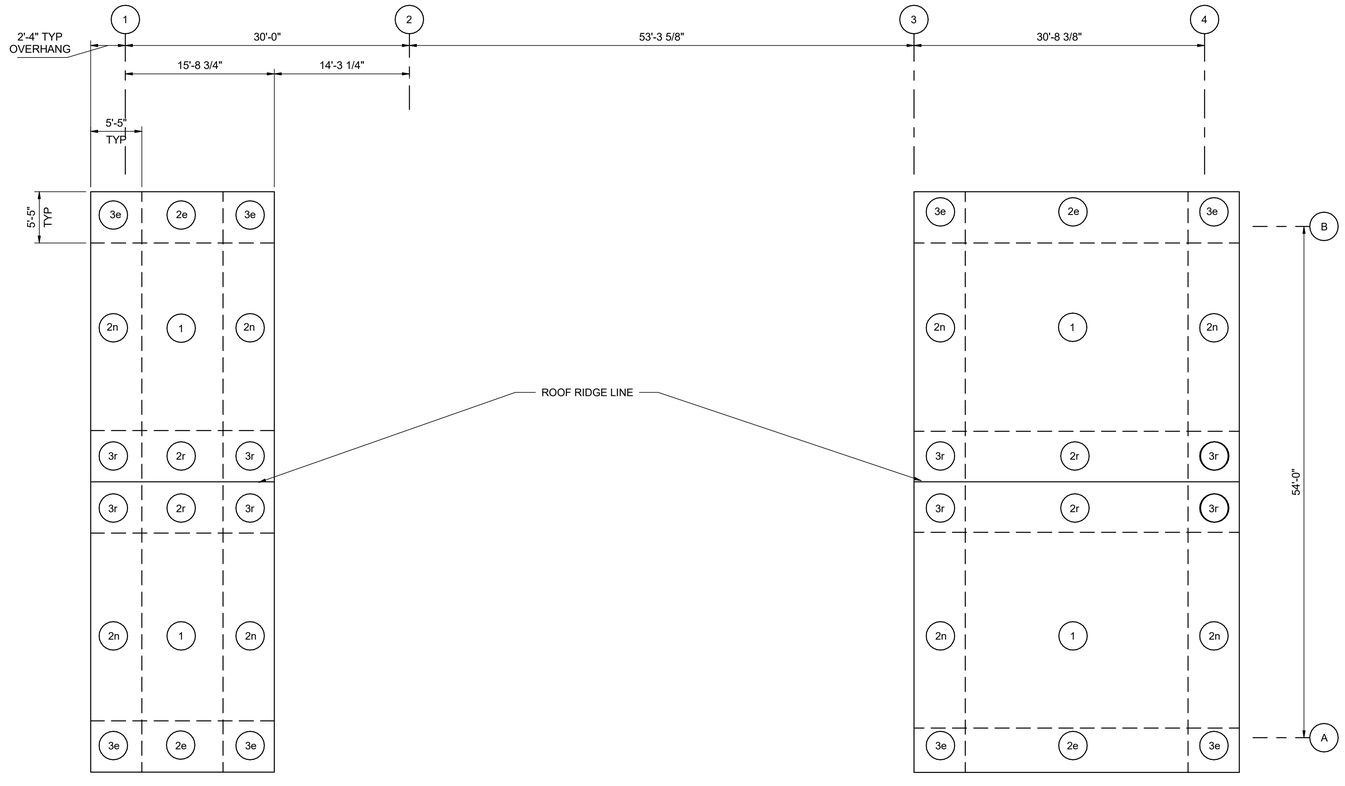
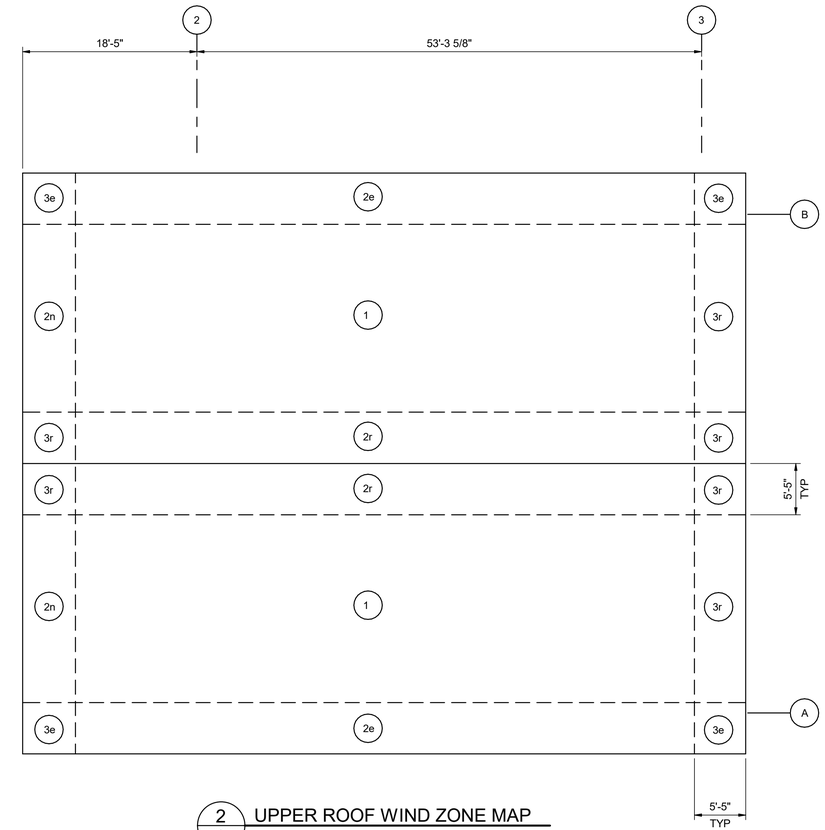
4/27/2026 9:14:25 PM

Registered Professional Engineer (Utah) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed: 04/27/2026 ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	C. MULDERICK DRAFTED: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	062 S601	WATER TREATMENT COLUMN AND FOOTING SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 70 OF 192



1 062S602 LOWER ROOFS WIND ZONE MAP
NOT TO SCALE



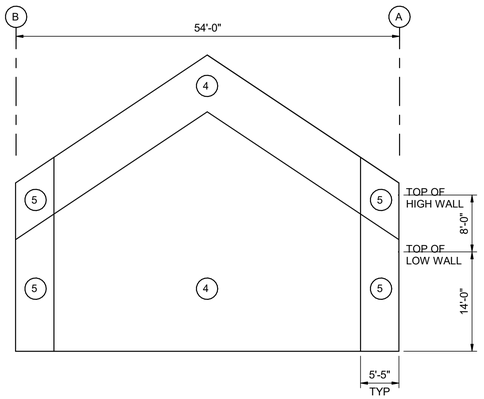
2 062S602 UPPER ROOF WIND ZONE MAP
NOT TO SCALE

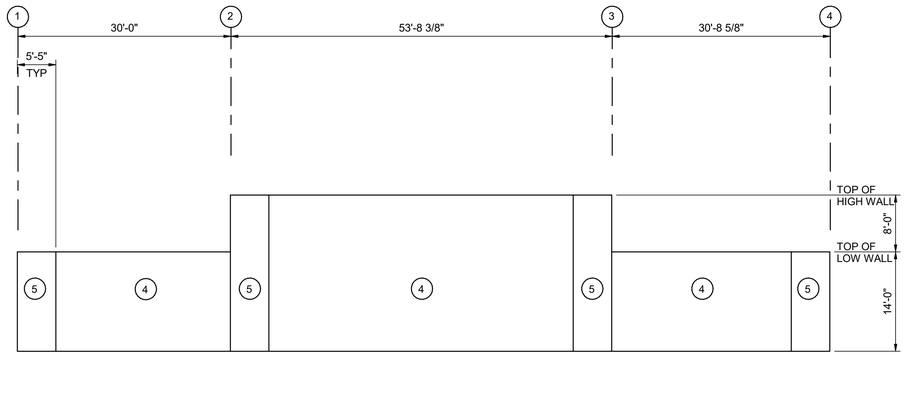
ULTIMATE WIND PRESSURES - COMPONENTS AND CLADDING																
ZONE / SQ. FT.	1		2e		2r		2n		3e		3r		4		5	
	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-
≥10	21	31	21	31	21	31	21	35	21	45	21	35	23	25	23	31
20	19	22	19	22	19	22	19	31	19	39	19	31	22	23	22	29
50	16	18	16	18	16	18	16	24	16	31	16	24	21	23	21	27
100	16	16	16	16	16	16	16	20	16	26	16	20	19	21	19	25
200	16	16	16	16	16	16	16	16	16	20	16	16	19	21	19	21
500<	16	16	16	16	16	16	16	16	16	16	16	16	17	19	17	19

NOTES:
REFER TO ASCE 7 FIGURES 30.3-1 AND 30.3-2D FOR PRESSURE DIAGRAMS.

ULTIMATE WIND PRESSURES - OVERHANGS												
ZONE / SQ. FT.	1		2e		2r		2n		3e		3r	
	+	-	+	-	+	-	+	-	+	-	+	-
≥10	-	31	-	31	-	31	-	35	-	52	-	41
20	-	22	-	22	-	22	-	31	-	45	-	36
50	-	18	-	18	-	18	-	24	-	36	-	27
100	-	16	-	16	-	16	-	20	-	29	-	23
200	-	16	-	16	-	16	-	16	-	23	-	18
500<	-	16	-	16	-	16	-	16	-	18	-	18

NOTES:
REFER TO ASCE 7 FIGURE 30.3-2D FOR PRESSURE DIAGRAM.





3 062S602 WALL WIND ZONE MAP
NOT TO SCALE

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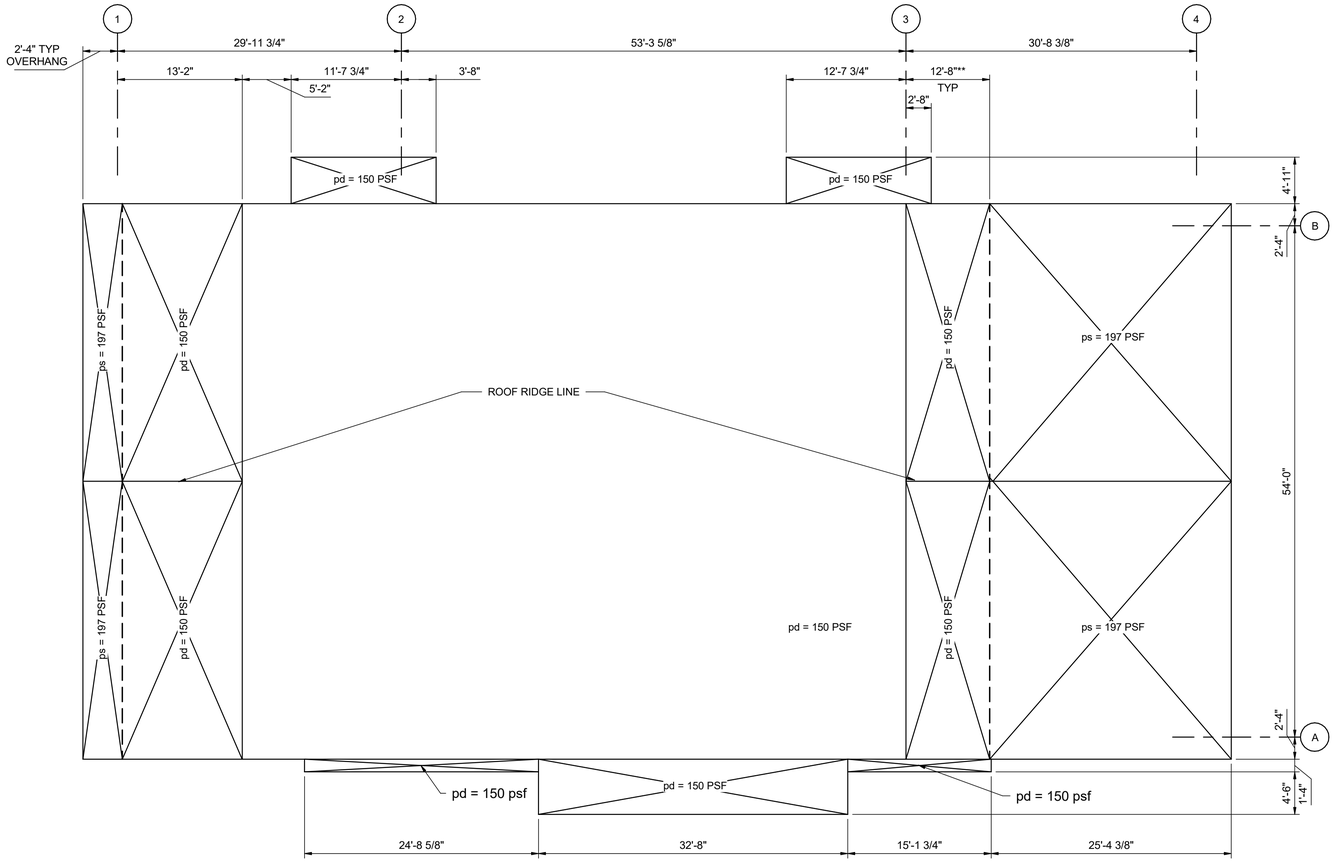
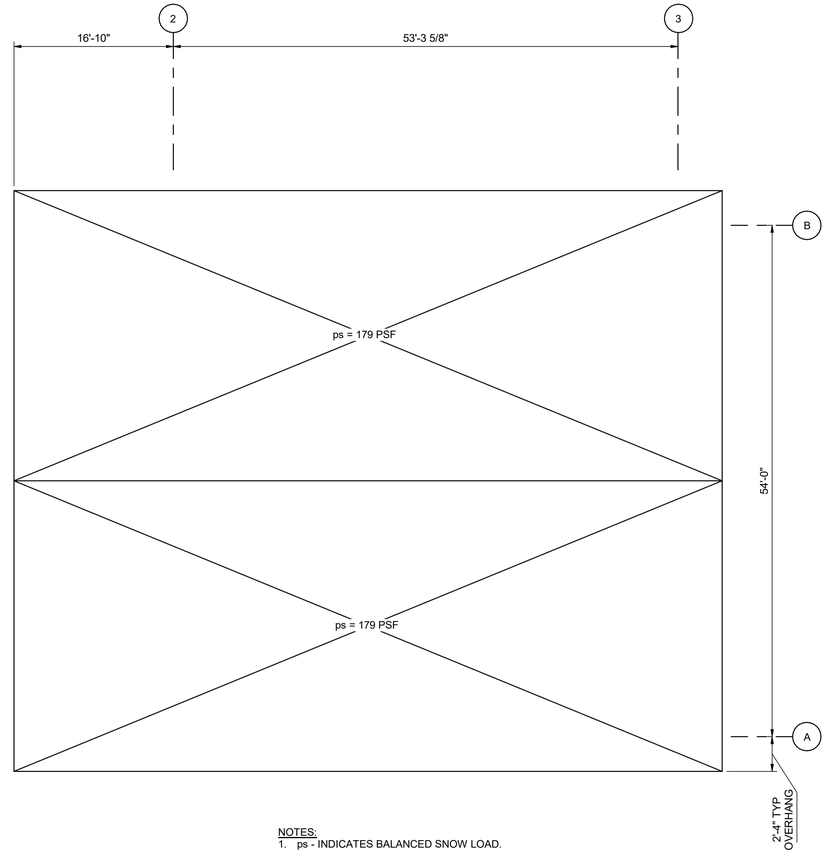
Registered Professional Engineer (Civil)
CERTIFICATE NO.
72480
COLIN MULDERICK
State of ARIZONA

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S602	TITLE OF SHEET WATER TREATMENT WIND LOADING DIAGRAMS - ROOFS AND WALLS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 71 OF 192



NOTES:
1. ** - INDICATES WIDTH OF SNOW DRIFT ZONES.
2. SNOW DRIFT LOADING (pd) SHALL BE SUPERIMPOSED ON BALANCED SNOW LOAD.
3. ps - INDICATES BALANCED SNOW LOAD.

1 / 062S603 LOWER ROOFS SNOW LOAD MAP
NOT TO SCALE



NOTES:
1. ps - INDICATES BALANCED SNOW LOAD.

2 / 062S603 UPPER ROOF SNOW LOAD MAP
NOT TO SCALE

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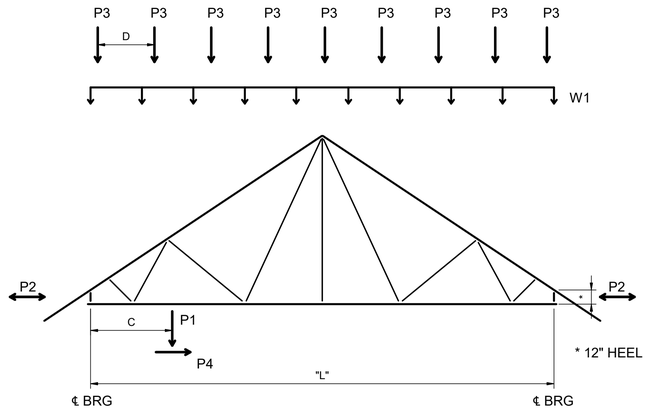
Professional Engineer (Civil) CERTIFICATE NO. 72480 COLIN MULDERICK State of Arizona

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 062 S603	TITLE OF SHEET WATER TREATMENT SNOW LOADING DIAGRAMS - ROOF NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 72 OF 192



JOIST LOAD DIAGRAM

JOIST LABEL	L	SJI LH JOIST DESIGNATION	DIM	LOAD	DEAD	SNOW	ROOF LIVE	LIVE NOTE 4	WIND	SEISMIC	OVERHANG FRAMING					REMARKS
											DEAD	SNOW	ROOF LIVE	LIVE NOTE 4	WIND	
J1	8'-4"	12-49-12 LH	-	W1	66 PLF	1388 PLF	80 PLF	-	NOTE 2	-	-	-	-	-	-	-
			-	P2	-	-	-	-	0.5 KIP	0.6 KIP	-	-	-	-	-	-
			D = 4'-0"	P3	-	-	-	-	-	-	0.5 KIP	6.3 KIP	0.6 KIP	-	-1.6 KIP	LOADING APPLIES AT OUTSIDE JOIST ONLY
J2	23'-7"	12-62-12 LH	-	W1	66 PLF	1388 PLF	80 PLF	-	NOTE 2	-	-	-	-	-	-	-
			-	P2	-	-	-	-	0.5 KIP	0.6 KIP	-	-	-	-	-	-
			D = 4'-0"	P3	-	-	-	-	-	-	0.5 KIP	6.3 KIP	0.6 KIP	-	-1.6 KIP	LOADING APPLIES AT OUTSIDE JOIST ONLY
J3	51'-8"	12-223-12 LH	-	W1	66 PLF	1388 PLF	80 PLF	-	NOTE 2	-	-	-	-	-	-	-
			-	P2	-	-	-	-	2.1 KIP	5.5 KIP	-	-	-	-	-	-
J4	51'-8"	12-223-12 LH		W1	66 PLF	1388 PLF	80 PLF	-	NOTE 2	-	-	-	-	-	-	-
			-	P2	-	-	-	-	2.1 KIP	5.5 KIP	-	-	-	-	-	-
			D = 4'-0"	P3	-	-	-	-	-	-	0.5 KIP	5.2 KIP	0.6 KIP	-	-1.4 KIP	-
J5	51'-8"	12-223-12 LH	-	W1	66 PLF	788 PLF	80 PLF	-	NOTE 2	-	-	-	-	-	-	-
			-	P2	-	-	-	-	2.1 KIP	5.5 KIP	-	-	-	-	-	-
J6	51'-8"	12-223-12 LH	-	W1	66 PLF	788 PLF	80 PLF	-	NOTE 2	-	-	-	-	-	-	-
			-	P2	-	-	-	-	2.1 KIP	5.5 KIP	-	-	-	-	-	-
			D = 4'-0"	P3	-	-	-	-	-	-	0.5 KIP	4.8 KIP	0.6 KIP	-	-1.4 KIP	-
J7	51'-8"	12-223-12 LH	-	W1	66 PLF	788 PLF	80 PLF	-	NOTE 2	-	-	-	-	-	-	-
			C = 4'-0"	P1	-	-	-	1.5 KIP	-	-	-	-	-	-	-	SEE PLAN FOR HOIST BEAM LENGTH
			-	P2	-	-	-	-	2.1 KIP	5.5 KIP	-	-	-	-	-1.4 KIP	-
			C = 4'-0"	P4	-	-	-	0.8 KIP	-	-	-	-	-	-		SEE PLAN FOR HOIST BEAM LENGTH
J8	51'-8"	12-223-12 LH	-	W1	66 PLF	1388 PLF	80 PLF	-	NOTE 2	-	-	-	-	-	-	-
			-	P2	-	-	-	-	2.1 KIP	5.5 KIP	-	-	-	-	-	-
J9	51'-8"	12-223-12 LH	-	W1	66 PLF	1388 PLF	80 PLF	-	NOTE 2	-	-	-	-	-	-	-
			-	P2	-	-	-	-	2.1 KIP	5.5 KIP	-	-	-	-	-	-
J10	51'-8"	12-223-12 LH	-	W1	66 PLF	788 PLF	80 PLF	-	NOTE 2	-	-	-	-	-	-	-
			-	P2	-	-	-	-	2.1 KIP	5.5 KIP	-	-	-	-	-	-
			D = 4'-0"	P3	-	-	-	-	-	-	0.5 KIP	4.8 KIP	0.6 KIP	-	-1.4 KIP	-

NOTES:

1. ALL LOADS SHOWN IN TABLE ARE UNFACTORED AND SHALL BE COMBINED PER THE REQUIREMENTS OF IBC 2024 AND ASCE 7-22.
2. WIND LOAD PRESSURES SHALL BE APPLIED TO THE JOIST TOP CHORD PER THE REQUIREMENTS OF ASCE 7-22 USING THE WIND DESIGN CRITERIA DEFINED ON SHEET 060S001 AND 062S602.
3. JOIST BOTTOM CHORD SHALL BE DESIGNED FOR AN ADDITIONAL POINT LOAD (DEAD LOAD) OF 100 LB TO OCCUR AT ANY LOCATION.
4. LOAD SHALL CONSIDER IMPACT LOAD FACTORS PER ASCE 7-22.

1 / 062S604 ROOF JOIST LOADING DIAGRAM
NOT TO SCALE

Registered Professional Engineer (Colo) CERTIFICATE NO. 72480 COLIN MULDERICK Date Signed: 05/07/26 HDR ENGINEERING, INC.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	C. MULDERICK	062 S604	WATER TREATMENT ROOF JOISTS LOADING TABLE AND DIAGRAM
	CADD: R. NELSON		NORTH RIM - GRAND CANYON NATIONAL PARK
	TECH. REVIEW: L. MENDOZA		
	DATE: MAY 2026		

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	73 OF 192

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GENERAL NOTES:

A. SEE SERIES 004 SHEETS FOR STANDARD DETAILS REFERENCED HEREIN.

B. SEE SHEET 004A509 FOR TYPICAL WALL TYPES TAGGED ON PLANS.

C. SEE SHEET 004A601 FOR DOOR AND FRAME SCHEDULES, ELEVATIONS AND NOTES.

D. SEE SHEET 004A602 AND SPECIFICATIONS FOR MATERIAL FINISH SCHEDULES AND NOTES.

E. SEE 062F SERIES SHEETS FOR CODE ANALYSIS, LIFE SAFETY PLANS AND FIRE EXTINGUISHER LOCATIONS.

F. PROVIDE IN-WALL BLOCKING IN WALLS FOR WALL-MOUNTED ITEMS, INCLUDING CABINETS, SHELVING, TOILET & BATH ACCESSORIES, & SIMILAR ITEMS.

G. ALL ITEMS NOTED ARE NEW UNLESS SPECIFICALLY IDENTIFIED AS EXISTING.

H. ALL DIMENSIONS ARE ACTUAL AND ARE TO FACE OF MASONRY, FACE OF METAL STUD, CENTERLINE OF COLUMN OR CENTERLINE OF BEAM UNLESS NOTED OTHERWISE. ALL DIMENSIONS ARE FOR BIDDING PURPOSES ONLY. ACTUAL FIELD DIMENSIONS SHALL BE VERIFIED PRIOR TO SUBMITTAL OF SHOP DRAWINGS, ORDERING RELATED MATERIALS AND PERFORMING DEMOLITION OR CONSTRUCTION WORK. ALL DIMENSIONS ALIGNING WITH OR RELATED TO EXISTING CONDITIONS ARE TO BE LAID OUT PRIOR TO COMMENCING WORK AND VERIFIED.

I. VERIFY MECHANICAL AND ELECTRICAL DRAWINGS FOR SCOPE AND INTERFACE. CONTRACTOR SHALL COORDINATE LOCATION FOR ALL MECHANICAL AND ELECTRICAL ITEMS WITH GENERAL CONSTRUCTION. REVIEW ANY DISCREPANCIES WITH THE CONTRACTING OFFICER PRIOR TO INSTALLATION AND/OR FABRICATION.

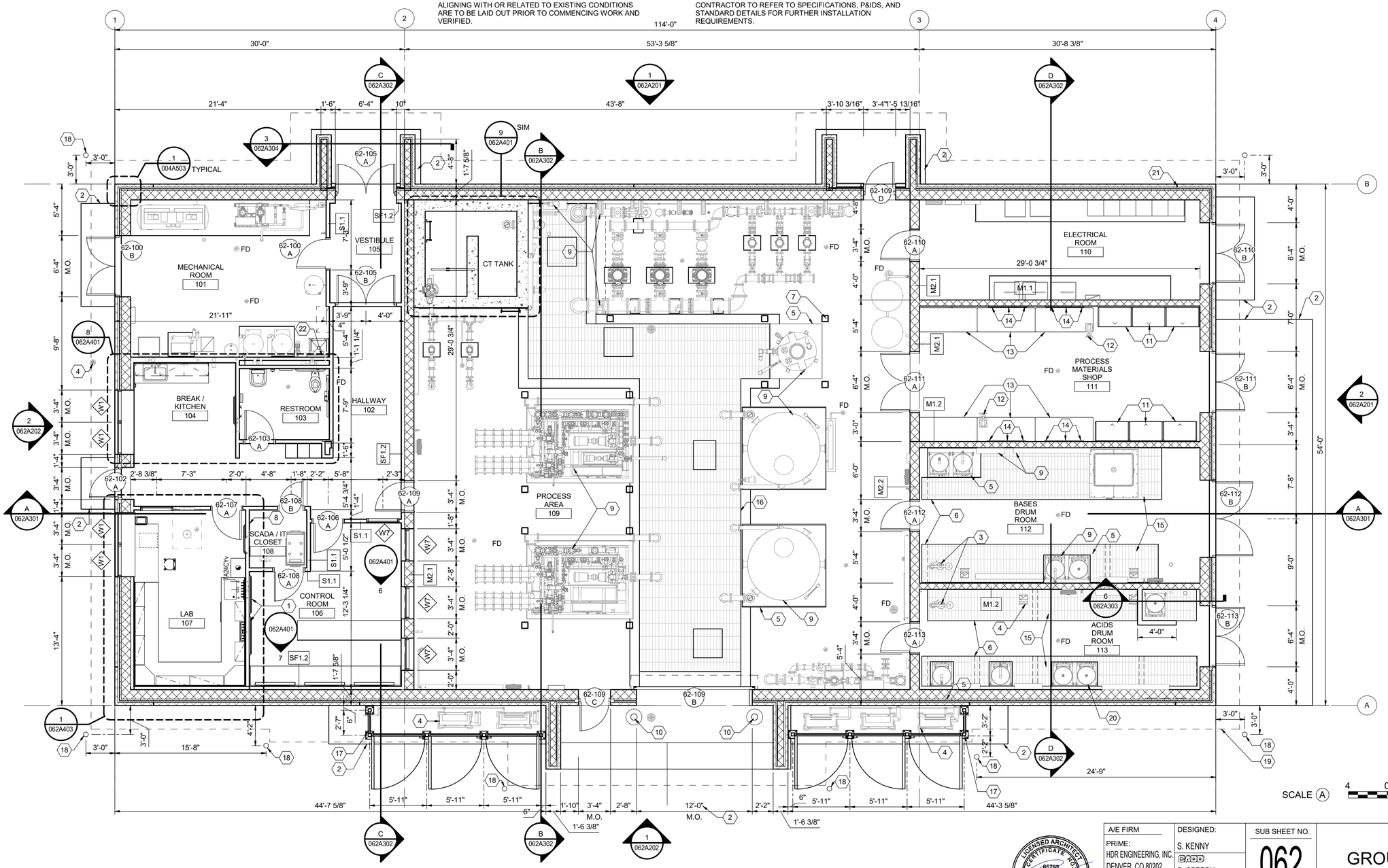
J. CONTRACTOR SHALL FIELD ROUTE PIPING / CONDUIT AS NECESSARY TO COORDINATE WITH EQUIPMENT LOCATIONS. SEE P&ID DRAWINGS FOR ALL PROCESSING PIPING FITTINGS, VALVES, AND OTHER PIPING APPURTENANCES AND LOCATIONS.

K. PLUMBING PIPES AND PROCESS PIPES SHALL NOT BE INSTALLED ABOVE ELECTRICAL PANELS OR CONTROL PANELS.

L. DRAWINGS ARE INTENDED TO DEPICT DESIGN INTENT; CONTRACTOR TO REFER TO SPECIFICATIONS, P&IDS, AND STANDARD DETAILS FOR FURTHER INSTALLATION REQUIREMENTS.

M. LOCATE INTERIOR CONTROL JOINTS WHERE SHOWN ON THE PLANS AND STANDARDS IDENTIFIED IN SPECIFICATIONS

N. ALL DIMENSIONS ALIGNING WITH OR RELATED TO EXISTING CONDITIONS ARE TO BE LAID OUT PRIOR TO COMMENCING WORK AND VERIFIED BY CONTRACTOR.



1 062A101 1/62A101 GROUND LEVEL FLOOR PLAN

KEYNOTES (#)

1 MARKERBOARD

2 CONCRETE STOOP; SEE STRUCTURAL. PROVIDE CHEMICAL FLOOR SEALER CS-3 ON ALL EXTERIOR STRUCTURAL STOOPS AND SLABS

3 EMERGENCY SHOWER/EYEWASH; SEE MECHANICAL

4 MECHANICAL EQUIPMENT; SEE MECHANICAL

5 CONCRETE EQUIPMENT PAD; SEE STRUCTURAL

6 FRP GRATING OVER CHEMICAL CONTAINMENT PIT

7 STEEL FRAMING; SEE STRUCTURAL

8 STATIC DISSIPATIVE FLOORING GROUNDING CONNECTION; SEE ELECTRICAL

9 PROCESS EQUIPMENT; SEE PROCESS

10 BOLLARD

11 METAL SHELVING

12 BENCH VISE

13 METAL WORK BENCH

14 PEGBOARD

15 COAT FLOOR AND WALLS OF CHEMICAL CONTAINMENT PIT WITH HPIC

16 METAL GRATING

17 EXTERIOR EQUIPMENT SCREEN; CONTRACTOR TO COORDINATE LOUVER UNRESTRICTED FREE AREA WITH MECHANICAL EQUIPMENT MANUFACTURER REQUIREMENTS

18 BOLLARD; SEE CIVIL DETAIL IN VOL. 0

19 LINE OF ROOF ABOVE

20 CHEMICAL DRUM; SEE PROCESS TYPICAL

21 MCJ

22 MOP RACK TA10A

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.05.01 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. KENNY	062 A101	WATER TREATMENT GROUND LEVEL PLAN
	DRAWN: D. SPERRY		
	TECH. REVIEW: R. MCKINLEY		
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	74 OF 192

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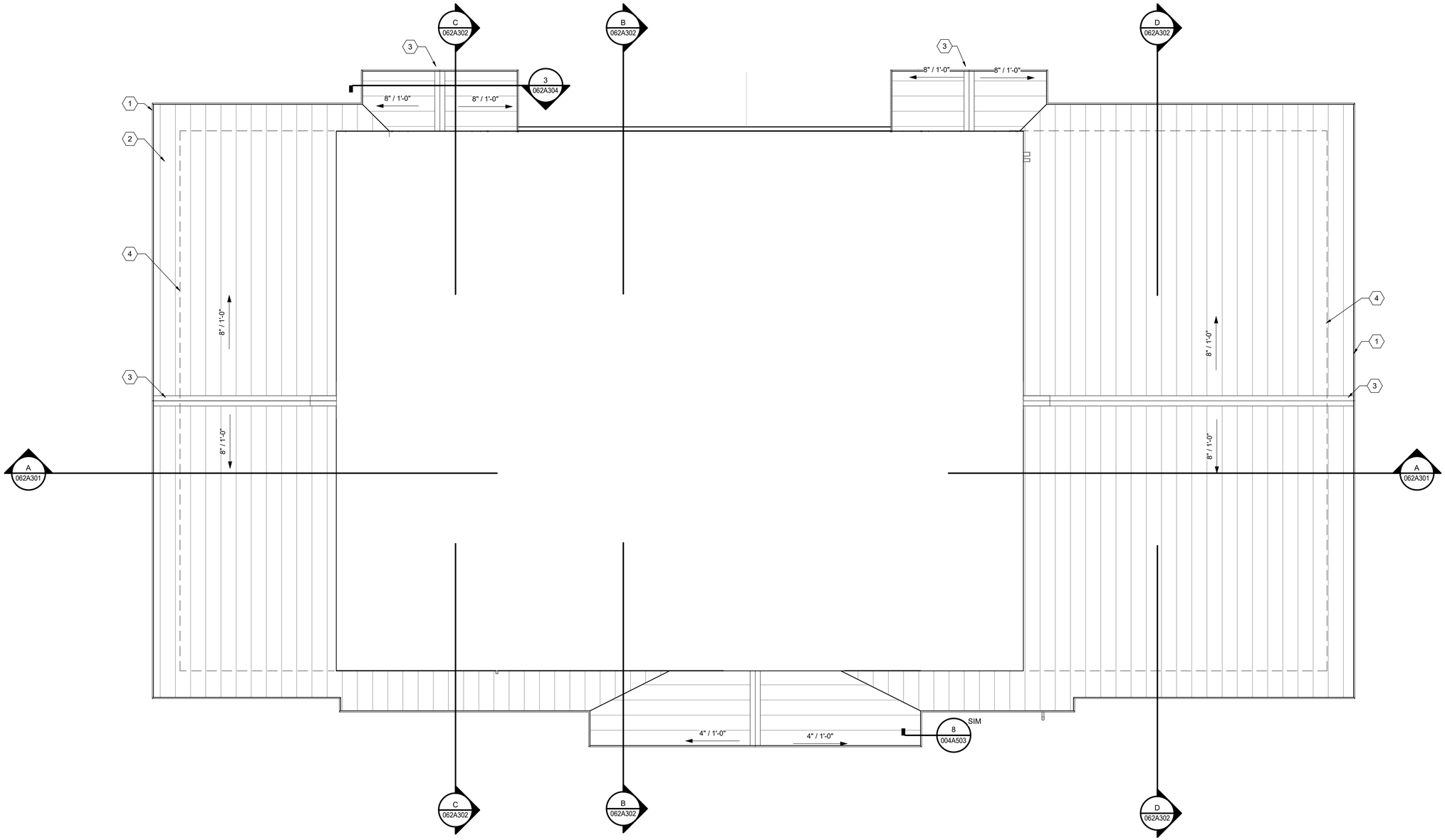

1 062A102 ROOF PLAN - LOWER ROOF

SCALE A 4 0 4 8 12 3/16" = 1'-0"

SCALE OF FEET

GENERAL ROOF PLAN NOTES:

A. SEE SERIES 004 SHEETS FOR STANDARD DETAILS REFERENCED HEREIN.

B. PROVIDE ICE AND WATER SHEILD PROTECTION AT ALL EAVES, VALLEYS, PENETRATIONS, AND ROOF EDGES TO 36" MIN AT EA SIDE.

C. COORDINATE ALL ROOF PENETRATIONS THAT REQUIRE FLASHING; SEE MECHANICAL.

D. PROVIDE ALL FLASHING, JOINTS, DRIPS, TRIM, CAPS, AND COPINGS REQUIRED AS A COMPONENT OF THE ROOF ASSEMBLY.

E. REFER TO ENGINEERING DISCIPLINE DOCUMENTS FOR ADDITIONAL EQUIPMENT, PIPE, CONDUIT, VALVES, ETC...

F. DRAWINGS ARE INTENDED TO DEPICT DESIGN INTENT; CONTRACTOR TO REFER TO SPECIFICATIONS, P&IDS, AND STANDARD DETAILS FOR FURTHER INSTALLATION REQUIREMENTS.

G. INSTALL ALL ROOFING ASSEMBLIES IN ACCORDANCE WITH PROJECT SPECIFICATIONS, DETAILS, AND MANUFACTURER'S STANDARD INSTALLATION REQUIREMENTS.

H. SEE ELECTRICAL FOR LIGHTING PROTECTION SYSTEMS WHERE REQUIRED.

I. UPPER ROOF SHOWN FOR CLARITY, SEE 062A104.

J. PROVIDE AND INSTALL LIGHTNING PROTECTION SYSTEM FOR NEW WWTP. PROVIDE CLASS 1 UL MASTER LABEL LIGHTNING PROTECTION SYSTEM IN ACCORDANCE WITH NFPA-780 AND THE LIGHTNING PROTECTION SPECIFICATION(26 41 13).

K. INSTALL CLASS -A ROOF ASSEMBLY PER 07 61 13.

KEYNOTES (#)

1 3/4" EXT. PLYWOOD FASCIA, COVER FRONT FACE WITH PRE-FINISHED STEEL TWO-PIECE WRAP

2 STANDING SEAM METAL ROOFING

3 PREFINISHED METAL ROOF RIDGE CAP

4 LINE OF WALL BELOW

LICENSED ARCHITECT CERTIFICATE NO. 85788 RICHARD MCKINLEY DATE SIGNED 04/24/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY DRAWN: D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A102	TITLE OF SHEET WATER TREATMENT ROOF PLAN - LOWER ROOF NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 75 OF 192

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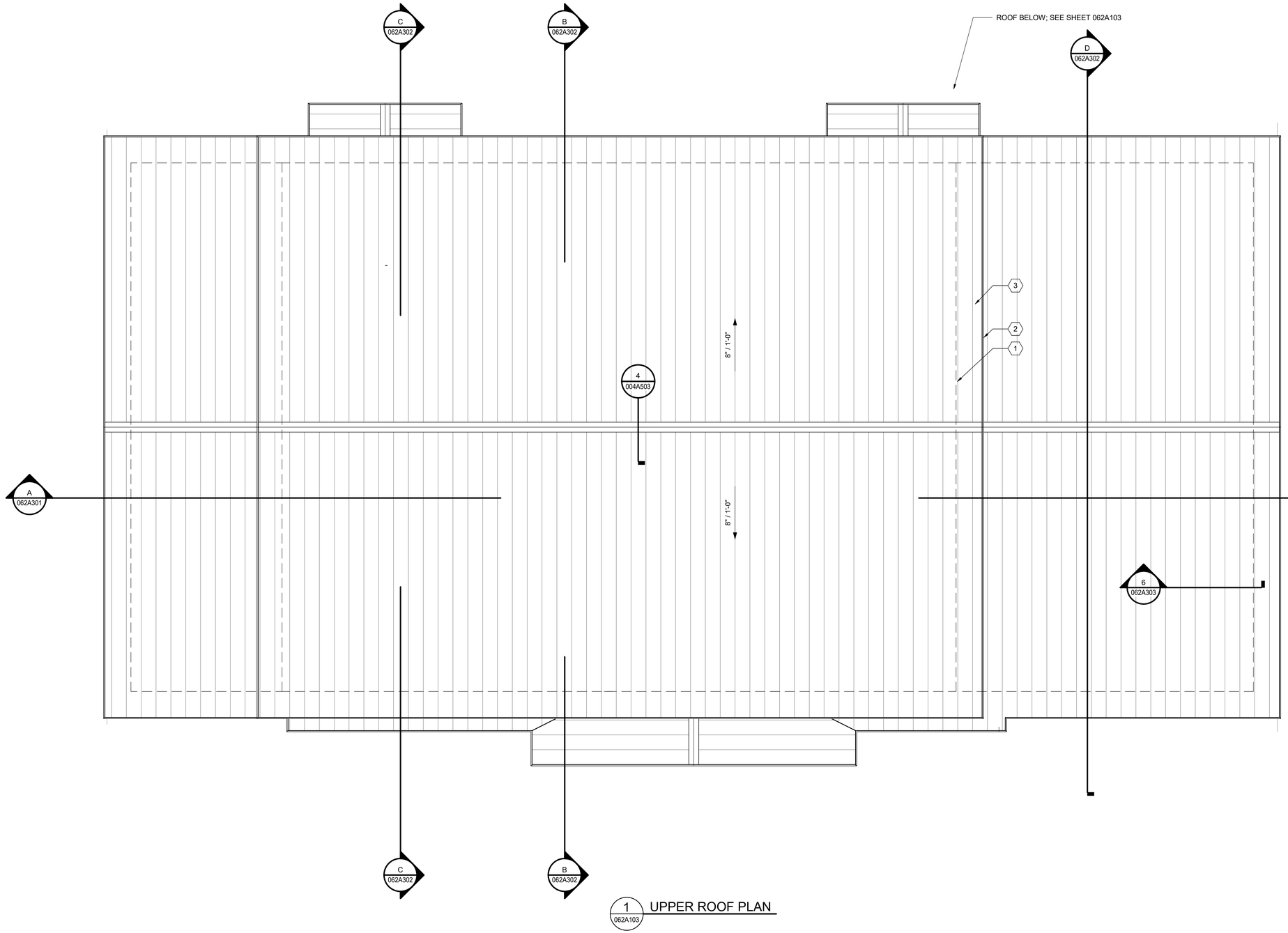

1 062A103 UPPER ROOF PLAN

GENERAL ROOF PLAN NOTES:

A. SEE SERIES 004 SHEETS FOR STANDARD DETAILS REFERENCED HEREIN.

B. PROVIDE ICE AND WATER SHEILD PROTECTION AT ALL EAVES, VALLEYS, PENETRATIONS, AND ROOF EDGES TO 36" MIN AT EA SIDE.

C. COORDINATE ALL ROOF PENETRATIONS THAT REQUIRE FLASHING; SEE MECHANICAL.

D. PROVIDE ALL FLASHING, JOINTS, DRIPS, TRIM, CAPS, AND COPINGS REQUIRED AS A COMPONENT OF THE ROOF ASSEMBLY.

E. REFER TO ENGINEERING DISCIPLINE DOCUMENTS FOR ADDITIONAL EQUIPMENT, PIPE, CONDUIT, VALVES, ETC...

F. DRAWINGS ARE INTENDED TO DEPICT DESIGN INTENT; CONTRACTOR TO REFER TO SPECIFICATIONS, P&IDS, AND STANDARD DETAILS FOR FURTHER INSTALLATION REQUIREMENTS.

G. INSTALL ALL ROOFING ASSEMBLIES IN ACCORDANCE WITH PROJECT SPECIFICATIONS, DETAILS, AND MANUFACTURER'S STANDARD INSTALLATION REQUIREMENTS.

H. SEE ELECTRICAL FOR LIGHTING PROTECTION SYSTEMS WHERE REQUIRED.

I. UPPER ROOF SHOWN FOR CLARITY, SEE 062A104.

J. PROVIDE AND INSTALL LIGHTNING PROTECTION SYSTEM FOR NEW WWTP. PROVIDE CLASS 1 UL MASTER LABEL LIGHTNING PROTECTION SYSTEM IN ACCORDANCE WITH NFPA-780 AND THE LIGHTNING PROTECTION SPECIFICATION(26 41 13).

K. INSTALL CLASS -A ROOF ASSEMBLY PER 07 61 13.

KEYNOTES (#)

1 LINE OF WALL BELOW

2 3/4" EXT. PLYWOOD FASCIA, COVER FRONT FACE WITH PRE-FINISHED STEEL TWO-PIECE WRAP

3 STANDING SEAM METAL ROOFING

SCALE (A) 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85788 RICHARD MCKINLEY DATE SIGNED 4/19/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY CADD D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A103	TITLE OF SHEET WATER TREATMENT ROOF PLAN - UPPER ROOF NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 76 OF 192

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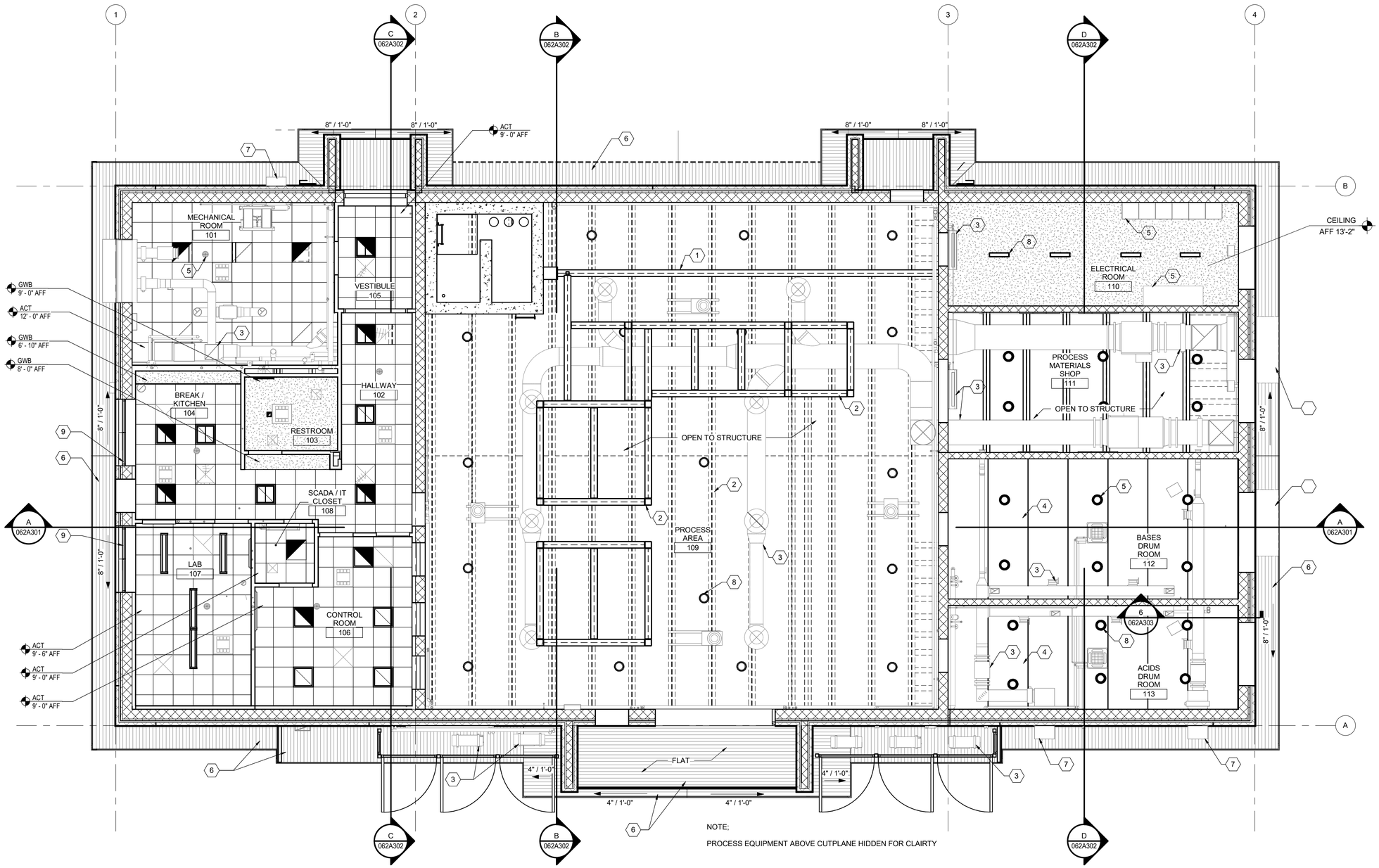

NOTE;

PROCESS EQUIPMENT ABOVE CUTPLANE HIDDEN FOR CLAIRTY

REFLECTED CEILING PLAN LEGEND

- ACOUSTICAL CEILING TILE
- GYPSUM BOARD CEILING
- METAL SOFFIT PANELING
- LIGHT FIXTURE
- EMERGENCY FIXTURE
- EXIT SIGN
- WALL MOUNTED LIGHT FIXTURE
- SUPPLY AIR DIFFUSER/GRILLE
- RETURN AIR GRILLE
- EXHAUST FAN GRILLE
- CEILING MOUNTED HEATER
- X'-X" CEILING HEIGHT ABOVE FINISH FLOOR

GENERAL NOTES:

A. REFER TO OTHER DISCIPLINES.

KEYNOTES (#)

ALUMINUM LOUVER W/ GALVANIZED SNOW HOOD; SEE DETAIL 11/004A506

1 MONORAIL, SEE STRUCTURAL

2 STEEL FRAMING; SEE STRUCTURAL

3 MECHANICAL EQUIPMENT; SEE MECHANICAL

4 HOLLOWCORE PLANK CEILING STRUCTURE; SEE STRUCTURAL

5 ELECTRICAL EQUIPMENT; SEE ELECTRICAL

6 1" PRE-FINISHED CONCEALED FASTENER STEEL SOFFIT PANELS WITH END TRIM

7 ALUMINUM LOUVER W/ GALVANIZED SNOW HOOD; SEE DETAIL 11/004A506

8 LIGHT FIXTURE; SEE ELECTRICAL

9 ROLLER WINDOW SHADE

1 / 062A104 REFLECTED CEILING PLAN

SCALE A 4 0 4 8 12 3/16" = 1'-0"

SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85783 RICHARD MCKINLEY DATE SIGNED 4/19/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY DRAWN: D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A104	TITLE OF SHEET WATER TREATMENT REFLECTED CEILING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 77 OF 192

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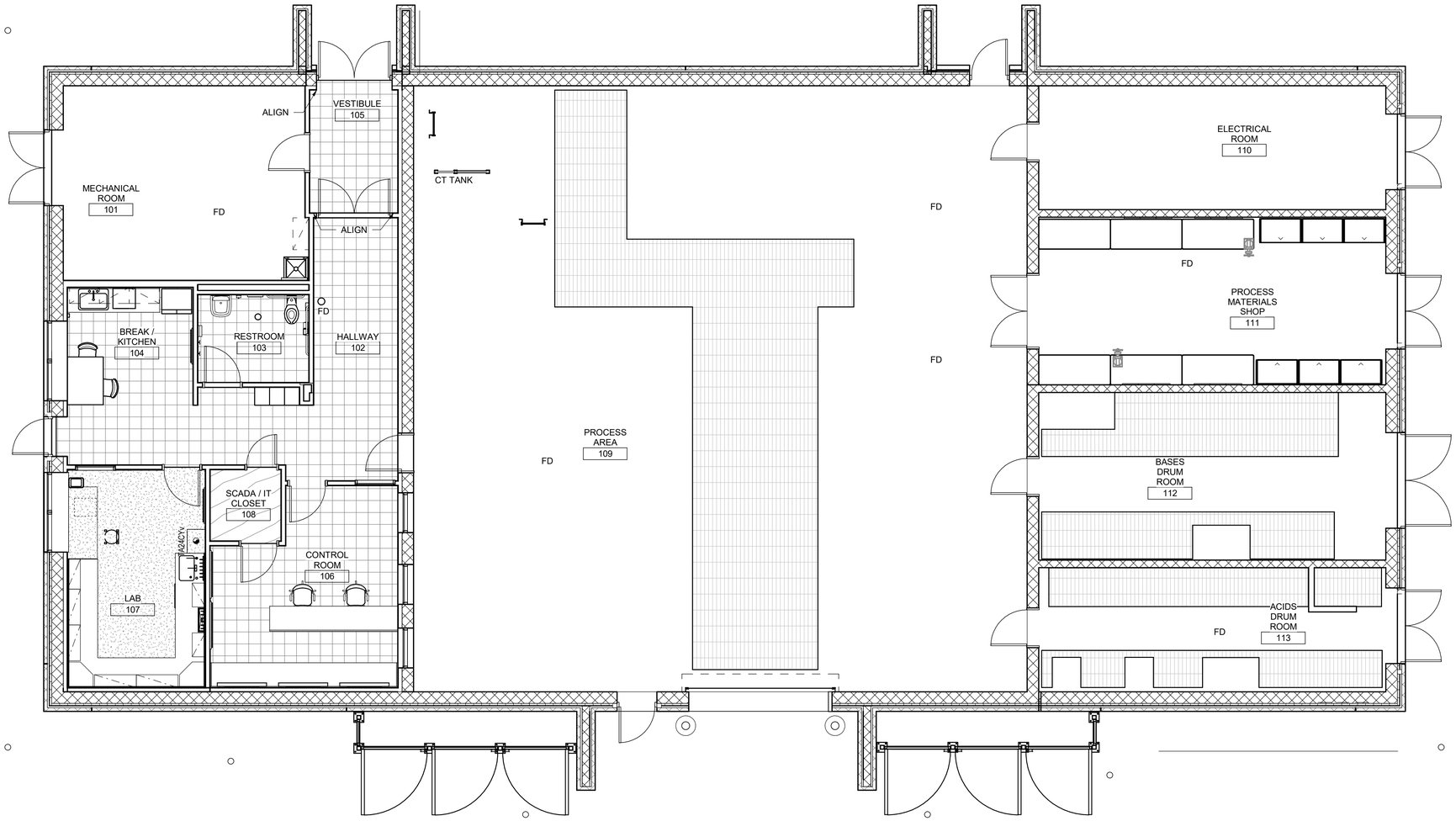

1 062A105 INTERIOR FINISH GROUND LEVEL FLOOR PLAN

SCALE (A)

NOTE: REFER TO SHEET 004A602 FOR ROOM FINISH SCHEDULE.

SCALE (B) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

SCALE (A) 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85783 RICHARD MCKINLEY DATE SIGNED ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY CADD D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A105	TITLE OF SHEET WATER TREATMENT INTERIOR FINISH PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 78 OF 192

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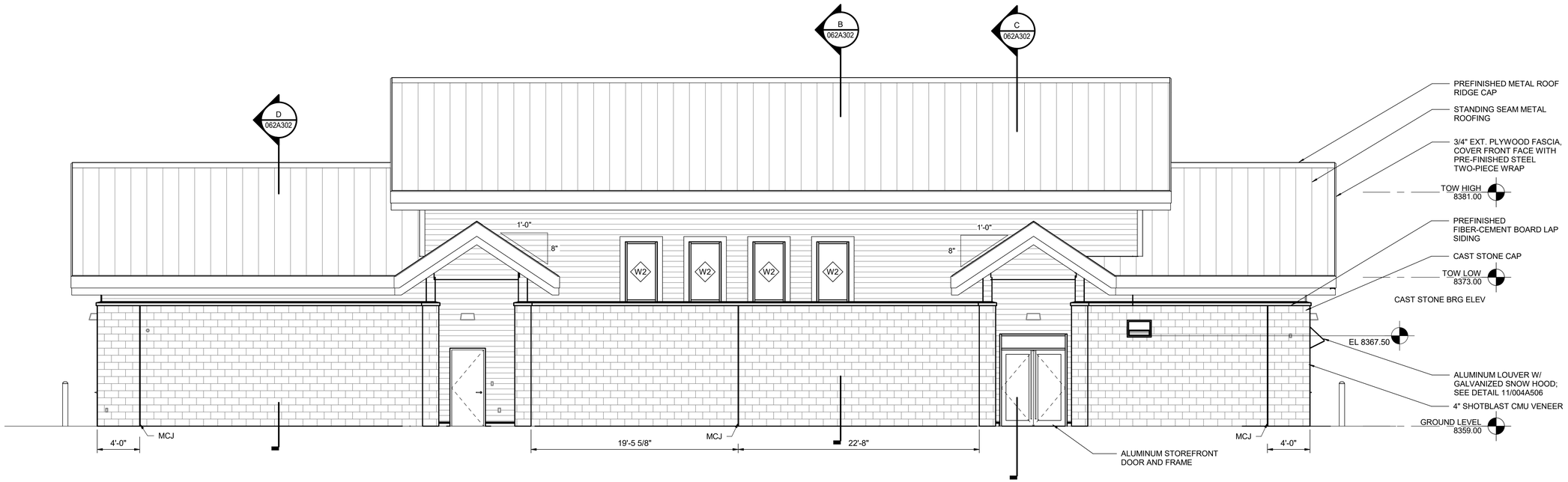
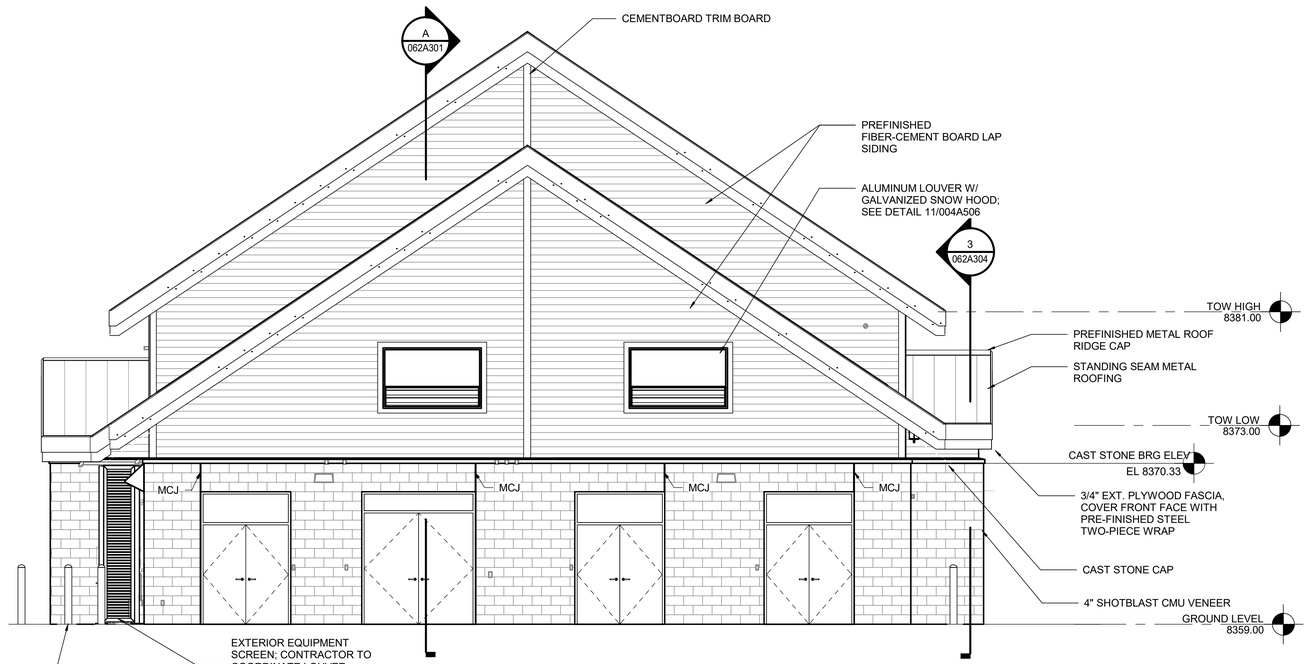


1 062A201 NORTH EXTERIOR ELEVATION



2 062A201 EAST EXTERIOR ELEVATION

SCALE A 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY STATE REGISTERED ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY CADD D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A201	TITLE OF SHEET WATER TREATMENT EXTERIOR ELEVATIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 79 OF 192

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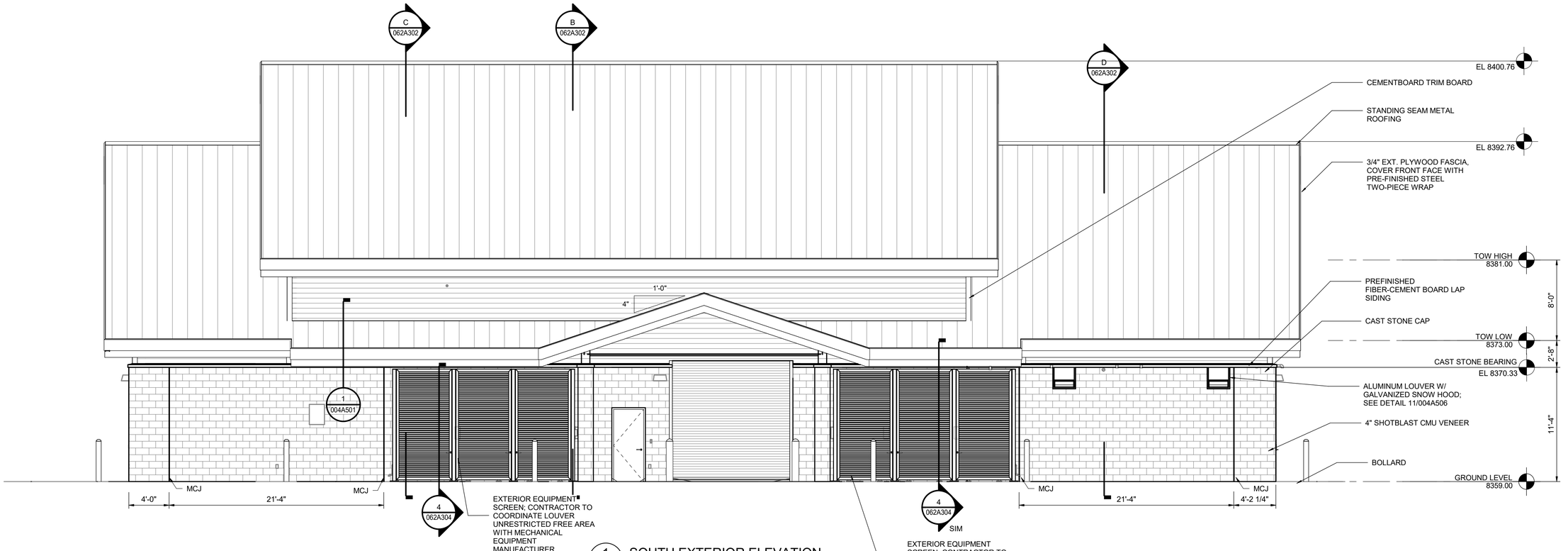
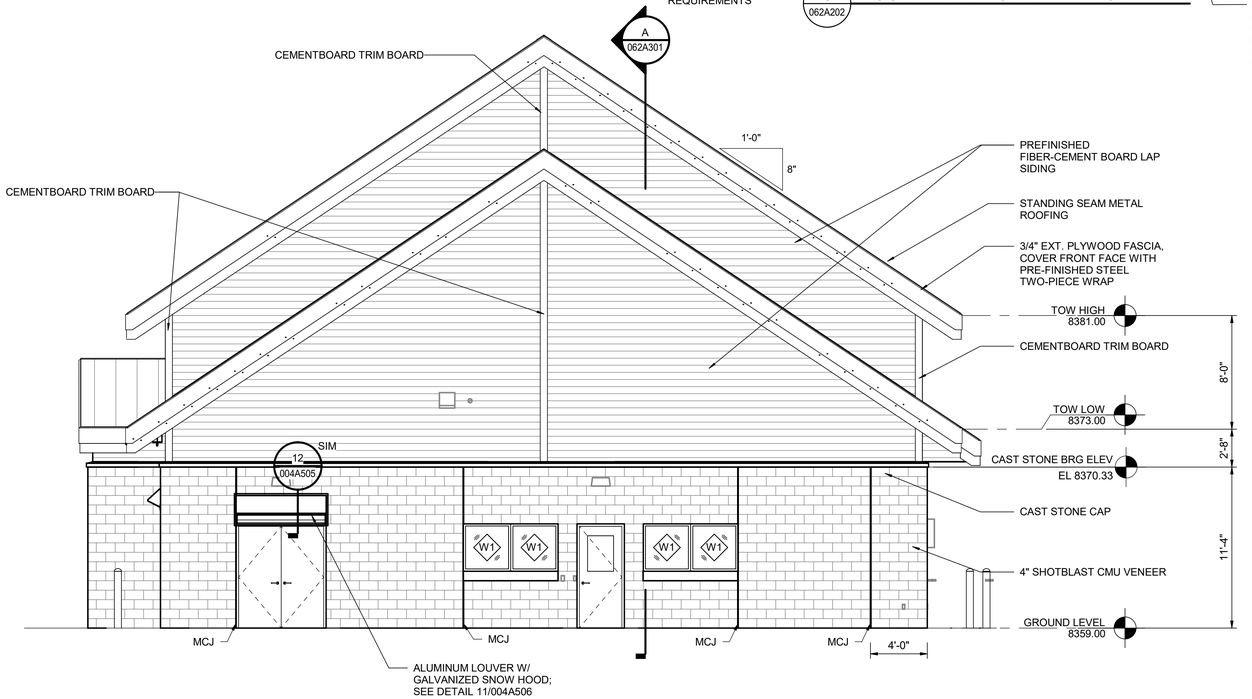


1 SOUTH EXTERIOR ELEVATION
062A202



2 WEST EXTERIOR ELEVATION
062A202

SCALE A 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY DRAWN: D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A202	TITLE OF SHEET WATER TREATMENT EXTERIOR ELEVATIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 80 OF 192

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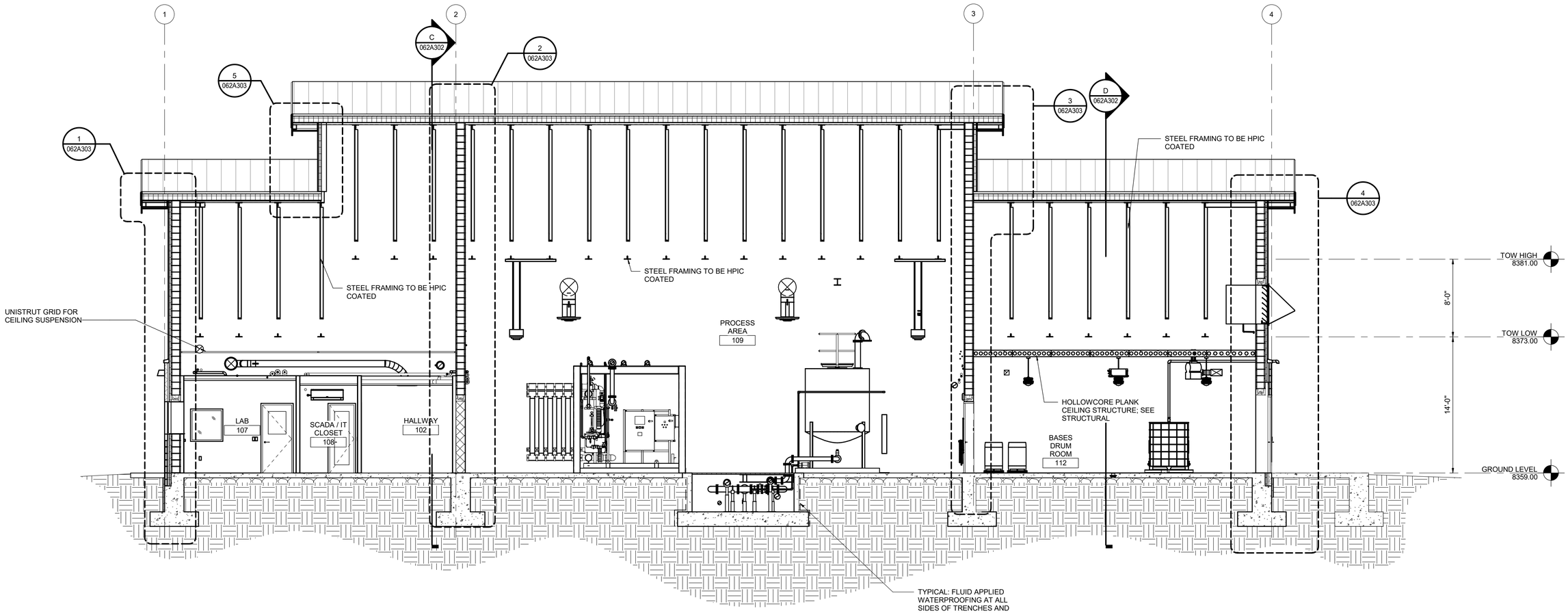

A 062A101 BUILDING SECTION - A

SCALE A 4 0 4 8 12 3/16" = 1'-0"

SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY CADD D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A301	TITLE OF SHEET WATER TREATMENT BUILDING SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 81 OF 192

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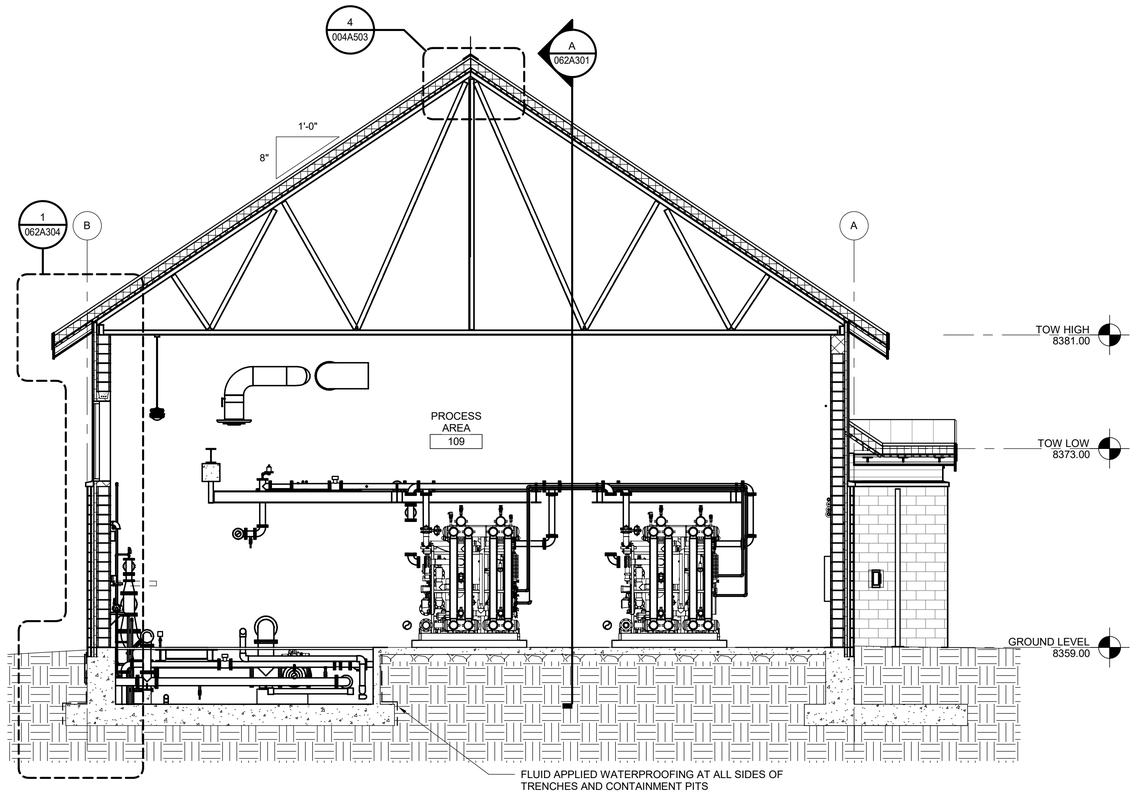
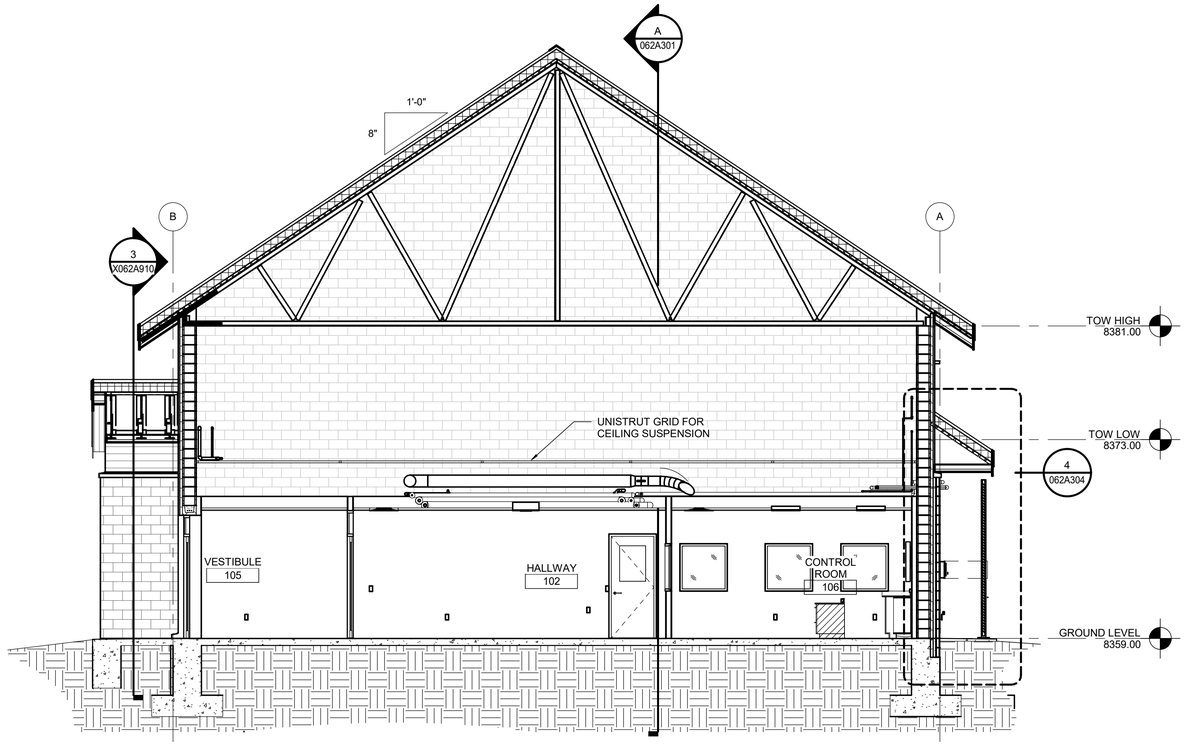
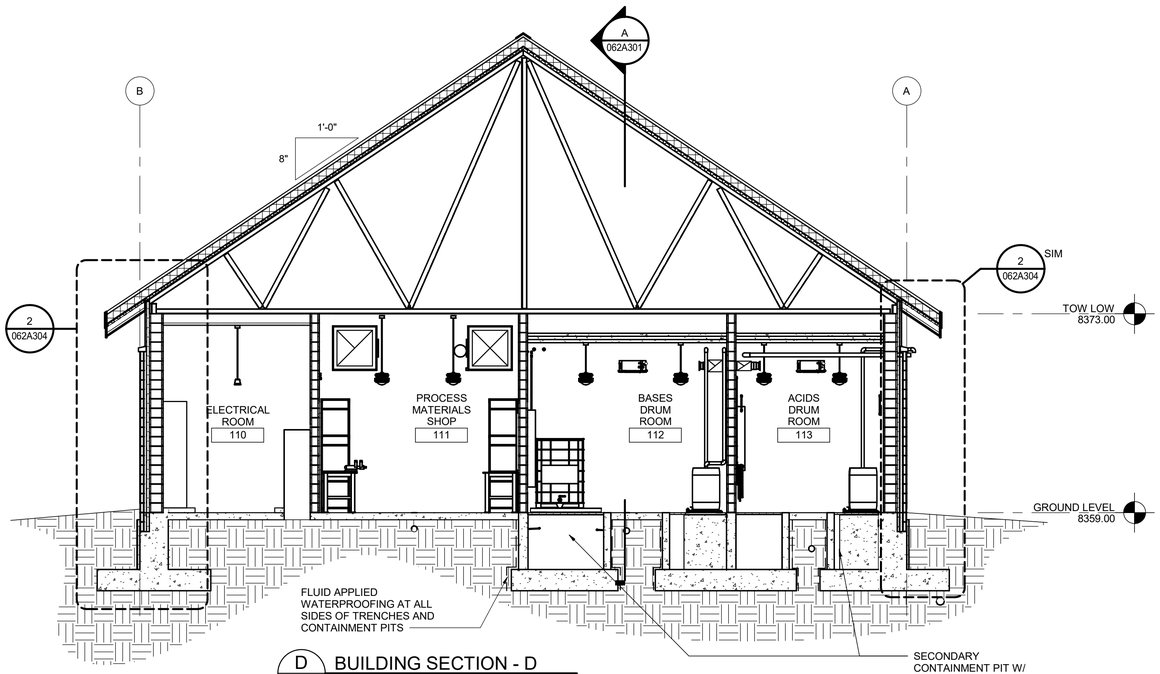


B 062A101 BUILDING SECTION - B



C 062A101 BUILDING SECTION - C



D 062A101 BUILDING SECTION - D

SCALE A 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.04.01 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY DRAWN: D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A302	TITLE OF SHEET WATER TREATMENT BUILDING SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 82 OF 192

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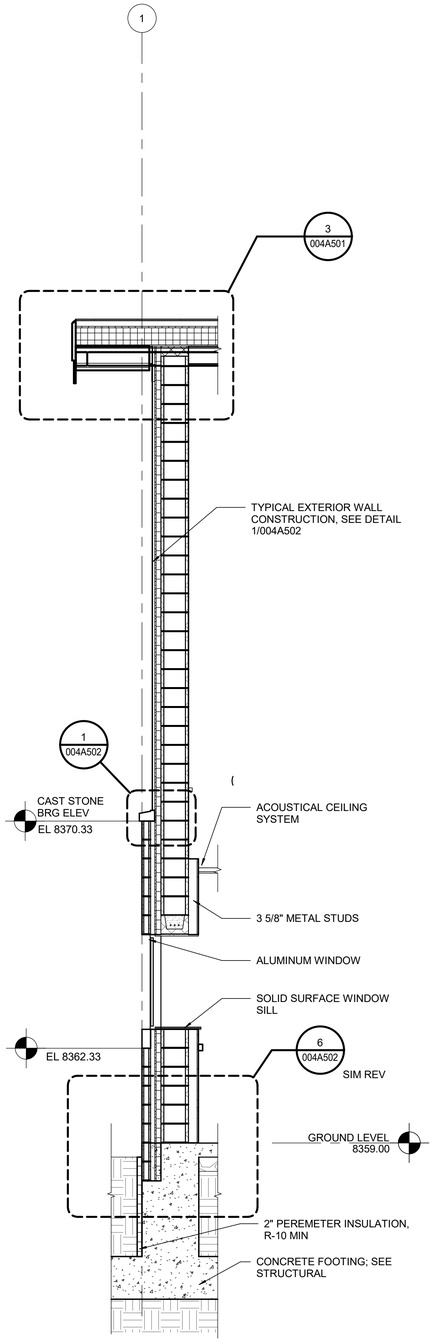
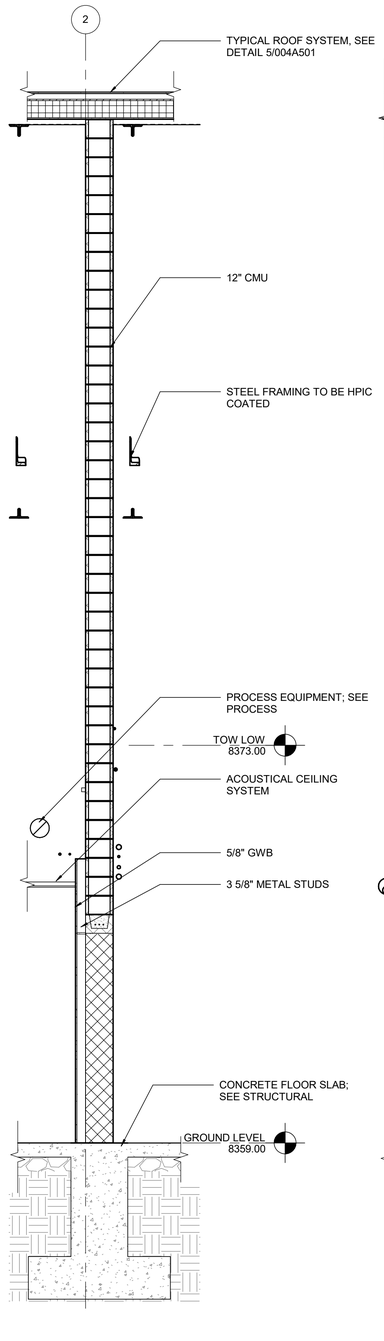


1 062A303 WALL SECTION
SCALE A



2 062A303 WALL SECTION
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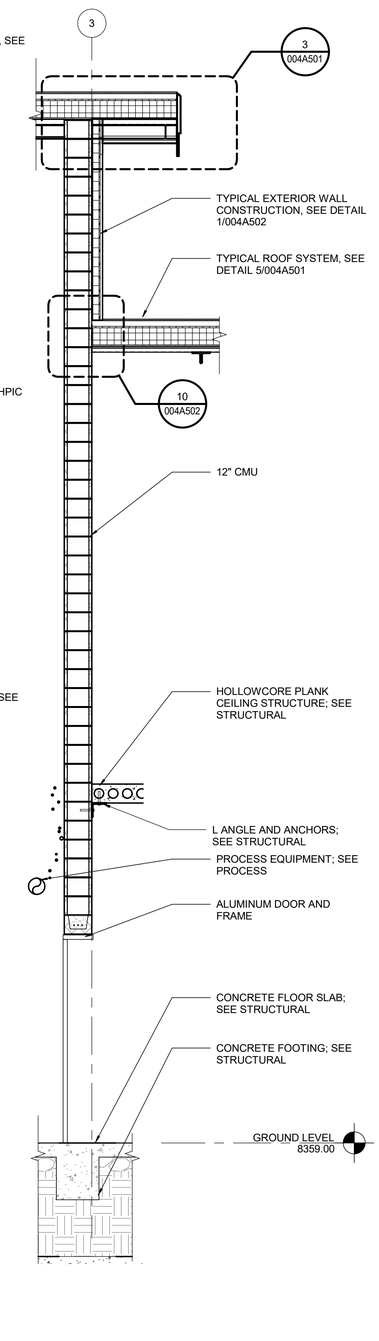

3 062A303 WALL SECTION
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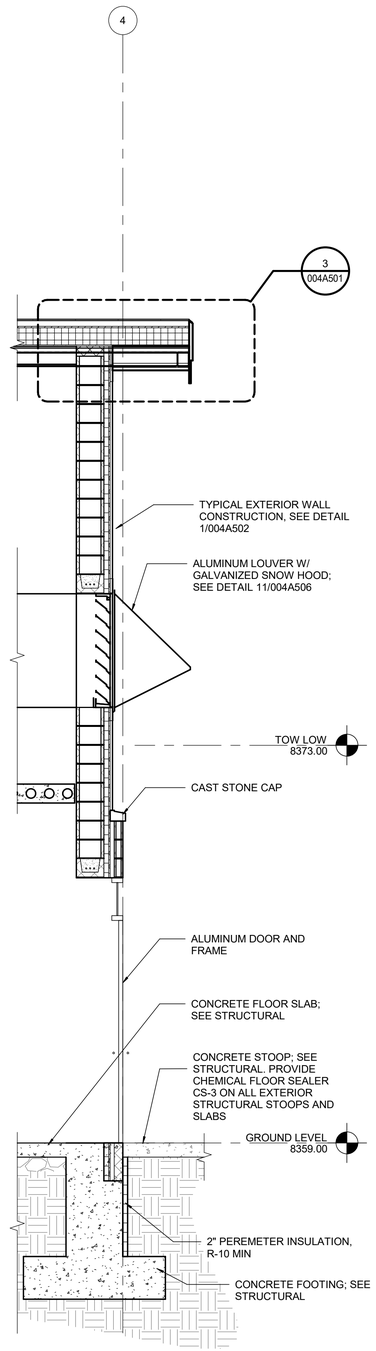
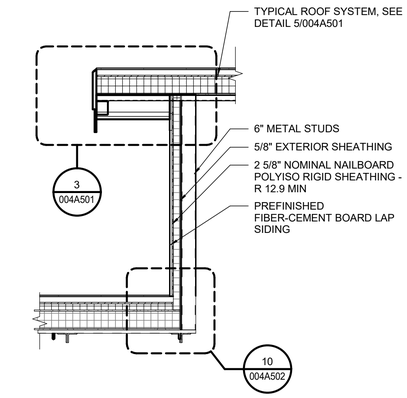
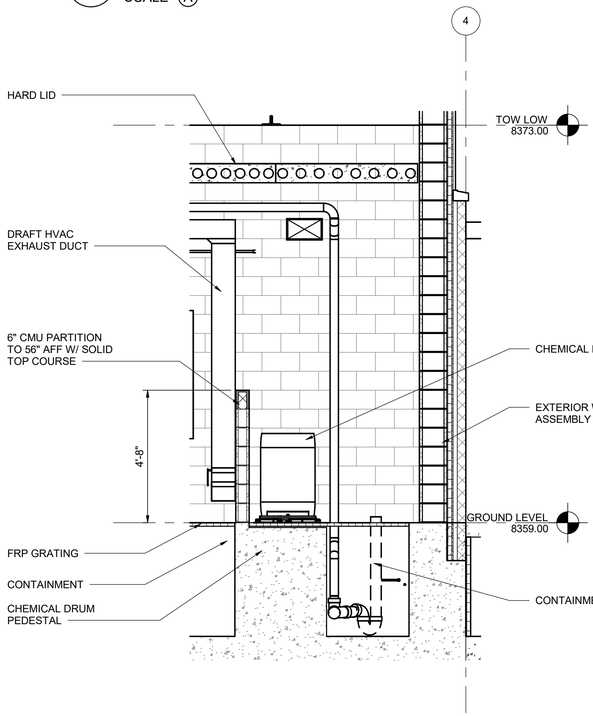


4 062A303 WALL SECTION
SCALE A



5 062A303 UPPER WALL SECTION
SCALE A



6 062A101 CHEMICAL SEPARATION WALL
SCALE A

SCALE A 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT
CERTIFICATE NO.
85785
RICHARD
MCKINLEY
DATE SIGNED 2026.05.01
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY DRAFTED: D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A303	TITLE OF SHEET WATER TREATMENT WALL SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	83 OF 192

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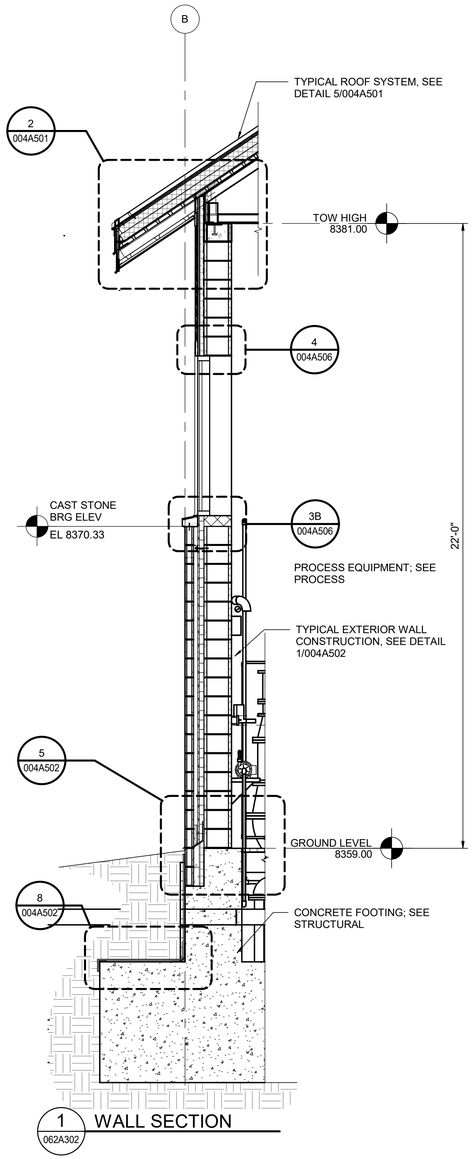

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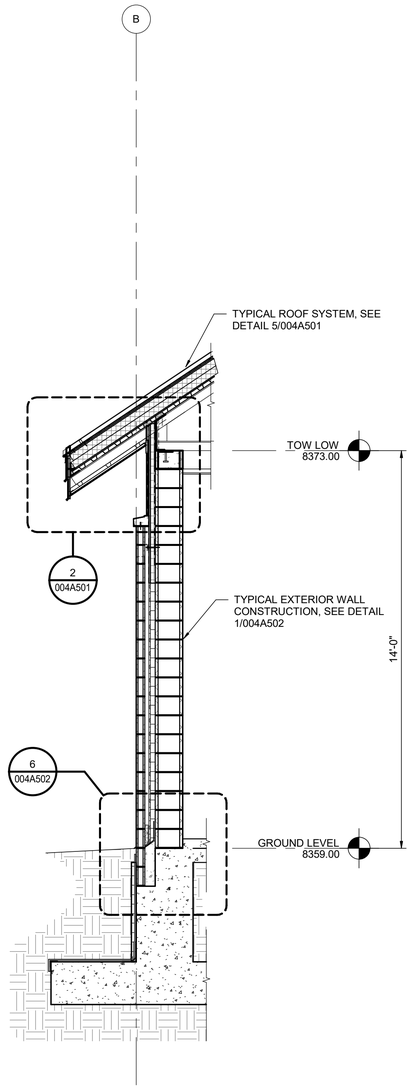

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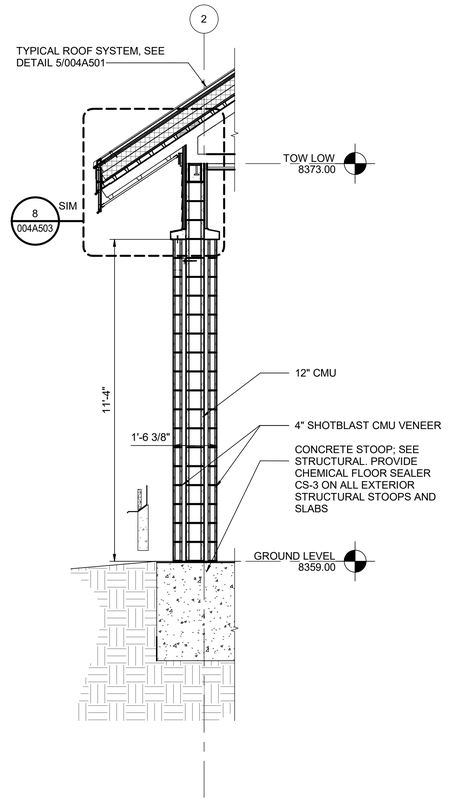

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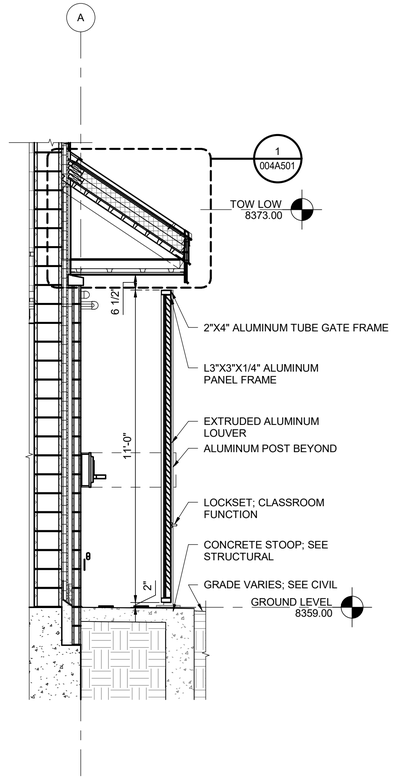

4 062A202 WALL SECTION

SCALE A 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85783 RICHARD MCKINLEY DATE SIGNED ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY CADD D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A304	TITLE OF SHEET WATER TREATMENT WALL SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 84 OF 192

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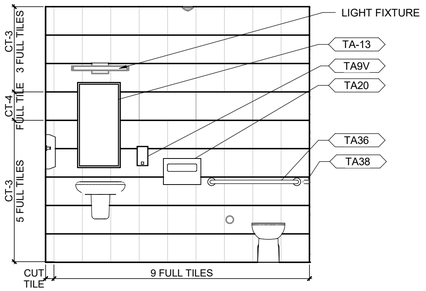
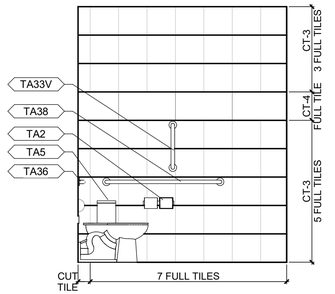
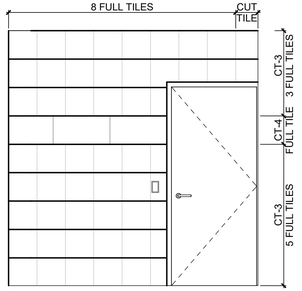


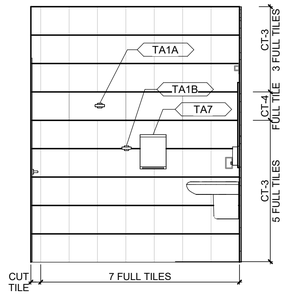
1 062A401 RESTROOM NORTH



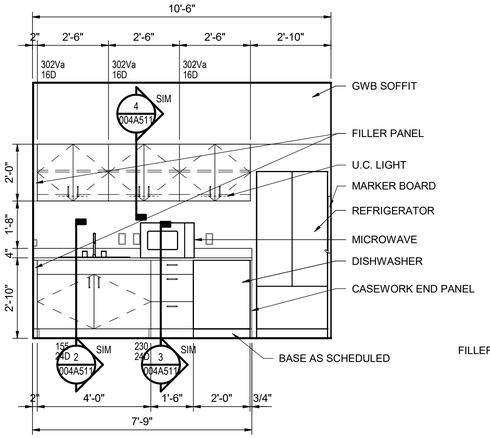
2 062A401 RESTROOM EAST



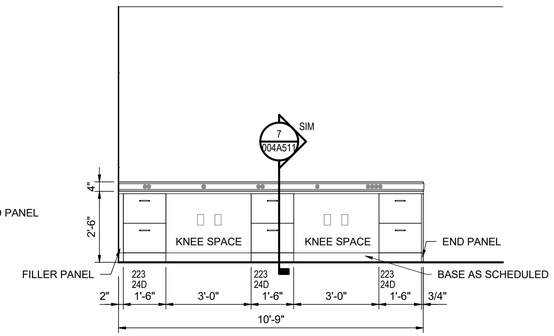
3 062A401 RESTROOM SOUTH



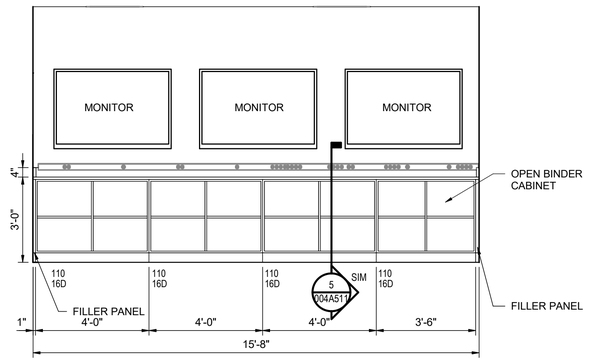
4 062A401 RESTROOM WEST



5 062A401 BREAKROOM 104



6 062A401 CONTROL ROOM 106 - ISLAND



7 062A401 CONTROL ROOM 106 - SOUTH

GENERAL NOTES

A. NUMBERS REFERENCED AT INTERIOR CASEWORK REFERENCES NORTH AMERICAN ARCHITECTURAL WOODWORK STANDARDS (NAAWS) CABINET DESIGN SERIES.

KEYNOTES (#)

1 METAL LOCKER

2 REFRIGERATOR

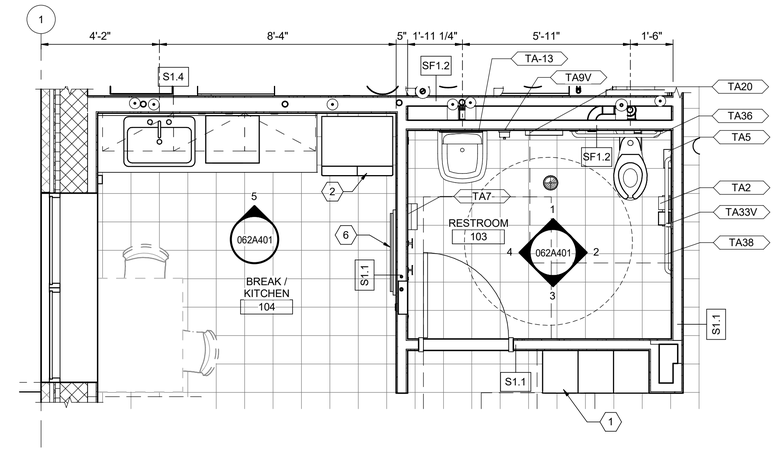
3 ALUMINUM LADDER BELOW WITH LADDER SAFETY EXTENTION POST AND LADDER SAFETY SYSTEM

4 RAILING SAFETY GATE

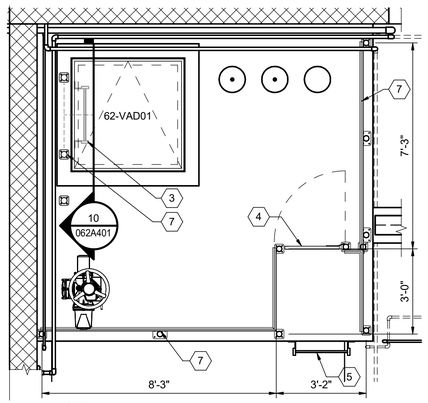
5 ALUMINUM LADDER WITH LADDER SAFETY EXTENTION POST AND LADDER SAFETY SYSTEM

6 MARKERBOARD

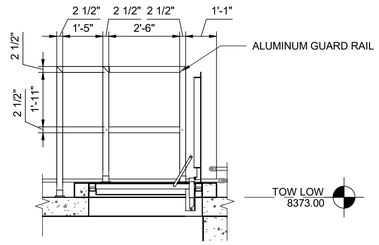
7 ALUMINUM GUARDRAIL



8 062A101 BREAKROOM & RESTROOM ENLARGED PLAN



9 062A401 CT TANK ENLARGED PLAN



10 062A401 CT TANK RAIL ELEVATION

SCALE (A) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY DRAWN: D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A401	TITLE OF SHEET WATER TREATMENT ENLARGED PLANS AND INTERIOR ELEVATIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 85 OF 192

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 4/19/26 ARIZONA U.S.A.

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_062_NWTP_A.rvt
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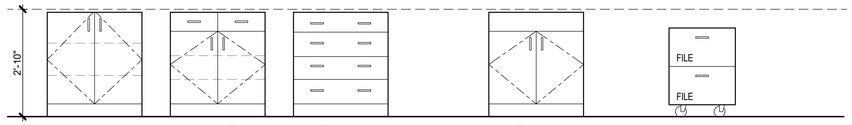
LABORATORY GENERAL NOTES

1. REFER TO ARCHITECTURAL FLOOR PLANS FOR ROOM DIMENSIONS, CONSTRUCTION TYPES, CONTROL AREAS, RATED WALLS AND FINISHES.
2. LOCATE/INSTALL BLOCKING/BACKING AT WALLS WHERE CASEWORK, SERVICE FIXTURES, SERVICE PANELS, CFCI OR OFCI LABORATORY EQUIPMENT IS PLACED/MOUNTED, PER SPECIFICATION SECTION 09 22 16.
3. LABORATORY FURNISHINGS CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS PRIOR TO FABRICATION.
4. LABORATORY CASEWORK SHALL BE PAINTED METAL, UNLESS NOTED OTHERWISE.
5. LABORATORY WORKSURFACES SHALL BE 34" AFF, UNLESS NOTED OTHERWISE.
6. LABORATORY WORKSURFACES SHALL BE EPOXY RESIN, UNLESS NOTED OTHERWISE.
7. OVERALL LENGTH OF TOPS SHALL BE DETERMINED BY MODULAR CASEWORK WIDTH AND DIMENSIONS AS INDICATED ON PLANS. SUCH LENGTHS SHALL REMAIN CONSTANT REGARDLESS OF SUCCESSFUL BIDDER'S STANDARDS. TOPS TYPICALLY SHALL OVERHANG 1" AT EACH END AND 1" FROM FRONT OF BASE CABINET. WHEN OVERALL DIMENSIONS ARE GIVEN, 1" OVERHANG IS INCLUDED. WHEN BASE CABINETS ARE LOCATED DIMENSIONALLY ON LABORATORY FURNISHINGS FLOOR PLANS, DIMENSIONS ARE TO FACE OF BASE CABINET, UNLESS NOTED OTHERWISE.
8. PROVIDE BACKSPLASHES AT ALL COUNTER EDGES ABUTTING WALLS, COLUMN FURRING, TALL STORAGE CABINETS, FUME HOODS AND TOP PENETRATIONS.
9. ALL EXPOSED BENCHTOP EDGES, BACKS OF SPLASHES, SHALL BE CAULKED, FILLED, SEALED AND FINISHED.
10. FREESTANDING TALL STORAGE CABINETS AND FREESTANDING HAZARDOUS MATERIALS CABINETS SHALL BE SEISMICALLY RESTRAINED; REFER TO SPECIFICATION SECTION 12 35 53.
11. FOR LABORATORY SERVICE FIXTURES, SERVICE TYPES, LOCATION AND ORDER, REFER TO THE LABORATORY FURNISHINGS DRAWINGS AND SPECIFICATIONS. SERVICE FIXTURES SHALL BE FURNISHED AND DELIVERED BY THE LABORATORY FURNISHINGS CONTRACTOR AND INSTALLED BY THE PLUMBING OR MECHANICAL PIPING CONTRACTOR. LABORATORY SINKS SHALL BE FURNISHED AND INSTALLED BY THE LABORATORY FURNISHINGS CONTRACTOR.
12. ELECTRICAL CONTRACTOR SHALL FURNISH AND INSTALL ALL ELECTRICAL/DATA RACEWAYS. FOR INSTALLATION, SEE ELECTRICAL AND LABORATORY FURNISHINGS DRAWINGS AND SPECIFICATIONS FOR QUANTITY, TYPE, LOCATION,AND MOUNTING HEIGHT.
13. ALL WALL-MOUNTED ELECTRICAL/DATA RACEWAY SHALL BE MOUNTED AT +45" AFF TO BOTTOM OF RACEWAY, UNLESS NOTED OTHERWISE.
14. UNDER-CABINET/SHELVING TASK LIGHTING TO BE HARD-WIRED, WHEN FURNISHED BY LABORATORY FURNISHINGS CONTRACTOR, SHALL BE INSTALLED BY ELECTRICAL CONTRACTOR.
15. ON LABORATORY EQUIPMENT ELEVATIONS, INDIVIDUAL WALL MOUNTED ELECTRICAL, DATA, VOICE OR OTHER DEVICES TYPICALLY NOT SHOWN. REFER TO BUILDING ELECTRICAL POWER, DATA/VOICE DRAWINGS FOR THOSE LOCATIONS.
16. PROVIDE TASK LIGHT BENEATH ALL WALL CABINETS WHEN LOCATED ABOVE WORKSURFACE, I.E. FIXED BENCH TOPS AND MOVABLE TABLES. DO NOT PROVIDE TASK LIGHT UNDER LOWER SHELF WHEN ABOVE EQUIPMENT OR EQUIPMENT ZONE.
17. ALL OFOI EQUIPMENT SHOWN DASHED, UNLESS NOTED OTHERWISE.
18. REFER TO RELATED SPECIFICATION SECTIONS INCLUDING, BUT NOT LIMITED TO:

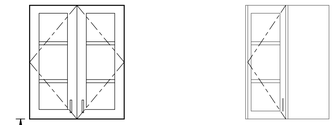
    - DIVISION 1
    - DIVISION 9
    - 11 53 00 LABORATORY EQUIPMENT
    - 11 53 43 LABORATORY SERVICE FITTINGS AND FIXTURES
    - 12 35 53 LABORATORY CASEWORK AND OTHER FURNISHINGS

BASE CABINETS - FIXED WITH FLUSH OVERLAY FRONTS | **BASE CABINET- SINK** | **MOBILE BASE CABINETS**

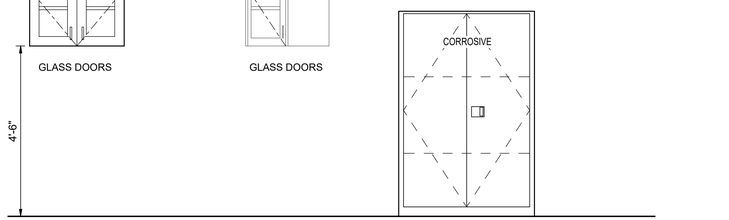


BAxxA BAxxB BAxxC BAxxS BMxxP

METAL WALL CABINETS - WITH FLUSH OVERLAY FRONTS



TALL CABINETS - SPECIALTY



WCxxB WCxxG TAxxAXv

- TAxxAXv
  - DENOTES VENTED CAB (WHEN SHOWN, OTHERWISE NON-VENTED)
  - CABINET TYPE: W = OXIDIZER, X = FLAMMABLE, Y = CORROSIVE
  - DENOTES CABINET HEIGHT: A = 35" HIGH, B = 44" HIGH, C = 65" HIGH

BASE CABINET NOMENCLATURE

B X xx X *

- * DENOTES CABINET LOCK
- DENOTES CABINET STYLE (SEE ADJACENT STYLE ELEVATIONS)
- DENOTES CABINET WIDTH
- DENOTES CABINET TYPE: S = STANDING HEIGHT (36"), A = ADA HEIGHT (34"), L = SITTING HEIGHT (30"), M = MOBILE CABINET, H = SUSPENDED CABINET
- DENOTES BASE CABINET

WALL CABINET NOMENCLATURE

W A xx A *

- * DENOTES CABINET LOCK
- DENOTES CABINET STYLE (SEE ADJACENT STYLE ELEVATIONS)
- DENOTES CABINET WIDTH
- DENOTES CABINET DEPTH & HEIGHT: A = 13"D x 24"H, B = 13"D x 30"H, C = 13"D x 36"H, D = 13"D x 48"H, E = 16"D x 24"H, F = 16"D x 30"H, G = 16"D x 36"H, H = 16"D x 48"H
- DENOTES WALL CABINET

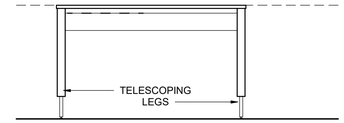
CASEWORK GENERAL NOTES:

1. REFER TO SHEET 061Q101 FOR LABORATORY GENERAL NOTES
2. DIMENSIONS ARE NOMINAL, COORDINATE VARIATIONS BASED UPON MANUFACTURER, SUBJECT TO ARCHITECT'S APPROVAL
3. LOCATE / INSTALL BLOCKING / BACKING AT WALLS WHERE CASEWORK IS PLACED / MOUNTED, PER SPECIFICATION SECTION 09110
4. INSTALL ALL FILLER AND END PANELS AS REQUIRED TO PROVIDE A COMPLETE INSTALLATION. FILLER AND END PANELS ARE NOT TAGGED ON PLANS
5. PROVIDE BACKSLPASHES AT ALL COUNTER EDGES ABUTTING WALLS, COLUMN FURRING AND FUME HOODS
6. AT ALL PENINSULA END-CAP CASEWORK, PROVIDE CONTINUOUS TOE KICK AT END AND SIDE ELEVATIONS OF END-CAP BASE CABINETS
7. ALL CASEWORK IN RADIOISOTOPE LABS SHALL HAVE ALL CABINET DOORS / DRAWERS EQUIPPED WITH KEYED LOCKS
8. PROVIDE EPOXY RESIN WORKSURFACES AT ALL FIXED / ADJUSTABLE HEIGHT TABLES, UNLESS NOTED OTHERWISE.
9. UNLESS NOTED OTHERWISE, FOR TALL STORAGE CABINETS UNITS IN ROOMS WITH INTEGRAL COVED SHEET FLOORING, INSTALL CABINETS AFTER FLOORING. CHAMFER REAR BACK EDGE OF BASE AT INTEGRAL FLOORING COVED SURFACE. COVER EXPOSED CABINET BASE SURFACES WITH RUBBER BASE PER ARCHITECTURAL FINISH SCHEDULE.
10. ALL CABINETS WITH THE SUFFIX ' * ' ARE TO HAVE LOCKS.

TABLE - ADJUSTABLE HEIGHT

(w/ SUSPENDED CASEWORK WHERE INDICATED ON PLANS- REF CASEWORK MENU FOR CABINET TYPE)

NOTE:
EPOXY WORKSURFACES AT ALL FIXED HT. & ADJUSTABLE HT. TABLES, UNLESS NOTED OTHERWISE.



AT S xx A

- AT
  - DENOTES ADJ. HT. TABLE
- S
  - HEIGHT (S = 36") (A = 34") (L = 30")
- xx
  - WIDTH
- A
  - DEPTH (A = 24") (B = 30") (C = 36") (D = 48") (E = 60")

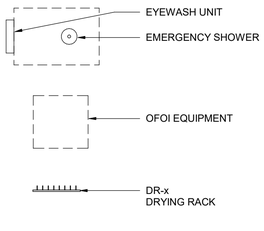
ENGINEERING COORDINATION SYMBOLS

- POINT EXHAUST FROM ABOVE
- TELEPHONE / DATA RECEPTACLE
- DATA RECEPTACLE
- TELEPHONE RECEPTACLE
- TELEPHONE / DATA RECEPTACLE AT CLNG
- INSTRUMENT REFERENCE GROUND
- DISCONNECT SWITCH

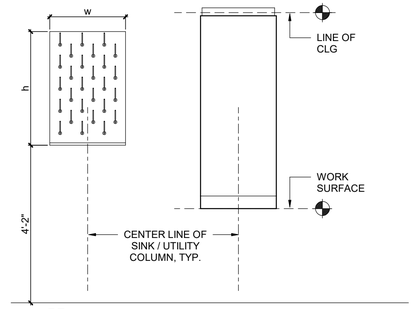
NORMAL	EMERGENCY DEDICATED	EMERGENCY + DEDICATED	
			DUPLEX RECEPTACLE AT WALL / RACEWAY
			QUAD RECEPTACLE AT WALL / RACEWAY
			SINGLE RECEPTACLE AT WALL / RACEWAY
		X	SPECIAL RECEPTACLE AT WALL / RACEWAY (208V, 3PH, 30A UNLESS OTHERWISE NOTED)
			DUPLEX RECEPTACLE AT CLNG / SVC PANEL
			QUAD RECEPTACLE AT CLNG / SVC PANEL

NOTE
1. RECEPTACLES AND RACEWAY AT 43" AFF TO CL UNO.
2. ELECTRICAL AND MECHANICAL SYMBOLS SHOWN FOR INFORMATION ONLY. REFER TO ELECTRICAL AND MECHANICAL DRAWINGS FOR DESIGN.

LABORATORY SYMBOLS



ACCESSORIES - DRYING RACK | **ACCESSORIES - UTILITY COLUMN**



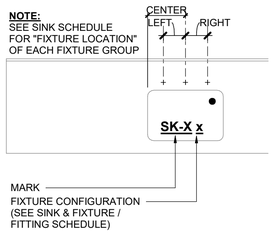
DR-x
(1 = 24"w x 30"h)
(2 = 24"w x 36"h)
(3 = 30"w x 30"h)
(4 = 30"w x 36"h)
(5 = 36"w x 36"h)

UC xx xx a(or)b

- STYLE
- WIDTH
- LENGTH
- TYPE: **a** = WALL MOUNTED, **b** = PENINSULA / BENCH MOUNTED

SINKS / FIXTURES

NOTE:
SEE SINK SCHEDULE FOR "FIXTURE LOCATION" OF EACH FIXTURE GROUP



SINK / SERVICE FIXTURE GENERAL NOTES:

1. REFER TO LABORATORY FURNISHINGS PLANS FOR LOCATIONS.
2. WALL MOUNTED FIXTURES - LOCATE CENTER LINE OF FIXTURE +42" AFF, TYP, WHERE OCCURS.
3. REFER TO SINK AND FIXTURE & FITTING SCHEDULES FOR ADDITIONAL INFORMATION.

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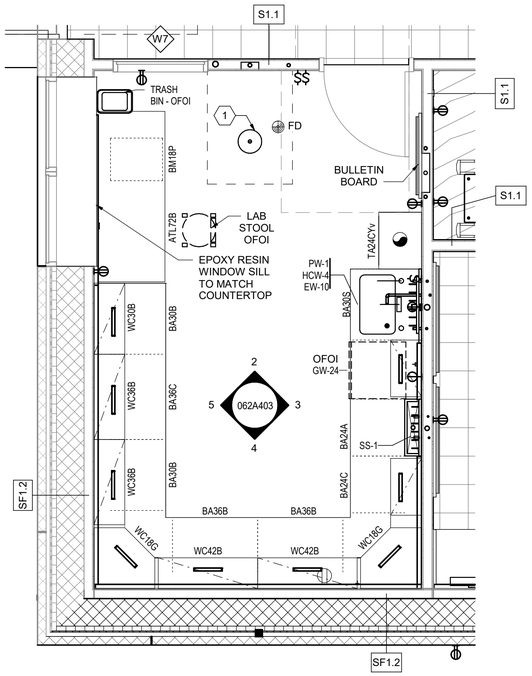
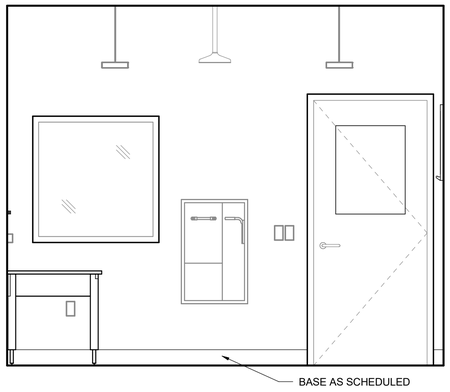
LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 4/20/2026 ARIZONA U.S.A.

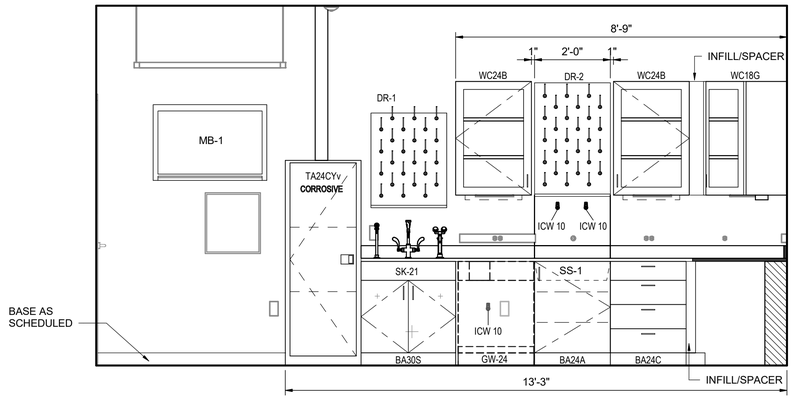
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: R. PRESCOTT DRAWN: R. PRESCOTT TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A402	TITLE OF SHEET WATER TREATMENT LABORATORY FURNISHINGS KEY SHEET NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 86 OF 192



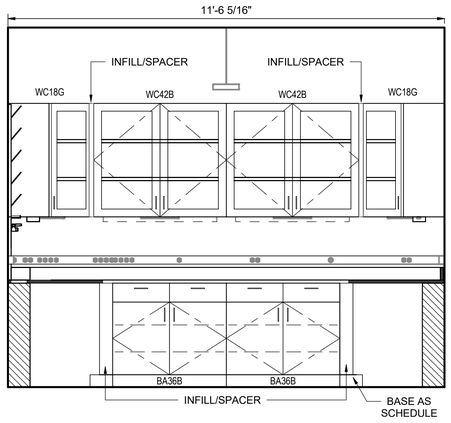
1 062A101 ENLARGED LABORATORY PLAN
SCALE (B)



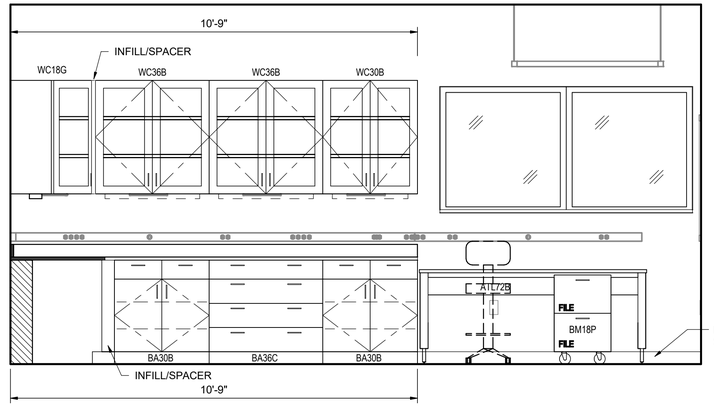
2 062A403 LAB #107 - NORTH WALL
SCALE (A)



3 062A403 LAB #107 - EAST WALL
SCALE (A)



4 062A403 LAB #107 - SOUTH WALL
SCALE (A)



5 062A403 LAB #107 - WEST WALL
SCALE (A)

LABORATORY FIXTURE & FITTINGS SCHEDULE (11 53 43)

SERVICE ID		SPECIFICATION MARK	SERVICE TYPE	FIXTURE TYPE	MOUNTING	FINISH	REMARKS
SERVICE MARK	ID						
DI		PW1	DEIONIZED WATER	PURE WATER FAUCET	DECK	CHROME	
EW		EW-10	POTABLE WATER (TEMPERED)	DRENCH HOSE DUAL HEAD	DECK		
HCW		HCW4	POTABLE HOT & COLD WATER	LABORATORY MIXING FAUCET	DECK	CHROME	WRIST BLADES; VACUUM BREAKER
ICW		ICW-10	INDUSTRIAL COLD WATER	LABORATORY SINGLE FAUCET	WALL	CHROME	

LABORATORY SINK SCHEDULE

MARK	DESCRIPTION	MANUFACTURER	MODEL	LENGTH (Left-Right)	WIDTH (Front-Back)	DEPTH (Top-Bottom)	FIXTURE TYPES			FIXTURE LOCATION			COMMENTS
							LEFT	CENTER/END	RIGHT	CENTER	LEFT	RIGHT	
SK-21	EPOXY RESIN DROP-IN SINK	DURCON	D55	2' - 1"	1' - 3"	0' - 10"	PW-1	HCW-4	EW-10	1' - 0 1/2"	0' - 10"	0' - 10"	
SS-1	EPOXY RESIN TROUGH SINK	DURCON		1'-10"	0' - 5"	0' - 5 1/2"							

SCALE (B) 2 0 2 4 6 — 3/8" = 1'-0"
SCALE OF FEET

SCALE (A) 2 0 2 4 — 1/2" = 1'-0"
SCALE OF FEET

LICENSED ARCHITECT CERTIFICATE NO. 85783 RICHARD MCKINLEY DATE SIGNED ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	R. PRESCOTT	062 A403	WATER TREATMENT LABORATORY FURNISHINGS
	CADD: R. PRESCOTT		
	TECH. REVIEW: R. MCKINLEY		
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	87 OF 192

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1 062A901 PERSPECTIVE VIEW
NOT TO SCALE

GENERAL NOTES:
1. PERSPECTIVE VIEW IS INTENDED TO SHOW BUILDING MASSING RELATIVE TO THE EXISTING EMERGENCY SERVICES BUILDING
2. SEE SHEET 061-A301 FOR BUILDING MATERIALS

LICENSED ARCHITECT CERTIFICATE NO. 85785 RICHARD MCKINLEY DATE SIGNED 2026.05.01 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. KENNY	062 A901	WATER TREATMENT 3D REPRESENTATION	113 182752
	DRAWN: D. SPERRY			PMIS/PKG NO. 318719
	TECH. REVIEW: R. MCKINLEY			SHEET 88 OF 192
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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1 062A902 PERSPECTIVE VIEW
NOT TO SCALE

GENERAL NOTES:
1. PERSPECTIVE VIEW IS INTENDED TO SHOW BUILDING MASSING RELATIVE TO THE EXISTING EMERGENCY SERVICES BUILDING
2. SEE SHEET 061-A301 FOR BUILDING MATERIALS

LICENSED ARCHITECT CERTIFICATE NO. 85783 RICHARD MCKINLEY DATE SIGNED ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. KENNY CADD D. SPERRY TECH. REVIEW: R. MCKINLEY DATE: MAY 2026	SUB SHEET NO. 062 A902	TITLE OF SHEET WATER TREATMENT 3D REPRESENTATION NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 89 OF 192

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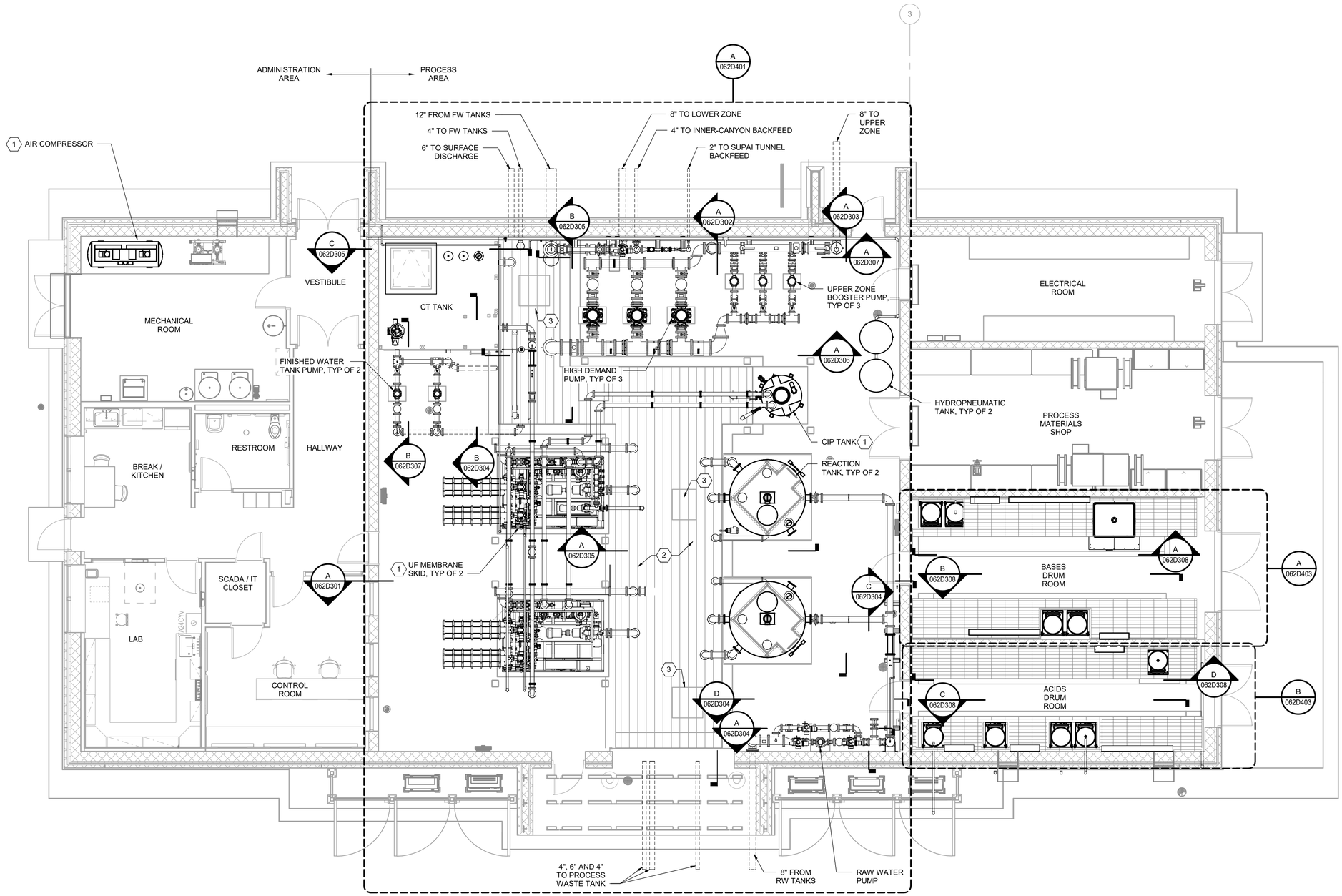
GENERAL NOTES

A. SMALL DIAMETER CHEMICAL, SAMPLE, AND COMPRESSED AIR PIPING NOT SHOWN FOR CLARITY. SEE ISOMETRIC ON 062D907.

KEY NOTES (X)

1. EQUIPMENT INCLUDED IN UF MEMBRANE SUPPLIER'S SCOPE.
2. PIPE TRENCH BELOW SLAB, SEE 062D401 FOR PIPING BELOW ALUMINUM PLANKS.
3. OPERATOR ACCESS POINT IN ALUMINUM PLANKS.



SCALE (A) 4 0 4 8 12 3/16" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Corp) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAFTED: D. DEATERLA TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2025	SUB SHEET NO. 062 D101	TITLE OF SHEET WATER TREATMENT GROUND LEVEL PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 90 OF 192

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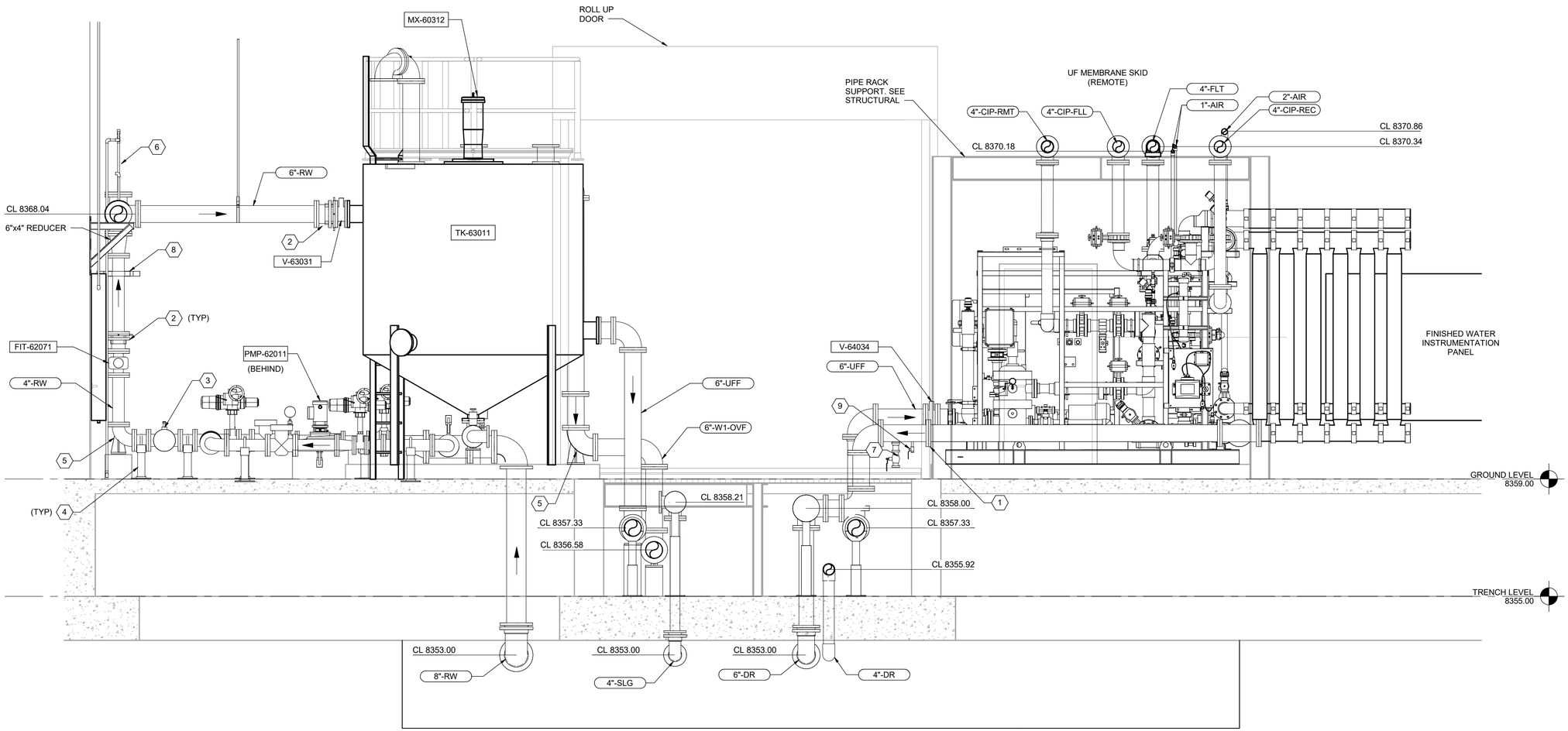

A 062D101 BUILDING SECTION
1/2" = 1'-0"

GENERAL NOTES

A. SEE SHEETS 005D501 AND 005D502 FOR STANDARD PIPE SUPPORT DETAILS.

B. SMALL DIAMETER CHEMICAL, SAMPLE, AND COMPRESSED AIR PIPING NOT SHOWN FOR CLARITY. SEE ISOMETRIC ON 062D907.

KEY NOTES (X)

1. PROVIDE FLANGE ON 6"-DR TO ALLOW FOR PIPE REMOVAL OVER PIPE TRENCH.
2. FLANGED COUPLING ADAPTER.
3. CONNECTION POINT FOR TEMPORARY RECYCLE LINE DURING PLANT STARTUP. SALVAGE VALVE TO GOVERNMENT AFTER STARTUP AND INSTALL BLIND FLANGE ON TEE.
4. FLOOR PIPE SUPPORT. SEE DETAIL 1/005D501
5. BASE ELBOW AND TEE PEDESTAL SUPPORT. SEE DETAIL 5/005D501
6. 1/2" COMBO AIR/VAC VALVE. SEE DETAIL 4/005D503.
7. 2" TAP AND BALL VALVE FOR DRAINING. SEE DETAIL 2/005D503.
8. RISER PIPE SUPPORT. SEE DETAIL 4/005D502.
9. 1/2" TAP AND MANUAL BLEED VALVE. SEE DETAIL 3/005D503.

SCALE (A) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: J. STUJENSKE TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D301	TITLE OF SHEET WATER TREATMENT REACTION TANK AND ULTRAFILTRATION SKID SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 91 OF 192

Autodesk Docs://10350545_NPS_GRCA_N_Rim_Utility_2026/10350545_062_NWTP_D.rvt
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GENERAL NOTES

A. SEE SHEETS 005D501 AND 005D502 FOR STANDARD PIPE SUPPORT DETAILS.

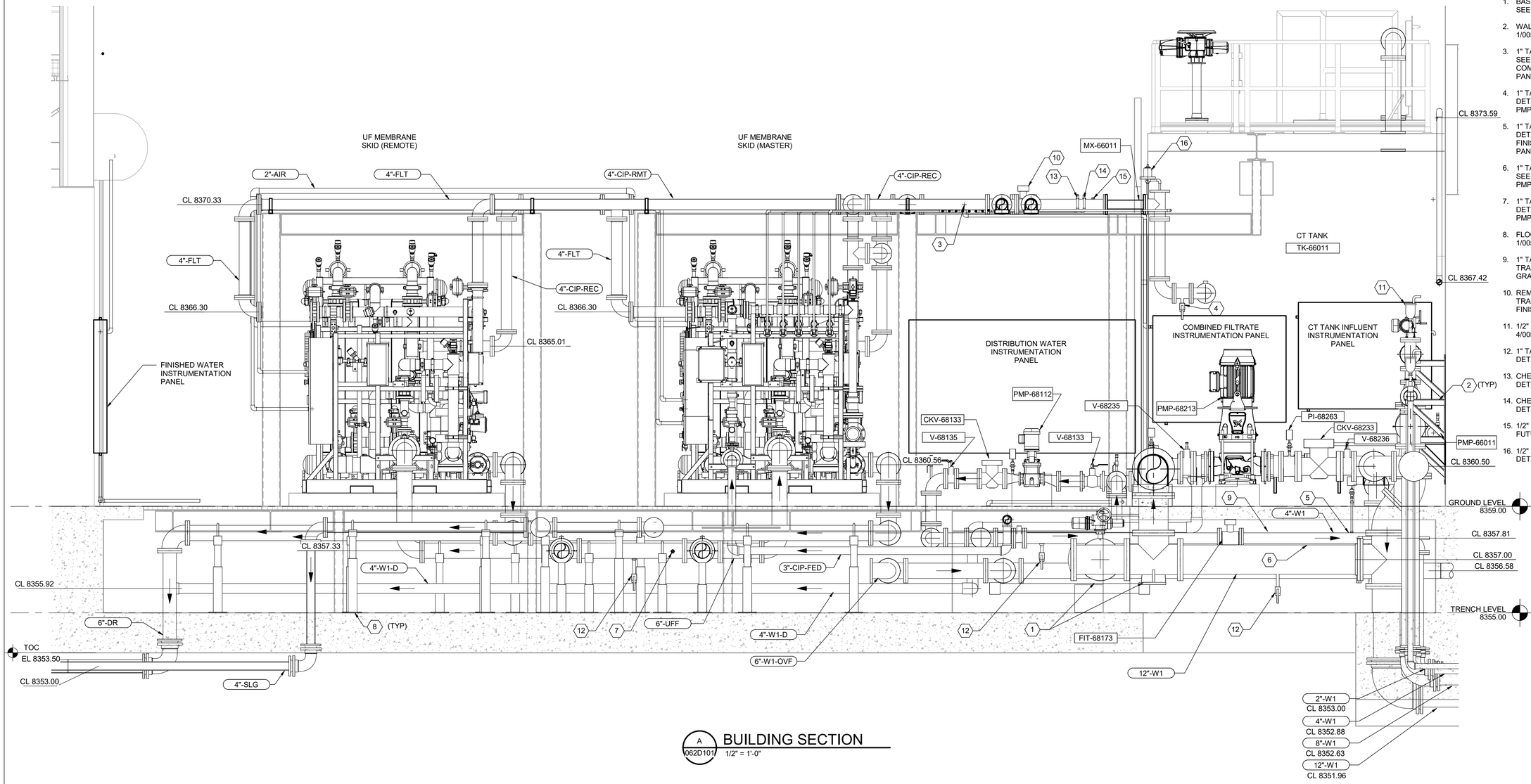
B. SMALL DIAMETER CHEMICAL, SAMPLE, AND COMPRESSED AIR PIPING NOT SHOWN FOR CLARITY. SEE ISOMETRIC ON 062D907.

C. FOR PIPE SLAB PENETRATIONS, SEE DETAIL 6/005D505.

D. FOR PIPES THROUGH FOOTING, SEE 003S507.

KEY NOTES (X)

1. BASE ELBOW AND TEE PEDESTAL SUPPORT. SEE DETAIL 5/005D501.
2. WALL BRACKET PIPE SUPPORT.SEE DETAIL 1/005D502.
3. 1" TAP AND BALL VALVE ON 4"-FLT PIPE. SEE DETAIL 2/005D503. ROUTE 1" SMP TO COMBINED FILTRATE INSTRUMENTATION PANEL. SEE 062D907 FOR ROUTING.
4. 1" TAP AND BALL VALVE ON 4"-FLT PIPE. SEE DETAIL 2/005D503. ROUTE 1" SMP TO PMP-68117. SEE 062D907 FOR ROUTING.
5. 1" TAP AND BALL VALVE ON 4"-W1 PIPE. SEE DETAIL 2/005D503. ROUTE 1" SMP TO FINISHED WATER INSTRUMENTATION PANEL. SEE 062D907 FOR ROUTING.
6. 1" TAP AND BALL VALVE ON 12"-W1 PIPE. SEE DETAIL 2/005D503. ROUTE 1" SMP TO PMP-66011. SEE 062D907 FOR ROUTING.
7. 1" TAP AND BALL VALVE ON 6"-UFF PIPE. SEE DETAIL 2/005D503. ROUTE 1" SMP TO PMP-64011. SEE 062D907 FOR ROUTING.
8. FLOOR PIPE SUPPORT. SEE DETAIL 1/005D501.
9. 1" TAP ON 4"-W1 FOR PIT-68161. MOUNT TRANSMITTER ON CT TANK WALL ABOVE GRATING. SEE DETAIL 3/005D503.
10. REMOTE MOUNT FLOW METER TRANSMITTER (FIT-64071) AT 4 FT ABOVE FINISHED FLOOR.
11. 1/2" AIR RELEASE VALVE. SEE DETAIL 4/005D503.
12. 1" TAP AND BALL VALVE FOR DRAINING. SEE DETAIL 2/005D503.
13. CHEMICAL INJECTOR FOR CAUSTIC. SEE DETAIL 5/005D503.
14. CHEMICAL INJECTOR FOR HYPO. SEE DETAIL 5/005D503.
15. 1/2" SPARE TAP WITH BALL VALVE. CAP FOR FUTURE.
16. 1/2" TAP AND MANUAL BLEED VALVE. SEE DETAIL 3/005D503.



A 062D101 BUILDING SECTION 1/2" = 1'-0"

SCALE (A) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Corp) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 4/15/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAFTED: J. STUJENSKE TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D302	TITLE OF SHEET WATER TREATMENT ULTRAFILTRATION SKIDS AND CT TANK SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 92 OF 192

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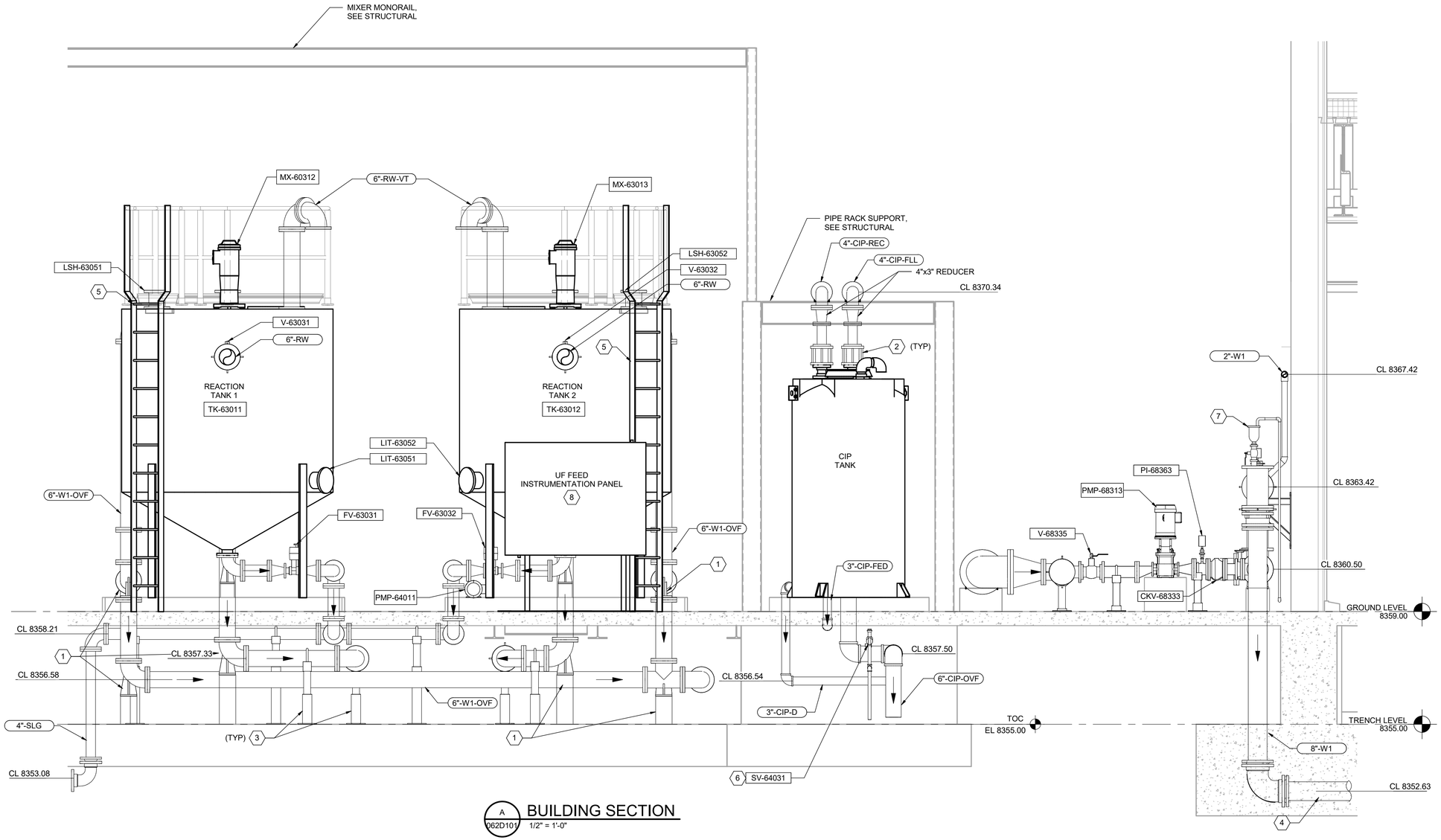

A 062D101 **BUILDING SECTION**
1/2" = 1'-0"

GENERAL NOTES

A. SEE SHEETS 005D501 AND 005D502 FOR STANDARD PIPE SUPPORT DETAILS.

B. SMALL DIAMETER CHEMICAL, SAMPLE, AND COMPRESSED AIR PIPING NOT SHOWN FOR CLARITY. SEE ISOMETRIC ON 062D907.

C. FOR PIPE SLAB PENETRATIONS, SEE DETAIL 6/005D505.

KEY NOTES (X)

1. BASE ELBOW AND TEE PEDESTAL SUPPORT. SEE DETAIL 5/005D501.
2. EXPANSION JOINT.
3. FLOOR PIPE SUPPORT. SEE DETAIL 1/005D501.
4. SEE STRUCTURAL FOR CONCRETE ENCASEMENT DETAIL AT FOOTING.
5. TANK LADDER AND RAILINGS DESIGNED BY TANK SUPPLIER. SEE 060S001 FOR STRUCTURAL DESIGN CRITERIA.
6. 1" DRAIN AND SOLENOID ON 3"-CIP-FED.
7. 2" COMBO AIR/VAC VALVE. SEE DETAIL 4/005D503.
8. FOR PANEL MOUNTING, SEE DETAIL 5/005D504.

SCALE (A) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 6/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: J. STUJENSKE TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D303	TITLE OF SHEET WATER TREATMENT REACTION TANK AND CIP TANK SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 93 OF 192

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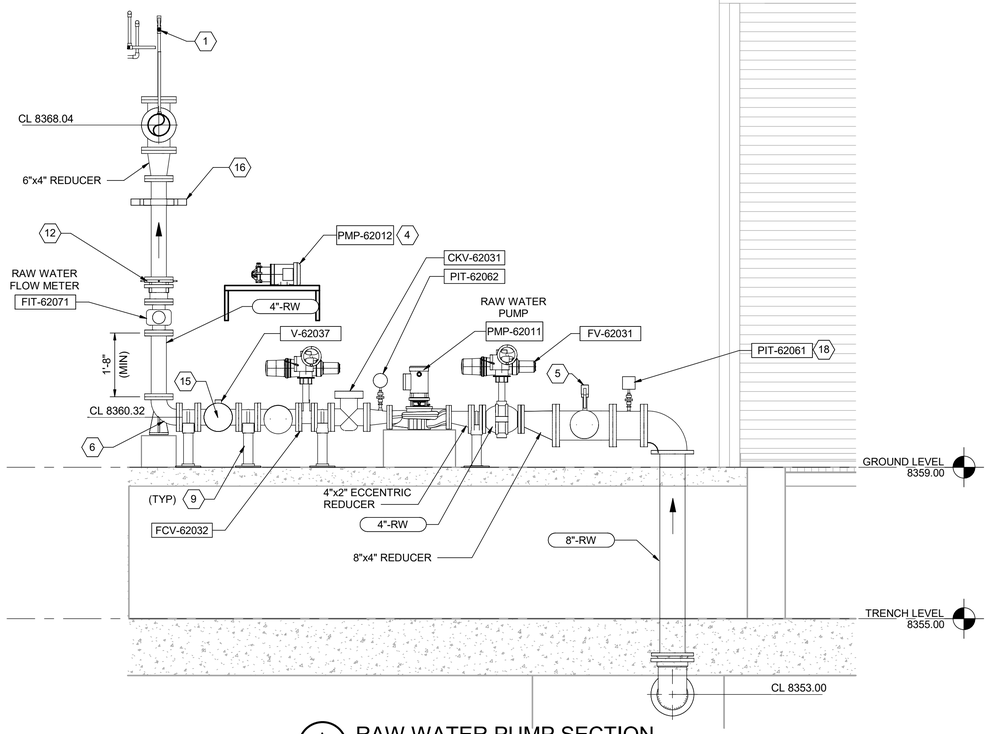
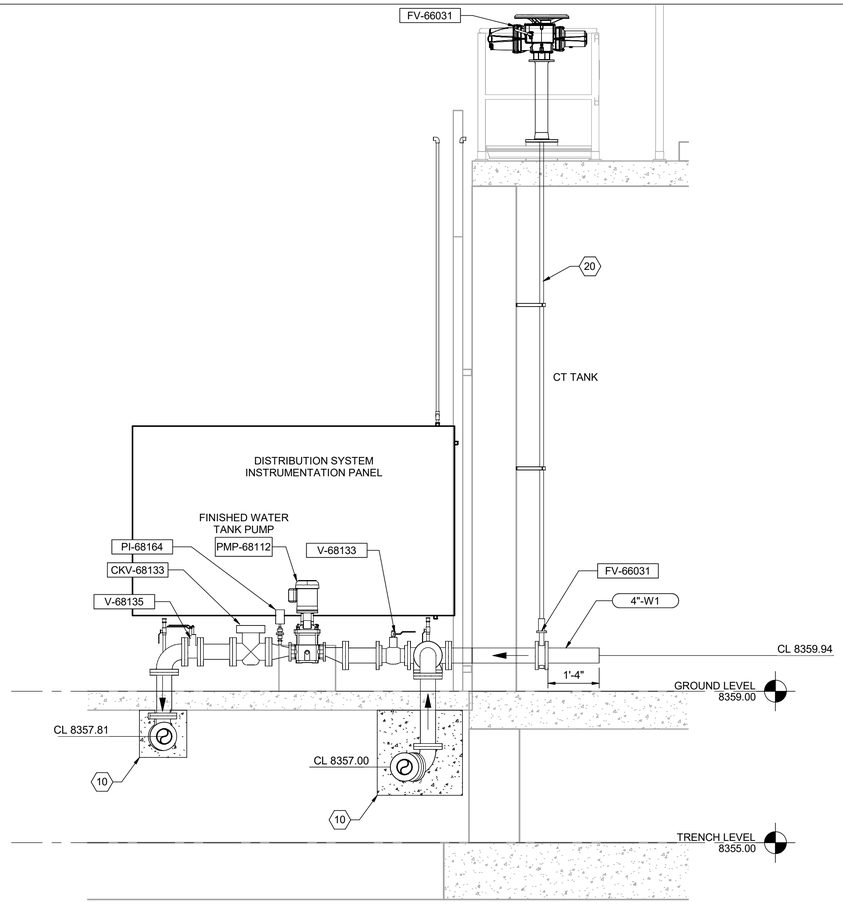
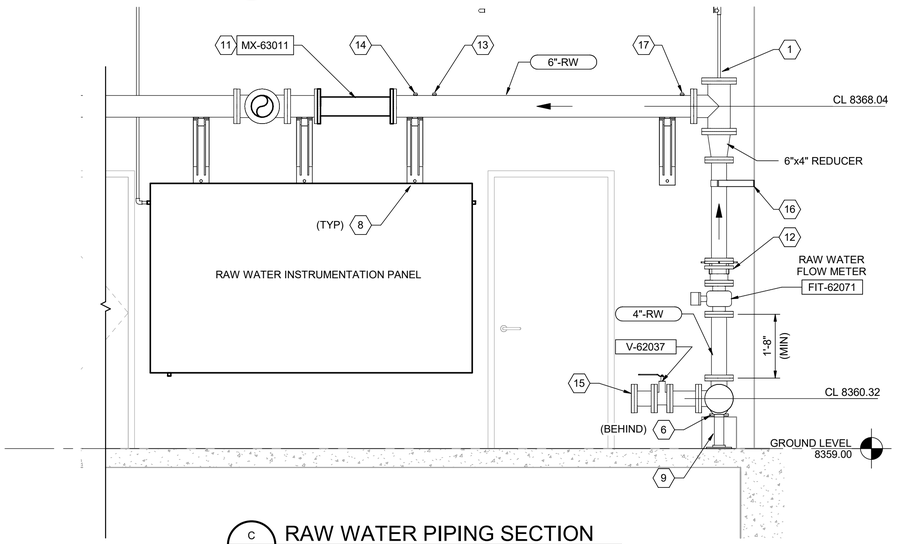


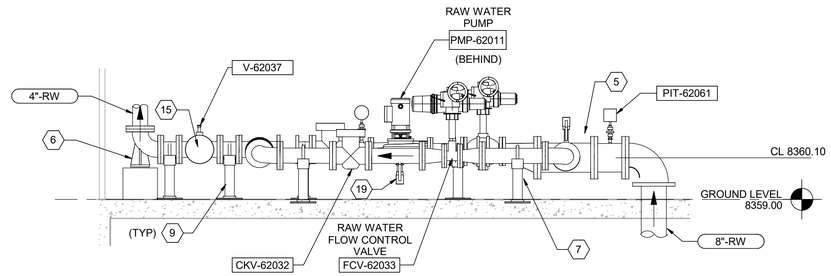
A 062D101 RAW WATER PUMP SECTION
1/2" = 1'-0"



B 062D101 FW TANK PUMP SECTION
1/2" = 1'-0"



C 062D101 RAW WATER PIPING SECTION
1/2" = 1'-0"



D 062D101 RAW WATER PIPING SECTION
1/2" = 1'-0"

GENERAL NOTES

A. SEE SHEETS 005D501 AND 005D502 FOR STANDARD PIPE SUPPORT DETAILS.

B. SMALL DIAMETER CHEMICAL, SAMPLE, AND COMPRESSED AIR PIPING NOT SHOWN FOR CLARITY. SEE ISOMETRIC ON 062D907.

C. FOR PIPE SLAB PENETRATIONS, SEE DETAIL 6/005D505.

D. FOR WALL PIPE PENETRATIONS, SEE DETAILS ON 005D505.

KEY NOTES (X)

1. 1/2" TAP AND MANUAL BLEED VALVE. SEE DETAIL 3/005D503.
2. NOT USED.
3. NOT USED.
4. SEE SHEET 062D906 FOR SAMPLE PIPE ROUTING TO/FROM PUMP.
5. 1" TAP AND BALL VALVE ON TOP OF 8" TEE. SEE DETAIL 3/005D503. ROUTE 1" SMP TO PMP-62012. SEE 062D907 FOR ROUTING.
6. BASE ELBOW AND TEE PEDESTAL SUPPORT. SEE DETAIL 5/005D501.
7. CONCRETE PEDESTAL SUPPORT. SEE DETAIL 3/005D501.
8. WALL BRACKET PIPE SUPPORT. SEE DETAIL 1/005D502.
9. FLOOR PIPE SUPPORT. SEE DETAIL 1/005D501.
10. ENCASE PIPE BELOW SLAB IN CONCRETE. SEE DETAIL 1/005D504.
11. STATIC MIXER, SEE SPEC SECTION 46 41 17.
12. FLANGED COUPLING ADAPTER.
13. CHEMICAL INJECTOR FOR COAG. SEE DETAIL 5/005D503.
14. 1/2" SPARE TAP WITH BALL VALVE. CAP FOR FUTURE.
15. CONNECTION POINT FOR TEMPORARY RECYCLE LINE DURING PLANT STARTUP. SALVAGE VALVE TO GOVERNMENT AFTER STARTUP AND INSTALL BLIND FLANGE ON TEE.
16. RISER PIPE SUPPORT. SEE DETAIL 4/005D502.
17. CHEMICAL INJECTOR FOR HYPO. SEE DETAIL 5/005D503.
18. 1" TAP ON 4"-RW FOR PIT-62062. SEE DETAIL 3/005D503.
19. 1" TAP AND BALL VALVE FOR DRAINING. SEE DETAIL 2/005D503.
20. EXTENDED STEM WITH GUIDES TO ELECTRIC ACTUATOR ABOVE. MOUNT FLOOR STAND ON 6" CONCRETE PAD WITH GASKET AT INTERFACE. CORE DRILL OPENING THROUGH PAD AND TANK SLAB.

SCALE (A) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Grp) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2025 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	M. HOWARD	062 D304	113 182752
	CADD J. STUJENSKE		PMIS/PKG NO. 10350545
	TECH. REVIEW: GOUGH/GLOVER		SHEET 94 OF 192
	DATE: MAY 2025		

TITLE OF SHEET
WATER TREATMENT
RAW WATER AND FINISHED WATER PIPING SECTIONS
NORTH RIM - GRAND CANYON NATIONAL PARK

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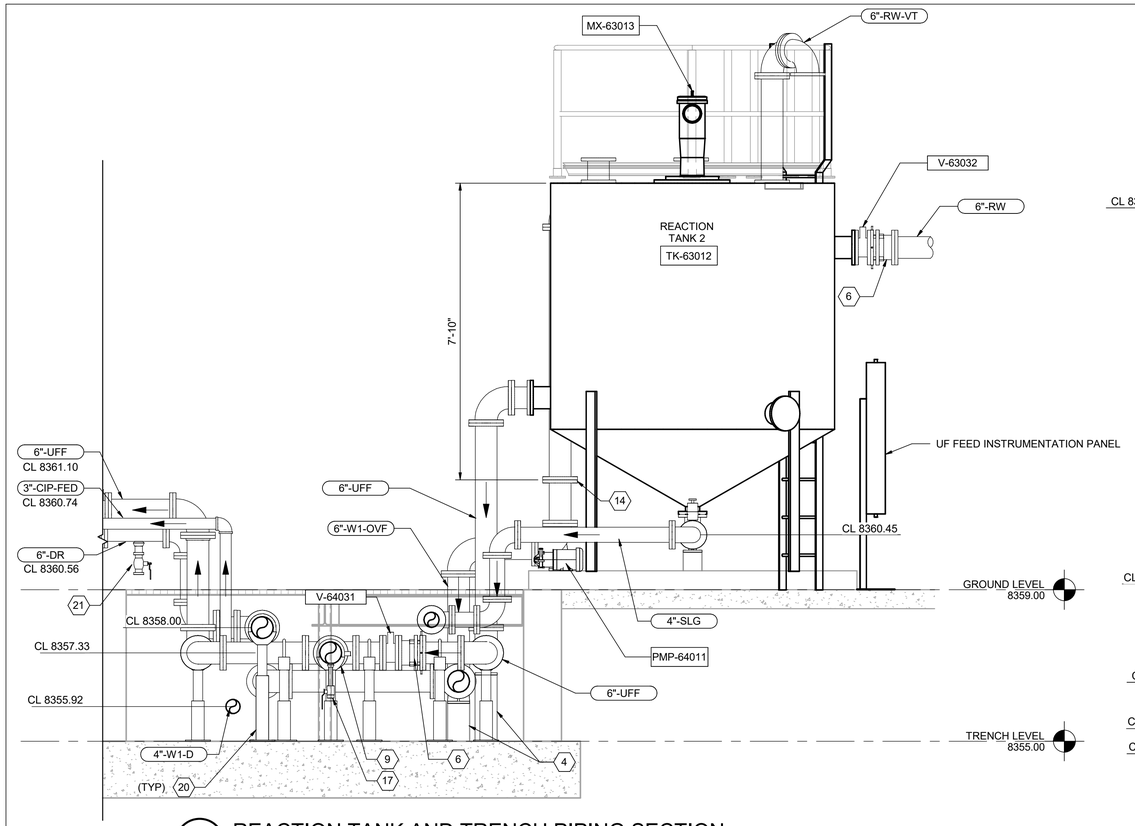
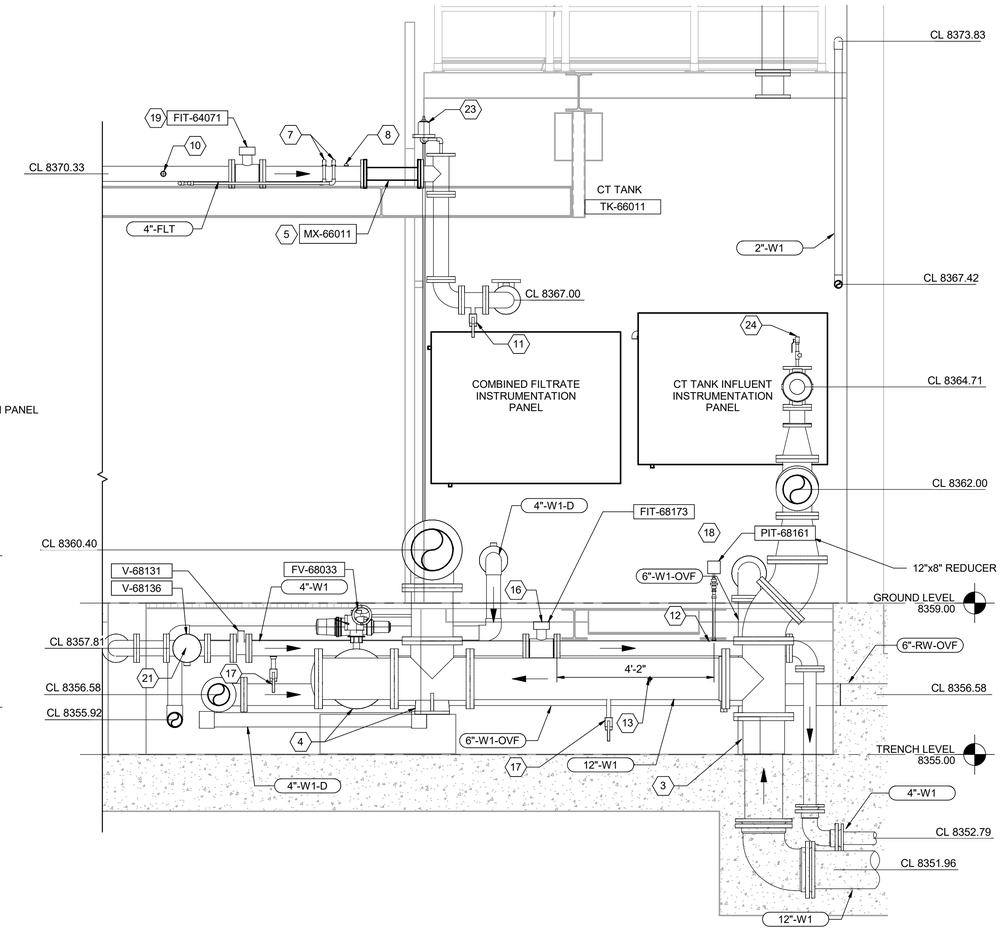
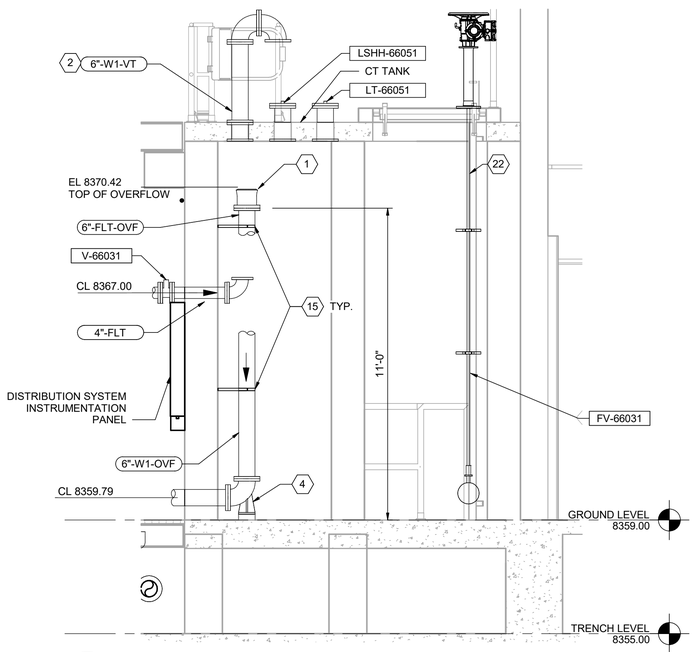


A 062D101 REACTION TANK AND TRENCH PIPING SECTION
1/2" = 1'-0"



B 062D101 CT TANK AND TRENCH PIPING SECTION
1/2" = 1'-0"



C 062D101 INTERNAL CT TANK SECTION
3/8" = 1'-0"

GENERAL NOTES

A. SEE SHEETS 005D501 AND 005D502 FOR STANDARD PIPE SUPPORT DETAILS.

B. SMALL DIAMETER CHEMICAL, SAMPLE, AND COMPRESSED AIR PIPING NOT SHOWN FOR CLARITY. SEE ISOMETRIC ON 062D907.

C. FOR PIPE SLAB PENETRATIONS, SEE DETAIL 6/005D505.

D. FOR WALL PIPE PENETRATIONS, SEE DETAILS ON 005D505.

KEY NOTES (X)

1. 6"x8" FLANGE x FLARE FITTING.
2. SEE DETAIL 4/005D504.
3. CONCRETE PEDESTAL SUPPORT. SEE DETAIL 3/005D501.
4. BASE ELBOW AND TEE PEDESTAL. SEE DETAIL 5/005D501.
5. STATIC MIXER, SEE SPEC SECTION 46 41 17.
6. FLANGED COUPLING ADAPTER (FCA).
7. CHEMICAL INJECTORS FOR HYPO AND CA. SEE DETAIL 5/005D503.
8. 1/2" SPARE TAP WITH BALL VALVE. CAP FOR FUTURE.
9. 1" TAP AND BALL VALVE ON 6"-UFF PIPE. SEE DETAIL 2/005D503. ROUTE 1" SMP TO PMP-64011. SEE 062D907 FOR ROUTING.
10. 1" TAP AND BALL VALVE ON 4"-FLT PIPE. SEE DETAIL 2/005D503. ROUTE 1" SMP TO COMBINED FILTRATE INSTRUMENTATION PANEL. SEE 062D907 FOR ROUTING.
11. 1" TAP AND BALL VALVE ON BOTTOM OF 4"-FLT PIPE. SEE DETAIL 2/005D503. ROUTE 1" SMP TO PMP-68117. SEE 062D907 FOR ROUTING.
12. 1" TAP AND BALL VALVE ON 4"-W1 PIPE. SEE DETAIL 2/005D503. ROUTE 1" SMP TO FINISHED WATER INSTRUMENTATION PANEL. SEE 062D907 FOR ROUTING.
13. 1" TAP AND BALL VALVE ON 12"-W1 PIPE. SEE DETAIL 2/005D503. ROUTE 1" SMP TO PMP-66011. SEE 062D907 FOR ROUTING.
14. TRANSITION FROM OVERFLOW PIPE PROVIDED BY TANK SUPPLIER TO CONTRACTOR PROVIDED PIPE.
15. RISER PIPE SUPPORT. SEE DETAIL 4/005D502.
16. REMOTE MOUNT FLOWMETER TRANSMITTER (FIT-68173) ABOVE GRATING.
17. 1" TAP AND BALL VALVE FOR DRAINING. SEE DETAIL 2/005D503.
18. 1" TAP FOR PIT-68161. MOUNT TRANSMITTER ON CT TANK WALL ABOVE GRATING. SEE DETAIL 3/005D503.
19. REMOTE MOUNT FLOW METER TRANSMITTER AT 4 FT ABOVE FINISHED FLOOR.
20. FLOOR PIPE SUPPORT. SEE DETAIL 1/005D501.
21. 1" TAP AND BALL VALVE FOR FLUSHING CONNECTION ON 6"-DR, TYP. BOTH SKIDS. SEE DETAIL 1/005D503.
22. EXTENDED STEM WITH GUIDES TO ELECTRIC ACTUATOR ABOVE, MOUNT FLOOR STAND ON 6" CONCRETE PAD WITH GASKET AT INTERFACE. CORE DRILL OPENING THROUGH PAD AND TANK SLAB.
23. 1/2" TAP AND MANUAL BLEED VALVE. SEE DETAIL 3/005D503.
24. 1/2" TAP AND MANUAL BLEED VALVE. SEE DETAIL 3/005D503.

SCALE (B) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

SCALE (A) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2025 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: J. STUJENSKE TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D305	TITLE OF SHEET WATER TREATMENT REACTION TANK AND CT TANK SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 95 OF 192

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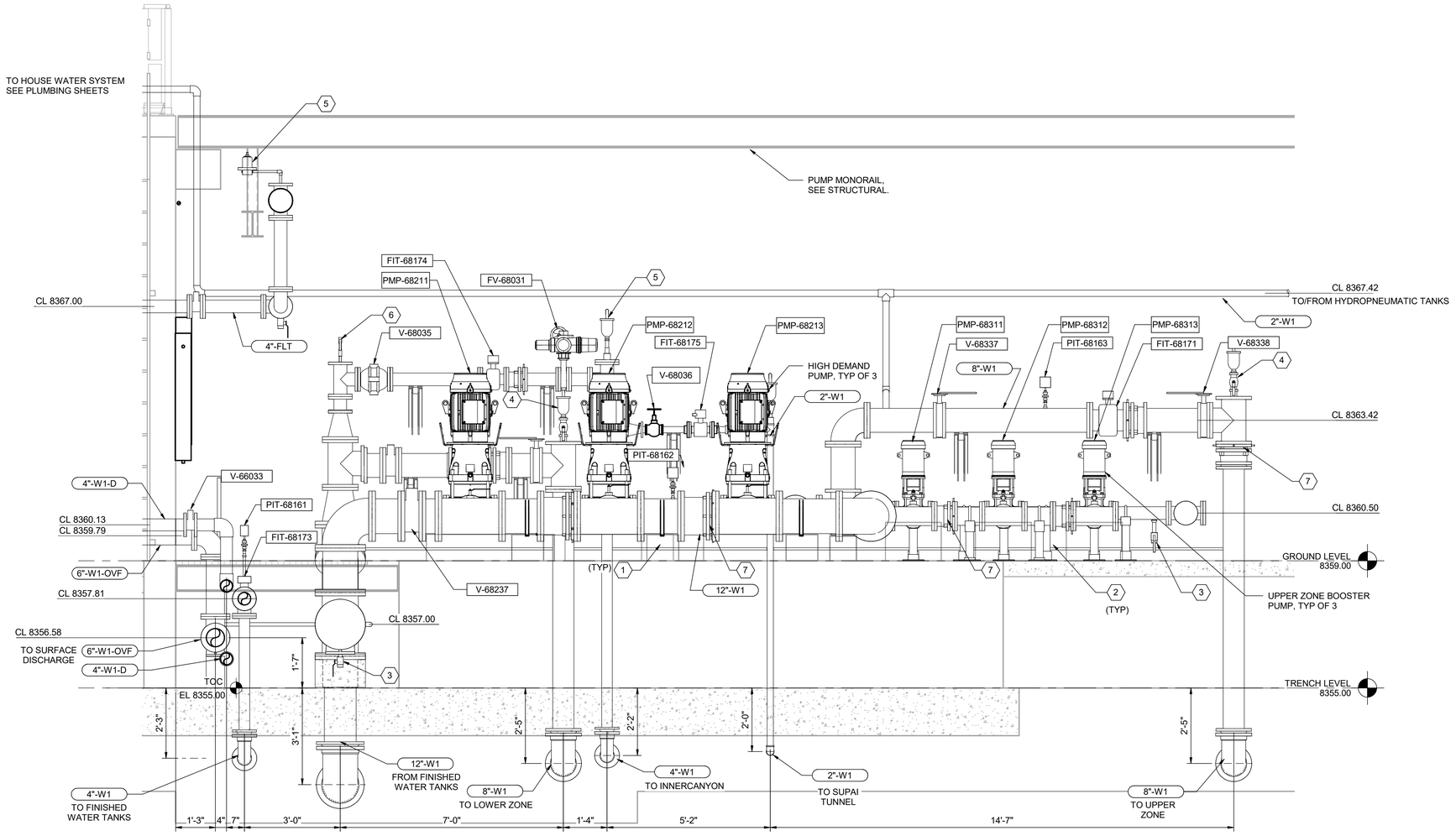

A 062D101 DISTRIBUTION PUMP SECTION
1/2" = 1'-0"

GENERAL NOTES

A. SEE SHEETS 005D501 AND 005D502 FOR STANDARD PIPE SUPPORT DETAILS.

B. SMALL DIAMETER CHEMICAL, SAMPLE, AND COMPRESSED AIR PIPING NOT SHOWN FOR CLARITY. SEE ISOMETRIC ON 062D907.

C. FOR PIPE SLAB PENETRATIONS, SEE DETAIL 6/005D505.

D. FOR WALL PIPE PENETRATIONS. SEE DETAILS ON 005D505.

E. FOR PIPES THROUGH FOOTING, SEE 003SD07.

KEY NOTES (X)

1. CONCRETE PEDESTAL SUPPORT. SEE DETAIL 3/005D501.
2. FLOOR PIPE SUPPORT. SEE DETAIL 1/005D501.
3. 1" TAP AND BALL VALVE FOR DRAINING. SEE DETAIL 2/005D503.
4. 2" COMBO AIR-VAC VALVE. SEE DETAIL 4/005D503.
5. 1/2" COMBO AIR-VAC VALVE. SEE DETAIL 4/005D503.
6. 1/2" TAP AND MANUAL BLEED VALVE. SEE DETAIL 3/005D503.
7. FLANGED COUPLING ADAPTER.

SCALE (A) 2 0 2 4
SCALE OF FEET
1/2" = 1'-0"

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2025 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: D. DEATERLA TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D306	TITLE OF SHEET WATER TREATMENT DISTRIBUTION PUMPING SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 96 OF 192

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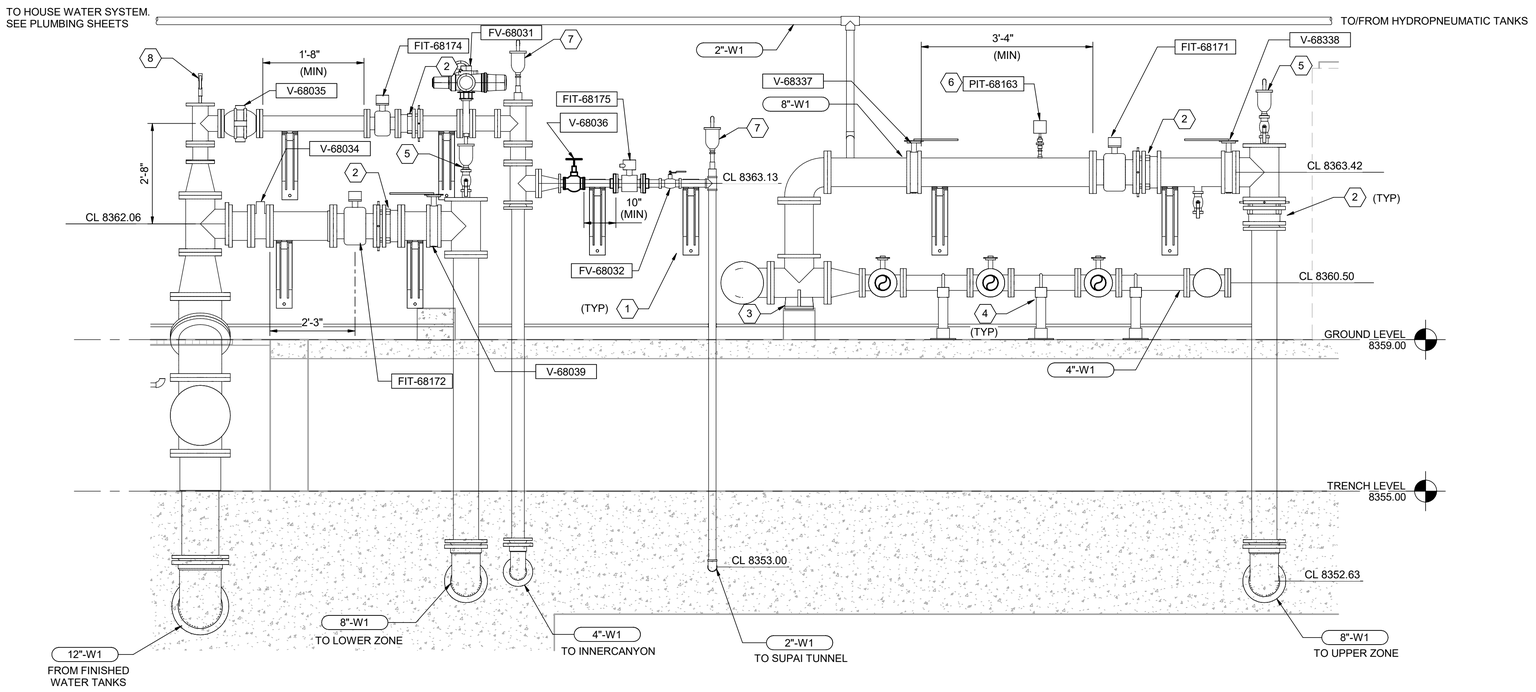
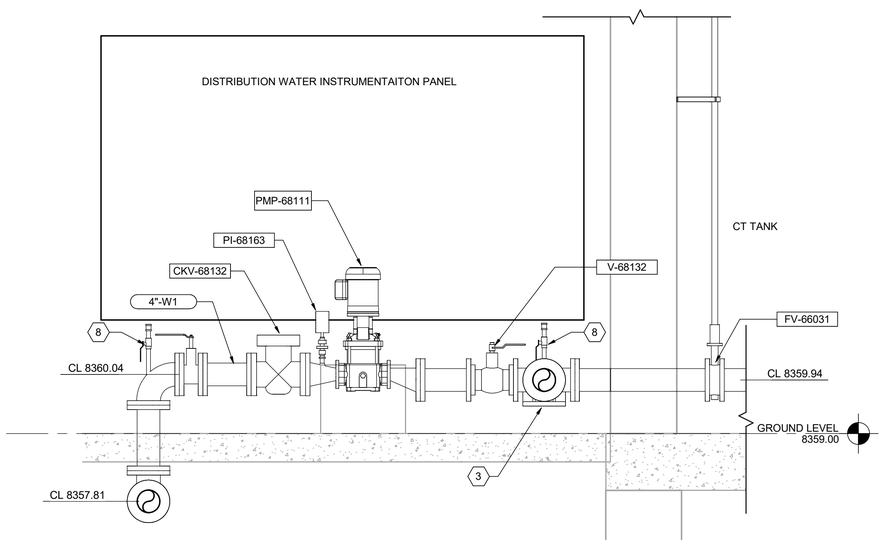


A 062D101 DISTRIBUTION PIPING SECTION
1/2" = 1'-0"



B 062D101 LOOKING AT WEST WALL
3/4" = 1'-0"

GENERAL NOTES

A. SEE SHEETS 005D501 AND 005D502 FOR STANDARD PIPE SUPPORT DETAILS.

B. SMALL DIAMETER CHEMICAL, SAMPLE, AND COMPRESSED AIR PIPING NOT SHOWN FOR CLARITY. SEE ISOMETRIC ON 062D907.

C. FOR PIPE SLAB PENETRATIONS, SEE DETAIL 6/005D505.

D. FOR PIPES THROUGH FOOTING, SEE 003SD07.

KEY NOTES (X)

1. WALL BRACKET PIPE SUPPORT. SEE DETAIL 1/005D502.
2. FLANGED COUPLING ADAPTER (FCA).
3. BASE ELBOW AND TEE PEDESTAL. SEE DETAIL 5/005D501.
4. FLOOR PIPE SUPPORT. SEE DETAIL 1/005D501.
5. 2" COMBO AIR/VAC VALVE. SEE DETAIL 4/000D503.
6. 1" TAP ON 8"-W1 FOR PIT-68163. SEE DETAIL 3/005D503.
7. 1/2" COMBO AIR/VAC VALVE. SEE DETAIL 4/005D503.
8. 1/2" TAP AND MANUAL BLEED VALVE. SEE DETAIL 3/005D503.

SCALE (A) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: D. DEATERLA TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D307	TITLE OF SHEET WATER TREATMENT DISTRIBUTION PIPING SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 97 OF 192

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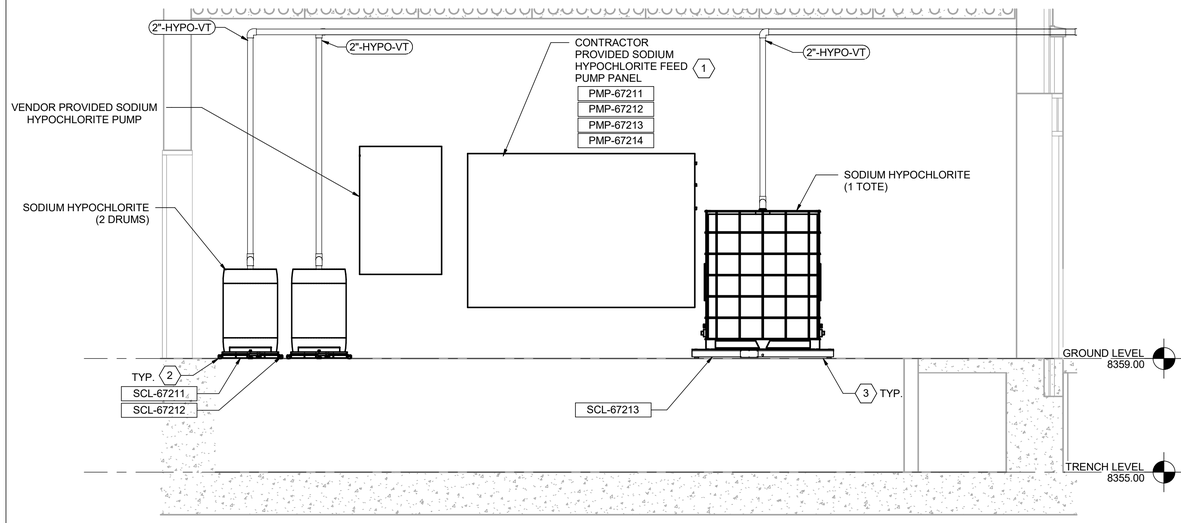
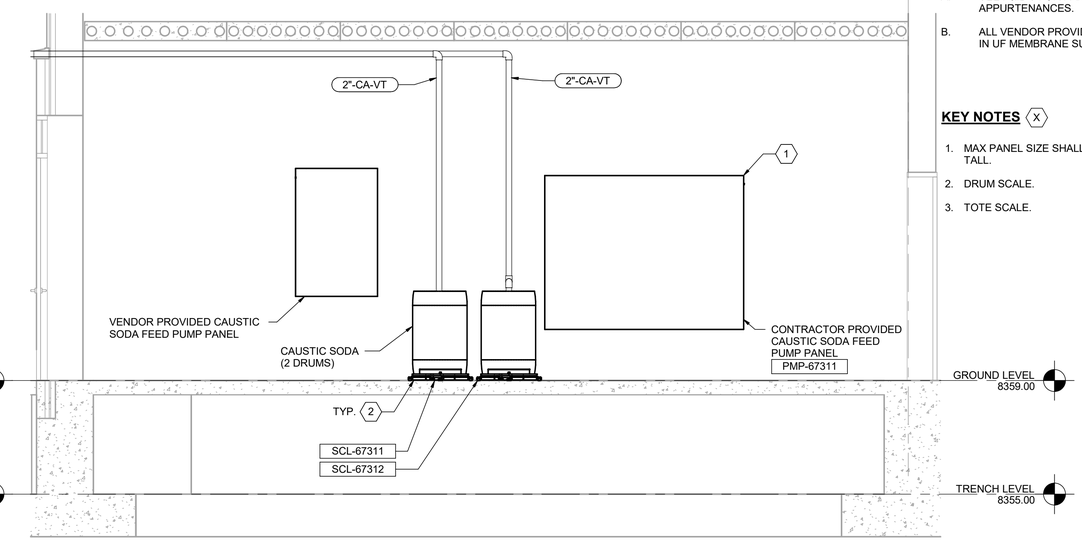
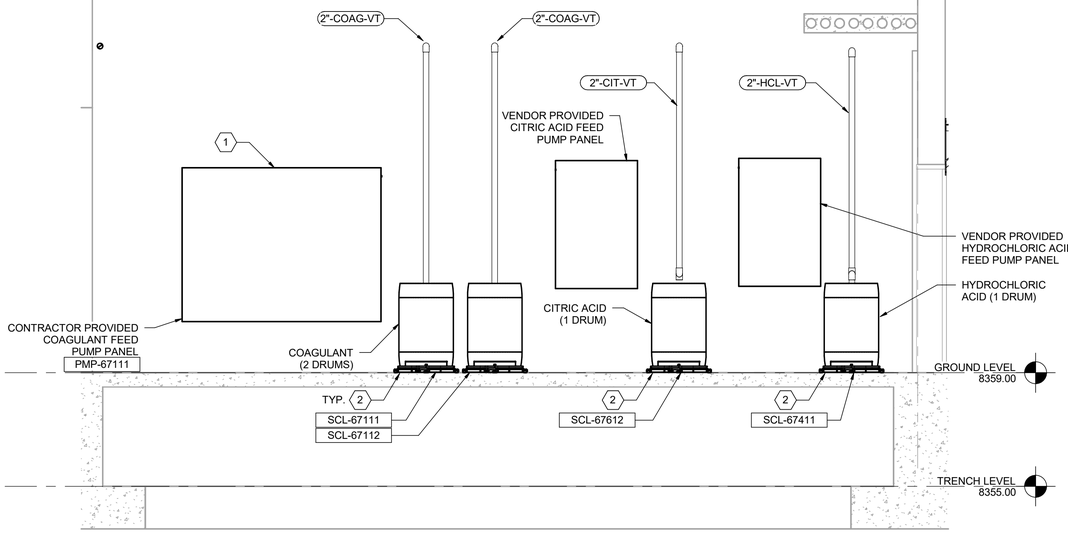


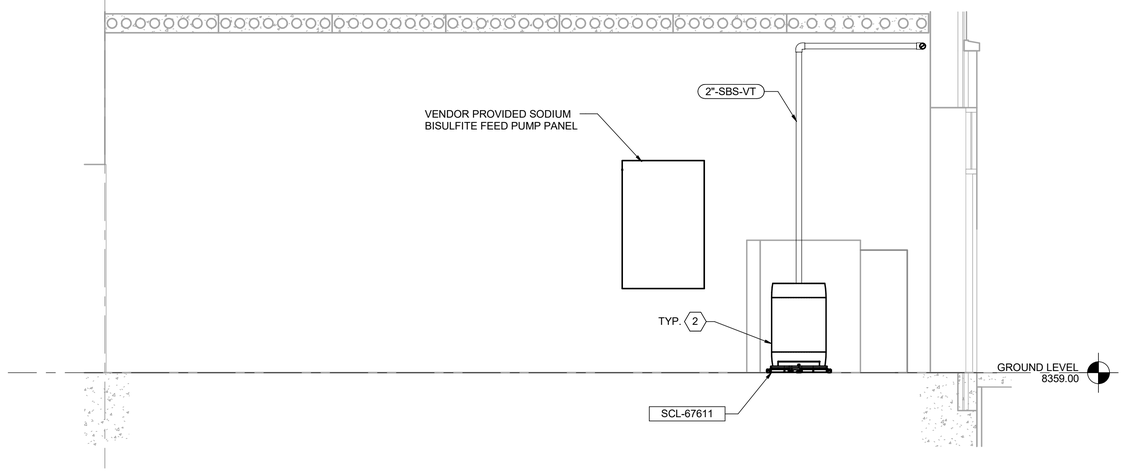
A 062D101 BASES CHEMICAL ROOM SECTION
3/8" = 1'-0"



B 062D101 BASES CHEMICAL ROOM SECTION
3/8" = 1'-0"



C 062D101 ACIDS CHEMICAL ROOM SECTION
3/8" = 1'-0"



D 062D101 ACIDS CHEMICAL ROOM SECTION
3/8" = 1'-0"

GENERAL NOTES

A. SEE P&ID'S FOR PANEL DETAILS AND APPURTENANCES.

B. ALL VENDOR PROVIDED PANELS INCLUDED IN UF MEMBRANE SUPPLIERS SCOPE.

KEY NOTES (X)

1. MAX PANEL SIZE SHALL BE 84" WIDE AND 66" TALL.
2. DRUM SCALE.
3. TOTE SCALE.

SCALE (A) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2025 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: J. STUJENSKE TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D308	TITLE OF SHEET WATER TREATMENT CHEMICAL ROOM SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 98 OF 192

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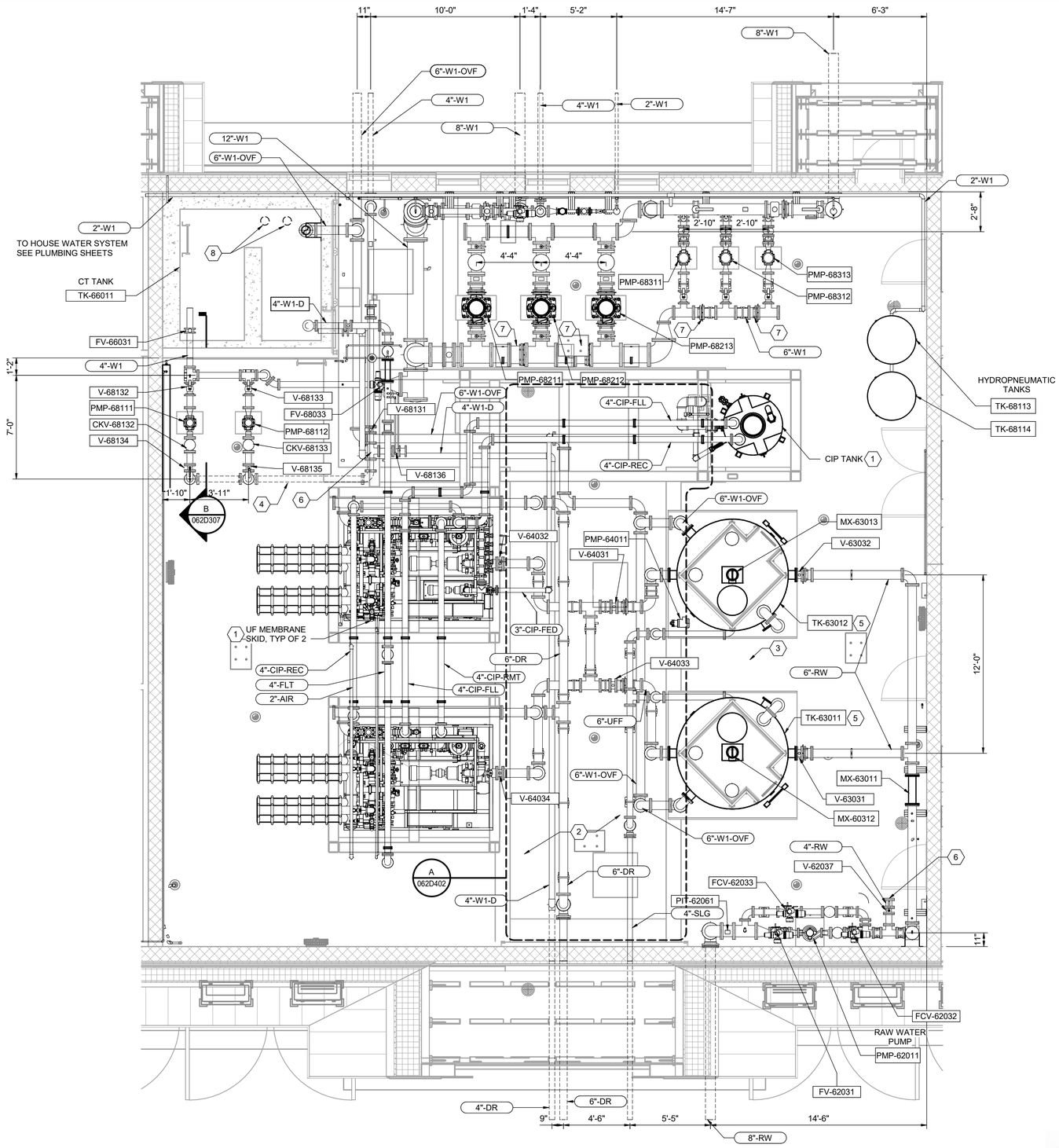

A 062D101 ENLARGED PLAN
1/4" = 1'-0"

GENERAL NOTES

A. SEE 062D101 FOR LOCATION OF SECTION CUTS.

B. SMALL DIAMETER CHEMICAL, SAMPLE, AND COMPRESSED AIR PIPING NOT SHOWN FOR CLARITY. SEE ISOMETRIC ON 062D907.

C. FOR WALL PIPE PENETRATIONS, SEE DETAILS ON 005D505.

D. FOR PIPE SLAB PENETRATIONS, SEE DETAIL 6/005D505.

E. SEE 062D906 FOR VALVE TAG LOCATIONS ON NORTH WALL PUMPS.

F. SEE 060D605 FOR PIPE SCHEDULE.

KEY NOTES (X)

1. EQUIPMENT INCLUDED IN UF MEMBRANE SUPPLIER'S SCOPE.
2. PIPE TRENCH BELOW SLAB. ALUMINUM PLANKING NOT SHOWN FOR CLARITY.
3. TANK LADDER AND RAILINGS DESIGNED BY TANK SUPPLIER. SEE 060S001 FOR STRUCTURAL DESIGN CRITERIA.
4. ENCASE PIPE BELOW SLAB IN CONCRETE. SEE DETAIL 1/005D504.
5. SEE 062D501 FOR REACTION TANK DETAILS.
6. CONNECTION POINT FOR TEMPORARY RECYCLE LINE DURING PLANT STARTUP. SALVAGE VALVE TO GOVERNMENT AFTER STARTUP AND INSTALL BLIND FLANGE ON TEE.
7. FLANGED COUPLING ADAPTER (FCA).
8. PIPE PENETRATIONS IN SLAB ABOVE FOR LEVEL INSTRUMENTS. LOCATE IN FIELD WITH MIN 12" BETWEEN CORES. SEE SHEETS 009Y502 AND 009H503 FOR MOUNTING DETAILS.

SCALE (A) 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 6/11/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: D. DEATERLA TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D401	TITLE OF SHEET WATER TREATMENT PROCESS FLOOR ENLARGED PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 99 OF 192

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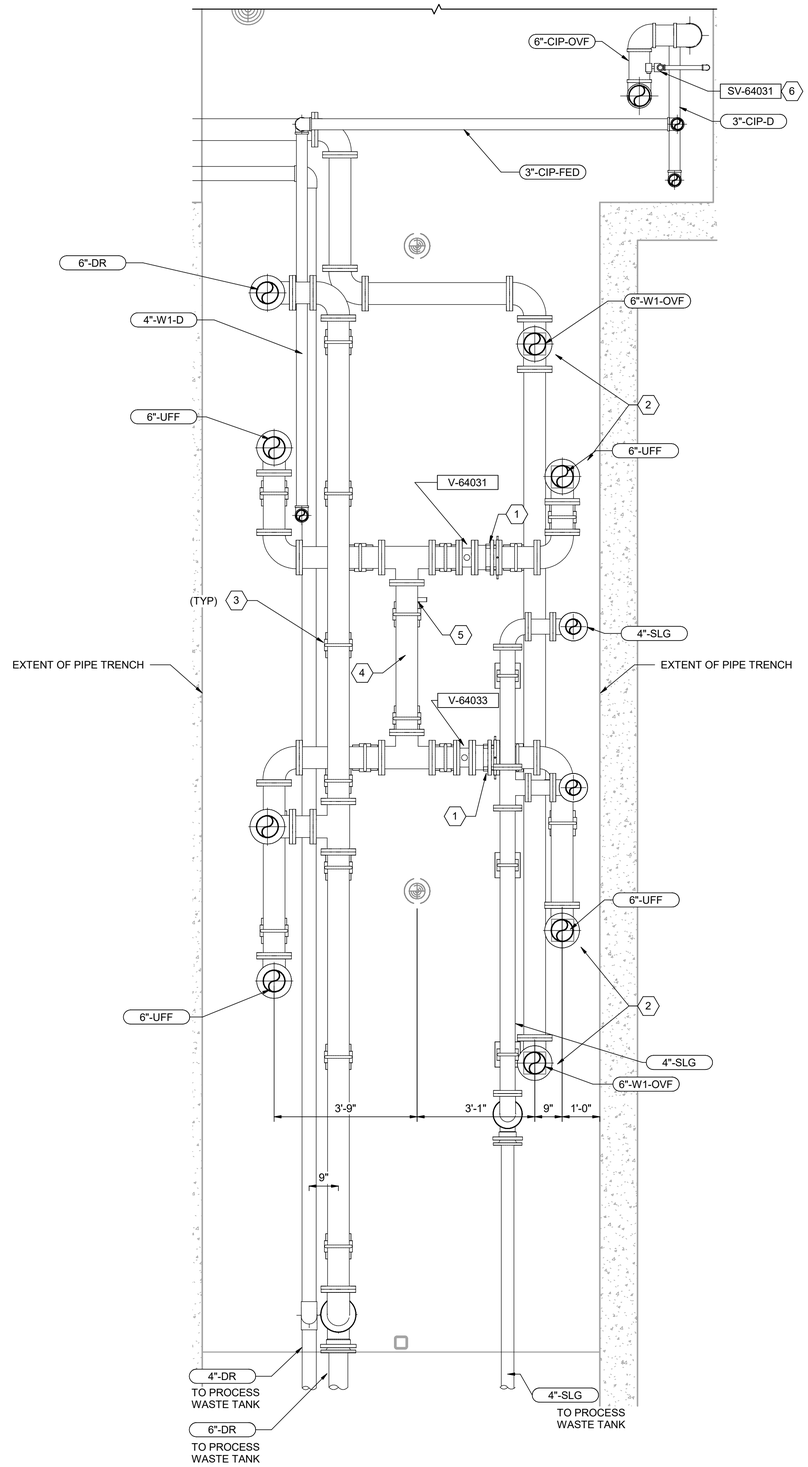

A 062D401 PIPE TRENCH ENLARGED PLAN

1/2" = 1'-0"

KEY NOTES (X)

1. FLANGED COUPLING ADAPTER.
2. BASE ELBOW. SEE DETAIL 5/005D501.
3. FLOOR PIPE SUPPORT, SEE DETAIL 1/005D501.
4. 1" TAP AND BALL VALVE AT BOTTOM OF PIPE FOR DRAINING. SEE DETAIL 2/005D503.
5. 1" TAP AND BALL VALVE ON 6"-UFF PIPE. SEE DETAIL 2/005D503. ROUTE 1" SMP TO PMP 64011.
6. 1" DRAIN AND SOLENOID ON 3"-CIP-FED. ROUTE TO FLOOR SINK DRAIN IN TRENCH.

SCALE (A) 2 0 2 4 1/2" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 4/15/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: D. DEATERLA TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D402	TITLE OF SHEET WATER TREATMENT PIPE TRENCH ENLARGED PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 100 OF 192

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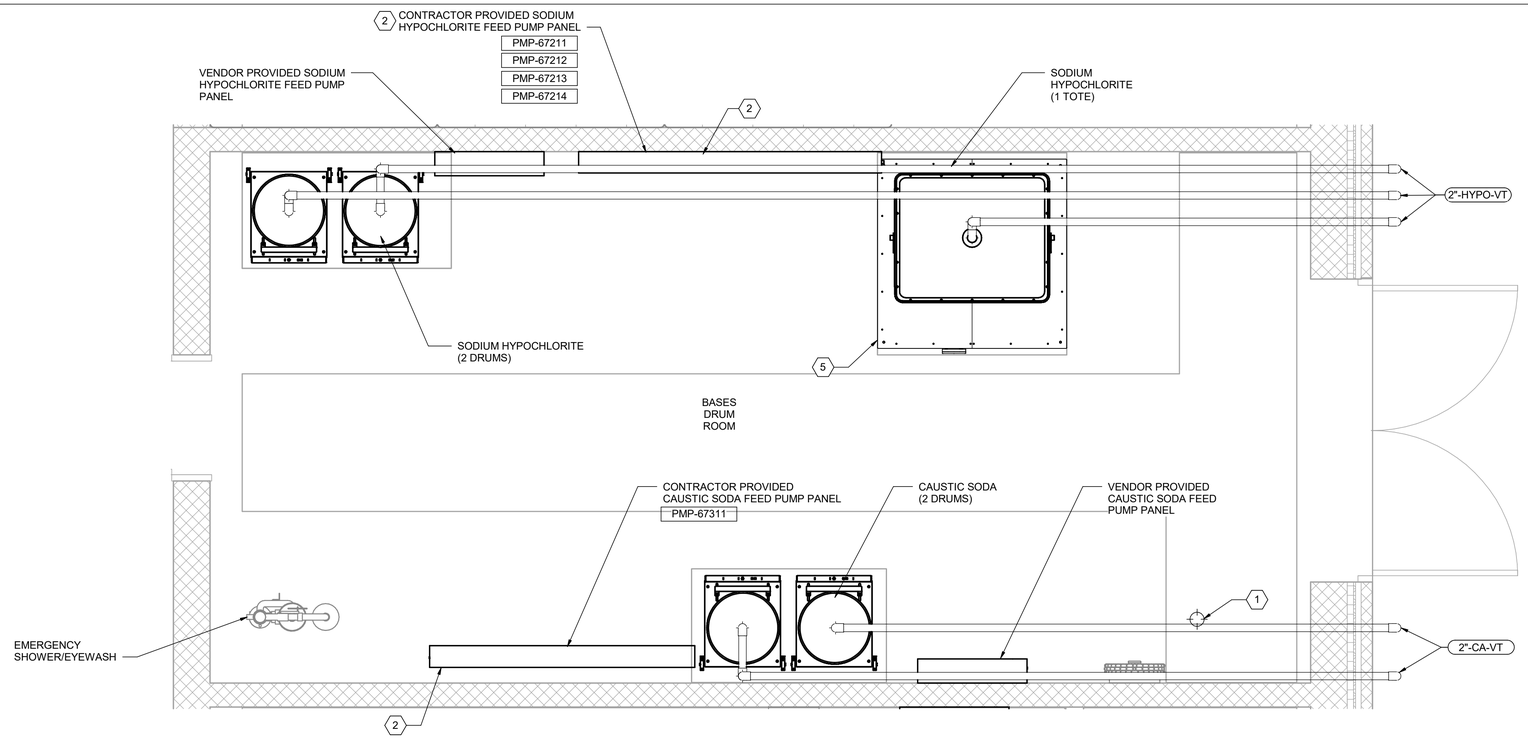
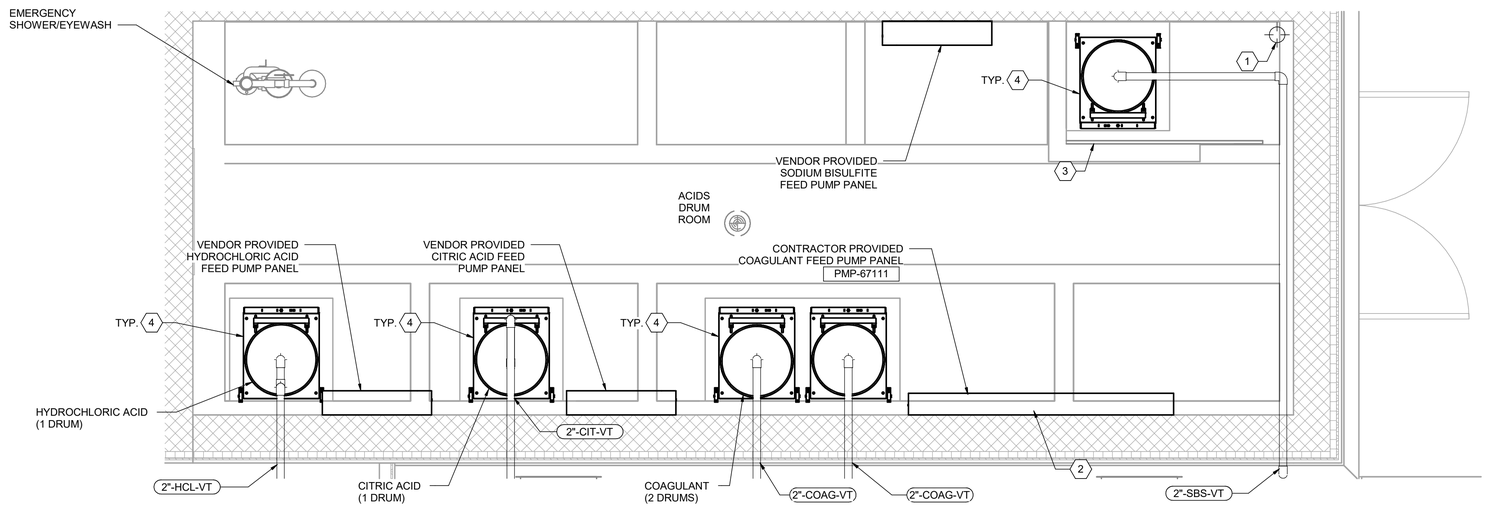


A 062D101 BASES DRUM ROOM ENLARGED PLAN

1/2" = 1'-0"



B 062D101 ACIDS DRUM ROOM ENLARGED PLAN

1/2" = 1'-0"

GENERAL NOTES

A. GRATING OVER CONTAINMENT SUMPS NOT SHOWN FOR CLARITY.

B. SMALL DIAMETER CHEMICAL, SAMPLE, AND COMPRESSED AIR PIPING NOT SHOWN FOR CLARITY. SEE ISOMETRIC ON 062D907.

C. ALL DRUM VENTS SHALL BE ROUTED INDIVIDUALLY AND BE TERMINATED 8' ABOVE FF (MIN) WITH 90-DEG ELBOW DOWN.

D. SEE SHEET 062D907 FOR ROUTING OF CHEMICAL FEED LINES IN PROCESS AREA.

E. ALL VENDOR PROVIDED PANELS INCLUDED IN UF MEMBRANE SUPPLIER'S SCOPE.

KEY NOTES (X)

1. 4" DRAIN FOR NEUTRALIZED CHEMICAL DISPOSAL TO WWTP. SEE SECTION 10 14 23 FOR SIGNAGE REQUIREMENT.
2. MAX PANEL SIZE SHALL BE 84" WIDE AND 66" TALL.
3. PROVIDE NON-COMBUSTIBLE REMOVABLE PARTITION WALL AROUND SODIUM BISULFITE DRUM. PARTITION MUST EXTEND 18" ABOVE TOP OF DRUM.
4. DRUM SCALE.
5. TOTE SCALE.

SCALE (A) 2 0 2 4 1/2" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: D. DEATERLA TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D403	TITLE OF SHEET WATER TREATMENT CHEMICAL DRUM ROOM ENLARGED PLANS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 101 OF 192

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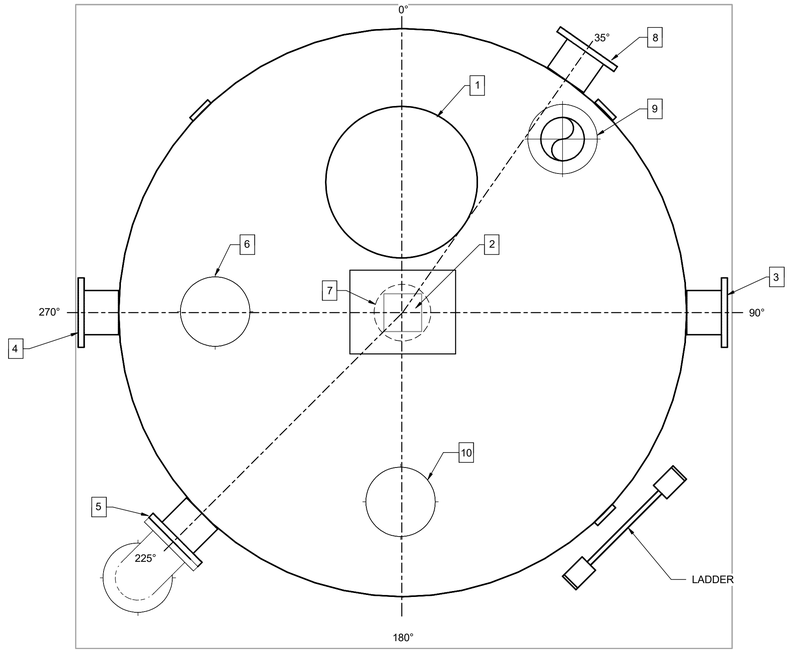


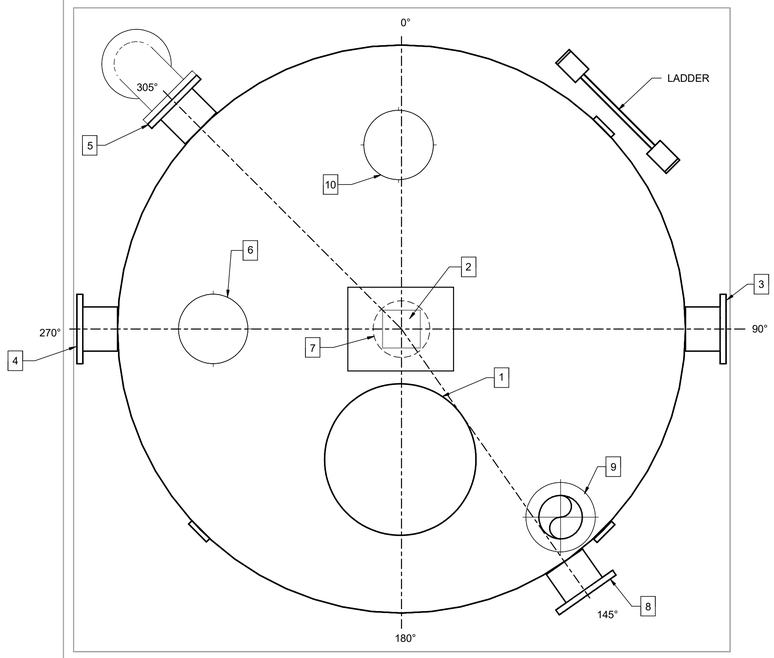
1 TK-63011 REACTION TANK OPENING SCHEDULE

1" = 1'-0"

ITEM	DESCRIPTION	VERTICAL LOCATION ON TANK	HORIZONTAL DISTANCE FROM CENTER OF TANK TO CENTERLINE OF ITEM (IF LOCATED ON TOP OF TANK)	COMMENT
1	24" MANWAY	TOP OF TANK	1'-4"	HINGED
2	OPENING IN TANK	TOP OF TANK	CENTER OF TANK (NOTE 2)	OPENING FOR MIXER
3	6" NOZZLE	4'-10" (NOTE 1)	N/A	INLET
4	6" NOZZLE	1' (NOTE 1)	N/A	OUTLET
5	6" NOZZLE	6'-1" (NOTE 1)	N/A	OVERFLOW (NOTE 3)
6	6" NOZZLE	TOP OF TANK	2'-5"	SPARE
7	4" FLANGE	BOTTOM OF TANK	1'-6"	SLUDGE
8	6" NOZZLE	5" (NOTE 1)	N/A	LIT
9	6" NOZZLE	TOP OF TANK	3'-1"	VENT
10	6" NOZZLE	TOP OF TANK	2'-6"	LEVEL SWITCH

NOTES:

1. DISTANCE FROM BOTTOM OF TANK STRAIGHT WALL TO CENTERLINE OF NOZZLE.
2. CONTRACTOR SHALL COORDINATE OPENING SIZE IN TANK AND MIXER STRUCTURAL SUPPORTS WITH MIXER SUPPLIER.
3. TANK SUPPLIER SHALL PROVIDE OVERFLOW PIPE AND ASSOCIATED BRACING OFF TANK WALL. SEE 062D305 FOR TRANSITION POINT.
4. GUARDRAILS AT TOP OF TANK NOT SHOWN.



2 TK-63012 REACTION TANK OPENING SCHEDULE

1" = 1'-0"

ITEM	DESCRIPTION	VERTICAL LOCATION ON TANK	HORIZONTAL DISTANCE FROM CENTER OF TANK TO CENTERLINE OF ITEM (IF LOCATED ON TOP OF TANK)	COMMENT
1	24" MANWAY	TOP OF TANK	1'-4"	HINGED
2	OPENING IN TANK	TOP OF TANK	CENTER OF TANK (NOTE 2)	OPENING FOR MIXER
3	6" NOZZLE	4'-10" (NOTE 1)	N/A	INLET
4	6" NOZZLE	1' (NOTE 1)	N/A	OUTLET
5	6" NOZZLE	6'-1" (NOTE 1)	N/A	OVERFLOW (NOTE 3)
6	6" NOZZLE	TOP OF TANK	2'-5"	SPARE
7	4" FLANGE	BOTTOM OF TANK	1'-6"	SLUDGE
8	6" NOZZLE	5" (NOTE 1)	N/A	LIT
9	6" NOZZLE	TOP OF TANK	3'-1"	VENT
10	6" NOZZLE	TOP OF TANK	2'-6"	LEVEL SWITCH

NOTES:

1. DISTANCE FROM BOTTOM OF TANK STRAIGHT WALL TO CENTERLINE OF NOZZLE.
2. CONTRACTOR SHALL COORDINATE OPENING SIZE IN TANK AND MIXER STRUCTURAL SUPPORTS WITH MIXER SUPPLIER.
3. TANK SUPPLIER SHALL PROVIDE OVERFLOW PIPE AND ASSOCIATED BRACING OFF TANK WALL. SEE 062D305 FOR TRANSITION POINT.
4. GUARDRAILS AT TOP OF TANK NOT SHOWN.

SCALE (A) 1 0 1 2 1" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed: 5/1/2026 ARIZONA U.S.A.

A/E FIRM	DESIGNED: M. HOWARD	SUB SHEET NO. 062 D501	TITLE OF SHEET: WATER TREATMENT REACTION TANK DETAILS
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DRAWN: J. STUJENSKE		DRAWING NO. 113 182752
	TECH. REVIEW: M. BEEBE		PMIS/PKG NO. 10350545
	DATE: MAY 2025		SHEET 102 OF 192

NORTH RIM - GRAND CANYON NATIONAL PARK

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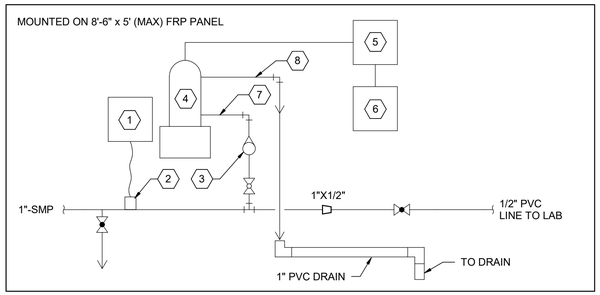
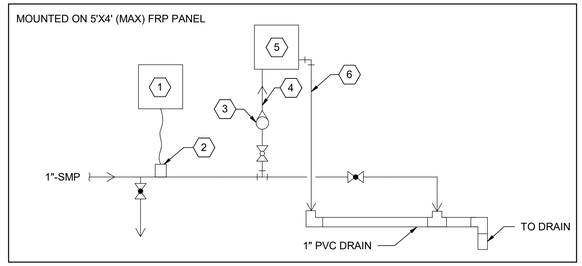


1. PH TRANSMITTER
2. PH PROBE INSTALLED AT 1/2" NPT THREADED CONNECTION
3. ROTAMETER
4. TURBIDIMETER
5. TURBIDIMETER PROBE MODULE
6. TURBIDIMETER DISPLAY MODULE
7. 1/4" OD POLYETHYLENE INLET TUBING
8. 1/4" OD POLYETHYLENE OUTLET TUBING

1 - RAW WATER INSTRUMENTATION PANEL

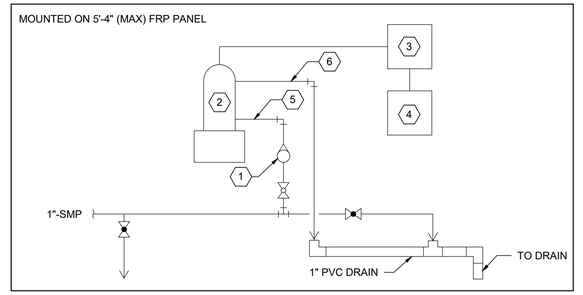
NOT TO SCALE



1. PH TRANSMITTER
2. PH PROBE INSTALLED AT 1/2" NPT THREADED CONNECTION
3. ROTAMETER
4. 1/4" OD POLYETHYLENE INLET TUBING
5. RESIDUAL CHLORINE ANALYZER
6. 1/4" OD POLYETHYLENE OUTLET TUBING

2 - UF FEED INSTRUMENTATION PANEL

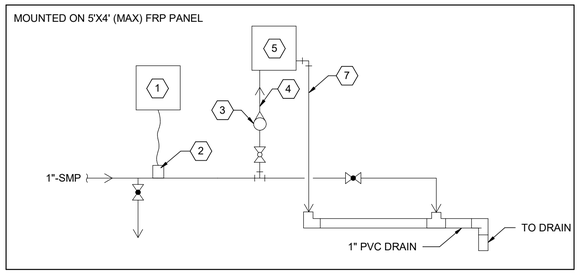
NOT TO SCALE



1. ROTAMETER
2. TURBIDIMETER
3. TURBIDIMETER PROBE MODULE
4. TURBIDIMETER DISPLAY MODULE
5. 1/4" OD POLYETHYLENE INLET TUBING
6. 1/4" OD POLYETHYLENE OUTLET TUBING

3 - COMBINED FILTRATE INSTRUMENTATION PANEL

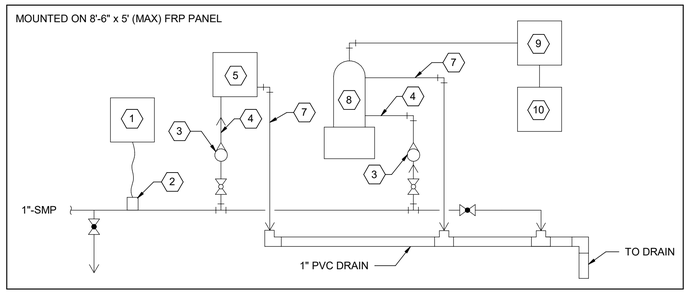
NOT TO SCALE



1. PH TRANSMITTER
2. PH PROBE INSTALLED AT 1/2" NPT THREADED CONNECTION
3. ROTAMETER
4. 1/4" OD POLYETHYLENE INLET TUBING
5. RESIDUAL CHLORINE ANALYZER
6. 1/4" OD POLYETHYLENE OUTLET TUBING

4 - CT TANK INFLUENT INSTRUMENTATION PANEL

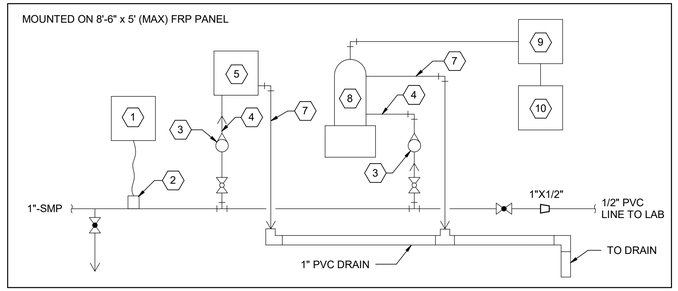
NOT TO SCALE



1. PH TRANSMITTER
2. PH PROBE INSTALLED AT 1/2" NPT THREADED CONNECTION
3. ROTAMETER
4. 1/4" OD POLYETHYLENE INLET TUBING
5. RESIDUAL CHLORINE ANALYZER
6. 1/4" OD POLYETHYLENE OUTLET TUBING
7. TURBIDIMETER
8. TURBIDIMETER PROBE MODULE
9. TURBIDIMETER DISPLAY MODULE

5 - FINISHED WATER INSTRUMENTATION PANEL

NOT TO SCALE



1. PH TRANSMITTER
2. PH PROBE INSTALLED AT 1/2" NPT THREADED CONNECTION
3. ROTAMETER
4. 1/4" OD POLYETHYLENE INLET TUBING
5. RESIDUAL CHLORINE ANALYZER
6. 1/4" OD POLYETHYLENE OUTLET TUBING
7. TURBIDIMETER
8. TURBIDIMETER PROBE MODULE
9. TURBIDIMETER DISPLAY MODULE

6 - DISTRIBUTION WATER INSTRUMENTATION PANEL

NOT TO SCALE

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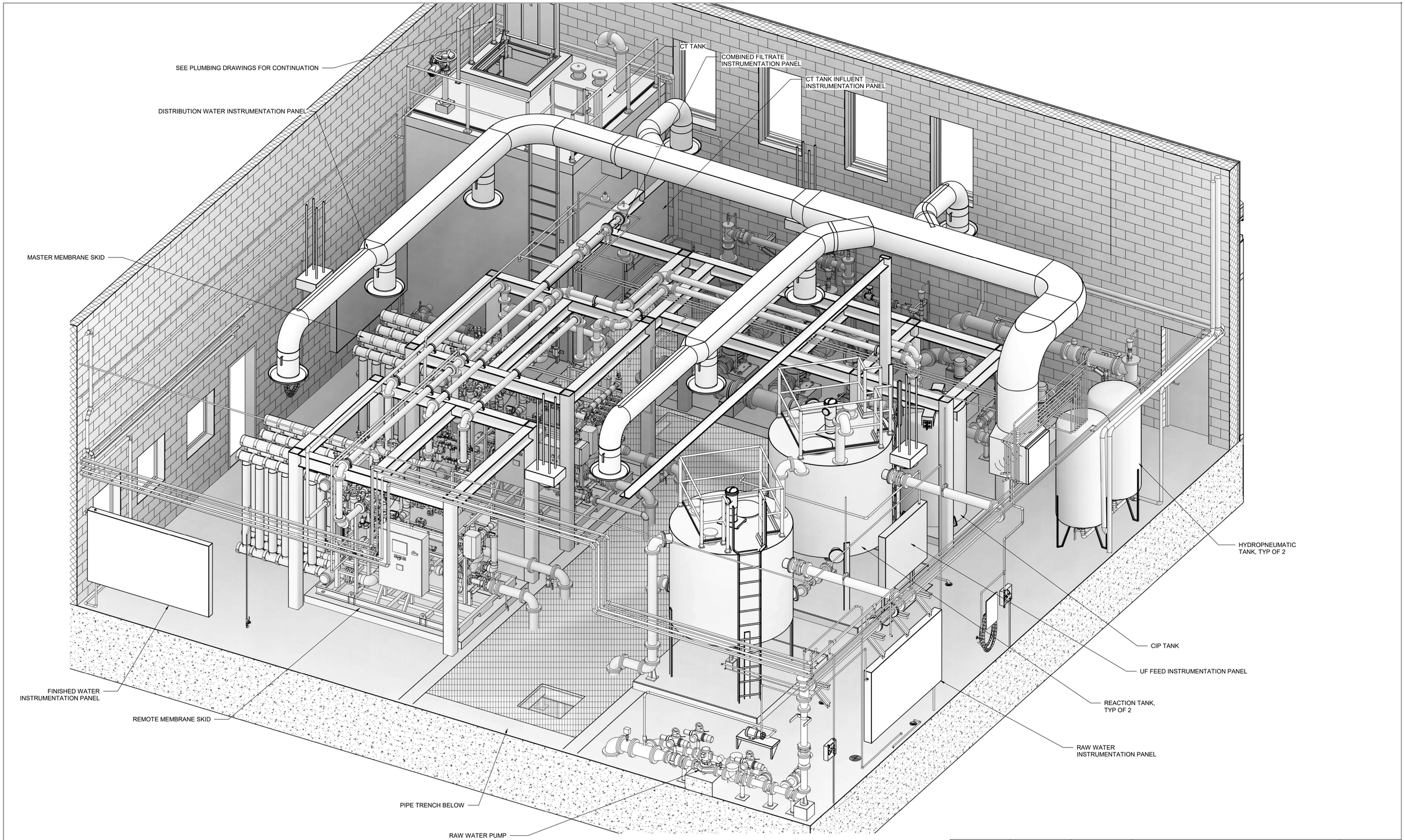
Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 4/15/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: J. STUJENSKE TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D502	TITLE OF SHEET WATER TREATMENT INSTRUMENTATION PANEL DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 103 OF 192



ISOMETRIC VIEW

NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD CADD D. DEATERLA TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D901	TITLE OF SHEET WATER TREATMENT PROCESS ROOM ISOMETRIC VIEW NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 104 OF 192

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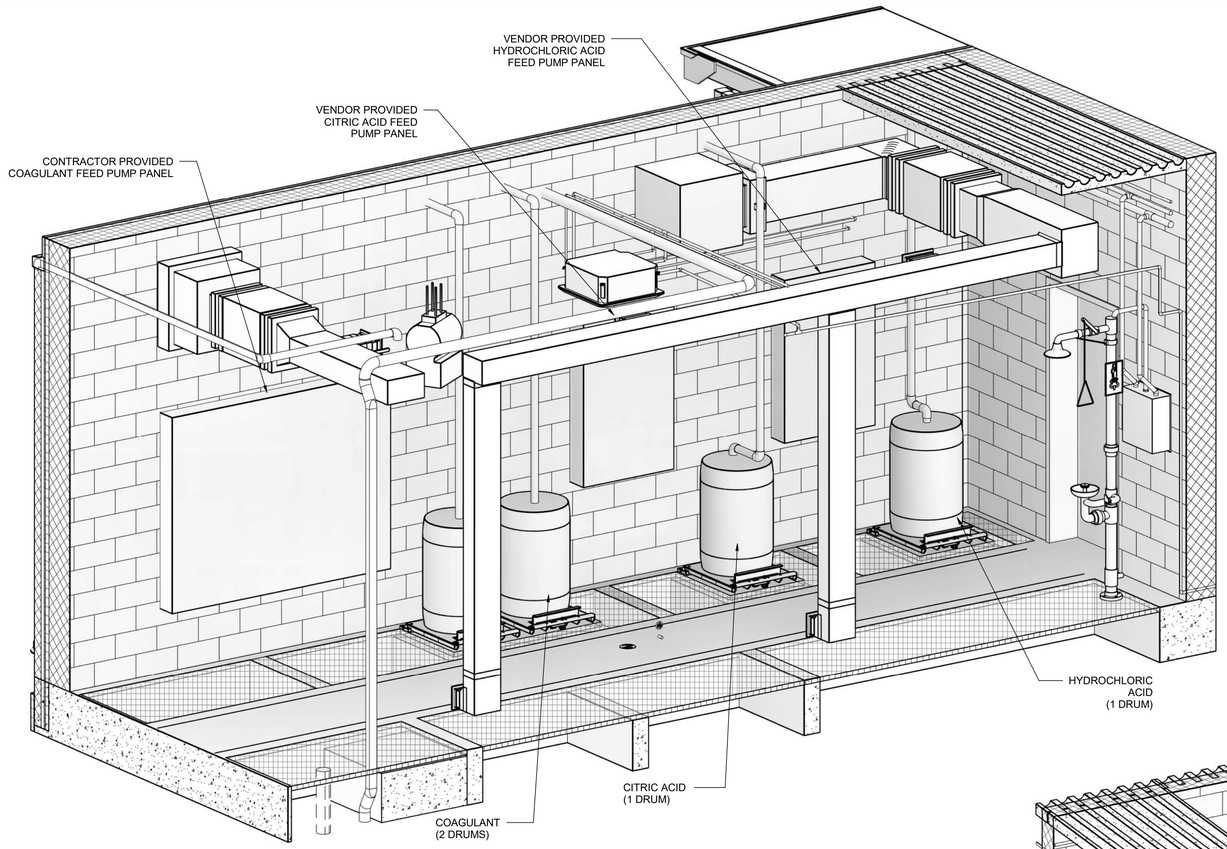
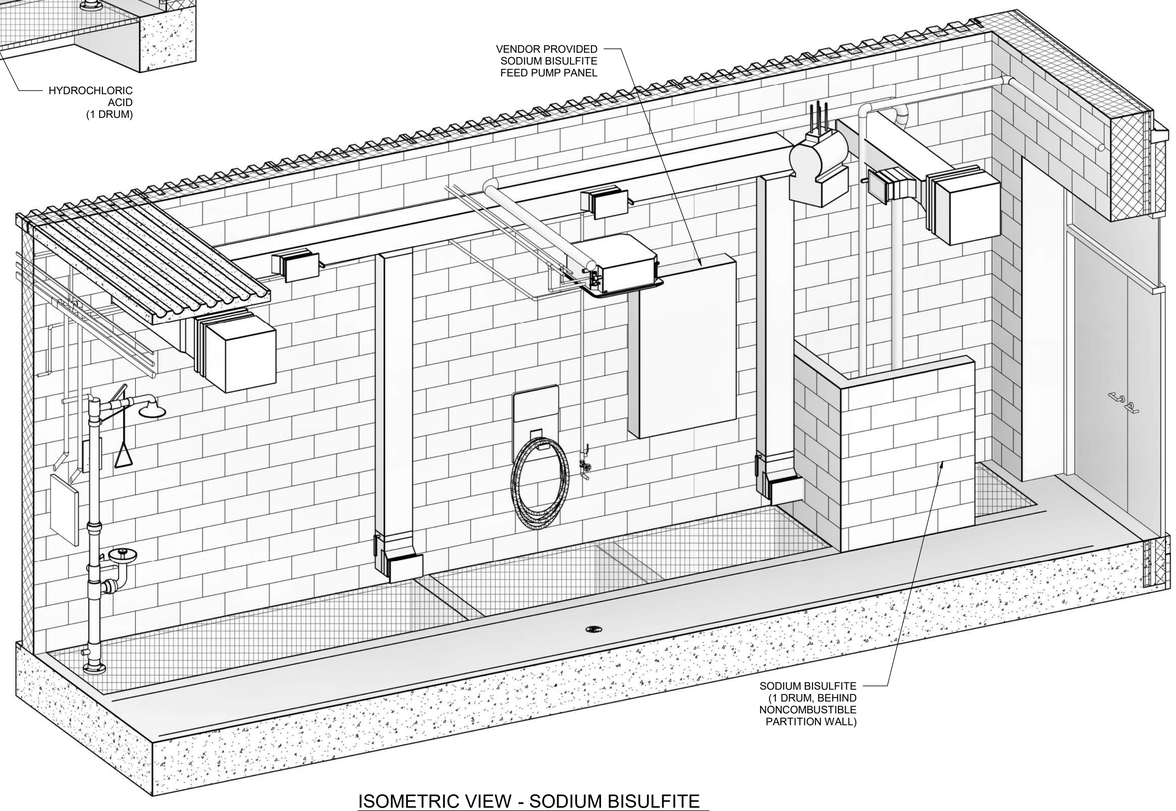


ISOMETRIC VIEW - ACIDS

NOT TO SCALE



ISOMETRIC VIEW - SODIUM BISULFITE

NOT TO SCALE

Registered Professional Engineer (Civil)
CERTIFICATE NO.
75827
JENNIFER
STILLMAN
Date Signed 5/1/2025
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD CADD D. DEATERLA TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D902	TITLE OF SHEET WATER TREATMENT ACIDS CHEMICAL ROOM ISOMETRIC VIEWS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 105 OF 192

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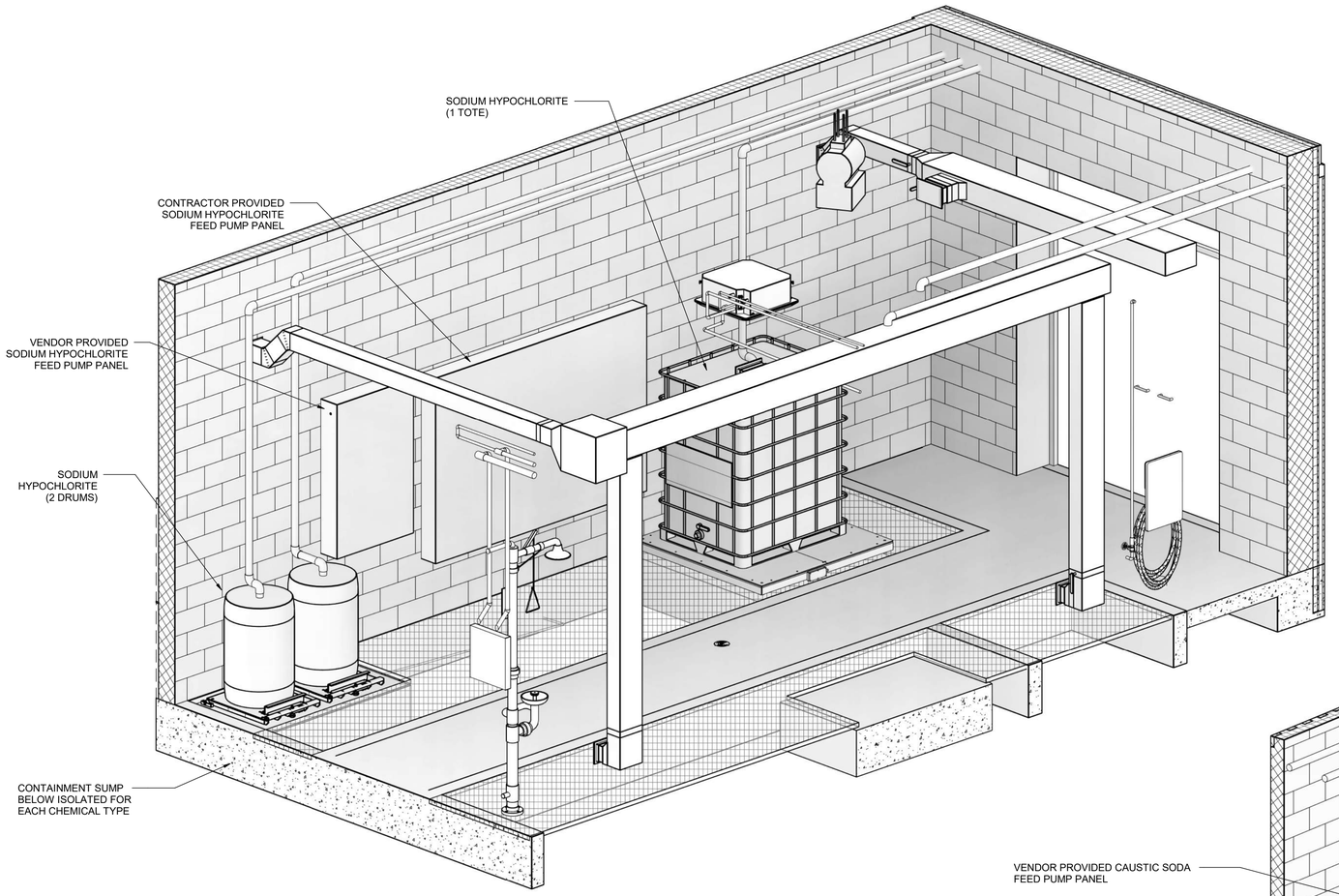
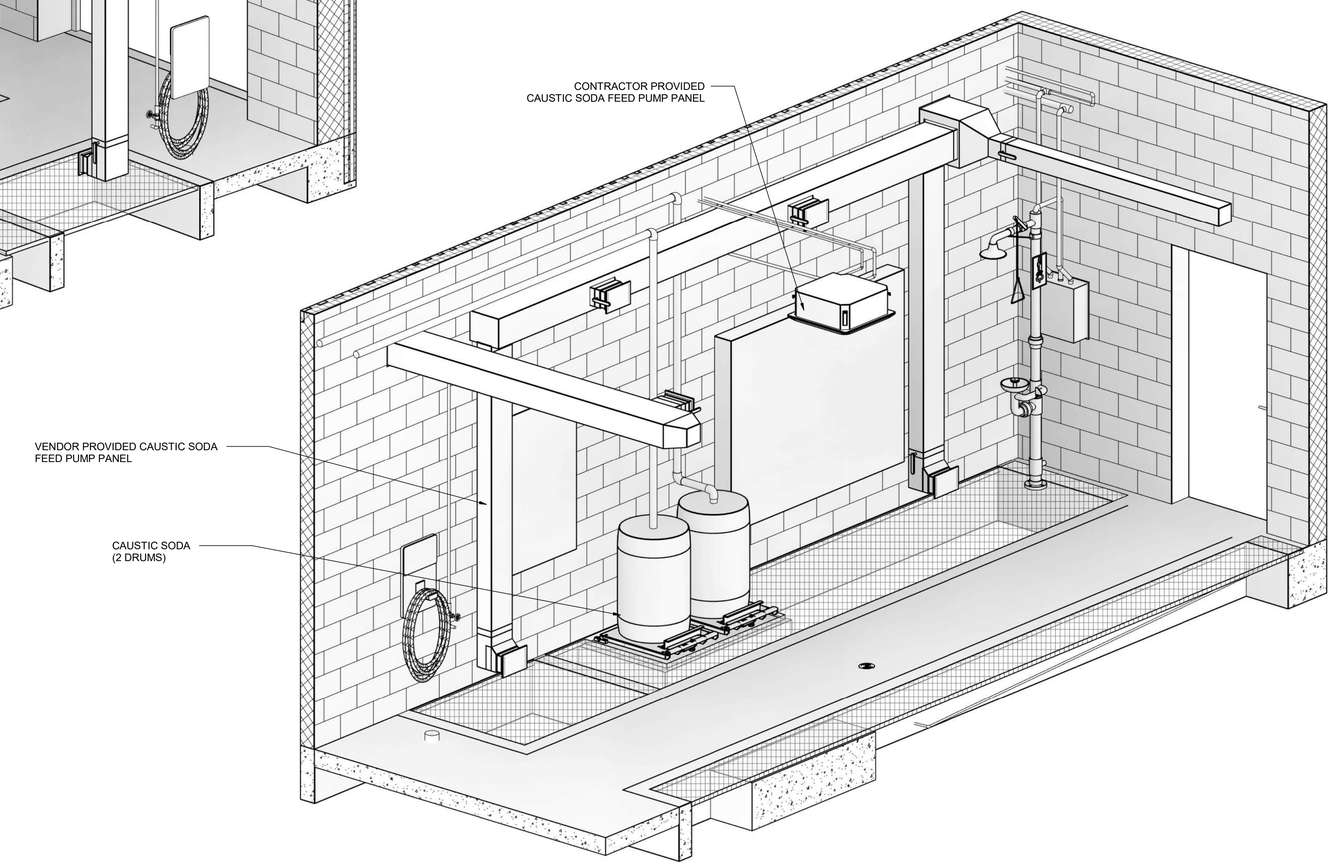


ISOMETRIC VIEW - SODIUM HYPOCHLORITE

NOT TO SCALE



ISOMETRIC VIEW - CAUSTIC

NOT TO SCALE

Registered Professional Engineer (Civil)
CERTIFICATE NO.
75827
JENNIFER
STILLMAN
Date Signed 4/15/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: D. DEATERLA TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D903	TITLE OF SHEET WATER TREATMENT BASES CHEMICAL ROOM ISOMETRIC VIEWS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 106 OF 192

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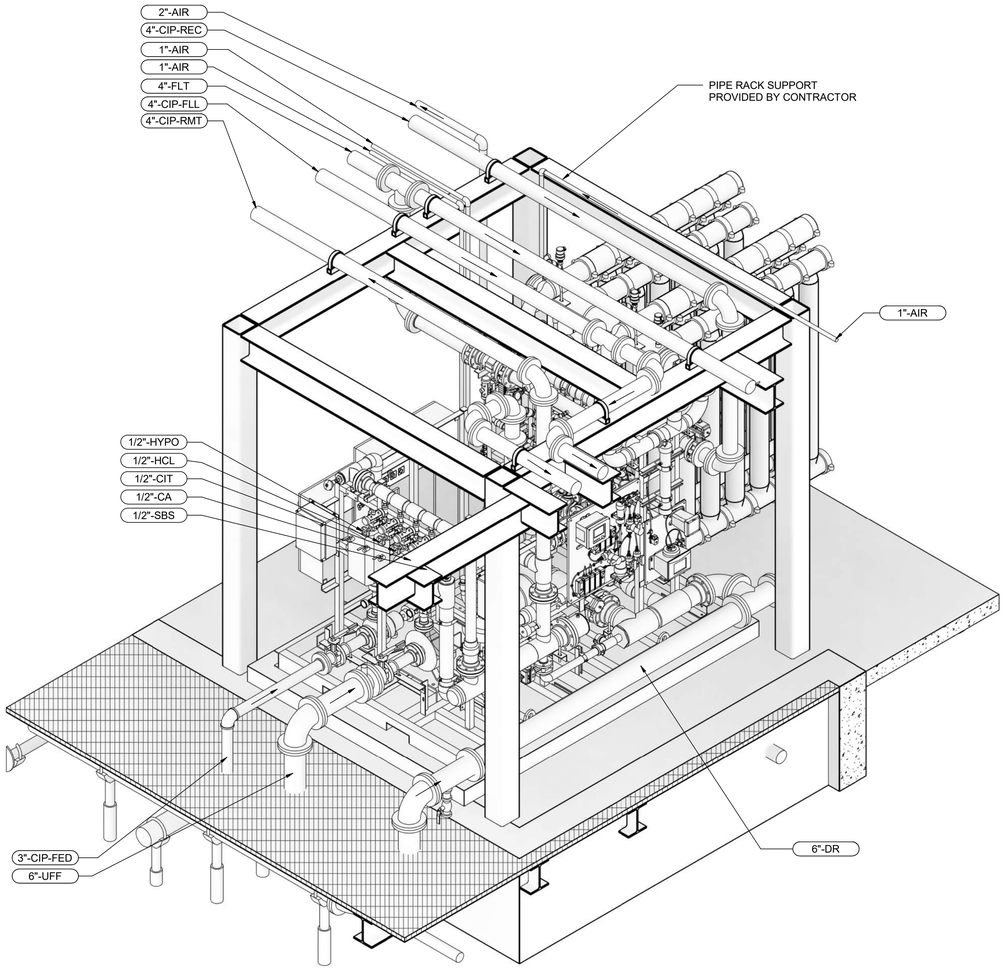
GENERAL NOTES

A. SOME CONTRACTOR-SUPPLIED PIPING SHOWN FOR CLARITY. REFERENCE UF MEMBRANE SUPPLIER'S GENERAL ARRANGEMENT DRAWINGS FOR PIPING CONNECTION POINTS AT SKID.



UF SKID 1 3D VIEW
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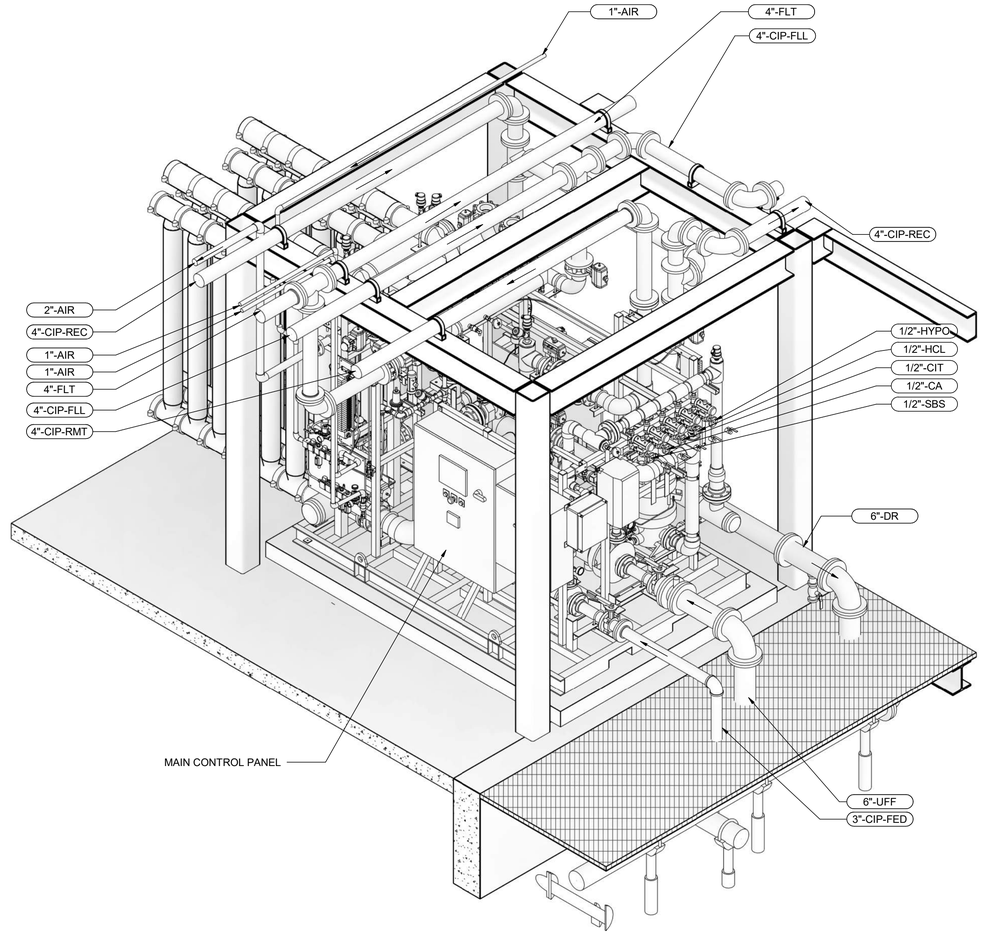

UF SKID 1 3D VIEW
NOT TO SCALE

Registered Professional Engineer (Civil)
CERTIFICATE NO.
75827
JENNIFER
STILLMAN
Date Signed 4/15/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAFTED: J. STUJENSKE TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D904	TITLE OF SHEET WATER TREATMENT PROCESS ROOM UF MASTER SKID ISOMETRICS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 107 OF 192

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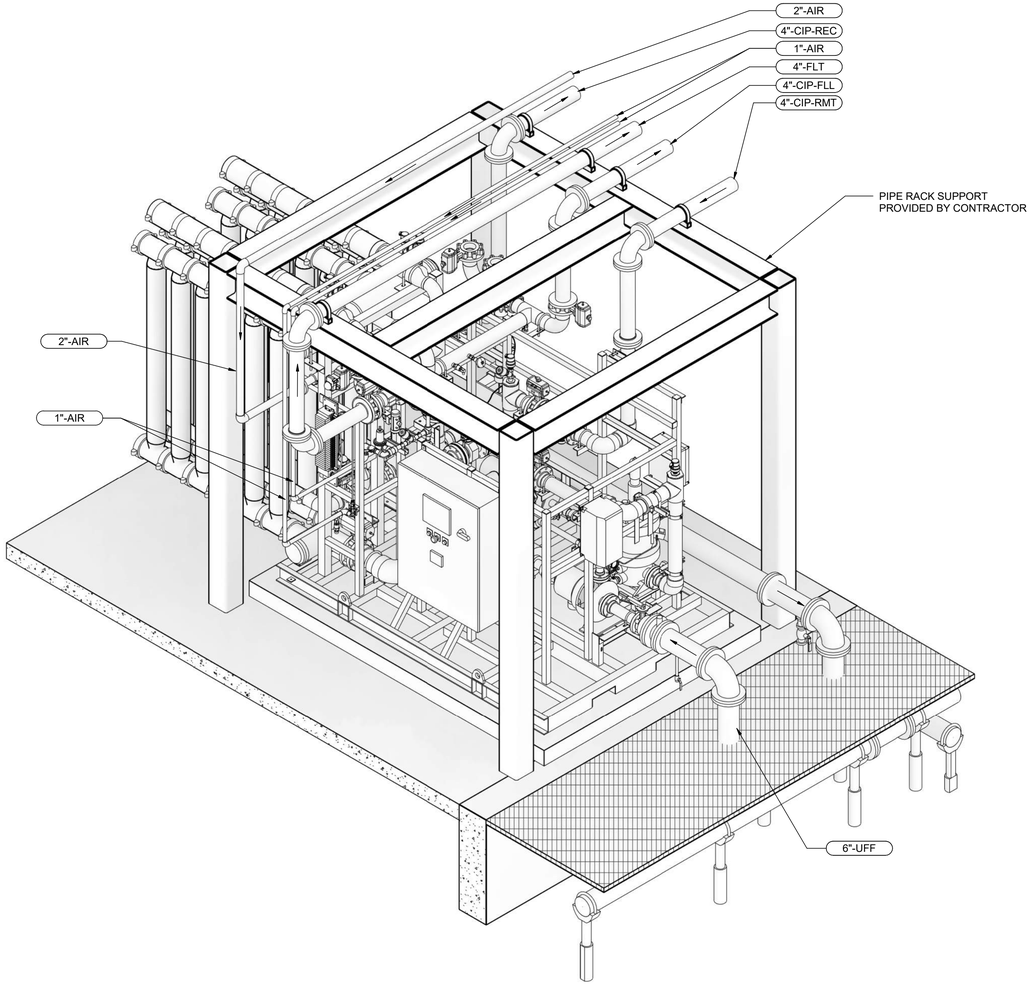
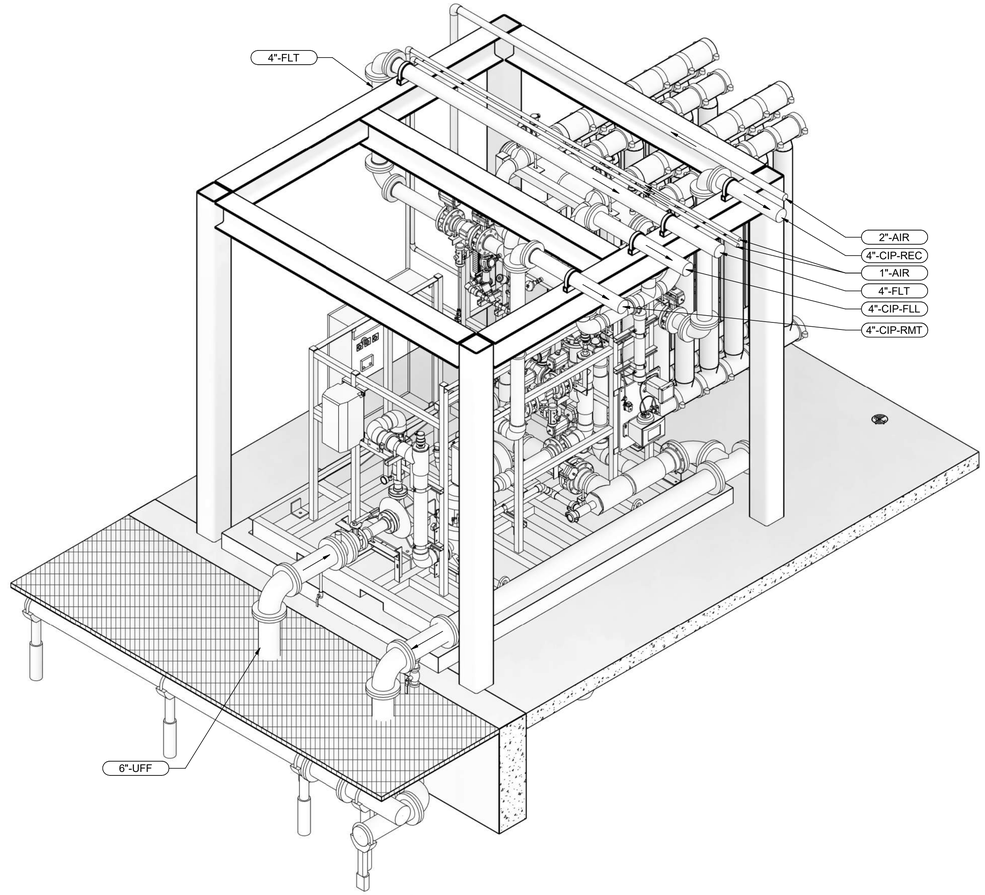
GENERAL NOTES

A. SOME CONTRACTOR-SUPPLIED PIPING SHOWN FOR CLARITY. REFERENCE UF MEMBRANE SUPPLIER'S GENERAL ARRANGEMENT DRAWINGS FOR PIPING CONNECTION POINTS AT SKID.



UF SKID 2 3D VIEW
NOT TO SCALE



UF SKID 2 3D VIEW
NOT TO SCALE

Registered Professional Engineer (Civil)
CERTIFICATE NO.
75827
JENNIFER STILLMAN
Date Signed 5/1/2025
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD CADD J. STUJENSKE TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D905	TITLE OF SHEET WATER TREATMENT PROCESS ROOM UF REMOTE SKID ISOMETRICS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 108 OF 192

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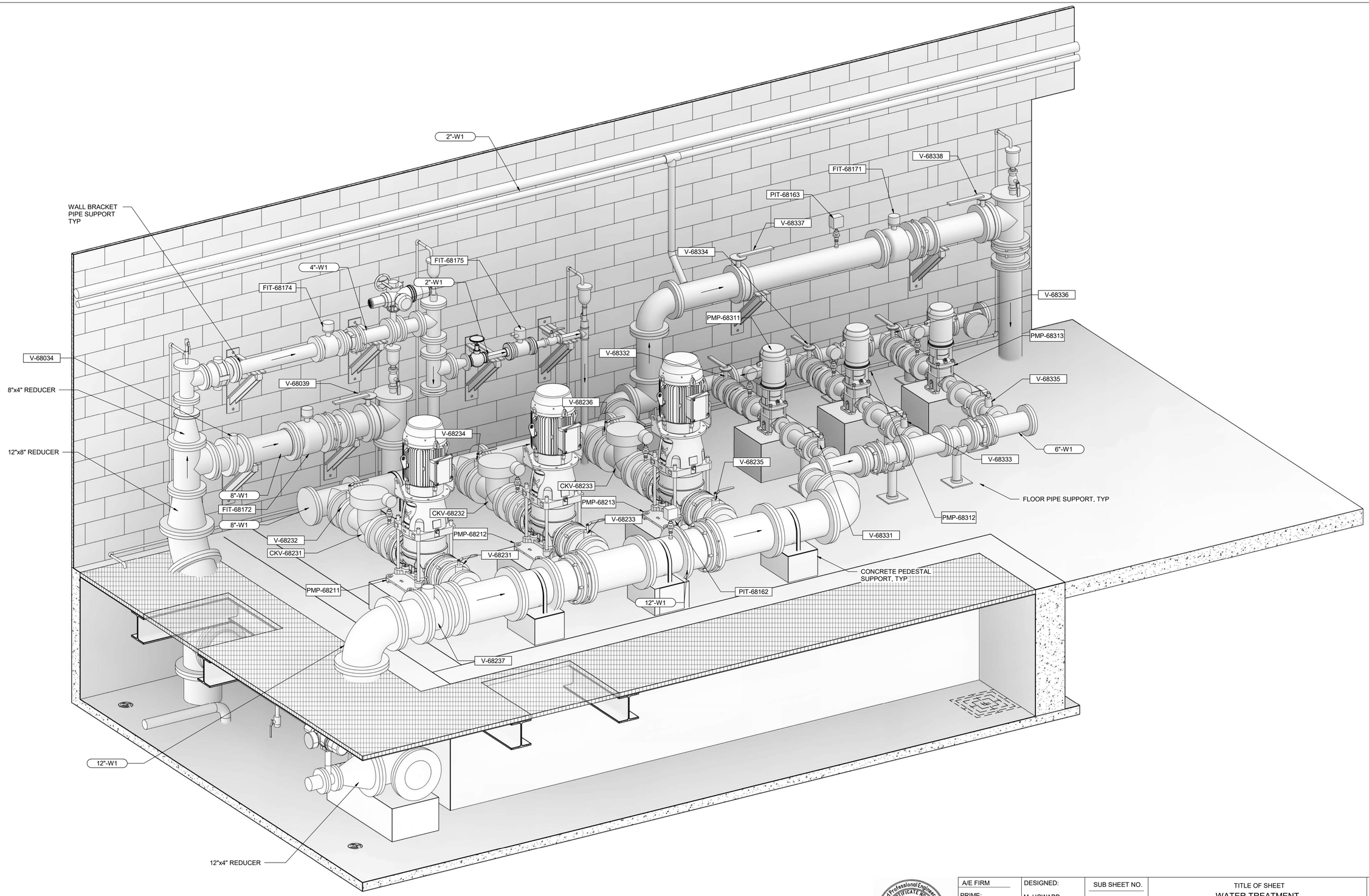

DISTRIBUTION PUMP 3D VIEW

NOT TO SCALE

Registered Professional Engineer (City) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: J. STUJENSKE TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D906	TITLE OF SHEET WATER TREATMENT PROCESS ROOM BOOSTER PUMPS ISOMETRIC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 109 OF 192

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GENERAL NOTES

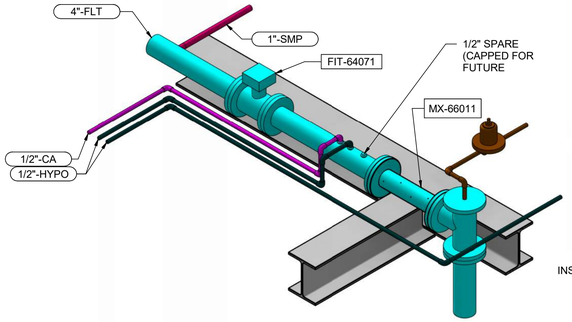
A. ALL CHEMICAL FEED LINES ROUTED ALONG WALLS SHALL BE HARD PIPED.

B. ALL CHEMICAL FEED LINES ROUTED ABOVE PERSONNEL WALKING PATHS SHALL BE TUBING ROUTED INSIDE 1.5" CONTAINMENT PIPE. SEE KEY NOTES ON THIS SHEET FOR LOCATIONS.

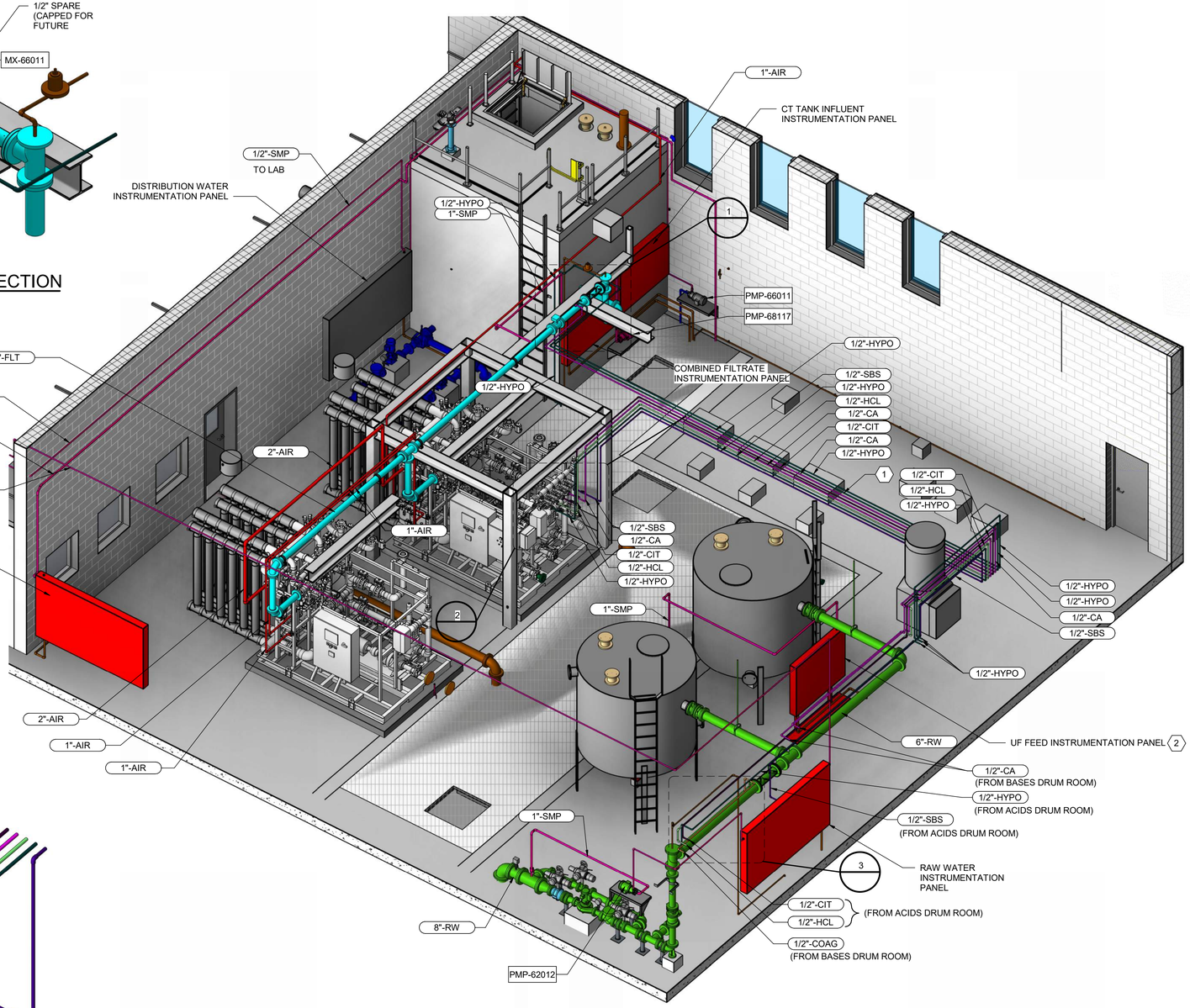
C. REFERENCE THE PIPE SCHEDULE ON SHEET 060D605 FOR MATERIAL TYPES OF FEED PIPING, TUBING, AND CONTAINMENT PIPING BY CHEMICAL TYPE.

KEY NOTES (X)

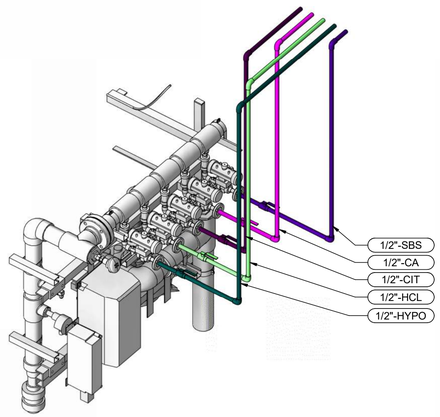
1. ALL CHEMICAL FEED LINES ROUTED ABOVE PERSONNEL WALKING PATH AT THIS LOCATION SHALL BE TUBING ROUTED INSIDE 1.5" CONTAINMENT PIPE FOR THE FULL DISTANCE BETWEEN WALLS AND/OR THE MEMBRANE SKIDS. LEAVE 2 FT OF COILED TUBING AT EACH END OF THE STRAIGHT CONTAINMENT PIPE RUN BEFORE TRANSITIONING TO HARD PIPE.
2. MOUNT INSTRUMENT PANEL PER DETAIL 5/005D504.



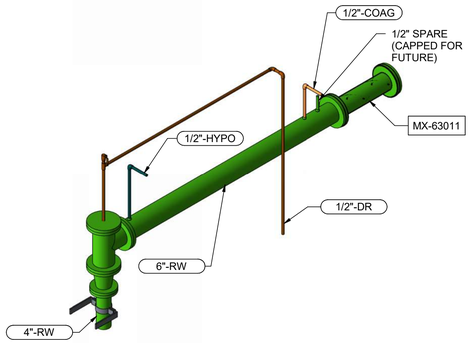
1 - FILTRATE SAMPLE INJECTION



COMPRESSED AIR, CHEMICAL, AND SAMPLE LINES 3D VIEW
NOT TO SCALE



2 - MEMBRANE SKID CHEMICAL INJECTION



3 - RW SAMPLE INJECTION

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 4/15/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD CADD J. STUJENSKE TECH. REVIEW: M. BEEBE DATE: MAY 2025	SUB SHEET NO. 062 D907	TITLE OF SHEET WATER TREATMENT COMPRESSED AIR, CHEMICAL, AND SAMPLE LINE ISOMETRIC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 110 OF 192

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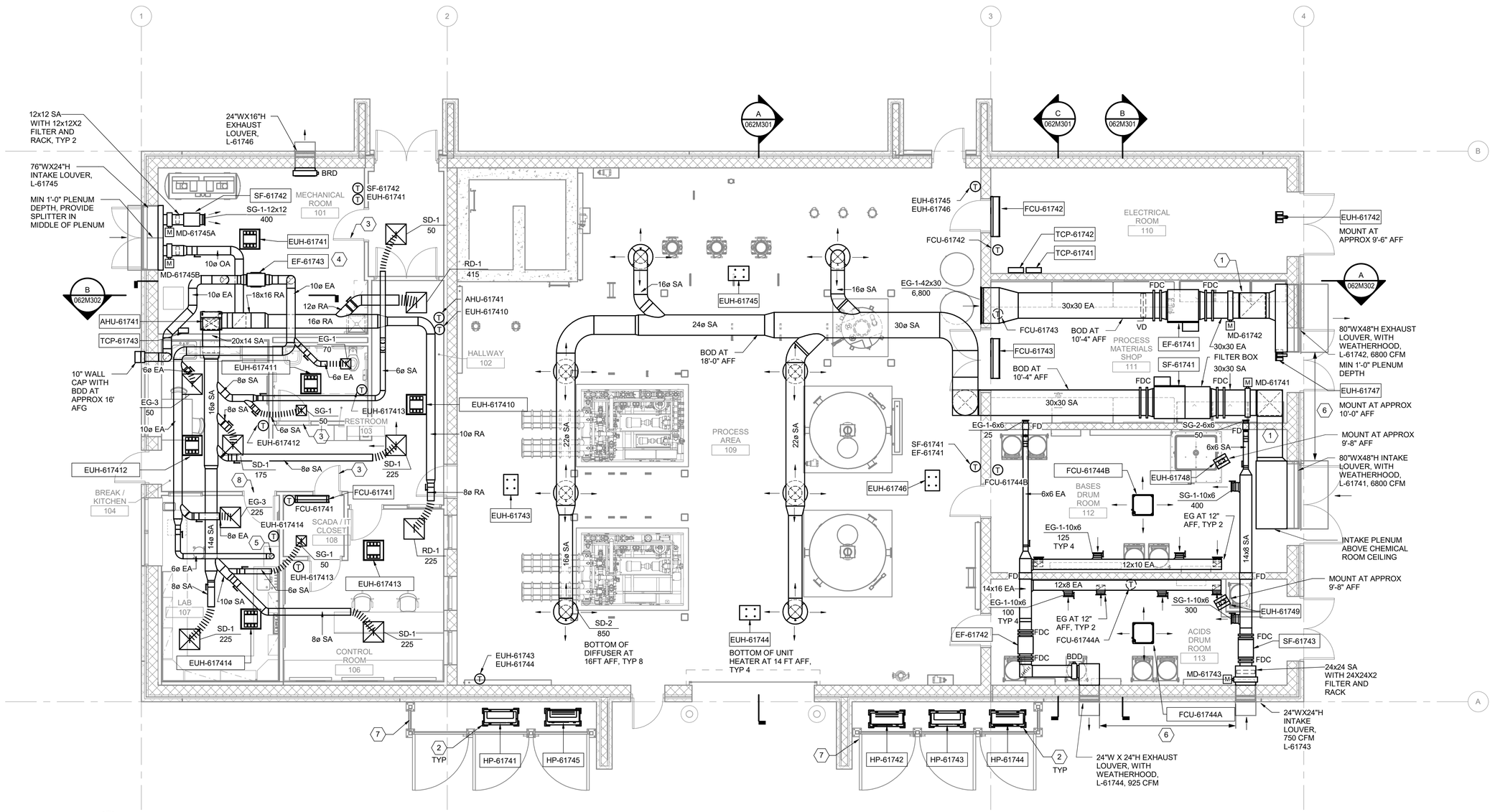

1 062M101 HVAC PLAN
SCALE (A)

KEYNOTES: (X)

1. INSULATE SUPPLY AND EXHAUST DUCTWORK BETWEEN FAN AND LOUVER, INCLUDING PLENUMS.
2. MOUNT HEAT PUMPS ON ELEVATED PLATFORM AT 3'-6" AFG. PROVIDE GALVANIZED STEEL SUPPORT STAND, DESIGNED BY CONTRACTOR, COAT WITH PRIMER, ANCHOR TO CONCRETE PAD. PROVIDE STAINLESS STEEL DRAIN PAN BENEATH HEAT PUMPS FOR DRAINAGE DURING DEFROST CYCLE.
3. UNCERCUT DOOR 3/4" TO ALLOW TRANSFER OF AIR.
4. EXHAUST FAN EXPOSED IN THE MECHANICAL ROOM, SUSPENDED BENEATH CEILING, APPROX 9'-0" AFF.
5. CONNECT DUCT TO CORROSIVES STORAGE CABINET, BALANCE TO APPROX 15 CFM.
6. 10 FT MIN BETWEEN INTAKE AND EXHAUST LOUVER.
7. LOUVERED ENCLOSURE. LOUVER UNRESTRICTED FREE AREA SHALL COMPLY WITH MECHANICAL EQUIPMENT MANUFACTURER REQUIREMENTS.
8. PROVIDE 14"X14" DOOR LOUVER FOR TRANSFER OF AIR.

SCALE (A) 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

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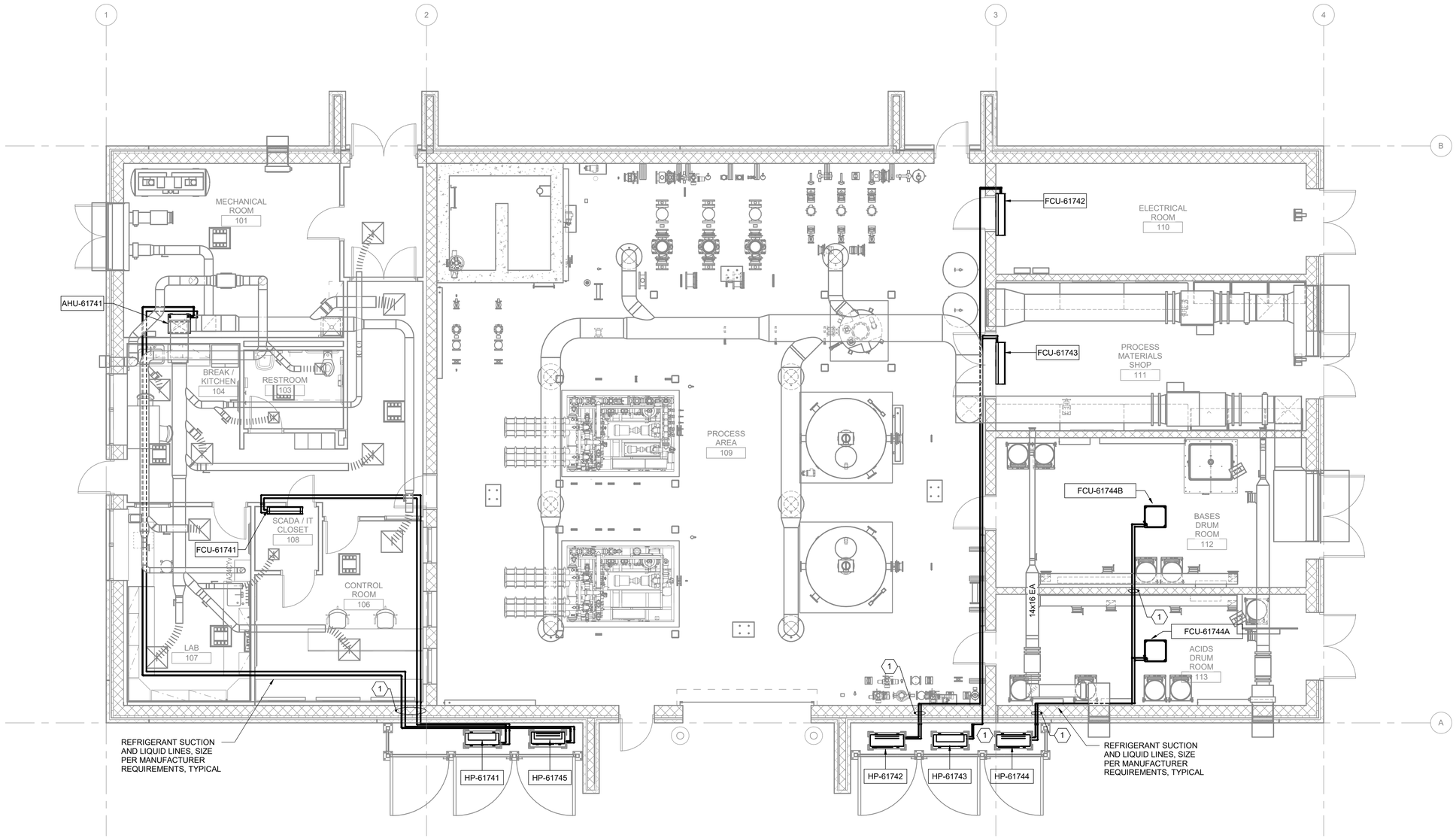
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 M101	TITLE OF SHEET WATER TREATMENT HVAC PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 111 OF 192



1 062M102 HVAC PIPING PLAN
SCALE A

KEYNOTES: X

1. REFER TO PROCESS DISCIPLINE 000 SERIES SHEETS FOR PIPE PENETRATION DETAILS.

SCALE A 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN CADD B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 M102	TITLE OF SHEET WATER TREATMENT HVAC PIPING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 112 OF 192

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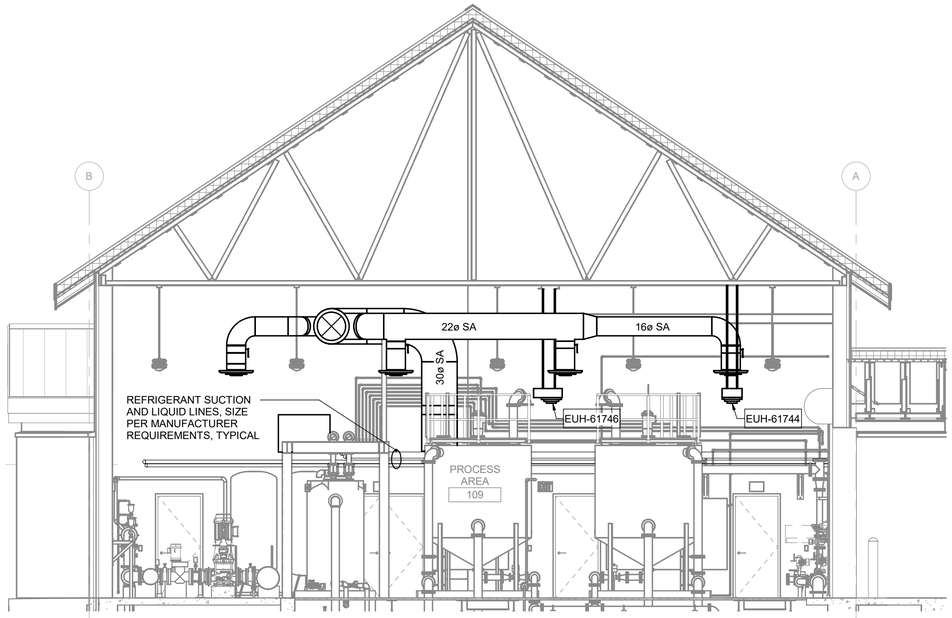
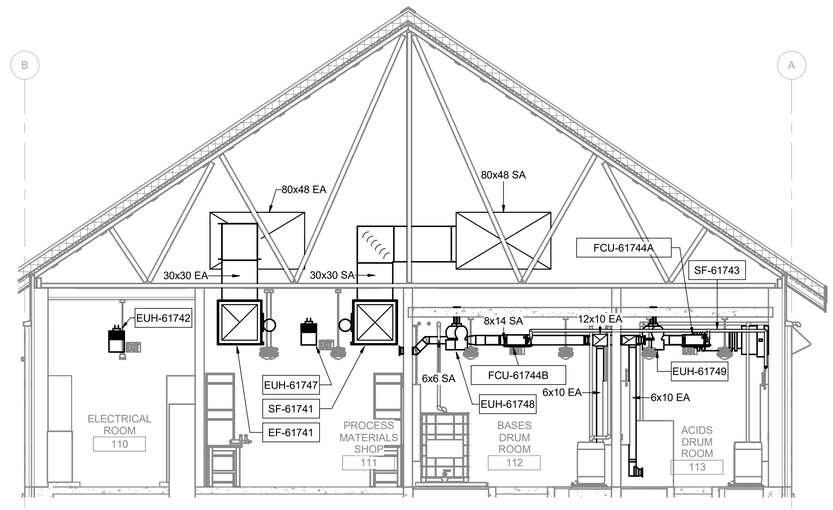
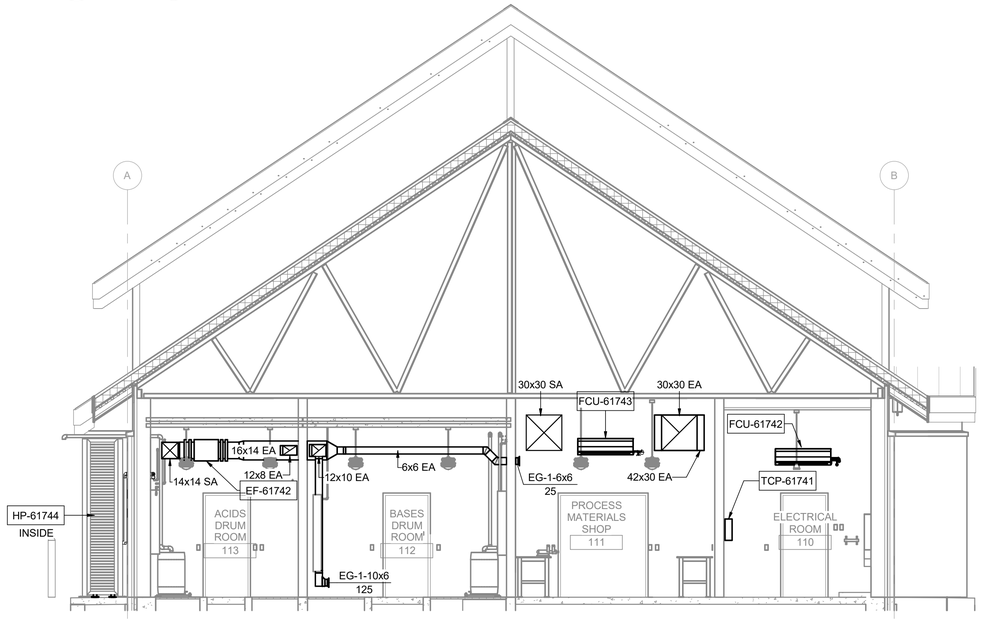


A 062M101 PROCESS AREA
SCALE A



B 062M101 NORTHEAST AREAS 1
SCALE A



C 062M101 NORTHEAST AREAS 2
SCALE A

SCALE A 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

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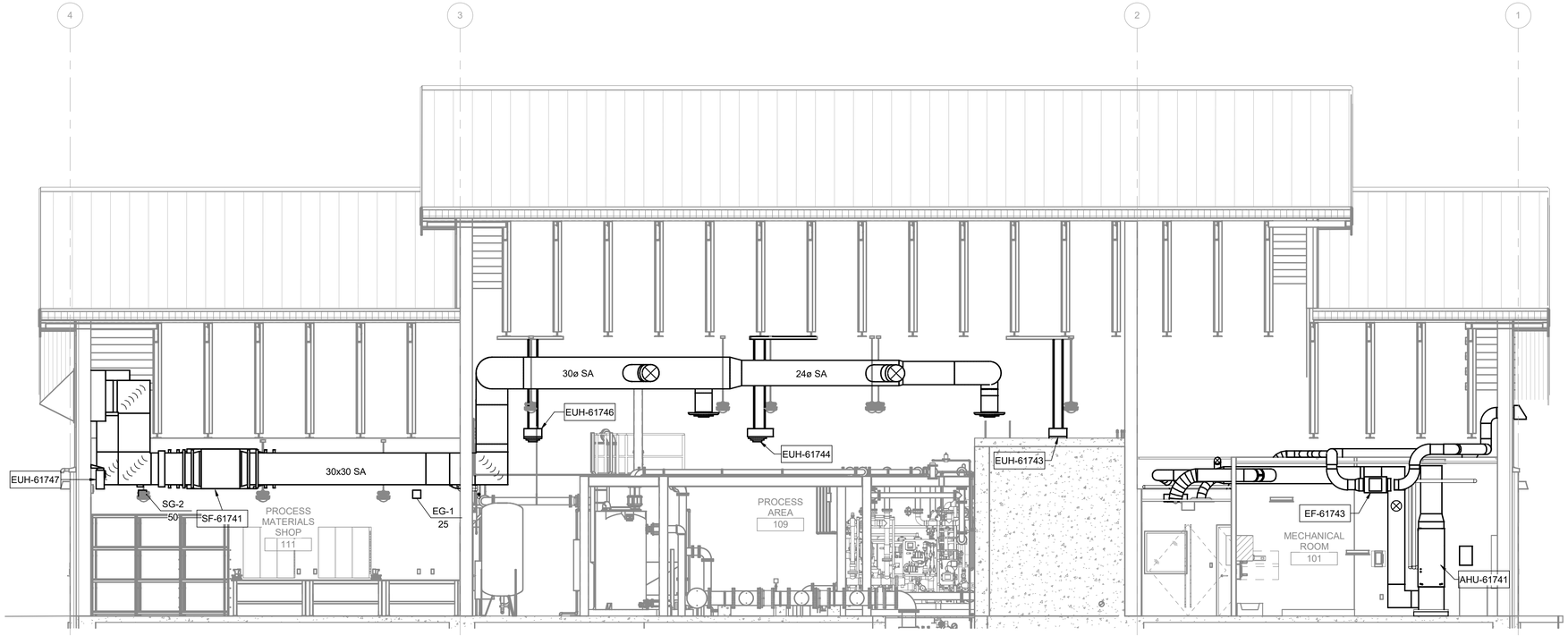
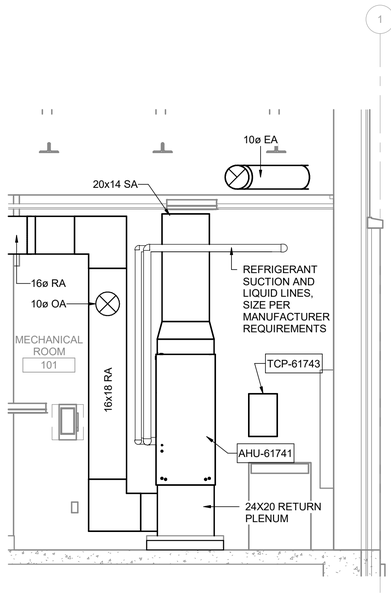
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 M301	TITLE OF SHEET WATER TREATMENT SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 / 182752
PMIS/PKG NO. 318719
SHEET 113 OF 192



A 062M101 PROCESS AREA SUPPLY AIR

SCALE (A)



B 062M101 MECHANICAL ROOM AHU

SCALE (B)

SCALE (B) 2 0 2 4 6 3/8" = 1'-0"

SCALE OF FEET

SCALE (A) 4 0 4 8 12 3/16" = 1'-0"

SCALE OF FEET

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A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 M302	TITLE OF SHEET WATER TREATMENT SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 114 OF 192

AIR HANDLING UNIT SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	MOUNT	TYPE	ASSOCIATED OUTDOOR UNIT	SUPPLY FAN		OUTSIDE AIR (CFM)	COOLING CAPACITY		HEATING		ELECTRICAL		UNIT WEIGHT (LBS)	NOTES
							SUPPLY AIRFLOW (CFM)	EXTERNAL SP (IN. W.G.)		TOTAL (BTUH)	SENSIBLE (BTUH)	HEAT PUMP CAPACITY (BTUH)	ELECTRICAL HEAT CAPACITY (KW)	V/HZ/PH	MCA		
AHU-61741	LG ZRNU423NBAA	WATER TREATMENT BLDG	ADMIN SPACE	VERTICAL, UPFLOW	HEAT PUMP	HP-61745	1,000	0.65	360	42,000	30,200	46,000	15	208-230/60/1	3.4	145	1-3

NOTES:

1) HEATING AND COOLING CAPACITIES LISTED ARE AT AHRI CONDITIONS.
2) PROVIDE ELECTRIC RESISTANCE HEAT FOR SUPPLEMENTAL HEAT AND FOR DEFROST CYCLE. A SEPARATE 208V/1PH POWER SUPPLY AND DISCONNECT ARE REQUIRED FOR ELECTRIC HEAT.
3) PROVIDE FILTER RACK WITH MERV-8 PLEATED FILTERS.

GENERAL NOTES:

A. MECHANICAL SCHEDULES PRESENT INFORMATION ON THE SALIENT CHARACTERISTICS OF MECHANICAL EQUIPMENT.

B. BRAND NAMES LISTED ARE CONSIDERED BASIS OF DESIGN. OR EQUAL SUBSTITUTIONS ARE ACCEPTABLE MEETING THE REQUIREMENTS OF 01 33 23 AND THE PRODUCT REQUIREMENTS LISTED IN THE ASSOCIATED TECHNICAL SPECIFICATIONS.

HEAT PUMP SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	INDOOR UNIT SERVED	TOTAL COOLING CAPACITY (BTUH)	TOTAL HEATING CAPACITY (BTUH)	REFRIGERANT	EER2 / SEER2	ELECTRICAL			OPERATING WEIGHT (LBS)	NOTES
									V/HZ/PH	MCA	MOCP		
HP-61741	LG KUSAE091A	WATER TREATMENT BLDG	IT ROOM	FCU-61741	9,000	10,900	R-32	12.0 / 18.0	208-230/60/1	10.0	15	56	1,2,4,5
HP-61742	LG ZRUM036GSS0	WATER TREATMENT BLDG	ELECTRICAL ROOM	FCU-61742	36,000	42,000	R-32	15.0 / 23.0	208-230/60/1	31.0	40	257	1-5
HP-61743	LG KUMXA181A	WATER TREATMENT BLDG	PROCESS MATERIALS	FCU-61743	18,000	22,000	R-32	13.5 / 20.0	208-230/60/1	22.7	30	148	1-5
HP-61744	LG ZRUM036GSS0	WATER TREATMENT BLDG	BASES DRUM ROOM, ACIDS DRUM ROOM	FCU-61744A/B	36,000	42,000	R-32	15.5 / 23.0	208-230/60/1	31.0	40	257	1-5
HP-61745	LG ZRUM048GSS0	WATER TREATMENT BLDG	ADMIN SPACE	AHU-61741	48,000	54,000	R-32	12.2 / 17.0	208-230/60/1	31.0	40	257	1-5

NOTES:

1) HEATING & COOLING CAPACITIES AND EFFICIENCIES ARE LISTED AT AHRI CONDITIONS AND SEA LEVEL ELEVATION.
2) PROVIDE VARIABLE SPEED INVERTER DUTY COMPRESSORS.
3) PROVIDE EXTREME HEAT TECHNOLOGY EQUAL TO "LG-RED" FOR 100% RATED HEATING CAPACITY DOWN TO 5°F.
4) PROVIDE FACTORY APPLIED CORROSION RESISTANT COATING ON CONDENSER COILS FOR CORROSION RESISTANCE.
5) PROVIDE MANUFACTURER'S INTEGRATED BASE PAN HEATER TO MINIMIZE FREEZING OF CONDENSATION.

SITE CONDITIONS

SITE LOCATION	GRAND CANYON NATIONAL PARK, NORTH RIM
	COCONINO COUNTY
ELEVATION ABOVE SEA LEVEL (FT)	ROARING SPRINGS PUMP HOUSE: 4870'
	WTF / WWTF: 8,300 FT
ASHRAE CLIMATE ZONE	5B (PER 2021 IECC)
OUTDOOR DESIGN TEMPERATURES	SUMMER (DB/WB): 90 / 56 °F
	WINTER (DB): -1 °F

NOTES:

1) WEATHER DATA IS SOURCED FROM THE 2021 ASHRAE FUNDAMENTALS ONLINE CLIMATE DATABASE FOR 0.4% (SUMMER) AND 99.6% (WINTER) CONDITIONS FOR GRAND CANYON NATIONAL PARK, ARIZONA.

DESIGN TEMPERATURES

AREA	SUMMER DB (°F)	WINTER DB (°F)
PROCESS AREAS	UNCONDITIONED	55
MECHANICAL EQUIPMENT ROOMS	UNCONDITIONED	55
ELECTRICAL ROOMS	80	55
IT ROOMS	75	70
OCCUPIED SPACES	75	70
ACIDS DRUM ROOM, BASES DRUM ROOM	75	70

FAN COIL UNIT SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	MOUNT	TYPE	ASSOCIATED OUTDOOR UNIT	SUPPLY AIRFLOW (CFM)	TOTAL COOLING CAPACITY (BTUH)	HEAT PUMP CAPACITY (BTUH)	ELECTRICAL		OPERATING WEIGHT (LBS)	NOTES
										V/HZ/PH	MCA		
FCU-61741	LG KNSAE091A	WATER TREATMENT BLDG	IT ROOM	WALL MOUNTED	HEAT PUMP	HP-61741	353	9,000	10,900	208-230/60/1	0.40	19	1
FCU-61742	LG ZRNU363SRAA	WATER TREATMENT BLDG	ELECTRICAL ROOM	WALL MOUNTED	HEAT PUMP	HP-61742	883	35,000	37,000	208-230/60/1	0.90	20	1
FCU-61743	LG ZRNU183SKSA	WATER TREATMENT BLDG	PROCESS MATERIALS	WALL MOUNTED	HEAT PUMP	HP-61743	370	19,000	21,500	208-230/60/1	0.52	29	1
FCU-61744A	LG ZRNU183TQAA	WATER TREATMENT BLDG	ACIDS DRUM ROOM	CEILING CASSETTE	HEAT PUMP	HP-61744	350	19,100	21,500	208-230/60/1	0.25	35	1
FCU-61744B	LG ZRNU183TQAA	WATER TREATMENT BLDG	BASES DRUM ROOM	CEILING CASSETTE	HEAT PUMP	HP-61744	350	19,100	21,500	208-230/60/1	0.25	32	1

NOTES:

1) PROVIDE FACTORY APPLIED CORROSION RESISTANT COATING ON COILS.

DIFFUSERS, GRILLES, REGISTERS

TAG NO.	BASIS OF DESIGN MAKE/MODEL	AREA SERVED	DESCRIPTION	FUCTION	MOUNT	MATERIAL	FINISH	NOTES
SG-1	TITUS 272FS	GENERAL	RECTANGULAR	SUPPLY	DUCT	ALUMINUM	WHITE	2,3,4,5
SD-1	TITUS MCD-AA	ADMIN SPACES, LABS	SQUARE	SUPPLY	CEILING	ALUMINUM	WHITE	1,3,5
SD-2	TITUS TMRA-AA	WTP PROCESS SPACE	ROUND	SUPPLY	DUCT	ALUMINUM	WHITE	1,3,4,5
RD-1	TITUS PAR-AA	ADMIN SPACES	SQUARE	RETURN	CEILING	ALUMINUM	WHITE	3
EG-1	TITUS 350ZFL	GENERAL	RECTANGULAR	EXHAUST	DUCT	ALUMINUM	WHITE	3
EG-3	TITUS PAR-AA	ADMIN SPACES	SQUARE	EXHAUST	CEILING	ALUMINUM	WHITE	3

NOTES:

1) DIFFUSERS SHALL HAVE 4-WAY THROW PATTERNS UNLESS OTHERWISE INDICATED ON THE PLANS.
2) PROVIDE TWO SETS OF BLADES FOR DOUBLE-DEFLECTION OF AIRFLOW.
3) MAXIMUM TOTAL PRESSURE DROP OF 0.12 INCHES WATER COLUMN ACROSS GRILLE OR DIFFUSER AT DESIGN AIRFLOW.
4) PROVIDE WITH INTEGRAL OPPOSED BLADE DAMPER, OR SEPARATE DUCT-MOUNTED DAMPER FOR BALANCING.
5) ADJUST BLADES DURING FINAL TEST AND BALANCE TO PROVIDE EVEN DISTRIBUTION OF AIRFLOW IN THE SPACES SERVED.

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A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

DESIGNED: A. PUTMAN
CADD: B. BAXTER
TECH. REVIEW: H. PEDRAZA
DATE: MAY 2026

SUB SHEET NO. 062 M601

TITLE OF SHEET
WATER TREATMENT
SCHEDULES

NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 115 OF 192

FAN SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	DESCRIPTION	AIRFLOW (CFM)	EXTERNAL SP (IN. WG.)	DRIVE TYPE	FAN RPM	ELECTRICAL			OPERATING WEIGHT (LBS)	NOTES
									V/HZ/PH	MOTOR HP	BHP		
SF-61741	GREENHECK BSQ-240	WATER TREATMENT BLDG	PROCESS SPACE	SQUARE INLINE CENTRIFUGAL FAN	6,800	1.5	BELT	1,118	480/60/3	5	3.10	295	1,2
SF-61742	GREENHECK SQ-90-VG	WATER TREATMENT BLDG	MECHANICAL ROOM	SQUARE INLINE CENTRIFUGAL FAN	400	0.25	DIRECT	1,428	115/60/1	1/6	0.04	34	1,2,5,6
SF-61743	GREENHECK SQ-9-M1-VG	WATER TREATMENT BLDG	ACIDS DRUM ROOM, BASES DRUM ROOM, PROCESS MATERIALS SHOP	SQUARE INLINE CENTRIFUGAL FAN	750	0.5	DIRECT	1,633	115/60/1	1/2	0.11	52	1,2,4,5,6
EF-61741	GREENHECK BSQ-240	WATER TREATMENT BLDG	PROCESS SPACE	SQUARE INLINE CENTRIFUGAL FAN	6,800	0.75	BELT	911	480/60/3	3	1.90	240	1
EF-61742	GREENHECK SQ-9-M1-VG	WATER TREATMENT BLDG	ACIDS DRUM ROOM, BASES DRUM ROOM, PROCESS MATERIALS SHOP	SQUARE INLINE CENTRIFUGAL FAN	925	0.5	DIRECT	1,708	115/60/1	1/2	0.14	52	1,3,4,5,6
EF-61743	GREENHECK SQ-98-VG	WATER TREATMENT BLDG	ADMIN SPACES	SQUARE INLINE CENTRIFUGAL FAN	360	0.6	DIRECT	1,627	115/60/1	1/4	0.12	50	1,4,5,6

NOTES:

1) PROVIDE INSULATED FAN HOUSING.
2) PROVIDE INLINE FILTER RACK WITH MERV-8 PLEATED FILTERS.
3) PROVIDE BACKDRAFT DAMPER.
4) PROVIDE PHENOLIC-BASED CORROSION RESISTANT COATING.
5) PROVIDE EC MOTOR WITH MOTOR-MOUNTED SPEED CONTROL DIAL.
6) PROVIDE WITH MANUFACTURER'S DISCONNECT SWITCH.

ELECTRIC UNIT HEATER SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	AIRFLOW (CFM)	HEATING CAPACITY (KW)	ELECTRICAL		WEIGHT (LBS)	NOTES
						V/HZ/PH	TOTAL AMPS		
EUH-61741	KING KDSRU4830-3	WATER TREATMENT BLDG	MECHANICAL ROOM	-	3.0	480/60/3	-	31	2,4,6
EUH-61742	INDEECO UHIR 240-U1050U	WATER TREATMENT BLDG	ELECTRICAL ROOM	350	5.0	480/60/3	6.2	25	1,5,6
EUH-61743	INDEECO UHIR 240-U2075U	WATER TREATMENT BLDG	PROCESS SPACE	650	7.5	480/60/3	9.39	40	2,3,4,6
EUH-61744	INDEECO UHIR 240-U2075U	WATER TREATMENT BLDG	PROCESS SPACE	650	7.5	480/60/3	9.39	40	2,3,4,6
EUH-61745	INDEECO UHIR 240-U2075U	WATER TREATMENT BLDG	PROCESS SPACE	650	7.5	480/60/3	9.39	40	2,3,4,6
EUH-61746	INDEECO UHIR 240-U2075U	WATER TREATMENT BLDG	PROCESS SPACE	650	7.5	480/60/3	9.39	40	2,3,4,6
EUH-61747	INDEECO UHIR 240-U2100U	WATER TREATMENT BLDG	MATERIALS SHOP	650	10.0	480/60/3	12.4	40	1,5,6
EUH-61748	INDEECO TRIAD 234-U11N-0090U	WATER TREATMENT BLDG	BASES DRUM ROOM	1450	9.0	480/60/3	12	56	2,5,6
EUH-61749	INDEECO TRIAD 234-U11N-0080U	WATER TREATMENT BLDG	ACIDS DRUM ROOM	700	7.5	480/60/3	11	56	2,5,6
EUH-617410	KING KDSRU4820-3	WATER TREATMENT BLDG	HALLWAY	-	2.0	480/60/3	-	31	2,4,6
EUH-617411	KING KDSRU4820-3	WATER TREATMENT BLDG	RESTROOM	-	2.0	480/60/3	-	31	2,4,6
EUH-617412	KING KDSRU4830-3	WATER TREATMENT BLDG	BREAK / KITCHEN	-	3.0	480/60/3	-	31	2,4,6
EUH-617413	KING KDSRU4830-3	WATER TREATMENT BLDG	CONTROL ROOM	-	3.0	480/60/3	-	31	2,4,6
EUH-617414	KING KDSRU4830-3	WATER TREATMENT BLDG	LAB	-	3.0	480/60/3	-	31	2,4,6

NOTES:

1) FURNISH WITH WALL MOUNTING KIT.
2) FURNISH WITH CEILING MOUNTING KIT.
3) FURNISH WITH MANUFACTURER'S RADIAL DIFFUSER.
4) PROVIDE WITH WALL-MOUNTED THERMOSTAT.
5) PROVIDE WITH UNIT-MOUNTED THERMOSTAT.
6) PROVIDE WITH MANUFACTURERS DISCONNECT SWITCH.

LOUVER SCHEDULE

TAG NO.	BUILDING SERVED	AREA SERVED	FUCTION	AIRFLOW (CFM)	LOUVER DIMENSIONS (IN)			NOTES
					WIDTH	HEIGHT	DEPTH	
L-61741	WATER TREATMENT BLDG	PROCESS SPACE	INTAKE	6,800	80	48	6	1,2
L-61742	WATER TREATMENT BLDG	PROCESS SPACE	EXHAUST	6,800	80	48	6	1,2
L-61743	WATER TREATMENT BLDG	ACIDS DRUM ROOM, BASES DRUM ROOM, PROCESS MATERIALS SHOP	INTAKE	725	24	24	6	1,2
L-61744	WATER TREATMENT BLDG	ACIDS DRUM ROOM, BASES DRUM ROOM, PROCESS MATERIALS SHOP	EXHAUST	925	24	24	6	1,2
L-61745	WATER TREATMENT BLDG	ADMIN SPACE	INTAKE	400	80	24	6	1,2
L-61746	WATER TREATMENT BLDG	MECHANICAL ROOM	EXAHUST	300	24	16	6	1,2

NOTES:

1) MINIMUM 50% FREE AREA.
2) MAX PRESSURE DROP OF 0.1 INCHES WATER COLUMN AT DESIGN AIRFLOW.

GENERAL NOTES:

A. MECHANICAL SCHEDULES PRESENT INFORMATION ON THE SALIENT CHARACTERISTICS OF MECHANICAL EQUIPMENT.

B. BRAND NAMES LISTED ARE CONSIDERED BASIS OF DESIGN. OR EQUAL SUBSTITUTIONS ARE ACCEPTABLE MEETING THE REQUIREMENTS OF 01 33 23 AND THE PRODUCT REQUIREMENTS LISTED IN THE ASSOCIATED TECHNICAL SPECIFICATIONS.

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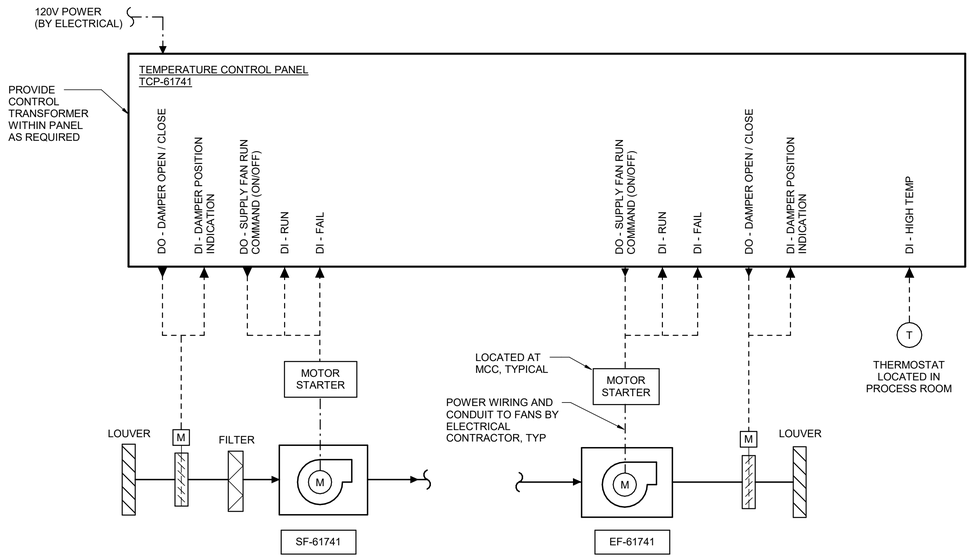
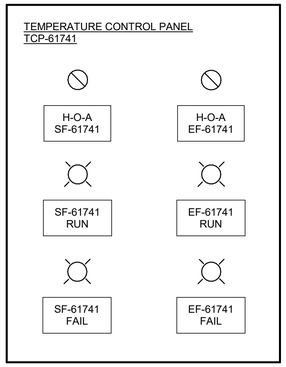
WORLDWIDE PROFESSIONAL ENGINEERING SERVICES
CERTIFICATE
86199
SIGNED
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 M602	TITLE OF SHEET WATER TREATMENT SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 / 182752
PMIS/PKG NO. 318719
SHEET 116 OF 192



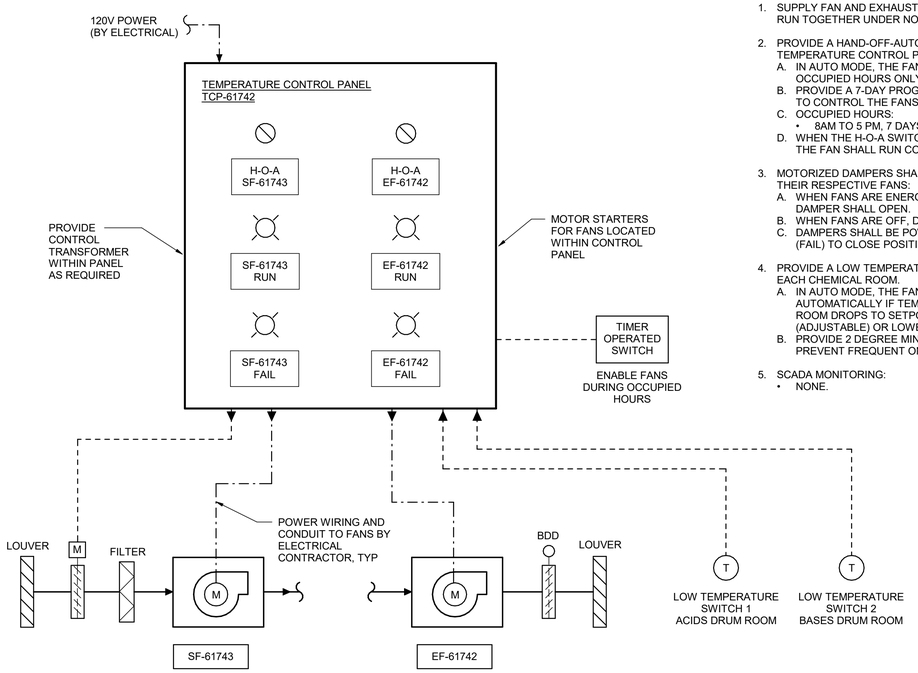


PANEL ELEVATION

CONTROL SEQUENCE: (PROCESS ROOM FANS)

1. SUPPLY FAN AND EXHAUST FAN ARE INTENDED TO RUN TOGETHER UNDER NORMAL OPERATION.
2. PROVIDE A HAND-OFF-AUTO (H-O-A) SWITCH AT THE TEMPERATURE CONTROL PANEL FOR EACH FAN.
   A. IN AUTO MODE, THE FANS SHALL BE THERMOSTAT CONTROLLED. WHEN ROOM TEMPERATURE RISES ABOVE SETPOINT (80°F, ADJUSTABLE) BOTH FANS SHALL RUN.
   B. WHEN ROOM TEMPERATURE IS BELOW SETPOINT, BOTH FANS SHALL REMAIN OFF.
   C. PROVIDE 2 DEGREE MINIMUM DEADBAND TO PREVENT FREQUENT CYCLING OF FANS.
   D. WHEN THE H-O-A SWITCH IS IN THE ON POSITION, THE FAN SHALL RUN CONTINUOUSLY.
3. MOTORIZED DAMPERS SHALL BE INTERLOCKED WITH BOTH FANS:
   A. WHEN EITHER FAN IS ENERGIZED, BOTH MOTORIZED DAMPERS SHALL AUTOMATICALLY OPEN.
   B. WHEN BOTH FANS ARE OFF, THE DAMPERS SHALL AUTOMATICALLY CLOSE.
   C. DAMPERS SHALL BE POWER TO OPEN, SPRING (FAIL) TO CLOSE POSITION.
4. SCADA MONITORING:
   - NONE

1 VENTILATION FANS - PROCESS AREA



CONTROL SEQUENCE: (CHEMICAL ROOM FANS)

1. SUPPLY FAN AND EXHAUST FANS ARE INTENDED TO RUN TOGETHER UNDER NORMAL OPERATION.
2. PROVIDE A HAND-OFF-AUTO (H-O-A) SWITCH AT THE TEMPERATURE CONTROL PANEL FOR EACH FAN.
   A. IN AUTO MODE, THE FAN SHALL RUN DURING OCCUPIED HOURS ONLY.
   B. PROVIDE A 7-DAY PROGRAMMABLE TIMER SWITCH TO CONTROL THE FANS.
   C. OCCUPIED HOURS:
      - 8AM TO 5 PM, 7 DAYS A WEEK (ADJUSTABLE).
   D. WHEN THE H-O-A SWITCH IS IN THE ON POSITION, THE FAN SHALL RUN CONTINUOUSLY.
3. MOTORIZED DAMPERS SHALL BE INTERLOCKED WITH THEIR RESPECTIVE FANS:
   A. WHEN FANS ARE ENERGIZED, MOTORIZED DAMPER SHALL OPEN.
   B. WHEN FANS ARE OFF, DAMPERS SHALL CLOSE.
   C. DAMPERS SHALL BE POWER TO OPEN, SPRING (FAIL) TO CLOSE POSITION.
4. PROVIDE A LOW TEMPERATURE SHUTOFF SWITCH IN EACH CHEMICAL ROOM.
   A. IN AUTO MODE, THE FANS SHALL SHUT OFF AUTOMATICALLY IF TEMPERATURE IN EITHER ROOM DROPS TO SETPOINT OF 65°F (ADJUSTABLE) OR LOWER.
   B. PROVIDE 2 DEGREE MINIMUM DEADBAND TO PREVENT FREQUENT ON/OFF CYCLING.
5. SCADA MONITORING:
   - NONE.

2 VENTILATION FANS - CHEMICAL ROOMS

CONTROL SEQUENCE:

1. EXHAUST FAN SERVING THE MECHANICAL ROOM:
   A. PROVIDE A HAND-OFF-AUTO (H-O-A) SWITCH AT THE TEMPERATURE CONTROL PANEL.
      a. IN AUTO MODE, THE FAN SHALL BE THERMOSTAT CONTROLLED. WHEN ROOM TEMPERATURE RISES ABOVE SETPOINT (85°F, ADJUSTABLE), THE FAN SHALL RUN.
      b. WHEN ROOM TEMPERATURE IS BELOW SETPOINT, FAN SHALL REMAIN OFF AND ASSOCIATED MOTORIZED DAMPER SHALL REMAIN CLOSED.
      c. PROVIDE 1-2 DEGREE DEADBAND TO PREVENT FREQUENT CYCLING OF FANS.
      d. WHEN THE H-O-A SWITCH IS IN THE HAND POSITION, THE FAN SHALL RUN CONTINUOUSLY.
   B. MOTORIZED DAMPERS SHALL BE INTERLOCKED WITH THE RESPECTIVE FANS:
      a. WHEN FAN IS ENERGIZED, MOTORIZED DAMPER SHALL OPEN.
      b. WHEN FAN IS OFF, DAMPER SHALL CLOSE.
      c. DAMPERS SHALL BE POWER TO OPEN, SPRING (FAIL) TO CLOSED POSITION.
2. EXHAUST FAN SERVING THE ADMIN SPACES:
   A. PROVIDE A HAND-OFF-AUTO (H-O-A) SWITCH AT THE TEMPERATURE CONTROL PANEL.
   B. IN AUTO MODE:
      a. PROVIDE A 7-DAY TIMER OPERATED SWITCH TO ENERGIZE THE FANS DURING OCCUPIED HOURS ONLY.
      b. OCCUPIED HOURS SHALL MATCH THE OCCUPIED HOURS OF THE AIR HANDLING UNIT SERVING THE ADMIN SPACES, SUCH THAT THE SUPPLY AND EXHAUST FANS RUN TOGETHER TO MAINTAIN AIR BALANCE WITHIN THE BUILDING.
      c. OUTSIDE OF OCCUPIED HOURS, THE FANS SHALL REMAIN OFF.
   C. IN HAND MODE, THE FAN SHALL RUN CONTINUOUSLY.
3. SCADA MONITORING:
   - NONE

3 VENTILATION FANS - MECHANICAL ROOM

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A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

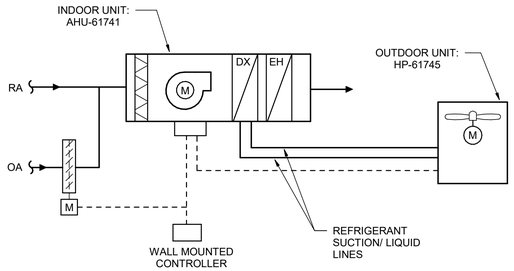
DESIGNED: A. PUTMAN
CADD: B. BAXTER
TECH. REVIEW: H. PEDRAZA
DATE: MAY 2026

SUB SHEET NO. 062 M701

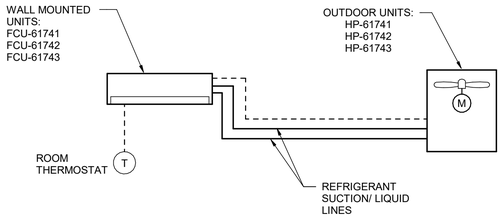
TITLE OF SHEET
WATER TREATMENT
HVAC CONTROLS

NORTH RIM - GRAND CANYON NATIONAL PARK

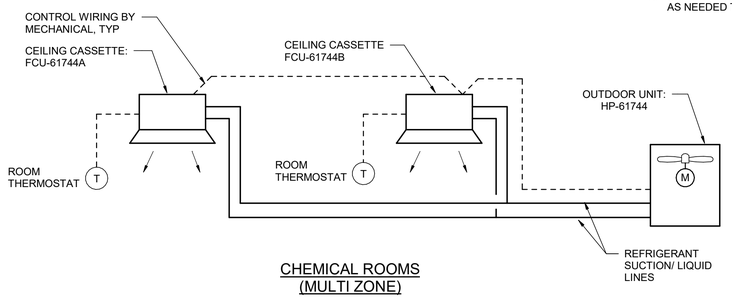
DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 117 OF 192



ADMINISTRATIVE AREA
(SINGLE ZONE)



ELECTRICAL ROOM / IT ROOM / PROCESS MTLS SHOP
(SINGLE ZONE)



CHEMICAL ROOMS
(MULTI ZONE)

1 SPLIT SYSTEM HEAT PUMPS

CONTROL SEQUENCES:

1. PROVIDE MANUFACTURER'S WALL MOUNTED CONTROLLER, WITH 7 DAY PROGRAMMABLE SCHEDULE, AND AUTOMATIC CHANGEOVER.
2. INDOOR TEMPERATURE SETPOINT SHALL BE AS FOLLOWS:

   A. ADMINISTRATIVE AREA, AND PROCESS MATERIALS SHOP:
   - OCCUPIED HOURS: MON-FRI, 8AM TO 5PM
   - COOLING SETPOINT: 75°F
   - HEATING SETPOINT: 70°F
   - COOLING SETPOINT: 80°F (OUTSIDE OF OCCUPIED HOURS)
   - HEATING SETPOINT: 65°F (OUTSIDE OF OCCUPIED HOURS)

   B. ELECTRICAL ROOMS:
   - 24 HRS A DAY, 7 DAYS A WEEK.
   - COOLING SETPOINT: 80°F
   - HEATING SETPOINT: 55°F

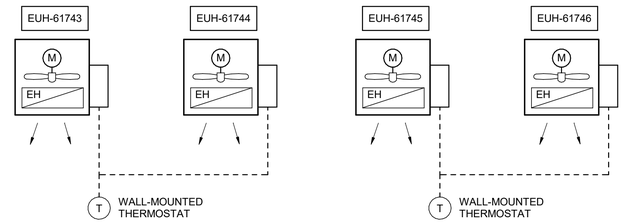
   C. IT ROOMS:
   - 24 HRS A DAY, 7 DAYS A WEEK.
   - COOLING SETPOINT: 75°F
   - HEATING SETPOINT: 70°F

   D. ACIDS DRUM ROOMS:
   - 24 HRS A DAY, 7 DAYS A WEEK.
   - COOLING SETPOINT: 75°F
   - HEATING SETPOINT: 70°F

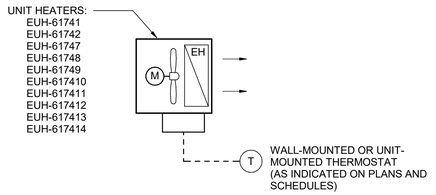
   E. BASES DRUM ROOMS:
   - 24 HRS A DAY, 7 DAYS A WEEK.
   - COOLING SETPOINT: 75°F
   - HEATING SETPOINT: 70°F

3. ADMIN AREA AHU:
   A. SUPPLY FAN SHALL RUN CONTINUOUSLY DURING OCCUPIED HOURS, AND MAY CYCLE ON/OFF OUTSIDE OF OCCUPIED HOURS.
   B. THE AHU SERVING ADMIN SPACE SHALL BE FURNISHED WITH ELECTRIC RESISTANCE HEAT INTEGRAL TO THE UNIT CABINET.
   C. ELECTRIC HEAT SHALL OPERATE DURING DEFROST CYCLE AND FOR SUPPLEMENTARY HEAT.
   D. IF THE HEAT PUPMP ALONE IS INSUFFICIENT TO MAINTAIN INDOOR TEMPERATURE, THE ELECTRIC RESISTANCE HEAT SHALL OPERATE AS A SECOND STAGE OF HEAT TO MAINTAIN INDOOR TEMPERATURE.
   E. COMPRESSOR SHALL CYCLE ON AND MODULATE CAPACITY AS NEEDED TO MAINTAIN INDOOR TEMPERATURE.
   F. MOTORIZED OUTSIDE AIR DAMPER:
      1. INTERLOCK WITH SUPPLY FAN OPERATION.
      2. WHEN FAN IS CALLED TO START, MOTORIZED DAMPER SHALL OPEN.
      3. DAMPER ACTUATOR SHALL BE POWER TO OPEN, AND SPRING (FAIL) TO THE CLOSED POSITION.

4. FAN COIL UNITS (CEILING CASSETTES AND WALL-MOUNTED UNITS):
   A. SUPPLY FANS AND COMPRESSORS SHALL CYCLE ON AND MODULATE AS NEEDED TO MAINTAIN SETPOINT.



SUSPENDED UNIT HEATERS - PROCESS ROOM



WALL-MOUNTED OR CEILING-MOUNTED UNIT HEATERS

CONTROL SEQUENCES:

1. GENERAL:
   A. NOTE THAT SOME SPACES ARE SERVED BY BOTH HEAT PUMPS AND ALSO ELECTRIC UNIT HEATERS. THE INTENT IS FOR THE ELECTRIC UNIT HEATERS TO SERVE AS SECONDARY HEAT IN THE EVENT THAT THE HEAT PUMPS ARE INSUFFICIENT TO MAINTAIN TEMPERATURE.
   B. THE TEMPERATURE SETPOINTS FOR THE UNIT HEATERS IN THESE AREAS IS INTENDED TO BE ~2°F LESS THAN THE HEAT PUMP SETPOINT, SUCH THAT THE HEAT PUMPS REMAIN AS THE FIRST STAGE OF HEAT.
   C. PROVIDE A 7-DAY PROGRAMMABLE WALL-MOUNTED THERMOSTAT FOR UNIT HEATERS, UNLESS OTHERWISE SPECIFIED TO BE PROVIDED WITH INTEGRAL UNIT-MOUNTED THERMOSTAT.

2. UNIT HEATER FAN AND ELECTRIC HEAT SHALL CYCLE ON/OFF AS REQUIRED TO MAINTAIN SPACE TEMPERATURE AT SETPOINT. HEATERS SHALL REMAIN OPERABLE 24 HOURS A DAY, 7 DAYS A WEEK.
3. UNIT HEATER SETPOINTS SHALL BE AS FOLLOWS:
   A. PROCESS ROOM: 55°F
   B. MECHANICAL ROOM: 55°F
   C. ELECTRICAL ROOM: 53°F
   D. CHEMICAL (ACIDS AND BASES) ROOMS: 68°F
   E. PROCESS MATERIALS SHOP: 68°F
   F. ADMINISTRATIVE AREAS: 68°F

4. PROCESS ROOM:
   A. NOTE THAT UNIT HEATERS IN THE PROCESS ROOM ARE INTENDED TO BE ZONED AS INDICATED, SUCH THAT MULTIPLE UNIT HEATERS CAN BE CONTROLLED FROM A SINGLE THERMOSTAT.

2 ELECTRIC UNIT HEATERS

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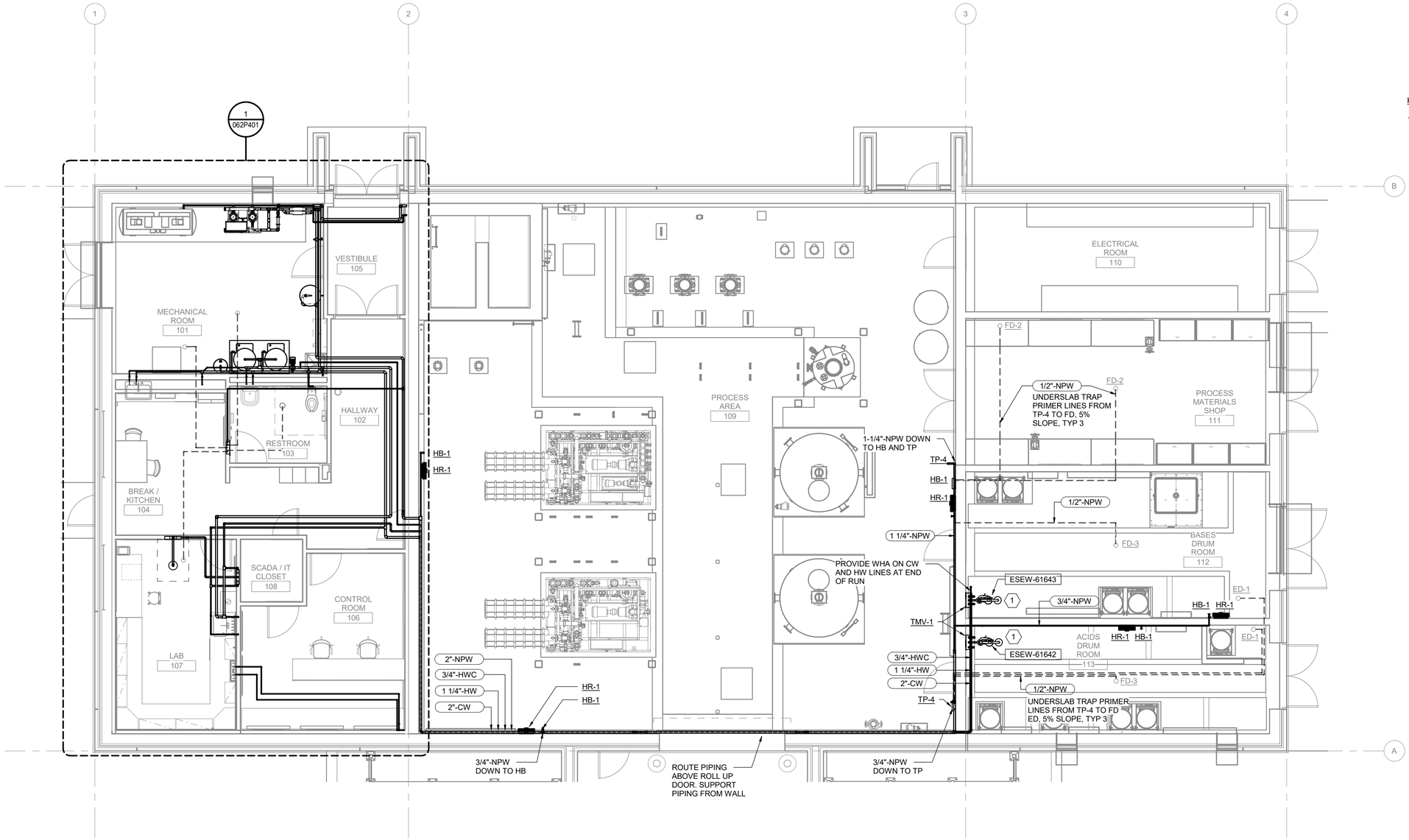
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN CADD B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 M702	TITLE OF SHEET WATER TREATMENT HVAC CONTROLS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 118 OF 192

GENERAL NOTES:

A. REFER TO PROCESS DISCIPLINE 000 SERIES SHEETS FOR PIPE PENETRATION DETAILS.

KEYNOTES: (X)

1. ESEW SHALL BE MOUNTED TO ON TOP OF SUMP GRATING. PROVIDE CONCRETE PEDESTAL WITHIN SUMP, AND ANCHOR ESEW INTO CONCRETE PEDESTAL.



1 062P101 DOMESTIC WATER OVERALL PLAN
SCALE (A)

SCALE (A) 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN CADD B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 P101	TITLE OF SHEET WATER TREATMENT WATER PIPING OVERALL PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 119 OF 192

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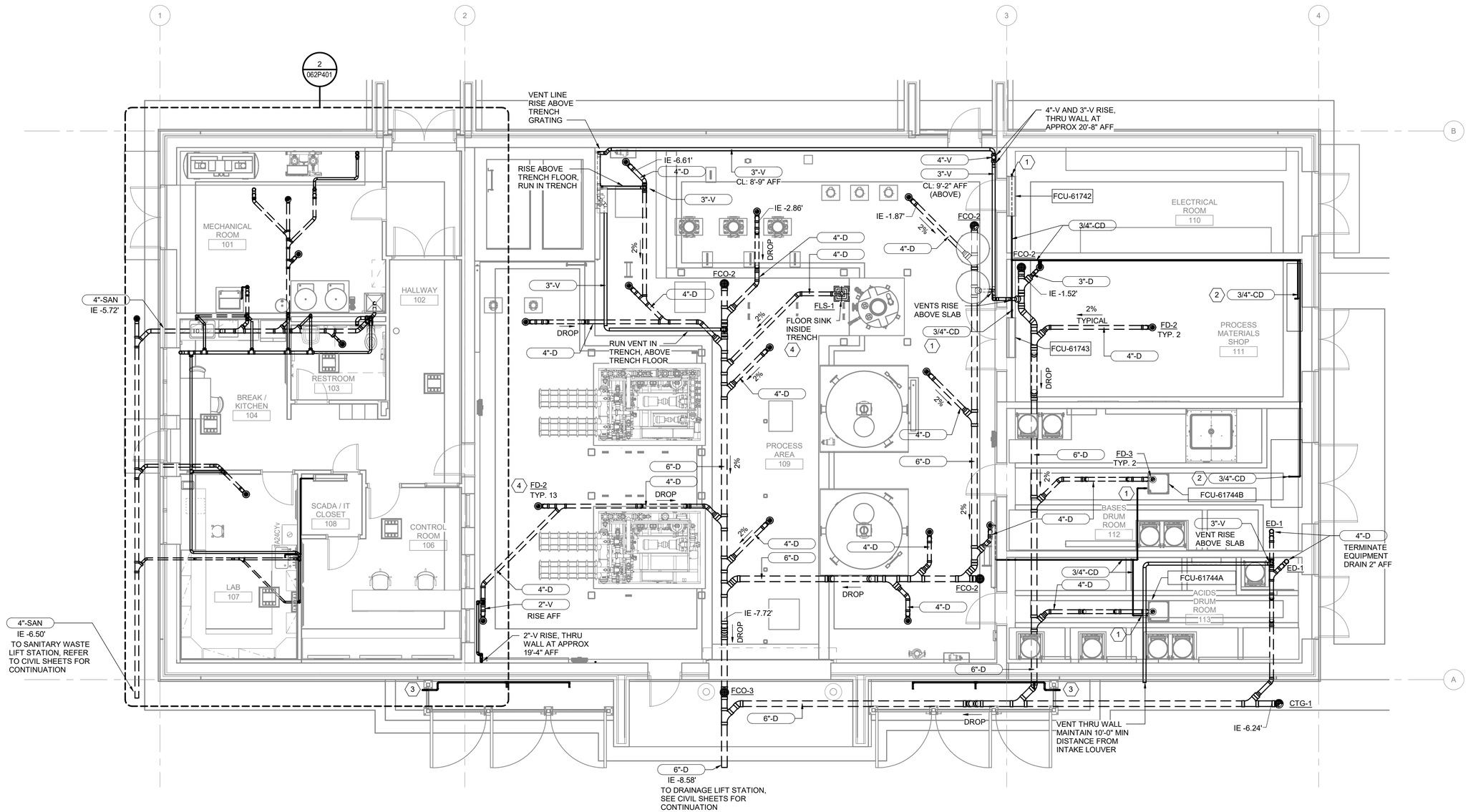

1 062P102 DRAINAGE AND VENT OVERALL PLAN

SCALE A

SCALE A 4 0 4 8 12 3/16" = 1'-0"

SCALE OF FEET

GENERAL NOTES:

A. REFER TO PROCESS DISCIPLINE 000 SERIES SHEETS FOR PIPE PENETRATION DETAILS.

KEYNOTES: (X)

1. CONDENSATE DRAIN FROM MECHANICAL EQUIPMENT, 2% SLOPE.
2. DRAIN FROM LOUVER PLENUM ABOVE, REFER TO HVAC SHEETS FOR LOCATION. PROVIDE BALL VALL ON DRAIN LINE IN ACCESSIBLE LOCATION, LEAVE CRACKED OPEN FOR DRAINAGE.
3. 1" CONDENSATE DRAIN FROM DRAIN PAN BENEATH HEAT PUMPS, TERMINATE OUTSIDE ENCLOSURE WITH ELBOW DOWN, 2" AFF.
4. PROVIDE TRAP SEAL DEVICES FOR ALL FLOOR DRAINS AND FLOOR SINKS IN THE PROCESS AREA NOT SERVED BY A TRAP PRIMER.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN CADD B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 P102	TITLE OF SHEET WATER TREATMENT DRAINAGE AND VENT OVERALL PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 120 OF 192

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_062_NWTP_MP.rvt
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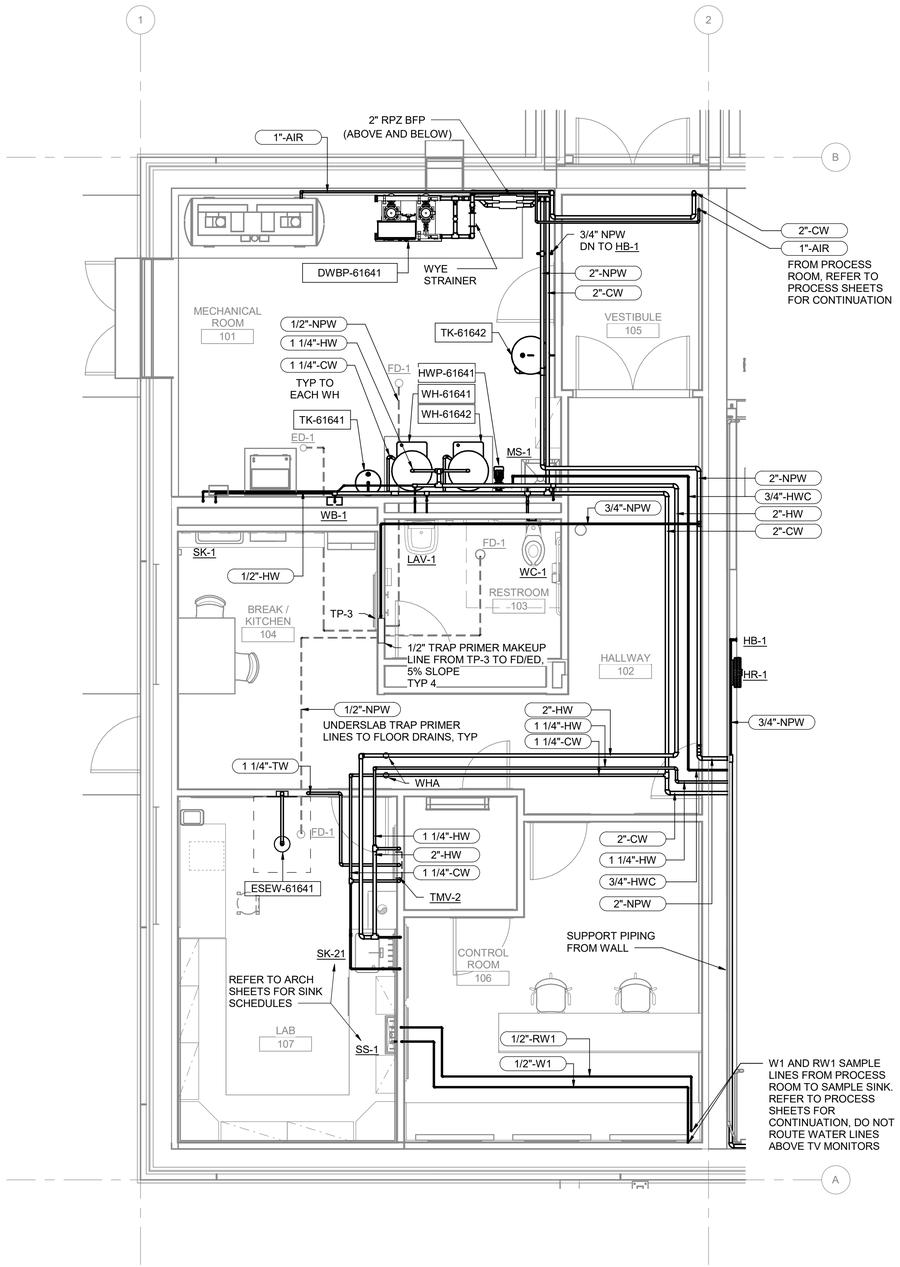
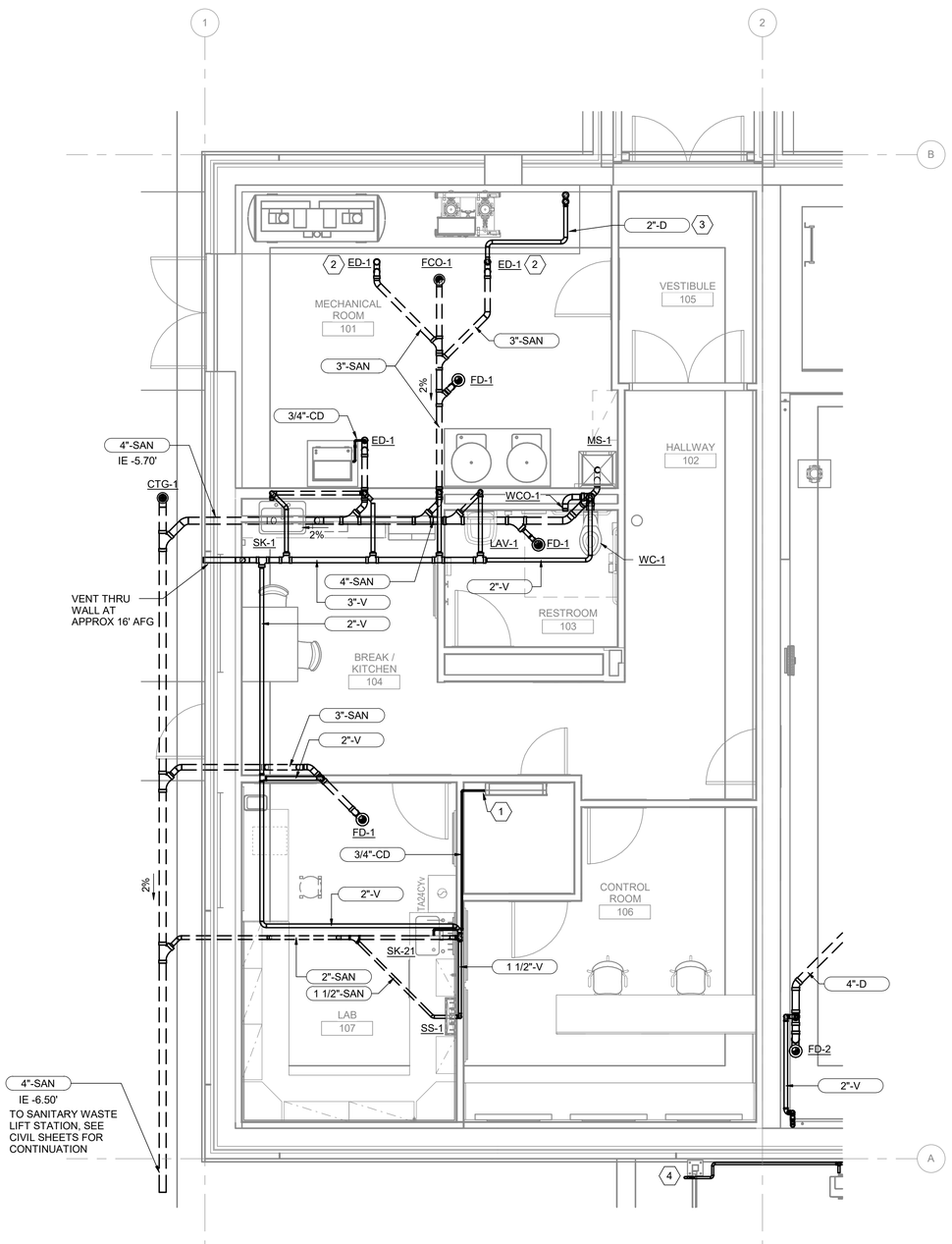


1 062P101 DOMESTIC WATER - ENLARGED PLAN
SCALE (A)



2 062P102 DRAINAGE AND VENT - ENLARGED PLAN
SCALE (A)

GENERAL NOTES:

A. REFER TO PROCESS DISCIPLINE 000 SERIES SHEETS FOR PIPE PENETRATION DETAILS.

KEYNOTES: (X)

1. CONDENSATE DRAIN FROM MECHANICAL EQUIPMENT, 2% SLOPE. CONNECT TO SINK TAILPIECE. SEE M-SHEETS FOR UNIT PLACEMENT.
2. EQUIPMENT DRAIN FOR COMPRESSOR CONDENSATE AND BACKFLOW PREVENTER DRAINS.
3. DRAIN FROM BFP AIR GAP FITTING TO EQUIPMENT DRAIN.
4. 1" CONDENSATE DRAIN FROM DRAIN PAN BENEATH HEAT PUMPS, TERMINATE OUTSIDE ENCLOSURE WITH ELBOW DOWN, 2" AFF.

SCALE (A) 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

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A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

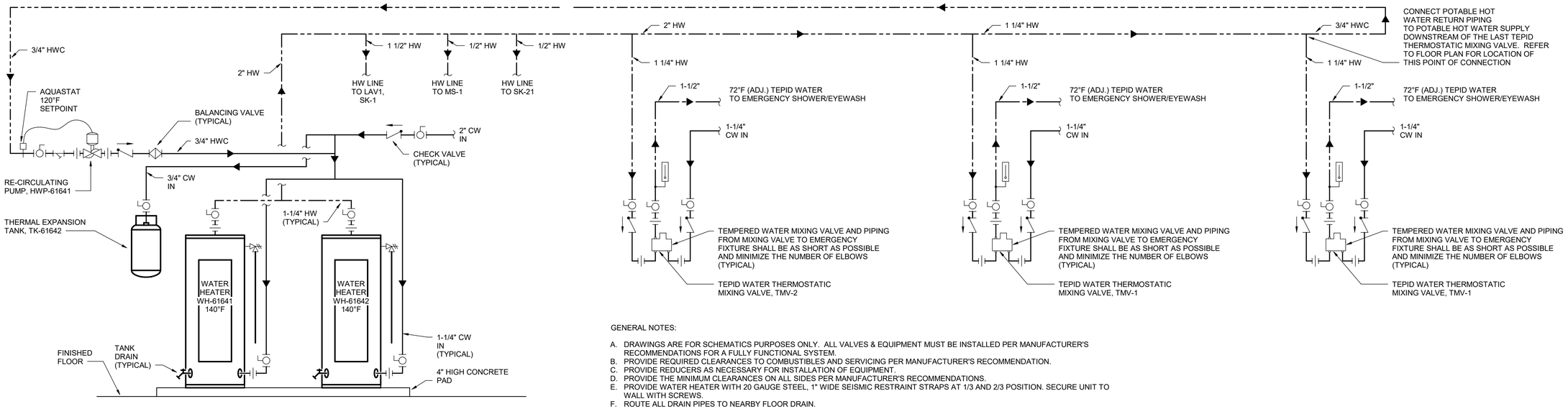
DESIGNED: A. PUTMAN
DRAWN: B. BAXTER
TECH. REVIEW: H. PEDRAZA
DATE: MAY 2026

SUB SHEET NO. 062 P401

TITLE OF SHEET
WATER TREATMENT
ENLARGED PLANS

NORTH RIM - GRAND CANYON NATIONAL PARK

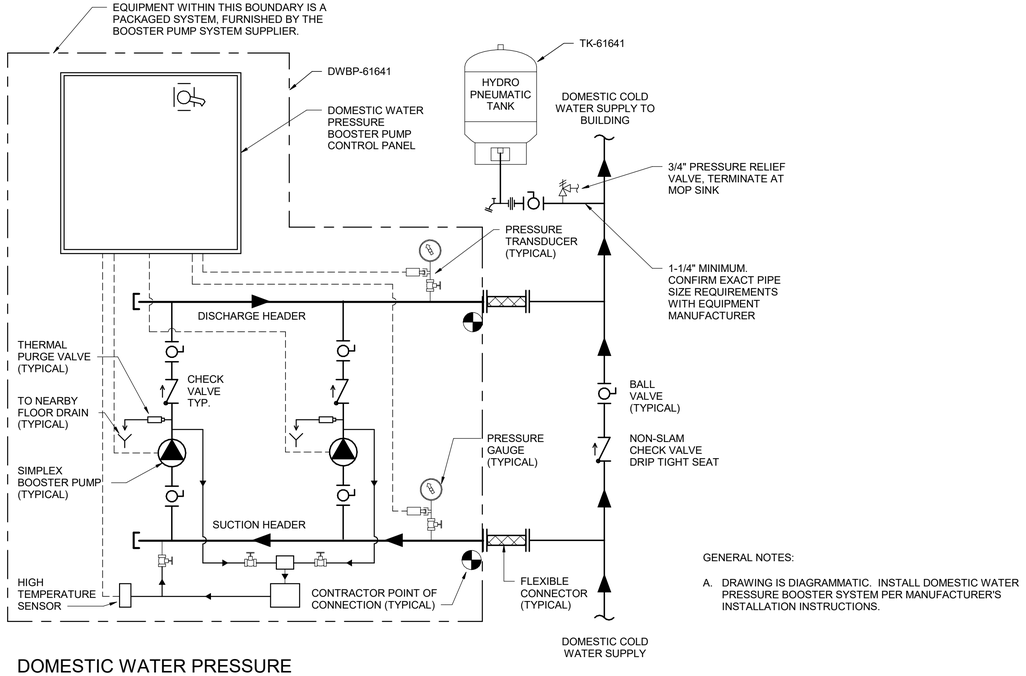
DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 121 OF 192



1 DOMESTIC HOT WATER PIPING DIAGRAM

GENERAL NOTES:

A. DRAWINGS ARE FOR SCHEMATICS PURPOSES ONLY. ALL VALVES & EQUIPMENT MUST BE INSTALLED PER MANUFACTURER'S RECOMMENDATIONS FOR A FULLY FUNCTIONAL SYSTEM.
B. PROVIDE REQUIRED CLEARANCES TO COMBUSTIBLES AND SERVICING PER MANUFACTURER'S RECOMMENDATION.
C. PROVIDE REDUCERS AS NECESSARY FOR INSTALLATION OF EQUIPMENT.
D. PROVIDE THE MINIMUM CLEARANCES ON ALL SIDES PER MANUFACTURER'S RECOMMENDATIONS.
E. PROVIDE WATER HEATER WITH 20 GAUGE STEEL, 1" WIDE SEISMIC RESTRAINT STRAPS AT 1/3 AND 2/3 POSITION. SECURE UNIT TO WALL WITH SCREWS.
F. ROUTE ALL DRAIN PIPES TO NEARBY FLOOR DRAIN.
G. INSTALL TEPID/TEMPERED WATER THERMOSTATIC MIXING VALVES PER MANUFACTURER'S RECOMMENDATIONS.
H. REFER TO STRUCTURAL PLANS FOR CONCRETE PAD DETAIL.



2 DOMESTIC WATER PRESSURE BOOSTING SYSTEM PIPING DIAGRAM

GENERAL NOTES:

A. DRAWING IS DIAGRAMMATIC. INSTALL DOMESTIC WATER PRESSURE BOOSTER SYSTEM PER MANUFACTURER'S INSTALLATION INSTRUCTIONS.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN CADD B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 P501	TITLE OF SHEET WATER TREATMENT DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 122 OF 192

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DOMESTIC ELECTRIC WATER HEATERS

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	LOCATION	KW	ELECTRICAL V/HZ/PH	RECOVERY RATE (GPH @ 90F RISE)	STORAGE CAPACITY (GAL)	WEIGHT (LBS)	NOTES
WH-61641	A.O. SMITH DRE-80-13.5	WATER TREATMENT BLDG	MECHANICAL ROOM	13.5	480/60/3	62	80	300	1,2
WH-61642	A.O. SMITH DRE-80-13.5	WATER TREATMENT BLDG	MECHANICAL ROOM	13.5	480/60/3	62	80	300	1,2

NOTES:

1) FURNISH AND INSTALL STRAPS TO SEISMICALLY SECURE UNIT TO WALL.

2) PROVIDE ELECTRICAL DISCONNECT SWITCH AT WATER HEATER.

TANK SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	LOCATION	TYPE	TANK VOLUME (GAL)	DIMENSIONS		CONNECTION SIZE (IN.)	WEIGHT (LBS)	NOTES
						DIAMETER (IN)	HEIGHT (IN)			
TK-61641	AMTROL ST-30VC-DD	WATER TREATMENT BLDG	MECHANICAL ROOM	DIAPHRAGM	16.5	15	25	3/4	48	-
TK-61642	AMTROL WX-448C	WATER TREATMENT BLDG	MECHANICAL ROOM	DIAPHRAGM	80	24	59	2	315	-

NOTES:

-

DOMESTIC HOT WATER RE-CIRCULATING PUMP SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	LOCATION	FLOW RATE (GPM)	TDH (FT)	PIPE CONNECTION SIZE (IN.)	VOLTAGE	HP	MOTOR RPM	FLA	WEIGHT (LBS)	NOTES
HWP-61641	BELL AND GOSSETT PL-36B	WATER TREATMENT BLDG	MECHANICAL ROOM	10	30	1-1/4	115/60/1	1/6	3300	2.1	13	1,2,3

NOTES:

1) PROVIDE WITH MANUFACTURER'S AQUASTAT AND AUTOMATIC TIMER.

2) ALL WETTED PARTS SHALL BE LEAD FREE FOR POTABLE WATER APPLICATION.

3) PROVIDE ELECTRICAL DISCONNECT SWITCH AT PUMP.

TEPID WATER MIXING VALVE SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	APPLICATION	MINIMUM FLOW (GPM)	DESIGN FLOW (GPM)	DESIGN PRESSURE DROP (PSI)	RATED OUTLET TEMPERATURE RANGE (F)	OUTLET TEMPERATURE SETTING (°F)	CONNECTION SIZE (IN)	NOTES
TMV-1	HAWS TWBS.SHE	EMERGENCY EYE WASH / SHOWER	1	23	<5	60 - 90	75	1-1/4"	1,2,6
TMV-2	HAWS TWBS.SHE	EMERGENCY EYE WASH / SHOWER	1	23	<5	60 - 90	75	1-1/4"	1,3,6
TMV-3	WATTS LFE480	SINKS	0.5	1.5	<5	80 - 120	110	1/2"	4,5,6
TMV-4	WATTS LFG480	BATHROOM LAVATORIES	0.25	0.5	<5	80 - 120	110	1/2"	4,5,6
TMV-5	HAWS 9201EW	LAB SINK EYE WASH	1	3.7	<5	60 - 90	75	1/2"	1,5,6

NOTES:

1) MIXING VALVE SHALL BE CERTIFIED PER ANSI Z358.1 FOR USE ON EMERGENCY FIXTURES.

2) PROVIDE SURFACE MOUNT STAINLESS STEEL CABINET.

3) PROVIDE RECESSED MOUNT STAINLESS STEEL CABINET.

4) MIXING VALVE SHALL COMPLY WITH ASSE 1070.

5) INSTALL MIXING VALVE WITHIN RECESSED CABINET WITH HINGED ACCESS DOOR, OR CONCEALED BENEATH SINK IN ACCESSIBLE LOCATION.

6) MIXING VALVES SHALL COMPLY WITH NSF 61/372 FOR USE IN POTABLE WATER APPLICATIONS.

DOMESTIC WATER PRESSURE BOOSTER SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	NUMBER OF PUMPS	FLOW RATE (GPM)		REQUIRED PRESSURE BOOST (PSI)	ELECTRICAL V/HZ/PH	MOTOR HP		MOTOR RPM	FLA	WEIGHT (LBS)	NOTES
					PER PUMP	TOTAL			PER PUMP	TOTAL HP				
DWBP-61641	FLOW THERM FMV2.1-1LH	WATER TREATMENT BLDG	MECHANICAL ROOM	2	35	70	25	480/60/3	1	2	3,500	4.0	740	1,2

NOTES:

1) PACKAGED VARIABLE SPEED DUPLEX PUMP SKID FOR POTABLE WATER APPLICATION, INCLUDING CONTROL PANEL, INTERCONNECTING PIPING, AND VALVES.

2) SPECIFIED UNDER SECTION 22 11 26.

GENERAL NOTES:

A. PLUMBING SCHEDULES PRESENT INFORMATION ON THE SALIENT CHARACTERISTICS OF PLUMBING EQUIPMENT.

B. BRAND NAMES LISTED ARE CONSIDERED BASIS OF DESIGN. OR EQUAL SUBSTITUTIONS ARE ACCEPTABLE MEETING THE REQUIREMENTS OF 01 33 23 AND THE PRODUCT REQUIREMENTS LISTED IN THE ASSOCIATED TECHNICAL SPECIFICATIONS.

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National Council of Examiners for Engineering and Surveying
CERTIFICATE
66199
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 P601	TITLE OF SHEET WATER TREATMENT SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 123 OF 192

PLUMBING FIXTURE SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	FIXTURE TYPE	CONNECTION SIZE (IN)				REMARKS
					WASTE	VENT	CW	HW	
ESEW-61641	HAWS 8355WCW	WATER TREATMENT FACILITY	LAB	RECESSED EMERGENCY SHOWER/ EYEWASH	-	-	-	1"	COMBINATION EMERGENCY SHOWER/EYEWASH, COMPLIANT WITH ANSI Z358.1. SHOWER: 20 GPM, EYEWASH: 3.7 GPM. PROVIDE STAINLESS STEEL RECESSED CABINET WITH PULL-TO OPEN EYE WASH. PROVIDE WITH MANUFACTURERS FLOW SWITCH FOR ALARM NOTIFICATION. PROVIDE WITH MIXING VALVE TMV-2.
ESEW-61642	HAWS 8336	WATER TREATMENT FACILITY	ACIDS ROOM	EMERGENCY SHOWER/ EYEWASH	-	-	-	1-1/4"	COMBINATION EMERGENCY SHOWER/EYEWASH, CORROSION RESISTANT, COMPLIANT WITH ANSI Z358.1. SHOWER: 20 GPM, EYEWASH: 3.7 GPM. PROVIDE WITH MANUFACTURERS FLOW SWITCH FOR ALARM NOTIFICATION. PROVIDE WITH MIXING VALVE TMV-1.
ESEW-61643	HAWS 8336	WATER TREATMENT FACILITY	BASES ROOM	EMERGENCY SHOWER/ EYEWASH	-	-	-	1-1/4"	COMBINATION EMERGENCY SHOWER/EYEWASH, CORROSION RESISTANT, COMPLIANT WITH ANSI Z358.1. SHOWER: 20 GPM, EYEWASH: 3.7 GPM. PROVIDE WITH MANUFACTURERS FLOW SWITCH FOR ALARM NOTIFICATION. PROVIDE WITH MIXING VALVE TMV-1.
FLS-1	JAY R SMITH 3200	WATER TREATMENT FACILITY	PROCESS AREAS	FLOOR SINK	4"	-	-	-	16 1/2" X 16 1/2" CAST IRON BODY, NICKEL BRONZE GRATE, 4" WASTE CONNECTION
FD-1	JAY R SMITH 2010	WATER TREATMENT FACILITY	RESTROOMS, MECH RM, LABORATORY, PROCESS MATERIALS SHOP	FLOOR DRAIN	3"	-	-	-	GALV CAST IRON BODY, 5" DIA NICKEL BRONZE STRAINER, ADJUSTABLE, TRAP PRIMER CONNECTION
FD-2	JAY R SMITH 2310	WATER TREATMENT FACILITY	PROCESS AREAS, PUMP ROOM	PROCESS FLOOR DRAIN	4"	-	-	-	GALV CAST IRON BODY, 8" DIA GALV CAST IRON STRAINER, ADJUSTABLE
FD-3	JAY R SMITH 9725	WATER TREATMENT FACILITY	CHEM ROOMS, PROCESS ROOMS	FLOOR DRAIN	3"	-	-	-	STAINLESS STEEL BODY, 8" DIA TYPE 316 STAINLESS STEEL STRAINER, ADJUSTABLE
ED-1	-	WATER TREATMENT FACILITY	PROCESS AREAS, MECH ROOMS	EQUIPMENT DRAIN	-	-	-	-	OPEN-ENDED DRAIN PIPE THROUGH FLOOR SLAB, SIZE AS NOTED ON PLAN SHEETS
FCO-1	JAY R SMITH 4123	WATER TREATMENT FACILITY	NON-PROCESS AREAS, MECH ROOMS, RESTROOMS	FLOOR CLEANOUT	4"	-	-	-	GALV CAST IRON BODY, NICKEL BRONZE TOP, BRONZE PLUG
FCO-2	JAY R SMITH 4237	WATER TREATMENT FACILITY	PROCESS AREAS	PROCESS FLOOR CLEANOUT	4"	-	-	-	GALV CAST IRON BODY, DUCTILE IRON TOP, BRONZE PLUG
FCO-3	JAY R SMITH 4223	WATER TREATMENT FACILITY	EXTERNAL	CLEANOUT TO GRADE	6"	-	-	-	HEAVY DUTY GALV CAST IRON BODY AND DUCTILE IRON TOP, BRONZE PLUG
CTG-1	JAY R SMITH 4263L	WATER TREATMENT FACILITY	-	CLEANOUT TO GRADE	4"	-	-	-	HEAVY DUTY GALV CAST IRON BODY AND DUCTILE IRON TOP, BRONZE PLUG
WCO-1	JAY R SMITH 4530	WATER TREATMENT FACILITY	-	WALL CLEANOUT	-	-	-	-	CAST IRON TEE, BRONZE PLUG, POLISHED BRONZE COVER FOR FINISHED AREAS
MS-1	FIAT MSB2424	WATER TREATMENT FACILITY	-	MOP SINK	3"	1-1/2"	1/2"	1/2"	24"X24"X10" ONE-PIECE MOLDED STONE MOP BASIN, STAINLESS STEEL STRAINER. FURNISH WITH WALL MOUNTED FAUCET, EQUAL TO CHICAGO 897 SERIES, WITH WALL BRACE, INDEPENDENT HOT AND COLD QUARTER-TURN LEVERS, INTEGRAL SUPPLY STOPS AND VACUUM BREAKER.
SK-1	JUST MANUFACTURING SLADA2131A65-J	WATER TREATMENT FACILITY	BREAK ROOMS	DROP-IN SINK (ADA COMPLIANT)	1-1/2"	1-1/2"	1/2"	1/2"	304 STAINLESS STEEL, DROP-IN SINK, SINGLE COMPARTMENT, 31" X 21-1/4" X 6-1/2". PROVIDE WITH CHICAGO 430 SERIES MANUAL FAUCET, SINGLE HOLE, SINGLE LEVER, 1.5 GPM LAMINAR FLOWRATE, PROVIDE MIXING VALVE TMV-3 AND CARTRIDGE TYPE WATER FITLER BELOW DECK.
SK-21	-	WATER TREATMENT FACILITY	LABORATORY	LAB SINK	1-1/2"	1-1/2"	1/2"	1/2"	REFER TO ARCHITECTURAL LAB FURNISHING SPECS FOR SINK DETAILS. PROVIDE DOMESTIC CW/HW FAUCET: CHICAGO LWM1 SERIES. PROVIDE PURE WATER FAUCET: CHICAGO LWS2 SERIES. PROVIDE DECK MOUNTED EMERGENCY EYEWASH: GUARDIAN G5022. PROVIDE MIXING VALVES BELOW DECK: TMV-3 FOR SINK FAUCET AND TMV-5 FOR EYE WASH. PROVIDE CARTRIDGE TYPE DEIONIZATION WATER FILTER BELOW DECK FOR PURE WATER FAUCET. PROVIDE ACID NEUTRALIZATION CARTRIDGE BELOW DECK FOR SINK WASTE.
LAV-1	SLOAN SS-3865	WATER TREATMENT FACILITY	RESTROOMS	LAVATORY (ADA COMPLIANT)	1-1/2"	1-1/2"	1/2"	1/2"	WALL HUNG, VITREOUS CHINA, 21-1/2" X 20-1/2". PROVIDE WITH CHICAGO FAUCET MODEL NUMBER 405-HZ, 8" CENTERS, TWO QUARTER-TURN LEVERS, 8" CENTERS, 0.5 GPM FLOWRATE, PROVIDE MIXING VALVE TMV-4 BELOW DECK.
WC-1	SLOAN ST-2229	WATER TREATMENT FACILITY	RESTROOMS	FLOOR MOUNTED WATER CLOSET (ADA COMPLIANT)	4"	2"	1"	-	FLOOR-MOUNTED, VITREOUS CHINA, TOP SPUD, ELONGATED BOWL. PROVIDE WITH SLOAN ROYAL 113 MANUAL FLUSH VALVE, 1.6 GALLONS PER FLUSH.
TP-3	PPP PTS-6	WATER TREATMENT FACILITY	-	TRAP PRIMER	-	-	3/4"	-	ELECTRONIC TRAP PRIMER, WALL MOUNTED ENCLOSURE 120V POWER CONNECTION, 24-HR TIMER OPERATED SOLENOID VAVLE, ANTISIPHON VACUUM BREAKER, 4 OR MORE DRAINS SERVED.
TP-4	PPP MPB-500	WATER TREATMENT FACILITY	-	TRAP PRIMER	-	-	1/2"	-	ELECTRONIC TRAP PRIMER, WALL MOUNTED ENCLOSURE 120V POWER CONNECTION, 24-HR TIMER OPERATED SOLENOID VAVLE, ANTISIPHON VACUUM BREAKER, 1 TO 4 DRAINS SERVED.
WB-1	MAINLINE ICE MAKER VALVE BOX	WATER TREATMENT FACILITY	BREAK ROOM	VALVE WALL BOX	-	-	1/2"	1/2"	RECESSED PLASTIC VALVE BOX WITH QUARTER TURN BRONZE ANGLE VALVE AND WATER HAMMER ARRESTOR FOR CW CONNECTION.
HB-1	-	WATER TREATMENT FACILITY	-	HOSE BIB	-	-	3/4"	-	GLOBE VALVE WITH MALE THREADED HOSE ADAPTER AND VACUUM BREAKER
HR-1	-	WATER TREATMENT FACILITY	-	HOSE REEL	-	-	3/4"	-	50FT INDUSTRIAL GRADE RUBBER HOSE, THREADED END CONNECTIONS

-

GENERAL NOTES:

A. PLUMBING SCHEDULES PRESENT INFORMATION ON THE SALIENT CHARACTERISTICS OF PLUMBING EQUIPMENT.

B. BRAND NAMES LISTED ARE CONSIDERED BASIS OF DESIGN. OR EQUAL SUBSTITUTIONS ARE ACCEPTABLE MEETING THE REQUIREMENTS OF 01 33 23 AND THE PRODUCT REQUIREMENTS LISTED IN THE ASSOCIATED TECHNICAL SPECIFICATIONS.

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A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

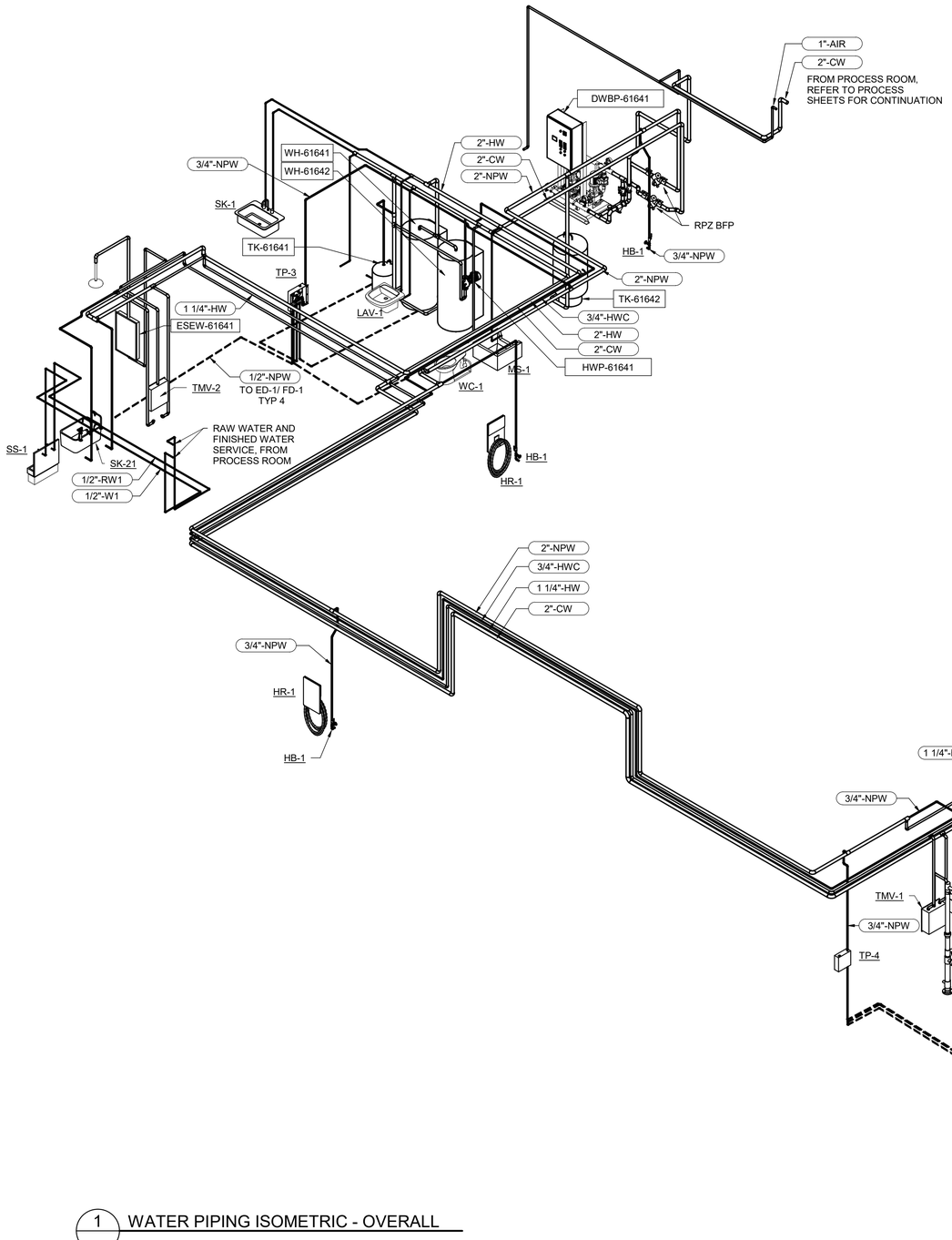
DESIGNED: A. PUTMAN
CADD: B. BAXTER
TECH. REVIEW: H. PEDRAZA
DATE: MAY 2026

SUB SHEET NO. 062 P602

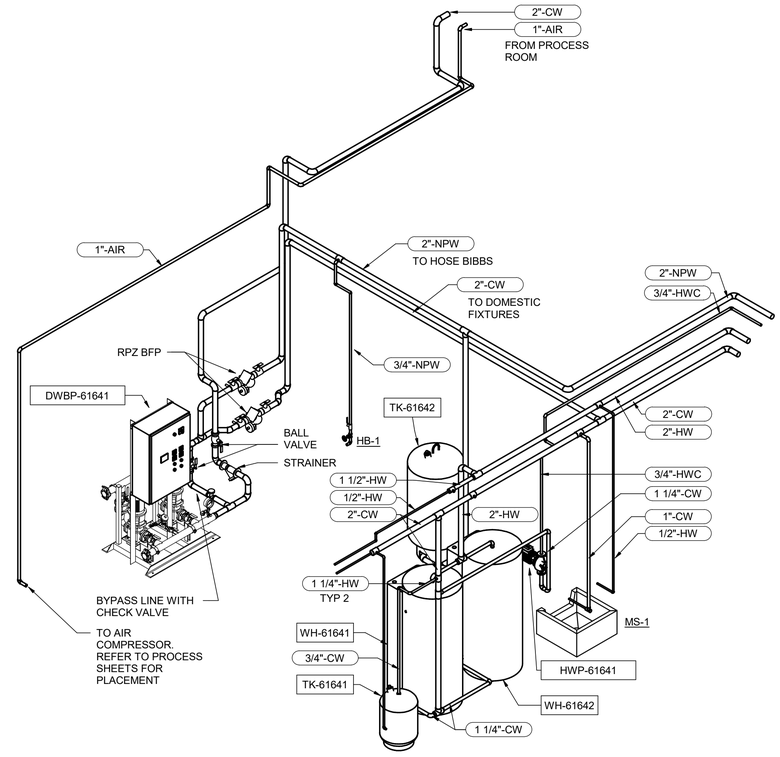
TITLE OF SHEET
WATER TREATMENT
SCHEDULES

NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 124 OF 192



1 WATER PIPING ISOMETRIC - OVERALL



2 WATER PIPING ISOMETRIC - MECHANICAL ROOM

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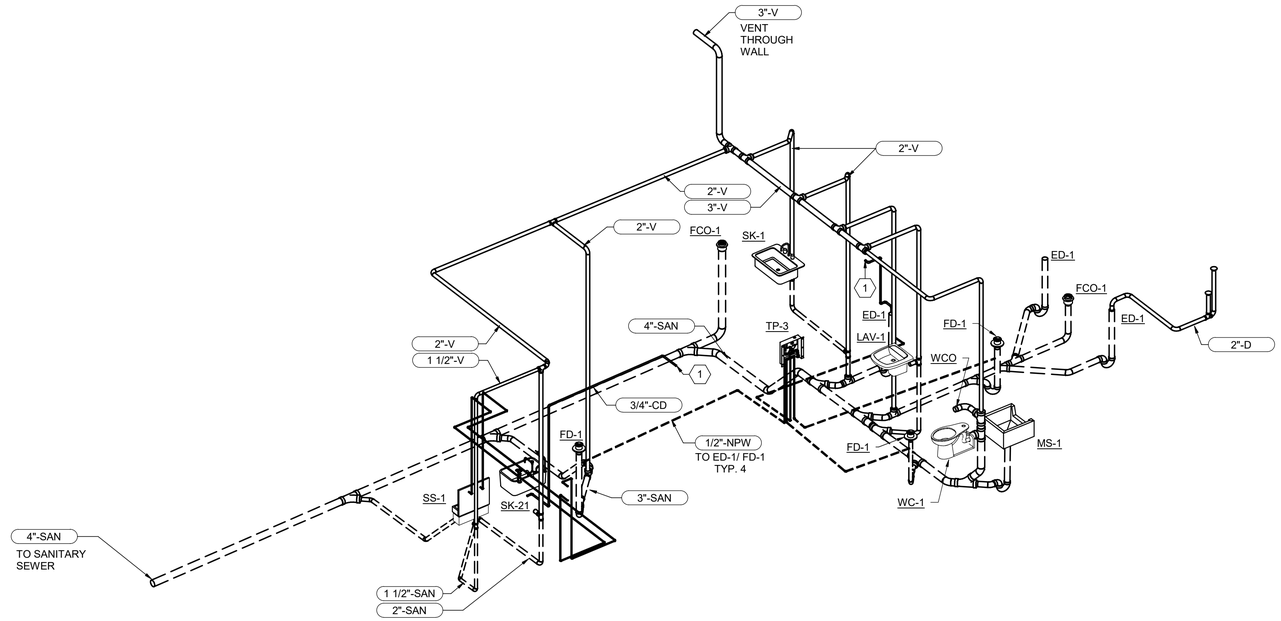
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 P901	TITLE OF SHEET WATER TREATMENT WATER PIPING ISOMETRICS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 125 OF 192

KEYNOTES: (X)

1. 3/4" CONDENSATE DRAIN FROM MECHANICAL EQUIPMENT. 2% SLOPE. SEE M SHEETS FOR UNIT PLACEMENT.
2. DRAIN FROM BFP AIR GAP FITTING TO EQUIPMENT DRAIN.



1 ADMIN AREA WASTE AND VENT ISOMETRIC

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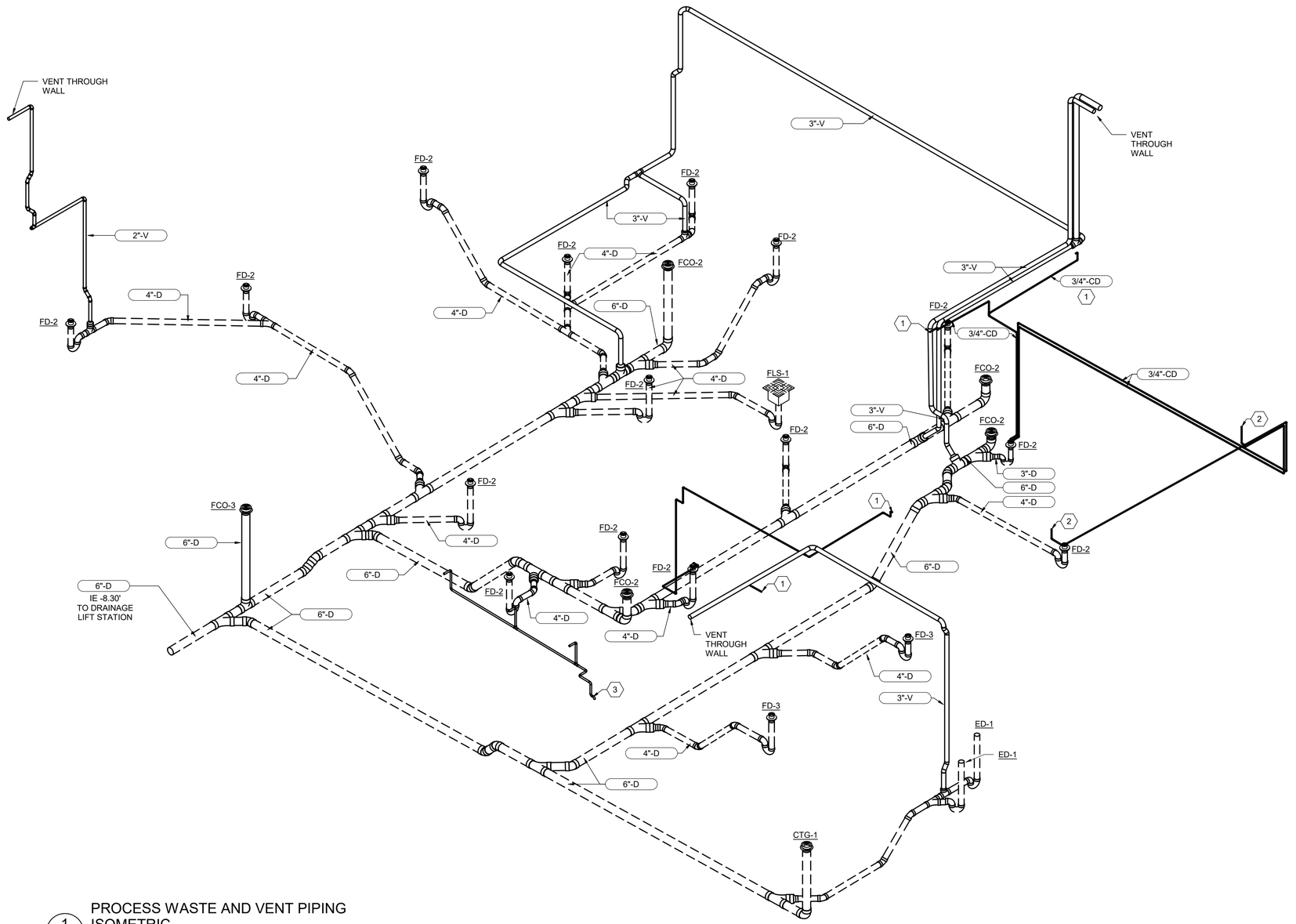
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 P902	TITLE OF SHEET WATER TREATMENT DOMESTIC WASTE AND VENT ISOMETRICS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 126 OF 192



1 PROCESS WASTE AND VENT PIPING ISOMETRIC

KEYNOTES: (X)

1. 3/4" CONDENSATE DRAIN FROM MECHANICAL EQUIPMENT. 2% SLOPE. SEE M SHEETS FOR UNIT PLACEMENT.
2. DRAIN FROM PLENUM DUCT, REFER TO HVAC SHEETS FOR LOCATION. PROVIDE BALL VALL ON DRAIN LINE IN ACCESSIBLE LOCATION, LEAVE CRACKED OPEN FOR DRAINAGE.
3. 1" CONDENSATE DRAIN FROM DRAIN PAN BENEATH HEAT PUMPS, TERMINATE OUTSIDE ENCLOSURE WITH ELBOW DOWN, 2" AFF.

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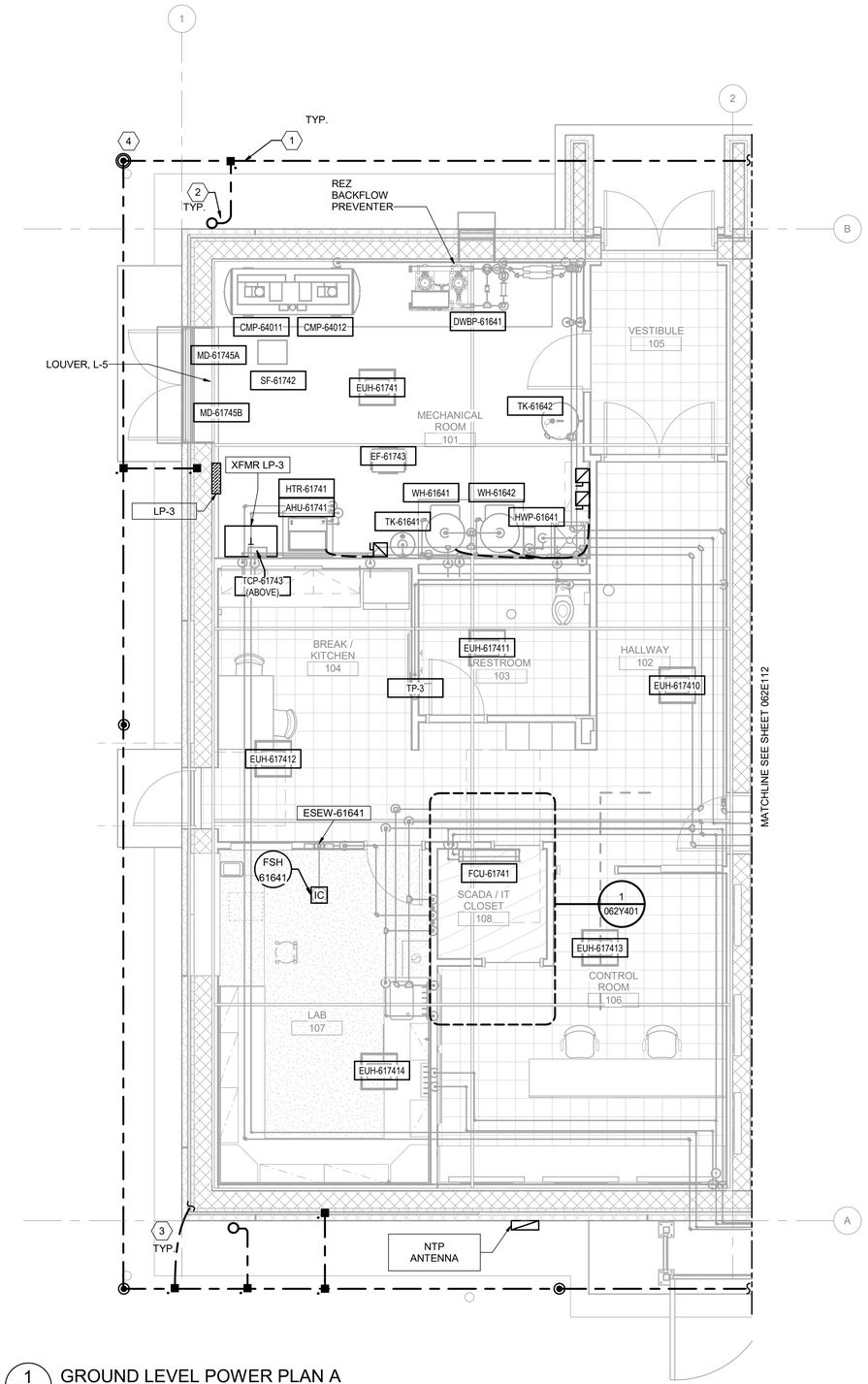
Professional Engineer Illinois CERTIFICATE 062-071199 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 062 P903	TITLE OF SHEET WATER TREATMENT PROCESS WASTE AND VENT ISOMETRICS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 127 OF 192



1 062E111 GROUND LEVEL POWER PLAN A

SCALE (A)

KEYNOTES (X)

1 #4/0 AWG SOFT-DRAWN BARE COPPER GROUND RING.

2 #4/0 AWG CONNECTION TO LIGHTNING PROTECTION SYSTEM DOWNLEAD. COORDINATE DOWNLEAD LOCATIONS WITH LIGHTNING PROTECTION SYSTEM DESIGNER.

3 #4/0 AWG BOND TO BUILDING, PIPING AND SYSTEMS PER NEC.

4 GROUND ROD WITH GROUNDING TEST WELL.

AREA CLASSIFICATION TYPE

CONTRACTOR SHALL PROVIDE AND INSTALL ALL APPLICABLE ELECTRICAL COMPONENTS, DEVICES,EQUIPMENT, ETC. PER THE AREA CLASSIFICATION TYPE LISTED FOR EACH ROOM TYPE INDICATED ON THIS SHEET. REFER TO ELECTRICAL SPECIFICATIONS FOR ADDITIONAL INFORMATION.

ROOM TYPE:	AREA CLASSIFICATION:
100 - MECHANICAL ROOM	INDOOR - WET
101 - LOCKER/LAUNDRY ROOM	INDOOR - WET
102 - CORRIDOR	INDOOR - DRY
103 - STORAGE	INDOOR - DRY
104 - RESTROOM	INDOOR - WET
105 -BREAK/KITCHEN	INDOOR - WET
106 - VESTIBULE	INDOOR - DRY
107 - LAB	INDOOR - DRY
109 - CONTROL ROOM	INDOOR - DRY
000 - EXTERIOR CANOPY	EXTERIOR - WET

NOTES:
1. STAINLESS STEEL MATERIALS NOT PERMITTED IN SODIUM HYPOCHLORITE AREAS. FRP OR PVC-COATED ELECTRICAL MATERIALS ARE REQUIRED.

SCALE (A) 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL CERTIFICATE NO. 58037 TRAVIS T. MOORE DATE SIGNED 1/14/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 062 E111	TITLE OF SHEET WATER TREATMENT GROUND LEVEL POWER PLAN - ADMIN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 128 OF 192

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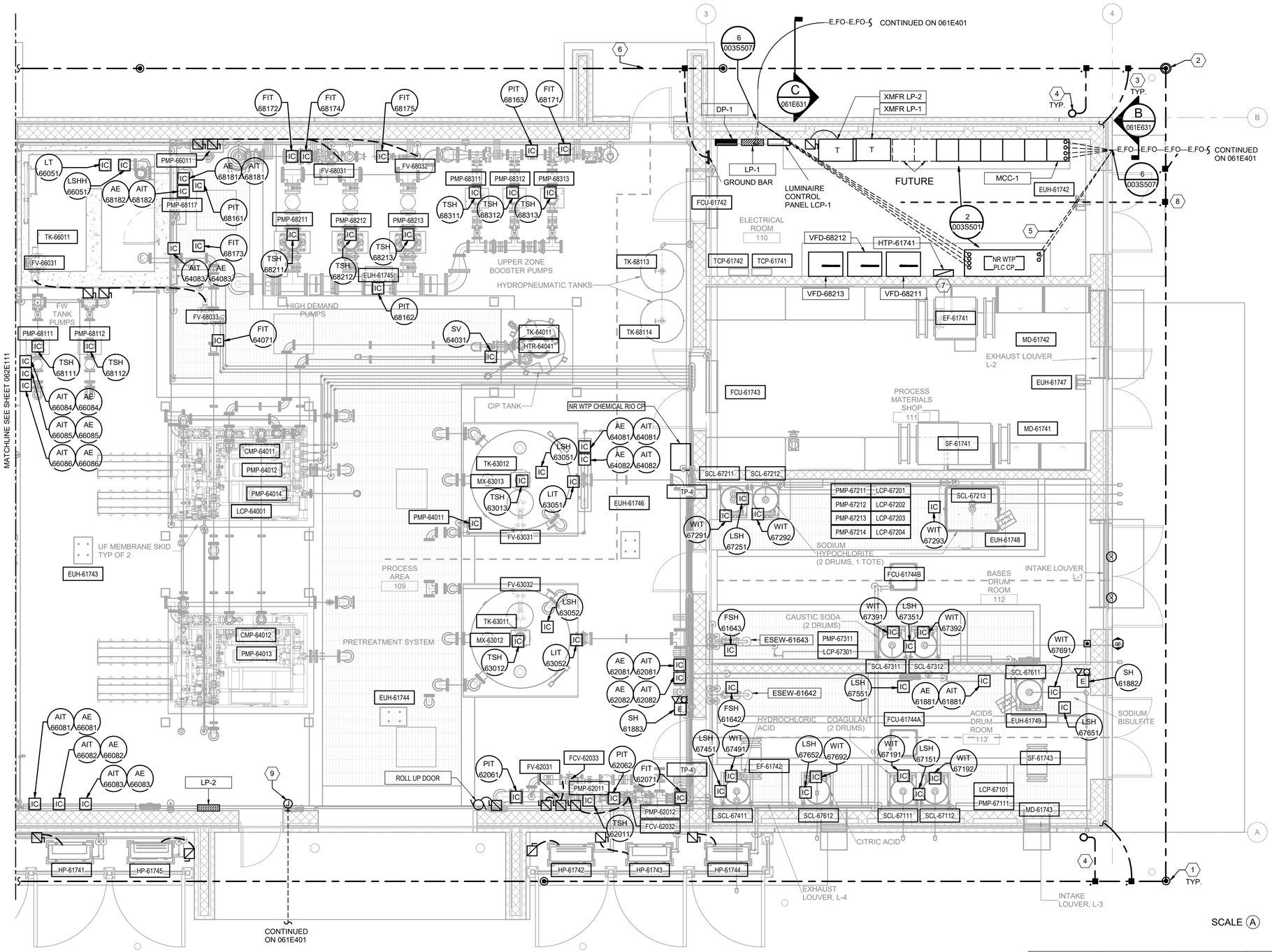

B 062E112 GROUND LEVEL POWER PLAN B

SCALE A

KEYNOTES X

1 10FT X 3/4INCH COPPER-CLAD STEEL GROUND ROD.

2 GROUND ROD WITH GROUNDING TEST WELL.

3 #4/0 AWG BOND TO BUILDING, PIPING AND SYSTEMS PER NEC.

4 #4/0 AWG CONNECTION TO LIGHTNING PROTECTION SYSTEM DOWNLEAD. COORDINATE DOWNLEAD LOCATIONS WITH LIGHTNING PROTECTION SYSTEM DESIGNER.

5 FIBEROPTIC LINE FROM DUCT BANK.

6 #4/0 AWG SOFT-DRAWN BARE COPPER GROUND RING.

7 THE DESIGN OF THE HEAT TRACE AND HEAT TRACE CONTROLLER IS DELEGATED TO THE CONTRACTOR. CONTRACTOR SHALL PROVIDE A CODE COMPLIANT SYSTEM MEETING THE REQUIREMENTS OF HEAT TRACE AND A CONTROLLER THAT IS BEING USED FOR A FIRE PROTECTION SYSTEM PER SPECIFICATION SECTION 40 41 13 AS WELL AS ALL LOCAL AND NATIONAL CODES.

8 CONCRETE-ENCASED ELECTRODE, 20FT MIN.

9 TRANSITION UNDERGROUND FOR PROCESS WASTE TANK INSTRUMENTATION. REFER TO SHEET 061E401 FOR SITE ROUTING. TO BE ROUTED UNDERGROUND IN 2 INCH SCHEDULE 80 PVC CONDUIT.

AREA CLASSIFICATION TYPE

CONTRACTOR SHALL PROVIDE AND INSTALL ALL APPLICABLE ELECTRICAL COMPONENTS, DEVICES,EQUIPMENT, ETC. PER THE AREA CLASSIFICATION TYPE LISTED FOR EACH ROOM TYPE INDICATED ON THIS SHEET. REFER TO ELECTRICAL SPECIFICATIONS FOR ADDITIONAL INFORMATION.

ROOM TYPE:	AREA CLASSIFICATION:
109 - PROCESS AREA	INDOOR - WET/CORROSIVE
110 - ELECTRICAL ROOM	INDOOR - DRY
111 - MAINTENANCE ROOM	INDOOR - WET/CORROSIVE
112 - CHEM DRUM RM. (NOTE 1)	INDOOR - WET/CORROSIVE
113 - ACIDS DRUM RM.	INDOOR - WET/CORROSIVE
000 - EXTERIOR CANOPY	OUTDOOR - WET

NOTES:
1. STAINLESS STEEL MATERIALS NOT PERMITTED IN SODIUM HYPOCHLORITE AREAS. FRP OR PVC-COATED ELECTRICAL MATERIALS ARE REQUIRED.

SCALE A 4 0 4 8 1/4" = 1'-0"

SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — Date Signed 8/11/2026 — ARIZONA, U.S.A.

A/E FIRM: PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 062 E112	TITLE OF SHEET: WATER TREATMENT GROUND LEVEL POWER PLAN - PROCESS NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 129 OF 192

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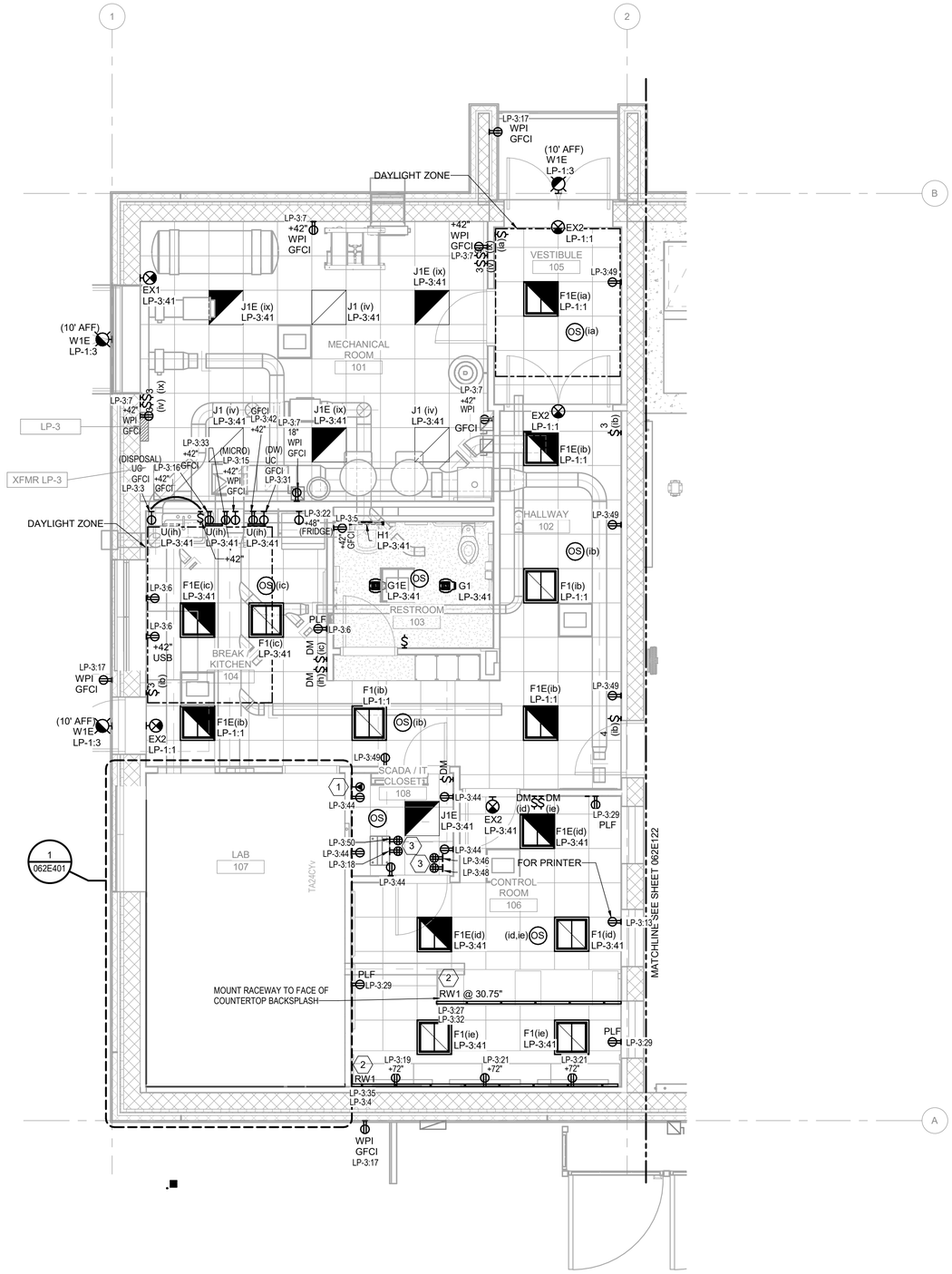

1 062E121 GROUND LEVEL LIGHTING AND RECEPTACLE PLAN - ADMIN

SCALE (A)

KEYNOTES (X)

1 PROVIDE GROUNDING JUNCTION BOX FROM ADJACENT RECEPTACLE. CONNECT GROUNDING STRAP FROM ESD FLOORING TO SYSTEM GROUND.

2 DUAL CHANNEL (ELECTRICAL/TELECOM) TWO-PIECE STEEL RACEWAY. COORDINATE AND PROVIDE PROPER MOUNTING WITH PROVIDED MILLWORK/FURNITURE. REFER TO RACEWAY SCHEDULE ON SHEET 062-E634 FOR ADDITIONAL INFORMATION.

3 RACK MOUNTED. COORDINATE FINAL LOCATION WITH SPECIAL SYSTEM CONTRACTOR.

GENERAL NOTES:

A. REFER TO SHEETS 062-E632 THROUGH 062-E634 FOR LUMINAIRE SCHEDULES. REFER TO SHEET 008-E503 FOR LUMINAIRE DETAILS AND SHEET 008-E504 FOR LUMINAIRE SCHEMATICS.

B. COORDINATE FINAL LOCATION OF LUMINAIRES WITH OTHER TRADES PRIOR TO ROUGH-IN. ADJUST LUMINAIRE LOCATIONS AND HEIGHT ACCORDINGLY TO AVOID ANY CONFLICT.

C. COORDINATE LOCATION OF LUMINAIRES WITH STRUCTURE. PROVIDE ADDITIONAL STRUCTURAL SUPPORTS FOR LUMINAIRES IN ORDER TO ACHIEVE REQUIRED MOUNTING HEIGHT.

D. EMERGENCY EXIT SIGNS AND EMERGENCY LUMINAIRE UNITS TO BE UNSWITCHED.

E. REFER TO POWER PLANS FOR AREA CLASSIFICATION TYPES.

F. REFER TO LUMINAIRE CONTROLS SEQUENCE SCHEDULE ON SHEET 062-E634 FOR ALL INTERIOR AND EXTERIOR SPACE/AREA TYPE LUMINAIRE CONTROLS REQUIREMENTS.

G. ALL BUILDING EXTERIOR LUMINAIRES TO BE CONTROLLED BY THE BUILDING LUMINAIRE CONTROL PANEL, UNLESS NOTED OTHERWISE.

PHOTOMETRIC AND POWER DENSITY STATISTICS - EXTERIOR

DESCRIPTION	AVERAGE	MAX./MIN.	LLF	W/FT2	TOTAL LINEAR FT.	W/LINEAR FT.
BUILDING ENTRIES DOORS	1 FC @ FF	5.0:1	0.78	N/A	39	11.67
LOADING DOCK	1 FC @ FF	5.0:1	0.78	0.20	N/A	N/A

NOTE:
1. CONTRACTOR TO SUBMIT WHOLE BUILDING (INDIVIDUAL INTERIOR ROOMS/ SPACES) AND PROJECT LIMITS (EXTERIOR BASE SITE, PARKING, ENTRANCES/ EXITS, CANOPIES, WALKWAYS, AND BUILDING FAÇADE) PHOTOMETRIC AND POWER DENSITY ANALYSIS FOR ALL SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS NOT LISTED ON THE LUMINAIRE SCHEDULE. SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS MUST MEET THE REQUIRED PHOTOMETRIC AND POWER DENSITY STATISTICS FOR ALL AREAS/ SPACES INDICATED ON THIS SHEET. INCREASING LUMINAIRE QUANTITIES SHOWN ON THIS SHEET, INCREASING THE LUMEN OUTPUT ON THE LUMINAIRE SCHEDULE, OR INCREASING THE VA (VOLTAGE AMPS) ON THE LUMINAIRE SCHEDULE FOR THE PHOTOMETRIC AND POWER DENSITY STATISTICS COMPLIANCE IS PROHIBITED.

PHOTOMETRIC AND POWER DENSITY STATISTICS - INTERIOR

DESCRIPTION	AVERAGE	MAX./MIN.	LIGHT LOSS FACTOR (LLF)	W/FT² (WHOLE BLDG)
101 - MECHANICAL ROOM	40 FC @ FF	3.0:1	0.82	0.60
102 - HALLWAY	15 FC @ FF	3.0:1	0.90	
103 - RESTROOM	15 FC @ FF	3.0:1	0.90	
104 - BREAK/ KITCHEN	40 FC @ 30" AFF	3.0:1	0.90	
105 - VESTIBULE	10 FC @ FF	2.0:1	0.90	
106 - CONTROL ROOM	40 FC @ 30" AFF	2.0:1	0.90	
108 - SERVER ROOM	40 FC @ 30" AFF	2.0:1	0.90	

NOTE:
1. CONTRACTOR TO SUBMIT WHOLE BUILDING (INDIVIDUAL INTERIOR ROOMS/ SPACES) AND PROJECT LIMITS (EXTERIOR BASE SITE, PARKING, ENTRANCES/ EXITS, CANOPIES, WALKWAYS, AND BUILDING FAÇADE) PHOTOMETRIC AND POWER DENSITY ANALYSIS FOR ALL SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS NOT LISTED ON THE LUMINAIRE SCHEDULE. SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS MUST MEET THE REQUIRED PHOTOMETRIC AND POWER DENSITY STATISTICS FOR ALL AREAS/ SPACES INDICATED ON THIS SHEET. INCREASING LUMINAIRE QUANTITIES SHOWN ON THIS SHEET, INCREASING THE LUMEN OUTPUT ON THE LUMINAIRE SCHEDULE, OR INCREASING THE VA (VOLTAGE AMPS) ON THE LUMINAIRE SCHEDULE FOR THE PHOTOMETRIC AND POWER DENSITY STATISTICS COMPLIANCE IS PROHIBITED.

SCALE (A) 4 0 4 8 1/4" = 1'-0"

SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE STATE OF COLORADO 4/28/2026 U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. MARTINEZ CADD S. BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 062 E121	TITLE OF SHEET WATER TREATMENT GROUND LEVEL LIGHTING & RECEPTACLE PLAN - ADMIN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 130 OF 192

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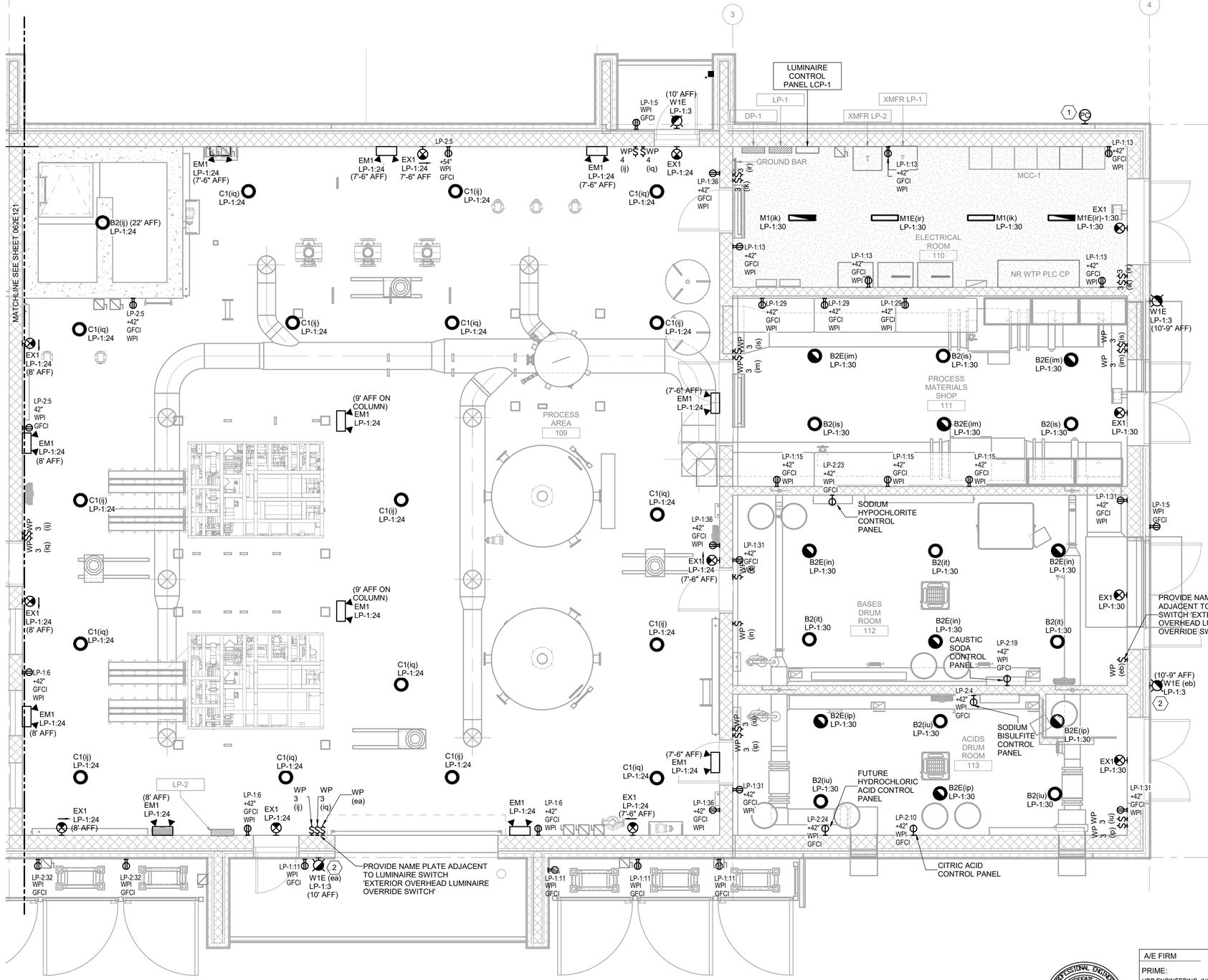

1 062E122 GROUND LEVEL ELECTRICAL LIGHTING AND RECEPTACLE PLAN - PROCESS

SCALE (A)

SCALE (A) 4 0 4 8 1/4" = 1'-0"

SCALE OF FEET

GENERAL NOTES:

A. REFER TO SHEETS 062-E632 THROUGH 062-E634 FOR LUMINAIRE SCHEDULES. REFER TO SHEET 008-E503 FOR LUMINAIRE DETAILS AND SHEET 008-E504 FOR LUMINAIRE SCHEMATICS.

B. COORDINATE FINAL LOCATION OF LUMINAIRES WITH OTHER TRADES PRIOR TO ROUGH-IN. ADJUST LUMINAIRE LOCATIONS AND HEIGHT ACCORDINGLY TO AVOID ANY CONFLICT.

C. COORDINATE LOCATION OF LUMINAIRES WITH STRUCTURE. PROVIDE ADDITIONAL STRUCTURAL SUPPORTS FOR LUMINAIRES IN ORDER TO ACHIEVE REQUIRED MOUNTING HEIGHT.

D. EMERGENCY EXIT SIGNS AND EMERGENCY LUMINAIRE UNITS TO BE UNSWITCHED.

E. REFER TO POWER PLANS FOR AREA CLASSIFICATION TYPES.

F. REFER TO LUMINAIRE CONTROLS SEQUENCE SCHEDULE ON SHEET 062-E634 FOR ALL INTERIOR AND EXTERIOR SPACE/AREA TYPE LUMINAIRE CONTROLS REQUIREMENTS.

G. ALL BUILDING EXTERIOR LUMINAIRES TO BE CONTROLLED BY THE BUILDING LUMINAIRE CONTROL PANEL, UNLESS NOTED OTHERWISE.

KEYNOTES (X)

1 PHOTOCELL FOR BUILDING EXTERIOR LUMINAIRE CONTROLS 'LCP-01'. MOUNT AT 9'-0" AFF, FACING NORTH. REFER TO DETAIL #1 ON SHEET 008-E504.

2 EXTERIOR LUMINAIRE CONTROLLED BY LUMINAIRE CONTROL RELAY PANEL 'LCP-01' AND LOCAL LOW-VOLTAGE OVERRIDE SWITCH DEVICE.

PHOTOMETRIC AND POWER DENSITY STATISTICS - EXTERIOR

DESCRIPTION	AVERAGE	MAX./MIN.	LLF	W/FT2	TOTAL LINEAR FT.	W/LINEAR FT.
BUILDING ENTRIES DOORS	1 FC @ FF	5.0:1	0.78	N/A	39	11.67
LOADING DOCK	1 FC @ FF	5.0:1	0.78	0.20	N/A	N/A

NOTE:
1. CONTRACTOR TO SUBMIT WHOLE BUILDING (INDIVIDUAL INTERIOR ROOMS/ SPACES) AND PROJECT LIMITS (EXTERIOR BASE SITE, PARKING, ENTRANCES/ EXITS, CANOPIES, WALKWAYS, AND BUILDING FAÇADE) PHOTOMETRIC AND POWER DENSITY ANALYSIS FOR ALL SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS NOT LISTED ON THE LUMINAIRE SCHEDULE. SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS MUST MEET THE REQUIRED PHOTOMETRIC AND POWER DENSITY STATISTICS FOR ALL AREAS/ SPACES INDICATED ON THIS SHEET. INCREASING LUMINAIRE QUANTITIES SHOWN ON THIS SHEET, INCREASING THE LUMEN OUTPUT ON THE LUMINAIRE SCHEDULE, OR INCREASING THE VA (VOLTAGE AMPS) ON THE LUMINAIRE SCHEDULE FOR THE PHOTOMETRIC AND POWER DENSITY STATISTICS COMPLIANCE IS PROHIBITED.

PHOTOMETRIC AND POWER DENSITY STATISTICS - INTERIOR

DESCRIPTION	AVERAGE	MAX./MIN.	LIGHT LOSS FACTOR (LLF)	W/FT² (WHOLE BLDG)
109 - PROCESS AREA	40 FC @ FF	3.0:1	0.82	0.60
110 - ELECTRICAL ROOM	40 FC @ 30" AFF	3.0:1	0.90	
111 - PROCESS MATERIALS SHOP	40 FC @ 30" FF	3.0:1	0.82	
112 - BASES DRUM ROOM	35 FC @ FF	3.0:1	0.82	
113 - ACIDS DRUM ROOM	35 FC @ FF	3.0:1	0.82	

NOTE:
1. CONTRACTOR TO SUBMIT WHOLE BUILDING (INDIVIDUAL INTERIOR ROOMS/ SPACES) AND PROJECT LIMITS (EXTERIOR BASE SITE, PARKING, ENTRANCES/ EXITS, CANOPIES, WALKWAYS, AND BUILDING FAÇADE) PHOTOMETRIC AND POWER DENSITY ANALYSIS FOR ALL SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS NOT LISTED ON THE LUMINAIRE SCHEDULE. SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS MUST MEET THE REQUIRED PHOTOMETRIC AND POWER DENSITY STATISTICS FOR ALL AREAS/ SPACES INDICATED ON THIS SHEET. INCREASING LUMINAIRE QUANTITIES SHOWN ON THIS SHEET, INCREASING THE LUMEN OUTPUT ON THE LUMINAIRE SCHEDULE, OR INCREASING THE VA (VOLTAGE AMPS) ON THE LUMINAIRE SCHEDULE FOR THE PHOTOMETRIC AND POWER DENSITY STATISTICS COMPLIANCE IS PROHIBITED.

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. MARTINEZ DRAWN: S. BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 062 E122	TITLE OF SHEET WATER TREATMENT GROUND LEVEL LIGHTING & RECEPTACLE PLAN - PROCESS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 131 OF 192

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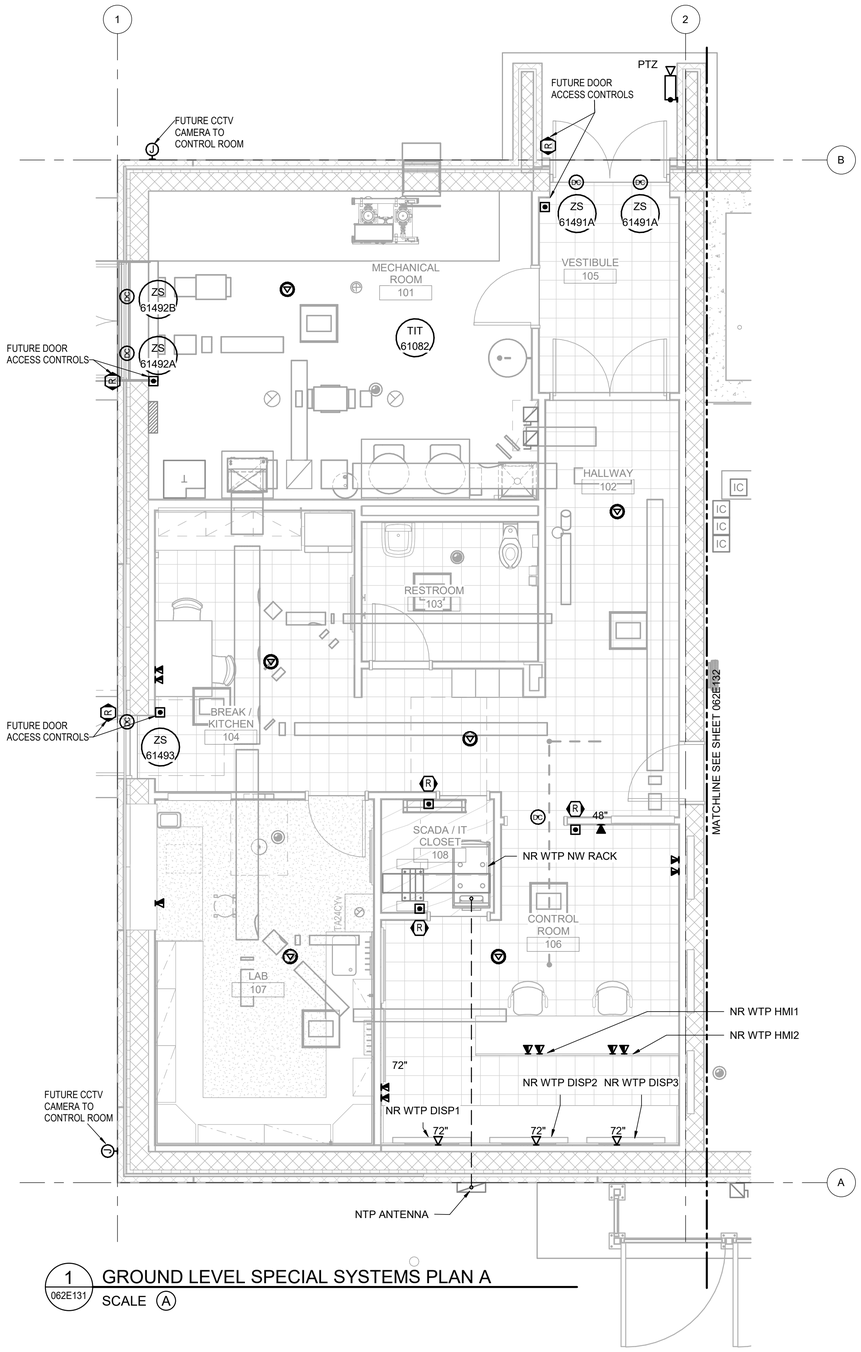

1 062E131 GROUND LEVEL SPECIAL SYSTEMS PLAN A

SCALE (A)

GENERAL NOTES:

A. REFER TO POWER PLANS FOR AREA CLASSIFICATION TYPES.

SCALE (A) 4 0 4 8 1/4" = 1'-0"

SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 2/4/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 062 E131	TITLE OF SHEET WATER TREATMENT GROUND LEVEL SPECIAL SYSTEMS PLAN - ADMIN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 132 OF 192

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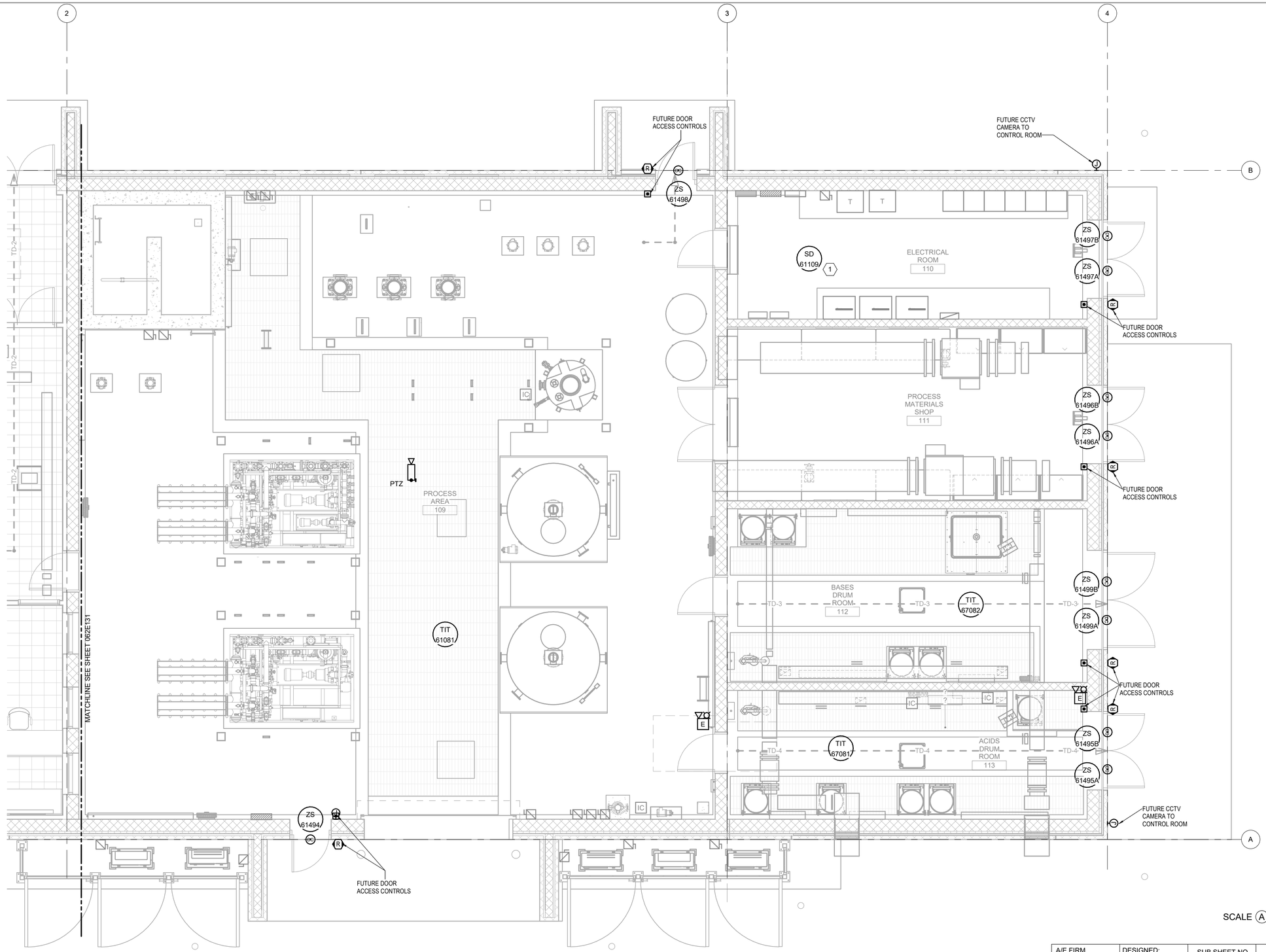

1 062E132 GROUND LEVEL SPECIAL SYSTEMS PLAN B

SCALE A

SCALE A 4 0 4 8 1/4" = 1'-0"

SCALE OF FEET

GENERAL NOTES:

A. REFER TO POWER PLANS FOR AREA CLASSIFICATION TYPES.

KEYNOTES: X

1. SMOKE DETECTORS ARE FOR ASSET PROTECTION ONLY. NOT REQUIRED BY CODE. PROVIDE A PHOTOELECTRIC SMOKE ALARM WITH HORN/STROBE OUTPUT AND FORM C RELAY CONTACT OUTPUT. CONTRACTOR TO PROVIDE 120VAC.

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — Date Signed 4/30/2026 — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 062 E132	TITLE OF SHEET WATER TREATMENT GROUND LEVEL SPECIAL SYSTEMS PLAN - PROCESS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 133 OF 192

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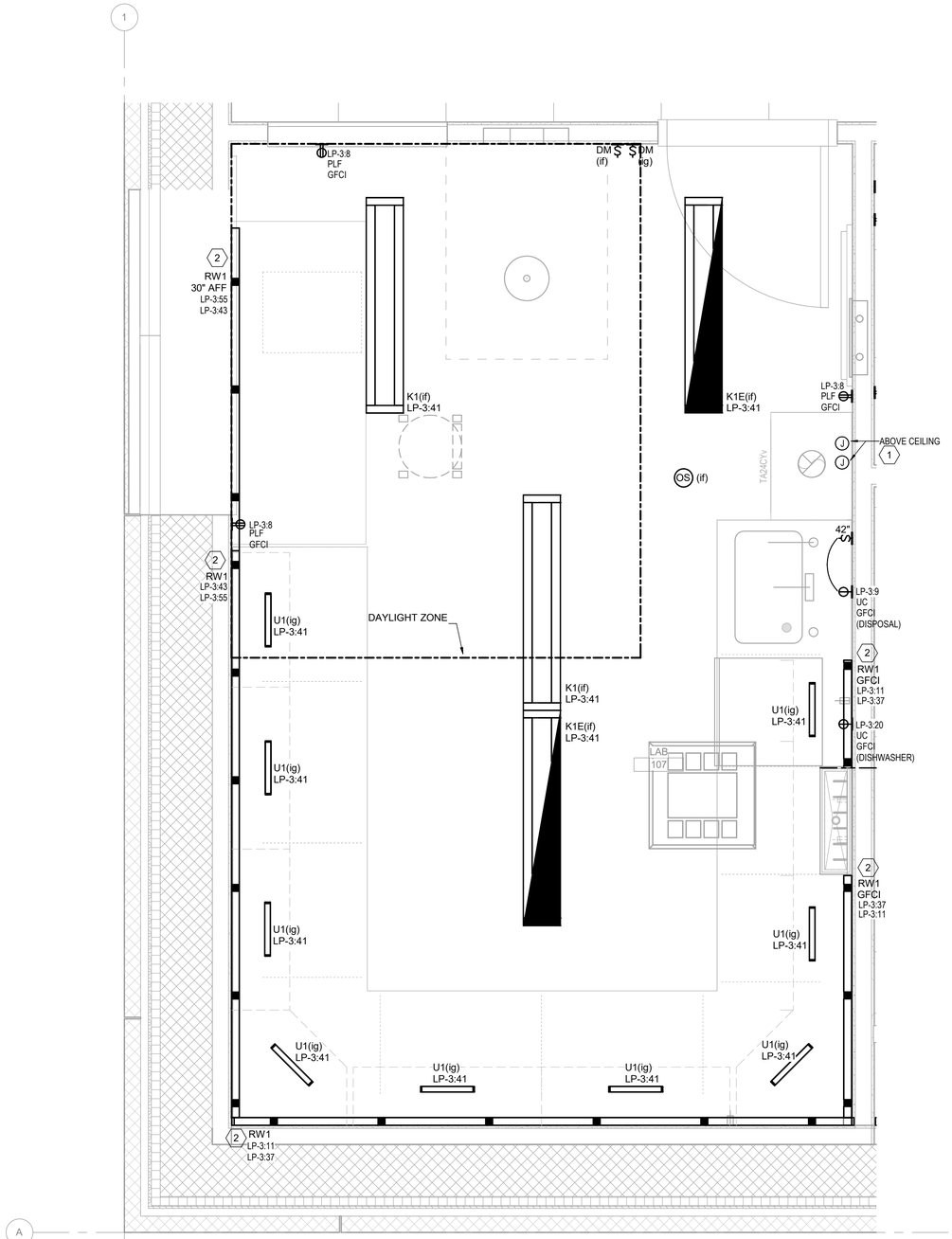

1 062E401 LIGHTING AND RECEPTACLE ENLARGED PLAN - LAB 107

SCALE A

GENERAL NOTES:

A. REFER TO SHEETS 062-E632 THROUGH 062-E634 FOR LUMINAIRE SCHEDULES. REFER TO SHEET 008-E503 FOR LUMINAIRE DETAILS AND SHEET 008-E504 FOR LUMINAIRE SCHEMATICS.

B. COORDINATE FINAL LOCATION OF LUMINAIRES WITH OTHER TRADES PRIOR TO ROUGH-IN. ADJUST LUMINAIRE LOCATIONS AND HEIGHT ACCORDINGLY TO AVOID ANY CONFLICT.

C. COORDINATE LOCATION OF LUMINAIRES WITH STRUCTURE. PROVIDE ADDITIONAL STRUCTURAL SUPPORTS FOR LUMINAIRES IN ORDER TO ACHIEVE REQUIRED MOUNTING HEIGHT.

D. EMERGENCY EXIT SIGNS AND EMERGENCY LUMINAIRE UNITS TO BE UNSWITCHED.

E. REFER TO POWER PLANS FOR AREA CLASSIFICATION TYPES.

F. REFER TO LUMINAIRE CONTROLS SEQUENCE SCHEDULE ON SHEET 062-E634 FOR ALL INTERIOR AND EXTERIOR SPACE/AREA TYPE LUMINAIRE CONTROLS REQUIREMENTS.

G. ALL BUILDING EXTERIOR LUMINAIRES TO BE CONTROLLED BY THE BUILDING LUMINAIRE CONTROL PANEL, UNLESS NOTED OTHERWISE.

KEYNOTES (X)

1 COORDINATE FINAL LOCATION OF FUTURE FUME HOOD WITH CONTRACTING OFFICER PRIOR TO ROUGH-IN.

2 DUAL CHANNEL (ELECTRICAL/TELECOM) TWO-PIECE STEEL RACEWAY. COORDINATE AND PROVIDE PROPER MOUNTING WITH PROVIDED MILLWORK/FURNITURE. REFER TO RACEWAY SCHEDULE ON SHEET 062-E634 FOR ADDITIONAL INFORMATION.

PHOTOMETRIC AND POWER DENSITY STATISTICS - INTERIOR

DESCRIPTION	AVERAGE	MAX./MIN.	LIGHT LOSS FACTOR (LLF)	W/FT² (WHOLE BLDG)
107 - LABORATORY	40 FC @ 30" AFF	2.0:1	0.90	

NOTE:
1. CONTRACTOR TO SUBMIT WHOLE BUILDING (INDIVIDUAL INTERIOR ROOMS/ SPACES) AND PROJECT LIMITS (EXTERIOR BASE SITE PARKING, ENTRANCES/ EXITS, CANOPIES, WALKWAYS, AND BUILDING FAÇADE) PHOTOMETRIC AND POWER DENSITY ANALYSIS FOR ALL SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS NOT LISTED ON THE LUMINAIRE SCHEDULE. SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS MUST MEET THE REQUIRED PHOTOMETRIC AND POWER DENSITY STATISTICS FOR ALL AREAS/ SPACES INDICATED ON THIS SHEET. INCREASING LUMINAIRE QUANTITIES SHOWN ON THIS SHEET, INCREASING THE LUMEN OUTPUT ON THE LUMINAIRE SCHEDULE, OR INCREASING THE VA (VOLTAGE AMPS) ON THE LUMINAIRE SCHEDULE FOR THE PHOTOMETRIC AND POWER DENSITY STATISTICS COMPLIANCE IS PROHIBITED.

SCALE A 1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO.
58037
TRAVIS T. MOORE
Date Signed 2/13/2026
ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	G. MARTINEZ	062 E401	WATER TREATMENT LIGHTING & RECEPTACLE ENLARGED PLAN
	DRAWN: S. BILD		
	TECH. REVIEW: P. DAVILA		NORTH RIM - GRAND CANYON NATIONAL PARK
	DATE: MAY 2026		

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	134 OF 192

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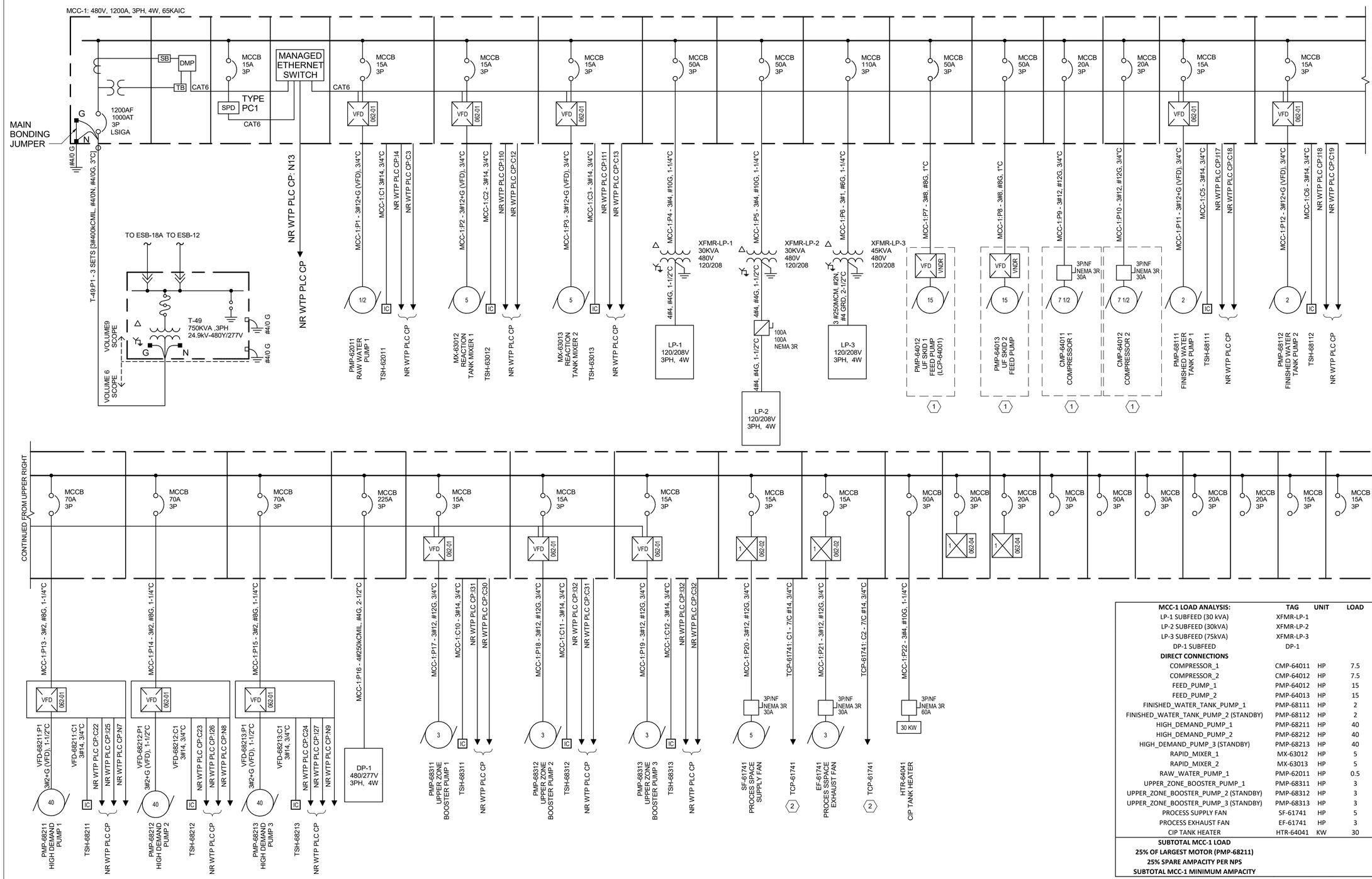

GENERAL NOTES:

A. ALL POWER CABLES USED TO CIRCUIT VFDS AND MOTORS SHALL BE VFD TYPE.

B. REFER TO SHEETS 062E606, 062E607, AND 062E608 FOR INSTRUMENT AND CONTROL WIRING INFORMATION.

KEYNOTES: ⟨X⟩

1. MANUFACTURER-PROVIDED EQUIPMENT, ACCESSORIES AND DEVICES MAY VARY BETWEEN MANUFACTURERS AND MAY NOT BE SHOWN COMPLETELY. COORDINATE WIRING AND INSTALLATION WITH SUPPLIED EQUIPMENT.
2. REFER TO SHEET 062M701 FOR HVAC CONTROL INFORMATION.

MCC-1 LOAD ANALYSIS:	TAG	UNIT	LOAD	AMPS @ 480V
LP-1 SUBFEED (30 kVA)	XFMR-LP-1			36.1
LP-2 SUBFEED (30kVA)	XFMR-LP-2			36.1
LP-3 SUBFEED (75kVA)	XFMR-LP-3			90.2
DP-1 SUBFEED	DP-1			180
DIRECT CONNECTIONS				
COMPRESSOR_1	CMP-64011	HP	7.5	11
COMPRESSOR_2	CMP-64012	HP	7.5	11
FEED_PUMP_1	PMP-64012	HP	15	21
FEED_PUMP_2	PMP-64013	HP	15	21
FINISHED_WATER_TANK_PUMP_1	PMP-68111	HP	2	3.4
FINISHED_WATER_TANK_PUMP_2 (STANDBY)	PMP-68112	HP	2	0
HIGH_DEMAND_PUMP_1	PMP-68211	HP	40	52
HIGH_DEMAND_PUMP_2	PMP-68212	HP	40	52
HIGH_DEMAND_PUMP_3 (STANDBY)	PMP-68213	HP	40	0
RAPID_MIXER_1	MX-63012	HP	5	7.6
RAPID_MIXER_2	MX-63013	HP	5	7.6
RAW_WATER_PUMP_1	PMP-62011	HP	0.5	1
UPPER_ZONE_BOOSTER_PUMP_1	PMP-68311	HP	3	4.8
UPPER_ZONE_BOOSTER_PUMP_2 (STANDBY)	PMP-68312	HP	3	0
UPPER_ZONE_BOOSTER_PUMP_3 (STANDBY)	PMP-68313	HP	3	0
PROCESS SUPPLY FAN	SF-61741	HP	5	7.6
PROCESS EXHAUST FAN	EF-61741	HP	3	4.8
CIP TANK HEATER	HTR-64041	KW	30	36
SUBTOTAL MCC-1 LOAD				**583.2**
25% OF LARGEST MOTOR (PMP-68211)				**13.0**
25% SPARE AMPACITY PER NPS				**149.1**
SUBTOTAL MCC-1 MINIMUM AMPACITY				**745.3**

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — ARIZONA, U.S.A.

A/E FIRM	DESIGNED: A. S-DELGADO	SUB SHEET NO. 062 E601	TITLE OF SHEET: WATER TREATMENT ELECTRIC MOTOR CONTROL CENTER ONE-LINE DIAGRAM
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DRAWN: A. S-DELGADO		NORTH RIM - GRAND CANYON NATIONAL PARK
	TECH. REVIEW: P. DAVILA		DRAWING NO. 113 / 182752
	DATE: AUG 2026		PMIS/PKG NO. 318719
			SHEET 135 OF 192

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	8	7	6	5	4	3	2	1
A	FUTURE	FUTURE	DP-1 480V DISTRIBUTION PANEL CB	CMP-64011 COMPRESSOR 1	LP-1 LIGHTING PANEL 1 CB	HTR-64041 CIP TANK HEATER CB	SPD	SPACE
B				CMP-64012 COMPRESSOR 2	LP-2 LIGHTING PANEL 2 CB	SF-61741 PROCESS SUPPLY FAN		600A MCB WITH CTs
C			PMP-68211 HIGH DEMAND PUMP 1 CB	PMP-68212 HIGH DEMAND PUMP 2 CB	PMP-68213 HIGH DEMAND PUMP 3 CB			
D			PMP-64012 UF SKID FEED PUMP #2 CB	PMP-64011 UF SKID FEED PUMP #1 CB	EF-61741 PROCESS EXHAUST FAN	SPARE CB	METER	
E			SPARE CB	SPARE CB		SPARE CB		
F			SPARE CB	SPARE CB	SPARE CB	SPARE CB	MANAGED ETHERNET SWITCH	
G			PMP-68113 FINISHED WATER PUMP 1 VFD	PMP-68111 FINISHED WATER PUMP 2 VFD	MX-63012 REACTION TANK 1 MIXER VFD	MX-63013 REACTION TANK 2 MIXER VFD	LP-3 LIGHTING PANEL 3 CB	
H							SPARE CB	
J							SPARE CB	
K			PMP-68112 UPPER ZONE BOOSTER PUMP 1 VFD	PMP-68114 UPPER ZONE BOOSTER PUMP 2 VFD	PMP-68115 UPPER ZONE BOOSTER PUMP 3 VFD	SPARE PROVISION UPPER ZONE BOOSTER PUMP 4	PMP-62011 RAW WATER PUMP 1 VFD	
L								
M								

1 / 062E602 MCC-1 ELEVATION

NOT TO SCALE

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REGISTERED PROFESSIONAL ENGINEER - ELECTRICAL - CERTIFICATE NO. 58037 TRAVIS T. MOORE - ARIZONA, U.S.A. - Date Signed 1 MAY 2026

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 062 E602	TITLE OF SHEET WATER TREATMENT MCC ELEVATIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 136 OF 192

BRANCH PANEL: DP-1

LOCATION: ELECTRICAL ROOM 110 | **VOLTS:** 480Y/277 | **A.I.C. RATING:** 65K
SUPPLY FROM: MCC-1 | **PHASES:** 3 | **MAINS TYPE:** MLO
MOUNTING: SURFACE | **WIRES:** 4 | **MAINS RATING:** 225 A
ENCLOSURE: NEMA 1 | **INTEGRAL SPD:** No | **SERVICE ENTRANCE LABEL:** No

WIRE & CONDUIT	CKT	CIRCUIT DESCRIPTION	CLASS	TRIP	P	A		B		C		P	TRIP	CLASS	CIRCUIT DESCRIPTION	CKT	WIRE & CONDUIT
	1					1000.0	4500.0									2	
3 #12, #12 G, 3/4"C	3	EUH-61741, RM 100	HVAC	20 A	3			1000.0	4500.0			3	20 A	Equipment	WH-61641, RM 101	4	3 #12, #12 G, 3/4"C
	5									1000.0	4500.0					6	
	7					4500.0	2500.0									8	
3 #12, #12 G, 3/4"C	9	WH-61642, RM 101	Equipment	20 A	3			4500.0	2500.0			3	20 A	HVAC	EUH-61745, RM 109	10	3 #12, #12 G, 3/4"C
	11									4500.0	2500.0					12	
	13					2500.0	2500.0									14	
3 #12, #12 G, 3/4"C	15	EUH-61743, RM 109	HVAC	20 A	3			2500.0	2500.0			3	20 A	HVAC	EUH-61746, RM 109	16	3 #12, #12 G, 3/4"C
	17									2500.0	2500.0					18	
	19					2500.0	2500.0									20	
3 #12, #12 G, 3/4"C	21	EUH-61744, RM 109	HVAC	20 A	3			2500.0	2500.0			3	20 A	HVAC	EUH-61749, RM 113	22	3 #12, #12 G, 3/4"C
	23									2500.0	2500.0					24	
	25					3000.0	1666.7									26	
3 #12, #12 G, 3/4"C	27	EUH-61748, RM 112	HVAC	20 A	3			3000.0	1666.7			3	20 A	HVAC	EUH-61742, RM 110	28	3 #12, #12 G, 3/4"C
	29									3000.0	1666.7					30	
	31					3333.3	942.0									32	
3 #12, #12 G, 3/4"C	33	EUH-61747, RM 111	HVAC	20 A	3			3333.3	942.0			3	20 A	HVAC	DWBP-61641, RM 101	34	3 #12, #12 G, 3/4"C
	35									3333.3	942.0					36	
	37					1110.0	666.7									38	
3 #12, #12 G, 3/4"C	39	PROCESS CONTROL VALVES, RM 109 (1)	PROC	20 A	3			1110.0	666.7			3	20 A	HVAC	EUH-617410, RM 102	40	3 #12, #12 G, 3/4"C
	41									1110.0	666.7					42	
	43					666.7	1000.0									44	
3 #12, #12 G, 3/4"C	45	EUH-617411, RM 103	HVAC	20 A	3			666.7	1000.0			3	20 A	HVAC	EUH-617412, RM 104	46	3 #12, #12 G, 3/4"C
	47									666.7	1000.0					48	
	49					1000.0	1000.0									50	
3 #12, #12 G, 3/4"C	51	EUH-617413, RM 106	HVAC	20 A	3			1000.0	1000.0			3	20 A	HVAC	EUH-617414, RM 107	52	3 #12, #12 G, 3/4"C
	53									1000.0	1000.0					54	
	55					0.0	0.0									56	
	57	SPARE	--	20 A	3			0.0	0.0			3	20 A	--	SPARE	58	
	59									0.0	0.0					60	
	61					0.0	0.0									62	
	63	SPARE	--	20 A	3			0.0	0.0			3	20 A	--	SPARE	64	
	65									0.0	0.0					66	
	67					0.0	0.0									68	
	69	SPARE	--	20 A	3			0.0	0.0			3	20 A	--	SPARE	70	
	71									0.0	0.0					72	
	73					0.0	0.0									74	
	75	SPARE	--	20 A	3			0.0	0.0			3	20 A	--	SPARE	76	
	77									0.0	0.0					78	
	79					0.0	0.0									80	
	81	SPARE	--	20 A	3			0.0	0.0			3	20 A	--	SPARE	82	
	83									0.0	0.0					84	
				TOTAL LOAD:		36885 VA		36885 VA		36885 VA							
				TOTAL AMPS:		133 A		133 A		133 A							

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND
HVAC	80326 VA	100.00%	80326 VA
PROC	3330 VA	100.00%	3330 VA
Equipment	27000 VA	100.00%	27000 VA

PANEL TOTALS	
TOTAL CONNECTED LOAD:	110656 VA
TOTAL ESTIMATED DEMAND:	110656 VA
TOTAL CONNECTED:	133 A
TOTAL ESTIMATED DEMAND	133 A
SPARE AMPERAGE:	25.00%
TOTAL ESTIMATED DEMAND + SPARE:	166 A

NOTES:
1. PROVIDE LOCAL DISCONNECT FOR EACH VALVE. DISCONNECT SWITCHES SHALL BE 3 PHASE,10 A, 600V, NEMA 3R RATED.

BRANCH PANEL: LP-1

LOCATION: ELECTRICAL ROOM 110 | **VOLTS:** 208Y/120 | **A.I.C. RATING:** 22
SUPPLY FROM: XMFR LP-1 | **PHASES:** 3 | **MAINS TYPE:** MCB
MOUNTING: SURFACE | **WIRES:** 4 | **MAINS RATING:** 100 A
ENCLOSURE: NEMA 1 | **INTEGRAL SPD:** No | **SERVICE ENTRANCE LABEL:** No

WIRE & CONDUIT	CKT	CIRCUIT DESCRIPTION	CLASS	TRIP	P	A		B		C		P	TRIP	CLASS	CIRCUIT DESCRIPTION	CKT	WIRE & CONDUIT
2 #10, #10 G, 3/4"C	1	LIGHTS RMS 102 & 105	LTG	20 A	1	201.0	500.0					1	20 A	Equipment	LUM CTRL PNL LCP-1, RM 110	2	2 #12, #12 G, 3/4"C
2 #10, #10 G, 3/4"C	3	LIGHTS,BUILDING EXTERIOR	LTG	20 A	1			256.0	5760.0			1	60 A	IC	NPS TELECOM PANEL	4	2 #6, #8 G, 2"C
2 #12, #12 G, 3/4"C	5	REC, EXTERIOR GFCI	REC	20 A	1					360.0	540.0	1	20 A	REC	REC, RM 109	6	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	7	FCU-61742, RM 110	HVAC	20 A	2	93.5	600.0					1	20 A	IC	NR WTP PLC CP, RM 110	8	2 #12, #12 G, 3/4"C
	9							93.5	180.0			1	20 A	HVAC	TCP-61742, RM 110	10	
2 #12, #12 G, 3/4"C	11	REC, EXTERIOR GFCI	REC	20 A	1					720.0	864.0	1	20 A	PROC	PMP-66011, RM 109	12	2 #12, #12G, 3/4"C
2 #12, #12 G, 3/4"C	13	REC, RM 110	REC	20 A	1	900.0	1176.0					1	20 A	PROC	PMP-64011, RM 109	14	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	15	REC, RM 111	REC	20 A	1			540.0	0.0							16	
2 #12, #12 G, 3/4"C	17	PMP-68117, RM 109	PROC	20 A	1					864.0	0.0	3	20 A	--	SPARE	18	3 #12, #12 G, 3/4"C
2 #10, #10 G, 2"C	19	HTP-61741, RM 110	HVAC	30 A(G)	1	2880.0	0.0									20	
	21	SPARE	--	20 A	1			0.0	500.0			1	20 A	IC	NR WTP PLC CP, RM 110	22	2 #12, #12 G, 3/4"C
	23	SPARE	--	20 A	1					0.0	2085.5	1	30 A	LTG	PROCESS AREA LIGHTS, RM 109	24	2 #8, #8 G,1"C
2 #12, #12 G, 3/4"C	25	TCP-61741, RM 110	HVAC	20 A	1	180.0	41.5					2	20 A	HVAC	FCU-61743, RM 111	26	2 #12, #12 G, 3/4"C
	27	SPARE	--	20 A	1			0.0	41.5							28	
2 #12, #12 G, 3/4"C	29	REC, RM 111	REC	20 A	1					540.0	650.1	1	20 A	LTG	LIGHTS RMS 110-113	30	2 #10, #10 G, 3/4"C
2 #12, #12 G, 3/4"C	31	REC, RMS 112 & 113	REC	20 A	1	720.0	0.0					1	20 A	--	SPARE	32	
	33	SPARE	--	20 A	1			0.0	0.0			1	20 A	--	SPARE	34	
	35	SPARE	--	20 A	1					0.0	540.0	1	20 A	REC	REC, RM 109	36	2 #12, #12 G, 3/4"C
	37	SPARE	--	20 A	1	0.0	0.0					1	20 A	--	SPARE	38	
	39	SPARE	--	20 A	1			0.0	0.0			1	20 A	--	SPARE	40	
	41	SPARE	--	20 A	1					0.0	0.0	1	20 A	--	SPARE	42	
				TOTAL LOAD:		7292 VA		7371 VA		7164 VA							
				TOTAL AMPS:		61 A		62 A		60 A							

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND
HVAC	3510 VA	100.00%	3510 VA
REC	4860 VA	100.00%	4860 VA
IC	6860 VA	100.00%	6860 VA
PROC	2904 VA	100.00%	2904 VA
Equipment	500 VA	100.00%	500 VA
LTG	3193 VA	100.00%	3193 VA

PANEL TOTALS	
TOTAL CONNECTED LOAD:	21827 VA
TOTAL ESTIMATED DEMAND:	21827 VA
TOTAL CONNECTED:	61 A
TOTAL ESTIMATED DEMAND	61 A
SPARE AMPERAGE:	25.00%
TOTAL ESTIMATED DEMAND + SPARE:	76 A

NOTES:
(G) NOTATION ON TRIP RATING INDICATES GROUND - FAULT PROTECTED BREAKER

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 8/12/2026 ARIZONA, U.S.A.

A/E FIRM	DESIGNED: A. S-DELGADO	SUB SHEET NO. 062 E603	TITLE OF SHEET: WATER TREATMENT ELECTRICAL PANELBOARD SCHEDULES
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DRAWN: A. S-DELGADO		NORTH RIM - GRAND CANYON NATIONAL PARK
	TECH. REVIEW: P. DAVILA		DRAWING NO. 113 182752
	DATE: AUG 2026		PMIS/PKG NO. 318719
			SHEET 137 OF 192

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8/12/2026 9:20:37 PM

BRANCH PANEL: LP-2

LOCATION: PROCESS AREA 109 | **VOLTS:** 208Y/120 | **A.I.C. RATING:** 22
SUPPLY FROM: XMFR LP-2 | **PHASES:** 3 | **MAINS TYPE:** MCB
MOUNTING: SURFACE | **WIRES:** 4 | **MAINS RATING:** 100 A
ENCLOSURE: NEMA 1 | **INTEGRAL SPD:** No | **SERVICE ENTRANCE LABEL:** No

WIRE & CONDUIT	CKT	CIRCUIT DESCRIPTION	CLASS	TRIP	P	A	B	C	P	TRIP	CLASS	CIRCUIT DESCRIPTION	CKT	WIRE & CONDUIT
2 #12, #12 G, 3/4"C	1	LCP-64001, UF SKID CNTRL PNL #1	PROC	20 A	1	676.0	696.0		1	20 A	PROC	COAGULANT MTRING PMP/PNL, RM 113	2	2 #12, #12 G, 3/4"C
2 #10, #10 G, 3/4"C	3	SODIUM HYPO FEED PMP PNL, RM 112	PROC	30 A	1	2784.0	180.0		1	20 A	PROC	SODIUM BISULFITE CIP PNL, RM 113	4	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	5	REC, RM 109	REC	20 A	1			540.0	696.0	1	20 A	PROC	CSTIC SODA MTRNG PMP/PNL, RM 112	6	2 #12, #12 G, 3/4"C
2 #8, #10 G, 3/4"C	7	HP-61742, EXTERIOR (4)	HVAC	40 A	2	2392.0	180.0			1	20 A	IC	NR WTP CHEMICAL RIO CP, RM 109	8	2 #12, #12 G, 3/4"C
	9							2392.0	180.0	1	20 A	PROC	CITRIC ACID CIP PNL, RM 113	10	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	11	FCU-61744A, RM 113	HVAC	20 A	2			26.0	26.0	2	20 A	HVAC	FCU-61744B, RM 112	12	2 #12, #12 G, 3/4"C
	13					26.0	26.0							14	
2 #10, #10 G, 3/4"C	15	HP-61743, EXTERIOR (3)	HVAC	30 A	2			1976.0	1176.0	1	20 A	PROC	PMP-62012 RAW WTR PMP/PNL, RM 109	16	2 #12, #12 G, 3/4"C
	17							1976.0	3224.0	2	40 A	HVAC	HP-61744, EXTERIOR (4)	18	2 #8, #10 G, 3/4"C
3 #12, #12 G, 3/4"C	19	CAUSTIC SODA CIP PNL, RM 112	PROC	20 A	1	180.0	3224.0							20	
	21	SPARE	--	20 A	1		0.0	24.0		1	20 A	MECH	TRAP PRIMER 4	22	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	23	SODIUM HYPO CIP PNL, RM 112	PROC	20 A	1			180.0	180.0	1	20 A	PROC	HYDROCHLORIC ACID CIP PNL, RM 113	24	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	25	EF-61742, RM 113	HVAC	20 A	1	1176.0	576.4							26	
	27	SPARE	--	20 A	1		0.0	576.4		3	20 A	Equipment	ROLL UP DOOR, RM 109	28	3 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	29	SF-61743, RM 113	HVAC	20 A	1			1176.0	576.4					30	
	31	SPARE	--	20 A	1	0.0	360.0			1	20 A	REC	REC, EXTERIOR GFCI	32	2 #12, #12 G, 3/4"C
	33	SPARE	--	20 A	1		0.0	0.0		1	20 A	--	SPARE	34	
	35	SPARE	--	20 A	1			0.0	0.0	1	20 A	--	SPARE	36	
	37	SPARE	--	20 A	1	0.0	0.0			1	20 A	--	SPARE	38	
	39	SPARE	--	20 A	1		0.0	0.0		1	20 A	--	SPARE	40	
	41	SPARE	--	20 A	1			0.0	0.0	1	20 A	--	SPARE	42	
				TOTAL LOAD:		9512 VA	9288 VA	8600 VA							
				TOTAL AMPS:		80 A	78 A	72 A							

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND
HVAC	17640 VA	100.00%	17640 VA
REC	900 VA	100.00%	900 VA
IC	180 VA	100.00%	180 VA
PROC	6928 VA	100.00%	6928 VA
MECH	24 VA	100.00%	24 VA
Equipment	1729 VA	100.00%	1729 VA

PANEL TOTALS	
TOTAL CONNECTED LOAD:	**27401 VA**
TOTAL ESTIMATED DEMAND:	**27401 VA**
TOTAL CONNECTED:	**76 A**
TOTAL ESTIMATED DEMAND	**76 A**
SPARE AMPERAGE:	**25.00%**
TOTAL ESTIMATED DEMAND + SPARE:	**95 A**

NOTES:
1. REFER TO CONTROL DIAGRAM 062-03 FOR 120 VOLT PUMP WIRING DIAGRAM. REFER TO MANUFACTURER INSTRUCTIONS FOR CIP PUMP WIRING.
2. REFER TO SHEETS 062M701 AND 062M702 FOR EXHAUST AND SUPPLY FAN CONTROL WIRING.
3. PROVIDE 30A, 3 PHASE, NON-FUSED NEMA 3R DISCONNECT SWITCH ADJACENT TO HEAT PUMP UNIT.
4. PROVIDE 60A, 3 PHASE, NON-FUSED NEMA 3R DISCONNECT SWITCH ADJACENT TO HEAT PUMP UNIT.

BRANCH PANEL: LP-3

LOCATION: MECHANICAL ROOM 101 | **VOLTS:** 208Y/120 | **A.I.C. RATING:** 22
SUPPLY FROM: XFMR LP-3 | **PHASES:** 3 | **MAINS TYPE:** MCB
MOUNTING: SURFACE | **WIRES:** 4 | **MAINS RATING:** 250 A
ENCLOSURE: NEMA 1 | **INTEGRAL SPD:** No | **SERVICE ENTRANCE LABEL:** No | **WIRING**

WIRE & CONDUIT	CKT	CIRCUIT DESCRIPTION	CLASS	TRIP	P	A	B	C	P	TRIP	CLASS	CIRCUIT DESCRIPTION	CKT	WIRE & CONDUIT
2 #12, #12 G, 3/4"C	1	EF-61743, RM 101	HVAC	20 A	1	264.0	528.0		1	20 A	HVAC	SF-61742, RM 101	2	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	3	REC, RM 104, KITCHEN, DISPOSAL	Equipment	20 A	1		1500.0	720.0	1	20 A	REC	REC, RM 106 RACEWAY	4	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	5	REC, REST ROOM 103	REC	20 A	1			180.0	540.0	1	20 A	REC	REC, RM 104, KITCHEN	6	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	7	REC, RM 101, MECHANICAL ROOM	REC	20 A	1	900.0	540.0		1	20 A	REC	REC, RM 107, LAB, WALL OUTLETS	8	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	9	REC, RM 107, LAB, DISPOSAL	Equipment	20 A	1		1500.0	7500.0	2	80 A	Equipment	HTR-61741, RM 101	10	2 #6, #8 G, 3/4"C
2 #12, #12 G, 3/4"C	11	REC, RM 107, LAB, RACEWAY	REC	20 A	1			1440.0	7500.0				12	
2 #12, #12 G, 3/4"C	13	REC, RM 106, PRINTER	REC	20 A	1	180.0	528.0		1	20 A	HVAC	HWP-61641, RM 101	14	2 #12, #12 G, 3/4"C
2#12, #12 G, 3/4"C	15	REC, RM 104, MICROWAVE	POWER	20 A	1		180.0	180.0	1	20 A	REC	REC, RM 104, ABOVE-COUNTER	16	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	17	REC, EXTERIOR GFCI	REC	20 A	1			540.0	1500.0	1	20 A	REC	RM 108, SCADA SERVER UPS #2	18	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	19	REC, RM 106, TV MONITORS	REC	20 A	1	180.0	1500.0			1	20 A	Equipment	LAB ROOM 107 GLASSWARE WASHER	20	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	21	REC, RM 106, TV MONITORS	REC	20 A	1		360.0	1500.0		1	20 A	Equipment	REC, RM 104, REFRIGERATOR	22	2 #12, #12 G, 3/4"C
2#12, #12 G, 3/4"C	23	FCU-61741, RM 108	HVAC	20 A	2			41.5	3224.0	2	40 A	HVAC	HP-61745, EXTERIOR (1)	24	2#8, #10 G, 3/4"C
	25					41.5	3224.0							26	
2 #12, #12 G, 3/4"C	27	REC, RM 106, CONTROL RM RACEWAY	REC	20 A	1		540.0	353.5		2	20 A	HVAC	AHU-61741, RM 101	28	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	29	REC, RM 106, CONTROL ROOM	REC	20 A	1			540.0	353.5					30	
2#12, #12 G, 3/4"C	31	REC, RM 104, DISHWASHER	Equipment	20 A	1	1500.0	360.0			1	20 A	REC	REC, RM 106 CONTROL RM RACEWAY	32	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	33	REC, RM 104, ABOVE-COUNTER	REC	20 A	1		180.0	180.0		1	20 A	HVAC	TCP-61743, RM 101	34	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	35	REC, RM 106 RACEWAY	REC	20 A	1			720.0	0.0	1	20 A	--	SPARE	36	
2#12, #12 G, 3/4"C	37	REC, RM 107, LAB, RACEWAY	REC	20 A	1	1260.0	1040.0			2	20 A	HVAC	HP-61741, EXTERIOR (1)	38	2 #12, #12 G, 3/4"C
	39	SPARE	--	20 A	1		0.0	1040.0						40	
2 #12, #12 G, 3/4"C	41	RMS 101, 103, 104, 106, 107, 108 LGTS	LTG	20 A	1			425.0	180.0	1	20 A	REC	REC, RM 104 ABOVE COUNTER	42	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	43	REC, RM 107 RACEWAY	REC	20 A	1	900.0	900.0			1	20 A	REC	REC, RM 108 SCADA CLOSET	44	2 #12, #12 G, 3/4"C
	45	SPARE	--	20 A	1		0.0	1500.0		1	20 A	REC	REC, RM 108 SCADA CLOSET	46	2 #12, #12 G, 3/4"C
	47	SPARE	--	20 A	1			0.0	1500.0	1	20 A	REC	REC, RM 108 SCADA CLOSET	48	2 #12, #12 G, 3/4"C
2 #12, #12 G, 3/4"C	49	REC, RMS 102, 105, HALLWAY	REC	20 A	1	720.0	1500.0			1	20 A	REC	RM 108, SCADA SERVER UPS #1	50	2 #12, #12 G, 3/4"C
	51	SPARE	--	20 A	1		0.0	0.0		1	20 A	--	SPARE	52	
	53	SPARE	--	20 A	1			0.0	0.0	1	20 A	--	SPARE	54	
2 #12, #12 G, 3/4"C	55	REC, RM 107 RACEWAY	REC	20 A	1	720.0	34.0			1	20 A	MECH	TRAP PRIMER 3	56	2 #12, #12 G, 3/4"C
	57	SPARE	--	20 A	1		0.0	0.0		1	20 A	--	SPARE	58	
	59	SPARE	--	20 A	1			0.0	0.0	1	20 A	--	SPARE	60	
	61	SPARE	--	20 A	1	0.0	0.0			1	20 A	--	SPARE	62	
	63	SPARE	--	20 A	1		0.0	0.0		1	20 A	--	SPARE	64	
	65	SPARE	--	20 A	1			0.0	0.0	1	20 A	--	SPARE	66	
	67	SPARE	--	20 A	1	0.0	0.0			1	20 A	--	SPARE	68	
	69	SPARE	--	20 A	1		0.0	0.0		1	20 A	--	SPARE	70	
	71	SPARE	--	20 A	1			0.0	0.0	1	20 A	--	SPARE	72	
	73	SPARE	--	20 A	1	0.0	0.0			1	20 A	--	SPARE	74	
	75	SPARE	--	20 A	1		0.0	0.0		1	20 A	--	SPARE	76	
	77	SPARE	--	20 A	1			0.0	0.0	1	20 A	--	SPARE	78	
	79	SPARE	--	20 A	1	0.0	0.0			1	20 A	--	SPARE	80	
	81	SPARE	--	20 A	1		0.0	0.0		1	20 A	--	SPARE	82	
	83	SPARE	--	20 A	1			0.0	0.0	1	20 A	--	SPARE	84	
				TOTAL LOAD:		16820 VA	17234 VA	18684 VA						
				TOTAL AMPS:		140 A	144 A	156 A						

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND
HVAC	10818 VA	100.00%	10818 VA
REC	18780 VA	76.62%	14390 VA
POWER	180 VA	100.00%	180 VA
MECH	34 VA	100.00%	34 VA
Equipment	22500 VA	100.00%	22500 VA
LTG	425 VA	100.00%	425 VA

PANEL TOTALS	
TOTAL CONNECTED LOAD:	**52737 VA**
TOTAL ESTIMATED DEMAND:	**48347 VA**
TOTAL CONNECTED:	**146 A**
TOTAL ESTIMATED DEMAND	**134 A**
SPARE AMPERAGE:	**10.00%**
TOTAL ESTIMATED DEMAND + SPARE:	**148 A**

NOTES:
1. PROVIDE 30A, 3 PHASE, NON-FUSED NEMA 3R DISCONNECT SWITCH ADJACENT TO HEAT PUMP UNIT.

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE Date signed 7/10/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO CADD A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 062 E604	TITLE OF SHEET WATER TREATMENT ELECTRICAL PANELBOARD SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 138 OF 192

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_062_NWTP_E.rvt
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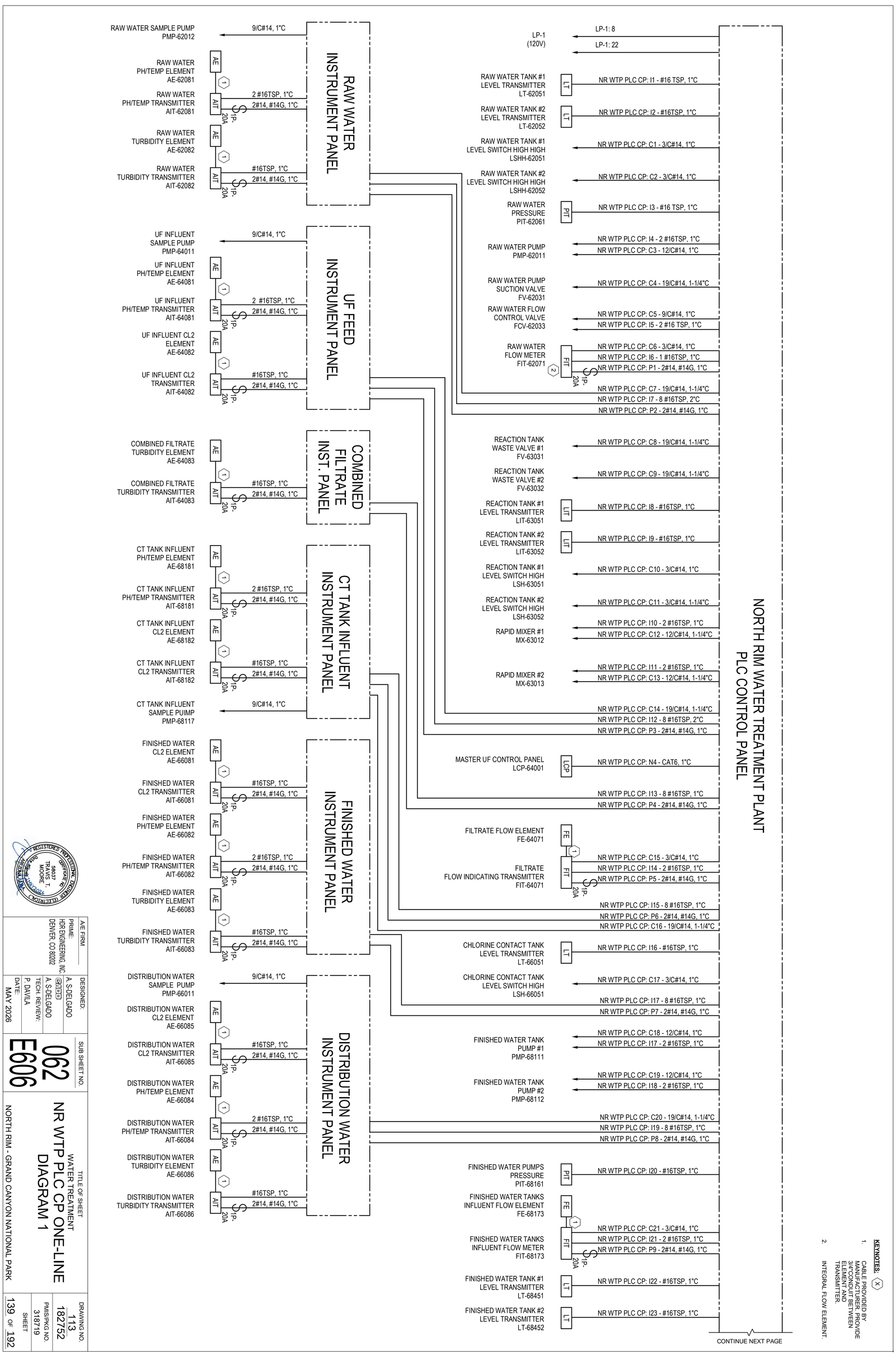
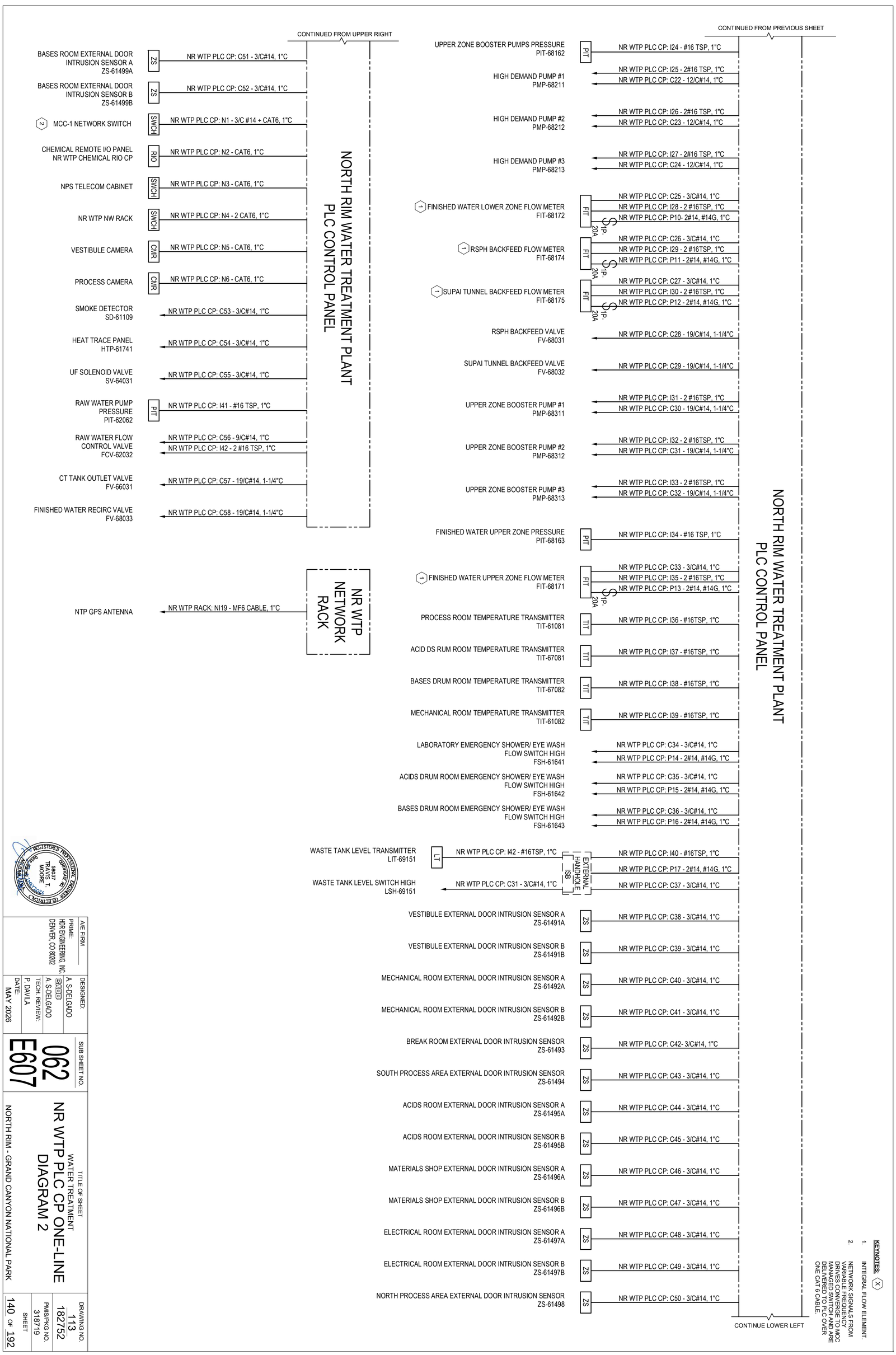


A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202
DESIGNED: A. S-DELGADO
DRAWN: A. S-DELGADO
TECH. REVIEW: P. DAVILA
DATE: MAY 2026
SUB SHEET NO. 062 E606
TITLE OF SHEET
WATER TREATMENT
NR WTP PLC CP ONE-LINE DIAGRAM 1
NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 139 OF 192



REGISTERED PROFESSIONAL ENGINEER
STATE OF COLORADO
58037
TRAVIS T. MOORE
ELECTRONICALLY SIGNED AND SEALED
A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202
DESIGNED:
A. S-DELGADO
DRAWN:
A. S-DELGADO
TECH. REVIEW:
P. DAVILA
DATE:
MAY 2026

SUB SHEET NO.
062
E607

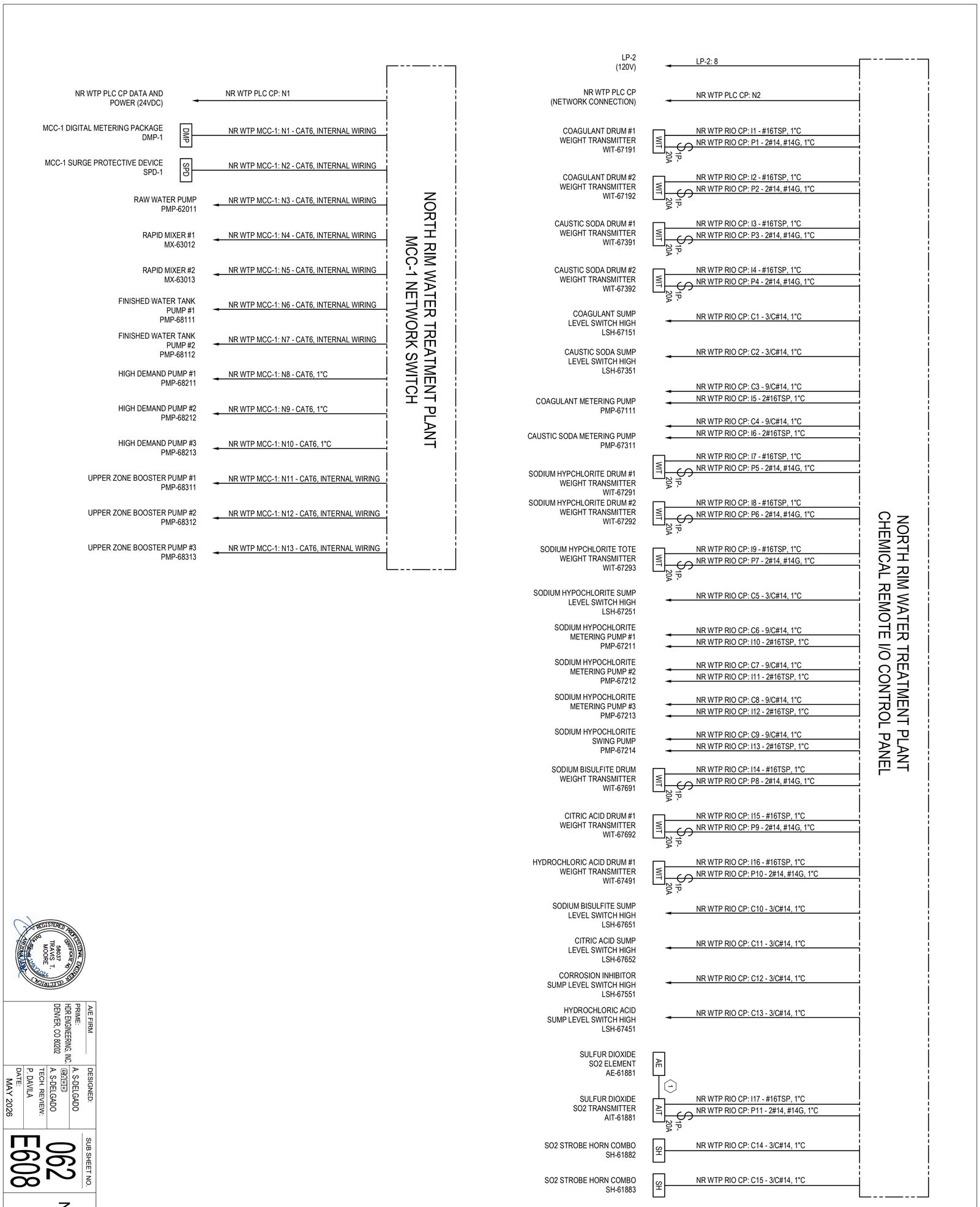
TITLE OF SHEET
WATER TREATMENT
NR WTP PLC CP ONE-LINE DIAGRAM 2

NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.
113
182752

PMIS/PKG NO.
318719

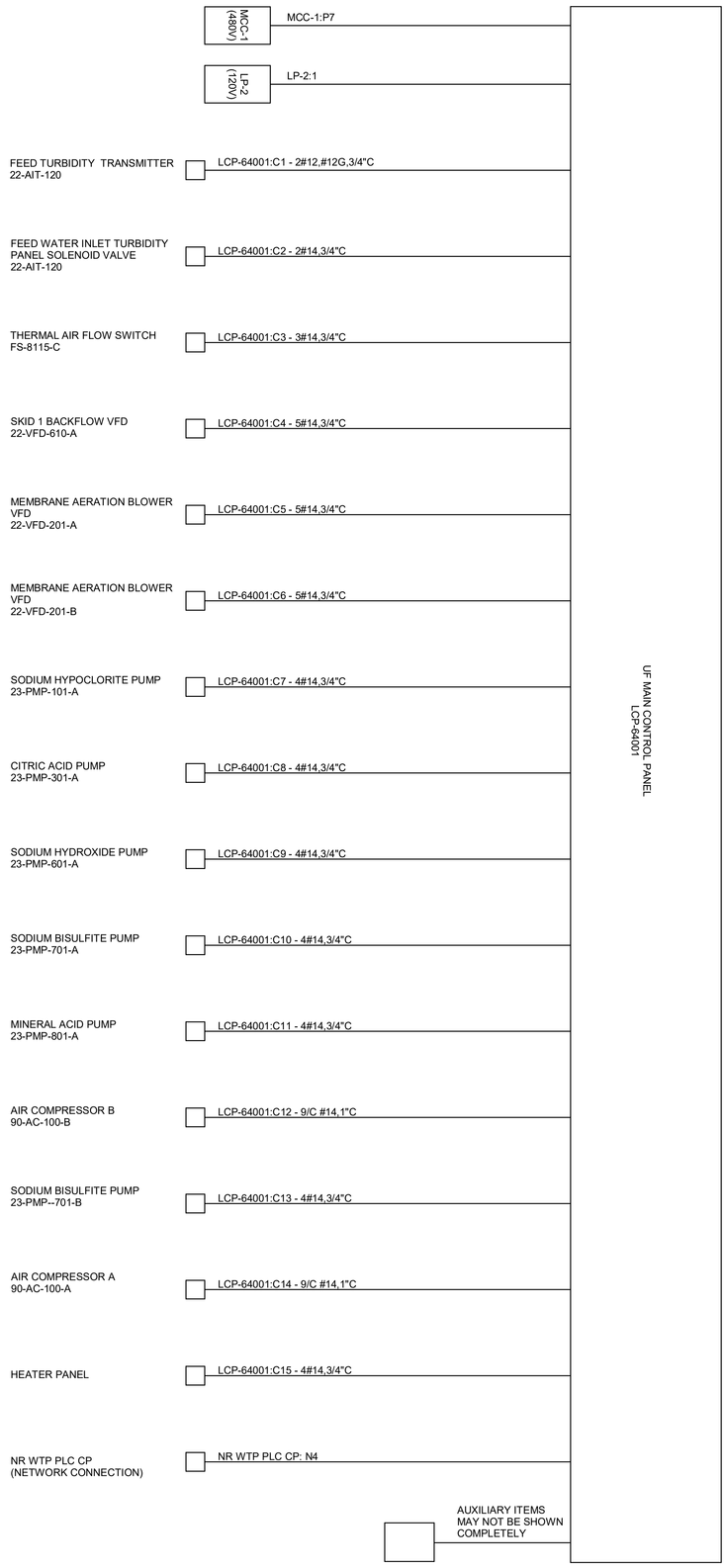
SHEET
140 OF 192



KEYNOTES:

1. CABLE PROVIDED BY MANUFACTURER PROVIDE 3/4"CONDUIT BETWEEN ELEMENT AND TRANSMITTER.

A/E FIRM	
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	
DESIGNED: A. S-DELGADO	
DRAWN: A. S-DELGADO	
TECH. REVIEW: P. DAVILA	
DATE: MAY 2026	
SUB SHEET NO. E608 062	
TITLE OF SHEET: WATER TREATMENT NR WTP CHEMICAL RIO CP ONE-LINE DIAGRAM	
NORTH RIM - GRAND CANYON NATIONAL PARK	
DRAWING NO. 113 182752	
PMIS/PKG NO. 318719	
SHEET 141 OF 192	



A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

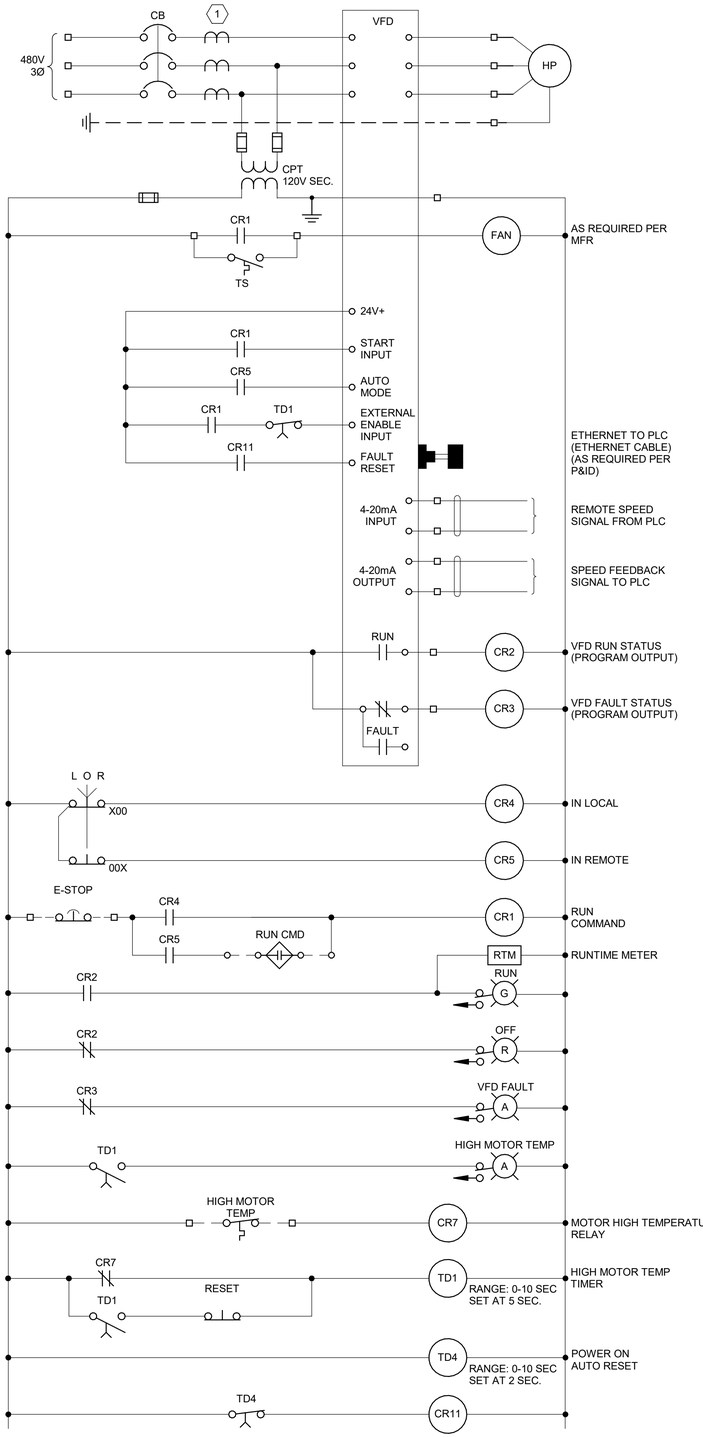
DESIGNED: A. S-DELGADO
DRAWN: A. S-DELGADO
TECH. REVIEW: P. DAVILA
DATE: MAY 2026

SUB SHEET NO. 062

E610

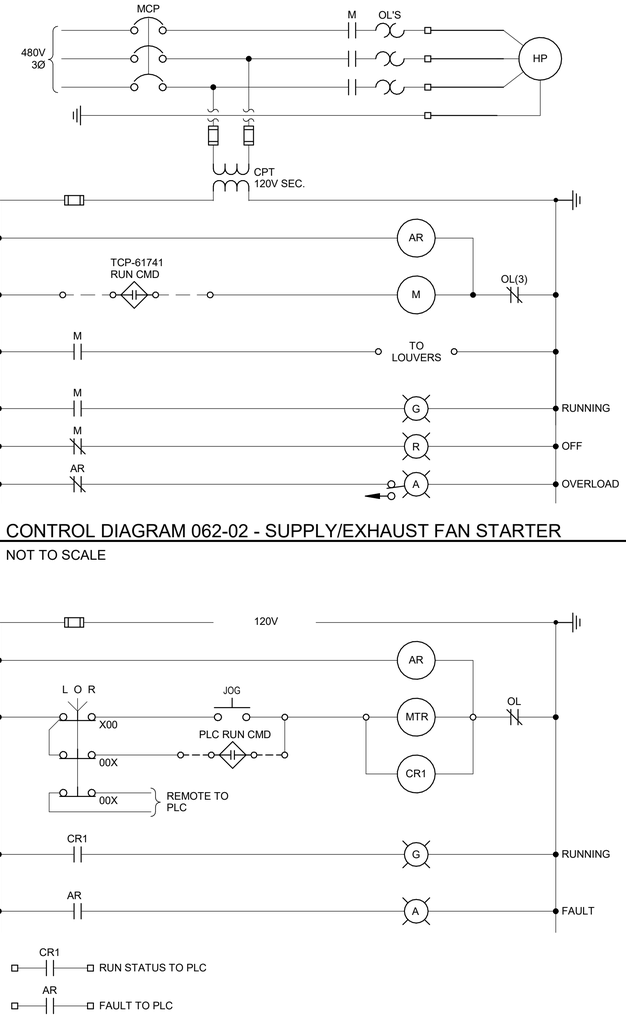
TITLE OF SHEET
WATER TREATMENT
ELECTRICAL
MISCELLANEOUS ONE-LINE DIAGRAMS
NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 142 OF 192



1 062E612 CONTROL DIAGRAM 062-01 - VARIABLE FREQUENCY DRIVE CONTROLS

NOT TO SCALE

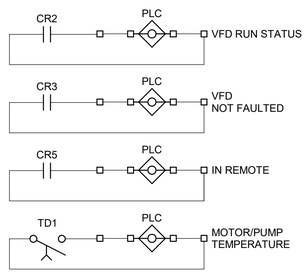


2 062E612 CONTROL DIAGRAM 062-02 - SUPPLY/EXHAUST FAN STARTER

NOT TO SCALE

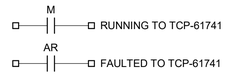
3 062E612 CONTROL DIAGRAM 062-03 - 120V PUMP STARTER

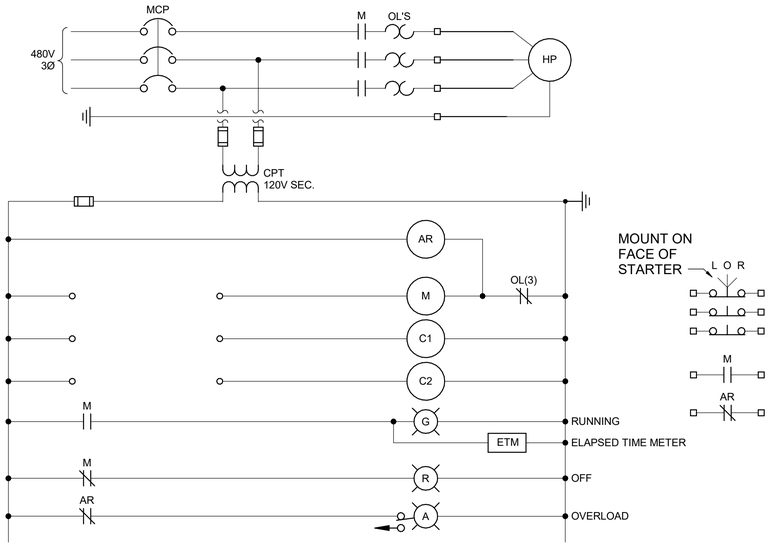
NOT TO SCALE



KEYNOTES (X)

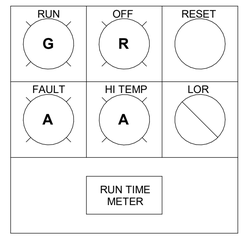
1. PROVIDE 5% INPUT REACTORS ON SEPARATELY-ENCLOSED VFDS. REFER TO SPECIFICATION 26 29 23 FOR SPECIFIC POWER QUALITY REQUIREMENTS.





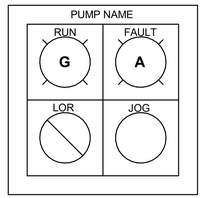
4 062E612 CONTROL DIAGRAM 062-04 - SPARE MOTOR STARTER

NOT TO SCALE



5 062E612 062-01 FACEPLATE LAYOUT

NOT TO SCALE



6 062E612 062-03 120V PUMP CNTRL BRD LAYOUT

NOT TO SCALE

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4/28/2026 7:43:49 AM

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed April 28, 2026 STATE OF ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 062 E612	TITLE OF SHEET WATER TREATMENT MOTOR CONTROL DIAGRAMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 143 OF 192

GENERAL NOTE:

A. LUMINAIRE SCHEDULE PRESENTS INFORMATION ON THE SALIENT CHARACTERISTIC OF THE LUMINAIRES. BRAND NAMES LISTED ARE CONSIDERED BASIS OF DESIGN. "OR EQUAL" SUBSTITUTIONS ARE ACCEPTABLE MEETING THE REQUIREMENTS OF 01 33 23 AND THE PRODUCT REQUIREMENTS LISTED IN THE ASSOCIATED TECHNICAL SPECIFICATIONS. ALL "OR EQUAL" SUBSTITUTED LUMINAIRES MUST COMPLY WITH THE PHOTOMETRIC AND POWER DENSITY STATISITICS (ALL INTERIOR AND EXTERIOR SPACES/ AREAS) INDICATED ON THE ELECTRICAL LIGHTING AND RECEPTACLE PLAN(S).

LUMINAIRE SCHEDULE

TAG	MANUFACTURER/ MODEL NUMBER	LAMPS							MOUNTING	HEIGHT	DESCRIPTION	IMAGE	NOTES
		TYPE	TEMP	LUMENS	CRI	BUG	VOLTAGE	VA					
B2/ B2E	HOLOPHANE "PETROLUX" #PXLW-5000LM-MDU5FR-120-40K-80CRI-PM-nPP16D-WGX-CR-DGXD WALL/STANCHION MOUNT OPTION: #P3US-DGXD WIREGUARD ACCESSORY: #WGPXLW-CR 'B2E' - PROVIDE WITH INTEGRAL EMERGENCY BATTERY BACK-UP: #E10WMCP OR APPROVED EQUAL	LED	4000K	4403	80	N/A	120	35	PENDANT OR WALL (TO BOTTOM OF FIXTURE)	9'-0" AFF (OR AS NOTED PER DRAWINGS)	COMPACT HEAVY-DUTY INDUSTRIAL LUMINAIRE WITH ROBUST CAST ALUMINUM ALLOY (0.6%CU CONTENT) HOUSING, STAINLESS STEEL TAMPER-RESISTANT HARDWARE, DISTORTION RESISTANT HOLLOW CORE SILICONE TYPE GASKET, GRAY CORROSION RESISTANT POLYESTER POWDER COAT FINISH, ENDURAL PRISMATIC BOROSILICATE GLASS REFRACTORS WITH UNIFORMITY AND GLARE CONTROL, MEDIUM FROSTED W/ UPLIGHT DISTRIBUTION OPTICS, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL 0-10V DIMMING CONTROLS, INTEGRAL nLIGHT CONTROL MODULE, HIGH OUTPUT LONG-LIFE LEDs, HIGH EFFICIENCY DRIVER, INTEGRAL OPTIC WIREGUARD, INTEGRAL 90 DEGREE UNIVERSAL MOUNTING BRACKET FOR WALL OR 1-1/2" NPT STANCHION MOUNTING, INTEGRAL 3/4" NPT PENDANT MOUNTING, WET LOCATION RATED.		
C1	HOLOPHANE "PETROLUX" #PXHW-15000LM-WDU-120-40K-80CRI-UNM-nPP16D-WGX-CR-DGXD WALL/STANCHION MOUNT OPTION: #P3US-DGXD WIREGUARD ACCESSORY: #WGPXHW-CR OR APPROVED EQUAL	LED	4000K	12841	80	N/A	120	106	PENDANT OR WALL (TO BOTTOM OF FIXTURE)	16'-0" AFF (OR AS NOTED PER DRAWINGS)	COMPACT HEAVY-DUTY INDUSTRIAL LUMINAIRE WITH ROBUST CAST ALUMINUM ALLOY (0.6%CU CONTENT) HOUSING, STAINLESS STEEL TAMPER-RESISTANT HARDWARE, DISTORTION RESISTANT HOLLOW CORE SILICONE TYPE GASKET, GRAY CORROSION RESISTANT POLYESTER POWDER COAT FINISH, ENDURAL PRISMATIC BOROSILICATE GLASS REFRACTORS WITH UNIFORMITY AND GLARE CONTROL, TYPE V WIDE HIGH ANGLE W/ UPLIGHT DISTRIBUTION OPTICS, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL 0-10V DIMMING CONTROLS, INTEGRAL nLIGHT CONTROL MODULE, HIGH OUTPUT LONG-LIFE LEDs, HIGH EFFICIENCY DRIVER, INTEGRAL OPTIC WIREGUARD, INTEGRAL 90 DEGREE UNIVERSAL MOUNTING BRACKET FOR WALL OR 1-1/2" NPT STANCHION MOUNTING, INTEGRAL 3/4" NPT PENDANT MOUNTING, WET LOCATION RATED.		
EM1	LITHONIA "INDURA" #EXTL-SP2200L-UVOLT-LTP-SDRT-EHO-SW-CW OR APPROVED EQUAL	LED	-	2200	-	N/A	120/277	22	WALL (TO BOTTOM OF FIXTURE)	9'-0" AFF (OR AS NOTED PER DRAWINGS)	INDUSTRIAL EMERGENCY LUMINAIRE UNIT WITH 5VA FLAME RATED/IMPACT-RESISTANT/SCRATCH-RESISTANT/CORROSION PROOF/ UV STABLE RESIN GRAY THERMOPLASTIC COMPACT NEMA 4X HOUSING, TOOL-LESS ACCESS FRONT COVER, UNIQUE TRACK AND SWIVEL FULLY ADJUSTABLE DUAL LED LAMP HEADS, INTEGRAL EXTRA HIGH-OUTPUT SEALED MAINTENANCE-FREE LITHIUM IRON PHOSPHATE EMERGENCY BATTERY PACK, INTEGRAL DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL SELF-DIAGNOSTIC TESTING, INTEGRAL MULTI-COLOR LED STATUS INDICATOR, INTEGRAL CEILING/WALL MOUNT, COLD WEATHER/ WET/ NEMA 4X LOCATION RATED.		2
EX1	LITHONIA #WLTE-W-(NOTE 1)-R-EL-TP-SD OR APPROVED EQUAL	LED	-	-	-	N/A	120/277	3	SURFACE, WALL, END, OR PENDANT (TO BOTTOM OF FIXTURE)	1'-0" ABOVE EGRESS DOOR FRAME/ CEILING (OR AS NOTED PER DRAWINGS)	INDUSTRIAL EXIT SIGN WITH IMPACT-RESISTANT SCRATCH-RESISTANT CORROSION-PROOF THERMOPLASTIC WHITE COMPACT HOUSING, RUGGED HEAVY-DUTY POLYCARBONATE CLEAR HOUSING LENS, TAMPER-RESISTANT SCREW HARDWARE, 6" HIGH WITH 3/4" STROKE RED LETTERS, UNIVERSAL REMOVABLE DIRECTIONAL CHEVRONS INDICATORS, INTEGRAL SEALED MAINTENANCE FREE NICKEL-CADMIUM EMERGENCY BATTERY PACK, INTEGRAL DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL SELF-DIAGNOSTIC TESTING, INTEGRAL UNIVERSAL (TOP, END, OR BACK) CANOPY MOUNT, WET LOCATION RATED.		1
EX2	LITHONIA #LQC-(NOTE 1)-R-ELN SWIVEL STEM KIT: ELA-US(NOTE 3) OR APPROVED EQUAL	LED	-	-	-	N/A	120/277	1	SURFACE, WALL, END, OR PENDANT (TO BOTTOM OF FIXTURE)	1'-0" ABOVE EGRESS DOOR FRAME/ CEILING (OR AS NOTED PER DRAWINGS)	EXIT SIGN WITH DIE-CAST ALUMINUM COMPACT HOUSING AND CANOPY, BRUSHED ALUMINUM FACEPLATE WITH CLEAR LACQUER FINISH, MATTE BLACK ELECTROSTATIC POLYMERIC TRIM, FULLY OVERLAPPING LIGHT SEAL, NO EXPOSED HARDWARE, 6" HIGH WITH 3/4" STROKE RED LETTERS, UNIVERSAL DIRECTIONAL CHEVRON INDICATORS, INTEGRAL SEALED MAINTENANCE-FREE NICKEL-CADMIUM EMERGENCY BATTERY PACK, INTEGRAL DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL SELF-DIAGNOSTIC TESTING, INTEGRAL UNIVERSAL (TOP, END OR BACK) CANOPY MOUNT, DAMP LOCATION RATED.		1,3
F1/ F1E	LITHONIA "BREEZ" #2BZL2-30L-EZB-LP840-N100 DRYWALL CEILING ADAPTER: #DGA22 'F1E' - PROVIDE WITH INTEGRAL EMERGENCY BATTERY BACK-UP: #E10WLCP OR APPROVED EQUAL	LED	4000K	3185	80	N/A	MVOLT	33	RECESSED	CEILING	2'X2' LOW-PROFILE RECESSED INDIRECT LUMINAIRE WITH COMPACT-DESIGN COLD-ROLLED STEEL CHANNEL HOUSING, TABBED AND SCREWED ON INWARD END PLATES, ROLLED AND COMPLETELY HEMMED EDGES, INTEGRAL T-BAR CLIPS MOUNTING BRACKETS, COLD ROLLED STEEL ENDS/END TRIM/ALUMINUM EXTRUSION SIDE FLANGES TYPE REFLECTOR SYSTEM WITH HIGHLY REFLECTIVE OPTICAL COMPOSITE COATING, WHITE POWDER COAT FINISH, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL 0-10V DIMMING CONTROLS, INTEGRAL nLIGHT CONTROL MODULE, SECONDARY OCCUPANCY/MOTION SENSOR CAPABILITY, HIGH-OUTPUT LONG-LIFE LEDs, HIGH-EFFICIENCY DRIVER, DRYWALL GRID MOUNTING BRACKET OPTION, DAMP LOCATION RATED.		
G1/ G1E	LITHONIA #LDN4SQ-40-15-LS4-AR-LD-120-EZ1-NPS80EZ 'G1E' - PROVIDE WITH INTEGRAL EMERGENCY BATTERY BACK-UP: #E10WCP OR APPROVED EQUAL	LED	4000K	1134	80	N/A	120	18	RECESSED	CEILING	4" SQUARE DOWNLIGHT LUMINAIRE WITH RUGGED/ GALVANIZED STEEL MOUNTING FRAME, TORSION SPRING BRACKETS, VERTICAL ADJUSTABLE MOUNTING BRACKETS, COMMERICAL BAR HANGERS WITH 3-3/4" TOTAL ADJUSTMENT, GALVANIZED STEEL JUNCTION BOX WITH BOTTOM-HINGED ACCESS COVER AND SPRING LATCHES, FOUR 1/2" KNOCKOUTS, 12"-24" JOIST SPACING, SELF-FLANGED ANODIZED MATTE CLEAR DIFFUSER/REFLECTOR, DIFFUSED OPTICAL LENS, INTEGRAL DIMMING TO 1% CONTROLS, INTEGRAL nLIGHT CONTROL MODULE, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, SECONDARY OCCUPANCY/MOTION SENSOR CAPABILITY, WET LOCATION RATED.		

NOTES:

1) REFER TO DRAWINGS FOR EXACT NUMBER OF FACES REQUIRED AT EACH EXIT SIGN LOCATION. FIELD ADJUST DIRECTIONAL CHEVRON ACCORDINGLY FOR PATH OF EGRESS.

2) COORDINATE FINAL AIMING ANGLE WITH CONTRACTING OFFICER.

3) FIELD COORDINATE EXACT LENGTH OF STEM REQUIRED FOR FINAL MOUNTING HEIGHT WITH CEILING STRUCTURE. PROVIDE EXACT STEM LENGTH REQUIRED FOR EACH LUMINAIRE.

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — Date Signed 4/28/2026 — ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	G. MARTINEZ	062 E632	WATER TREATMENT LUMINAIRE SCHEDULES	113 182752
	CADD: S. BILD			PMIS/PKG NO. 318719
	TECH. REVIEW: P. DAVILA			SHEET 144 OF 192
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_062_NWTP_E.rvt
4/28/2026 7:43:50 AM

GENERAL NOTE:

A. LUMINAIRE SCHEDULE PRESENTS INFORMATION ON THE SALIENT CHARACTERISTIC OF THE LUMINAIRES. BRAND NAMES LISTED ARE CONSIDERED BASIS OF DESIGN. "OR EQUAL" SUBSTITUTIONS ARE ACCEPTABLE MEETING THE REQUIREMENTS OF 01 33 23 AND THE PRODUCT REQUIREMENTS LISTED IN THE ASSOCIATED TECHNICAL SPECIFICATIONS. ALL "OR EQUAL" SUBSTITUTED LUMINAIRES MUST COMPLY WITH THE PHOTOMETRIC AND POWER DENSITY STATISITICS (ALL INTERIOR AND EXTERIOR SPACES/ AREAS) INDICATED ON THE ELECTRICAL LIGHTING AND RECEPTACLE PLAN(S).

LUMINAIRE SCHEDULE

TAG	MANUFACTURER/ MODEL NUMBER	LAMPS							MOUNTING	HEIGHT	DESCRIPTION	IMAGE	NOTES
		TYPE	TEMP	LUMENS	CRI	BUG	VOLTAGE	VA					
H1	LITHONIA #FMVCSLS-24IN-MVOLT-40K-90CRI-BN PROVIDE nLIGHT POWER PACK 'PP' (SEE LUMINAIRE CONTROLS SCHEDULE) OR APPROVED EQUAL	LED	4000K	1304	80	N/A	MVOLT	18	WALL (TO BOTTOM OF FIXTURE)	6" ABOVE VANITY MIRROR FRAME (OR AS NOTED PER DRAWINGS)	2' LOW-PROFILE LENSED LINEAR SQUARE VANITY LUMINAIRE WITH DIE-CAST ALUMINUM HOUSING/END CAPS/REMOVABLE CANOPY-JUNCTION BOX COVER, BRUSHED NICKEL FINISH, LINEAR WHITE EXTRUDED ACRYLIC DIFFUSER LENS, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL 0-10V DIMMING CONTROLS, INTEGRAL nLIGHT CONTROL MODULE, SECONDARY OCCUPANCY/MOTION SENSOR CAPABILITY, HIGH-OUTPUT LONG-LIFE LEDs, HIGH-EFFICIENCY DRIVER, DAMP LOCATION RATED.		
J1/J1E	LITHONIA #2WRTL-G-L24-5000LM-IAW-AFL-MD-MVOLT-EZ1-40K-80CRI-SPD-NLTAIR2-WH DRYWALL CEILING FLANGE KIT: #2XRTL-XLF-L24-WH-FLANGE-KIT 'J1E' - PROVIDE WITH INTEGRAL EMERGENCY BATTERY PACK-UP: #IE10WLCP OR APPROVED EQUAL	LED	4000K	5081	80	N/A	MVOLT	41	RECESSED	CEILING	2'x2'x4" LOW-PROFILE FLAT PANEL LUMINAIRE WITH RUGGED SEAMLESS LEAK-PROOF SOLID SILICONE GASTKETED ALUMINUM FRAME, 20-GAUGE COLD ROLLED STEEL HOUSING, INTEGRAL T-BAR CLIPS FOR 15/16" OR 9/16" T-GRID CEILING, WHITE POLYESTER POWDER COAT FINISH, IMPACT-MODIFIED ACRYLIC SATIN WHITE FULLY LUMINOUS WHITE DIFFUSED LENS, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL 0-10V DIMMING TO 1% CONTROLS, INTEGRAL nLIGHT CONTROL MODULE, SECONDARY OCCUPANCY/MOTION SENSOR CAPABILITY, HIGH-OUTPUT LONG-LIFE LEDs, HIGH-EFFICIENCY DRIVER, DRYWALL CEILING FLANGE MOUNTING BRACKET, IK10 IMPACT RATED, WET LOCATION RATED.		
K1/ K1E	LITHONIA #GRD-LLP-(NOTE 2)FT-MSL(NOTE2)-80CRI-40K-ID800LMF-80/20-MIN1-NLIGHT-120-SCT-(NOTE 4)-C210-DU 'K1E' - PROVIDE WITH INTEGRAL EMERGENCY BATTERY BACK-UP: #E10WLCP OR APPROVED EQUAL	LED	4000K	3980	80	N/A	MVOLT	20	PENDANT (TO BOTTOM OF FIXTURE)	8'-0" AFF AND MINIMUM 18" FROM CEILING SURFACE (OR AS NOTED PER DRAWINGS)	8-1/4"x1-3/4"x4' LOW-PROFILE LENSED INDIRECT/ DIRECT LINEAR LUMINAIRE WITH DIE-CAST RECTANGULAR COLD-ROLLED STEEL HOUSING, FINE TEXTURED WHITE FINISH, MECHANICALLY ATTACHED (NO EXPOSED FASTENERS) END CAPS, FROSTED DIFFUSED LENS, 80% UP/20% DOWN INTENSITY RATIO, INTEGRAL VERTICAL JUNCTION BOX AND WIREWAY, INTEGRAL MATCHING CANOPY AT SUPPORT, INTEGRAL DUST COVER, INTEGRAL nLIGHT DIMMING CONTROLS, INTEGRAL 0-10V DIMMING TO 0.1% DRIVER, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, SECONDARY OCCUPANCY SENSOR CAPABILITY, INTEGRAL T-BAR CEILING MOUNTING BRACKET OR INTEGRAL HARD CEILING HORIZONTAL JUNCTION BOX (SEE NOTE 4), INTEGRAL 48" ADJUSTABLE SUSPENSION SUPPORTS, DRY LOCATION RATED.		2,4
M1/ M1E	HOLOPHANE #EMX-L24-6000LM-FGFL-WD-MVOLT-GZ10-40K-80CRI-SPD20KV-RMBK-DGXD-CR WALL ANGLE MOUNT: #EMXRMBA MONOPOINT PENDANT MOUNT: PMP4X PROVIDE nLIGHT POWER PACK 'PP' (SEE LUMINAIRE CONTROLS SCHEDULE) 'M1E' - PROVIDE WITH INTEGRAL EMERGENCY BATTERY BACK-UP: #E10WLCP OR APPROVED EQUAL	LED	4000K	4697	80	N/A	MVOLT	40	PENDANT, SURFACE, OR WALL (TO BOTTOM OF FIXTURE)	10' AFF OR CEILING STRUCTURE (OR AS NOTED PER DRAWINGS)	2' LOW-PROFILE LENSED INDUSTRIAL STRIP LUMINAIRE WITH COMPACT-DESIGN CAST & EXTRUDED ALUMINUM HOUSING & COVER, 18-8 STAINLESS STEEL HARDWARE, INTEGRAL MOLDED SILICON GASKET, 3/4" THREADED NPT END-CAPS, ENHANCED CORROSION RESISTANT DURABLE GREY POWDER COAT FINISH, 0.125" THICK FLAT FROSTED GLASS LENS, WIDE DISTRIBUTION OPTICS, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL 0-10V DIMMING CONTROLS, INTEGRAL nLIGHT CONTROL MODULE, HIGH OUTPUT LONG-LIFE LEDs, HIGH-EFFICIENCY DRIVER, INTEGRAL RIGID MOUNTING BRACKET, WALL ANGLED (22.5°,45°,90°) RIGID MOUNTING BRACKET OPTION, MONOPOINT PENDANT MOUNTING BRACKET OPTION, IP67/ WET/ NEMA 4X LOCATION RATED.		
U1	JUNO LIGHTING #UCES-12IN-SWW6-90CRI-WH-M6 ACCESSORIES: (NOTE 3) OR APPROVED EQUAL	LED	4000K	492	90	N/A	120	7	SURFACE (TO BOTTOM OF FIXTURE)	CENTER OF UNDER CABINET (OR AS NOTED PER DRAWINGS)	1"x 2-1/2"x12" LOW-PROFILE LENSED UNDERCABINET LUMINAIRE WITH DURABLE STEEL HOUSING, WHITE FINISH, WHITE END CAPS, FROSTED ACRYLIC DIFFUSER, CONTINUOUS ROW LONG-LIFE LEDs, HIGH-EFFICIENCY DRIVER, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, INTERGRAL DIRECT-WIRE REAR ACCESS PLATE/KNOCKOUT, INTEGRAL DAISY CHAIN SIDE ACCESS KNOCKOUT, INTEGRAL COLOR TEMP SWITCH, DAMP LOCATION RATED.		3
W1/ W1E	LITHONIA "D-SERIES" #DSXW1-LED-P1-AMBLW-AMCRI-T2S-MVOLT-SRW-PIR1FC3V-VG-DDBXD 'W1E' - PROVIDE WITH INTEGRAL EMERGENCY BATTERY BACK-UP: #E20WC OR APPROVED EQUAL	LED	TRUE AMBER PHOSPHOR (595nm)	613	-	B0-U0-G0	MVOLT	10	WALL (TO BOTTOM OF LUMINAIRE)	A.F.F. AS NOTED PER DRAWINGS	SMALL EXTERIOR WALL PACK LUMINAIRE WITH TWO-PIECE DIE-CAST ALUMINUM HOUSING, INTEGRAL HEAT SINKS FINS, ONE-PIECE SOILD SILICONE GASKET, DARK BRONZE ZINC-INFUSED TGIC THERMOSET POWDER COAT FINISH, FULL "CUT-OFF" (NIGHTTIME FRIENDLY) VISUAL COMFORT TYPE II SHORT DISTRIBUTION OPTICS, INTEGRAL PRECISION-MOLDED ACRYLIC LENSES, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, HIGH OUTPUT LONG-LIFE LEDs, HIGH EFFICIENCY DRIVER, INTEGRAL LENS VANDAL GUARD, INTEGRAL MOTION/AMBIENT SENSOR ENABLED AT 1 FC, WET LOCATION RATED.		1

NOTES:

1) LUMINAIRES TO BE CONTROLLED VIA THE FACILITY PHOTOCELL/ TIME CLOCK/ RELAY-CONTACTOR LIGHTING CONTROL PANEL. REFER TO LIGHTING CONTROL SCHEMATIC DETAIL #1 ON SHEET 008-E504.

2) REFER TO DRAWINGS FOR EXACT ROW LENGTH. PROVIDE INTERNAL JOINERS TO ACHIEVE ROW LENGTH AS SHOWN. OPTIMIZE MANUFACTURER'S CONFIGURATION OPTIONS FOR FEWEST SEGMENTS AND MOUNTING DROP LOCATIONS.

3) PROVIDE ALL REQUIRED LINKING CORDS, CONNECTORS, SPLICE BOX, POWER SOURCE, MOUNTING HARDWARE, ETC. FOR A COMPLETE INSTALLATION.

4) FOR T-BAR CEILING MOUNTING, PROVIDE WITH #F1A/48A-MCS. FOR HARD CEILING MOUNTING, PROVIDE WITH #F2/48A-MCSJ. REFER TO ARCHITECTURAL REFLECTED CEILING PLAN FOR EXACT CEILING TYPE.

REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 4/28/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. MARTINEZ CADD: S. BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 062 E633	TITLE OF SHEET WATER TREATMENT LUMINAIRE SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 145 OF 192

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GENERAL NOTE

A. LUMINAIRE CONTROL SCHEDULES AND RACEWAY SCHEDULE PRESENTS INFORMATION ON THE SALIENT CHARACTERISTIC OF THE LUMINAIRE CONTROLS AND RACEWAYS. BRAND NAMES LISTED ARE CONSIDERED BASIS OF DESIGN. "OR EQUAL" SUBSTITUTIONS ARE ACCEPTABLE MEETING THE REQUIREMENTS OF 01 33 23 AND THE PRODUCT REQUIREMENTS LISTED IN THE ASSOCIATED TECHNICAL SPECIFICATIONS.

LUMINAIRE CONTROLS DEVICE SCHEDULE

TAG	MANUFACTURER/ MODEL NUMBER	VOLTAGE	MOUNTING	HEIGHT	DESCRIPTION	IMAGE	NOTES
OS	ACUITY nLIGHT #NCM-PDT-9-RJB-AR-20M-LT OR APPROVED EQUAL	CAT-5e	SURFACE	CEILING	DUAL TECHNOLOGY (MICROPHONICS AND DIGITAL PASSIVE INFRARED) CEILING/SURFACE MOUNTED OCCUPANCY SENSOR WITH RUGGED HIGH-IMPACT INJECTION-MOLDED WHITE PLASTIC HOUSING, SNAP-LOCKS COVER PLATE, nLIGHT ENABLED, SMALL MOTION 360 DEGREE COVERAGE, DAYLIGHT SENSING, INTEGRAL ADJUSTABLE OCCUPANCY SINGLE TIME DELAY/PHOTOCELL SET-POINTS, INTEGRAL LED STATUS INDICATOR, INTEGRAL REMOTELY UPGRADEABLE FIRMWARE, LOW TEMP/ HIGH HUMIDITY/DAMP LOCATION RATED.		1
S	ACUITY nLIGHT #NPODM-WH-LT OR APPROVED EQUAL	CAT-5e	SURFACE, RECESSED (OR AS NOTED ON DRAWINGS)	AT 42" A.F.F. (OR AS NOTED ON DRAWINGS)	ON/OFF WALL MOUNTED nLIGHT-ENABLED WALL SWITCH WITH RUGGED HIGH-IMPACT INJECTION-MOLDED WHITE PLASTIC HOUSING, WHITE WALL PLATE, INTEGRAL LED STATUS INDICATOR, INTEGRAL SOFT-CLICK PUSH-BUTTON SINGLE CHANNEL CONTROL, INTEGRAL REMOTELY UPGRADEABLE FIRMWARE, LOW TEMP/ DAMP LOCATION RATED.		1
SDM	ACUITY nLIGHT #NPODM-DX-WH-LT OR APPROVED EQUAL	CAT-5e	SURFACE, RECESSED (OR AS NOTED ON DRAWINGS)	AT 42" A.F.F. (OR AS NOTED ON DRAWINGS)	ON/OFF WALL MOUNTED nLIGHT-ENABLED DIMMABLE WALL SWITCH WITH RUGGED HIGH-IMPACT INJECTION-MOLDED WHITE PLASTIC HOUSING, WHITE WALL PLATE, INTEGRAL LED STATUS INDICATOR, INTEGRAL SOFT-CLICK PUSH-BUTTON SINGLE CHANNEL CONTROL, INTEGRAL REMOTELY UPGRADEABLE FIRMWARE, LOW TEMP/ DRY LOCATION RATED.		1
PLF	ACUITY nLIGHT #NPP20PL-LT OR APPROVED EQUAL	120-277V/ 15VDC/ CAT-5e	SURFACE	CEILING/ STRUCTURE	PLUG LOAD CONTROL RELAY POWER PACK WITH RUGGED HIGH-IMPACTED INJECTION-MOLDED BLUE HOUSING, nLIGHT ENABLED, INTEGRAL 120V SIDE SLOT POWER INPUT/OUTPUT, INTEGRAL 15VDC/40mA PER RJ-45 PORT OUTPUT, INTEGRAL EXTERNAL FAULT PROTECTION, INTEGRAL LATCHING RELAY SWITCHES LINE VOLTAGE LOAD, INTEGRAL AUTO ON MODE, INTEGRAL PROGRAMMING BUTTON, INTEGRAL REMOTELY UPGRADABLE FIRMWARE, LOW TEMP/ PLENUM LOCATION RATED.		1
PP	ACUITY nLIGHT #NPP16-DS-EFP-LT OR APPROVED EQUAL	120-277V/ CAT-5e/ 0-10V	SURFACE	CEILING/ STRUCTURE	nLIGHT-ENABLED RELAY POWER PACK WITH RUGGED HIGH-IMPACTED INJECTION-MOLDED WHITE HOUSING, INTEGRAL 0-10V SIDE SLOT DIMMING OUTPUT, INTEGRAL EXTERNAL FAULT PROTECTION, INTEGRAL SELF-CONTAINED RELAY SWITCHES LINE VOLTAGE LOAD, INTEGRAL AUTO ON MODE, INTEGRAL PROGRAMMING BUTTON, INTEGRAL REMOTELY UPGRADABLE FIRMWARE, LOW TEMP/ PLENUM LOCATION RATED.		1

NOTE:
1) COMPLY WITH SELECTED OCCUPANCY/MOTION SENSOR, WALL SWITCH, AND POWER RELAY PACK MANUFACTURER'S INSTALLATION AND FINAL PLACEMENT REQUIREMENTS PRIOR TO ROUGH-IN. APPLICABLE FOR FOR NON-PROCESS AREAS ONLY. REFER TO SPECIFICATIONS FOR LUMINAIRE CONTROL COMPONENTS REQUIRED FOR PROCESS AREAS.

RACEWAY SCHEDULE

TAG	MANUFACTURER/ MODEL NUMBER	POWER & DATA OUTLET		MOUNTING	HEIGHT	DESCRIPTION	NOTES
		120V NEMA 5-20R DUPLEX RECEPTACLE	DATA OUTLET				
RW1	LEGRAND WIREMOLD 4000 SERIES DUAL CHANNEL MULTI-SERVICE TWO-PIECE STEEL RACEWAY BASE ACCESSORY: #G4000B-10 19.5" PRE-CUT COVER ACCESSORY: #G4000C195 STEEL DIVIDER & DIVIDER CLIPS ACCESSORIES: #G4000D & #4001DA WIRE CLIPS ACCESSORY: #G4000WC GALVANIZED STEEL PLATED COUPLING ACCESSORY: #4001A DIVIDED ENTRANCE END FITTING ACCESSORY: #G4010DFO BLANK END PLATE ACCESSORY: #G4010B TWO-GANG OVERLAPPING RECTANGULAR OPENING & MODULAR FURNITURE COVER PLATE ACCESSORY: #G4047RF OR APPROVED EQUAL	1 @ 24" O.C.	SEE PLANS	SURFACE TO WALL (TO BOTTOM OF RACEWAY)	3'-3" A.F.F. (OR AS NOTED PER DRAWINGS)	1.63" X 4.75" X (NOTE 4) DUAL CHANNEL MULTI-SERVICE TWO-PIECE STEEL RACEWAY WITH DURABLE SCRATCH-RESISTANT SCUFFCOAT BAKED ENAMEL ALUMINUM FINISH, .050" GALVANIZED STEEL BASE WITH TWO 9/32" DIA. KNOCKOUTS 18" ON-CENTER, LARGE CROSS SECTION WIRING AREA, INTEGRAL 19.5" PRE-CUT COVER FOR MOUNTING DEVICES 24" ON-CENTER, .040" GALVANIZED STEEL (ONE SIDE PAINTED BUFF PRIMER) DIVIDER AND DIVIDER CLIPS, INTEGRAL PLATED WIRE CONDUCTORS CLIPS, INTEGRAL GALVANIZED STEEL PLATED JOINING COUPLING, INTEGRAL DIVIDED ENTRANCE FITTING WITH REMOVABLE DIVIDER/ 1-1/4" BEND RADIUS CONTROLLED INSERT/ CONCENTRIC 3/4", 1", 1-1/4" END AND BOTTOM KNOCKOUTS, INTEGAL TWO-GANG OVERLAPPING COVER PLATES WITH RECTANGULAR OPENING FOR GFCI/ SURGE/ DECOR-STYLE DEVICES AND COMBINATION MODULAR FURINTURE-2 KEYSTONE TWIST OUTS, INTEGRAL BLANK END PLATES.	1,2,3

NOTES:
1) ONE CHANNEL DEDICATED FOR POWER CABLES AND ONE CHANNEL DEDICATED FOR DATA CABLES
2) RECEPTACLES TO INCLUDE USB JACKS. PROVIDE GFCI DUPLEX RECEPTACLES WHERE REQUIRED PER NEC AND SPECIFICATIONS.
3) MULTIPLE BRANCH CIRCUITS ROUTED THROUGH THE POWER CHANNEL PORTION OF THE RACEWAY TO RECEPTACLES. EQUALLY DIVIDE THE QUANTITY OF RECEPTACLES PER EACH BRANCH CIRCUIT INDICATED ON THE DRAWINGS.
4) LENGTH AS SHOWN ON THE ELECTRICAL PLANS AND ARCHITECTURAL INTERIOR SECTION ELEVATIONS.

LUMINAIRE CONTROLS SEQUENCE SCHEDULE

SPACE/ AREA TYPE	SEQUENCE OF OPERATIONS (NO DAYLIGHT HARVESTING)	LIGHTING RELAY PANEL	DAYLIGHT HARVESTING - ILLUMINANCE REQUIREMENTS (FC SETPOINTS)
RESTROOM	OS - OCCUPANT ENTERS THE SPACE AND THE LUMINAIRES AUTOMATICALLY TURN ON TO 100% OF FULL LUMINAIRE OUTPUT. UPON SENSING VACANCY AFTER 20 MINUTES, THE LUMINAIRES AUTOMATICALLY TURN OFF.	N/A	N/A
HALLWAY	OS - OCCUPANT ENTERS THE SPACE AND THE LUMINAIRES AUTOMATICALLY TURN ON TO 100% OF FULL LUMINAIRE OUTPUT. UPON SENSING VACANCY AFTER 20 MINUTES, THE LUMINAIRES AUTOMATICALLY TURN OFF.	LIGHTING CIRCUITS TO BE ROUTED THROUGH THE DIGITAL LIGHTING CONTROL RELAY PANEL FOR DIGITAL PHOTOCELL, TIME CLOCK, OR SCHEDULED ON/OFF CONTROL. PROVIDE LOCAL LOW-VOLTAGE MANUAL OVERRIDE SWICH(S) FOR 2 HOUR MAXIMUM OVERRIDE.	N/A
VESTIBULE	OS - OCCUPANT ENTERS THE SPACE AND THE LUMINAIRES AUTOMATICALLY TURN ON TO 100% OF FULL LUMINAIRE OUTPUT. UPON SENSING VACANCY AFTER 20 MINUTES, THE LUMINAIRES AUTOMATICALLY TURN OFF.	LIGHTING CIRCUITS TO BE ROUTED THROUGH THE DIGITAL LIGHTING CONTROL RELAY PANEL FOR DIGITAL PHOTOCELL, TIME CLOCK, OR SCHEDULED ON/OFF CONTROL. PROVIDE LOCAL LOW-VOLTAGE MANUAL OVERRIDE SWICH(S) FOR 2 HOUR MAXIMUM OVERRIDE.	10 FC
ELECTRICAL ROOM, MECHANICAL ROOM, PROCESS AREA, MATERIALS SHOP, BASES DRUM ROOM, ACIDS DRUM ROOM	MANUAL ON/ MANUAL OFF BY LOCAL BI-LEVEL CONTROL DEVICES. OCCUPANT ENTERS THE SPACE AND MANUALLY TURN ON TO 50% OF FULL LUMINAIRES OR 100% OF FULL LUMINAIRES IN THE SPACE.	N/A	N/A
CONTROLS ROOM, SERVER ROOM	OS - OCCUPANT ENTERS THE SPACE AND THE LUMINAIRES AUTOMATICALLY TURN ON TO 50% OF FULL LUMINAIRE OUTPUT AND DEDICATED PLUG LOAD FULL CONTROL 'PLF' RECEPTACLES WILL TURN ON. OCCUPANT CAN MANUALLY TURN ON THE LUMINIARES TO FULL LUMINAIRE OUTPUT BY LOCAL CONTROL DEVICE. UPON SENSING VACANCY AFTER 20 MINUTES, THE LUMINAIRES AND 'PLF' RECEPTACLES AUTOMATICALLY TURN OFF.	N/A	N/A
STORAGE ROOM	OS - OCCUPANT ENTERS THE SPACE AND THE LUMINAIRES AUTOMATICALLY TURN ON TO 50% OF FULL LUMINAIRE OUTPUT. OCCUPANT CAN MANUALLY TURN ON THE LUMINIARES TO FULL LUMINAIRE OUTPUT BY LOCAL CONTROL DEVICE. UPON SENSING VACANCY AFTER 20 MINUTES, THE LUMINAIRES AUTOMATICALLY TURN OFF.	N/A	N/A
LABORATORY ROOM	OS - OCCUPANT ENTERS THE SPACE AND THE LUMINAIRES AUTOMATICALLY TURN ON TO 50% OF FULL LUMINAIRE OUTPUT AND DEDICATED PLUG LOAD FULL CONTROL 'PLF' RECEPTACLES WILL TURN ON. OCCUPANT CAN MANUALLY TURN ON THE LUMINIARES TO FULL LUMINAIRE OUTPUT BY LOCAL CONTROL DEVICE. UPON SENSING VACANCY AFTER 20 MINUTES, THE LUMINAIRES AND 'PLF' RECEPTACLES AUTOMATICALLY TURN OFF.	N/A	40 FC
BREAK ROOM	OS - OCCUPANT ENTERS THE SPACE AND THE LUMINAIRES AUTOMATICALLY TURN ON TO 50% OF FULL LUMINAIRE OUTPUT AND DEDICATED PLUG LOAD FULL CONTROL 'PLF' RECEPTACLES WILL TURN ON. OCCUPANT CAN MANUALLY TURN ON THE LUMINIARES TO FULL LUMINAIRE OUTPUT BY LOCAL CONTROL DEVICE. UPON SENSING VACANCY AFTER 20 MINUTES, THE LUMINAIRES AND 'PLF' RECEPTACLES AUTOMATICALLY TURN OFF.	N/A	40 FC
EXTERIOR WALL PACKS	INTEGRAL - OCCUPANT ENTERS THE SPACE AND THE LUMINAIRES AUTOMATICALLY TURN ON TO 100% OF FULL LUMINAIRE OUTPUT. UPON SENSING VACANCY AFTER 15 MINUTES, THE LUMINAIRES AUTOMATICALLY TURN DOWN TO 30% OF FULL LUMINAIRE OUTPUT.	LIGHTING CIRCUITS TO BE ROUTED THROUGH THE DIGITAL LIGHTING CONTROL RELAY PANEL FOR DIGITAL PHOTOCELL, TIME CLOCK, OR SCHEDULED ON/OFF CONTROL. PROVIDE LOCAL LOW-VOLTAGE MANUAL OVERRIDE SWICH(S) FOR 2 HOUR MAXIMUM OVERRIDE.	N/A

GENERAL NOTES:

1) REFER TO LIGHTING PLANS AND SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.

2) FOR AREAS INDICATED ON THE LIGHTING PLANS TO HAVE DAYLIGHT HARVESTING REQUIREMENTS, THE SEQUENCE OF OPERATIONS FOR THE OCCUPANCY/VANCY SENSOR SHALL BE AS INDICATED FOR THE SPACE/AREA TYPE IN ADDITION TO THE AUTOMATIC DIMMING CONTROL REQUIREMENTS. PROGRAMMING OF THE DAYLIGHT HARVESTING CONTROLS SHALL UTILIZE THE ILLUMINANCE SETPOINTS (FC) INDICATED ON THIS SCHEDULE.

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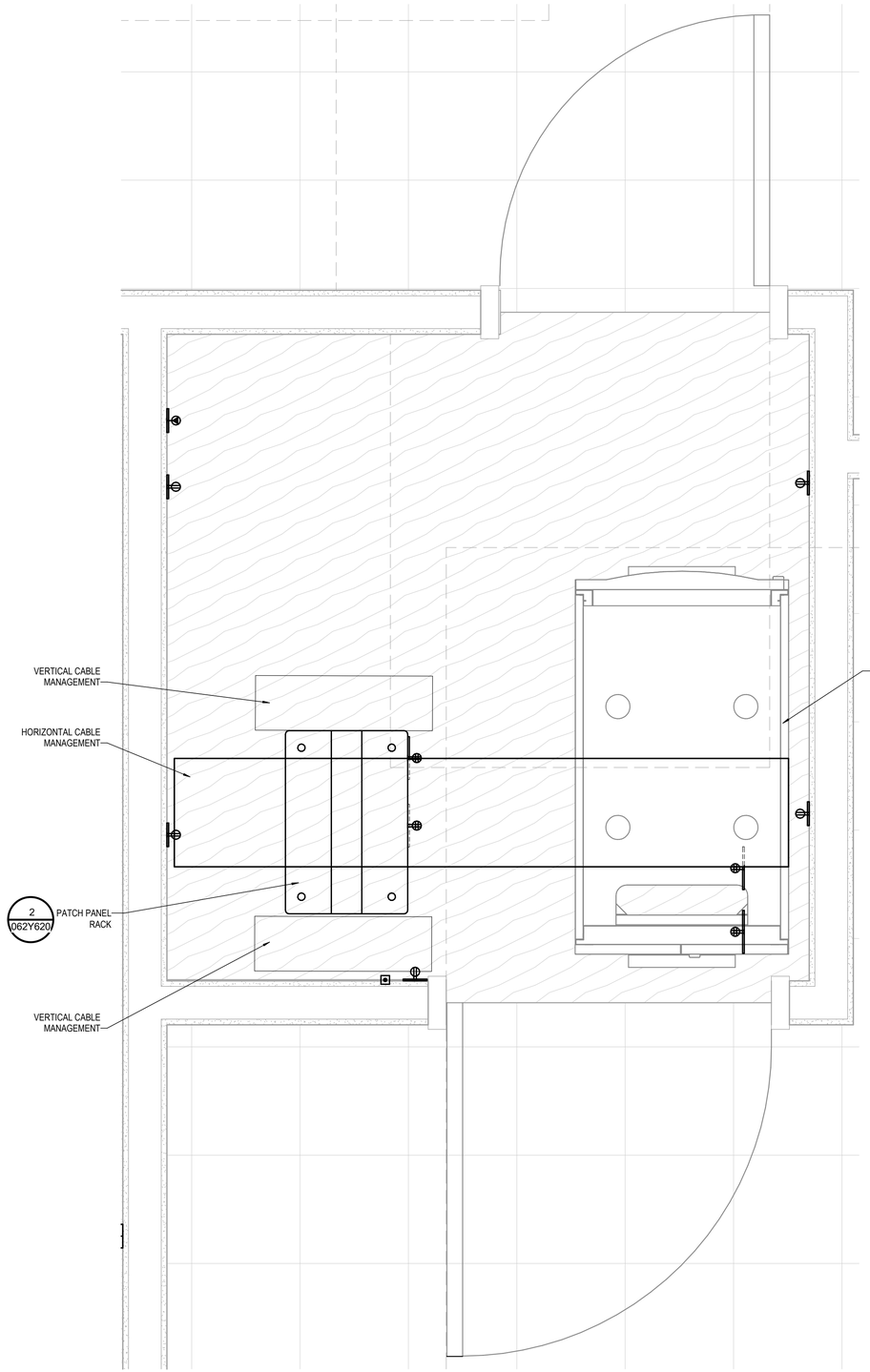
A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

DESIGNED: G. MARTINEZ
DRAWN: S. BILD
TECH. REVIEW: P. DAVILA
DATE: MAY 2026

SUB SHEET NO. 062 E634

TITLE OF SHEET
WATER TREATMENT
LUMINAIRE SCHEDULES
NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 146 OF 192



1 062E111 SCADA/IT ROOM - ENLARGED ROOM

SCALE (B)

SCALE (B) 12 6 0 12

1-1/2" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Certified) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD CADD S. BILD TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 062 Y401	TITLE OF SHEET WATER TREATMENT SERVER ROOM ENLARGED PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 147 OF 192

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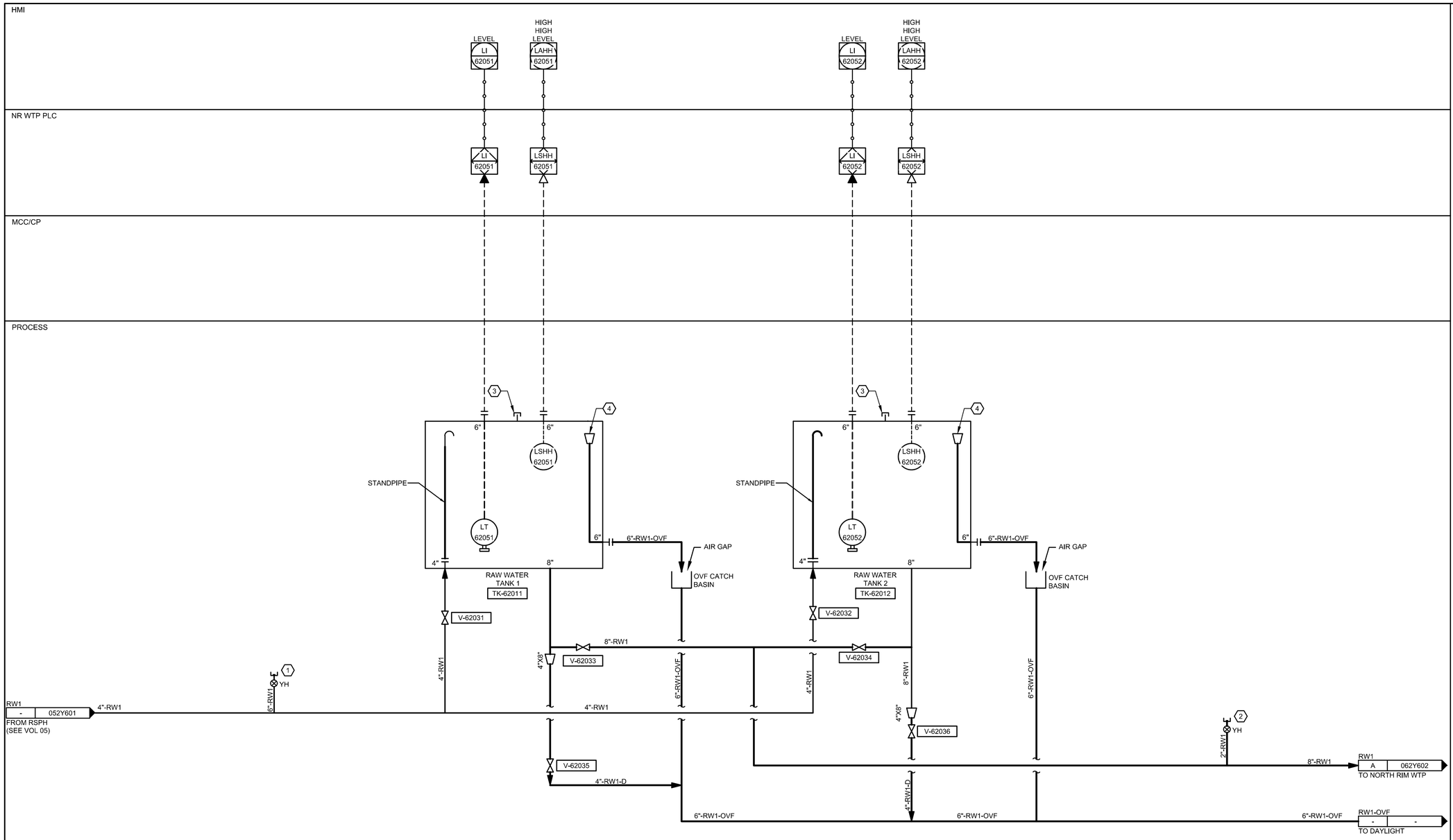

KEYNOTES: (X)

1. YARD HYDRANT WITH CAM 6" LOCK CONNECTION TO ALLOW FOR FLUSHING OF RAW WATER LINE FROM ROARING SPRINGS PUMPHOUSE.
2. YARD HYDRANT WITH 2" CAM LOCK CONNECTION TO ALLOW FOR TRUCK FILLING FOR PARK WILDLIFE AND VEGETATION PROJECTS.
3. TANK VENTS SIZED BY TANK SUPPLIER.
4. 6"X8" FLANGE BY FLARE FITTING.

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD J. STILLMAN DRAWN: R.D. CLARK TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 062 Y601	TITLE OF SHEET NORTH RIM WTP RAW WATER TANKS P&ID NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 148 OF 192

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4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 – NPS GRCA N. RIM UTILITIES\PID DWG\062Y602.dwg

HMI

NR WTP PLC

MCC/CP

PROCESS

PRESS PI 62061

RUN CMD XC 62012 | RUN STATUS NI 62012 | FAULT UA 62012 | REMOTE HSR 62012

REMOTE HSR 62031 | OPEN ZSO 62031 | CLOSED ZSC 62031 | OPEN CMD XVO 62031 | CLOSE CMD XVC 62031 | FAULT UA 62031

RUN CMD XC 62011 | RUN STATUS NI 62011 | FAULT UA 62011 | REMOTE HSR 62011 | SPEED CMD SC 62011 | SPEED SI 62011 | HIGH MTR TEMP TAH 62011 | MULTI DATA UU 62011

POSITION REMOTE HSR 62033 | POSITION CMD ZC 62033 | POSITION ZI 62033 | OPEN ZSO 62033 | CLOSED ZSC 62033 | FAULT UA 62033

FLOW FI 62071 | TOTAL FLOW FQI 62071

PH AI 62081 | TEMP TI 62081

TURB AI 62082

I/I

G 62011 / PB 62011 RUN, RESET, HS 62011 LOR, I, RTM

VFD 480V (2)

X4:
G-RUN
R-OFF
A-FAULT
A-HIGH TEMP

X2:
PIT-62061
PIT-62062

X2:
FCV-62033
FCV-62032

(5) 120V LCP-62012
A 62012 FAULT | G 62012 RUN
HS 62012 LOR | PB 62012 JOG

PI 62061

M

RAW SAMPLE WATER PUMP
PMP-62012

1"-SMP

PIT 62061

RW1 A 062Y601 FROM RAW WATER TANKS

8"-RW1

1"-SMP

8"-RW1

8"X4"

480V M FV-62031

4"X3"

TSH 62011

M

RAW WATER PUMP
PMP-62011

3"X4"

PIT 62062

480V M

CKV-62031

RAW WATER PUMP DISCHARGE FLOW CONTROL VALVE
FCV-62032

(1) 4"-RW1

V-62037

4"-RW1

FIT 62071 120V

M

RAW WATER FLOW METER

4"X6"

1/2"-CAV

(6)

6"-RW1

RW1 C 062Y603 TO REACTION TANKS

4"X8"

480V M

3"X4"

4"-RW1

4"X3"

FCV-62033

RAW WATER FLOW CONTROL VALVE

CKV-62032

1"-RW-DR

(4)

120V AIT 62082 TURB

AE 62082

1"-SMP-DR

FD

120V AIT 62081 PH/TEMP

AE 62081

FI 62071

1/4"-SMP

1"X1/4"

1"X1/2"

1/2"-SMP

SMP - -

TO SAMPLE SINK IN LAB
SEE P SERIES DRAWINGS

1/2"-SMP

RAW WATER INSTRUMENTATION PANEL

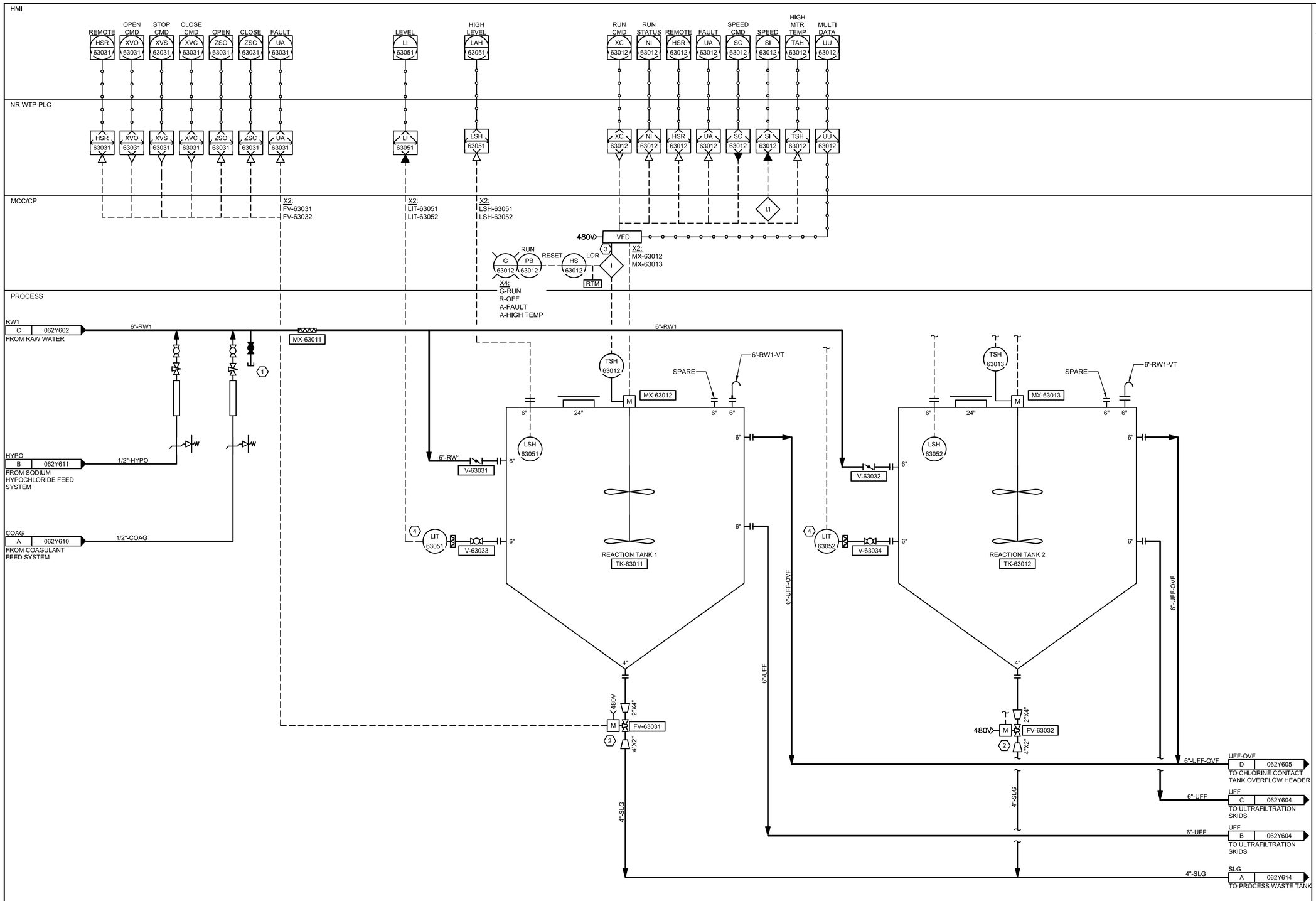
KEYNOTES: (X)

1. 4" ISOLATION VALVE WITH BLIND FLANGE. REQUIRED FOR CONNECTION OF TEMPORARY RECYCLE LINE DURING PLANT START-UP. FOLLOWING START-UP, REPLACE TAPPED BLIND FLANGE WITH SOLID BLIND FLANGE.
2. SEE CONTROL DRAWING 1/062E612.
3. SIGNAL ISOLATORS TO BE INSTALLED INSIDE PLC CONTROL PANEL AND POWERED WITH 24VDC, TYPICAL FOR ALL.
4. SEE INSTRUMENTATION PANEL DETAIL ON 062D502.
5. SEE CONTROL DRAWING 3/062E612.
6. 1/2" MANUAL BLEED VALVE FOR AIR DURING DRAINING/FILLING.

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM	
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	
DESIGNED: J. BRAINARD, J. STILLMAN	
CADD: R.D. CLARK	
TECH. REVIEW: GOUGH/GLOVER	
DATE: MAY 2026	
SUB SHEET NO.	062 Y602
TITLE OF SHEET	WATER TREATMENT RAW WATER METERING AND PUMPING P&ID
NORTH RIM - GRAND CANYON NATIONAL PARK	
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	149 OF 192



KEYNOTES: (X)

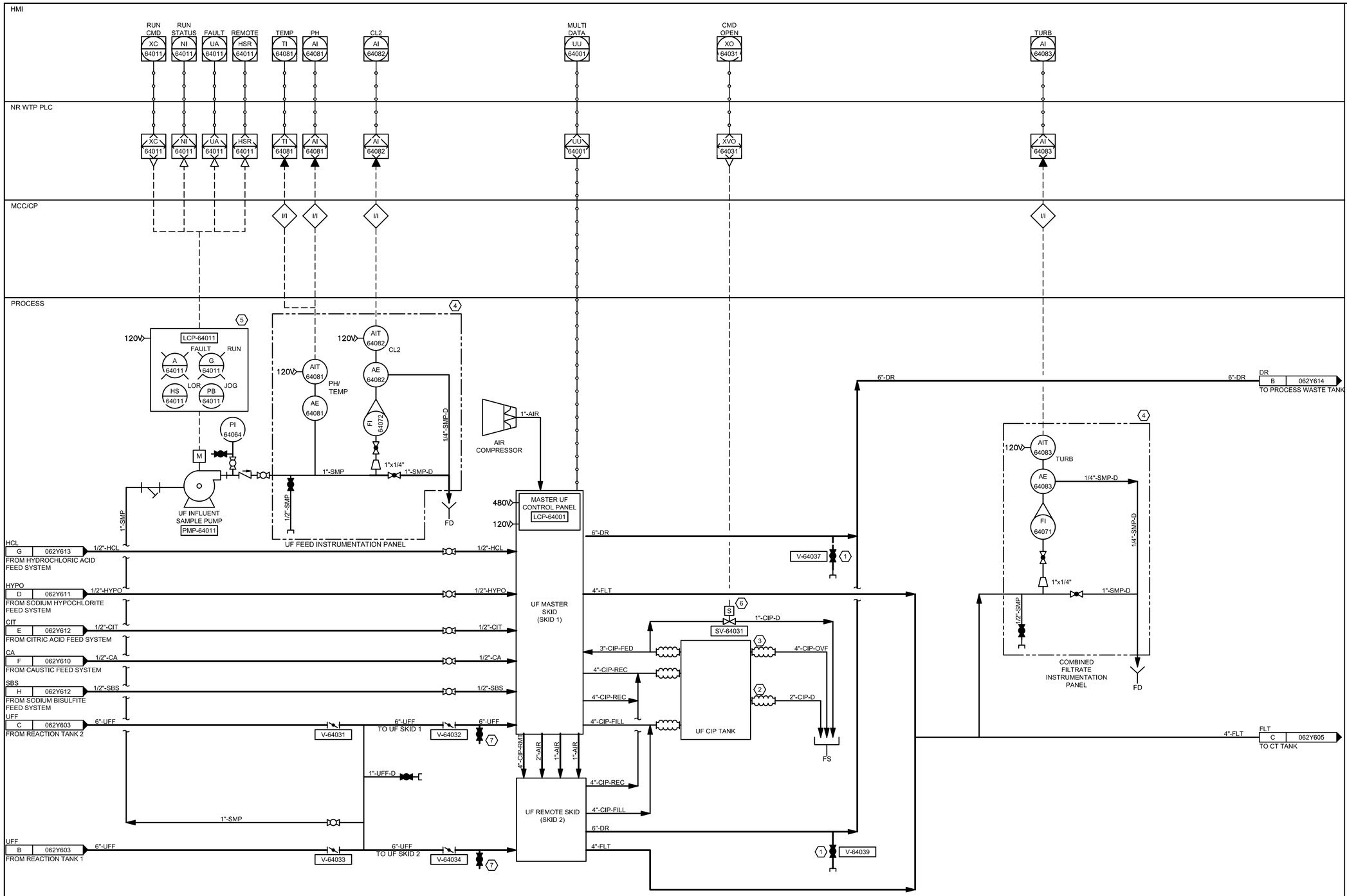
1. 1/2" SPARE CHEMICAL INJECTION POINT. CAP FOR FUTURE.
2. SLUDGE FLOW CONTROL VALVE.
3. SEE CONTROL DRAWING 1/062E612.
4. PROVIDE TANK MOUNTED INSTRUMENT WITH INTEGRAL DIAPHRAGM.

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 4/27/26 ARIZONA, U.S.A.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 4/27/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD J. STILLMAN DRAWN: R.D. CLARK TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 062 Y603	TITLE OF SHEET WATER TREATMENT REACTION TANKS P&ID NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 150 OF 192

4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 - NPS GRCA N. RIM UTILITIES\PID DWG\062Y603.dwg



KEYNOTES: (X)

1. TAP 6" PIPE FOR 2" FLUSHING CONNECTION LINE, INCLUDE 2" BALL VALVE.
2. MANUAL CIP TANK DRAIN. NEUTRALIZED CIP WASTE FROM AN AUTOMATIC CIP CLEAN WILL DISCHARGE AT THE 6" DR CONNECTION ON UF SKIDS.
3. CONTRACTOR RESPONSIBLE FOR SUPPLY OF EXPANSION JOINTS AT TANK NOZZLES ON VENDOR PROVIDED CIP TANK.
4. SEE INSTRUMENTATION PANEL DETAIL ON 062D502.
5. SEE CONTROL DRAWING 3/062E612.
6. CONTRACTOR RESPONSIBLE FOR FURNISHING AND INSTALLATION OF DRAIN LINE WITH SOLENOID VALVE ON CIP FEED LINE TO UF MASTER SKID.
7. 1/2" MANUAL BLEED VALVE FOR AIR DURING DRAINING/FILLING.

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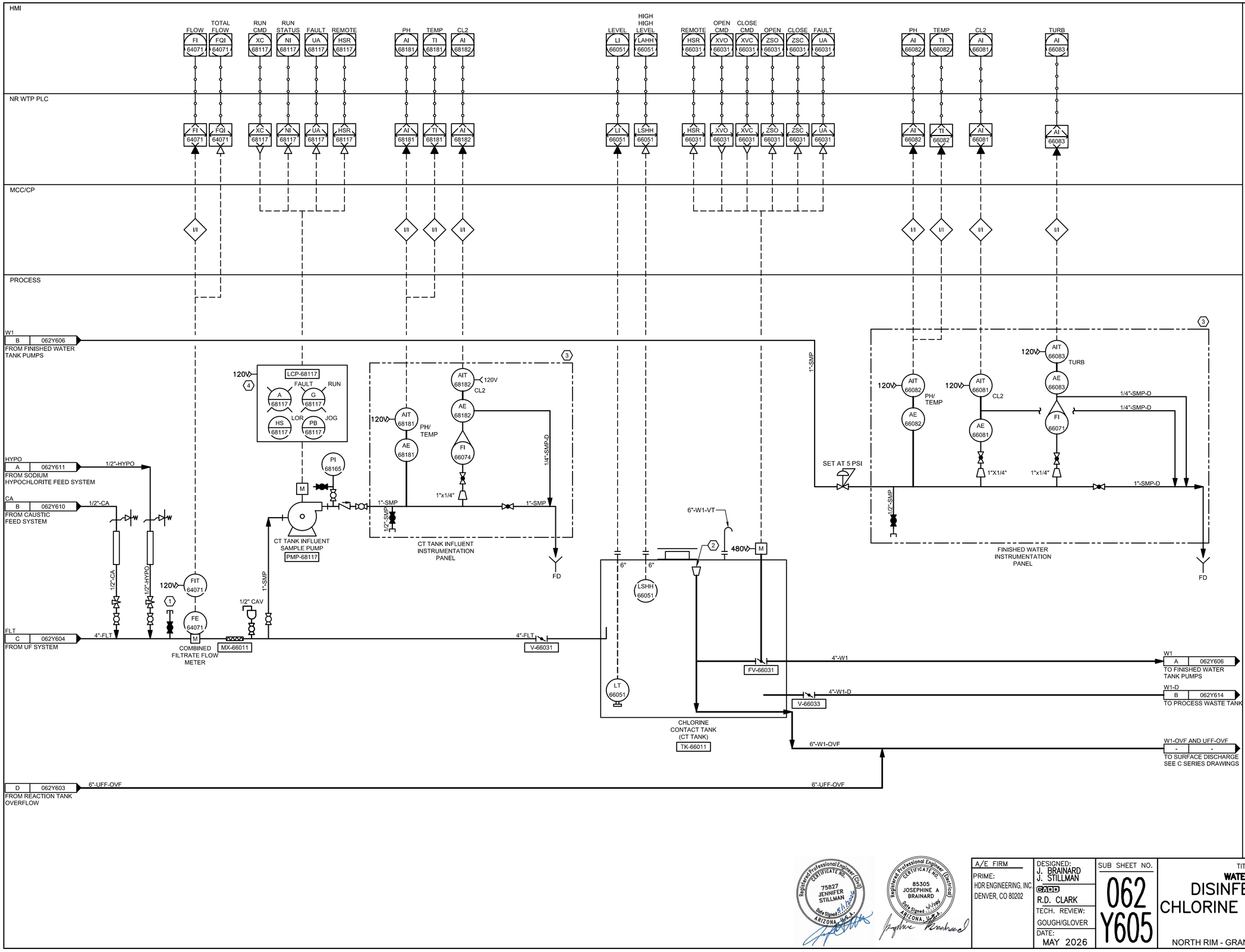

KEYNOTES: (X)

1. 1/2" SPARE CHEMICAL INJECTION POINT. CAP FOR FUTURE.
2. 6"X8" FLANGE BY FLARE FITTING.
3. SEE INSTRUMENTATION PANEL DETAIL ON 062D502.
4. SEE CONTROL DRAWING 3/062E612.

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HMI

NR WTP PLC

MCC/CP

PROCESS

RUN CMD XC 66011 | RUN STATUS NI 66011 | FAULT UA 66011 | REMOTE HSR 66011

RUN CMD XC 68111 | RUN STATUS NI 68111 | REMOTE HSR 68111 | FAULT UA 68111 | SPEED CMD SC 68111 | SPEED SI 68111 | HIGH MTR TEMP TAH 68111 | MULTI DATA UU 68111

PH AI 66084 | TEMP TI 66084 | CL2 AI 66085 | TURB AI 66086

FLOW FI 68173 | TOTAL FLOW FQI 68173 | PRESS PI 68161

480V VFD (1)

RUN G 68111 PB 68111 | RESET | LOR HS 68111

I

RTM

X2
PMP-68111
PMP-68112

X4:
G-RUN
R-OFF
A-FAULT
A-HIGH TEMP

W1 A 062Y605
FROM CT TANK

4"-W1

TSH 68111 M

PI 68163

2"X4"

4"X2"

4"-W1 (5) V-68132

FW TANK PUMP 1
PMP-68111

CKV-68132 V-68134

TSH 68112 M

PI 68164

2"X4"

4"X2"

4"-W1 V-68133

FW TANK PUMP 2
PMP-68112

CKV-68133 V-68135

4"-W1 (5)

4"-W1 (4) 4"-W1 V-68136

4"-W1 V-68131

1"-W1-D

120V LCP-66011 (3)
FAULT A 66011 | RUN G 66011
LOR HS 66011 | JOG PB 66011

120V FIT 68173 M
FINISHED WATER TANKS INFLUENT FLOW METER

PIT 68161

4"-W1

W1 A 062Y607
TO FINISHED WATER TANKS

1"-W1

W1 B 062Y605
TO FINISHED WATER SAMPLE PANEL

(2)

120V AIT 66084 PH/TEMP
AE 66084

120V AIT 66085 CL2
AE 66085 1/4"-SMP-D
FI 66073
1"X1/4"

120V AIT 66086 TURB
AE 66086 1/4"-SMP-D
FI 66072
1"x1/4" 1"X1/2"
1"-SMP-DR

FD

DISTRIBUTION WATER INSTRUMENTATION PANEL

W1 B 062Y608
FROM FINISHED WATER TANKS

HYPO. C 062Y611
FROM SODIUM HYPOCHLORITE FEED SYSTEM

1/2"-HYPO

1"-SMP

4"-W1

M

PI 66064

DISTRIBUTION WATER INSTRUMENTATION PANEL SAMPLE PUMP
PMP-66011

1"-SMP

1/2"-SMP

1/2"-SMP SMP - -
TO SAMPLE SINK IN LAB
SEE P SERIES DRAWINGS

W1 D 062Y608
FROM FINISHED WATER TANKS

4"-W1

KEY NOTES: (X)

1. SEE CONTROL DRAWING 1/062E612.
2. SEE INSTRUMENTATION PANEL DETAIL ON 062D502.
3. SEE CONTROL DRAWING 3/062E612.
4. 4" ISOLATION WITH BLIND FLANGE REQUIRED FOR CONNECTION OF TEMPORARY RECYCLE LINE DURING PLANT STARTUP. SALVAGE VALVE TO GOVERNMENT AND INSTALL BLIND FLANGE ON TEE.
5. 1/2" MANUAL BLEED VALVE FOR AIR DURING DRAINING/FILLING.

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Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD J. STILLMAN DRAWN: R.D. CLARK TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 062 Y606	TITLE OF SHEET WATER TREATMENT FINISHED WATER TANK PUMPS P&ID NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 153 OF 192	

4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 - NPS GRCA N. RIM UTILITIES\PID DWG\0621606.dwg

HMI

NR WTP PLC

MCC/CP

PROCESS

LEVEL LI 68451

LEVEL LI 68452

LI 68451

LI 68452

W1 FROM NRPH

4"-W1

V-68031

V-68032

FW A 062Y606 FROM WTP

4"-W1

8"-W1

2 V-68534

3

1 V-68532

8"-W1

W1 TO NRPH

12"-W1

8"X12"

12"-W1

W1 B 062Y608 TO WTP

FINISHED WATER TANK A 2MG TK-68411

6"-W1-OVF

4"-W1

8"-W1

LT 68451

2 V-68533

4"-W1

V-68431

8"-W1

V-68433

6"-W1-D

V-68435

TANK 1 VALVE VAULT

4"-W1-D

DR DR

8"-W1

FINISHED WATER TANK B 2MG TK-68412

3 3

12"-W1

V-68438

6"-W1-OVF

4"-W1

LT 68452

2 V-68535

4"-W1

V-68432

8"-W1-REC

V-68437

8"-W1

V-68434

8"-W1-D

V-68436

TANK 2 VALVE VAULT

4"-W1-D

8"x12"

2 V-68536

2 V-68537

DR DR

KEY NOTES: (X)

1. ISOLATION VALVE AND PIPING TO BE CAPPED AND REMOVED AFTER CONSTRUCTION OF WATER TREATMENT PLANT IS COMPLETE.
2. LINE STOP VALVE.
3. DOTTED LINES INDICATE PIPING TO BE REMOVED, SEE SEQUENCING PLAN FOR TIMING OF REMOVAL.

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4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 – NPS GRCA N. RIM UTILITIES\PID DWG\062Y607.dwg

HMI

NR WTP PLC

MCC/CP

PROCESS

PRESS PI 68162 | RUN CMD XC 68211 | RUN STATUS NI 68211 | REMOTE HSR 68211 | FAULT UA 68211 | SPEED CMD SC 68211 | SPEED SI 68211 | HIGH MTR TEMP TAH 68211 | MULTI DATA UU 68211

FLOW FI 68174 | TOTAL FLOW FQI 68174 | REMOTE HSR 68031 | OPEN CMD XVO 68031 | STOP CMD XVS 68031 | CLOSE CMD XVC 68031 | OPEN ZSO 68031 | CLOSE ZSC 68031 | FAULT UA 68031

NR WTP PLC: PI 68162 | XC 68211 | NI 68211 | HSR 68211 | UA 68211 | SC 68211 | SI 68211 | TSH 68211 | UU 68211

NR WTP PLC: FI 68174 | FQI 68174 | HSR 68031 | XVO 68031 | XVS 68031 | XVC 68031 | ZSO 68031 | ZSC 68031 | UA 68031

I/I

480V VFD 1

G 68211 RUN PB 68211 RESET HS 68211 LOR RTM

X3:
PMP-68211
PMP-68212
PMP-68213

X4:
G-RUN
R-OFF
A-FAULT
A-HIGH TEMP

X3:
FIT-68174
FIT-68175
FIT-68172

X3:
FV-68031
FV-68032
FV-68033

W1 B 062Y607 FROM FINISHED WATER TANKS

120V FIT 68174 M — COMBINED BACKFEED FLOW METER

V-68035

480V M FV-68031

2"X4"

2"-W1 V-68036 — 120V FIT 68175 M — BACKFEED TO SUPAI TUNNEL FLOW METER

480V M FV-68032

1/2" CAV — 4"-W1 — W1 - - BACKFEED TO INNER-CANYON SEE VOL 05

1/2" CAV — 2"-W1 — W1 - - BACKFEED TO SUPAI TUNNEL SEE VOL 03

120V FIT 68172 M — LOWER ZONE DISTRIBUTION FLOW METER

2" CAV — V-68039 — 8"-W1 — W1 - - TO NORTH RIM DISTRIBUTION SYSTEM LOW ZONE SEE C SERIES DRAWINGS

W1 B 062Y606 TO DISTRIBUTION WATER SAMPLE PANEL

8"X4"

12"X8"

8"-W1 — V-68034

1"-W1

V-68237

8"-W1 — V-68231 — TSH 68211 M — HIGH DEMAND PUMP 1 PMP-68211 — PI 68261 — CKV-68231 — V-68232 — 8"-W1

8"-W1 — V-68233 — TSH 68212 M — HIGH DEMAND PUMP 2 PMP-68212 — PI 68262 — CKV-68232 — V-68234 — 8"-W1

1"-W1-D

8"-W1 — V-68235 — TSH 68213 M — HIGH DEMAND PUMP 3 PMP-68213 — PI 68263 — CKV-68233 — V-68236 — 8"-W1

8"-W1 — 8"-W1 — 1"-W1-D

12"-W1

12"-W1

PIT 68162

8"-W1 — W1 A 062Y609 TO COMBINED FW DISCHARGE

12"X6"

6"-W1 — 6"-W1 — W1 B 062Y609 TO UPPER ZONE BOOSTER PUMPS

480V M FV-68033

4"-W1 — 4"-W1 — W1 D 062Y606 TO FINISHED WATER TANK PUMPS

KEYNOTES: (X)

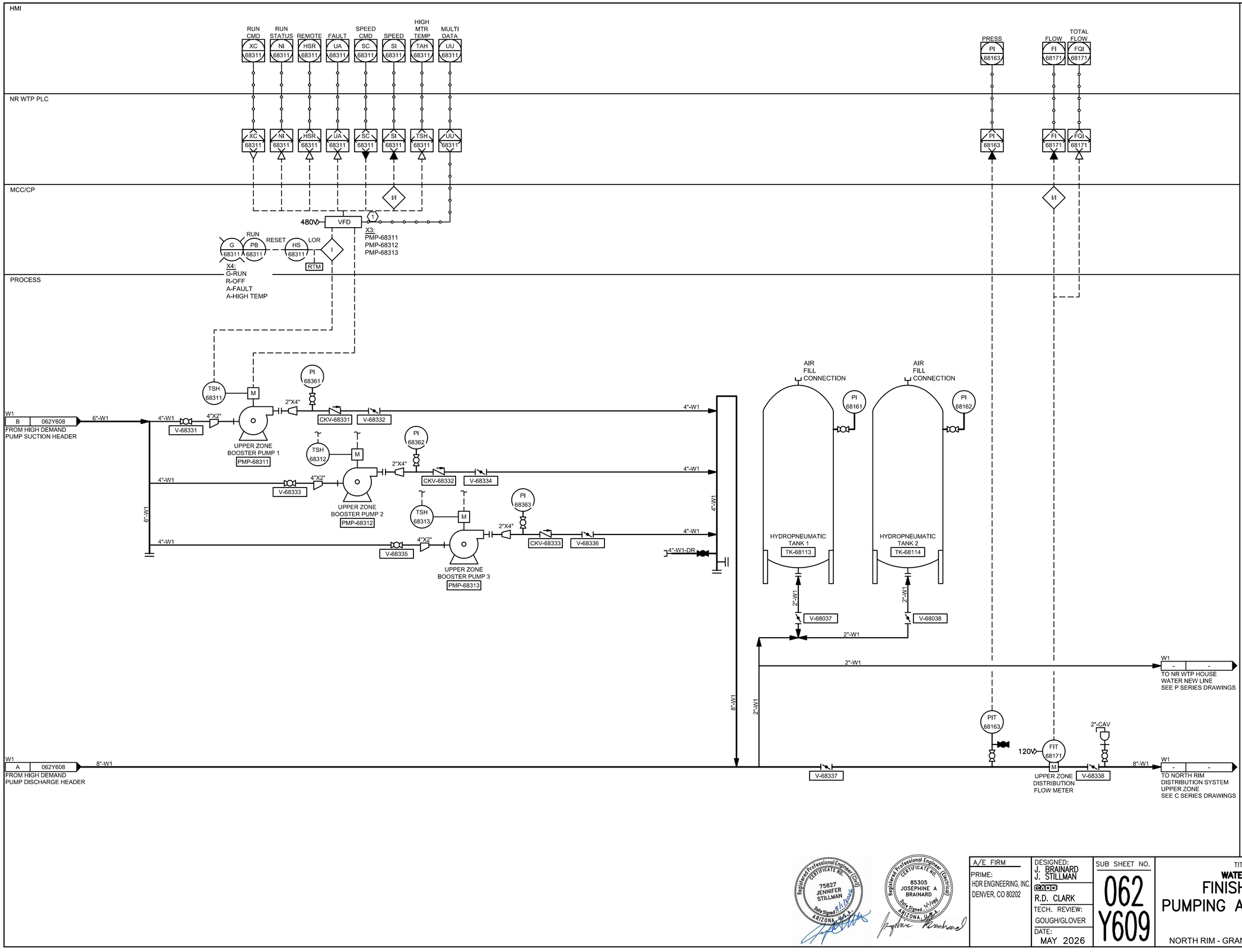
1. SEE CONTROL DRAWING 1/062E612.
2. 1/2" MANUAL BLEED VALVE FOR AIR DURING DRAINING/FILLING.

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/11/26 ARIZONA, U.S.A.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/11/26 ARIZONA, U.S.A.

A/E FIRM	
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	
DESIGNED: J. BRAINARD J. STILLMAN	
CADD: R.D. CLARK	
TECH. REVIEW: GOUGH/GLOVER	
DATE: MAY 2026	
SUB SHEET NO.	062 Y608
TITLE OF SHEET	WATER TREATMENT FINISHED WATER METERING AND PUMPING 1 P&ID
	NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	155 OF 192

4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 – NPS GRCA N. RIM UTILITIES\PID DWG\062Y608.dwg



KEYNOTES: (X)

1. SEE CONTROL DRAWING 1/062E612.

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4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 – NPS GRCA N. RIM UTILITIES\PID DWG\062Y609.dwg

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HMI

WEIGHT WI 67191 | HIGH LEVEL LAH 67151 | RUN CMD XC 67111 | RUN STATUS NI 67111 | REMOTE HSR 67111 | FAULT UA 67111 | SPEED CMD SC 67111 | SPEED SI 67111

NR WTP CHEMICAL RIO

WI 67191 | LSH 67151 | XC 67111 | NI 67111 | HSR 67111 | UA 67111 | SC 67111 | SI 67111

MCC/CP

I/I | I/I

X4:
WIT-67191
WIT-67192
WIT-67391
WIT-67392

X2:
LSH-67151
LSH-67351

X2:
PMP-67111
PMP-67311

PROCESS

COAGULANT FEED SYSTEM

VENT (3) | VENT (3)

2"-COAG-VT | 1"X2" | 2"-COAG-VT | 1"-COAG-VT

RIGID FLEX TUBING | RIGID FLEX TUBING

1"-VT

CALIBRATION COLUMN

LCP-67101
HS 67111 LOR (2)
120V M

PI 67161

SET AT 20 PSI

1/2"-COAG

COAGULANT METERING PUMP PMP-67111

PUMP AND PIPING PANEL

FLEX TUBING RIGID | RIGID FLEX TUBING

120V WIT 67191

WEIGHT SCALE 55 GALLON DRUMS SCL-67111

FLEX TUBING RIGID (4)

120V WIT 67192

WEIGHT SCALE 55 GALLON DRUMS SCL-67112

LSH 67151

CONTAINMENT SUMP

CAUSTIC FEED SYSTEM

VENT (3) | VENT (3)

1/2"-COAG | 1"X2" 2"-CA-VT | 2"-CA-VT | 1"X2" 2"-CA-VT

RIGID FLEX TUBING | RIGID FLEX TUBING

1"-VT

CALIBRATION COLUMN

LCP-67301
HS 67311 LOR (2)
120V M

PI 67361

SET AT 20 PSI

1/2"-CA

CAUSTIC SODA METERING PUMP PMP-67311

PUMP AND PIPING PANEL

FLEX TUBING RIGID | RIGID FLEX TUBING

120V WIT 67391

WEIGHT SCALE 55 GALLON DRUMS SCL-67311

(1) CIP PUMP PANEL (SUPPLIED & CONTROLLED BY MFR)

120V

FLEX TUBING RIGID (4)

120V WIT 67392

WEIGHT SCALE 55 GALLON DRUMS SCL-67312

LSH 67351

CONTAINMENT SUMP

1/2"-COAG → COAG A 062Y603 TO RAW WATER LINE UPSTREAM OF REACTION TANKS

1/2"-CA → CA B 062Y605 TO FILTRATE LINE UPSTREAM OF CHLORINE CONTACT TANK

CA F 062Y604 TO UF MEMBRANE MASTER SKID

KEYNOTES: (X)

1. SEE VEOLIA REFERENCE DRAWINGS FOR FURTHER INFORMATION.
2. LOR SWITCH IS INTEGRAL TO THE FEED PUMPS.
3. VENT LINES ROUTED TO BUILDING EXTERIOR.
4. ONLY ONE DRUM ONLINE AND CONNECTED TO PUMP AT A TIME. TUBING AND WAND SWITCHED TO SECOND DRUM WHEN FIRST DRUM IS EMPTY.

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4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 – NPS GRCA N. RIM UTILITIES\PID DWG\062Y611.dwg

HMI

NR WTP CHEMICAL RIO

MCC/CP

PROCESS

WEIGHT WI 67293

WEIGHT WI 67291

HIGH LEVEL LAH 67251

RUN CMD XC 67211

RUN STATUS NI 67211

REMOTE HSR 67211

FAULT UA 67211

SPEED SI 67211

SPEED CMD SC 67211

WI 67293

WI 67291

LSH 67251

XC 67211

NI 67211

HSR 67211

UA 67211

SI 67211

SC 67211

I/I

I/I

I/I

X2:
WIT-67291
WIT-67292
4

X3:
PMP-67211
PMP-67212
PMP-67213
PMP-67214

3

2"-HYPO-VT

1"-HYPO

1/2"-HYPO

RIGID

FLEX TUBING

FLEX TUBING

RIGID

120V

WIT 67293

WEIGHT SCALE
300 GALLON TOTE
SCL-67213

WIT 67291

4

WEIGHT SCALE
55 GALLON DRUM
SCL-67211

LSH 67251

CONTAINMENT SUMP

LCP-67201
LOR
HS 67211
2
M

SODIUM HYPOCHLORITE METERING PUMP 1
PMP-67211

PI 67261

SET @ 20 PSI

LCP-67202
LOR
HS 67212
2
M

SODIUM HYPOCHLORITE METERING PUMP 2
PMP-67212

PI 67262

SET @ 20 PSI

LCP-67203
LOR
HS 67213
2
M

SODIUM HYPOCHLORITE METERING PUMP 3
PMP-67213

PI 67263

SET @ 30 PSI

LCP-67204
LOR
HS 67214
2
M

SODIUM HYPOCHLORITE SWING PUMP
PMP-67214

PI 67264

SET @ 30 PSI

PUMP AND PIPING PANEL

1

CIP PUMP PANEL
(SUPPLIED & CONTROLLED BY MFR)

120V

HYPO A 062Y605
TO FILTRATE LINE UPSTREAM OF CHLORINE CONTACT TANK

HYPO B 062Y603
TO RAW WATER LINE UPSTREAM OF REACTION TANKS

HYPO C 062Y606
TO FINISHED WATER LINE UPSTREAM OF FINISHED WATER TANK PUMPS

HYPO D 062Y604
TO UF MEMBRANE MASTER SKID

SODIUM HYPOCHLORITE SYSTEM

KEY NOTES: (X)

1. SEE VEOLIA REFERENCE DRAWINGS FOR FURTHER INFORMATION.
2. LOR SWITCH IS INTEGRAL TO THE FEED PUMPS.
3. VENT LINES ROUTED TO BUILDING EXTERIOR.
4. SODIUM HYPOCHLORITE FEED SYSTEM TO CONTAIN TWO 55-GALLON DRUMS AND ONE 300-GALLON TOTE. ONLY ONE DRUM SHOWN FOR REFERENCE.

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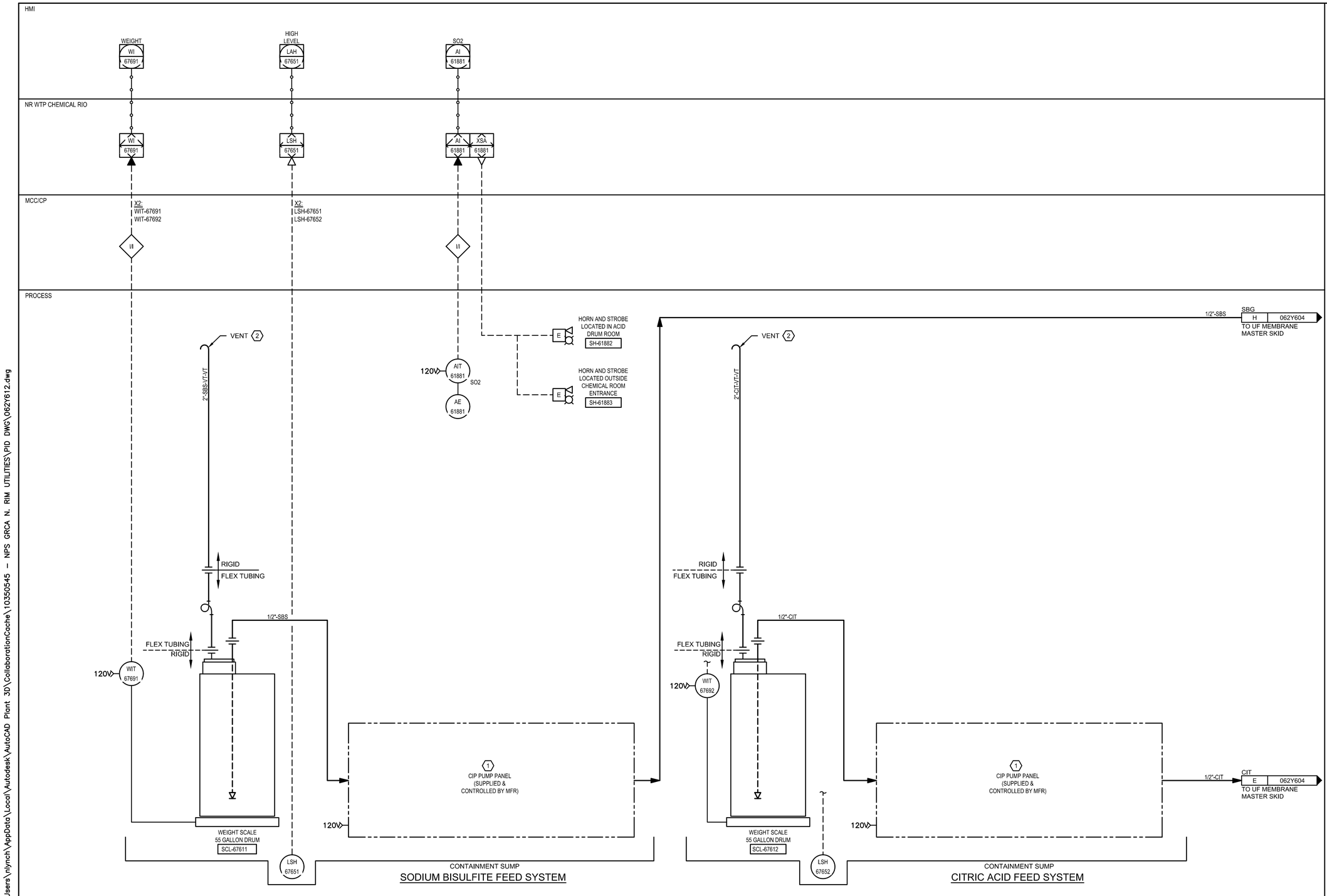
DESIGNED: J. BRAINARD J. STILLMAN	
DRAWN: R.D. CLARK	
TECH. REVIEW: GOUGH/GLOVER	
DATE: MAY 2026	
SUB SHEET NO.	062 Y611
TITLE OF SHEET	WATER TREATMENT SODIUM HYPOCHLORITE SYSTEM P&ID
	NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	158 OF 192



KEY NOTES: (X)

1. SEE VEOLIA REFERENCE DRAWINGS FOR FURTHER INFORMATION.
2. VENT LINES ROUTED TO BUILDING EXTERIOR.

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/11/2026 ARIZONA, U.S.A.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/11/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD J. STILLMAN DRAWN: R.D. CLARK TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 062 Y612	TITLE OF SHEET WATER TREATMENT SODIUM BISULFITE & CITRIC ACID SYSTEMS P&ID NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 159 OF 192

4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 - NPS GRCA N. RIM UTILITIES\PID DWG\062Y612.dwg

HMI

WEIGHT
WI
67491

HIGH LEVEL
LAH
67451

NR WTP CHEMICAL RIO

WI
67491

LSH
67451

MCC/CP

I/I

X2:
LSH-67451
LSH-67551

PROCESS

VENT (2)

2"-HCL-VT

RIGID

FLEX TUBING

1/2"-HCL

FLEX TUBING

RIGID

120V

WIT
67491

WEIGHT SCALE
55 GALLON DRUM
SCL-67411

(1)
CIP PUMP PANEL
(SUPPLIED &
CONTROLLED BY
MFR)

120V

LSH
67451

CONTAINMENT SUMP

HYDROCHLORIC ACID FEED SYSTEM

1/2"-HCL

HCL
G
062Y604
TO UF MEMBRANE
MASTER SKID

LSH
67551

CONTAINMENT SUMP

CORROSION INHIBITOR SYSTEM

CHEMICAL FEED PUMP SYSTEM FOR CORROSION INHIBITOR IS NOT INCLUDED IN PROJECT. CONTAINMENT SUMP IS INCLUDED FOR FUTURE USE.

KEYNOTES: (X)

1. SEE VEOLIA REFERENCE DRAWINGS FOR FURTHER INFORMATION.
2. VENT LINES ROUTED TO BUILDING EXTERIOR.

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 4/27/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD J. STILLMAN CADD R.D. CLARK TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 062 Y613	TITLE OF SHEET **WATER TREATMENT** HYDROCHLORIC ACID & CORROSION INHIBITOR SYSTEMS P&ID NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 160 OF 192

4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 – NPS GRCA N. RIM UTILITIES\PID DWG\062Y613.dwg

HMI

NR WTP PLC

MCC/CP

PROCESS

HIGH HIGH LEVEL
LAHH
69151

LEVEL
LI
69151

LSHH
69151

LI
69151

ISR
1

ISB
1

LSHH
69151

LT
69151

SAN
- | -
4"-SAN
DOMESTIC WASTE
SEE P SERIES DRAWINGS

SLG
A | 062Y603
4"-SLG
FROM REACTION TANKS

DR
B | 062Y604
6"-DR
FROM ULTRAFILTRATION BACKWASH & CIP

W1-D
B | 062Y605
4"-W1-D
FROM CT TANK DRAIN

- | -
6"
FLOOR DRAINS
SEE P SERIES DRAWINGS

PROCESS WASTE TANK
TK-69111

6"-SAN

SANITARY MANHOLE

6"-SAN
SAN
- | -
TO NEW MANHOLE
SEE C SERIES DRAWINGS

KEY NOTES: (X)

1. INTRINSICALLY SAFE BARRIERS AND RELAYS TO BE INSTALLED WHERE THIS SIGNAL WIRES ENTER THE WTP BUILDING.

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/11/2026 ARIZONA, U.S.A.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/11/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD J. STILLMAN CADD R.D. CLARK TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 062 Y614	TITLE OF SHEET WATER TREATMENT PROCESS WASTE TANK P&ID NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 161 OF 192

4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 - NPS GRCA N. RIM UTILITIES\PID DWG\062Y614.dwg

GENERAL NOTES:

A. REFER TO MECHANICAL AND PLUMBING PLAN SHEETS FOR LOCATION OF DEVICE.

KEYNOTES: (X)

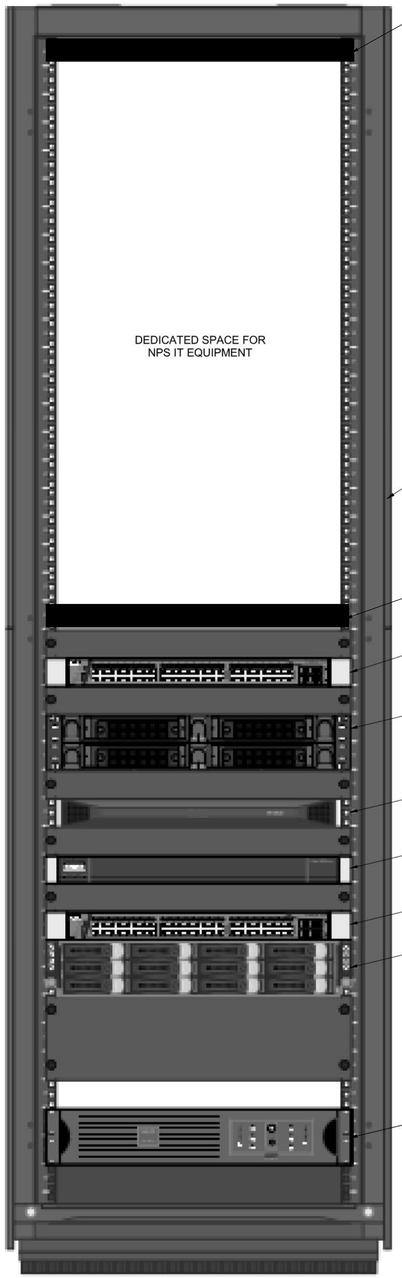
1. SMOKE DETECTORS ARE FOR ASSET PROTECTION ONLY. NOT FOR FIRE & LIFE SAFETY. CONTRACTOR TO PROVIDE 120VAC DUCT TYPE SMOKE DETECTOR.

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 4/27/26 ARIZONA, U.S.A.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 4/27/26 ARIZONA, U.S.A.

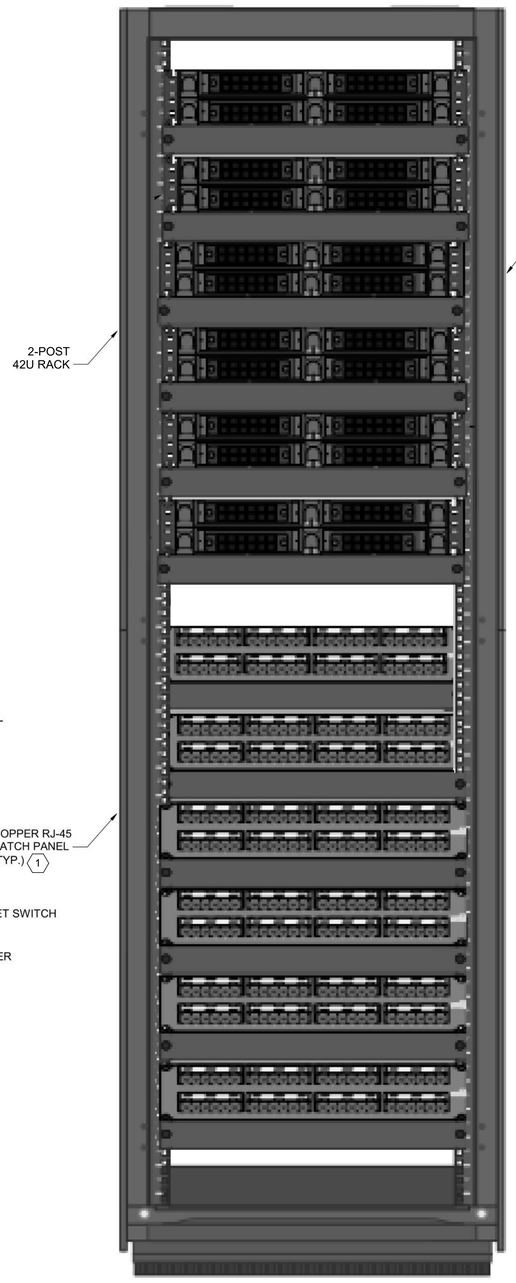
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD J. STILLMAN CADD R.D. CLARK TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 062 Y615	TITLE OF SHEET WATER TREATMENT MISCELLANEOUS SYSTEMS P&ID NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 162 OF 192	

4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 - NPS GRCA N. RIM UTILITIES\PID DWG\062Y615.dwg



A 062-Y620 **NORTH RIM WTP NETWORK RACK**

NO SCALE



B 062-Y620 **NORTH RIM WTP CAT 6/FIBER PATCH PANEL(S) RACK**

NO SCALE

KEYNOTES X

1. QUANTITY OF FIBER OPTIC PATCH PANELS AND COPPER RJ45 PATCH PANELS QUANTITY AS REQUIRED.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 04/Feb/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD CADD R.D. CLARK TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 062 Y620	TITLE OF SHEET WATER TREATMENT WTP NETWORK RACK NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 163 OF 192

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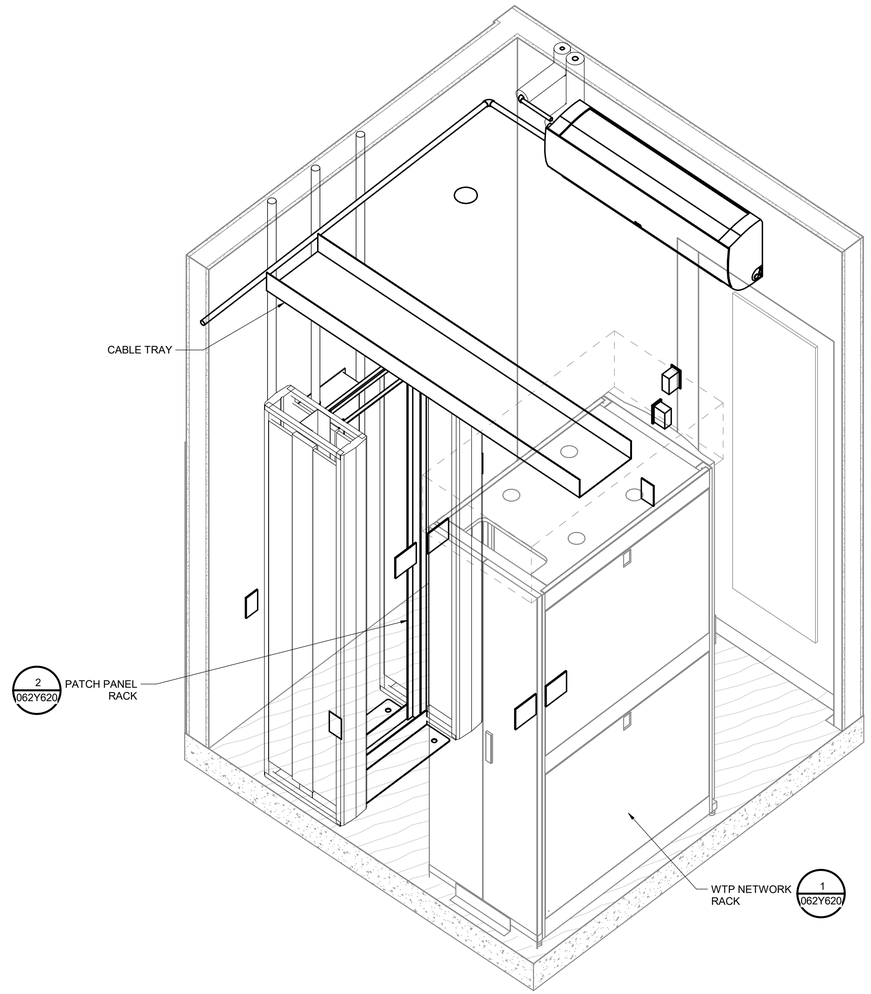

Registered Professional Engineer (Electrical)
CERTIFICATE NO.
85305
JOSEPHINE A
BRAINARD
Date Signed 4/1/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD CADD S. BILD TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 062 Y901	TITLE OF SHEET WATER TREATMENT SERVER ROOM ISOMETRIC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 164 OF 192

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GENERAL NOTES

A. REFER TO SHEETS 062S001 FOR GENERAL STRUCTURAL NOTES.

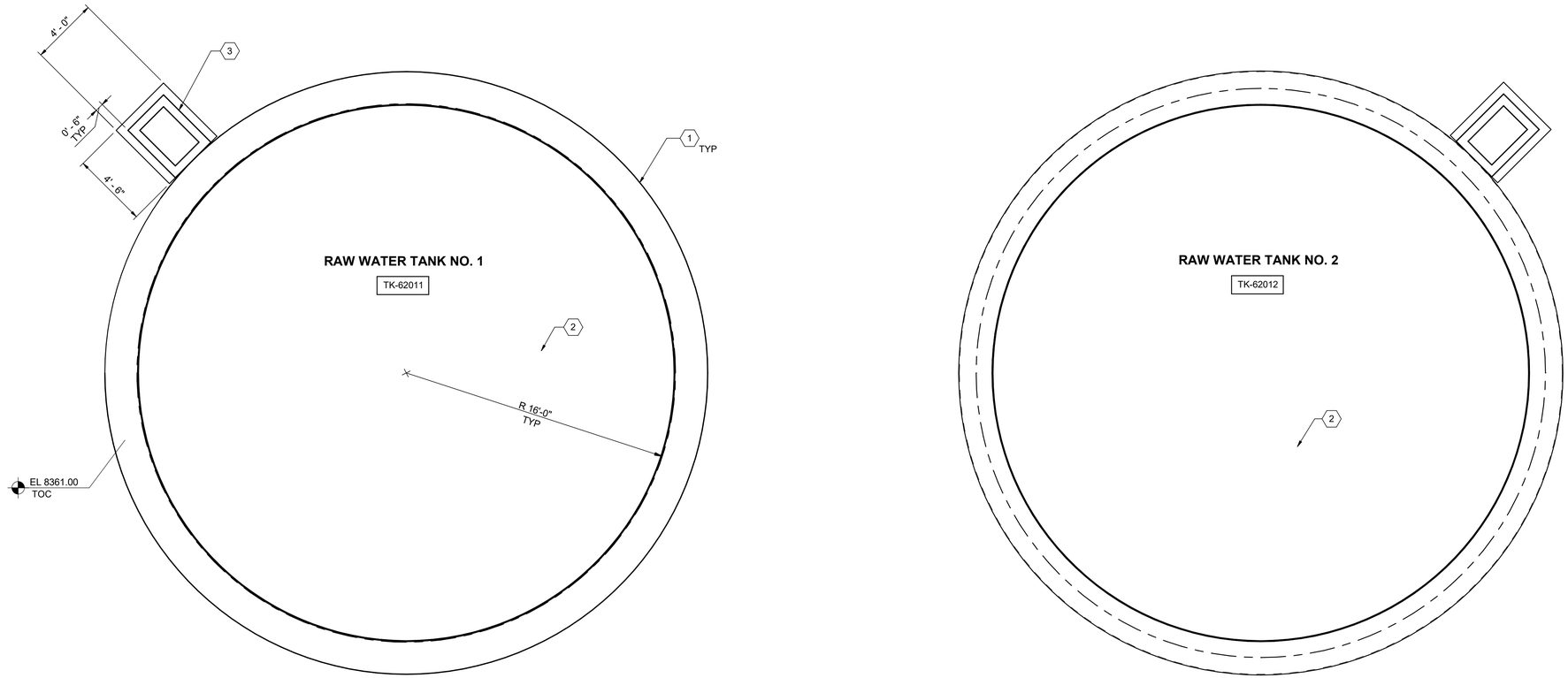
B. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING, EQUIPMENT, ANCHORS, CONDUITS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTURAL ELEMENTS THAT MAY NOT BE SHOWN.

KEYNOTES

1 CONCRETE RINGWALL.

2 2" MINIMUM SAND CUSHION ON BACKFILL AND COMPACTED SUBGRADE.

3 PRECAST BELOW GRADE SUMP



1 / 063S101 RAW WATER TANK UNDERSLAB FLOOR PLAN
SCALE (A)

SCALE (A) 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Corp) CERTIFICATE NO. 72480 COLIN MULDERICK State of Arizona

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 063 S101	TITLE OF SHEET RAW WATER TANKS UNDERSLAB FLOOR PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 165 OF 192

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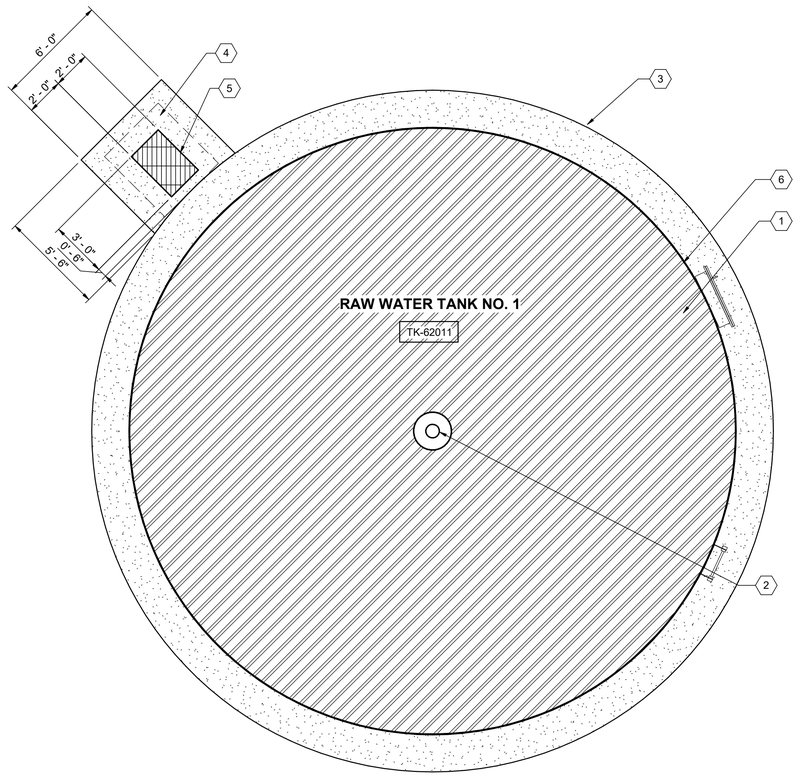

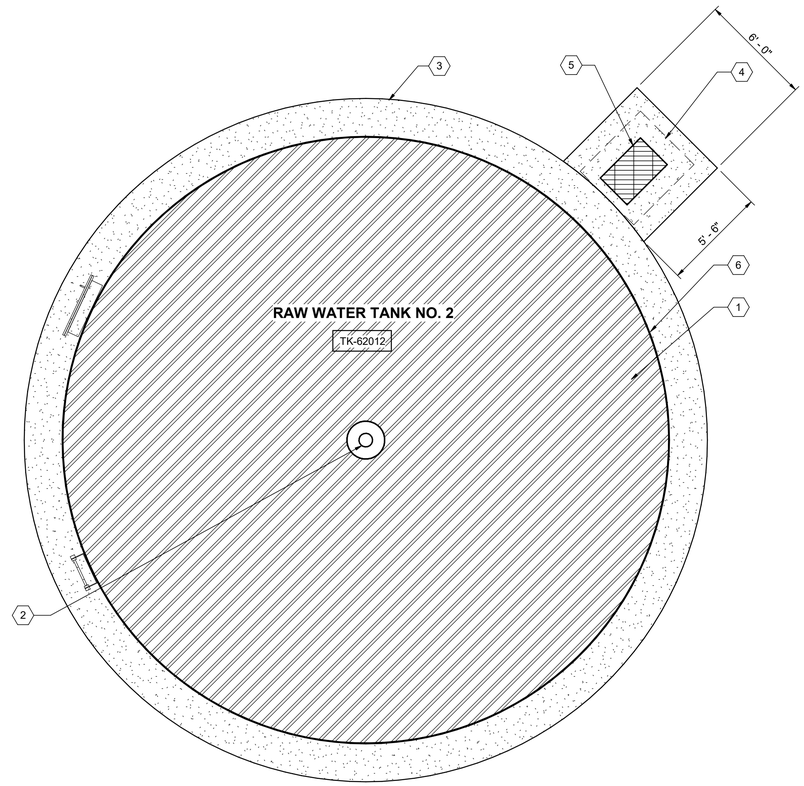

1 063S102 RAW WATER TANK ABOVE SLAB FLOOR PLAN
SCALE (A)

SCALE (A) 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

GENERAL NOTES

A. REFER TO SHEETS 062S001 FOR GENERAL STRUCTURAL NOTES.

B. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING, EQUIPMENT, ANCHORS, CONDUITS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTURAL ELEMENTS THAT MAY NOT BE SHOWN.

KEYNOTES

1 TANK STEEL FLOOR PLATE BY RAW WATER TANK SUPPLIER.

2 INTERIOR SUPPORT COLUMN BY RAW WATER TANK SUPPLIER.

3 CONCRETE RINGWALL.

4 PRECAST BELOW GRADE SUMP

5 SUMP PIT GRATING. SEE VOLUME 00 TYPICAL DETAILS FOR GRATING SUPPORT DETAILS.

6 STEEL PLATE TANK SHELL BY RAW WATER TANK SUPPLIER.

Registered Professional Engineer (Not) CERTIFICATE NO. 72480 COLIN MULDERICK Data Support Solutions ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. MULDERICK DRAWN: R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	SUB SHEET NO. 063 S102	TITLE OF SHEET RAW WATER TANKS ABOVE SLAB FLOOR PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 166 OF 192

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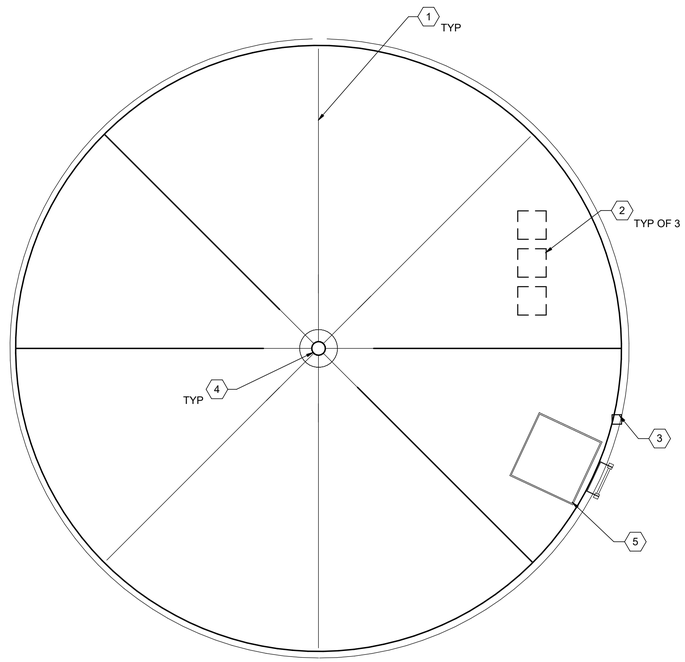

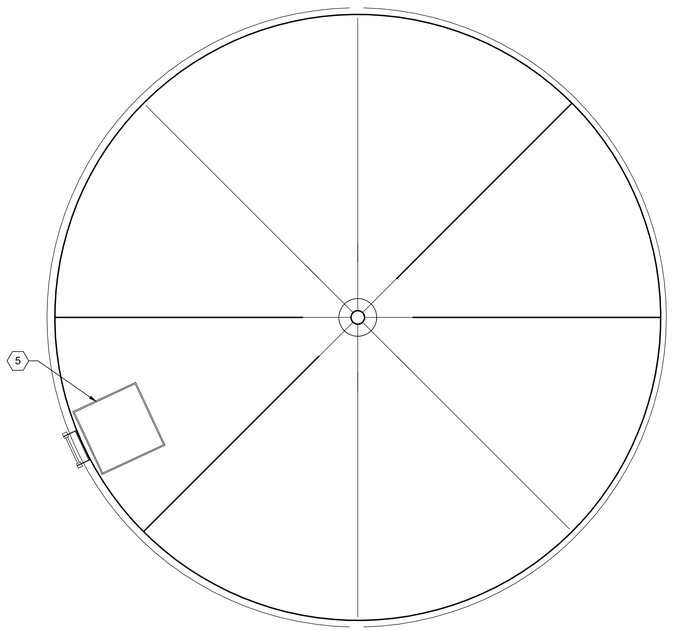

1 063S103 RAW WATER TANK DOME ROOF PLAN

SCALE (A)

GENERAL NOTES

A. REFER TO SHEETS 062S001 FOR GENERAL STRUCTURAL NOTES.

B. COORDINATE THE STRUCTURAL DRAWINGS WITH ALL OTHER DISCIPLINE DRAWINGS FOR PIPING, EQUIPMENT, ANCHORS, CONDUITS, AND ANY OTHER ITEMS ATTACHED TO OR EMBEDDED IN STRUCTURAL ELEMENTS THAT MAY NOT BE SHOWN.

KEYNOTES

1 TANK ROOF RAFTERS BY RAW WATER TANK SUPPLIER.

2 ANTENNA MAST BASE SUPPORTS.

3 PROVIDE 5,000-LBS RATE EYEBOLT ATATCHED TO LANDING FRAMING.

4 INTERIOR SUPPORT COLUMN BY RAW WATER TANK SUPPLIER.

5 ACCESS PLATFORM.

SCALE (A) 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Corp)
CERTIFICATE NO.
72480
COLIN MULDERICK
Date Signed: 05/01/2026
ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	C. MULDERICK CADD R. NELSON TECH. REVIEW: L. MENDOZA DATE: MAY 2026	063 S103	RAW WATER TANKS DOME ROOF PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 10350545 SHEET 167 OF 192

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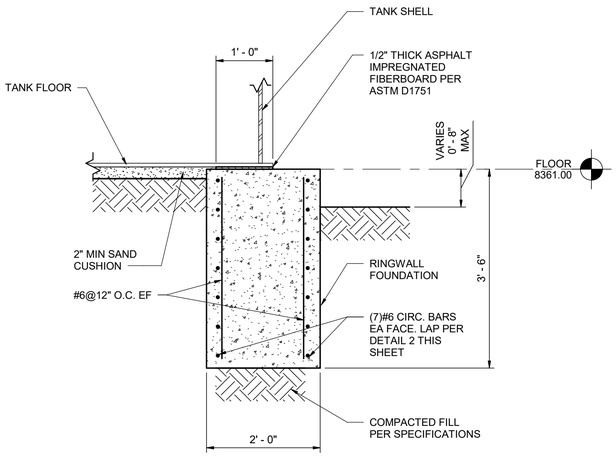
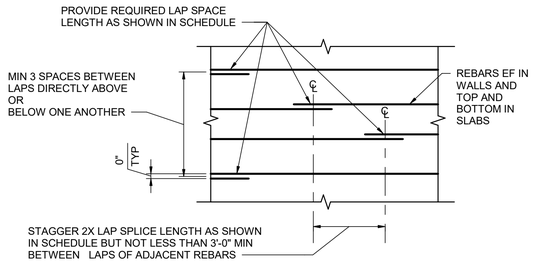


1 063S301 SECTION THRU RINGWALL FOUNDATION

SCALE (A)



NOTES:

1. LAP AND STAGGER ALL HORIZONTAL CIRCULAR WALL REINFORCING PER THIS DETAIL, INCLUDING ONE LAP STAGGER BETWEEN INSIDE FACE LAPS AND OUTSIDE FACE LAPS.

2 063S301 RINGWALL HORIZONTAL WALL REINFORCING STAGGERED LAP

NO SCALE

SCALE (A) 1 0 1 2 3 3/4" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Corp) CERTIFICATE NO. 72480 COLIN MULDERICK State Of Arizona

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	C. MULDERICK	063 S301	RAW WATER TANKS STRUCTURAL SECTIONS	113 182752
	CADD R. NELSON			PMIS/PKG NO. 10350545
	TECH. REVIEW: L. MENDOZA			SHEET 168 OF 192
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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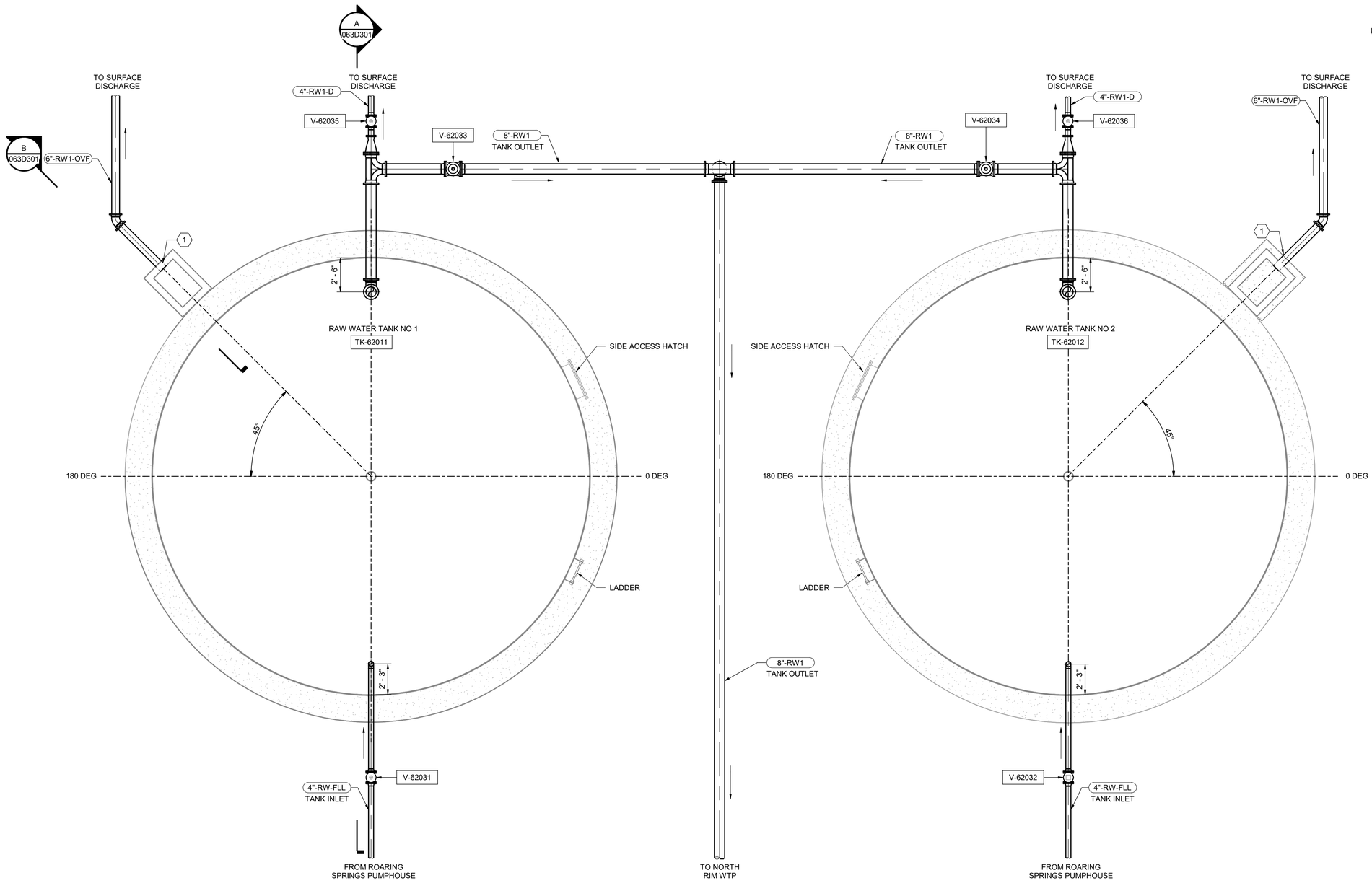
GENERAL NOTES:

A. SEE CIVIL SHEETS FOR CONTINUATION OF PIPING.

KEYNOTES:

1. FOR PIPE WALL PENETRATIONS. SEE SHEET 005D505.



063 D101 RAW WATER TANK UNDERSLAB PIPING PLAN

1/4" = 1'-0"

SCALE A 4 0 4 8 1/4" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: J. STUJENSKE TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 063 D101	TITLE OF SHEET RAW WATER TANKS UNDERSLAB PIPING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 169 OF 192

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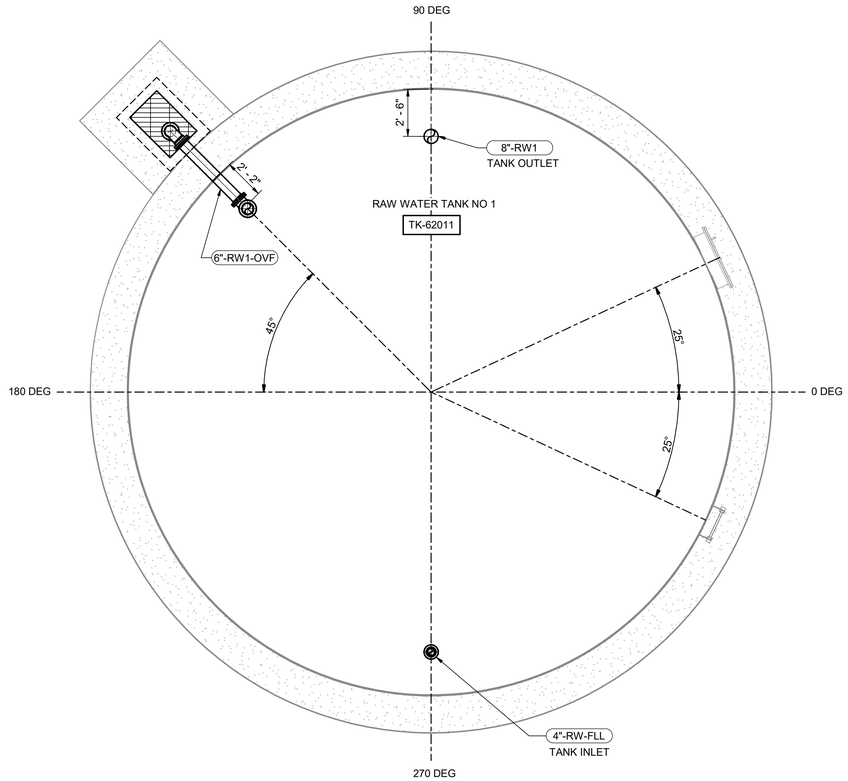

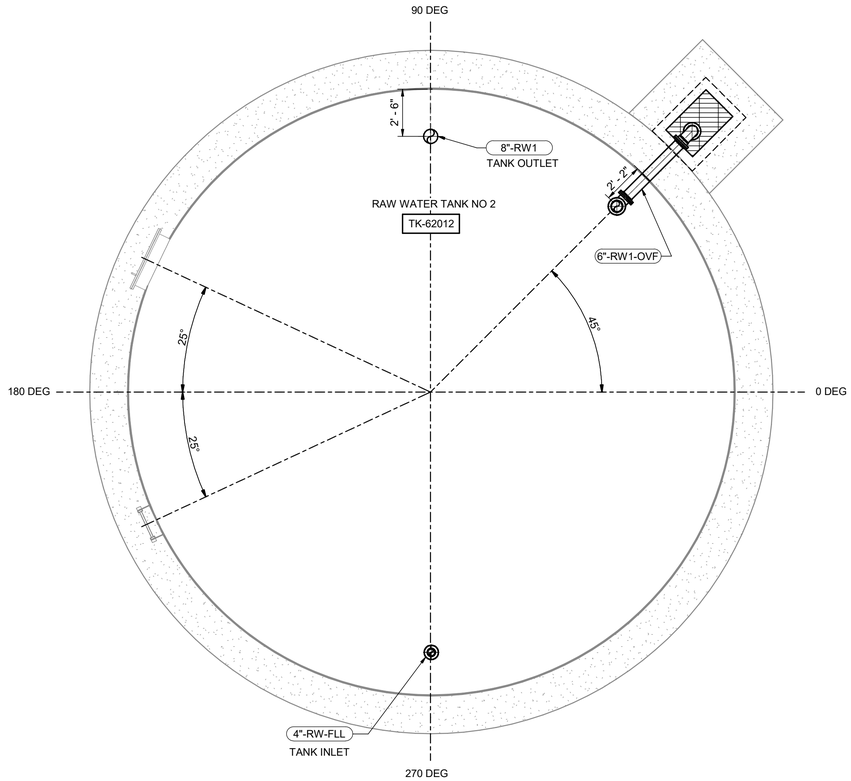

063 / D102 RAW WATER TANK ABOVE SLAB PIPING PLAN

1/4" = 1'-0"

SCALE (A) 4 0 4 8 1/4" = 1'-0"

SCALE OF FEET

Registered Professional Engineer (Corp) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD CADD J. STUJENSKE TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 063 D102	TITLE OF SHEET RAW WATER TANKS ABOVE SLAB PIPING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 170 OF 192

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_063_RWT_U.rvt
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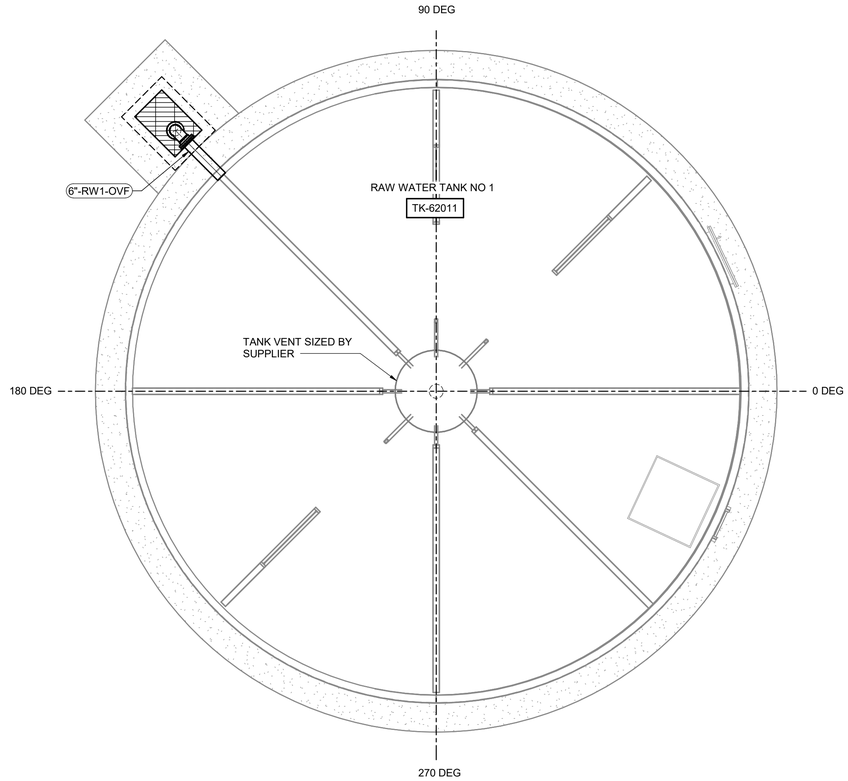
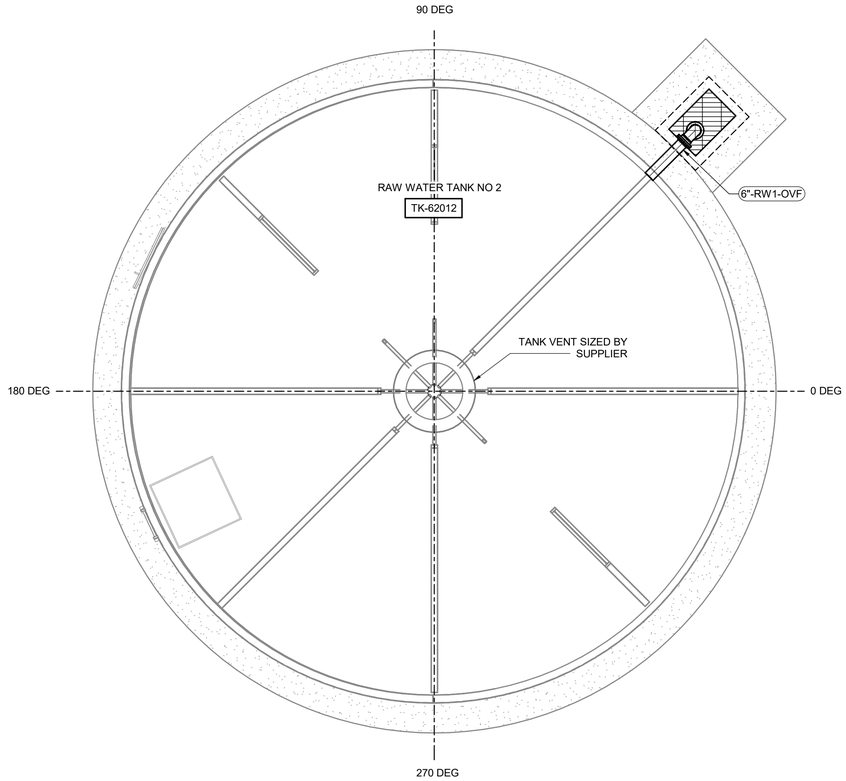
GENERAL NOTES:

A. PROVIDE ROOF PENETRATIONS AND ACCESS TO LEVEL INSTRUMENTS. SEE ELECTRICAL SHEETS AND SHEET 062Y601.





063 D103 RAW WATER TANK DOME ROOF PLAN
1/4" = 1'-0"

SCALE A 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

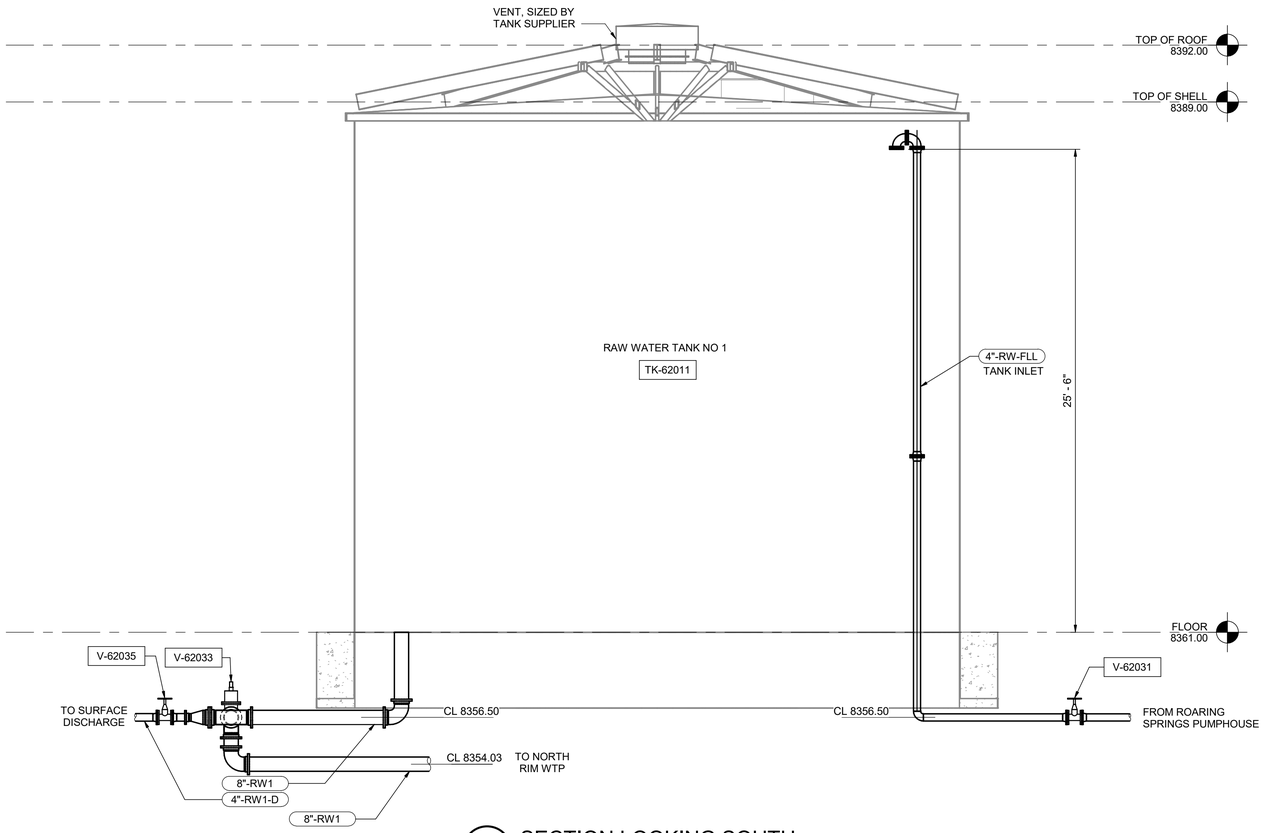
Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: J. STUJENSKE TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 063 D103	TITLE OF SHEET RAW WATER TANKS DOME ROOF PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 171 OF 192

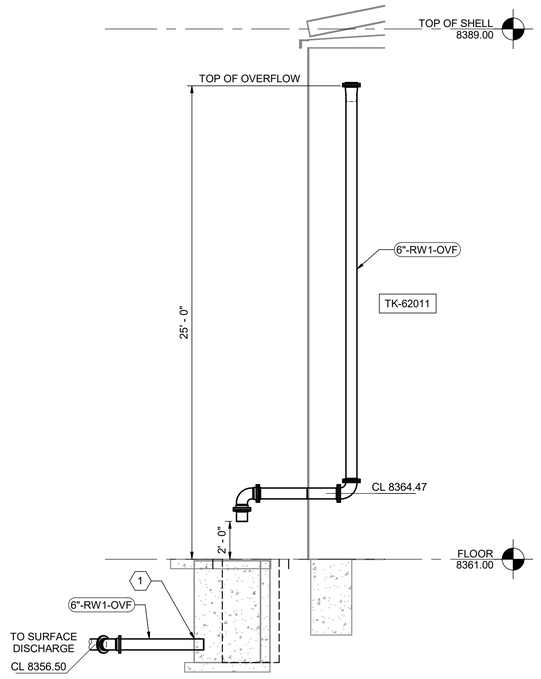
Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_063_RWT_U.rvt
4/28/2026 8:05:02 AM

KEYNOTES:

1. FOR PIPE WALL PENETRATIONS. SEE SHEET 005D505.



A 063D101 SECTION LOOKING SOUTH
1/4" = 1'-0"



B 063D101 OVERFLOW SECTION
1/4" = 1'-0"

SCALE A 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 6/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: J. STUJENSKE TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 063 D301	TITLE OF SHEET RAW WATER TANKS SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 172 OF 192

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_063_RWT_U.rvt
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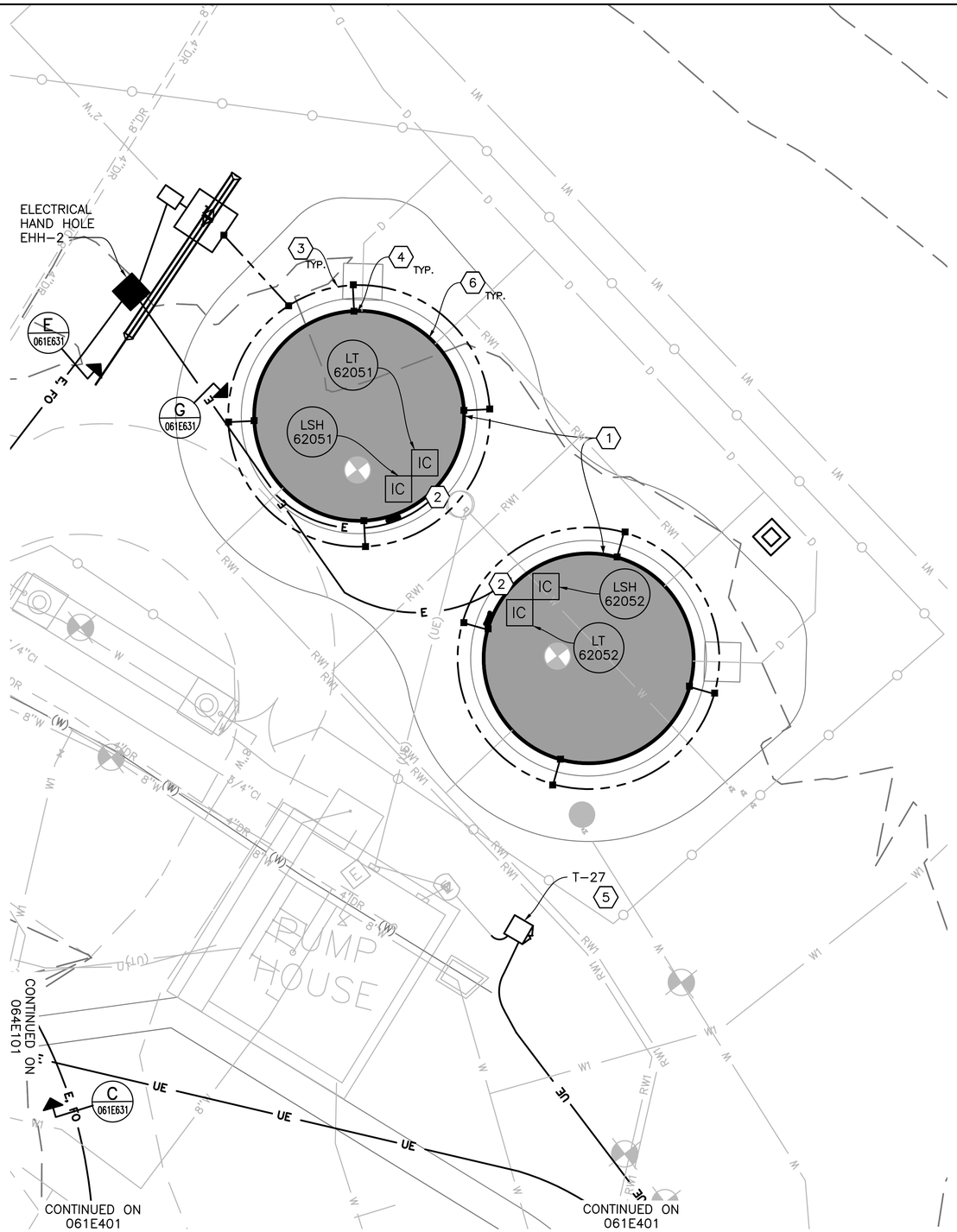


GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. ELECTRICAL DEMO IS SHOWN ON SHEET 061X101.

KEY NOTES: (X)

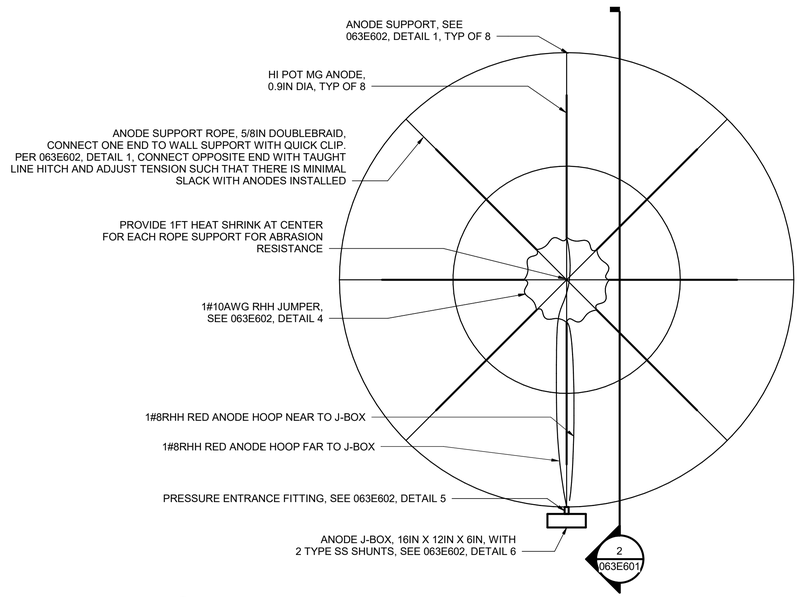
1. RAW WATER TANKS.
2. PROVIDE NEMA 4X JUNCTION BOX ON UNISTRUT SUPPORT STAND AT THIS LOCATION FOR INSTRUMENTATION CONNECTIONS.
3. #4/0AWG SOFT-DRAWN BARE COPPER GROUND RING.
4. #4/0 BOND TO TANKS.
5. REPLACE EXIST TRANSFORMER T-27 WITH 45 KVA, 24.9GY/14.4KV - 208GY/120V PADMOUNT TRANSFORMER. INSTALL EQUIPMENT VAULT AND PAD. SET APPROXIMATELY 35 FEET NORTH EAST OF EXIST LOCATION AND AT MINIMUM 5 FT AWAY FROM PUMP HOUSE BUILDING EXTENT. TERMINATE MEDIUM VOLTAGE POWER CABLE FROM ESB-18A WITH 200 AMP LOADBREAK ELBOWS. LOCATE AND RECONNECT EXIST SECONDARY SERVICE CONDUCTOR TO PUMP HOUSE. IF EXISTING SECONDARY SERVICE CONDUCTOR CANNOT BE REUSED, RECONNECT USING 4/C #4/0 ARMORED CABLE.
6. LIGHTNING PROTECTION REQUIRED ON NEW RAW WATER TANKS. REFER TO SPECIFICATION SECTION 26 41 13 FOR LIGHTNING PROTECTION REQUIREMENTS.

10 0 10 20
SCALE OF FEET
1" = 10'

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO. 58037
TRAVIS T. MOORE
Date Signed 1/4/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO DRAWN: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 063 E101	TITLE OF SHEET RAW WATER TANKS ABOVE SLAB FLOOR PLAN RW TANKS NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 173 OF 192

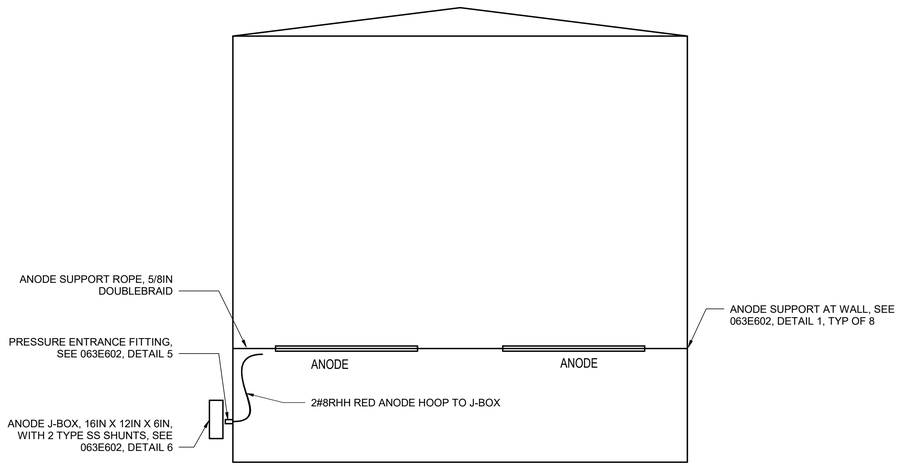
8/13/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10414450_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\ELEC\063-E101.dwg



1 063E601 RAW WATER TANK CATHODIC PROTECTION PLAN

NO SCALE

RAW WATER TANK MATERIALS SCHEDULE						
#	ITEM NAME	LOCATION	DETAIL	# REQUIRED	SIZE	DESCRIPTION
1	ANODE MOUNTING CLIP					3/8IN DIA MILD STEEL ROD
	A	OUTER WALL	1	8	3 5/8 IN	
2	QUICK LINK					5/16IN STEEL 2,000LB WLL
	A	OUTER WALL	1	8		
3	ROPE					
	A	WALL TO FAR WALL		8	32 FT	5/8IN DOUBLE BRADED POLYESTER RIGGING WHITE. LENGTH BETWEEN SUPPORTS ADD MIN 3FT EACH END FOR KNOTS. CONFIRIM ACTUAL TANK DIMENSIONS BEFORE CONSTRUCTING ROPE SUPPORTS
4	ANODE					
	A	OUTER		8	15 FT 1 IN 5.2 LB EACH ANODE, 41.3 LB TOTAL WEIGHT	GA-MG-R900 HI POTENTIAL MAGNESIUM EXTRUDED 0.9IN DIAMETER 0.516LB/FT WITH 0.135IN STEEL CORE, LENGTH OF ANODE MATERIAL PROVIDE 3 5/8IN EXPOSED STEEL CORE EACH END
5	WIRE					
	A	ANODE TO ANODE JUMPERS				
		OUTTER ANODE		16	3 FT	#10AWG RHH, STRANDED COPPER WITH BLACK INSULATION
		SPLIT BOLT		8		HIGH STRENGTH BRONZE ALLOY
	B	ANODE TO J-BOX				
		ANODE HOOP NEAR		1	25 FT	#10AWG RHH, STRANDED COPPER WITH RED INSULATION
		ANODE HOOP FAR		1	35 FT	
6	REFERENCE ELECTRODE		3	2		
7	SPLICE KIT			8		THERMO-SHRINK UNDERGROUND COPPER BUTT SPLICE KIT, 2 TO 8 AWG
8	PRESSURE ENTRANCE FITTING			2		
9	J-BOX	PER DETAIL	6	1		



2 063E601 RAW WATER TANK CATHODIC PROTECTION SECTION

NO SCALE

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 2/18/2026 ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. SMITH	063 E601	RAW WATER TANKS CATHODIC PROTECTION PLAN AND SECTION	113 182752
	CADD: A. S-DELGADO			PMIS/PKG NO. 10350545
	TECH. REVIEW: P. DAVILA			SHEET 174 OF 192
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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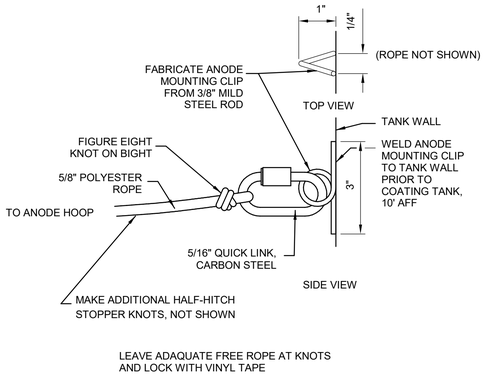
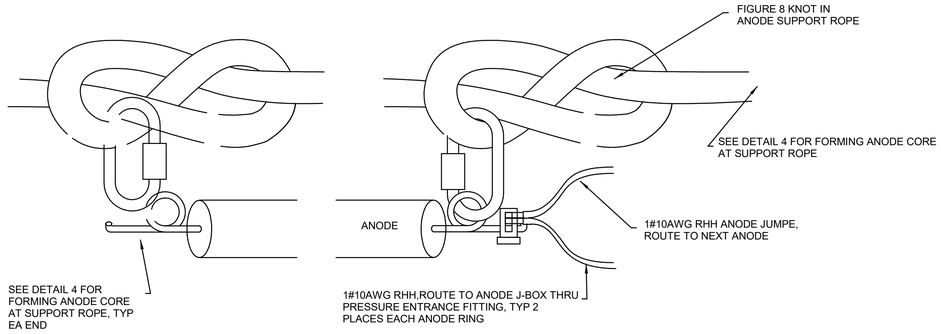
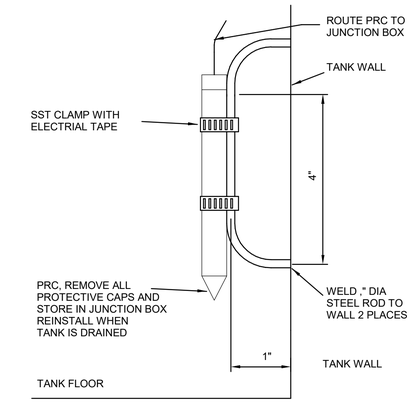


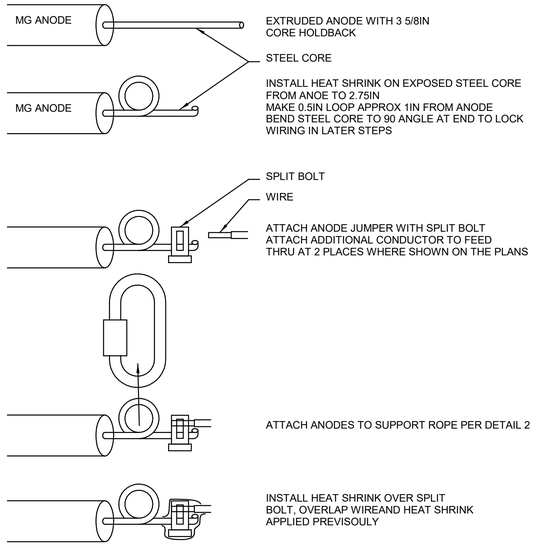
1 063E602 ANODE ROPE MOUNTING CLIP
NO SCALE



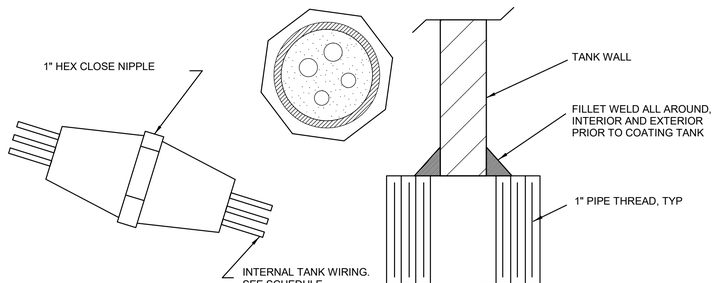
2 063E602 ANODE SUPPORT AT ANODE ROPE
NO SCALE



3 063E602 WALL MOUNT REFERENCE ELECTRODE
NO SCALE



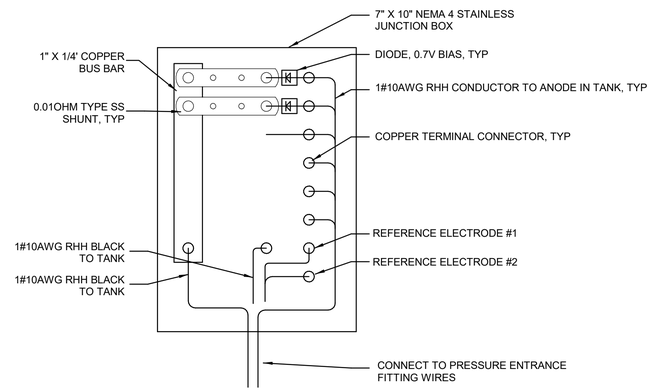
4 063E602 ANODE CORE LOOP
NO SCALE



NOTES:
INSTALL PIPE PLUGS BEFORE COATING, REMOVE AND INSTALL ENTRANCE FITTING/WIRES BEFORE FILLING TANK.
INSTALL & SECURELY TIGHTEN 1" PRESSURE FITTING INTO 1" STEEL COUPLING USING THREAD SEALANT TO RENDER A WATERTIGHT CONNECTION.
CONNECT TO INTERNAL WIRING WITH A SPLIT BOLT AND SPLICE KIT
MAKE CONNECTIONS TO EXTERNAL WIRING WITH WIRE NUTS IN ENCLOSURE

PRESSURE ENTRANCE FITTING SCHEDULE		
CONDUCTOR	CONNECTION	WIRE CLR
1	ANODE OUTTER	RED
2	ANODE INNER	RED
3	REFERENCE ELECTRODE WALL	YELLOW
4	REFERENCE ELECTRODE CENTER	ORANGE

5 063E602 PRESSURE ENTRANCE FITTING
NO SCALE



6 063E602 ANODE JUNCTION BOX
NO SCALE

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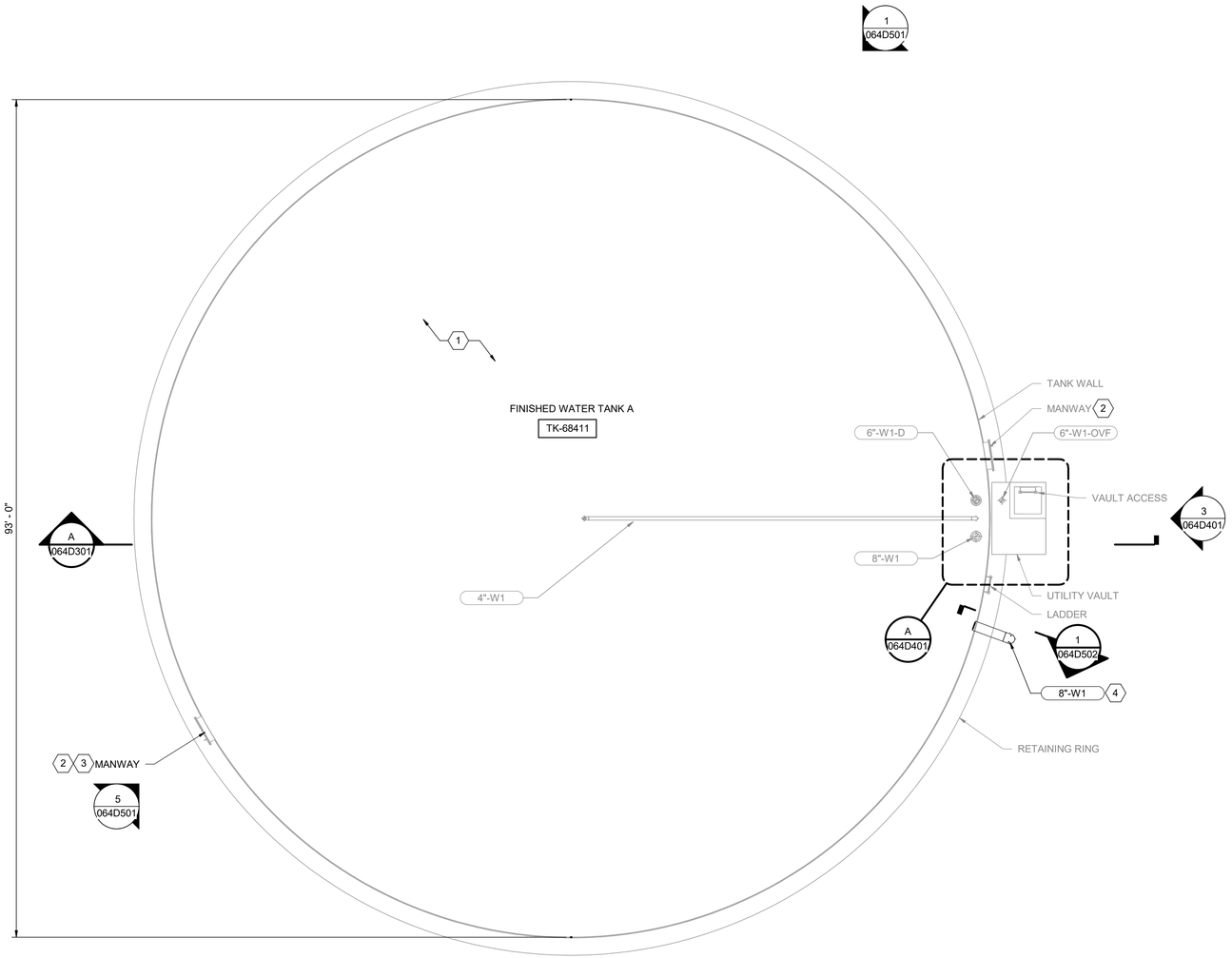
REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 2/13/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. SMITH DRAWN: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 063 E602	TITLE OF SHEET RAW WATER TANKS CATHODIC PROTECTION DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 175 OF 192



1 - TANK A FLOOR PLAN
1/8" = 1'-0"

SCALE A 8 0 8 16 1/8" = 1'-0"
SCALE OF FEET

GENERAL NOTES

A. REMOVE ONE TANK FROM SERVICE AT A TIME TO PERFORM ALL IMPROVEMENTS TO FW TANKS. SEE SHEETS 064D901 THROUGH 064D904 FOR SEQUENCING PLAN.

B. DRAIN TANK TO DAYLIGHT USING EXISTING DRAIN LINE. SEE 061C105 FOR DRAIN OUTLET LOCATION. PROVIDE DECHLORINATION ON TANK DISCHARGE AND COORDINATE WITH COR ON TIMING AND RATE OF DISCHARGE. AFTER TANK IS EMPTY, PROVIDE SPECIALIST TO PERFORM TANK CONDITION ASSESSMENT PER REQUIREMENTS OF SECTION 43 41 15.

C. COAT INTERIOR AND EXTERIOR OF TANK. SEE SECTION 43 41 15 FOR DETAILS. NOTE EXTERIOR ROOF COATING CONTAINS LEAD.

KEY NOTES (X)

1. REMOVE UP TO 6" OF SANDY SILT SEDIMENT FROM TANK FLOOR.
2. REPLACE SEAL ON 24" SIDE ACCESS MANWAY. SEE DETAIL 5/064D501 FOR EXISTING CONDITION.
3. LOCATION IS APPROXIMATE. VERIFY IN FIELD.
4. CUT IN NEW 8"-W1 OUTLET PIPE.

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D101	TITLE OF SHEET FINISHED WATER TANKS TANK A FLOOR PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 176 OF 192

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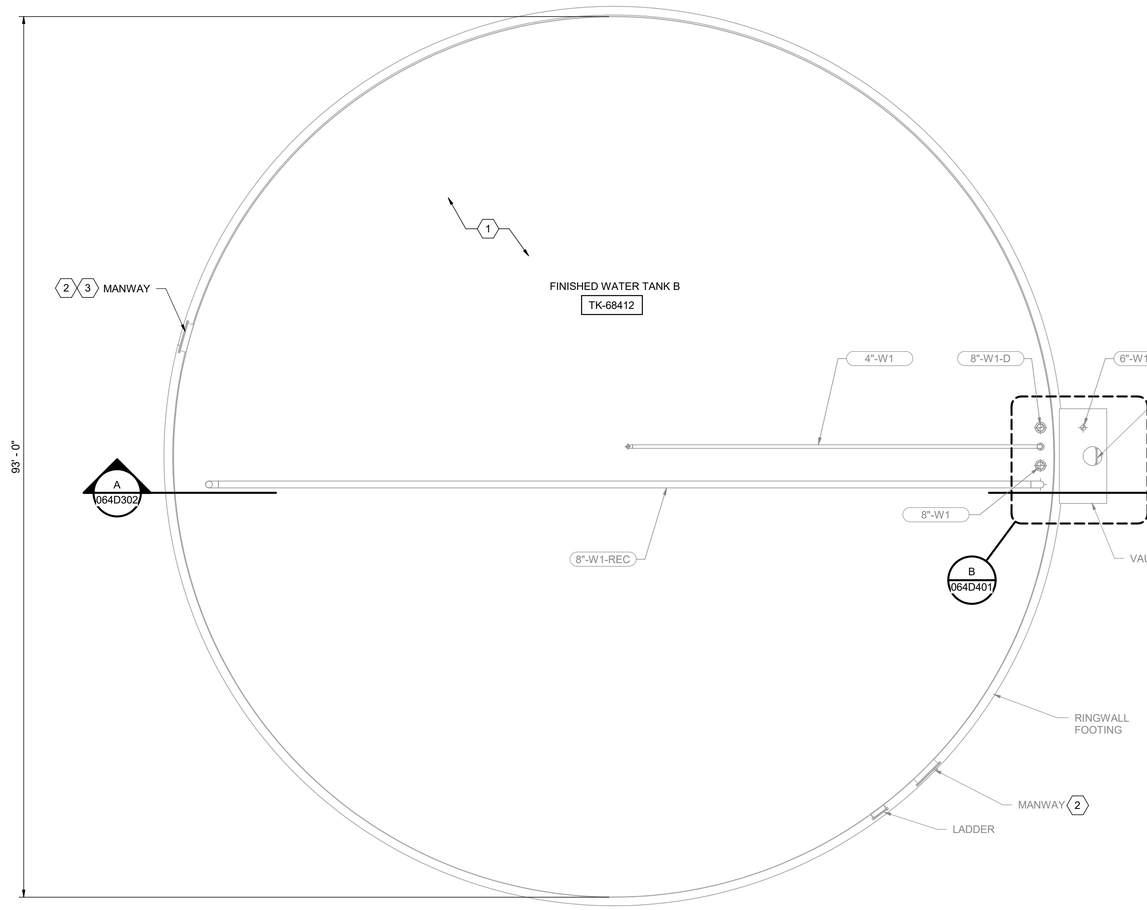

1 TANK B FLOOR PLAN
1/8" = 1'-0"

GENERAL NOTES

A. REMOVE ONE TANK FROM SERVICE AT A TIME TO PERFORM ALL IMPROVEMENTS TO FW TANKS. SEE SHEETS 064D901 THROUGH 064D904 FOR SEQUENCING PLAN.

B. DRAIN TANK TO DAYLIGHT USING EXISTING DRAIN LINE. SEE 061C105 FOR DRAIN OUTLET LOCATION. PROVIDE DECHLORINATION ON TANK DISCHARGE AND COORDINATE WITH COR ON TIMING AND RATE OF DISCHARGE. AFTER TANK IS EMPTY, PROVIDE SPECIALIST TO PERFORM TANK CONDITION ASSESSMENT PER REQUIREMENTS OF SECTION 43 41 15.

C. COAT INTERIOR AND EXTERIOR OF TANK. SEE SECTION 43 41 15 FOR DETAILS. NOTE EXTERIOR ROOF COATING CONTAINS LEAD.

KEY NOTES (X)

1. REMOVE UP TO 6" OF SANDY SILT SEDIMENT FROM TANK FLOOR.
2. REPLACE SEAL ON 24" SIDE ACCESS MANWAY. SEE DETAIL 5/064D501 FOR EXISTING CONDITION.
3. LOCATION IS APPROXIMATE. VERIFY IN FIELD.

SCALE (A) 8 0 8 16 1/8" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD CADD: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D102	TITLE OF SHEET FINISHED WATER TANKS TANK B FLOOR PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 177 OF 192

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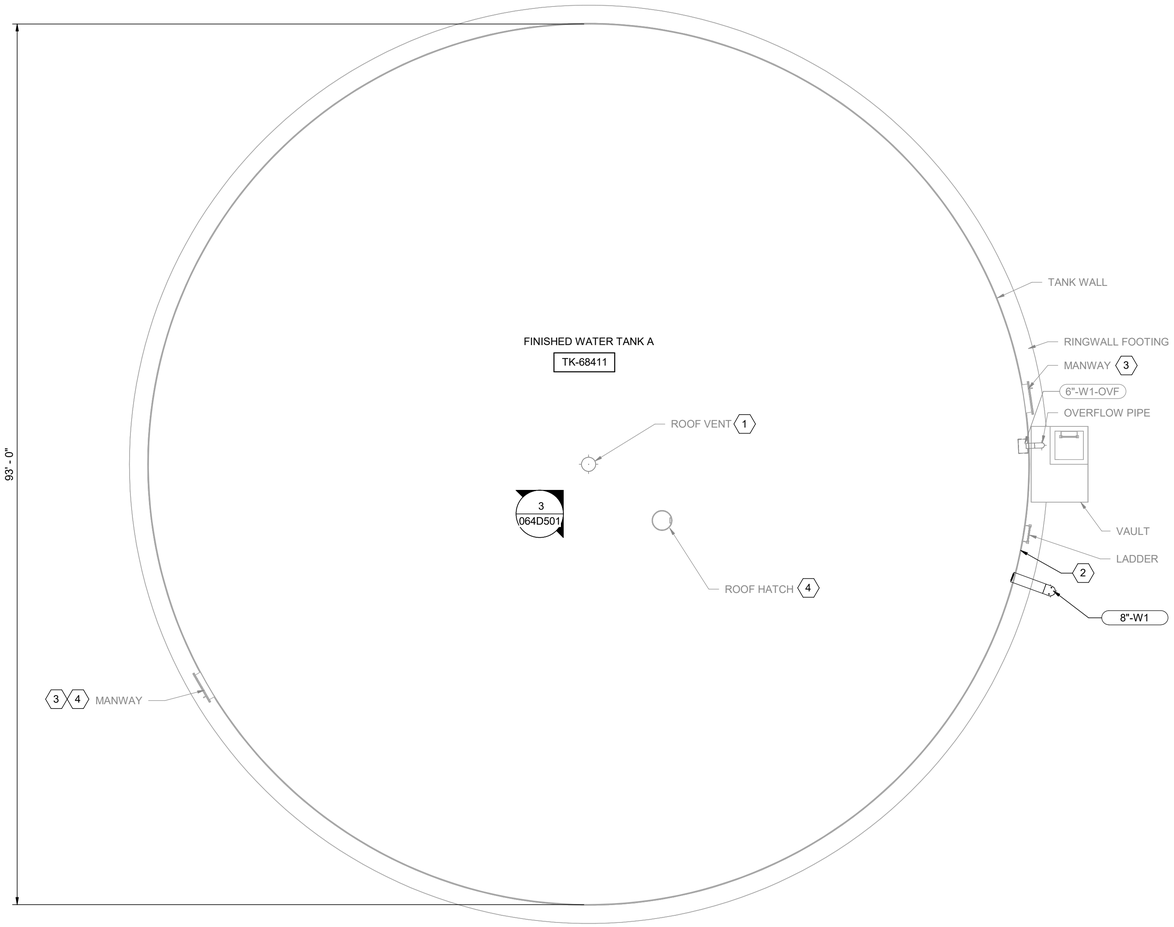

1 / - TANK A ROOF PLAN
1/8" = 1'-0"

GENERAL NOTES

A. REMOVE ONE TANK FROM SERVICE AT A TIME TO PERFORM ALL IMPROVEMENTS TO FW TANKS. SEE SHEETS 064D901 THROUGH 064D904 FOR SEQUENCING PLAN.

B. DRAIN TANK TO DAYLIGHT USING EXISTING DRAIN LINE. SEE 061C105 FOR DRAIN OUTLET LOCATION. PROVIDE DECHLORINATION ON TANK DISCHARGE AND COORDINATE WITH COR ON TIMING AND RATE OF DISCHARGE. AFTER TANK IS EMPTY, PROVIDE SPECIALIST TO PERFORM TANK CONDITION ASSESSMENT PER REQUIREMENTS OF SECTION 43 41 15.

C. COAT INTERIOR AND EXTERIOR OF TANK. SEE SECTION 43 41 15 FOR DETAILS. NOTE EXTERIOR ROOF COATING CONTAINS LEAD.

KEY NOTES (X)

1. REPLACE VENT. SEE DETAIL 3/064D501 FOR EXISTING VENT CONDITION. RAW WATER TANK SUPPLIER TO PROVIDE NEW VENT DESIGN. SEE SECTION 43 41 15 FOR NEW VENT REQUIREMENTS.
2. REPLACE EXTERNAL VISUAL LEVEL INDICATOR. SEE DETAIL 4/064D501 FOR EXISTING CONDITION. SEE SECTION 43 41 15 FOR NEW LEVEL INDICATOR REQUIREMENTS. COORDINATE LOCATION OF NEW INDICATOR WITH COR.
3. REPLACE SEAL ON 24" SIDE ACCESS MANWAY. SEE DETAIL 5/064D501 FOR EXISTING CONDITION.
4. LOCATION IS APPROXIMATE. VERIFY IN FIELD.

SCALE (A) 8 0 8 16 1/8" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D103	TITLE OF SHEET FINISHED WATER TANKS TANK A ROOF PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 178 OF 192

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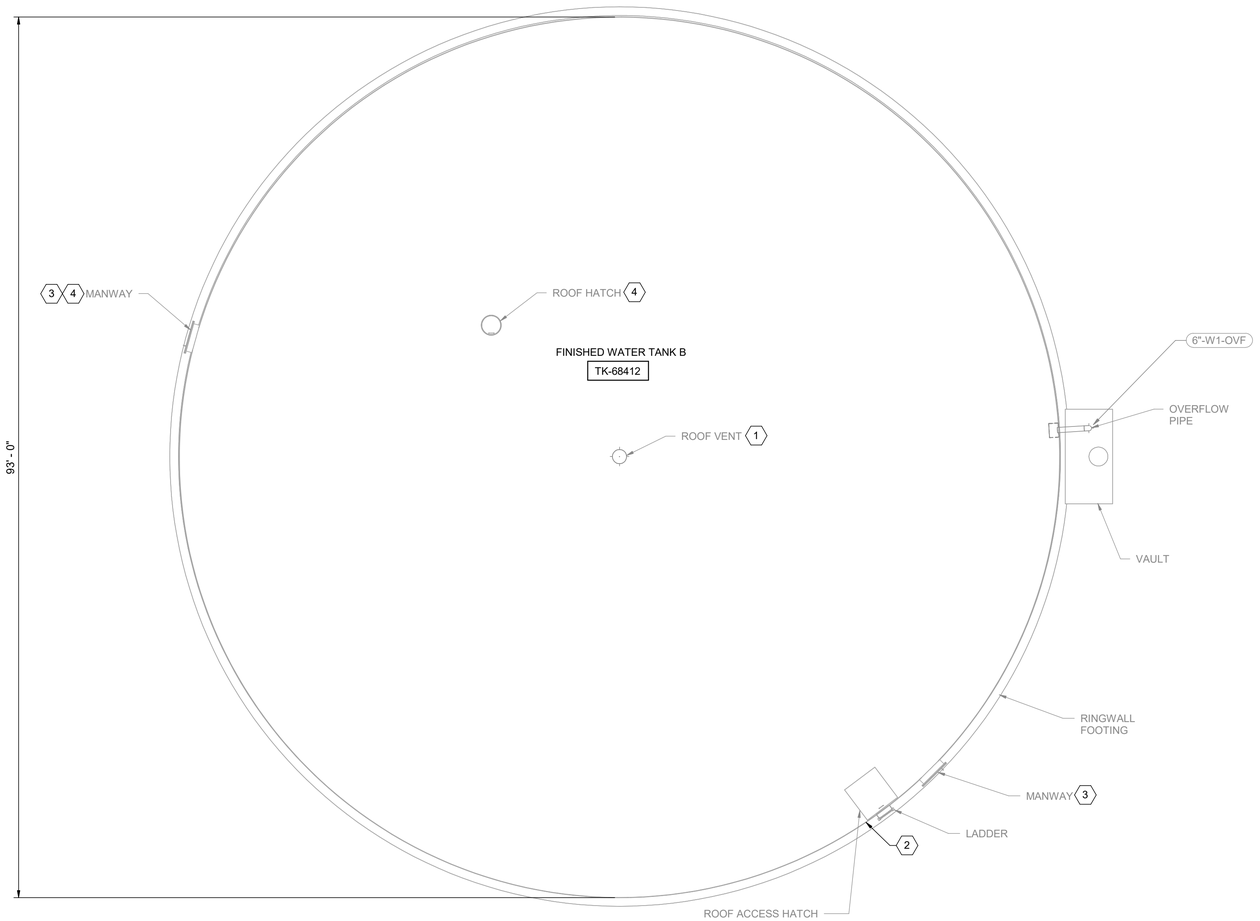

1 / - TANK B ROOF PLAN
1/8" = 1'-0"

GENERAL NOTES

A. REMOVE ONE TANK FROM SERVICE AT A TIME TO PERFORM ALL IMPROVEMENTS TO FW TANKS. SEE SHEETS 064D901 THROUGH 064D904 FOR SEQUENCING PLAN.

B. DRAIN TANK TO DAYLIGHT USING EXISTING DRAIN LINE. SEE 061C105 FOR DRAIN OUTLET LOCATION. PROVIDE DECHLORINATION ON TANK DISCHARGE AND COORDINATE WITH COR ON TIMING AND RATE OF DISCHARGE. AFTER TANK IS EMPTY, PROVIDE SPECIALIST TO PERFORM TANK CONDITION ASSESSMENT PER REQUIREMENTS OF SECTION 43 41 15.

C. COAT INTERIOR AND EXTERIOR OF TANK. SEE SECTION 43 41 15 FOR DETAILS. NOTE EXTERIOR ROOF COATING CONTAINS LEAD.

KEY NOTES (X)

1. REPLACE VENT. SEE DETAIL 3/064D501 FOR EXISTING VENT CONDITION. RAW WATER TANK SUPPLIER TO PROVIDE NEW VENT DESIGN. SEE SECTION 43 41 15 FOR NEW VENT REQUIREMENTS.
2. REPLACE EXTERNAL VISUAL LEVEL INDICATOR. SEE DETAIL 4/064D501 FOR EXISTING CONDITION. SEE SECTION 43 41 15 FOR NEW LEVEL INDICATOR REQUIREMENTS. COORDINATE LOCATION OF NEW INDICATOR WITH COR.
3. REPLACE SEAL ON 24" SIDE ACCESS MANWAY. SEE DETAIL 5/064D501 FOR EXISTING CONDITION.
4. LOCATION IS APPROXIMATE. VERIFY IN FIELD.

SCALE (A) 8 0 8 16 1/8" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D104	TITLE OF SHEET FINISHED WATER TANKS TANK B ROOF PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 179 OF 192

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GENERAL NOTES

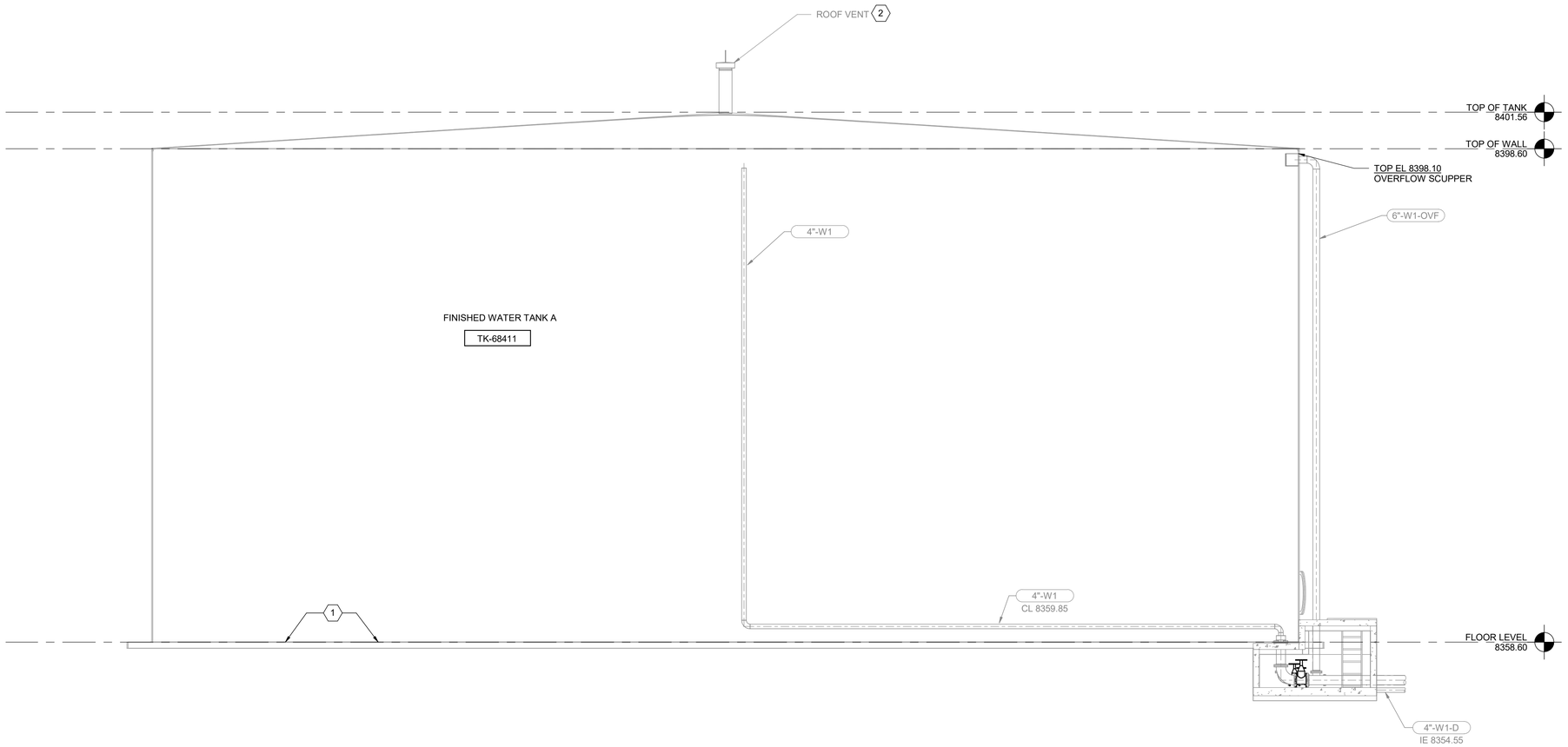
A. REMOVE ONE TANK FROM SERVICE AT A TIME TO PERFORM ALL IMPROVEMENTS TO FW TANKS. SEE SHEETS 061D601 THROUGH 064D605 FOR SEQUENCING PLAN.

B. DRAIN TANK TO DAYLIGHT USING EXISTING DRAIN LINE. SEE 061C105 FOR DRAIN OUTLET LOCATION. PROVIDE DECHLORINATION ON TANK DISCHARGE AND COORDINATE WITH COR ON TIMING AND RATE OF DISCHARGE. AFTER TANK IS EMPTY, PROVIDE SPECIALIST TO PERFORM TANK CONDITION ASSESSMENT PER REQUIREMENTS OF SECTION 43 41 15.

C. COAT INTERIOR AND EXTERIOR OF TANK. SEE SECTION 43 41 15 FOR DETAILS. NOTE EXTERIOR ROOF COATING CONTAINS LEAD.

KEY NOTES (X)

1. REMOVE UP TO 6" OF SEDIMENT FROM TANK FLOOR.
2. REPLACE VENT. SEE DETAIL 3/064D501 FOR EXISTING VENT CONDITION. RAW WATER TANK SUPPLIER TO PROVIDE NEW VENT DESIGN. SEE SECTION 43 41 15 FOR NEW VENT REQUIREMENTS.
3. SEE DETAIL 1/064D502.



A 064D101 TANK A SECTION
3/16" = 1'-0"

SCALE (B) 4 0 4 8 1/4" = 1'-0"
SCALE OF FEET

SCALE (A) 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D301	TITLE OF SHEET FINISHED WATER TANKS TANK A SECTION NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 / 182752
PMIS/PKG NO. 318719
SHEET 180 OF 192

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GENERAL NOTES

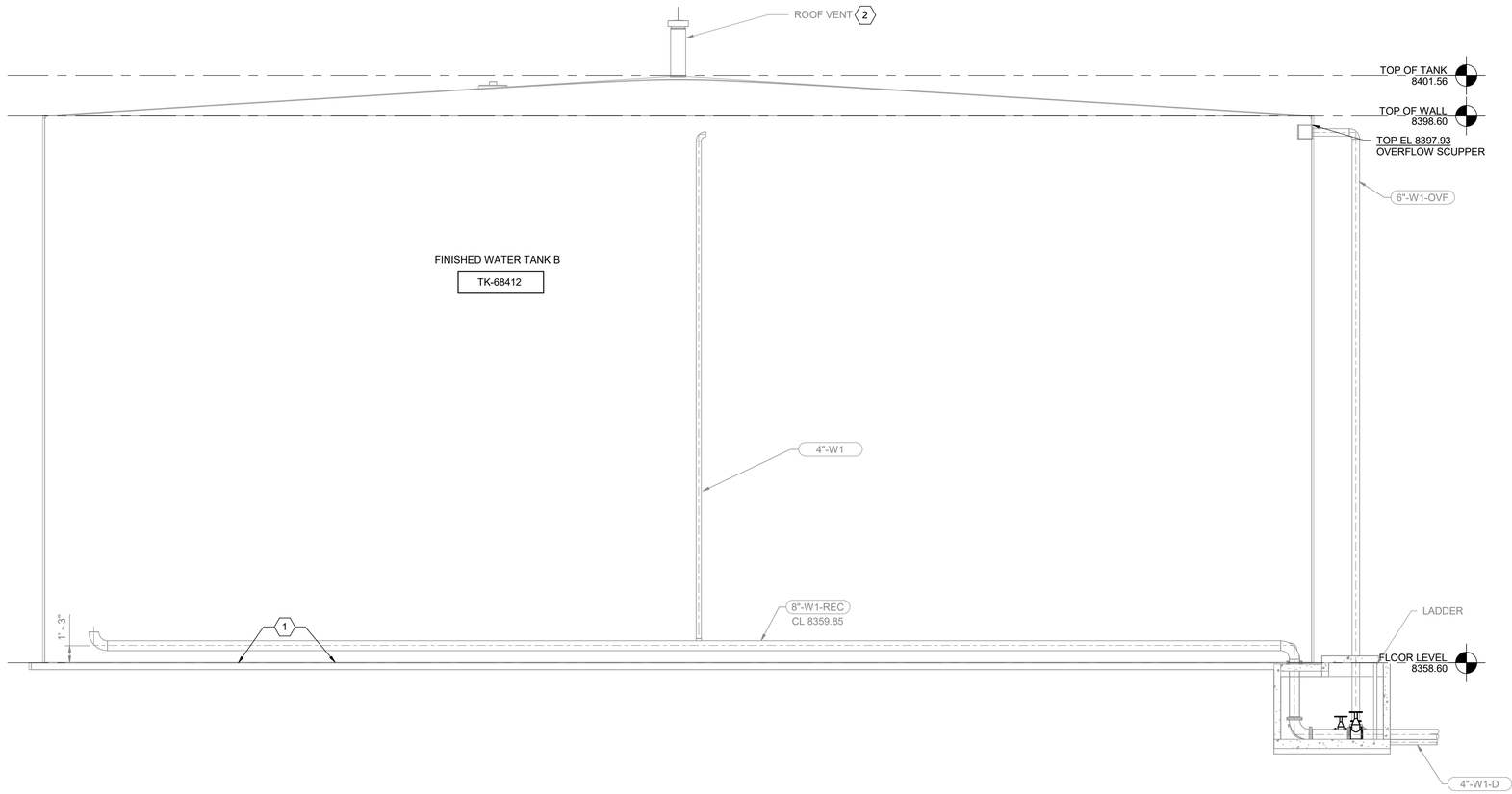
A. REMOVE ONE TANK FROM SERVICE AT A TIME TO PERFORM ALL IMPROVEMENTS TO FW TANKS. SEE SHEETS 064D901 THROUGH 064D904 FOR SEQUENCING PLAN.

B. DRAIN TANK TO DAYLIGHT USING EXISTING DRAIN LINE. SEE 061C105 FOR DRAIN OUTLET LOCATION. PROVIDE DECHLORINATION ON TANK DISCHARGE AND COORDINATE WITH COR ON TIMING AND RATE OF DISCHARGE. AFTER TANK IS EMPTY, PROVIDE SPECIALIST TO PERFORM TANK CONDITION ASSESSMENT PER REQUIREMENTS OF SECTION 43 41 15.

C. COAT INTERIOR AND EXTERIOR OF TANK. SEE SECTION 43 41 15 FOR DETAILS. NOTE EXTERIOR ROOF COATING CONTAINS LEAD.

KEY NOTES (X)

1. REMOVE UP TO 6" OF SEDIMENT FROM TANK FLOOR.
2. REPLACE VENT. SEE DETAIL 3/064D501 FOR EXISTING VENT CONDITION. RAW WATER TANK SUPPLIER TO PROVIDE NEW VENT DESIGN. SEE SECTION 43 41 15 FOR NEW VENT REQUIREMENTS.



A 064D102 TANK B SECTION
3/16" = 1'-0"

SCALE (A) 4 0 4 8 12 3/16" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D302	TITLE OF SHEET FINISHED WATER TANKS TANK B SECTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 181 OF 192

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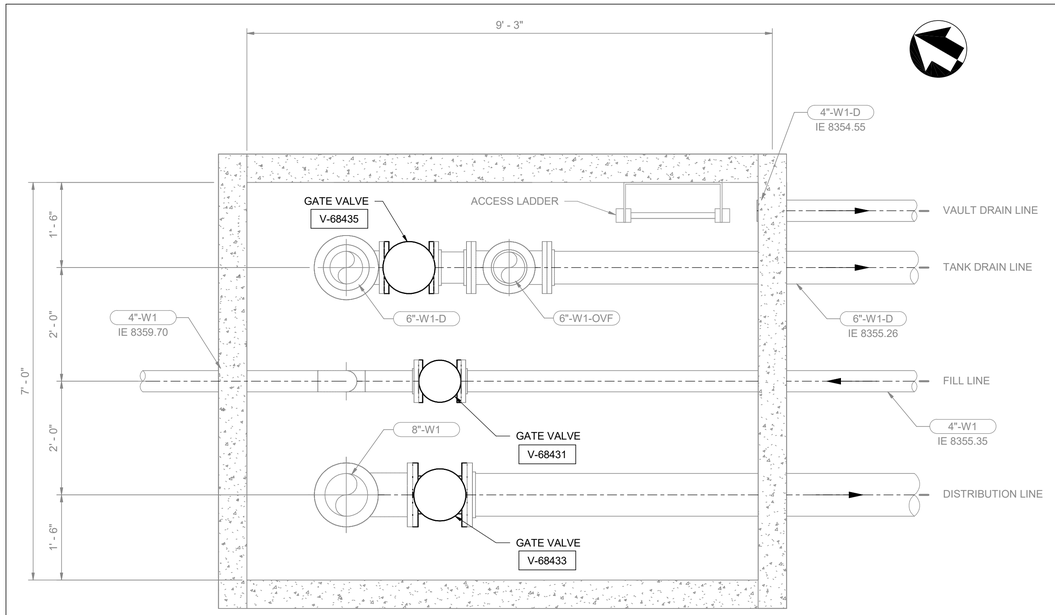
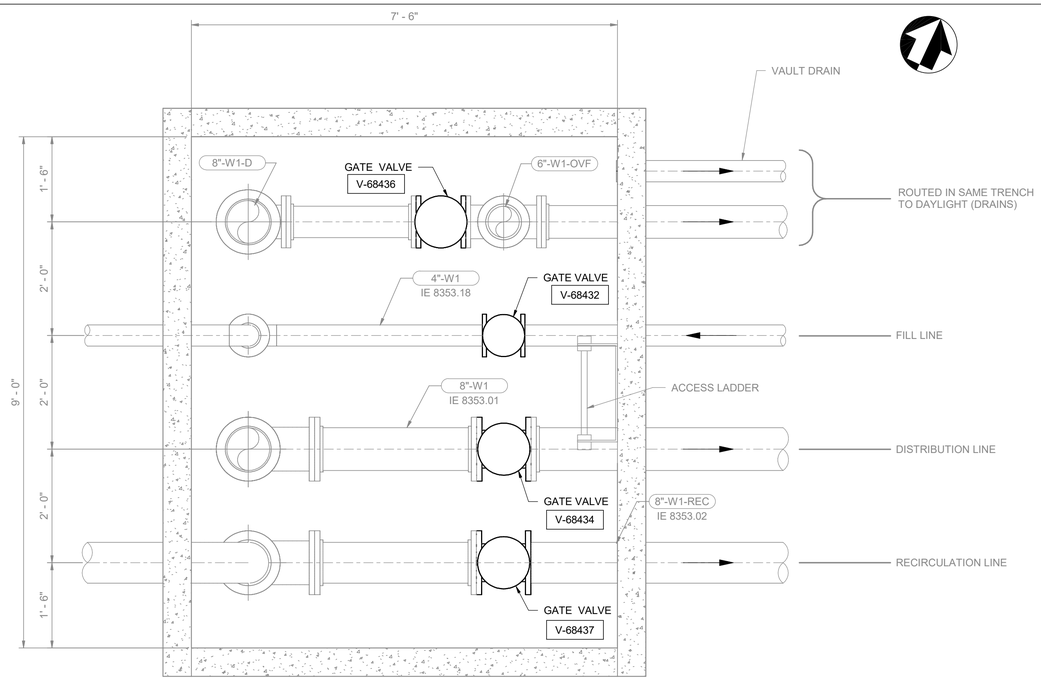


A 064D401 TANK A ENLARGED VAULT PLAN (TK-68411)
3/4" = 1'-0"



B 064D401 TANK B ENLARGED VAULT PLAN (TK-68412)
3/4" = 1'-0"

GENERAL NOTES

A. REPLACE ALL EXISITNG VALVES IN TANK A AND TANK B VALVE VAULTS IN KIND.

B. SEE SHEETS 064D901 - 064D904 FOR CONSTRUCTION SEQUENCING.

3 064D401 PHOTO
NOT TO SCALE

4 064D401 PHOTO
NOT TO SCALE

SCALE A 1 0 1 2 3
3/4" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D401	TITLE OF SHEET FINISHED WATER TANKS ENLARGED VAULT PLANS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 182 OF 192

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_064_FWT_U.rvt
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1 064D501 TANK A
NOT TO SCALE

2 064D501 TANK B
NOT TO SCALE

3 064D501 TANK VENT, TYP
NOT TO SCALE

4 064D501 LEVEL INDICATOR, TYP
NOT TO SCALE

5 064D501 SIDE ACCESS MANWAY, TYP
NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D501	TITLE OF SHEET FINISHED WATER TANKS DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 183 OF 192

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_064_FWT_U.rvt
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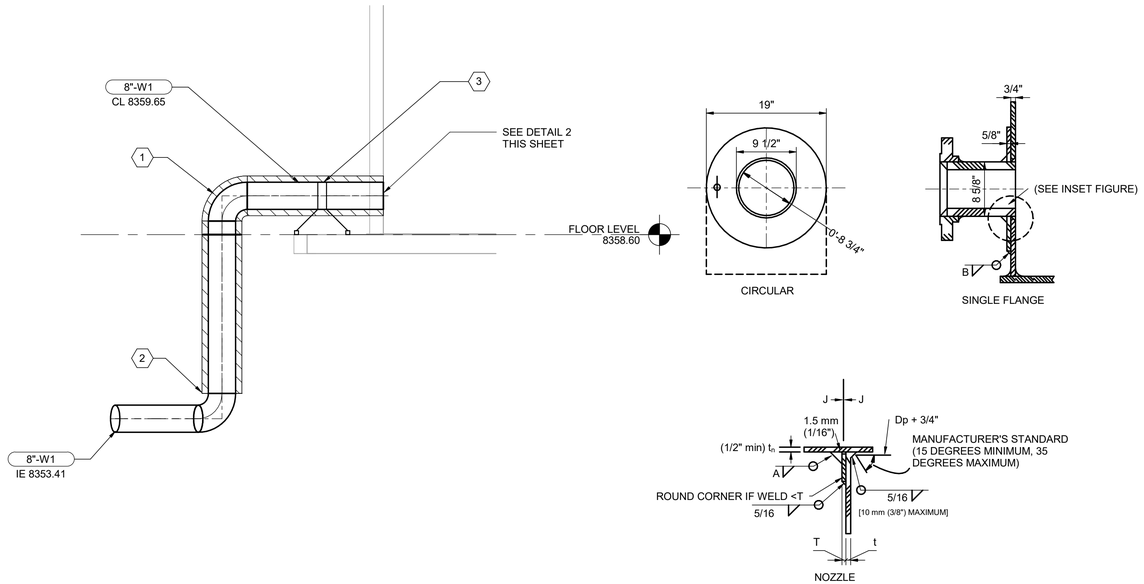

1 064D502 FINISHED WATER TANK REINFORCEMENT PLATE
NOT TO SCALE

2 064D502 FINISHED WATER TANK REINFORCEMENT PLATE
NOT TO SCALE

GENERAL NOTES

A. SEE ELECTRICAL SHEETS FOR HEAT TRACE CONNECTIONS.

KEY NOTES (X)

1. INSULATED PIPE WITH UL LISTED HEAT TRACE WIRE AND ALUMINUM JACKET. INSULATION AND HEAT TRACE IN ACCORDANCE WITH SECTION 40 41 13.
2. PROVIDE END SEAL TO PREVENT MOISTURE INTRUSION FROM PIPE INSULATION.
3. 4-WAY BRACE DESIGNED AND INSTALLED BY TANK SUPPLIER IN ACCORDANCE WITH NFPA 13 AND PROJECT SEISMIC DESIGN CATEGORY. INSTALL WITHIN 24 INCHES OF VERTICAL RISER PIPE.

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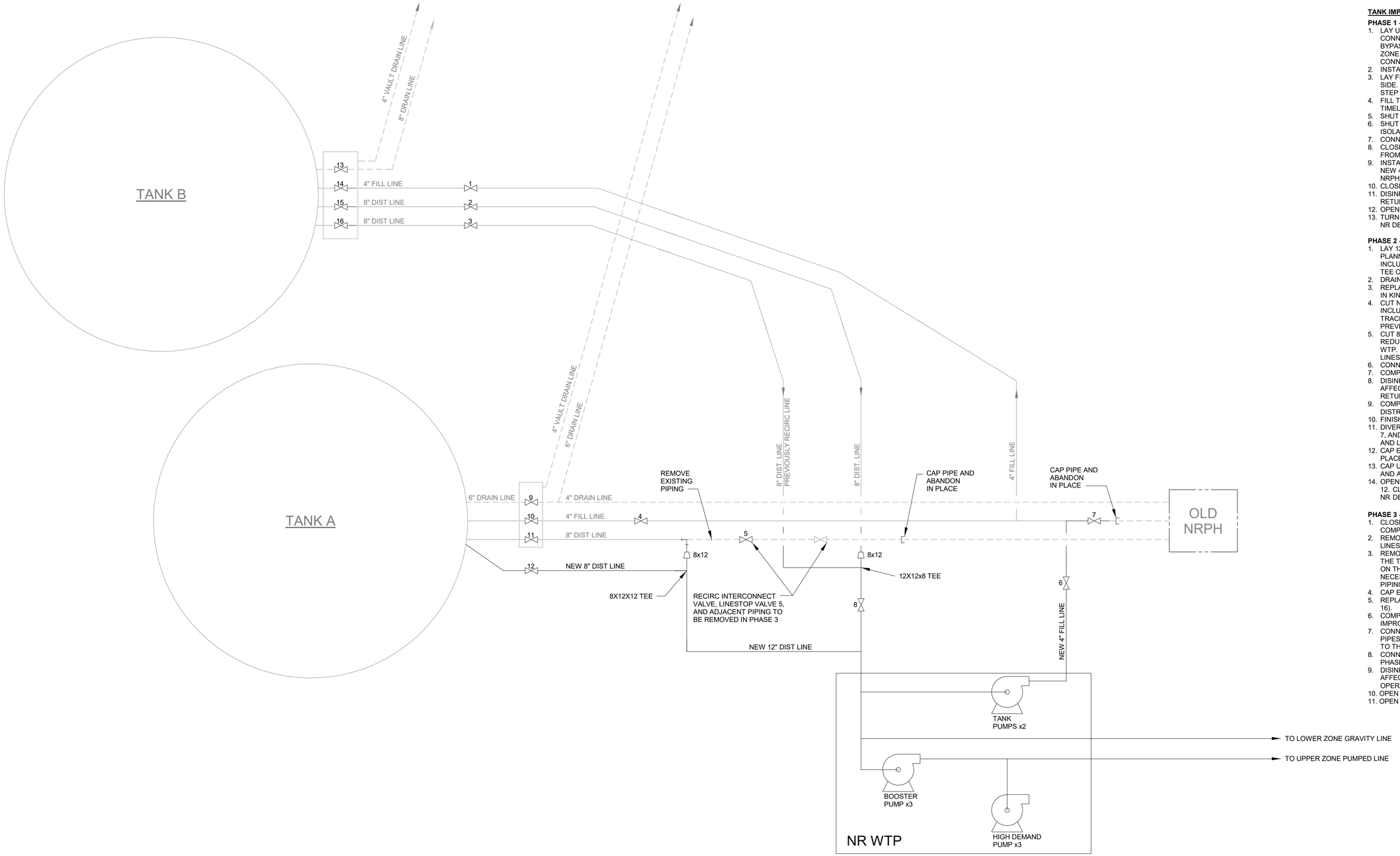
Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D502	TITLE OF SHEET FINISHED WATER TANKS DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 184 OF 192



GENERAL NOTES

A. SEQUENCING PLAN IS NOT TO SCALE. REFER TO CIVIL SHEETS FOR LOCATION OF PIPING AND CONNECTIONS.
B. TANK VAULTS AND VAULT DRAINS ARE SHOWN ON THIS SHEET ONLY FOR CLARITY.

TANK IMPROVEMENTS PHASE PLAN

PHASE 1 - NEW TANK FILL LINE

1. LAY UPPER ZONE BYPASS PIPING AND VALVES AND CONNECT TO UPPER ZONE. REMOVE EXISTING BYPASSED UPPER ZONE PIPING. LAY NEW UPPER ZONE PIPING FROM WTP BUT DO NOT OPEN CONNECTION. (SEE CIVIL).
2. INSTALL LINE STOP VALVES 1-5.
3. LAY FILL PIPING- INCLUDE NEW 4" VALVE (6) ON WTP SIDE. DO NOT CONNECT TO EXIST FILL LINE UNTIL STEP 7.
4. FILL TANKS TO CAPACITY AND ESTABLISH STORAGE TIMELINE.
5. SHUT OFF PUMPING FROM RSPH.
6. SHUT LINESTOP VALVES 1 AND 4. CLOSE FILL LINE ISOLATION VALVE IN OLD NRPH.
7. CONNECT NEW FILL PIPING TO EXISTING FILL PIPING.
8. CLOSE NEW 4" FILL VALVE (6) ON NEW 4" FILL LINE FROM WTP CONNECTION.
9. INSTALL AND OPEN NEW 4" FILL LINE VALVE (7) NEAR NEW 4" FILL LINE FROM WTP CONNECTION ON THE NRPH SIDE OF THE FILL LINE.
10. CLOSE LINESTOP VALVE 5.
11. DISINFECT NEW AND EXISTING FILL LINES BEFORE RETURNING TO OPERATION.
12. OPEN LINESTOP VALVE 1 AT TANK B.
13. TURN RSPH PUMPING BACK ON AND USE TANK B FOR NR DEMAND.

PHASE 2 - TANK A IMPROVEMENTS

1. LAY 12" PIPING FROM WTP TO TANK A AND TO THE PLANNED TANK B CONNECTION FOR PHASE 3 SET-UP INCLUDING A CLOSED NEW 12" VALVE (8). ADD 8"X12" TEE ON THE END OF THE LINE.
2. DRAIN TANK A COMPLETELY.
3. REPLACE VALVES IN TANK A VALVE VAULT (9, 10, & 11) IN KIND.
4. CUT NEW OPENING IN TANK FOR NEW 8" DIST LINE INCLUDING NEW 8" VALVE (12) AND INSTALL HEAT TRACE TO THE EXPOSED SECTION OF LINE TO PREVENT WINTER FREEZING.
5. CUT 8" EXISTING DIST LINE AND INSTALL ELBOW AND REDUCER TO CONNECT TO NEW 12" DIST LINE TO THE WTP. EXISTING PIPING BETWEEN ELBOW AND LINESTOP VALVE 5 TO BE REMOVED.
6. CONNECT NEW 8" DIST LINE TO NEW 12" FW LINE.
7. COMPLETE ALL ADDITIONAL REHAB WORK FOR TANK A.
8. DISINFECT ALL NEW PIPING AND EXISTING PIPING AFFECTED BY TANK A IMPROVEMENTS BEFORE RETURNING TO OPERATION.
9. COMPLETE NEW PIPING CONNECTION TO LOWER DISTRIBUTION SYSTEM.
10. FINISH AND COMMISSION WTP.
11. DIVERT RSPH FLOW TO NEW RW TANKS, CLOSE VALVE 7, AND OPEN VALVE 6. OPEN CONNECTIONS TO UPPER AND LOWER DISTRIBUTION SYSTEMS.
12. CAP EXISTING FILL LINE FROM NRPH AND ABANDON IN PLACE.
13. CAP UNUSED SECTION OF UPPER ZONE BYPASS LINE AND ABANDON IN PLACE.
14. OPEN LINESTOP VALVE 4 AND TANK A VALVES 10, 11, 12. CLOSE LINESTOP VALVE 1. USE TANK A TO SUPPLY NR DEMAND.

PHASE 3 - TANK B IMPROVEMENTS

1. CLOSE LINESTOP VALVES 2 AND 3, AND DRAIN TANK B COMPLETELY.
2. REMOVE RECIRCULATION INTERCONNECT VALVE AND LINESTOP VALVE 5.
3. REMOVE EXISTING DISTRIBUTION PIPING BETWEEN THE TWO VALVES AND AS MUCH DISTRIBUTION PIPING ON THE NRPH SIDE OF THE RECIRC VALVE AS NECESSARY TO MAKE ROOM FOR THE NEW DIST PIPING AND CONNECTIONS.
4. CAP EXISTING 8" DIST LINE GOING TO OLD NRPH.
5. REPLACE VALVES IN TANK B VALVE VAULT (13, 14, 15, & 16).
6. COMPLETE ALL NON-PIPING RELATED TANK B IMPROVEMENTS.
7. CONNECT EXISTING 8" DIST AND 8" RECIRC TO NEW 8" PIPES. INSTALL 8"X12" REDUCER ON LINE CONNECTED TO THE EXISTING TANK B DIST LINE.
8. CONNECT BOTH LINES TO NEW 12" FW LINE FROM PHASE 2 AT 8"X12" TEE.
9. DISINFECT ALL NEW PIPING AND EXISTING PIPING AFFECTED BY PHASE 3 BEFORE RETURNING TO OPERATION.
10. OPEN NEW 12" VALVE (8) FROM PHASE 2.
11. OPEN VALVES 1, 2, 3, 14, 15, AND 16.

Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D901	TITLE OF SHEET FINISHED WATER TANKS SEQUENCING PLAN OVERVIEW NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 185 OF 192

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GENERAL NOTES

A. SEQUENCING PLAN IS NOT TO SCALE. REFER TO CIVIL SHEETS FOR LOCATION OF PIPING AND CONNECTIONS.
B. TANK VAULTS AND VAULT DRAINS NOT SHOWN FOR CLARITY.

TANK IMPROVEMENTS PHASE PLAN

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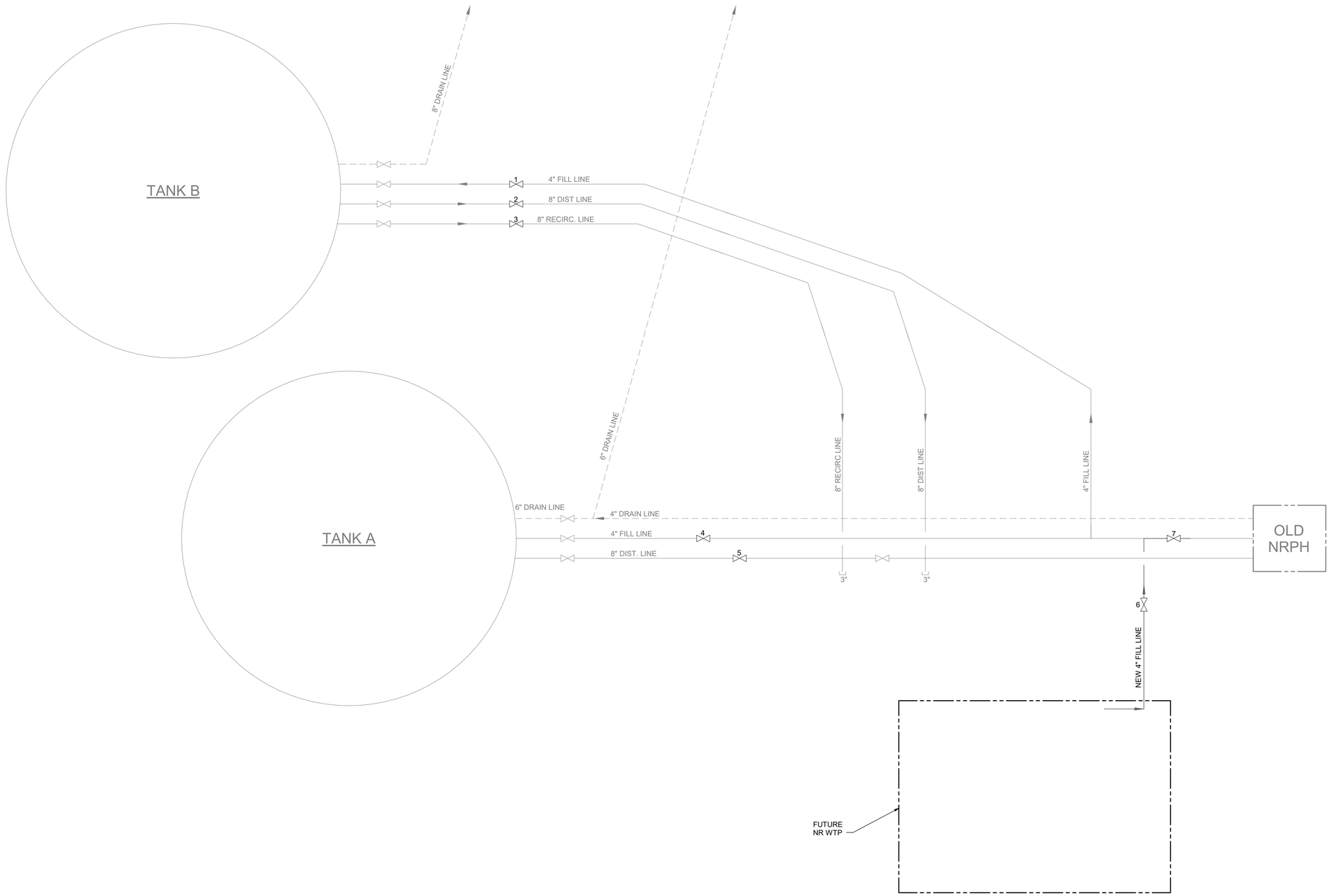
5. SHUT OFF PUMPING FROM RSPH.
6. SHUT LINESTOP VALVES 1 AND 4. CLOSE FILL LINE ISOLATION VALVE IN OLD NRPH.
7. CONNECT NEW FILL PIPING TO EXISTING FILL PIPING.
8. CLOSE NEW 4" FILL VALVE (6) ON NEW 4" FILL LINE FROM WTP CONNECTION.
9. INSTALL AND OPEN NEW 4" FILL LINE VALVE (7) NEAR NEW 4" FILL LINE FROM WTP CONNECTION ON THE NRPH SIDE OF THE FILL LINE.
10. CLOSE LINESTOP VALVE 5.
11. DISINFECT NEW AND EXISTING FILL LINES BEFORE RETURNING TO OPERATION.
12. OPEN LINESTOP VALVE 1 AT TANK B.
13. TURN RSPH PUMPING BACK ON AND USE TANK B FOR NR DEMAND.



Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D902	TITLE OF SHEET FINISHED WATER TANKS SEQUENCING PLAN PHASE 1 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 186 OF 192	

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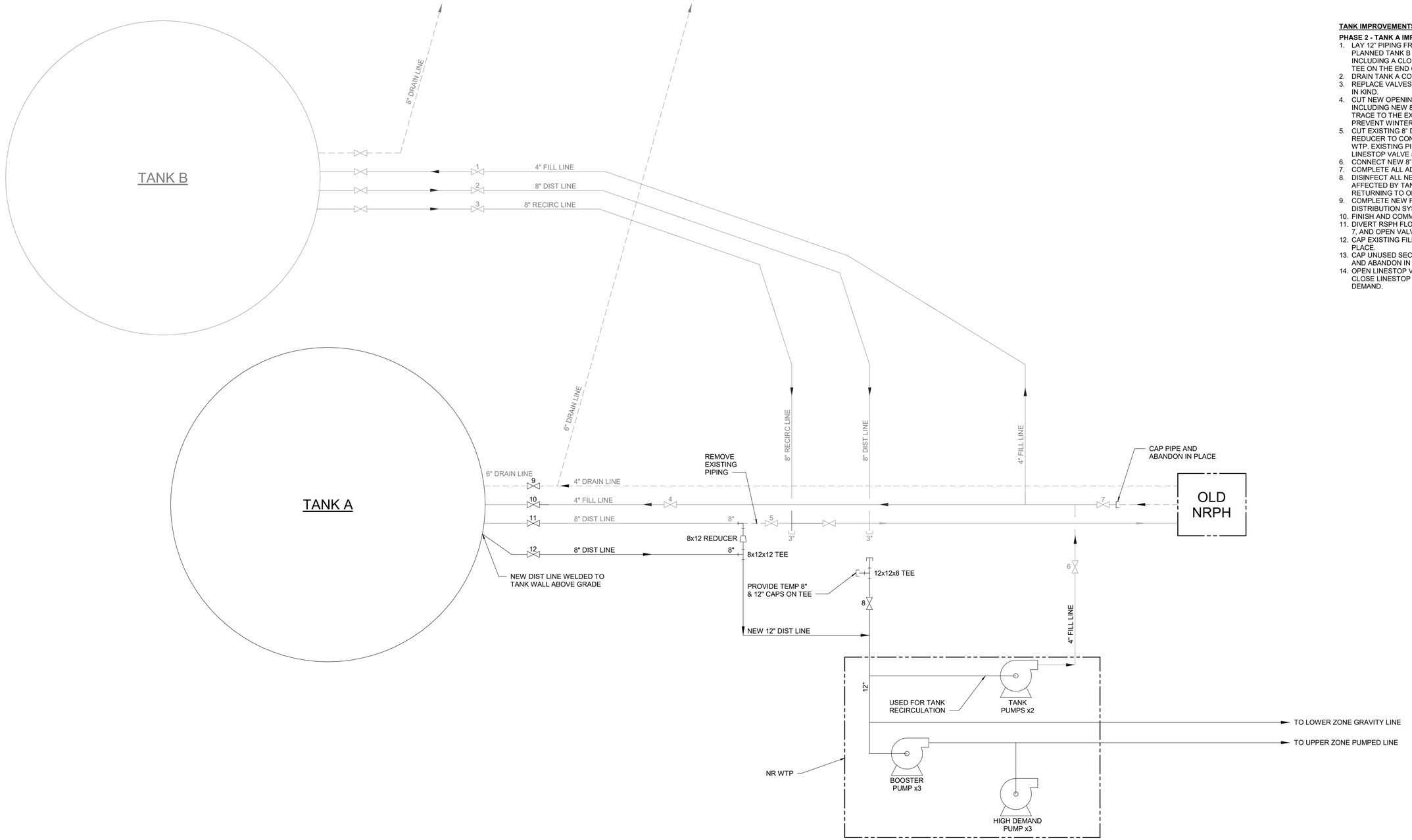
GENERAL NOTES

A. SEQUENCING PLAN IS NOT TO SCALE. REFER TO CIVIL SHEETS FOR LOCATION OF PIPING AND CONNECTIONS.
B. TANK VAULTS AND VAULT DRAINS NOT SHOWN FOR CLARITY.
C. PREVIOUS PHASES SHOWN IN GREY AS EXISTING.

TANK IMPROVEMENTS PHASE PLAN

PHASE 2 - TANK A IMPROVEMENTS

1. LAY 12" PIPING FROM WTP TO TANK A AND TO THE PLANNED TANK B CONNECTION FOR PHASE 3 SET-UP INCLUDING A CLOSED NEW 12" VALVE (8). ADD 8"X12" TEE ON THE END OF THE LINE.
2. DRAIN TANK A COMPLETELY.
3. REPLACE VALVES IN TANK A VALVE VAULT (9, 10, & 11) IN KIND.
4. CUT NEW OPENING IN TANK FOR NEW 8" DIST LINE INCLUDING NEW 8" VALVE (12) AND INSTALL HEAT TRACE TO THE EXPOSED SECTION OF LINE TO PREVENT WINTER FREEZING.
5. CUT EXISTING 8" DIST LINE AND INSTALL ELBOW AND REDUCER TO CONNECT TO NEW 12" DIST LINE TO THE WTP. EXISTING PIPING BETWEEN ELBOW AND LINESTOP VALVE 5 TO BE REMOVED.
6. CONNECT NEW 8" DIST LINE TO NEW 12" FW LINE.
7. COMPLETE ALL ADDITIONAL REHAB WORK FOR TANK A.
8. DISINFECT ALL NEW PIPING AND EXISTING PIPING AFFECTED BY TANK A IMPROVEMENTS BEFORE RETURNING TO OPERATION.
9. COMPLETE NEW PIPING CONNECTION TO LOWER DISTRIBUTION SYSTEM.
10. FINISH AND COMMISSION WTP.
11. DIVERT RSPH FLOW TO NEW RW TANKS, CLOSE VALVE 7, AND OPEN VALVE 6.
12. CAP EXISTING FILL LINE FROM NRPH AND ABANDON IN PLACE.
13. CAP UNUSED SECTION OF UPPER ZONE BYPASS LINE AND ABANDON IN PLACE.
14. OPEN LINESTOP VALVE 4 AND TANK A VALVES 10, 11 12. CLOSE LINESTOP VALVE 1. USE TANK A TO SUPPLY NR DEMAND.



Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD DRAWN: K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D903	TITLE OF SHEET FINISHED WATER TANKS SEQUENCING PLAN PHASE 2 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 187 OF 192	

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GENERAL NOTES

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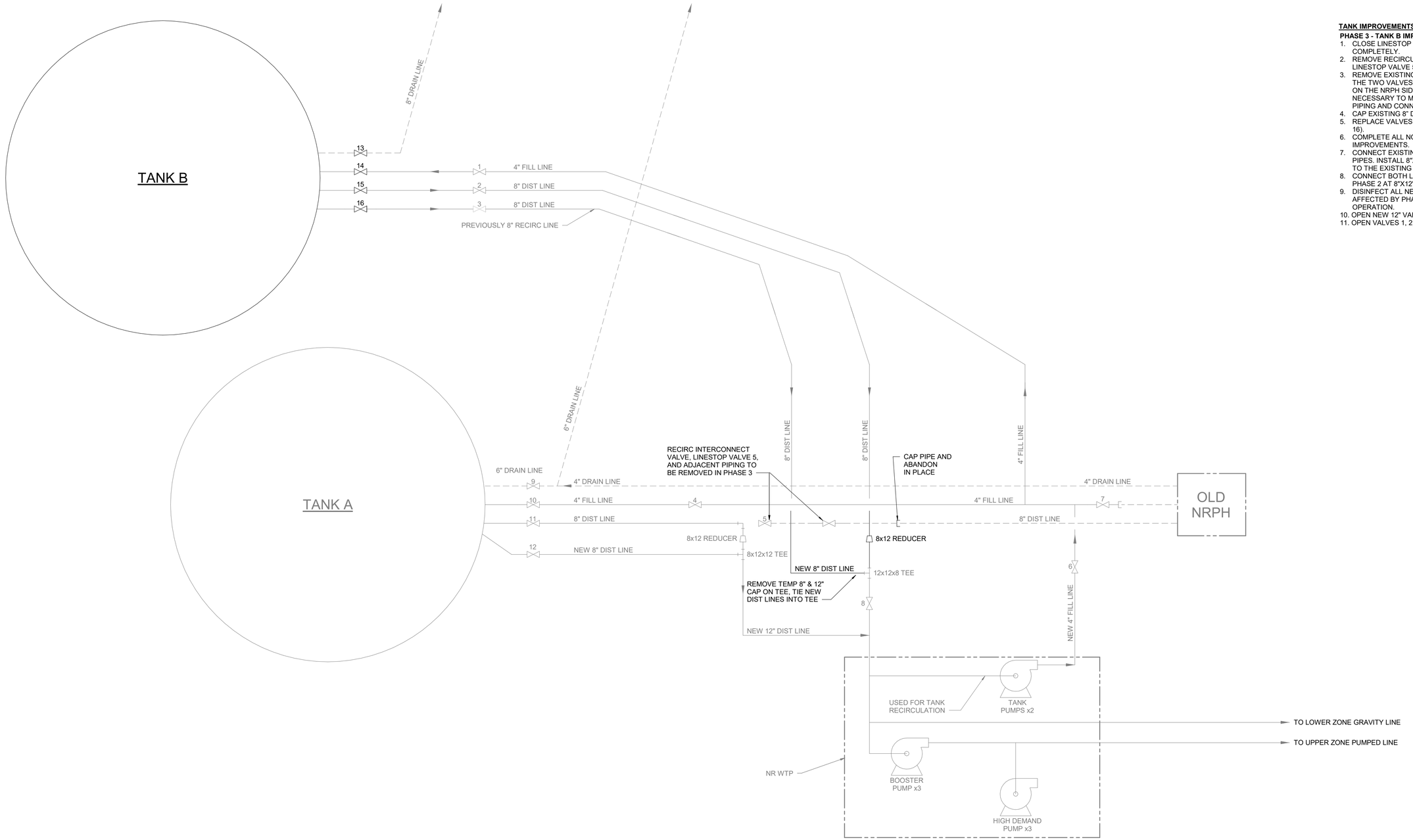
B. TANK VAULTS AND VAULT DRAINS NOT SHOWN FOR CLARITY.
C. PREVIOUS PHASES SHOWN IN GREY AS EXISTING.

TANK IMPROVEMENTS PHASE PLAN

PHASE 3 - TANK B IMPROVEMENTS

1. CLOSE LINESTOP VALVES 2 AND 3, AND DRAIN TANK B COMPLETELY.
2. REMOVE RECIRCULATION INTERCONNECT VALVE AND LINESTOP VALVE 5.
3. REMOVE EXISTING DISTRIBUTION PIPING BETWEEN THE TWO VALVES AND AS MUCH DISTRIBUTION PIPING ON THE NRPH SIDE OF THE RECIRC VALVE AS NECESSARY TO MAKE ROOM FOR THE NEW DIST PIPING AND CONNECTIONS.
4. CAP EXISTING 8" DIST LINE GOING TO OLD NRPH.
5. REPLACE VALVES IN TANK B VALVE VAULT (13, 14, 15, & 16).
6. COMPLETE ALL NON-PIPING RELATED TANK B IMPROVEMENTS.
7. CONNECT EXISTING 8" DIST AND 8" RECIRC TO NEW 8" PIPES. INSTALL 8"X12" REDUCER ON LINE CONNECTED TO THE EXISTING TANK B DIST LINE.
8. CONNECT BOTH LINES TO NEW 12" FW LINE FROM PHASE 2 AT 8"X12" TEE.
9. DISINFECT ALL NEW PIPING AND EXISTING PIPING AFFECTED BY PHASE 3 BEFORE RETURNING TO OPERATION.
10. OPEN NEW 12" VALVE (8) FROM PHASE 2.
11. OPEN VALVES 1, 2, 3, 14, 15, AND 16.



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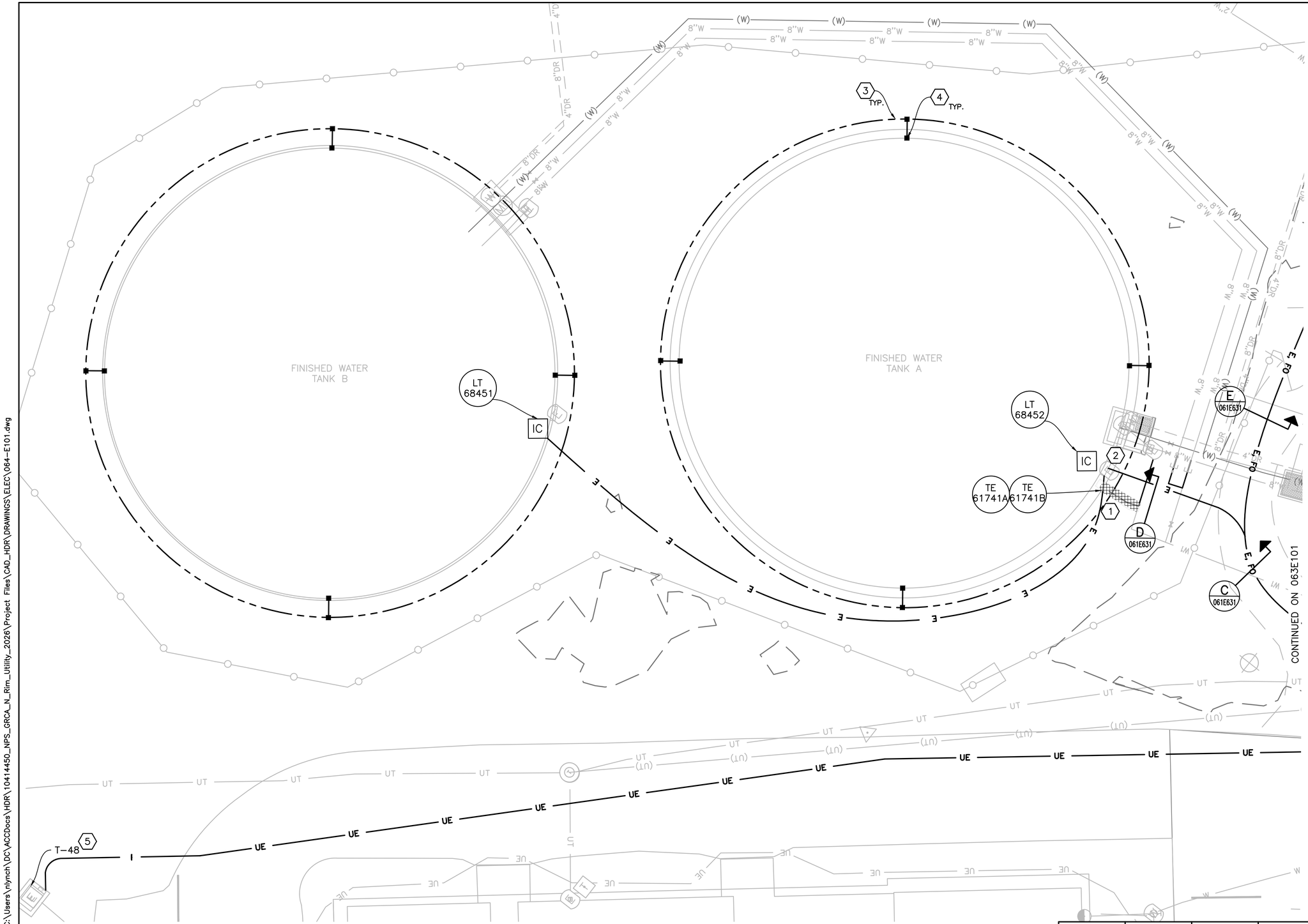
Registered Professional Engineer (Civil) CERTIFICATE NO. 75827 JENNIFER STILLMAN Date Signed 5/1/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HOWARD CADD K. CARSON TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 064 D904	TITLE OF SHEET FINISHED WATER TANKS SEQUENCING PLAN PHASE 3 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 188 OF 192



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. ELECTRICAL DEMO IS SHOWN ON SHEET 061X101.

KEY NOTES: (X)

1. PROVIDE HEAT TRACE CABLE, LOW TEMPERATURE SENSOR, AND HIGH TEMPERATURE SENSOR AT PIPING IN THE HATCHED AREA. APPLY HEAT TRACE TO THE EXPOSED PORTION OF THE NEW ABOVE-GROUND 8IN DISTRIBUTION LINE, AND 5FT BELOW GROUND. (SEE DETAIL 1/064D502.) THE HEAT TRACE CONTROLLER IS TO BE INSTALLED AT THE LOCATION SHOWN ON 062 E112. THE DESIGN OF THE HEAT TRACE AND CONTROLLER IS DELEGATED TO THE CONTRACTOR. CONTRACTOR SHALL PROVIDE A CODE-COMPLIANT SYSTEM MEETING THE REQUIREMENTS OF HEAT TRACE AND A CONTROLLER THAT IS BEING USED FOR A FIRE PROTECTION SYSTEM PER SECTION 40 41 13 AS WELL AS NATIONAL CODES.

2. PROVIDE NEMA 4X JUNCTION BOX ON UNISTRUT SUPPORT STAND AT THIS LOCATION FOR HEAT TRACE CABLE AND INSTRUMENTATION CONNECTIONS. ROUTE CABLE FROM TANK B TO JUNCTION BOX IN DIRECT-BURIED, 2 INCH PVC-80 CONDUIT.

3. #4/0AWG SOFT-DRAWN BARE COPPER GROUND RING.

4. #4/0 BOND TO TANKS.

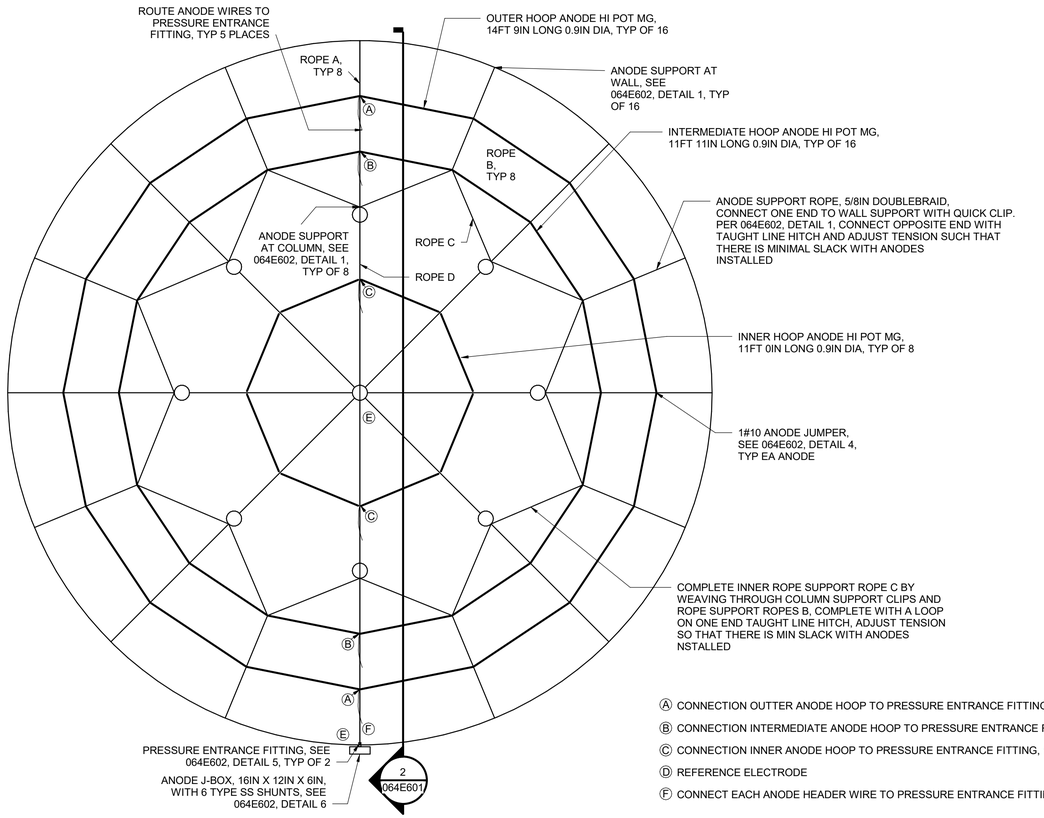
5. AT EXIST TRANSFORMER T-48, TERMINATE 25KV (3/C) 4/0 ALUMINUM ARMORED POWER CABLE FROM ESB-18A WITH 200AMP LOAD BREAK ELBOWS.

10 0 10 20 1" = 10'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO. 58037
TRAVIS T. MOORE
Date Signed 1/4/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. S-DELGADO CADD A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 064 E101	TITLE OF SHEET FINISHED WATER TANKS ELECTRICAL OVERALL SITE PLAN NORTH RIM – GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 189 OF 192

8/13/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10414450_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\ELEC\064-E101.dwg



Ⓐ CONNECTION OUTTER ANODE HOOP TO PRESSURE ENTRANCE FITTING,1#10AWG RHH RED

Ⓑ CONNECTION INTERMEDIATE ANODE HOOP TO PRESSURE ENTRANCE FITTING, 1#10AWG RHH BLACK

Ⓒ CONNECTION INNER ANODE HOOP TO PRESSURE ENTRANCE FITTING, 1#10AWG RHH BLUE

Ⓓ REFERENCE ELECTRODE

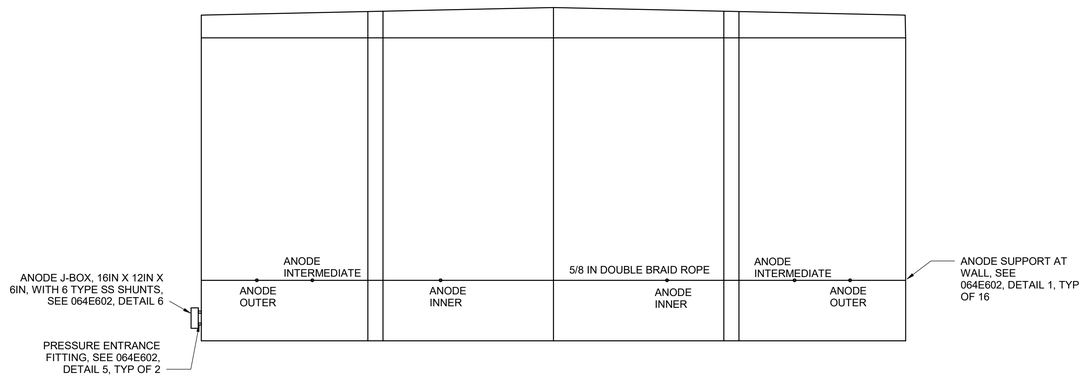
Ⓕ CONNECT EACH ANODE HEADER WIRE TO PRESSURE ENTRANCE FITTING WIRE, SEE 064E602, DETAIL 5

FINISHED WATER TANK MATERIALS SCHEDULE

#	ITEM NAME	LOCATION	DETAIL	# REQUIRED	SIZE	DESCRIPTION
1	ANODE MOUNTING CLIP					3/8IN DIA MILD STEEL ROD
	A	OUTTER WALL	1	16	3 5/8 IN	
	B	COLUMNS OUTER ANODES	1	8	3 5/8 IN	
	C	COLUMN INNER ANODES	1	8	3 5/8 IN	
	D	CENTER COLUMN INNER ANODES	1	8	3 5/8 IN	
2	QUICK LINK					5/16IN STEEL 2,000LB WLL
	A	OUTTER WALL	1	16		
	B	COLUMNS OUTER ANODES	1	8		
	C	COLUMN INNER ANODES	1	8		
	D	CENTER COLUMN INNER ANODES	1	8		
3	ROPE					5/8IN DOUBLE BRADED POLYESTER RIGGING WHITE. CONFIRIM ACTUAL TANK DIMENSIONS BEFORE CONSTRUCTING ROPE SUPPORTS
	A	OUTTER ANODE SUPPORT STRINGER TO INNER COLUMNS		8	22 FT	LENGTH BETWEEN SUPPORTS, ADD MIN 2 FT EACH END FOR KNOTS.
	B	OUTTER ANODE SUPPORT TO HOOP		8	14 FT 8 IN	
	C	OUTER ANODE HOOP		1	120 FT 8 IN	ADD 15FT EA END FOR KNOTS AND EASE OF RIGGING
	D	INNER SUPPORT COLUMN TO CENTER COLUMN		8	21 FT 6 IN	LENGTH BETWEEN SUPPORTS, ADD MIN 2 FT EACH END FOR KNOTS
4	ANODE					GA-MG-R900 HI POTENTIAL MAGNESIUM EXTRUDED 0.9IN DIAMETER 0.516LB/FT WITH 0.135IN STEEL CORE, LENGTH OF ANODE MATERIAL PROVIDE 3 5/8IN EXPOSED STEEL CORE EACH END. TOTAL WEIGHT 271.07 LBS
	A	OUTER		16	15 FT 1 IN 7.8 LB ANODE 124.5 LB HOOP	
	B	INRERMEDIATE		16	12 FT 3 IN 6.3 LB ANODE 101.1 LB HOOP	
	C	INNER		8	11 FT 5.7 LB ANODE 45 LB HOOP	
5	WIRE					
	A	ANODE TO ANODE JUMPERS				
		OUTTER ANODE		16	6 IN	#10AWG RHH, STRANDED COPPER WITH BLACK INSULATION
		INTERMEDIATE ANODE		16	6 IN	#10AWG RHH, STRANDED COPPER WITH BLACK INSULATION
		INNER ANODE		8	6 IN	#10AWG RHH, STRANDED COPPER WITH BLACK INSULATION
		SPLIT BOLT		80		HIGH STRENGTH BRONZE ALLOY
	B	ANODE TO J-BOX				
		OUTTER ANODE HOOP		1	20 FT	#10AWG RHH, STRANDED COPPER WITH RED INSULATION
				1	110 FT	
	C	INTERMEDIATE ANODE HOOP		1	30 FT	#10AWG RHH, STRANDED COPPER WITH BLACK INSULATION
				1	115 FT	
	D	INNER ANODE HOOP		1	50 FT	#10AWG RHH, STRANDED COPPER WITH BLUE INSULATION
				1	85 FT	
6	REFERENCE ELECTRODE		3	2		
7	SPLICE KIT			8		THERMO-SHRINK UNDERGROUND COPPER BUTT SPLICE KIT, 2 TO 8 AWG
8	PRESSURE ENTRANCE FITTING			2		
9	J-BOX	PER DETAIL	6	1		

1 064E601 FINISHED WATER TANK CATHODIC PROTECTION PLAN

NO SCALE



2 064E601 FINISHED WATER TANK CATHODIC PROTECTION SECTION

NO SCALE

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REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO
58037
TRAVIS T. MOORE
Date Signed 4/28/2026
ARIZONA, U.S.A.

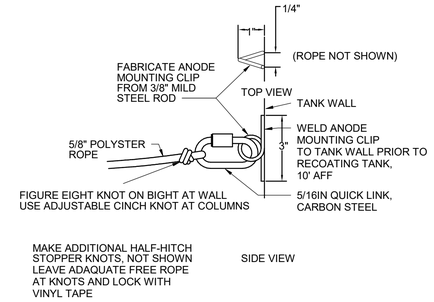
A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

DESIGNED:
A. SMITH
DRAWN:
A. S-DELGADO
TECH. REVIEW:
P. DAVILA
DATE:
MAY 2026

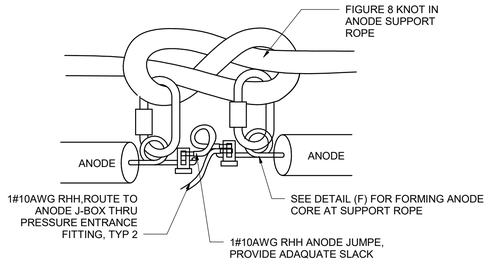
SUB SHEET NO.
064
E601

TITLE OF SHEET
FINISHED WATER TANKS
CATHODIC PROTECTION PLAN AND SECTION
NORTH RIM - GRAND CANYON NATIONAL PARK

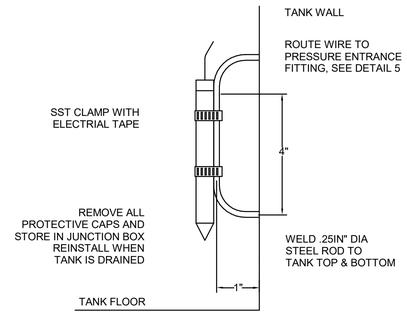
DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 190 OF 192



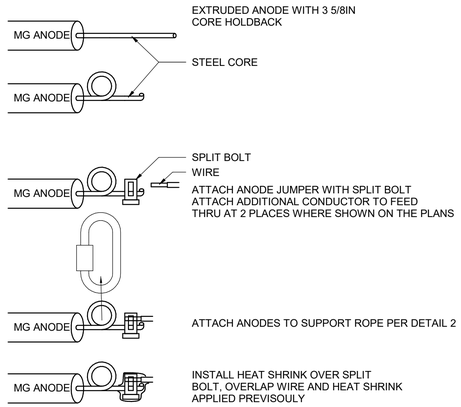
1 064E602 ANODE ROPE MOUNTING CLIP
NO SCALE



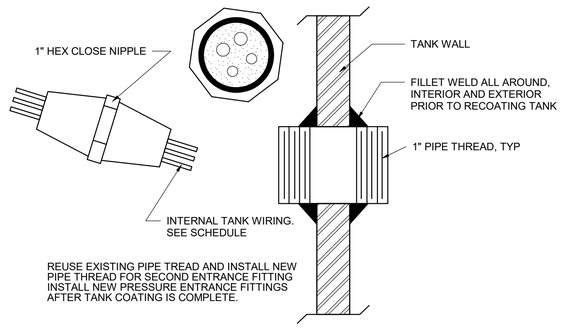
2 064E602 ANODE SUPPORT AT ROPE
NO SCALE



3 064E602 WALL MOUNT REFERENCE ELECTRODE
NO SCALE



4 064E602 ANODE CORE LOOP
NO SCALE



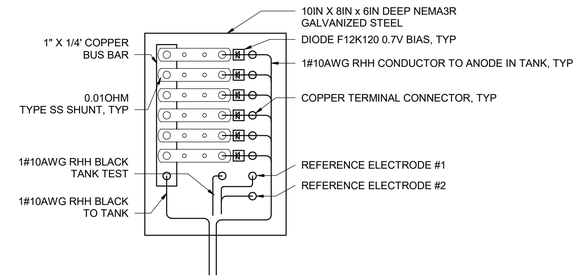
NOTES:
INSTALL PIPE PLUGS BEFORE COATING, REMOVE AND INSTALL ENTRANCE FITTING/WIRES BEFORE FILLING TANK.

INSTALL & SECURELY TIGHTEN 1" PRESSURE FITTING INTO 1" STEEL COUPLING USING THREAD SEALANT TO RENDER A WATERTIGHT CONNECTION.

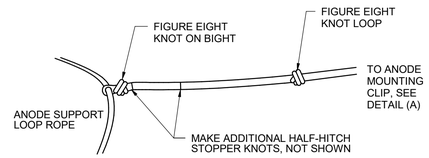
CONNECT TO INTERNAL WIRING WITH A SPLIT BOLT AND SPLICE KIT
MAKE CONNECTIONS TO EXTERNAL WIRING WITH WIRE NUTS IN ENCLOSURE

PRESSURE ENTRANCE FITTING SCHEDULE		
FITTING #1		
CONDUCTOR	CONNECTION	WIRE CLR
1	OUTER ANODE #1	RED
2	OUTER ANODE #2	RED
3	INTERMEDIATE ANODE #1	YELLOW
4	INTERMEDIATE ANODE #2	ORANGE
FITTING #2		
CONDUCTOR	CONNECTION	WIRE CLR
1	INNER ANODE #1	RED
2	INNER ANODE #2	RED
3	REFERENCE ELECTRODE WALL	YELLOW
4	REFERENCE ELECTRODE CENTER	ORANGE

5 064E602 PRESSURE ENTRANCE FITTING
NO SCALE



6 064E602 ANODE JUNCTION BOX
NO SCALE

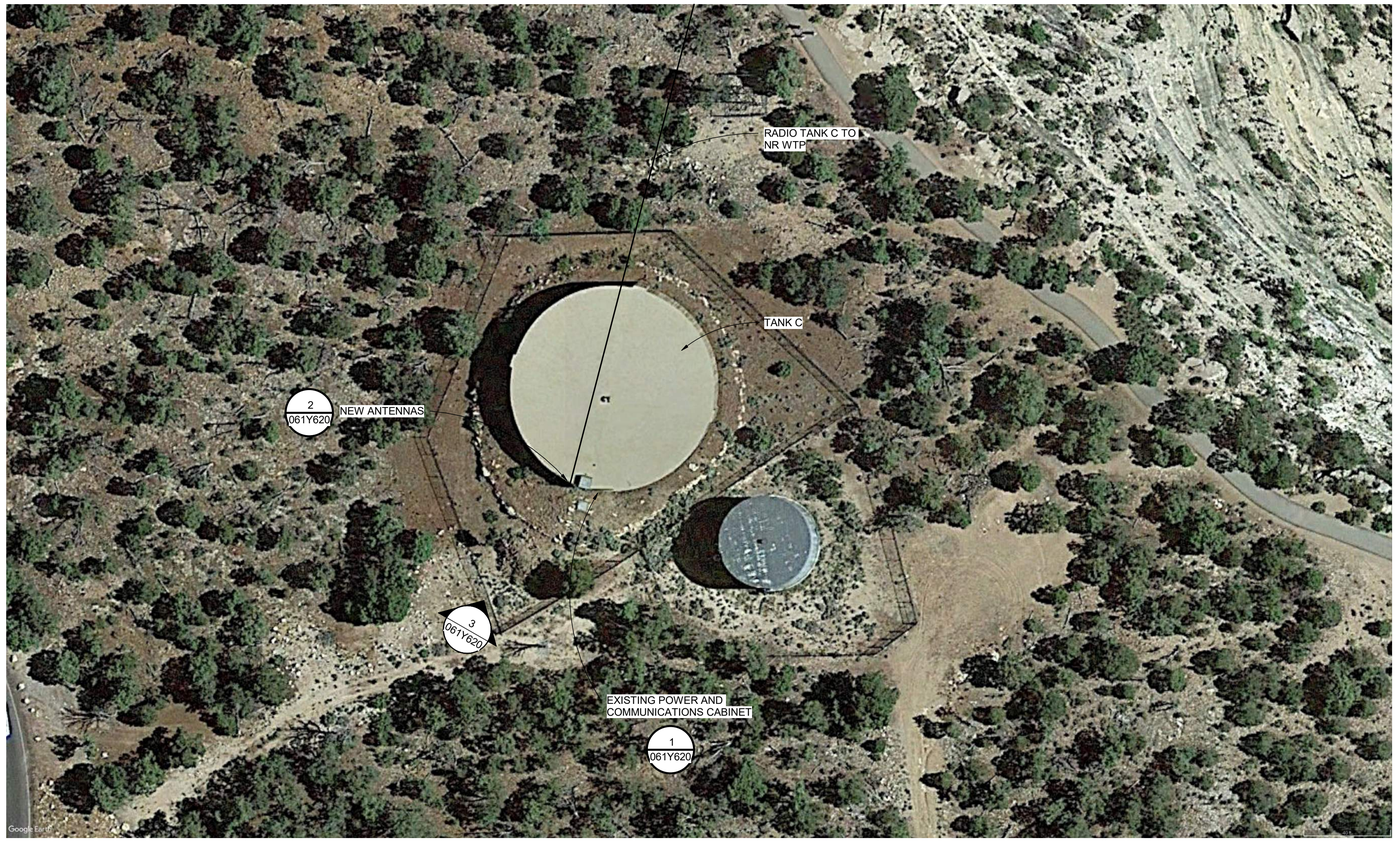


7 064E602 ANODE ROPE SUPPORT AT HOOP
NO SCALE

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_064_FWT_U.rvt
4/28/2026 2:08:27 PM

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO.
58037
TRAVIS T. MOORE
Date Signed 4/28/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. SMITH DRAWN: A. S-DELGADO TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 064 E602	TITLE OF SHEET FINISHED WATER TANKS CATHODIC PROTECTION DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 191 OF 192

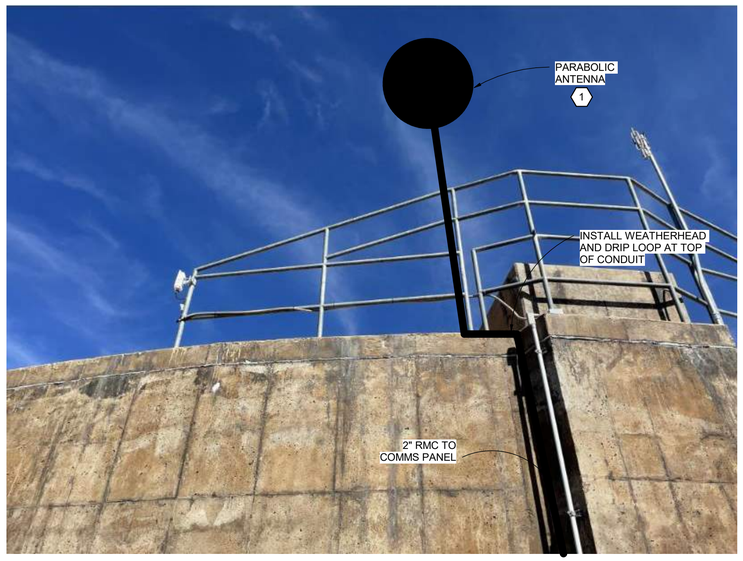


N

KEYNOTES

1

1 061Y620 **POWER AND COMMS CABINET**
NOT TO SCALE



2 061Y620 **ANTENNA AND CONDUIT LOCATIONS**
NOT TO SCALE

3 061Y620 **TANK C ELEVATION VIEW**
NOT TO SCALE

Registered Professional Engineer (Certified)
CERTIFICATE NO.
85305
JOSEPHINE A BRAINARD
Date Signed 5/1/26
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD CADD R.D. CLARK TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 065 Y501	TITLE OF SHEET RAW WATER TANKS SOUTH RIM TANK C COMMUNICATIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 10350545 SHEET 192 OF 192

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